

Gasoline Profiles and Impact of Ethanol Blending in Europe

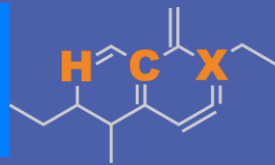
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**U.S. GRAINS &
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Ethanol Blending in Europe

There are important fuel quality and environmental impact of vehicle emission challenges in the Region.

- The use of ethanol improves gasoline quality and creates flexibility in gasoline production.
- Ethanol use is a cost-effective way to increase gasoline octane and to replace more expensive gasoline components.
- Ethanol contributes to transport decarbonization and air quality improvement.
- There are opportunities across Europe to increase the ethanol blend level and implement new policies on the use of gasoline-ethanol blends.



Twenty eight countries with potential and additional use of ethanol were studied: 1) Gasoline market profiles; 2) Optimization of gasoline blends with ethanol and, 3) Environmental impact of gasolines blended with ethanol.

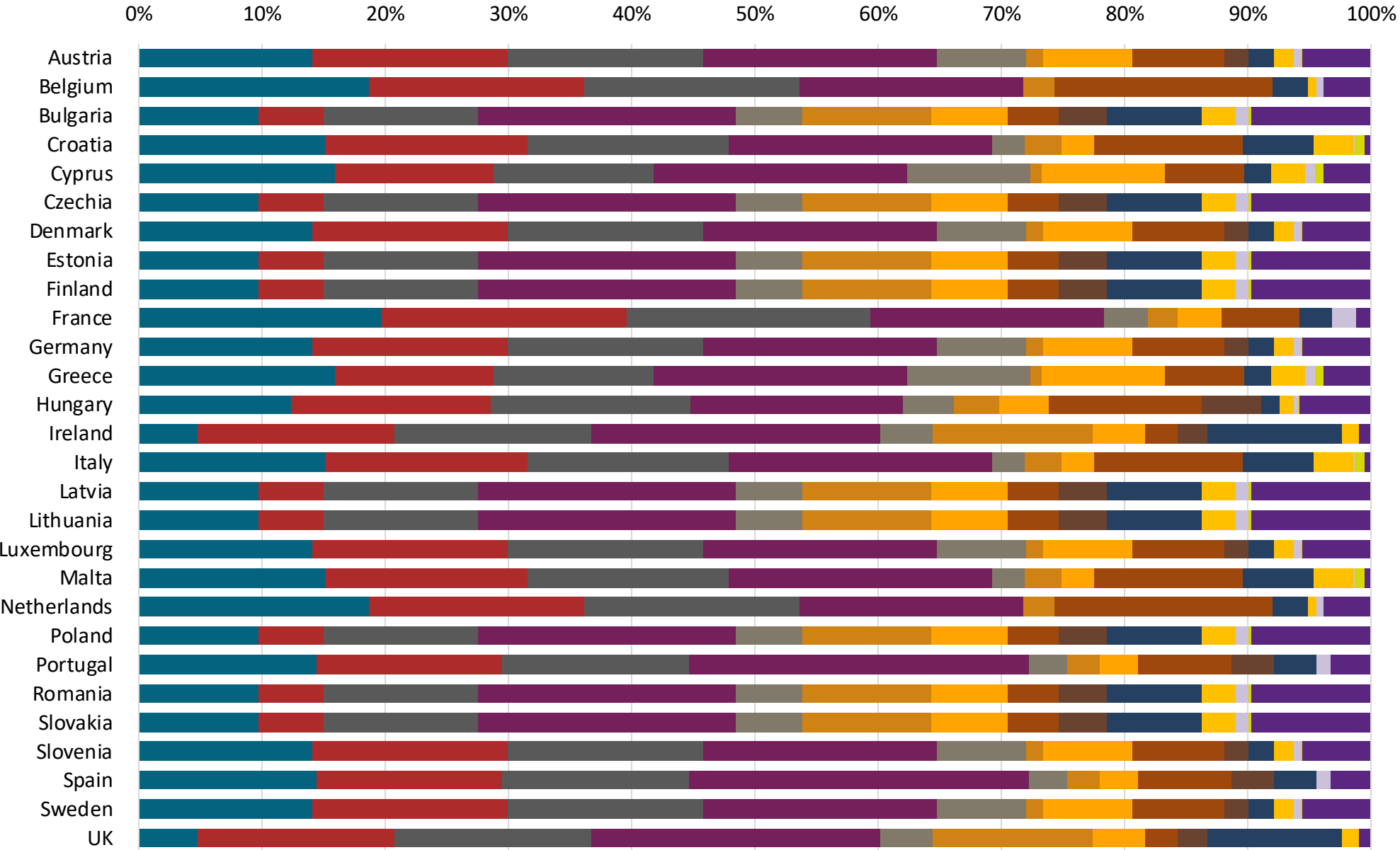
Gasoline Component Blending in Europe



Gasoline is a blend of a base gasoline and other components. Each component provides different properties to the final blend, for example, isomerates, alkylates and butanes increase the octane. The components commonly used in Europe are:

- Catalytic Gasoline
- Light Primary Naphta
- Heavy Primary Naphta
- Reformate
- Isopentane
- Alkylate
- Isomerate
- Hydrocracked Gasoline
- Coker Naphta
- n-Butane
- MTBE
- ETBE
- TAME
- Aromatics

Source: Faro90



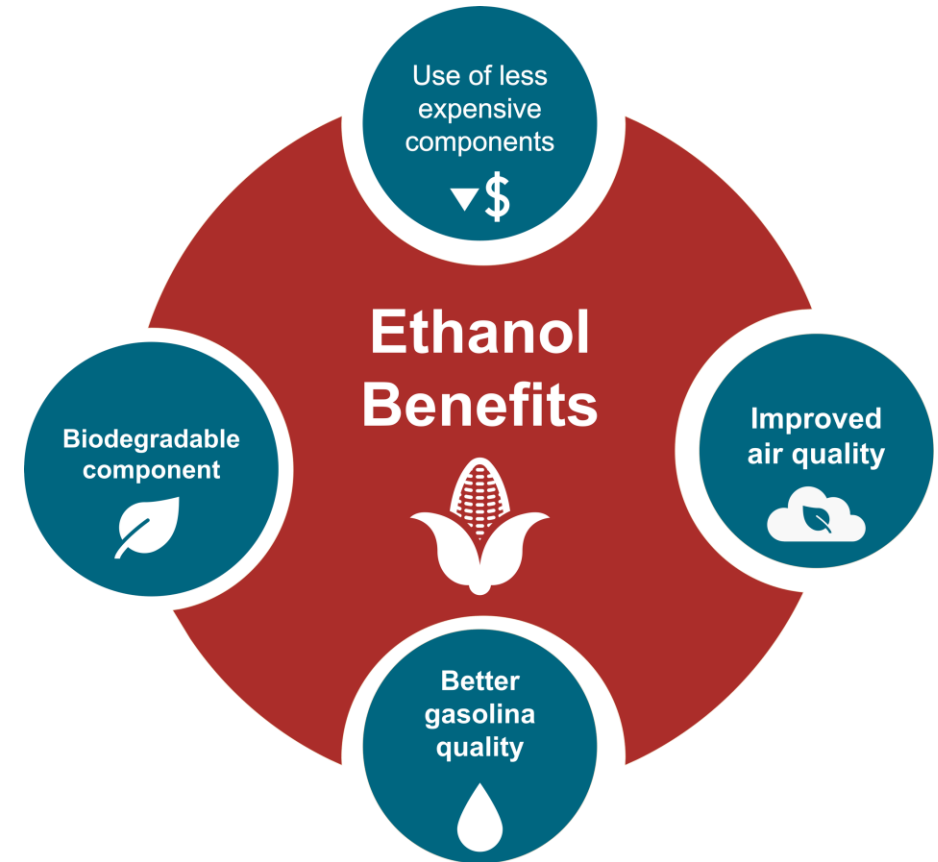
Gasoline Blending Optimization

In some parts of the world, ethanol is added to gasoline as a blending component. The advantages of ethanol include that it is a renewable fuel made of biomass; that it is an octane booster that helps to dilute sulfur; and that it allows the fulfillment of environmental objectives. To determine the optimal components to be blended with ethanol, a **blending model** was used. This model selects the components to add in the gasoline/ethanol blend based on:

- Components prices,
- Properties each component affects,
- Quality parameters by country, and
- Component availability by country.

Through iterations, the model obtains the %v/v of the components to be blended with 10%, 15%, 20%, 25% and 30% of ethanol, in such a way that the final blend complies with the required properties of a finished gasoline by country.

The blending model uses gasoline component spot average prices 2024 and provides fuel prices that do not include country distribution costs, local taxes and subsidies and import or gas station margins.



Vehicle Emission Impact for Ethanol Gasoline Blending

The model used in this analysis takes as a reference the [International Vehicle Emissions Model \(IVE\)](#).

The model uses the Base Emission Rates from IVE model, as well as its Adjustment Factors based on:

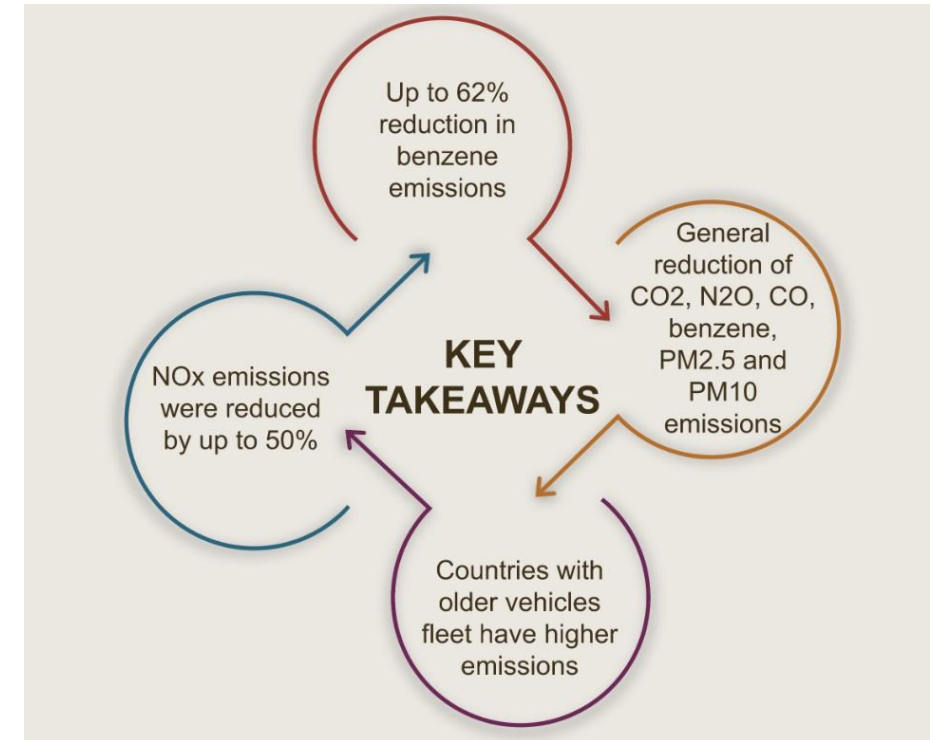
- Vehicle technology (cars, trucks, buses, motorcycles),
- Vehicle fleet average age,
- Average traveled distance per vehicle by country, as well as
- Geographical and climatic conditions (altitude, humidity, temperature).

Emissions of criteria pollutants, toxic pollutants, and greenhouse gases (GHG) were calculated and calibrated with emission inventories, using real gasoline quality data. The reduction rates for gasoline/ethanol blends were obtained from various sources (IPCC, US Grains, among others).

Emission estimations for different pollutants for gasoline and gasoline/ethanol blends (10%, 15%, 20%, 25% and 30% ethanol) were determined using the IVE Model. A comparison between the results and the European (Euro 6) requirements is made. Results are also compared with real emissions of the United States vehicle fleet*.

*Source: Bureau of transportation statistics.

Main Results



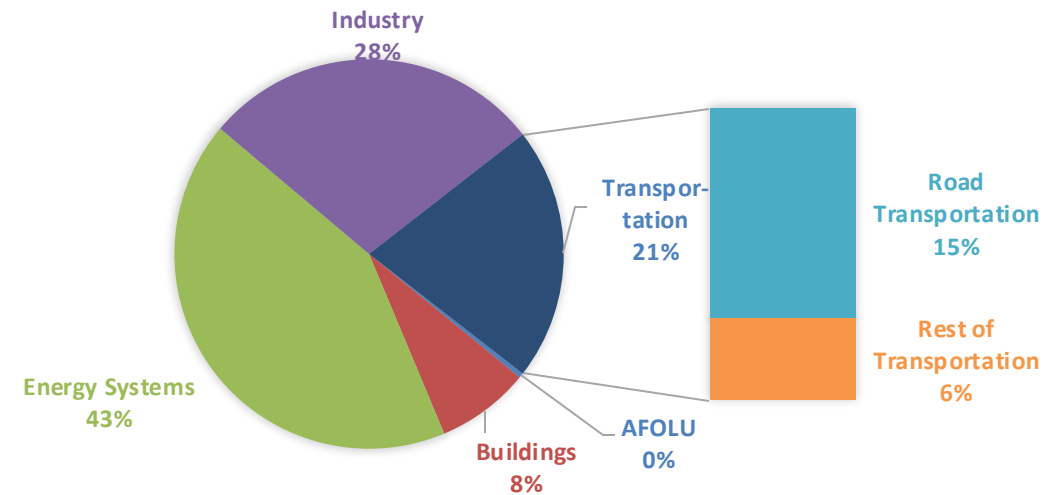
Life Cycle GHG Emissions

The Global Agenda for Climate Action calls for countries, cities and regions, businesses, and members of civil society around the world to achieve a low-carbon and climate-resilient economies in support of the Paris Agreement. Road Transportation accounts for 15% of total GHG emissions (IPCC,2023).

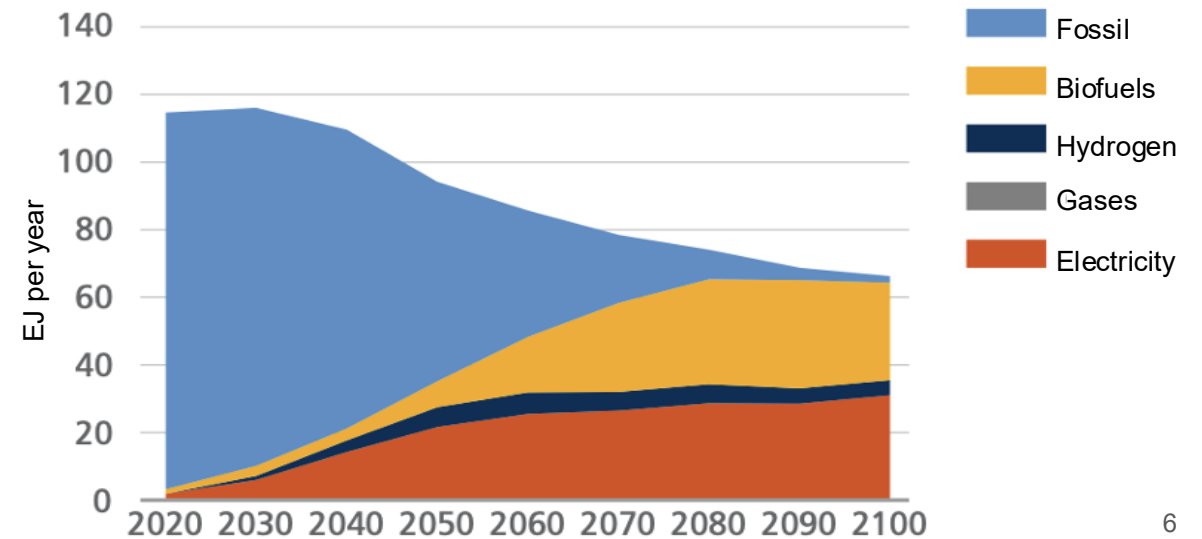
Ethanol gasoline blends are an internationally adopted practice that offer significant short-term benefits. Biofuels are an important solution to achieve reduction targets in the transportation sector.

Country GHG emission reductions are calculated using **RED II/III** and **GREET** life cycle emission methodologies and the **International Vehicular Emissions Model (IVE)**. Results are compared to the current country targets for 2035. Current 2024 country emission are those published by **IPCC**.

Current GHG Global Emissions Distribution



IPCC Transport Sector Sustainable Development Goals and climate policies Scenario, 2023



- | | | |
|-------------|-----------------|--------------------|
| 1. Austria | 11. Germany | 21. Poland |
| 2. Belgium | 12. Greece | 22. Portugal |
| 3. Bulgaria | 13. Hungary | 23. Romania |
| 4. Croatia | 14. Ireland | 24. Slovakia |
| 5. Cyprus | 15. Italy | 25. Slovenia |
| 6. Czechia | 16. Latvia | 26. Spain |
| 7. Denmark | 17. Lithuania | 27. Sweden |
| 8. Estonia | 18. Luxembourg | 28. United Kingdom |
| 9. Finland | 19. Malta | |
| 10. France | 20. Netherlands | |

Ethanol Gasoline Blending in Austria

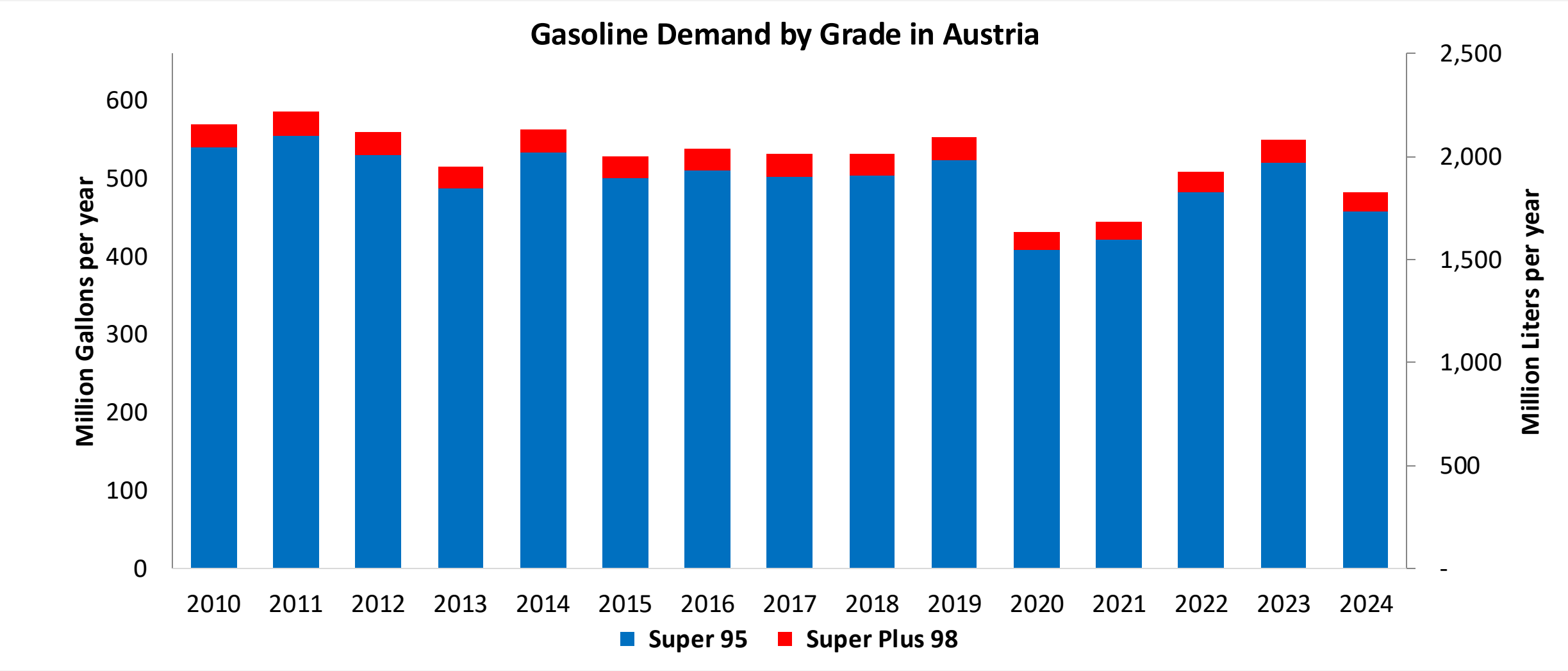


In 2024, gasoline consumption in Austria was 1,829 million liters (483 million gallons) and growth rate was -2.7%. Gasoline production and net imports were 2,317 and -392 million liters (612 and -103 million gallons) correspondingly. Austria complies with the European gasoline standards and offers two main grades: RON 95 and RON 98) with an average octane of 95.2 RON.

In 2024, ethanol consumption in Austria was 185 million liters (49 million gallons) representing 10.1% of gasoline demand. Ethanol production and Net Imports were 297 and -112 million liters (49 and -30 million gallons) correspondingly. Ethanol sources are Corn 75%, Cereals 15%, and Sugar Beet 10%.

Source: eurostat

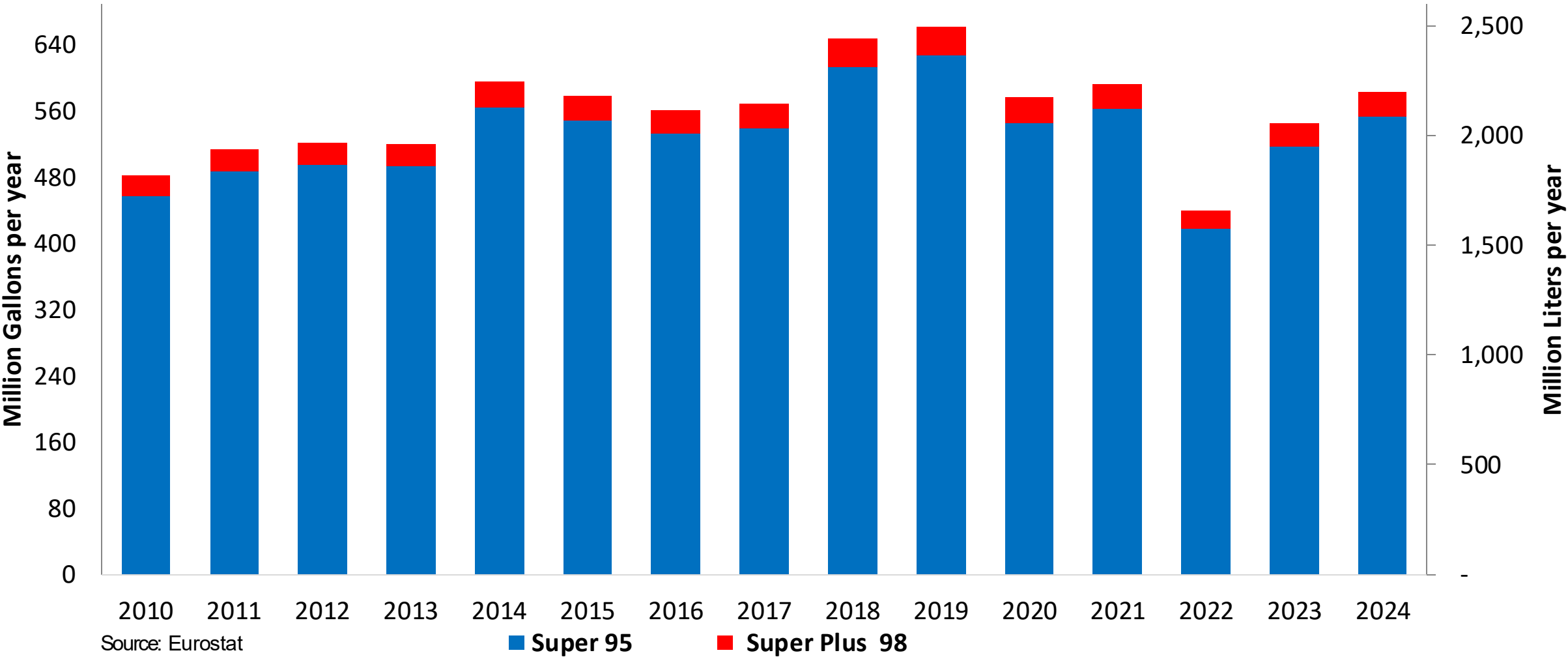
Gasoline Demand in Austria



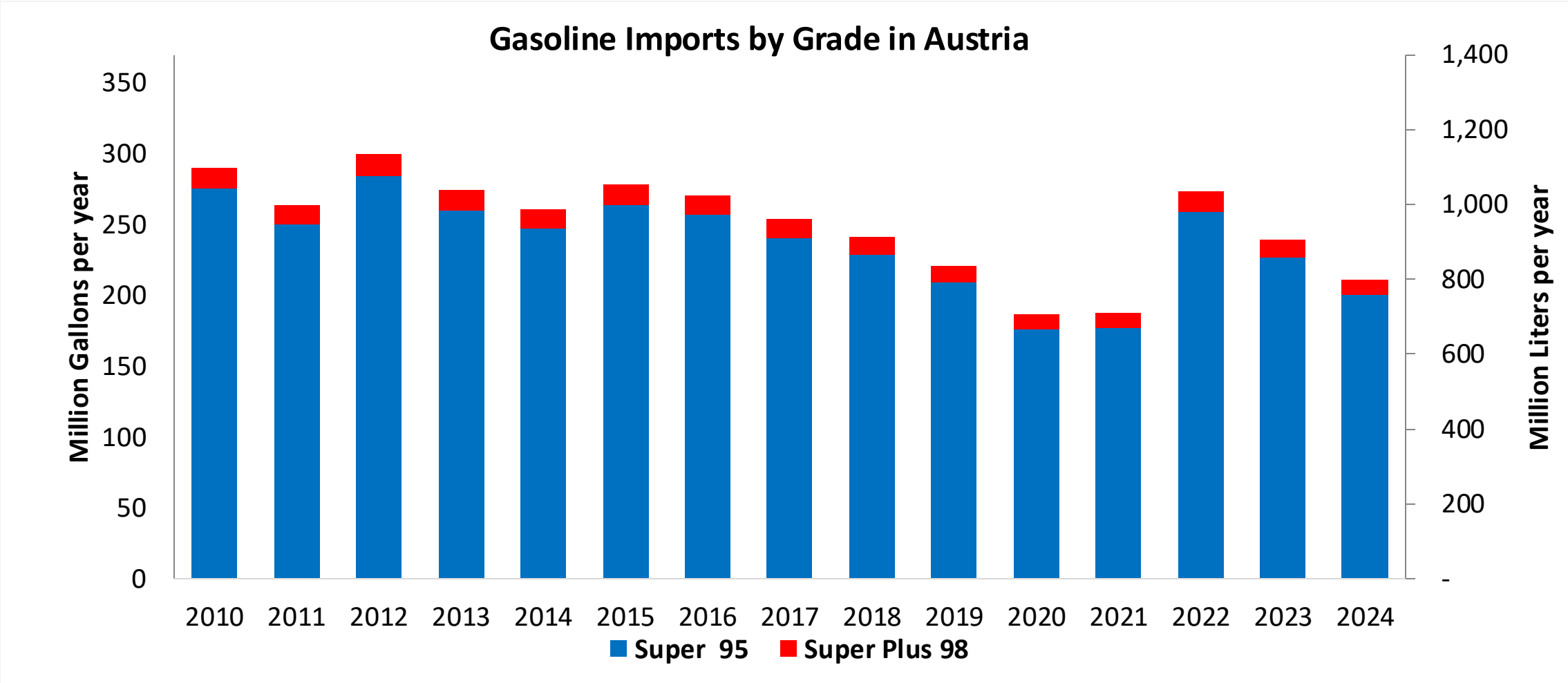
Source: Eurostat

Gasoline Production in Austria

Gasoline Production by Grade in Austria

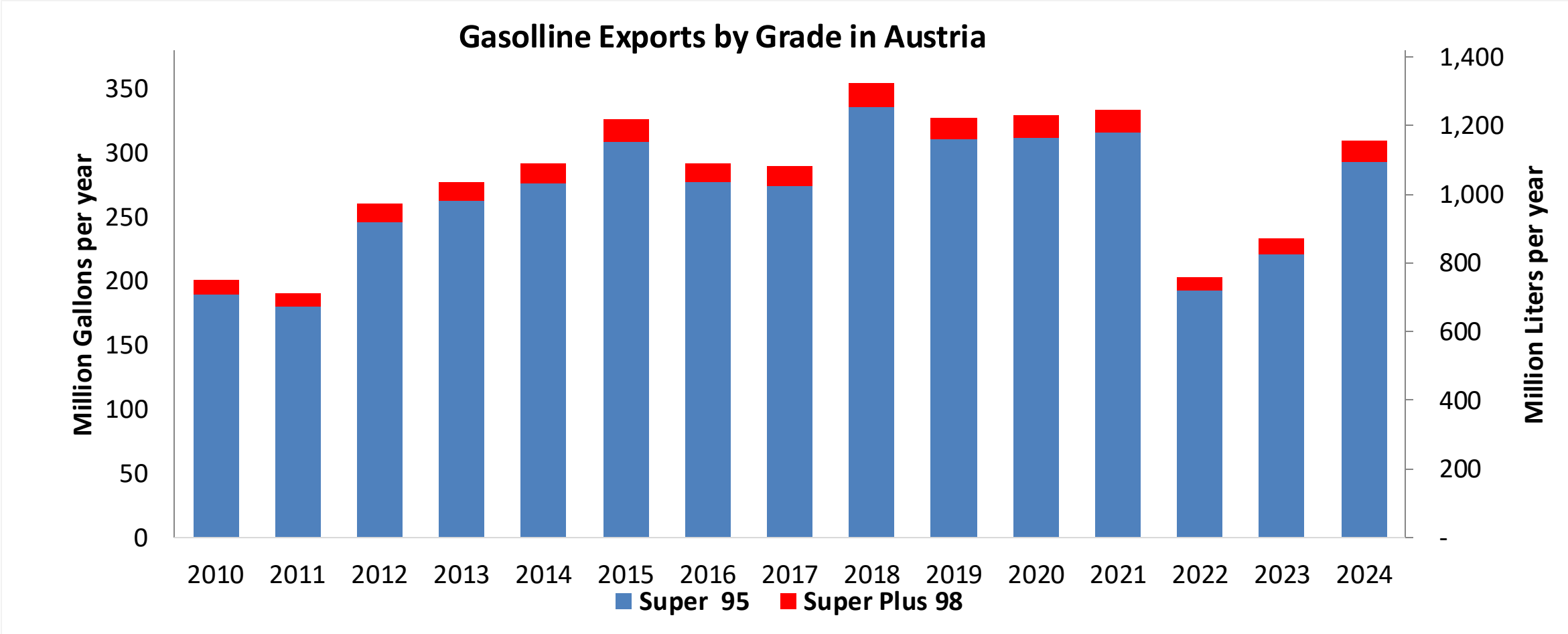


Gasoline Imports in Austria



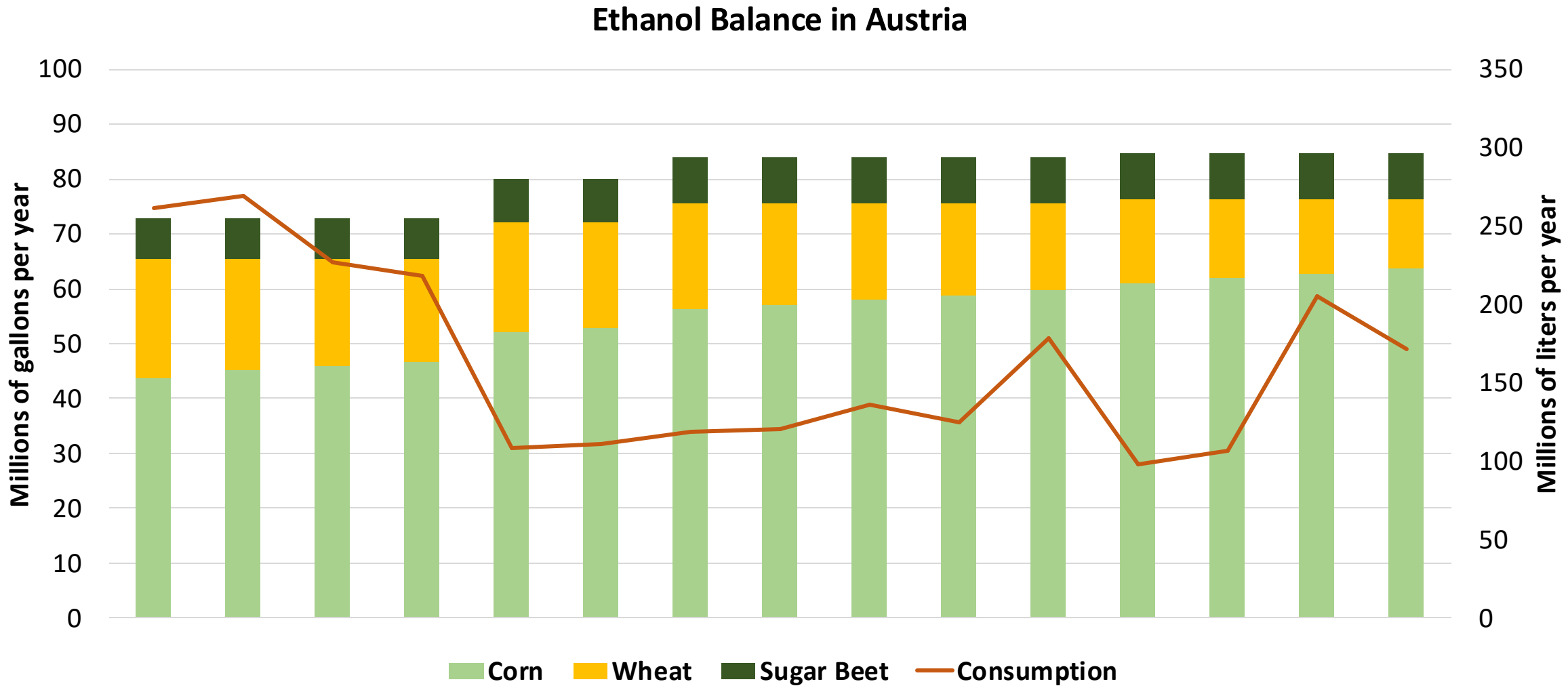
Source: Eurostat

Gasoline Exports in Austria



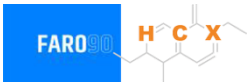
Source: Eurostat

Ethanol Balance in Austria



Source: Eurostat

Gasoline Quality in Austria

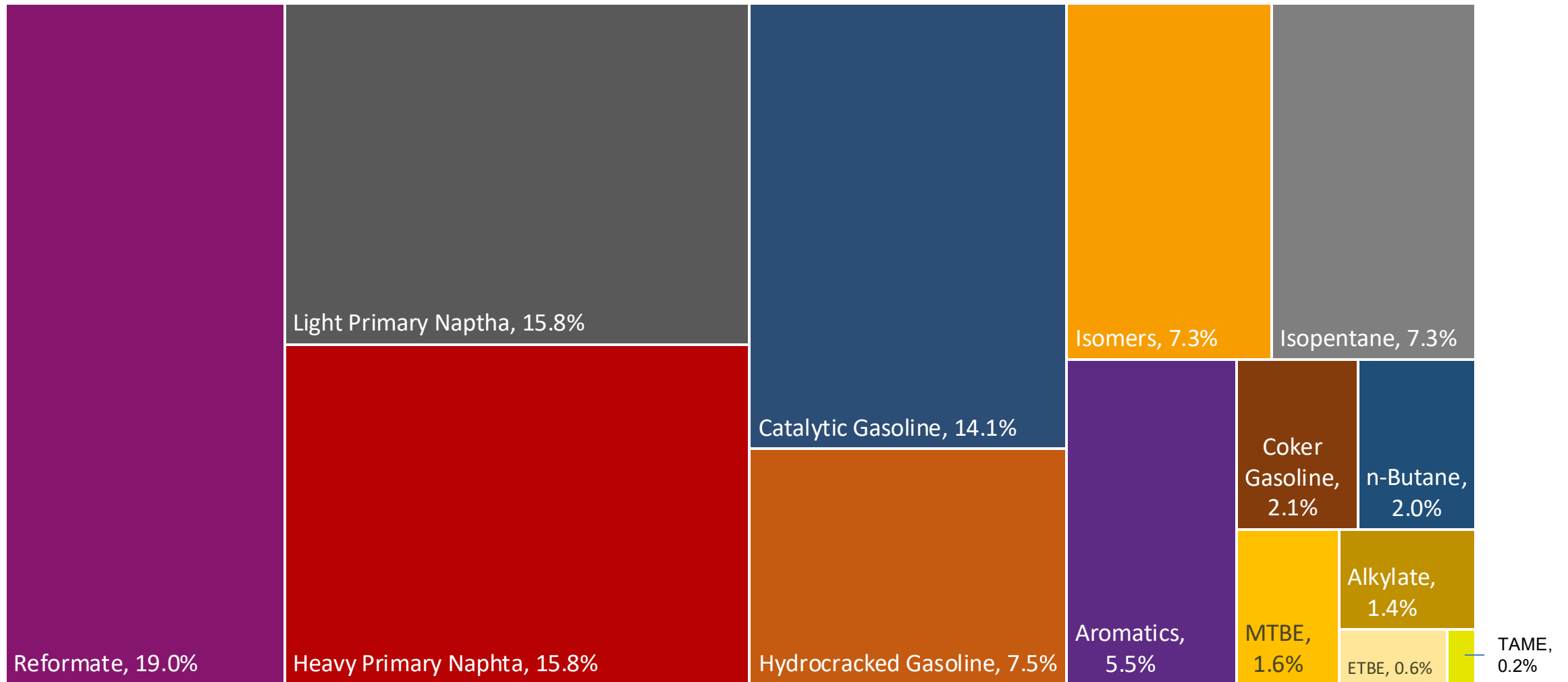


Name	EN 228:2025 (Euro 6); Country Level Fuel Ordinances							
Implementation Date	2025							
Applicability	All EU Contries + UK							
Grade	RON 95 E5		RON 95 E10		RON 98 E5		RON 98 E10	
	Min	Max	Min	Max	Min	Max	Min	Max
RON	95.0		95.0		98.0		98.0	
MON	> 85	> 88	> 85	> 88	> 85	> 88	> 85	> 88
AKI								
Benzene (%vol)	< 1							
Aromatics (% vol)	< 35							
Olefins (%vol)	< 18							
Lead (g/L)	< 0.005							
Manganese (mg/L)	< 2,0							
Sulfur (ppm)	10.0							
Oxygen (%w)*	< 2,7	< 3,7	< 2,7	< 3,7	< 2,7	< 3,7	< 2,7	< 3,7
Ethanol (%vol)	< 5	< 10	< 5	< 10	< 5	< 10	< 5	< 10
PVR 37.8°C (psi)	<> 45 - 100 kPa in 6 Volatility Classes							
Overall Biofuels (%cal)	-							
Advanced Biofuels (%cal)	1.0							
Bioethanol in Gasoline Sales (%cal)	3.4							
GHG Emission Reduction (%)	7.5							

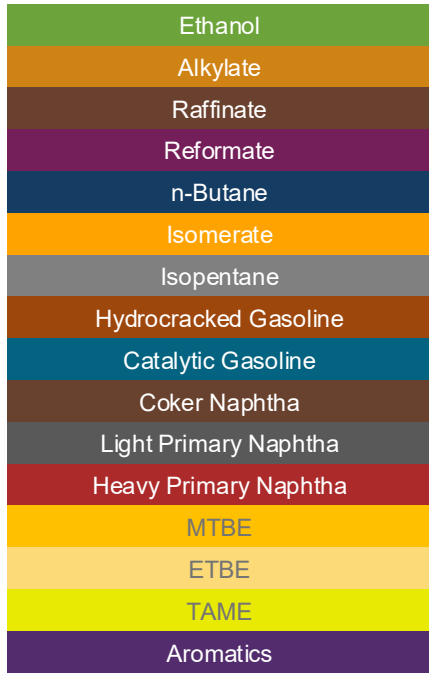
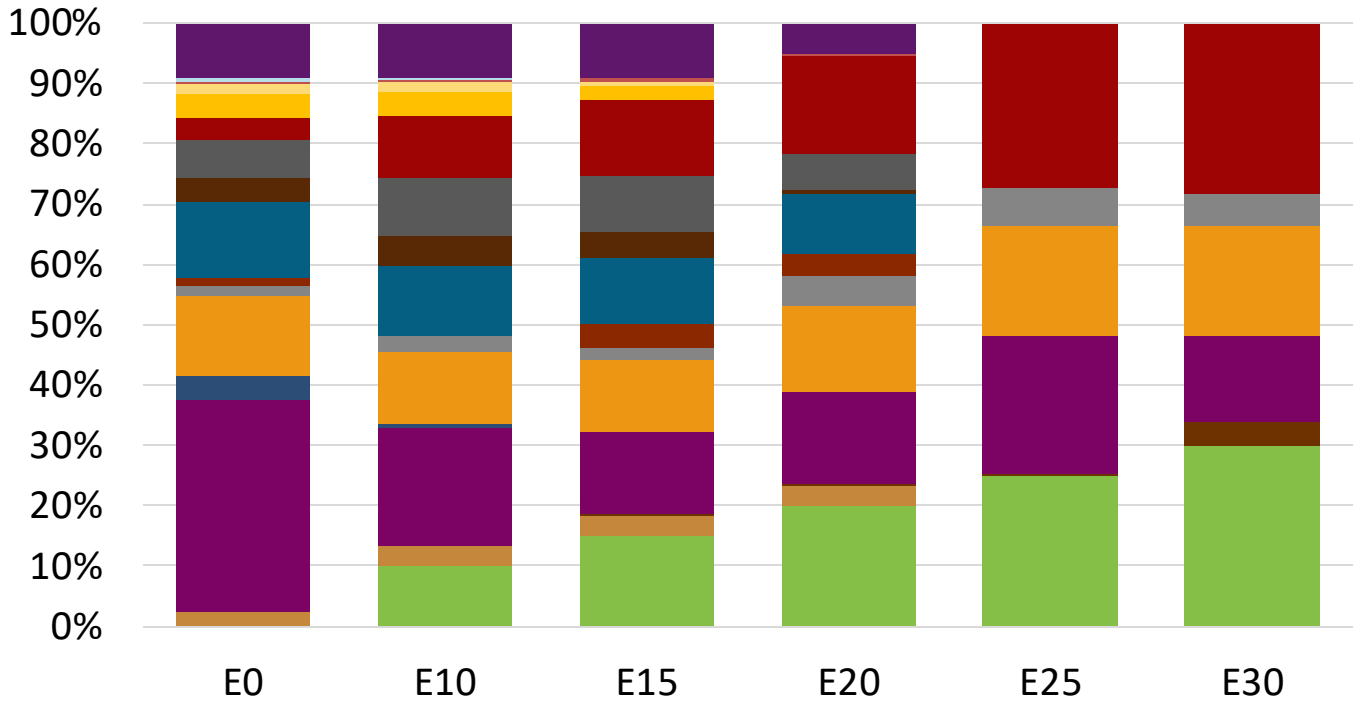
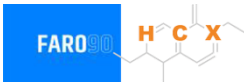
* Methanol 3%, Iso-propyl alcohol 12%, Ter-butyl alcohol 15%, Iso-butyl alcohol 15%, Ethers +5 Carbons 22%, Other Oxygenates 15%

Source: European Commision

Austria Available Blending Components



Austria – Gasoline 95 – Constant Octane

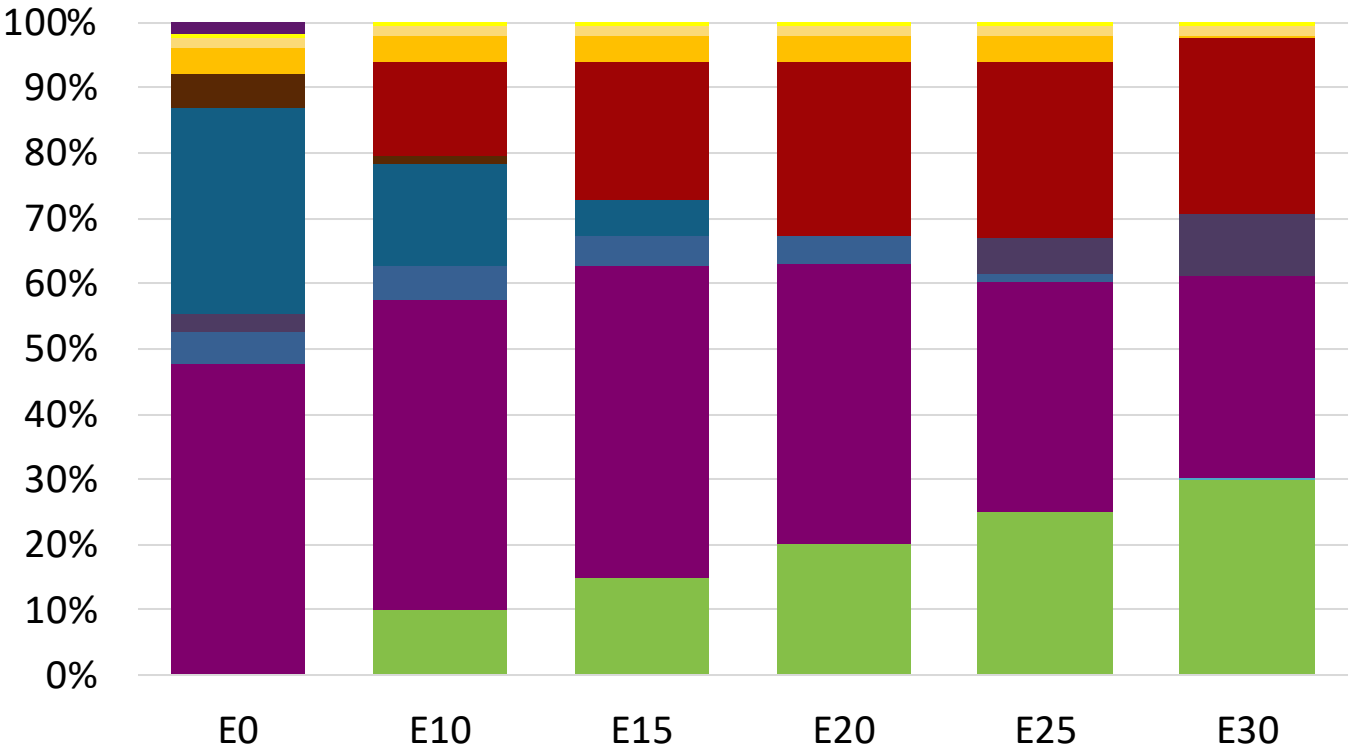
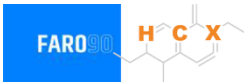


Average CIF 2024
Prices

Octane (RON)	95.0	95.0	95.0	95.0	95.1	95.6
Price (US\$/gal)	\$ 2.272	\$ 2.144	\$ 2.083	\$ 2.002	\$ 1.880	\$ 1.814

Source: Faro90

Austria - Gasoline 98 - Constant Octane

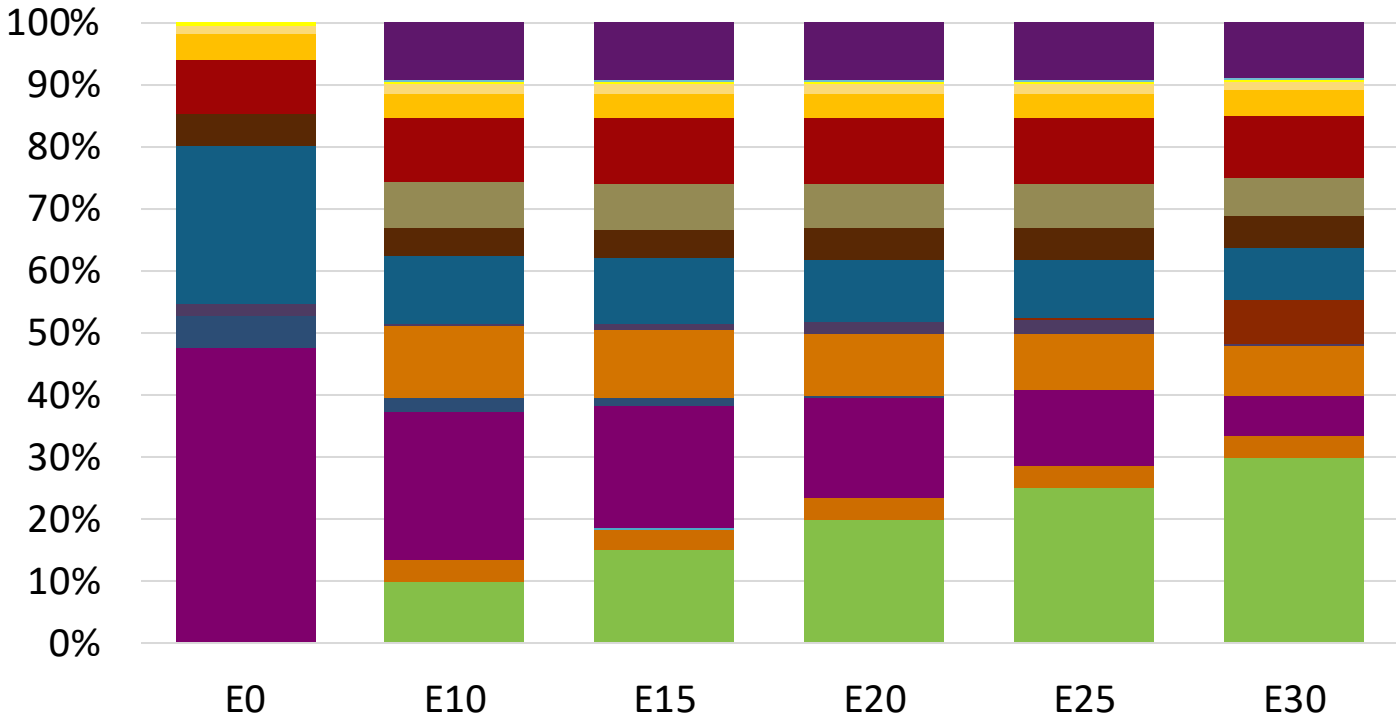
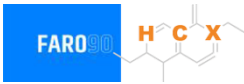


Average CIF 2024
Prices

Octane (RON)	98.0	98.0	98.0	98.0	98.0	98.0
Price (US\$/gal)	\$ 2.347	\$ 2.218	\$ 2.146	\$ 2.069	\$ 1.992	\$ 1.925

Source: Faro90

Austria – Gasoline 95 – Increased Octane

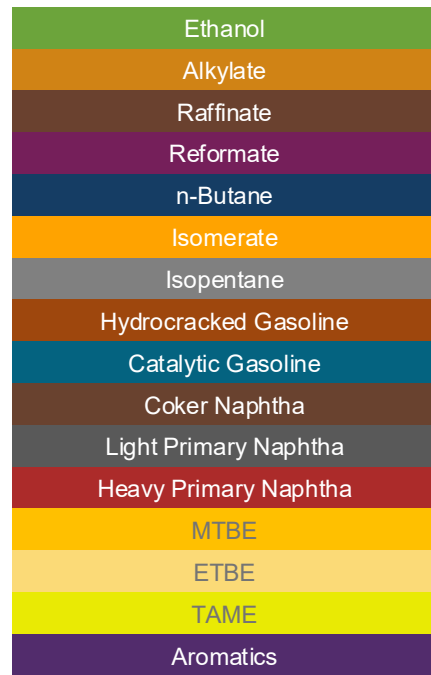
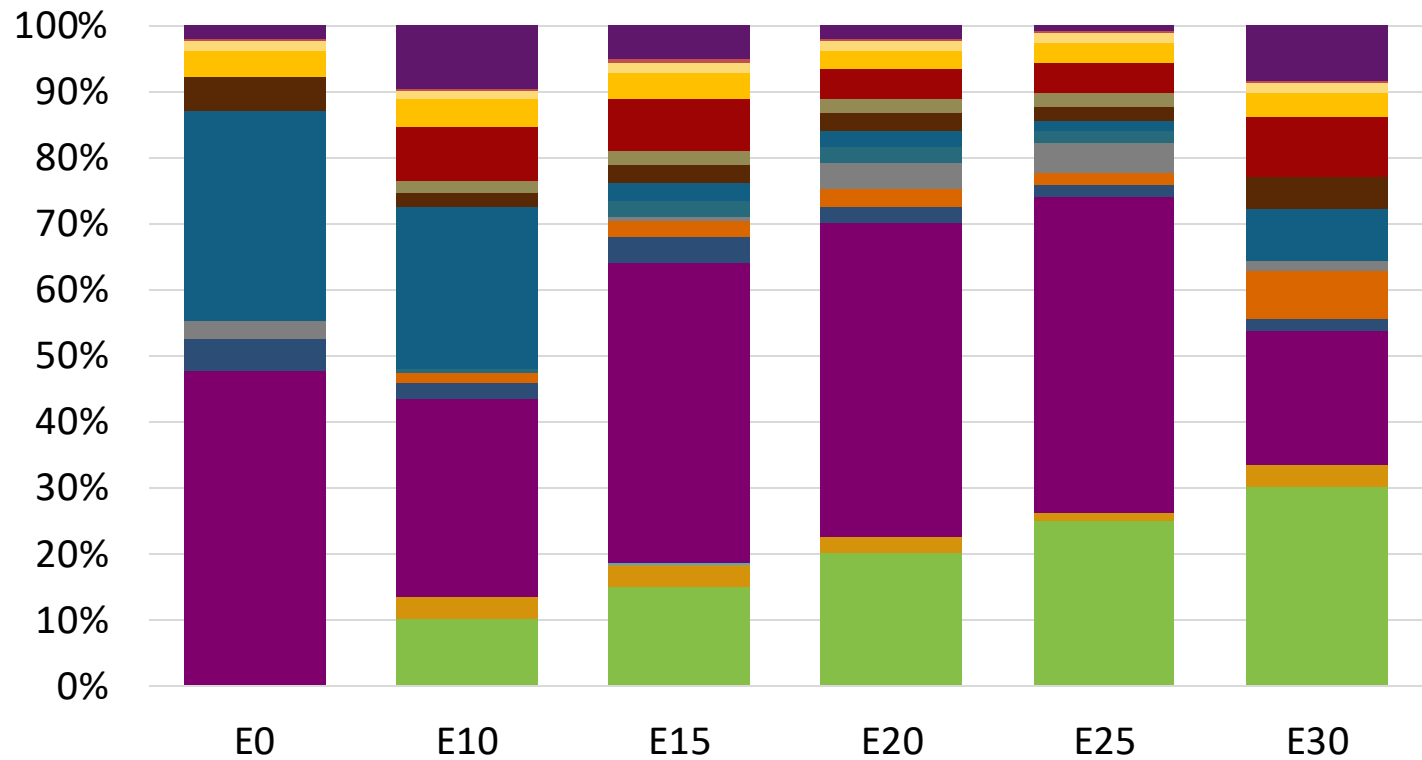


Average CIF 2024
Prices

Octane (RON)	95.0	96.5	97.9	99.4	100.8	102.0
Price (US\$/gal)	\$ 2.209	\$ 2.209	\$ 2.209	\$ 2.209	\$ 2.209	\$ 2.209

Source: Faro90

Austria – Gasoline 98 – Increased Octane

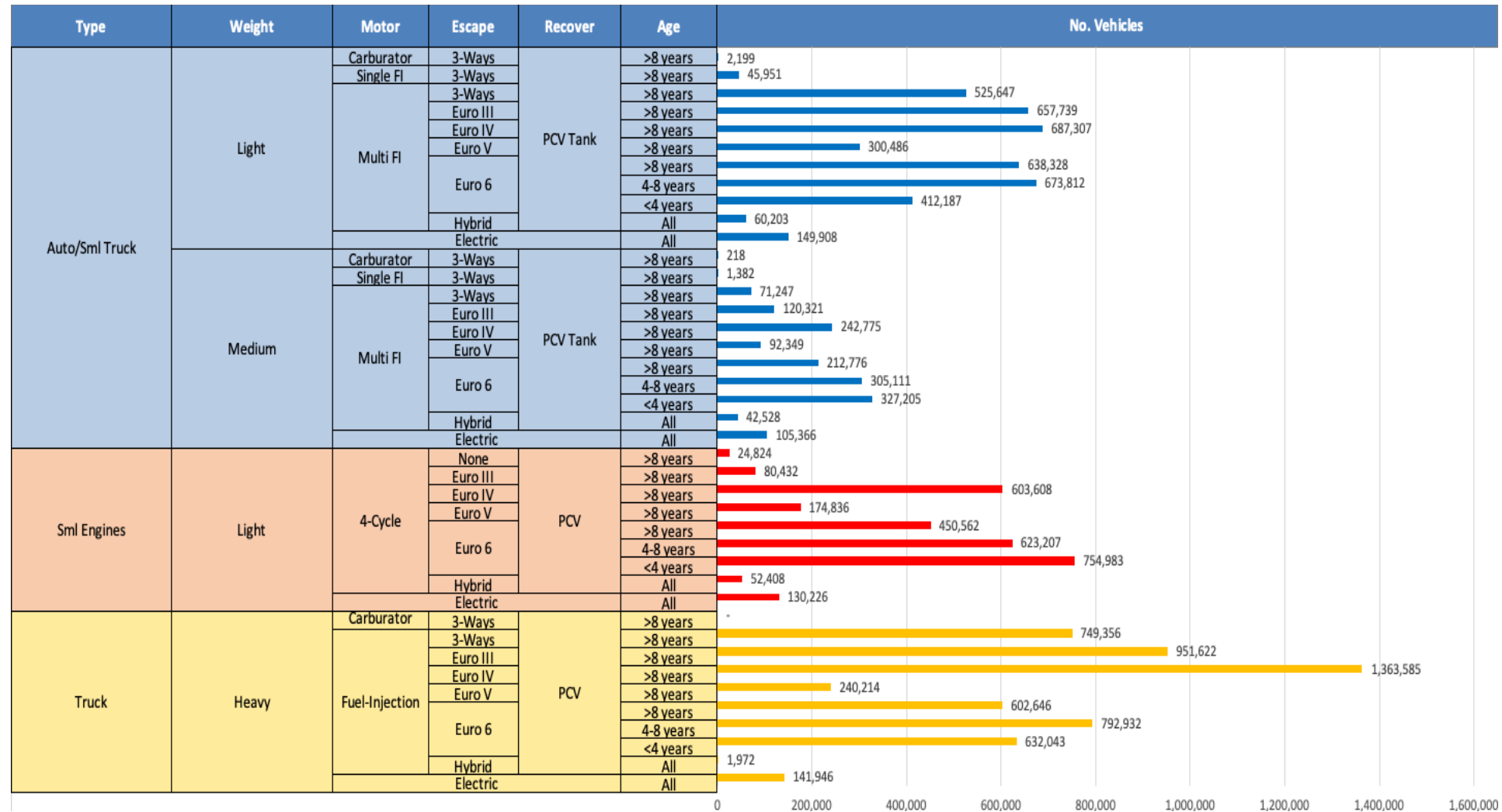


Average CIF 2024
Prices

Octane (RON)		98.0	99.2	101.3	102.6	104.0	104.7
Price (US\$/gal)	\$	2.347	\$ 2.347	\$ 2.347	\$ 2.347	\$ 2.347	\$ 2.347

Source: Faro90

Gasoline Vehicle Fleet - Austria

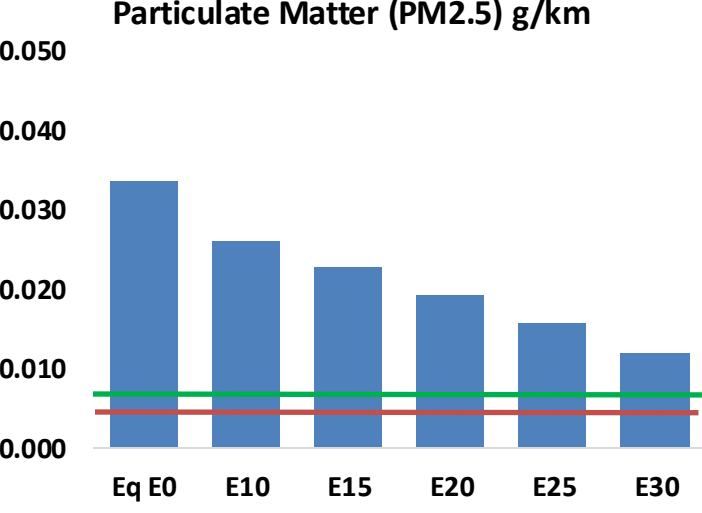
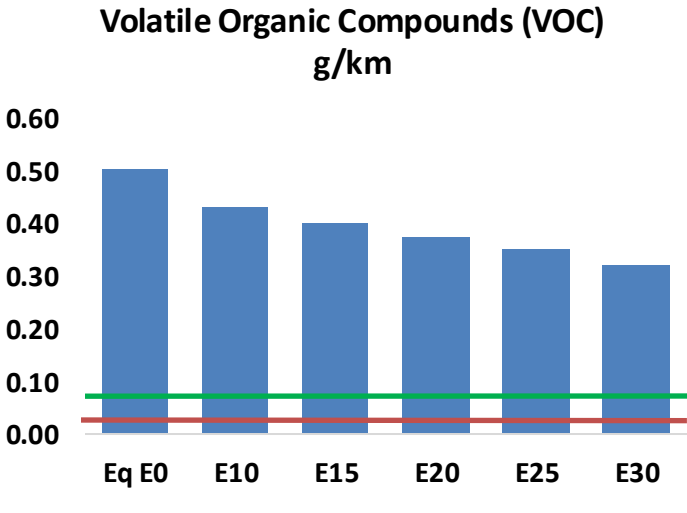
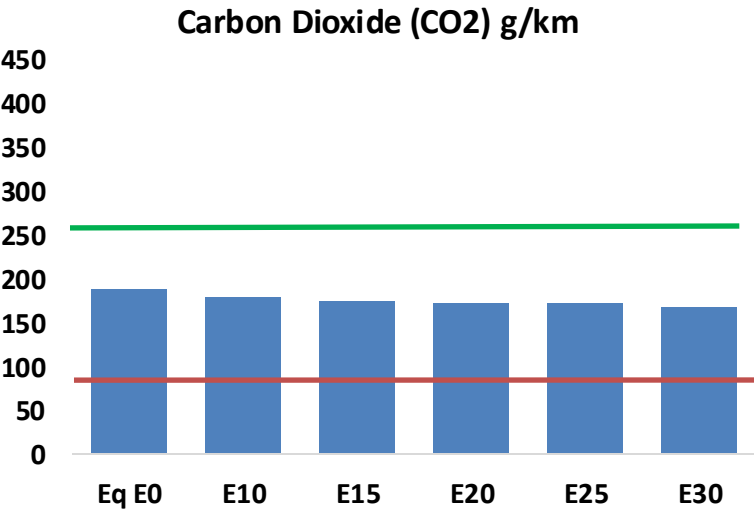
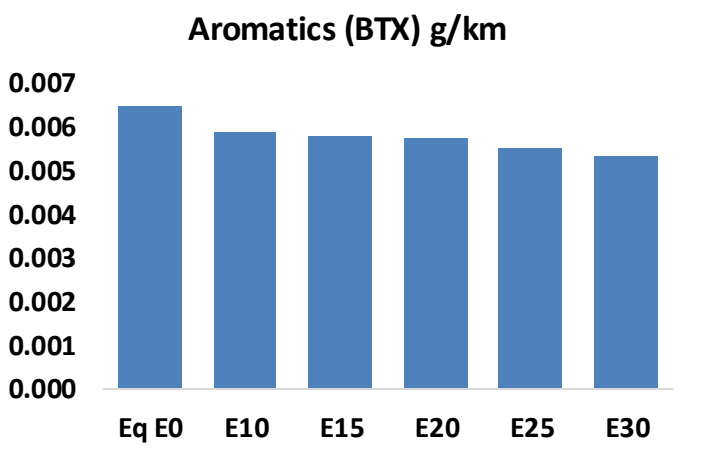
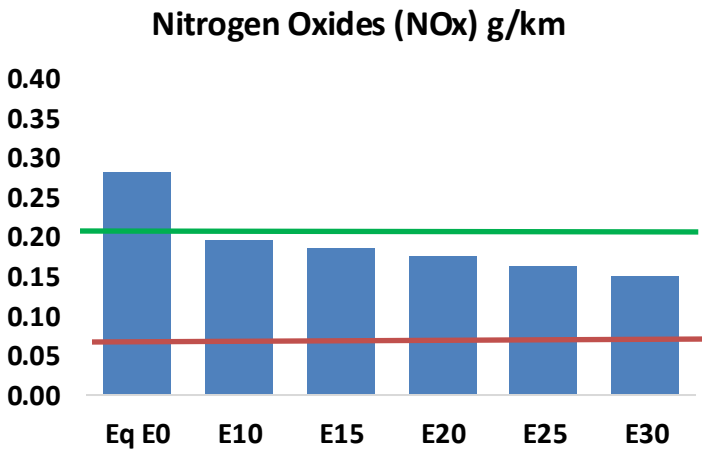
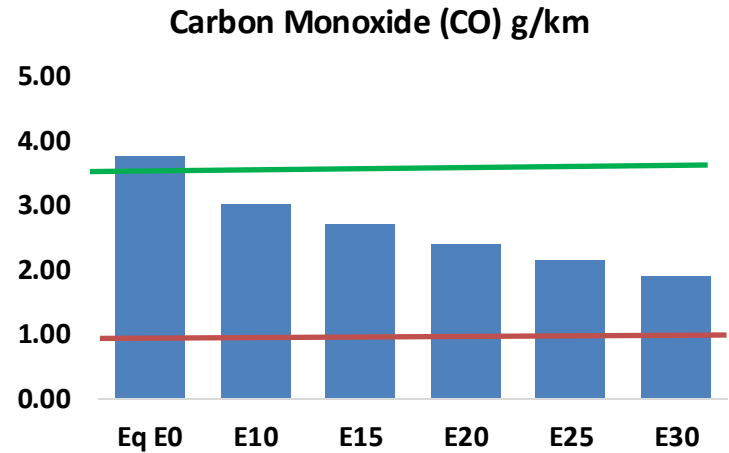
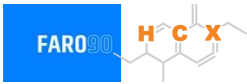


Vehicle Fleet: 14,046,44
Gasoline Vehicle Fleet: 3,251,450

Source: Eurostat, ACEA

Austria – Vehicular Emissions

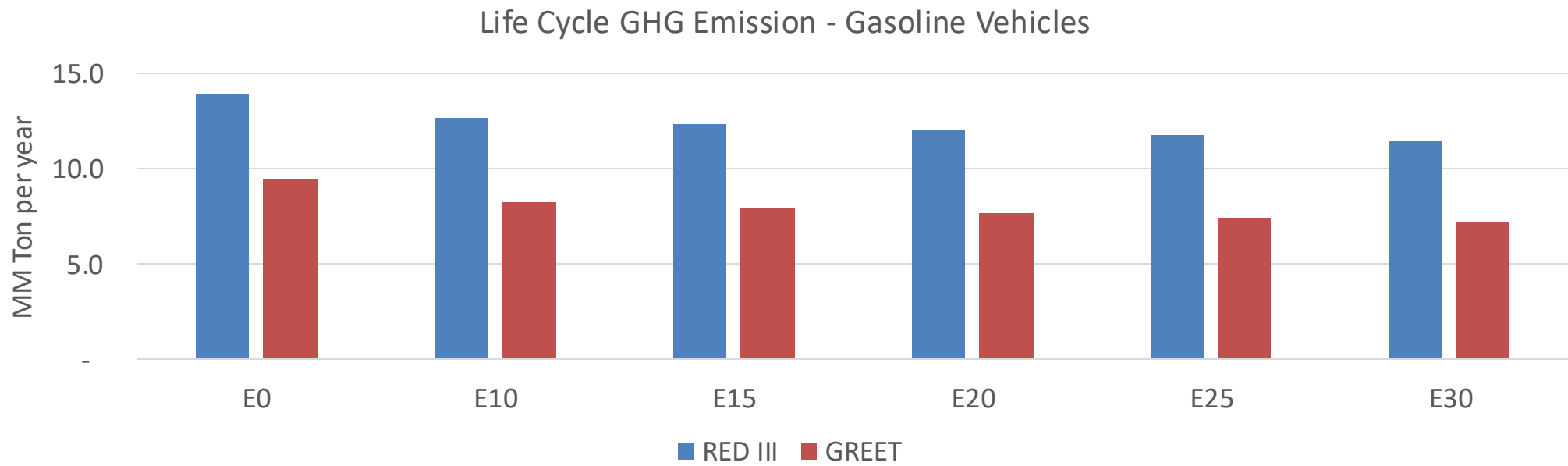
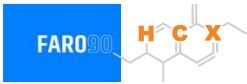
TIER III BIN 125 United States
Euro 6 Standard



Austria – Vehicular Emissions

Vehicular Fleet Emissions (kg/day)	Eq E0	E5	E10	E15	E20	E25	E30	Reduction (%)	
								E10	E20
CO	315,891	285,353	254,815	228,570	201,597	180,687	158,890	-19%	-36%
CO2	15,777,119	15,371,421	14,988,263	14,686,920	14,537,804	14,452,599	14,186,196	-5%	-8%
VOC	42,267	39,233	36,199	33,755	31,314	29,444	26,927	-14%	-26%
NOx	23,733	17,800	16,613	15,658	14,767	13,784	12,745	-30%	-38%
SOx	178	165	151	140	129	117	105	-15%	-28%
NH3	5,682	5,626	5,569	5,596	5,659	5,703	5,740	-2%	0%
N2O	251	249	248	250	255	258	261	-1%	2%
CH4	9,822	9,822	9,822	9,969	10,186	10,372	10,549	0%	4%
Lead	-	-	-	-	-	-	-	0%	0%
Butadiene	313	297	281	270	259	251	239	-10%	-17%
Acetaldehyde	819	940	1,379	2,100	2,861	3,270	3,863	68%	249%
Formaldehyde	2,974	2,892	3,371	3,958	4,146	4,587	4,994	13%	39%
Benzene	545	523	496	488	484	463	448	-9%	-11%
PM2.5	1,014	900	787	683	580	471	357	-22%	-43%
PM10	1,813	1,610	1,407	1,222	1,037	842	638	-22%	-43%

Austria – Gasoline Vehicle Fleet Life Cycle GHG Emissions



Country Emissions 2024 (MMT/año)	67.5
Target @ 2035 (MMT/year)	38.0
Delta	29.5

%Reduction vs E0	E0	E10	E15	E20	E25	E30
GREET 2024	0.0%	12.9%	16.1%	18.6%	21.0%	24.2%
RED II/III	0.0%	9.3%	11.7%	13.6%	15.4%	17.8%

% Target@2035	E0	E10	E15	E20	E25	E30
GREET 2024	0.0%	4.1%	5.1%	6.0%	6.7%	7.7%
RED II/III	0.0%	4.4%	5.5%	6.4%	7.3%	8.4%

Source: Greet, RED II/III, climateactiontracker.org, F90

Ethanol Gasoline Blending in Belgium



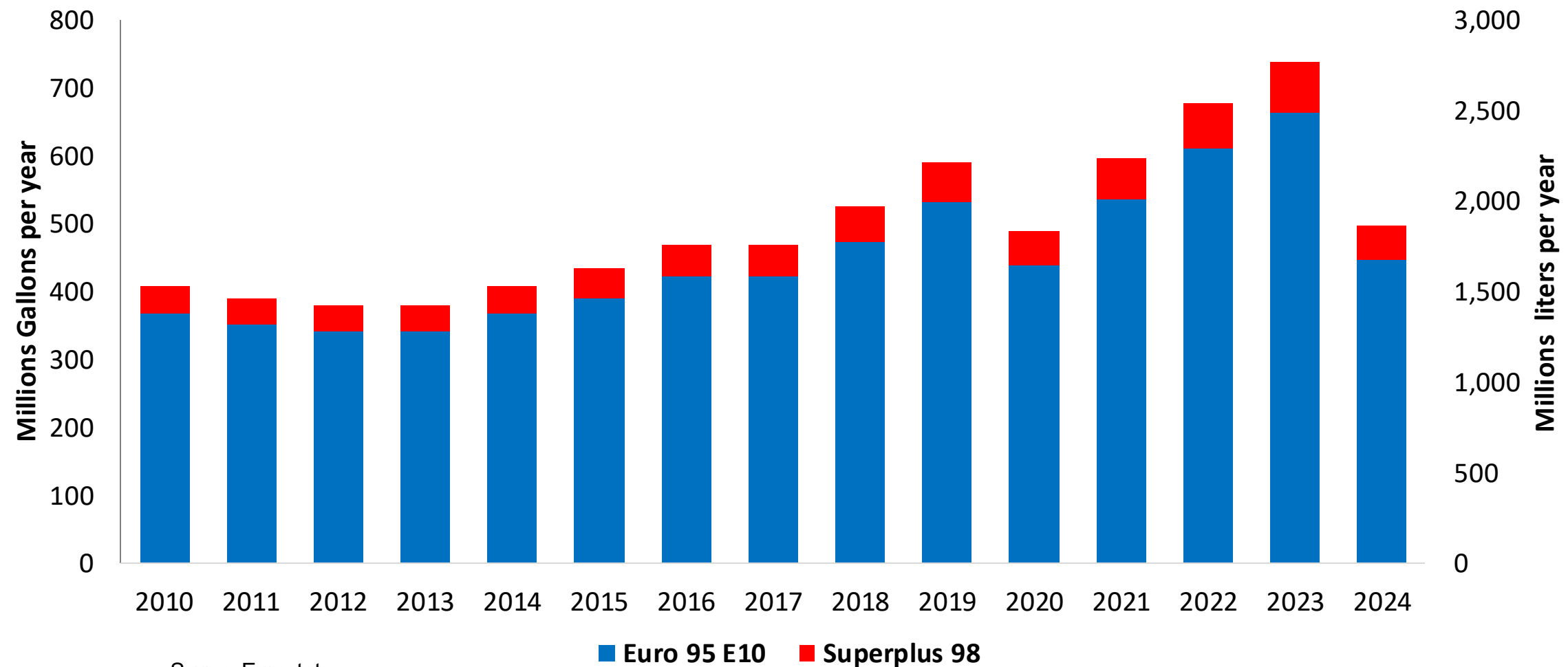
In 2024, gasoline consumption in Belgium was 1,865 million liters (493 million gallons) and growth rate was -3.4%. Gasoline production and net imports were 5,257 and -4,825 million liters (1,389 and -1,275 million gallons) correspondingly. Belgium complies with the European gasoline standards and offers two main grades: RON 95 and RON 98) with an average octane of 95.3 RON.

In 2024, ethanol consumption in Belgium was 239 million liters (63 million gallons) representing 12.8% of gasoline demand. Ethanol production and Net Imports were 554 and -315 million liters (63 and -83 million gallons) correspondingly. Ethanol sources are Sugar Cane 20% and Cereals 80%.

Source: Eurostat, European Commission

Gasoline Demand in Belgium

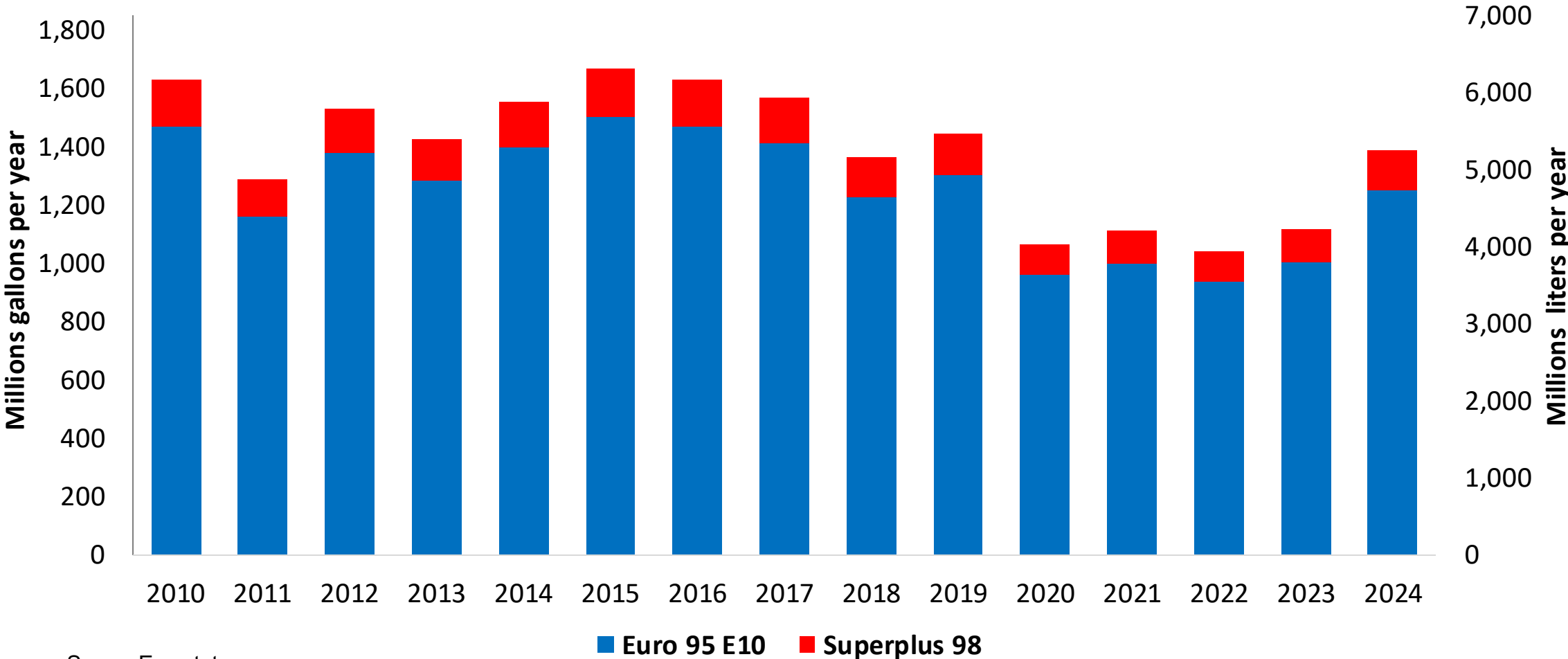
Gasoline Demand by Grade in Belgium



Source: Eurostat

Gasoline Production in Belgium

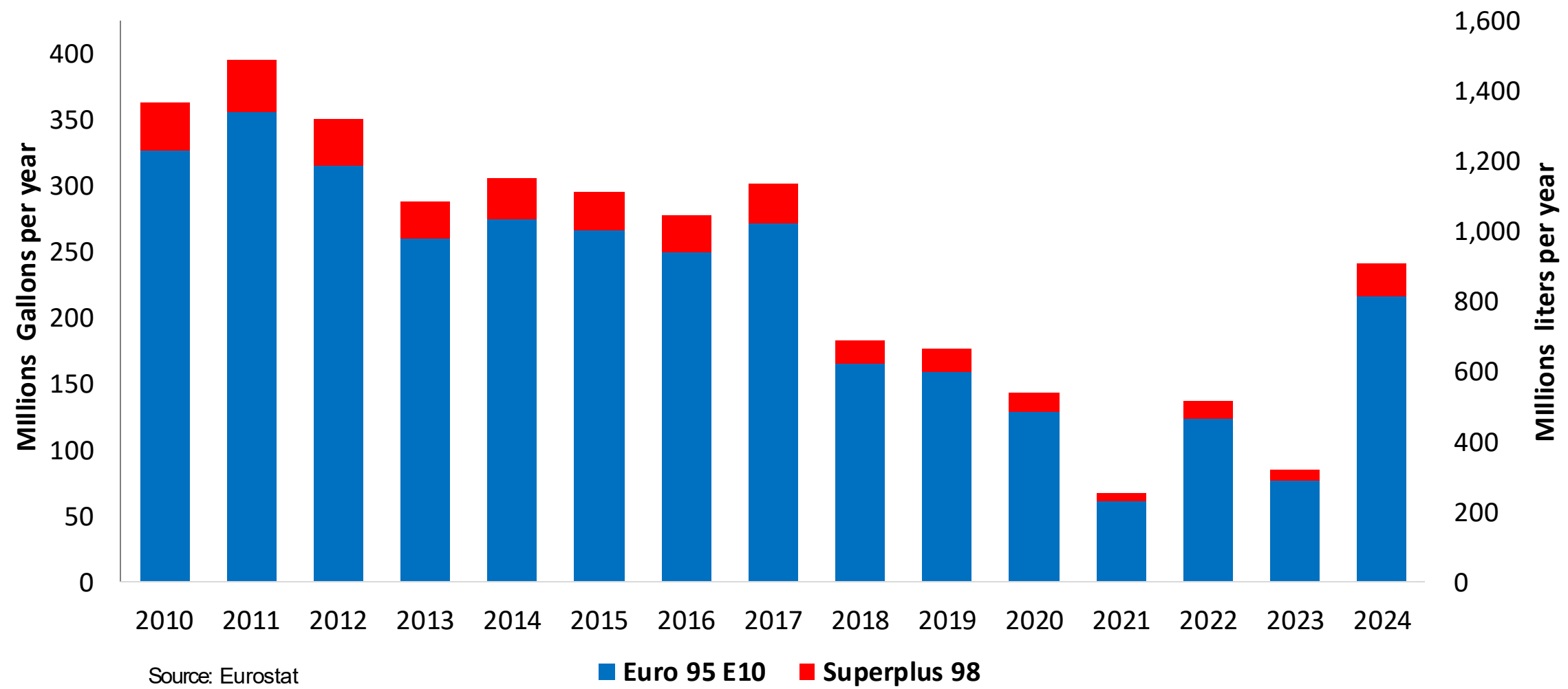
Gasoline Production by Grade in Belgium



Source: Eurostat

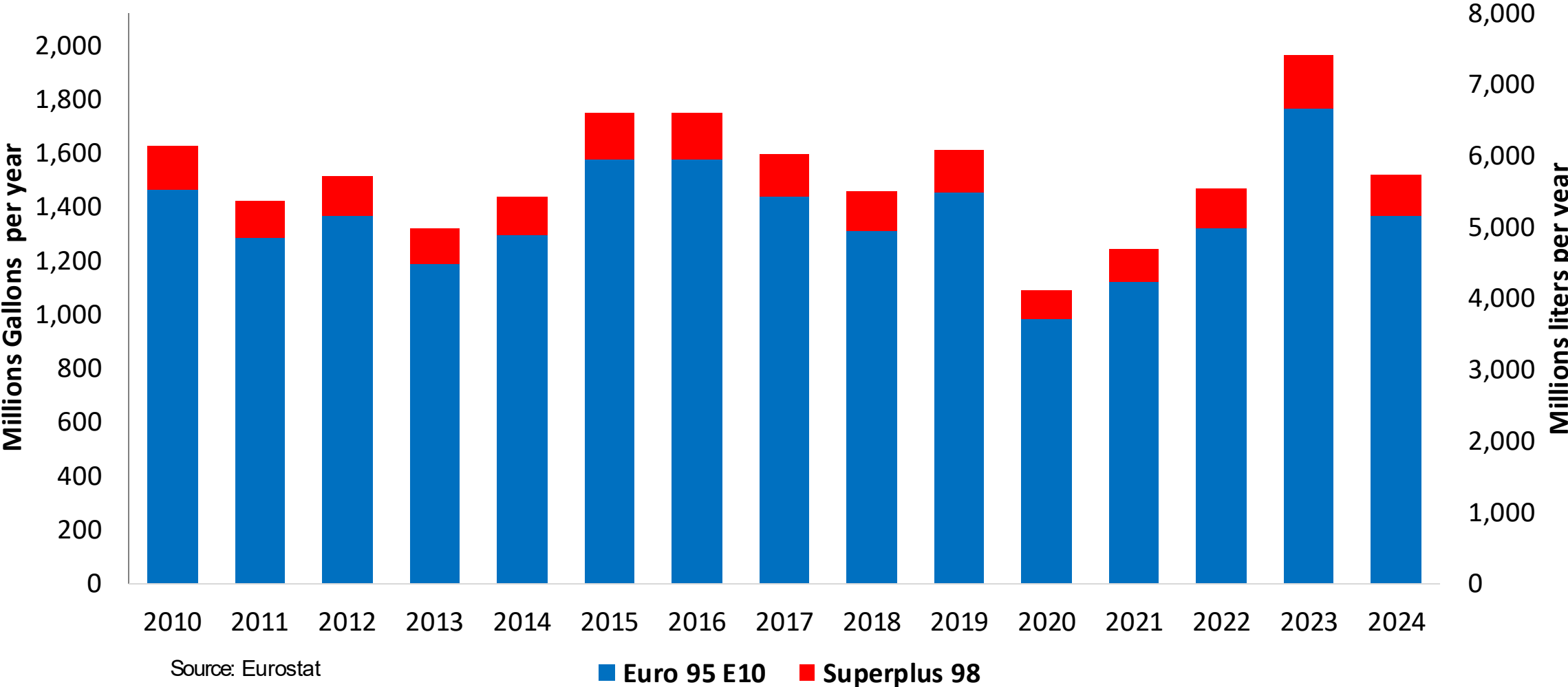
Gasoline Imports in Belgium

Gasoline Imports by Grade in Belgium



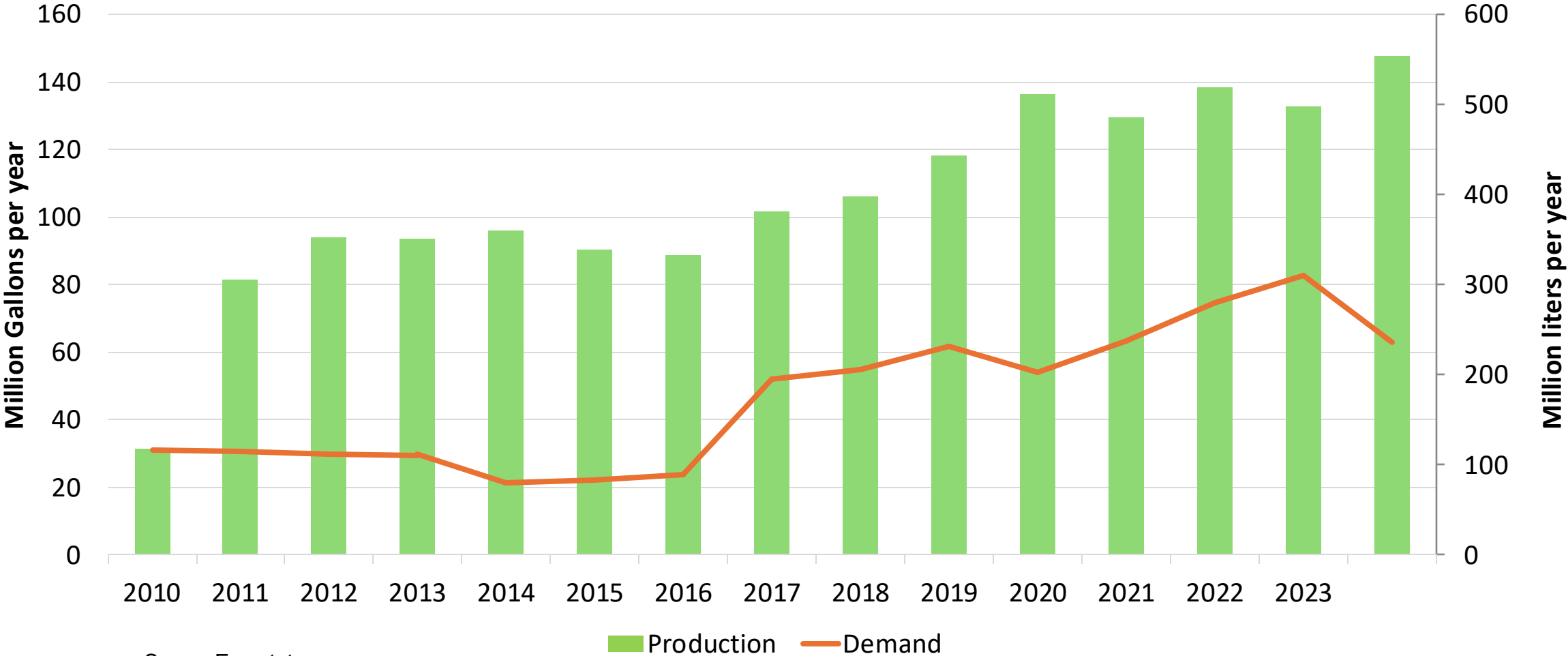
Gasoline Exports in Belgium

Gasoline Exports by Grade in Belgium



Ethanol Balance in Belgium

Ethanol Balance in Belgium



Source: Eurostat

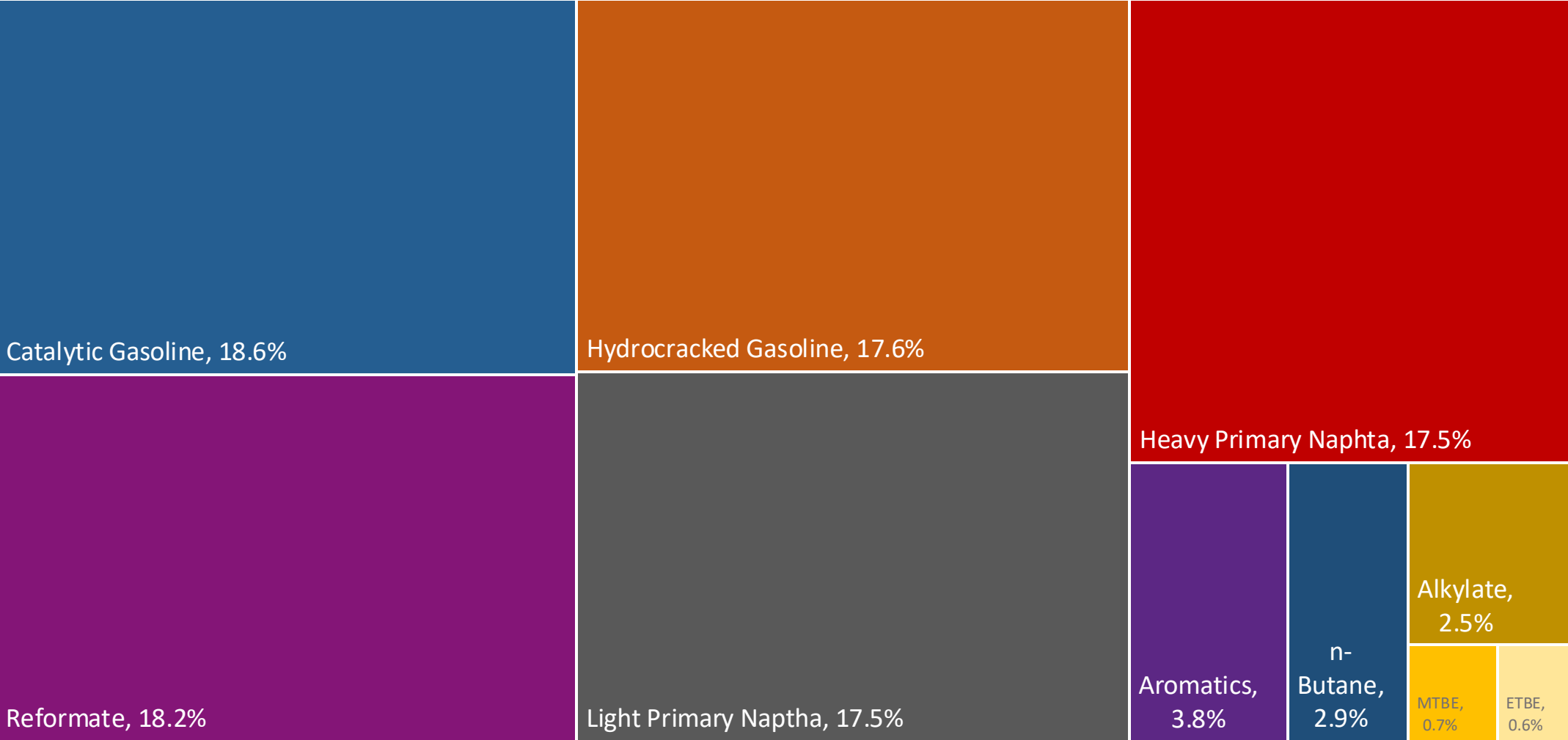
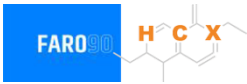
Gasoline Quality in Belgium

Name	EN 228:2025 (Euro 6); Country Level Fuel Ordinances							
Implementation Date	2025							
Applicability	All EU Contries + UK							
Grade	RON 95 E5		RON 95 E10		RON 98 E5		RON 98 E10	
	Min	Max	Min	Max	Min	Max	Min	Max
RON	95.0		95.0		98.0		98.0	
MON	> 85	> 88	> 85	> 88	> 85	> 88	> 85	> 88
AKI								
Benzene (%vol)	< 1							
Aromatics (% vol)	< 35							
Olefins (%vol)	< 18							
Lead (g/L)	< 0.005							
Manganese (mg/L)	< 2,0							
Sulfur (ppm)	10.0							
Oxygen (%w)*	< 2,7	< 3,7	< 2,7	< 3,7	< 2,7	< 3,7	< 2,7	< 3,7
Ethanol (%vol)	< 5	< 10	< 5	< 10	< 5	< 10	< 5	< 10
PVR 37.8°C (psi)	<> 45 - 100 kPa in 6 Volatility Classes							
Overall Biofuels (%cal)	10.5							
Advanced Biofuels (%cal)	-							
Bioethanol in Gasoline Sales (%cal)	5.7							
GHG Emission Reduction (%)	-							

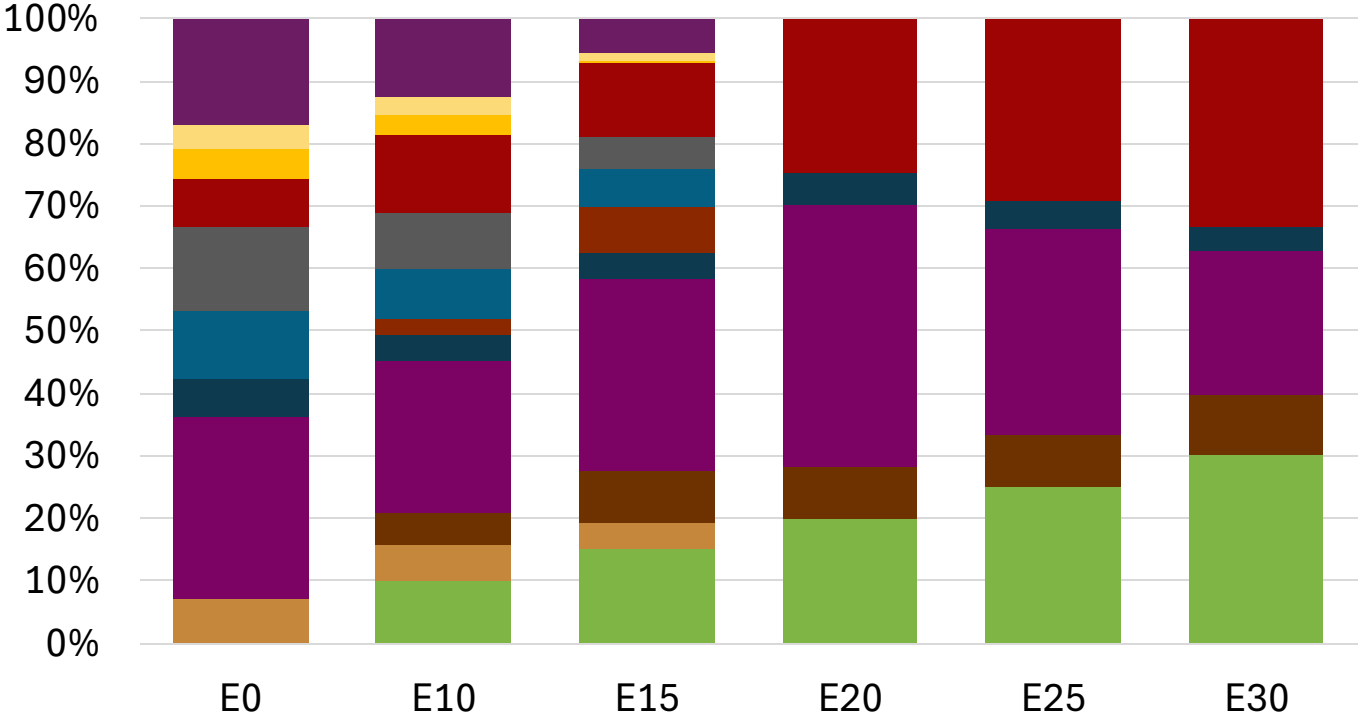
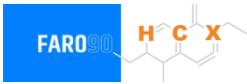
* Methanol 3%, Iso-propyl alcohol 12%, Ter-butyl alcohol 15%, Iso-butyl alcohol 15%, Ethers +5 Carbons 22%, Other Oxygenates 15%

Source: European Commision

Belgium Available Blending Components



Belgium – Gasoline 95 – Constant Octane

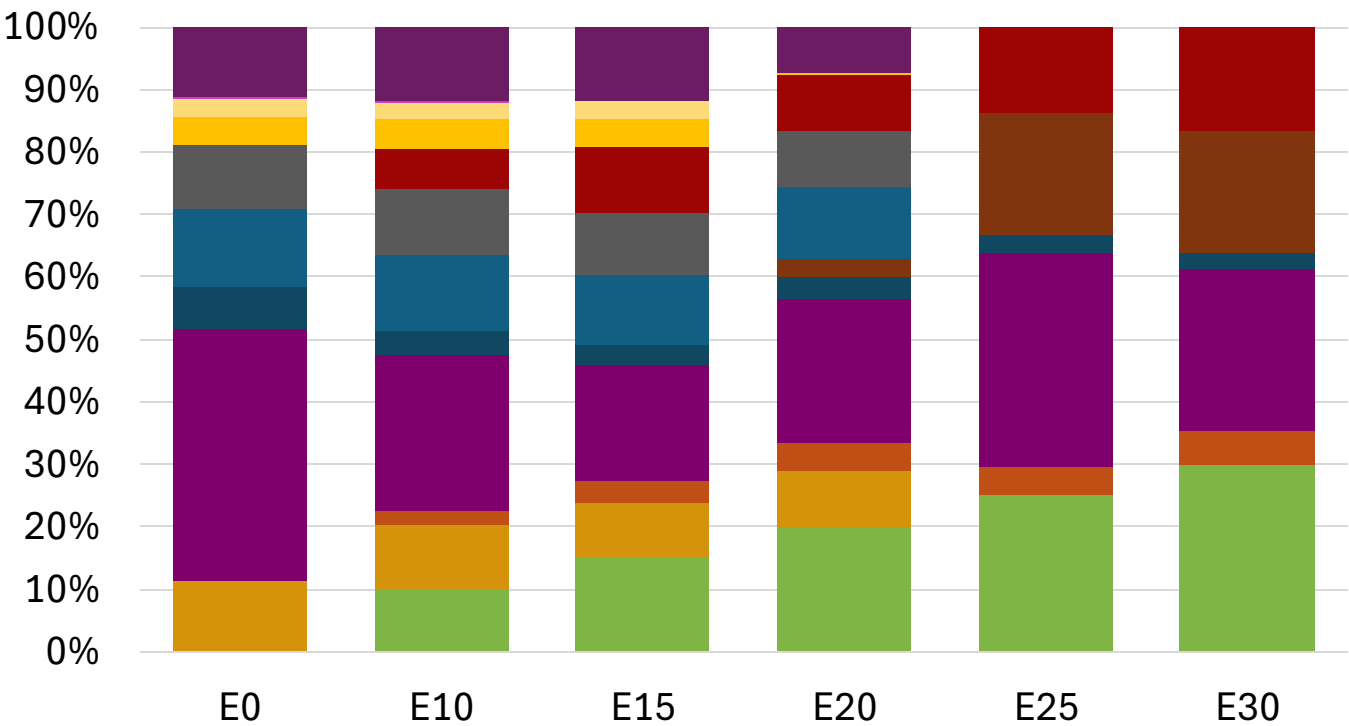
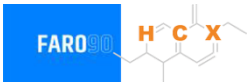


Average CIF 2024
Prices

Octane (RON)	95.0	95.0	95.0	95.1	95.3	95.3
Price (US\$/gal)	\$ 2.281	\$ 2.147	\$ 2.072	\$ 1.951	\$ 1.875	\$ 1.780

Source: Faro90

Belgium - Gasoline 98 - Constant Octane

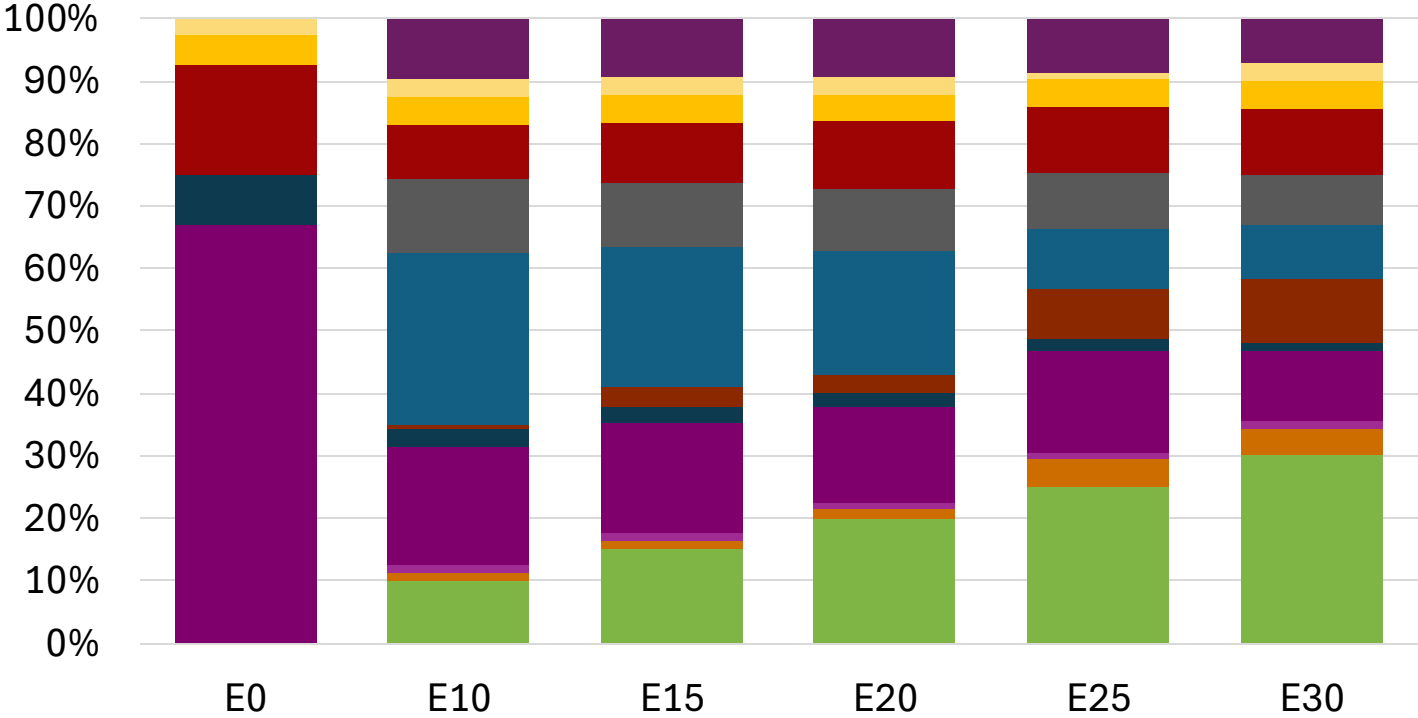
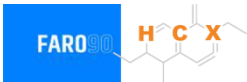


Average CIF 2024 Prices

Octane (RON)	98.0	98.0	98.0	98.0	98.1	98.5
Price (US\$/gal)	\$ 2.409	\$ 2.290	\$ 2.216	\$ 2.164	\$ 2.057	\$ 1.994

Source: Faro90

Belgium – Gasoline 95 – Increased Octane

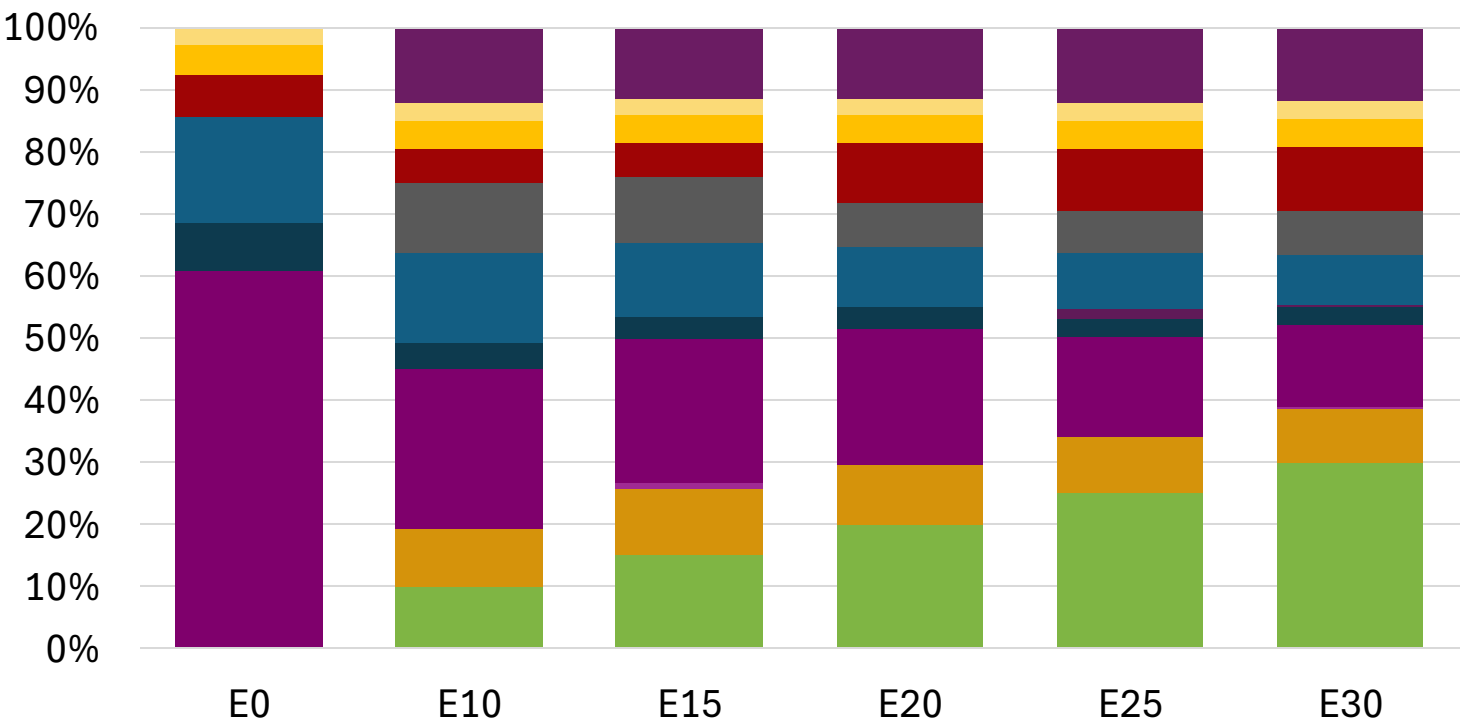


Average CIF 2024 Prices

Octane (RON)	95.0	96.5	97.9	99.4	100.5	102.1
Price (US\$/gal)	\$ 2.193	\$ 2.193	\$ 2.193	\$ 2.193	\$ 2.193	\$ 2.193

Source: Faro90

Belgium – Gasoline 98 – Increased Octane

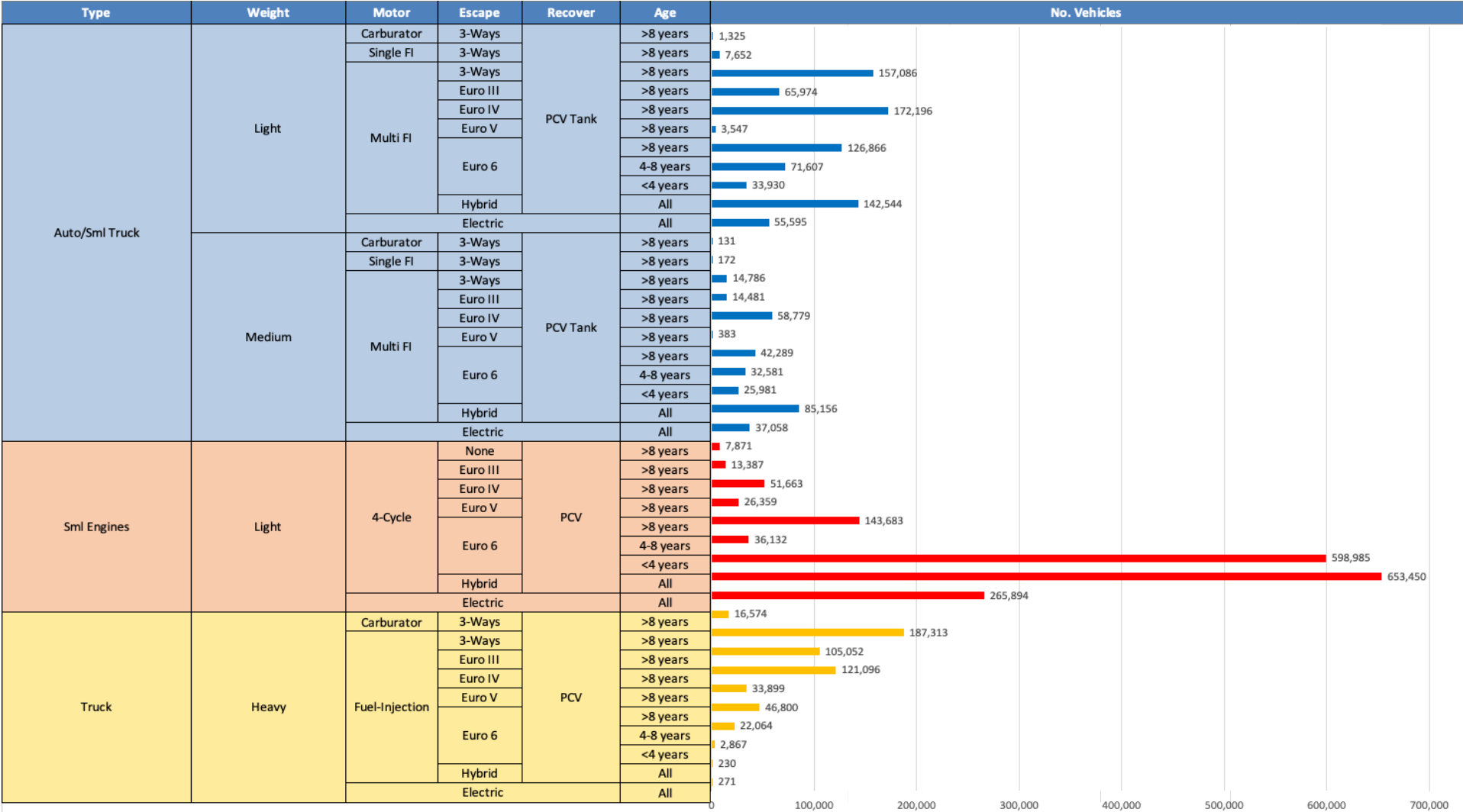


Average CIF 2024
Prices

Octane (RON)		98.0	98.8	100.1	101.6	102.9	104.2
Price (US\$/gal)	\$	2.329	\$ 2.329	\$ 2.329	\$ 2.329	\$ 2.329	\$ 2.329

Source: Faro90

Gasoline Vehicle Fleet - Belgium

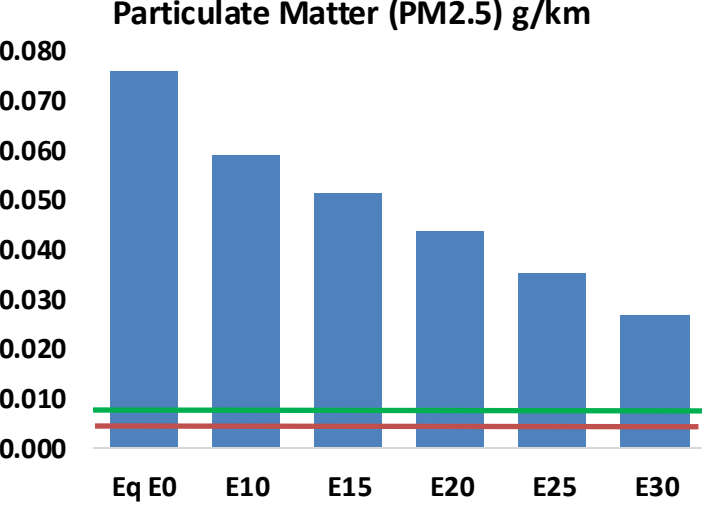
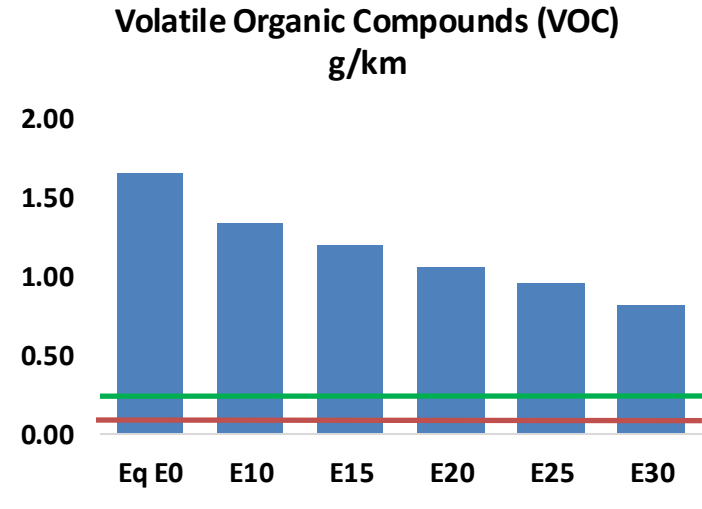
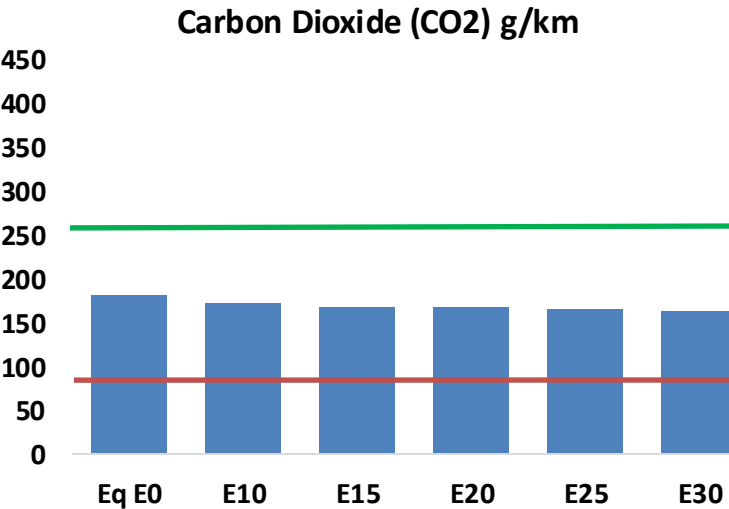
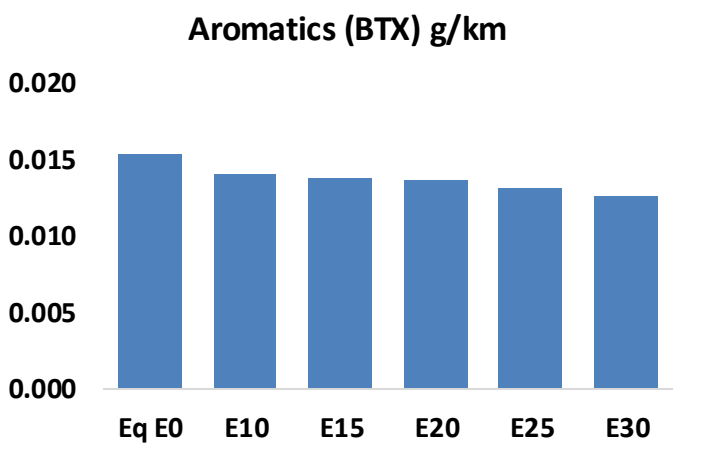
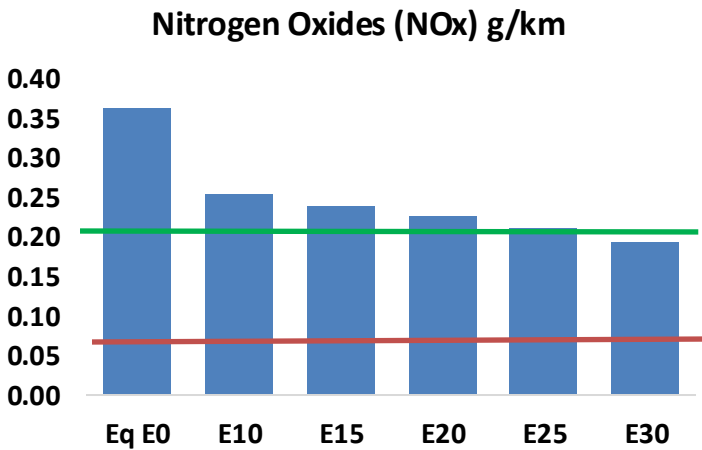
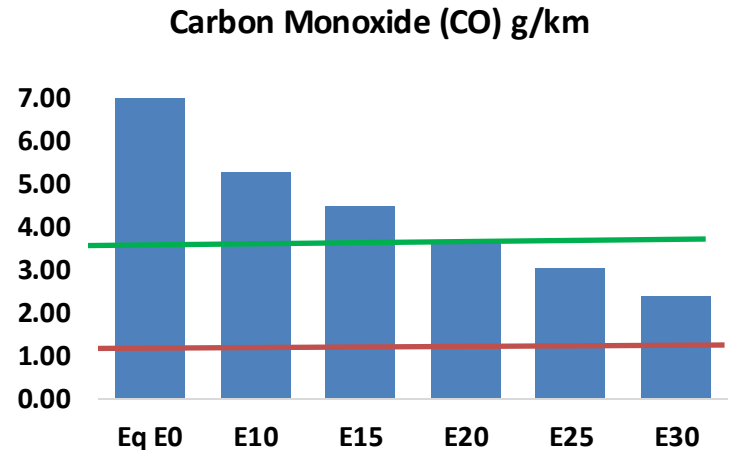
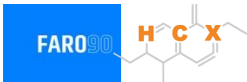


Vehicle Fleet: 3,483,706
Gasoline Vehicle Fleet: 2,367,805

Source: Eurostat, ACEA

Belgium – Vehicular Emissions

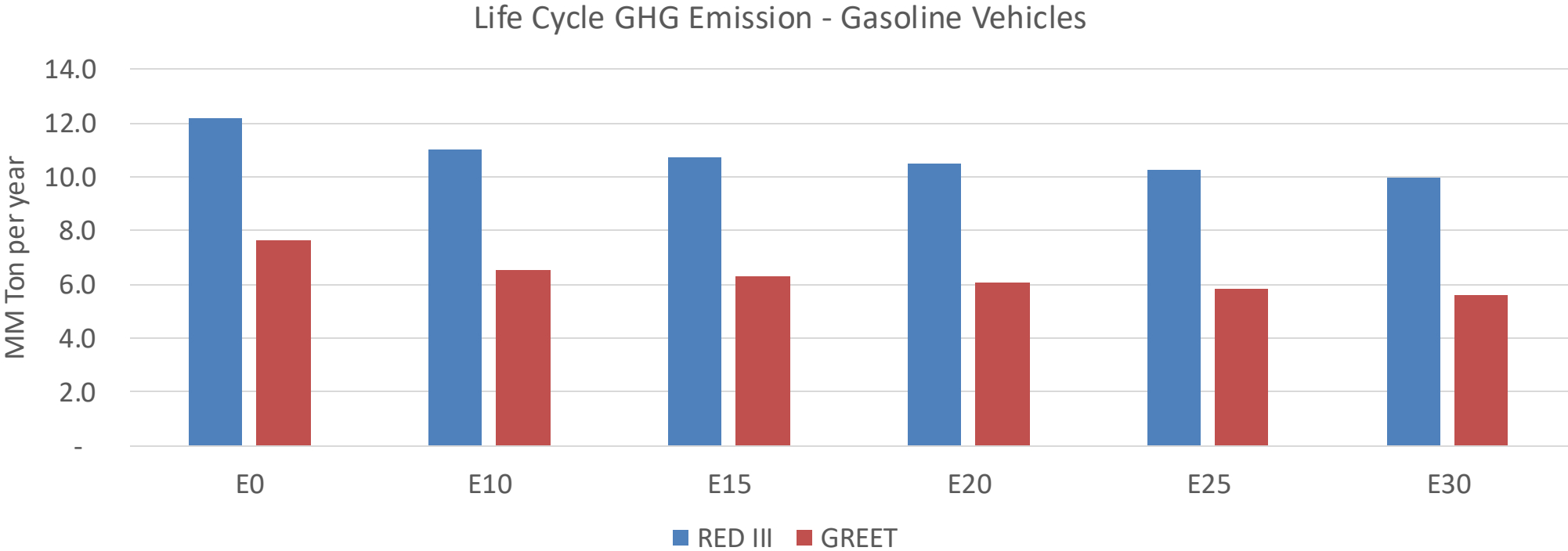
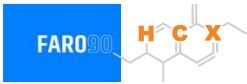
TIER III BIN 125 United States
Euro 6 Standard



Belgium – Vehicular Emissions

Vehicular Fleet Emissions (kg/day)	Eq E0	E5	E10	E15	E20	E25	E30	Reduction (%)	
								E10	E20
CO	424,560	371,491	318,422	270,948	221,338	182,637	143,912	-25%	-48%
CO2	10,967,384	10,685,366	10,419,015	10,209,538	10,105,881	10,047,600	9,862,394	-5%	-8%
VOC	99,964	90,398	80,831	72,624	64,196	57,665	49,129	-19%	-36%
NOx	21,975	16,481	15,382	14,498	13,673	12,763	11,801	-30%	-38%
SOx	124	115	105	97	90	81	73	-15%	-28%
NH3	4,858	4,810	4,762	4,785	4,839	4,877	4,908	-2%	0%
N2O	207	205	205	206	210	212	215	-1%	2%
CH4	19,986	19,986	19,986	20,285	20,725	21,105	21,465	0%	4%
Lead	-	-	-	-	-	-	-	0%	0%
Butadiene	403	375	347	324	302	285	261	-14%	-25%
Acetaldehyde	1,575	1,808	2,653	4,040	5,504	6,290	7,432	68%	249%
Formaldehyde	6,299	6,124	7,138	8,382	8,781	9,715	10,575	13%	39%
Benzene	934	897	850	837	830	793	768	-9%	-11%
PM2.5	706	627	547	475	404	328	248	-22%	-43%
PM10	3,897	3,461	3,024	2,626	2,230	1,809	1,371	-22%	-43%

Belgium – Gasoline Vehicle Fleet Life Cycle GHG Emissions



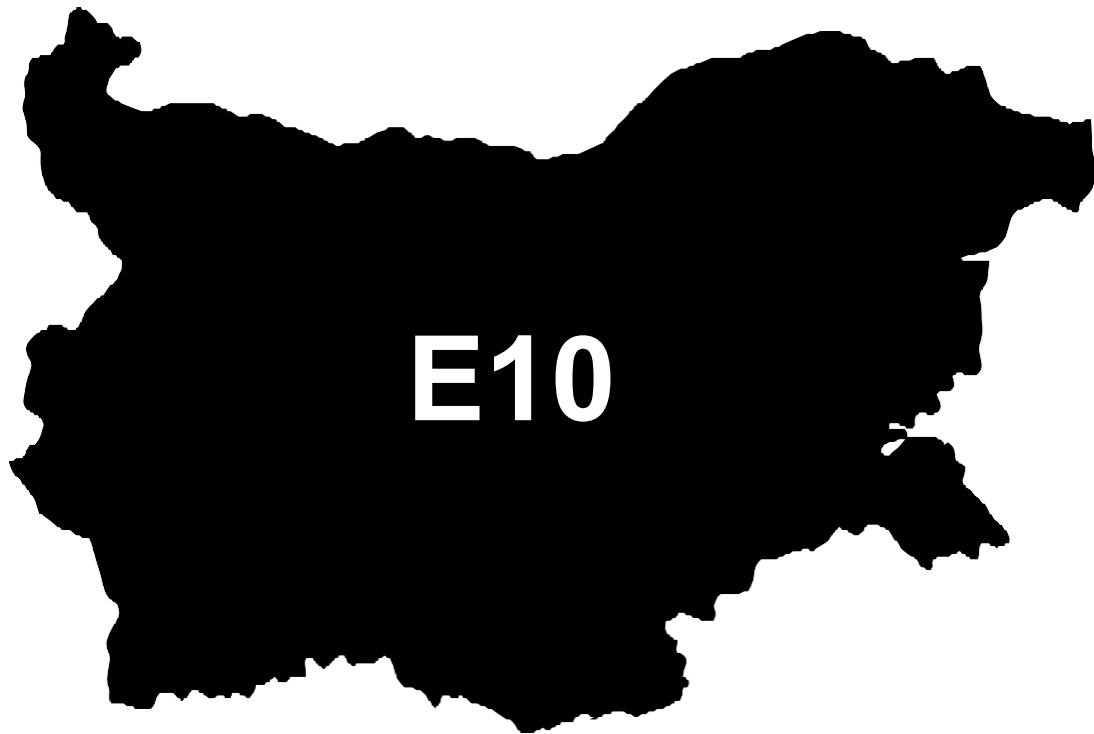
Country Emissions 2024 (MMT/año)	105.6
Target @ 2035 (MMT/year)	59.4
Delta	46.1

%Reduction vs E0	E0	E10	E15	E20	E25	E30
GREET 2024	0.0%	14.3%	17.7%	20.5%	23.3%	26.8%
RED II/III	0.0%	9.5%	12.0%	14.0%	15.9%	18.3%

% Target@2035	E0	E10	E15	E20	E25	E30
GREET 2024	0.0%	2.4%	2.9%	3.4%	3.8%	4.4%
RED II/III	0.0%	2.5%	3.2%	3.7%	4.2%	4.8%

Source: Greet, RED II/III, IPCC, climateactiontracker.org, F90

Ethanol Gasoline Blending in Bulgaria

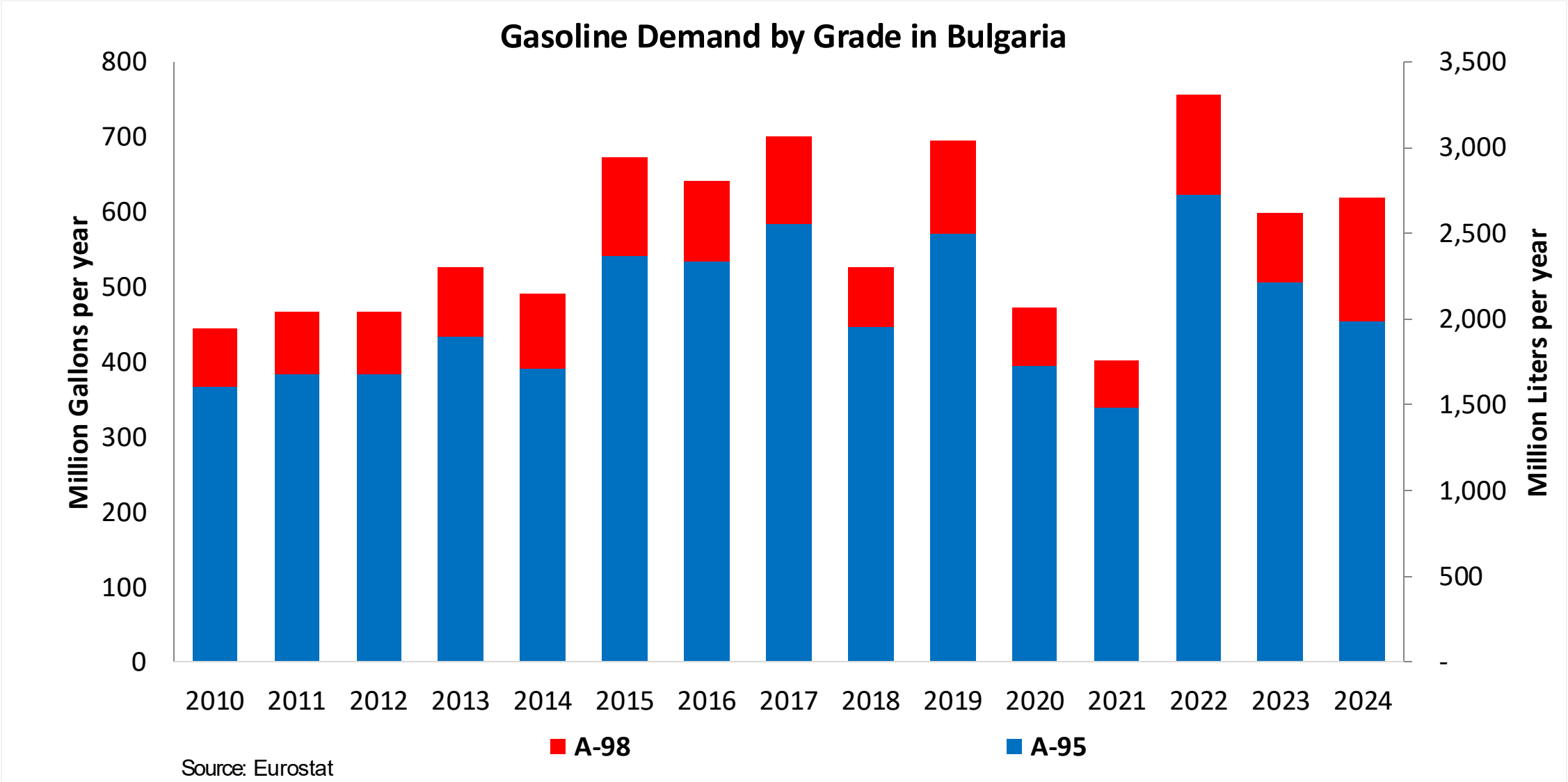


In 2024, gasoline consumption in Bulgaria was 2,342 million liters (619 million gallons) and growth rate was -2.3%. Gasoline production and net imports were 2,199 and 143 million liters (581 and 38 million gallons) correspondingly. Bulgaria complies with the European gasoline standards and offers two main grades: RON 95 and RON 98) with an average octane of 95.5 RON.

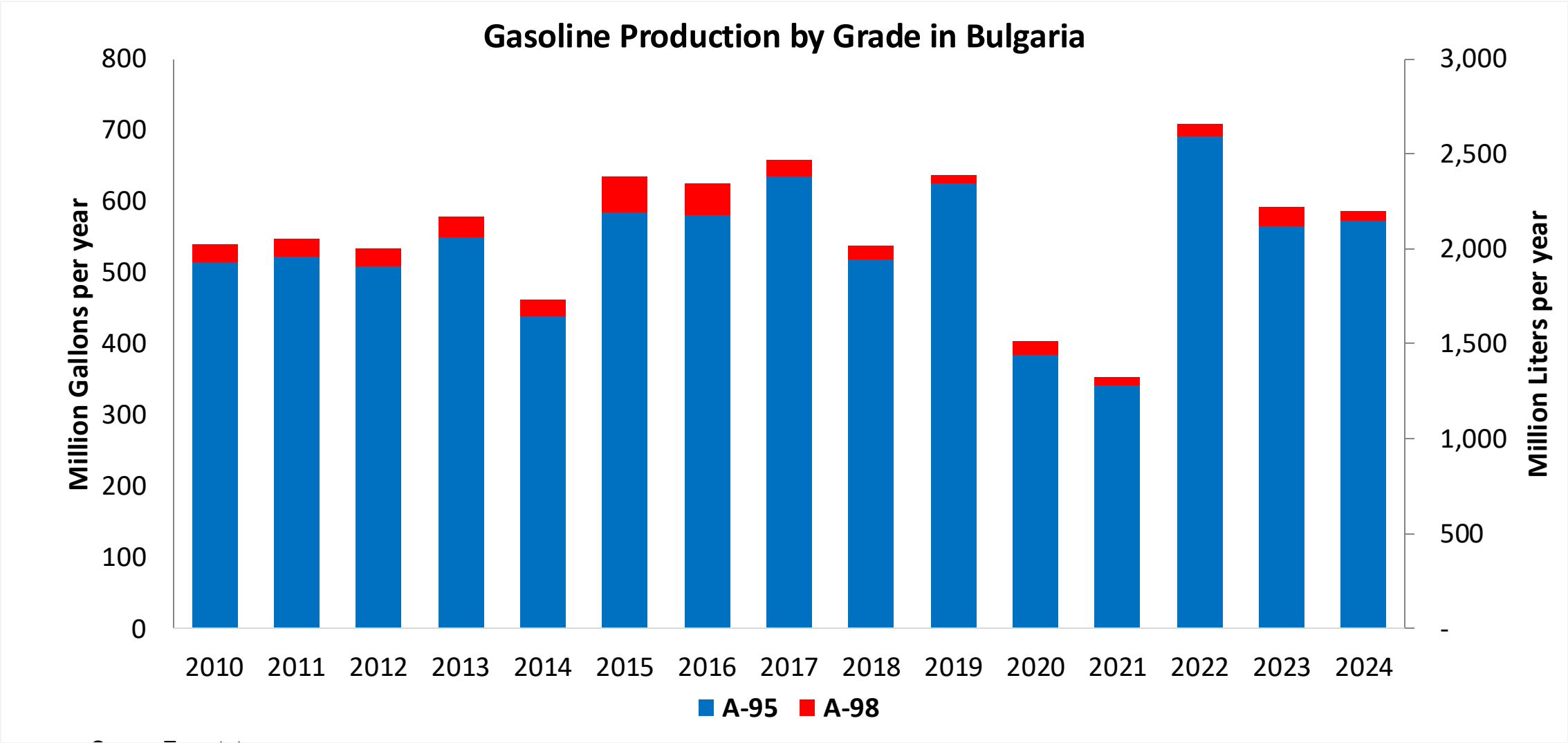
In 2024, ethanol consumption in Bulgaria was 54 million liters (14 million gallons) representing 2.3% of gasoline demand. Ethanol production and Net Imports were 40 and 14 million liters (14 and 4 million gallons) correspondingly. Ethanol sources are Corn 45% and Cereals 55.

Source: Eurostat, European Commission

Gasoline Demand in Bulgaria

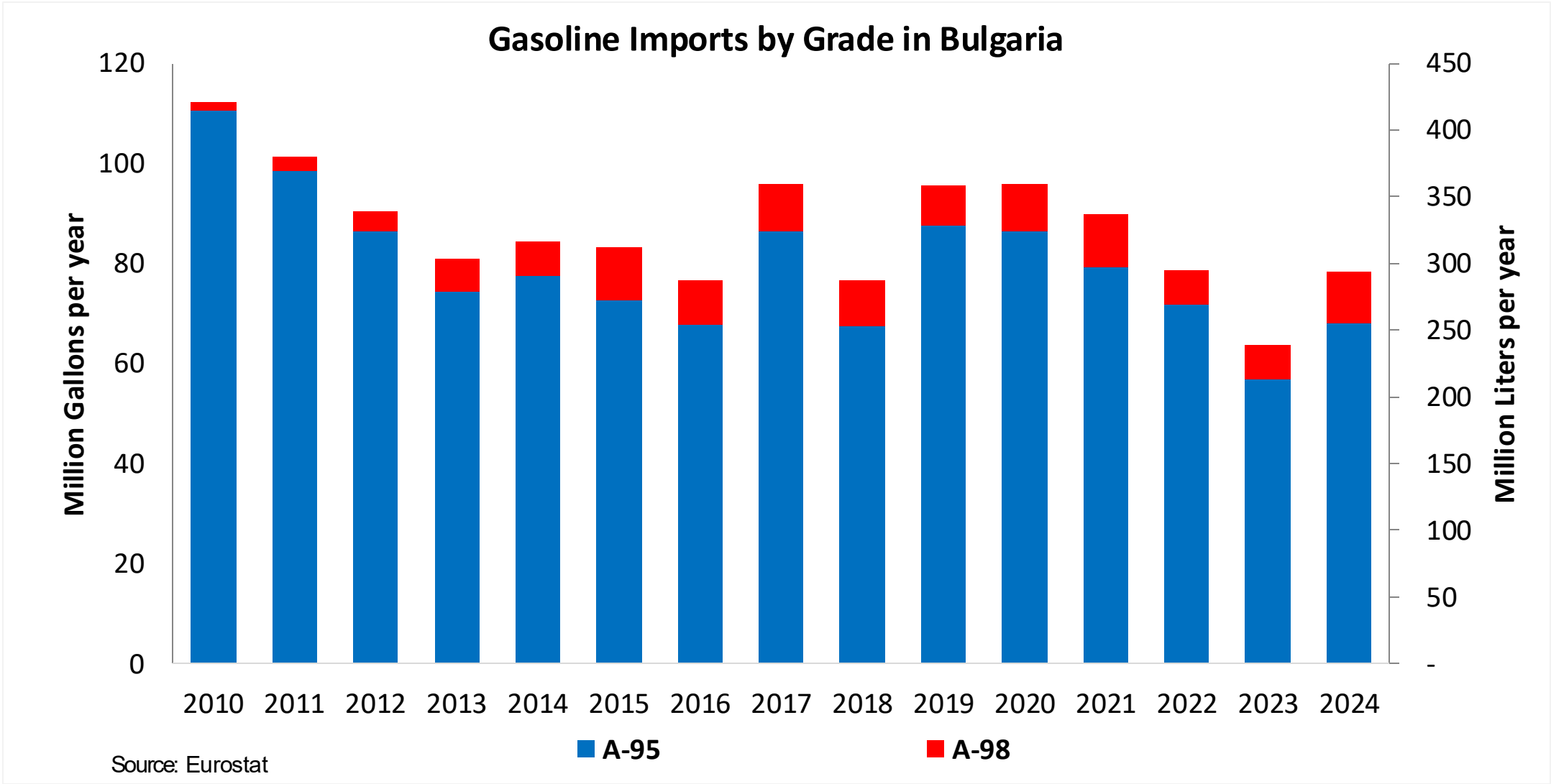


Gasoline Production in Bulgaria

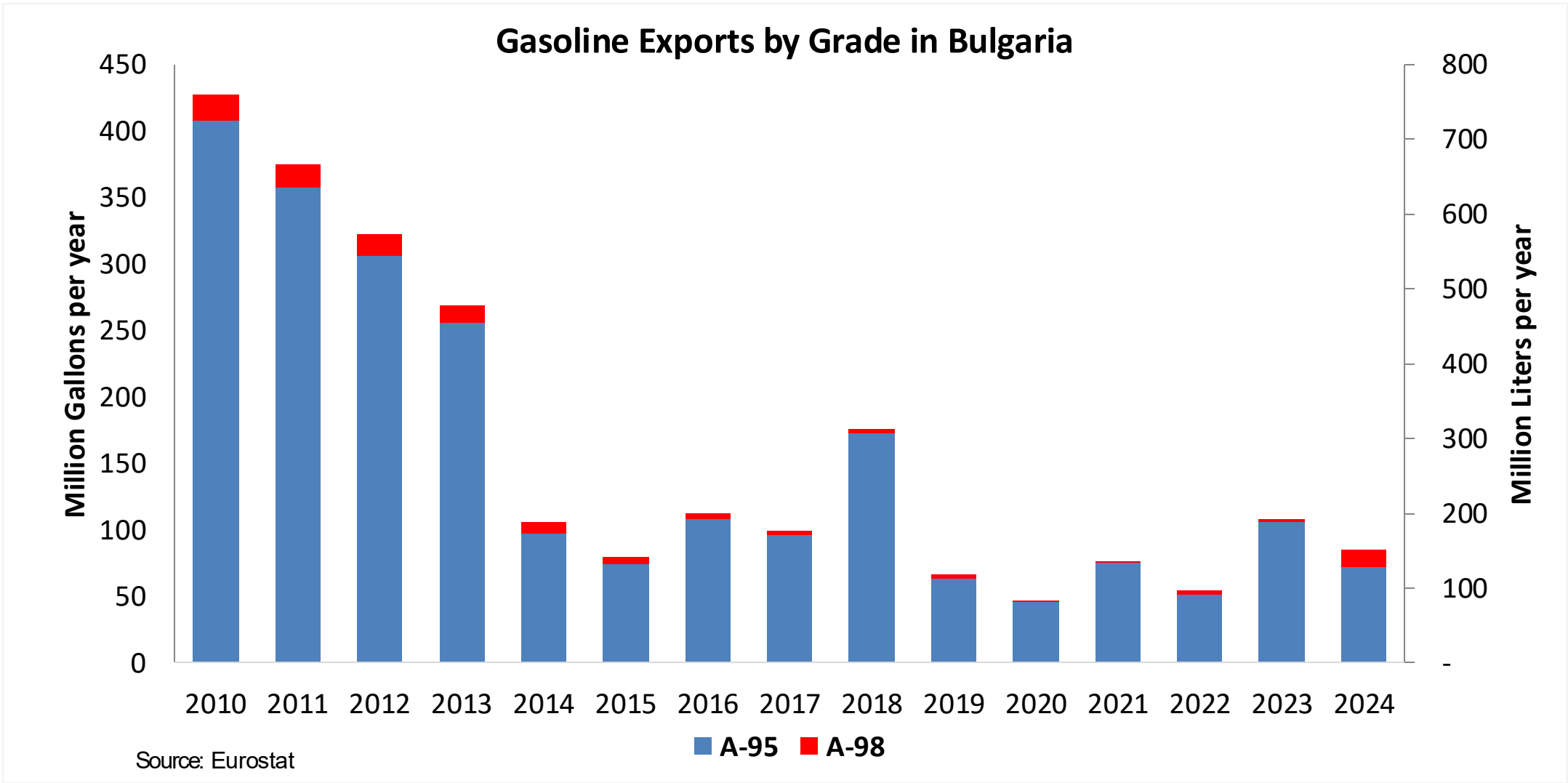


Source: Eurostat

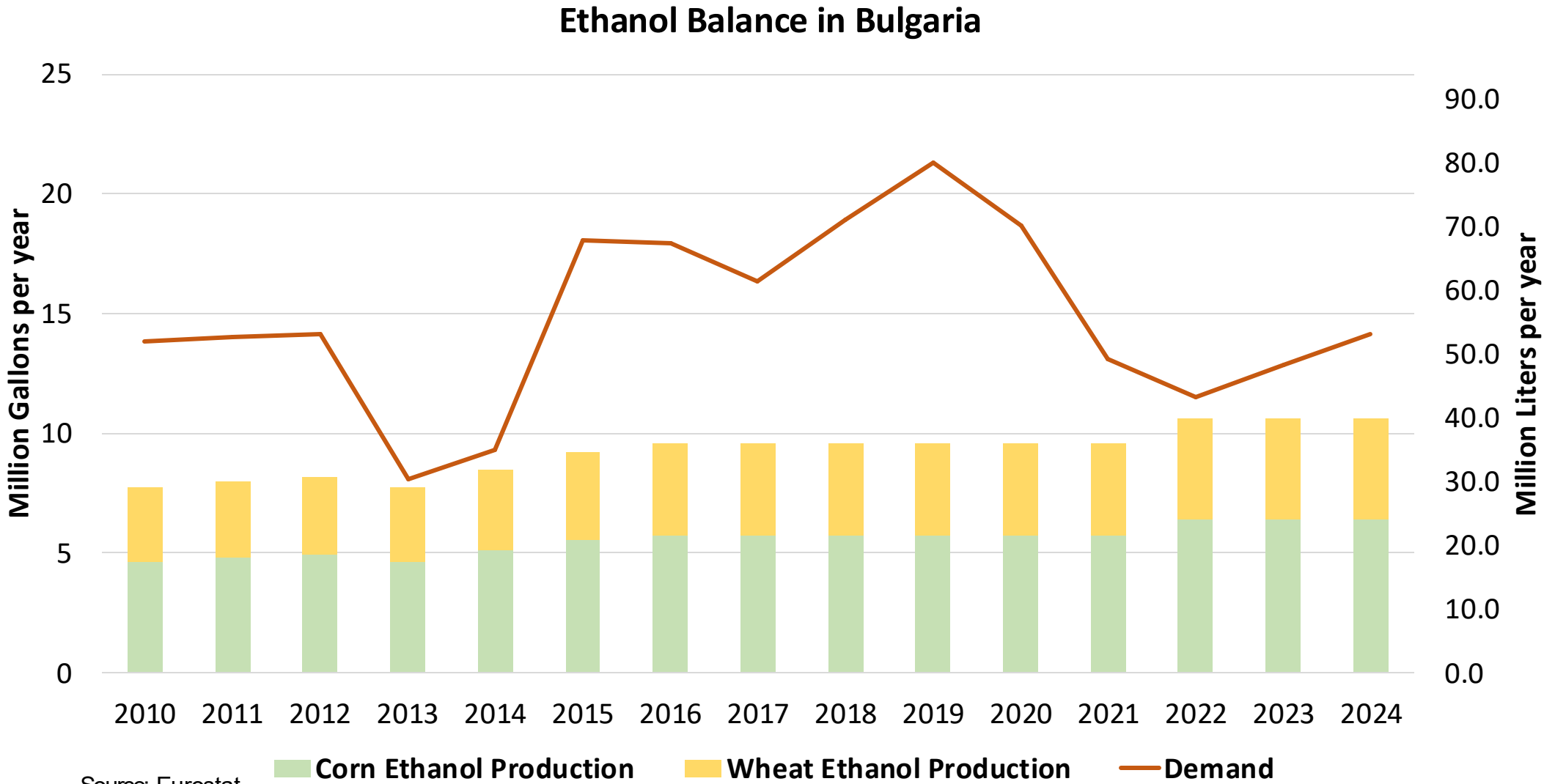
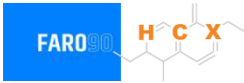
Gasoline Imports in Bulgaria



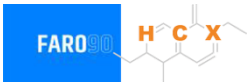
Gasoline Exports in Bulgaria



Ethanol Balance in Bulgaria



Gasoline Quality in Bulgaria

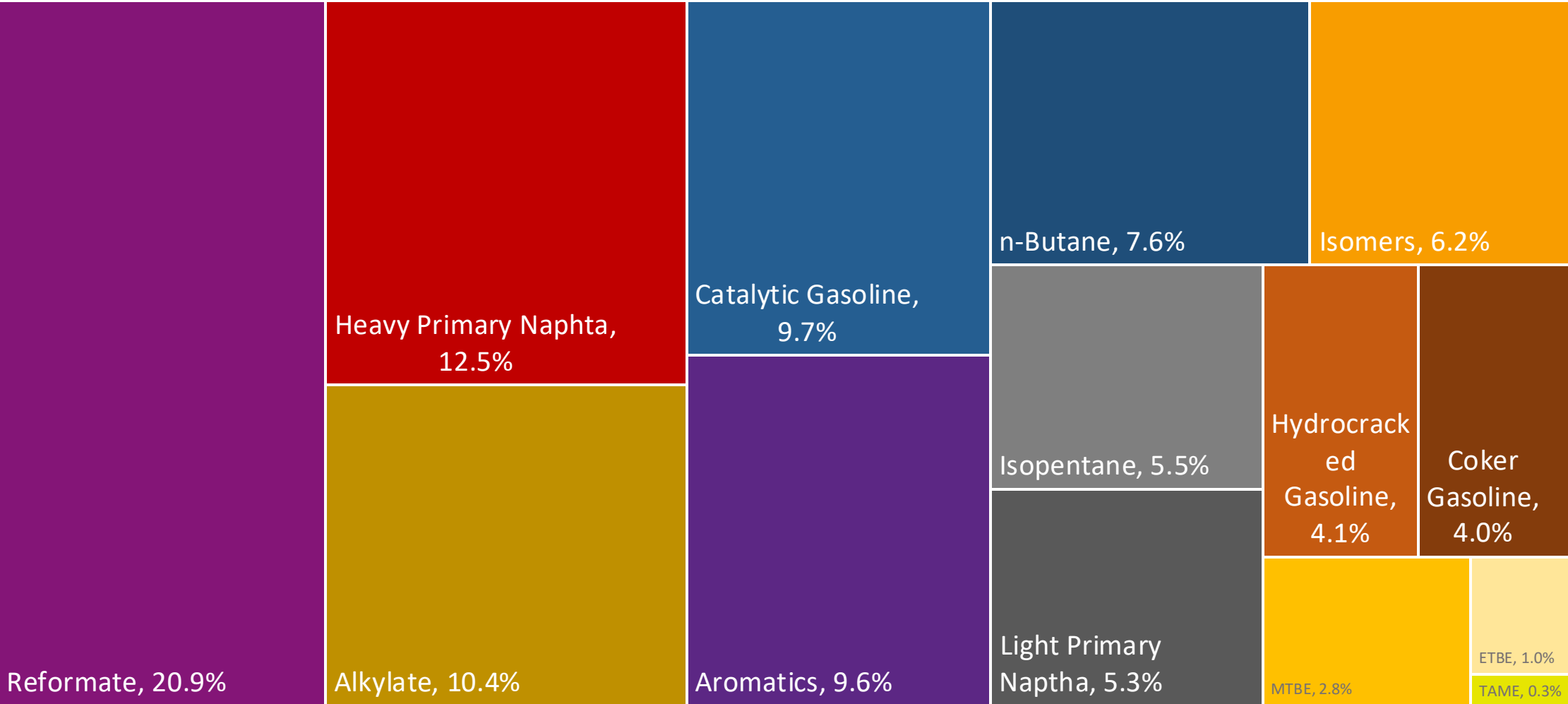
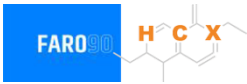


Name	EN 228:2025 (Euro 6); Country Level Fuel Ordinances							
Implementation Date	2025							
Applicability	All EU Contries + UK							
Grade	RON 95 E5		RON 95 E10		RON 98 E5		RON 98 E10	
	Min	Max	Min	Max	Min	Max	Min	Max
RON	95.0		95.0		98.0		98.0	
MON	> 85	> 88	> 85	> 88	> 85	> 88	> 85	> 88
AKI								
Benzene (%vol)	< 1							
Aromatics (% vol)	< 35							
Olefins (%vol)	< 18							
Lead (g/L)	< 0.005							
Manganese (mg/L)	< 2,0							
Sulfur (ppm)	10.0							
Oxygen (%w)*	< 2,7	< 3,7	< 2,7	< 3,7	< 2,7	< 3,7	< 2,7	< 3,7
Ethanol (%vol)	< 5	< 10	< 5	< 10	< 5	< 10	< 5	< 10
PVR 37.8°C (psi)	<> 45 - 100 kPa in 6 Volatility Classes							
Overall Biofuels (%cal)	-							
Advanced Biofuels (%cal)	1.0							
Bioethanol in Gasoline Sales (%cal)	-							
GHG Emission Reduction (%)	-							

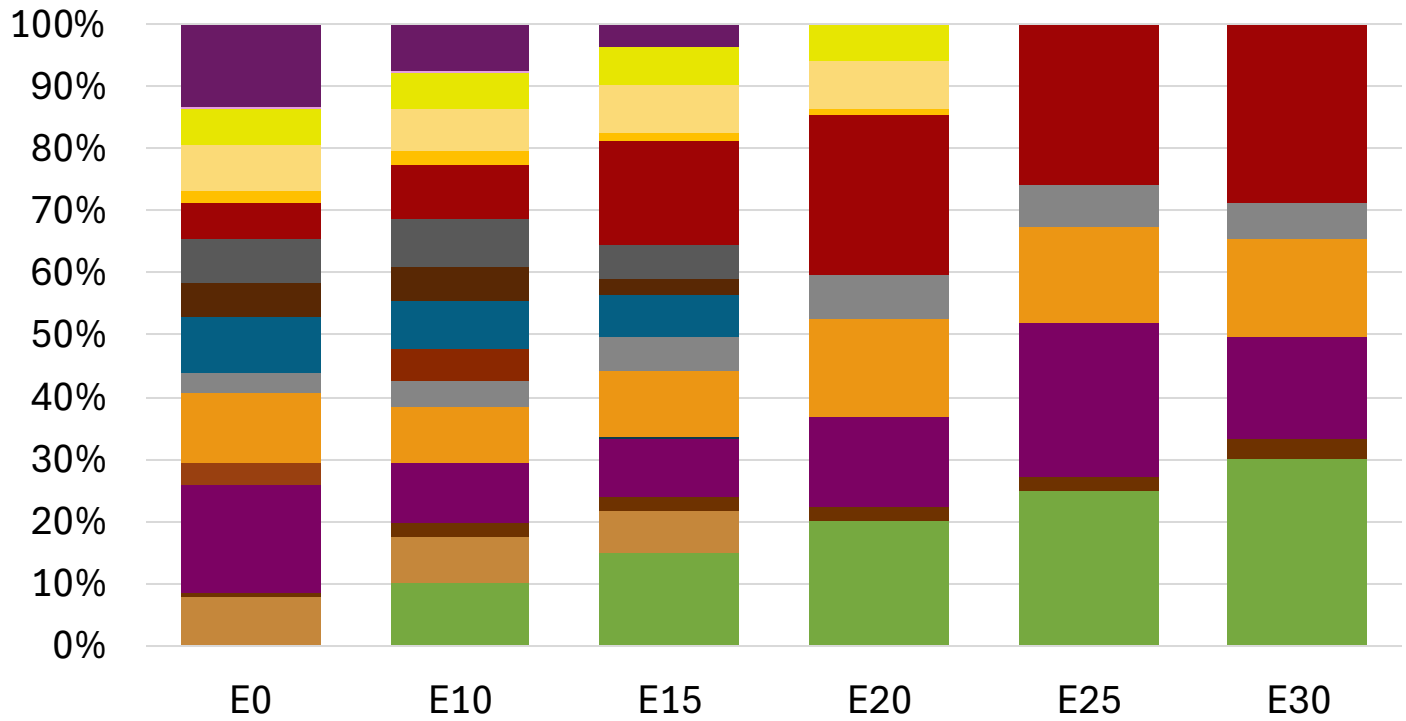
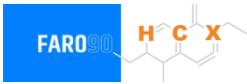
* Methanol 3%, Iso-propyl alcohol 12%, Ter-butyl alcohol 15%, Iso-butyl alcohol 15%, Ethers +5 Carbons 22%, Other Oxygenates 15%

Source: European Commision

Bulgaria Available Blending Components



Bulgaria – Gasoline 95 – Constant Octane

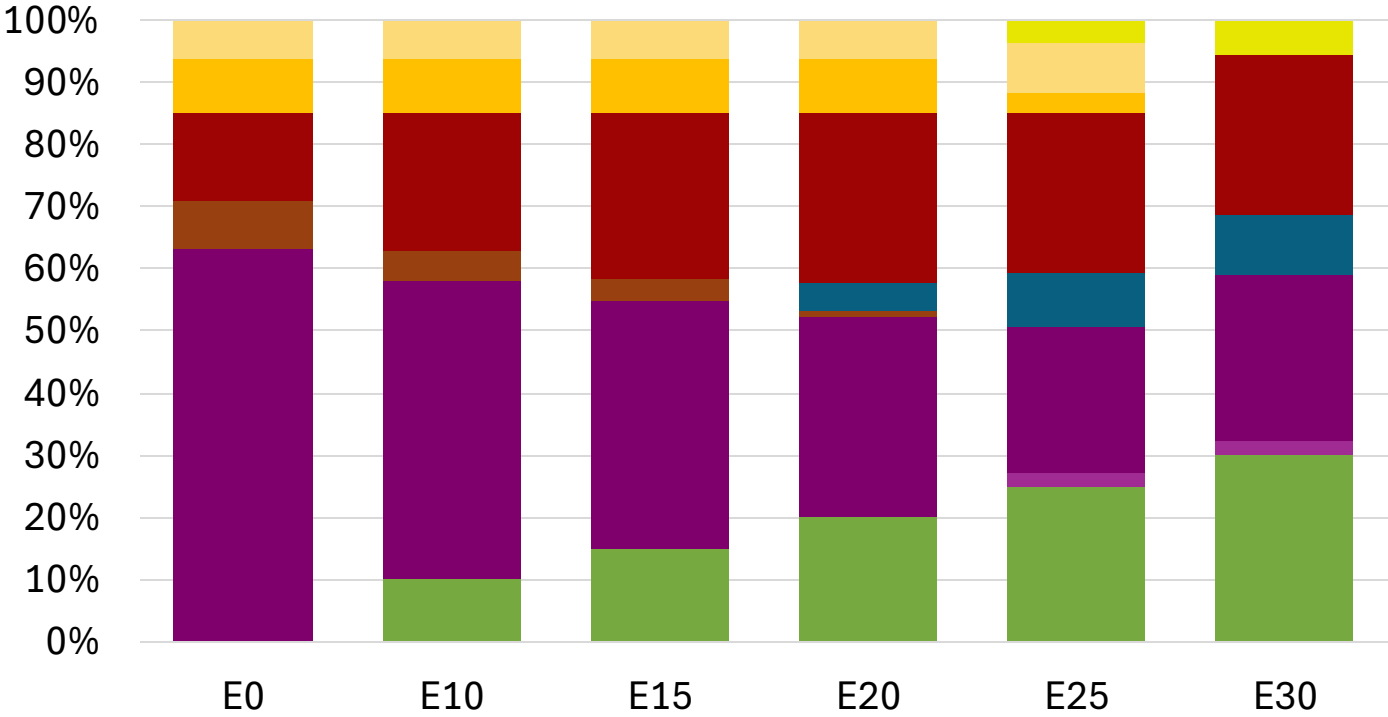
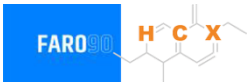


Average CIF 2024 Prices

Octane (RON)	95.0	95.0	95.0	95.0	95.0	95.8
Price (US\$/gal)	\$ 2.294	\$ 2.156	\$ 2.056	\$ 1.941	\$ 1.895	\$ 1.842

Source: Faro90

Bulgaria - Gasoline 98 - Constant Octane



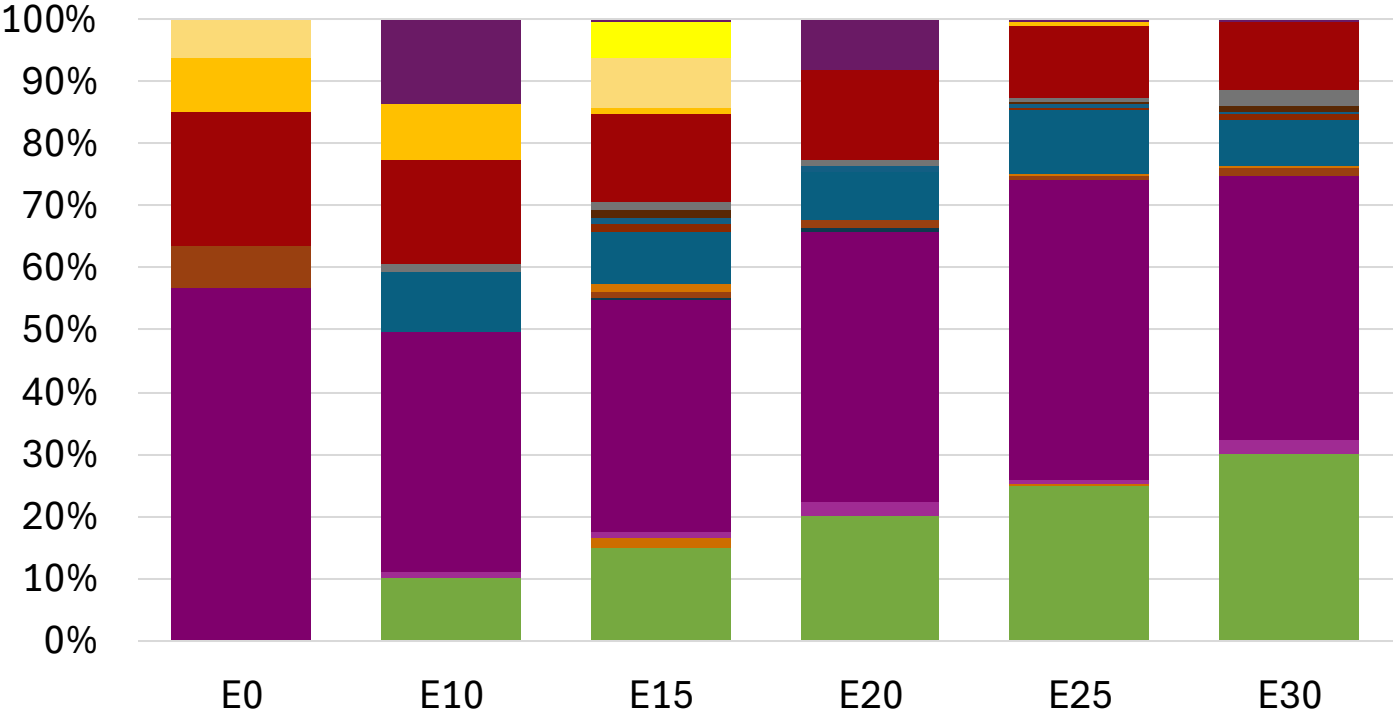
- Ethanol
- Alkylate
- Raffinate
- Reformate
- n-Butane
- Isomerate
- Isopentane
- Hydrocracked Gasoline
- Catalytic Gasoline
- Coker Naphtha
- Light Primary Naphtha
- Heavy Primary Naphtha
- MTBE
- ETBE
- TAME
- Aromatics

Average CIF 2024
Prices

Octane (RON)	98.0	98.0	98.0	98.0	98.0	98.0
Price (US\$/gal)	\$ 2.314	\$ 2.197	\$ 2.128	\$ 2.056	\$ 1.991	\$ 1.941

Source: Faro90

Bulgaria – Gasoline 95 – Increased Octane

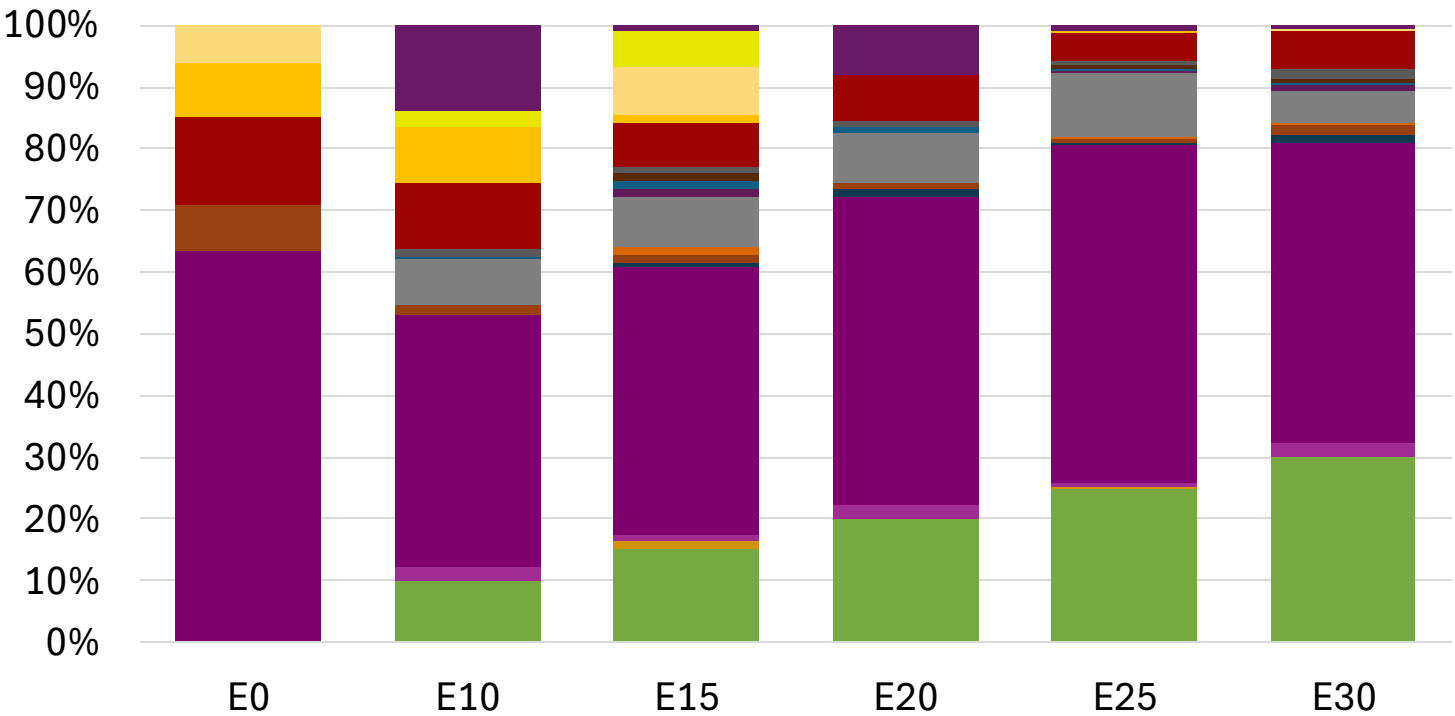


Average CIF 2024
Prices

Octane (RON)	95.0	96.2	98.3	98.7	100.5	101.8
Price (US\$/gal)	\$ 2.182	\$ 2.182	\$ 2.182	\$ 2.182	\$ 2.182	\$ 2.182

Source: Faro90

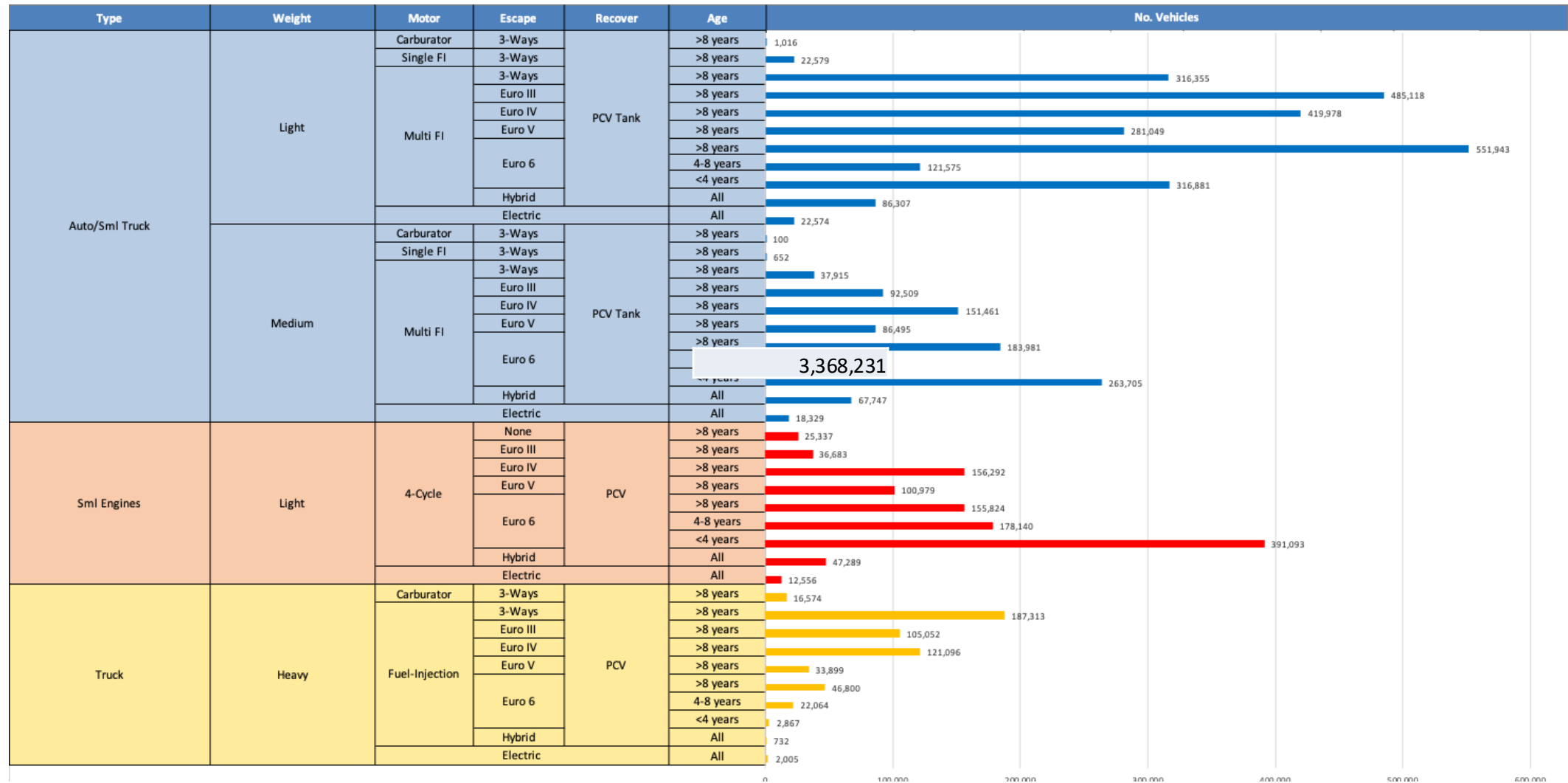
Bulgaria – Gasoline 98 – Increased Octane



Average CIF 2024
Prices

Source: Faro90

Gasoline Vehicle Fleet - Bulgaria

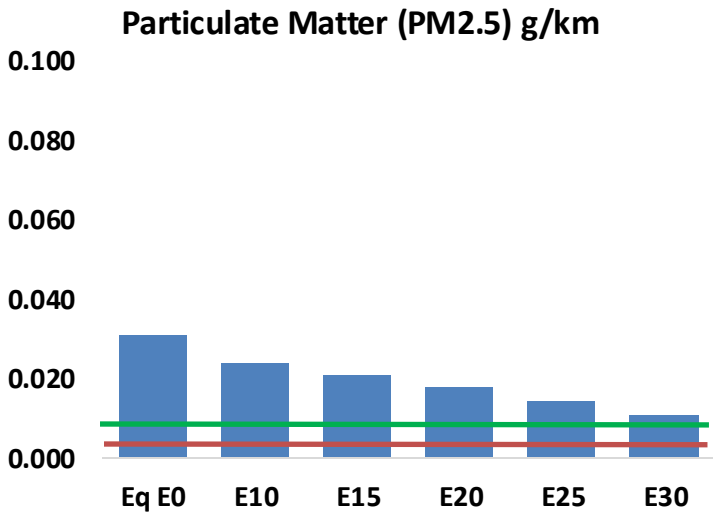
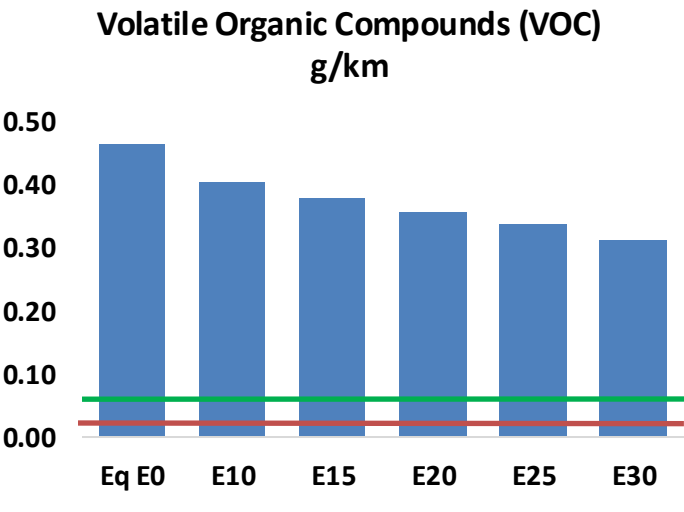
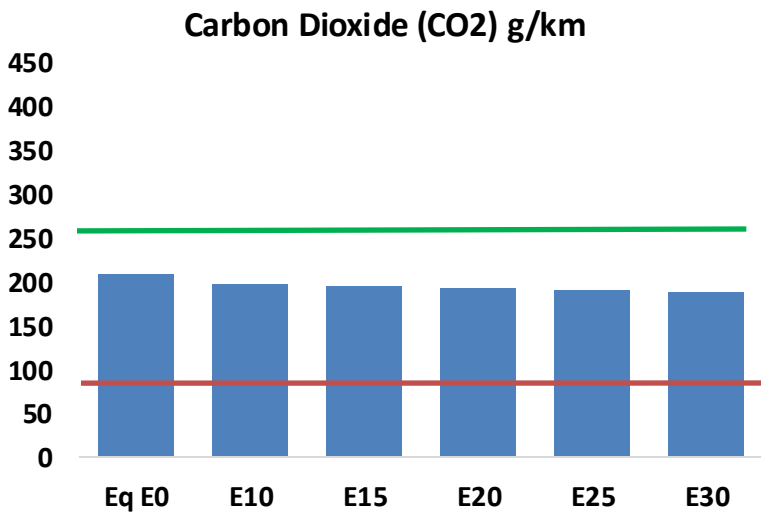
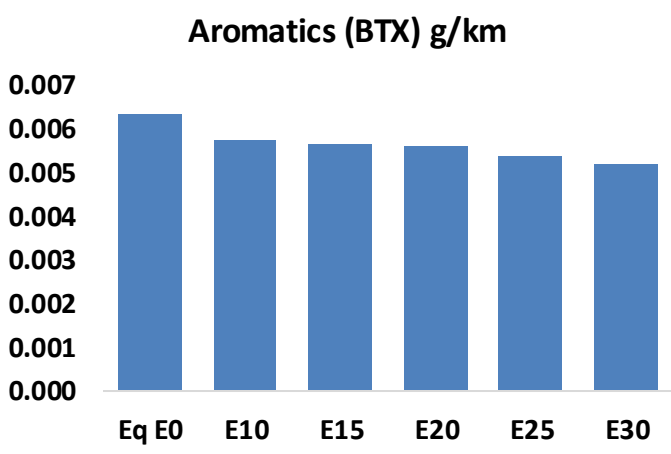
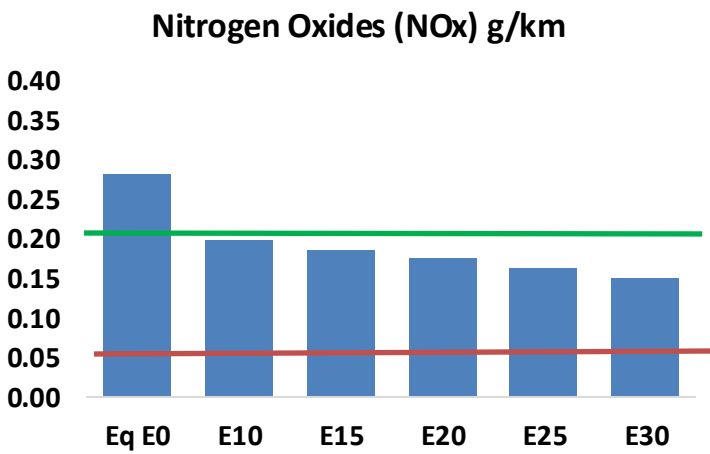
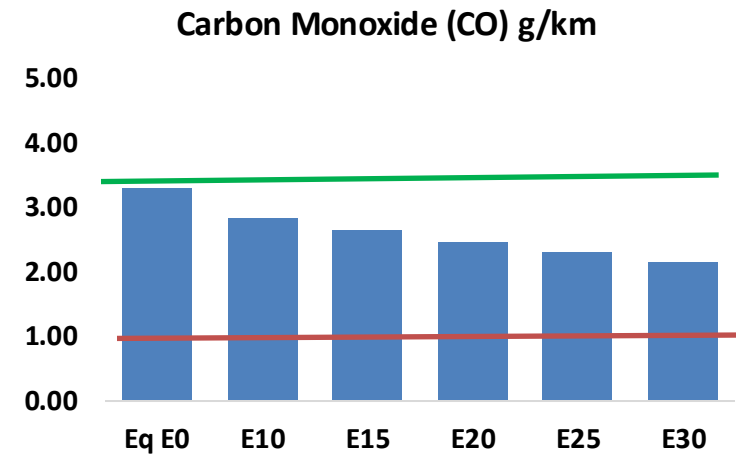
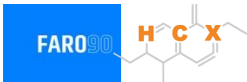


Vehicle Fleet: 5,239,744
Gasoline Vehicle Fleet: 3,368,231

Source: Eurostat, ACEA

Bulgaria – Vehicular Emissions

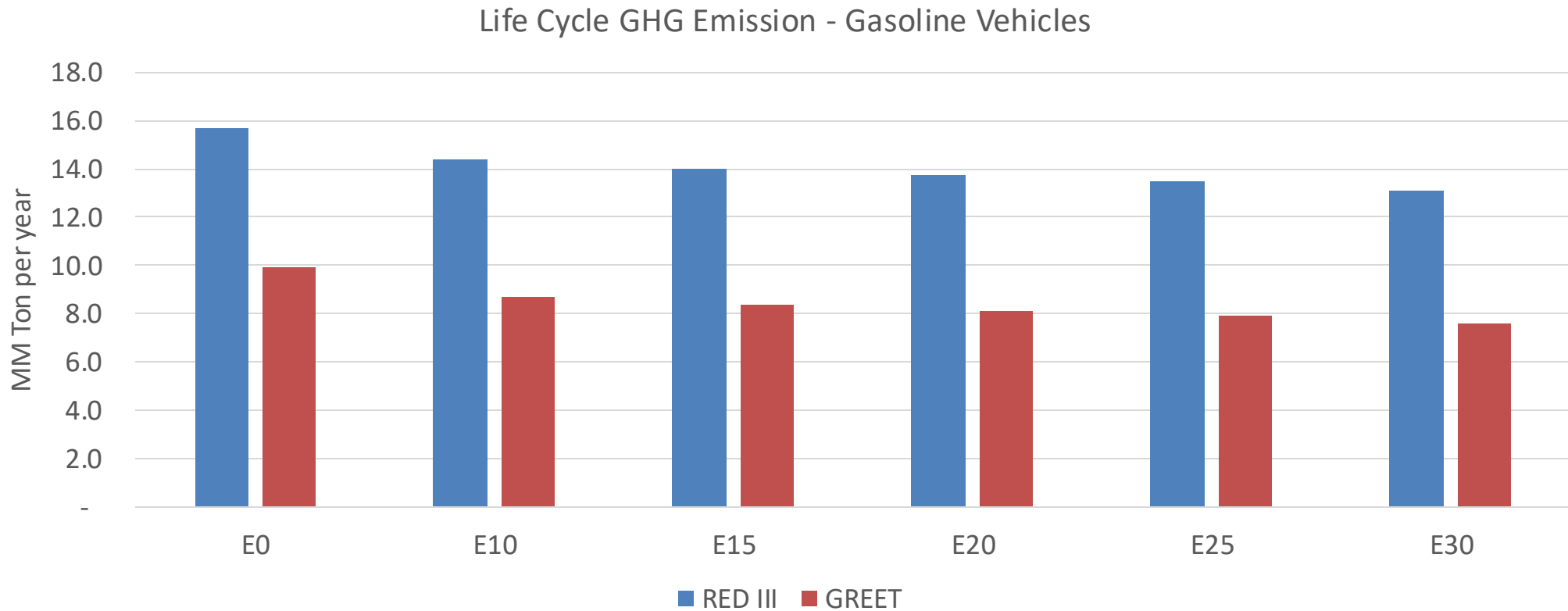
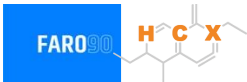
TIER III BIN 125 United States
Euro 6 Standard



Bulgaria – Vehicular Emissions

Vehicular Fleet Emissions (kg/day)	Eq E0	E5	E10	E15	E20	E25	E30	Reduction (%)	
								E10	E20
CO	263,559	244,735	225,910	210,765	195,653	184,072	171,172	-14%	-26%
CO2	16,709,105	16,279,442	15,873,649	15,554,505	15,396,581	15,309,889	15,027,684	-5%	-8%
VOC	37,225	34,766	32,308	30,372	28,460	27,000	25,014	-13%	-24%
NOx	22,636	16,977	15,845	14,934	14,084	13,147	12,156	-30%	-38%
SOx	199	184	169	156	144	131	117	-15%	-28%
NH3	5,918	5,859	5,800	5,828	5,894	5,940	5,978	-2%	0%
N2O	304	302	301	302	309	312	316	-1%	2%
CH4	8,123	8,123	8,123	8,245	8,423	8,578	8,724	0%	4%
Lead	-	-	-	-	-	-	-	0%	0%
Butadiene	271	261	252	247	243	240	235	-7%	-10%
Acetaldehyde	701	804	1,180	1,797	2,449	2,798	3,306	68%	249%
Formaldehyde	2,445	2,377	2,771	3,254	3,409	3,772	4,106	13%	39%
Benzene	507	487	461	454	450	431	417	-9%	-11%
PM2.5	1,033	917	801	696	591	480	363	-22%	-43%
PM10	1,437	1,276	1,115	968	822	667	506	-22%	-43%

Bulgaria – Gasoline Vehicle Fleet Life Cycle GHG Emissions



Country Emissions 2024 (MMT/año)	15.3
Target @ 2035 (MMT/year)	3.1
Delta	12.1

%Reduction vs E0	E0	E10	E15	E20	E25	E30
GREET 2024	0.0%	12.5%	15.7%	18.2%	20.7%	23.9%
RED II/III	0.0%	8.4%	10.6%	12.4%	14.2%	16.4%

% Target@2035	E0	E10	E15	E20	E25	E30
GREET 2024	0.0%	10.2%	12.8%	14.9%	16.9%	19.6%
RED II/III	0.0%	10.8%	13.7%	16.1%	18.3%	21.1%

Source: Greet, RED II/III, IPCC, climateactiontracker.org, F90

Ethanol Gasoline Blending in Croatia

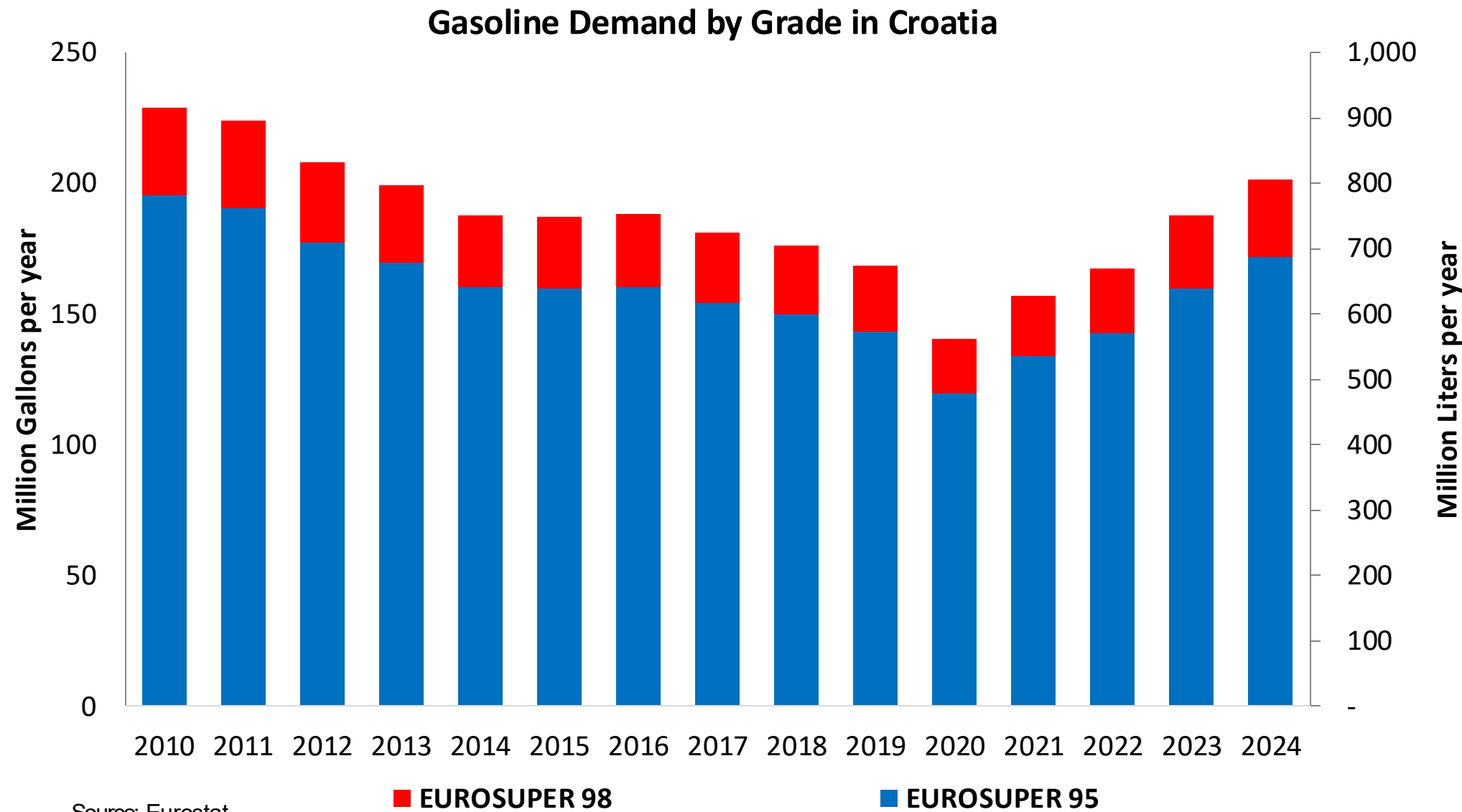


In 2024, gasoline consumption in Croatia was 763 million liters (202 million gallons) and growth rate was 3.7%. Gasoline production and net imports were 709 and 82 million liters (187 and 22 million gallons) correspondingly. Croatia complies with the European gasoline standards and offers two main grades: RON 95 and RON 98) with an average octane of 95.3 RON.

Croatia does not produce or consumes bioethanol in a commercial level. There is a small production of generic ethanol that is exported in minimal quantities to Italy, Slovenia, Austria and Bosnia.

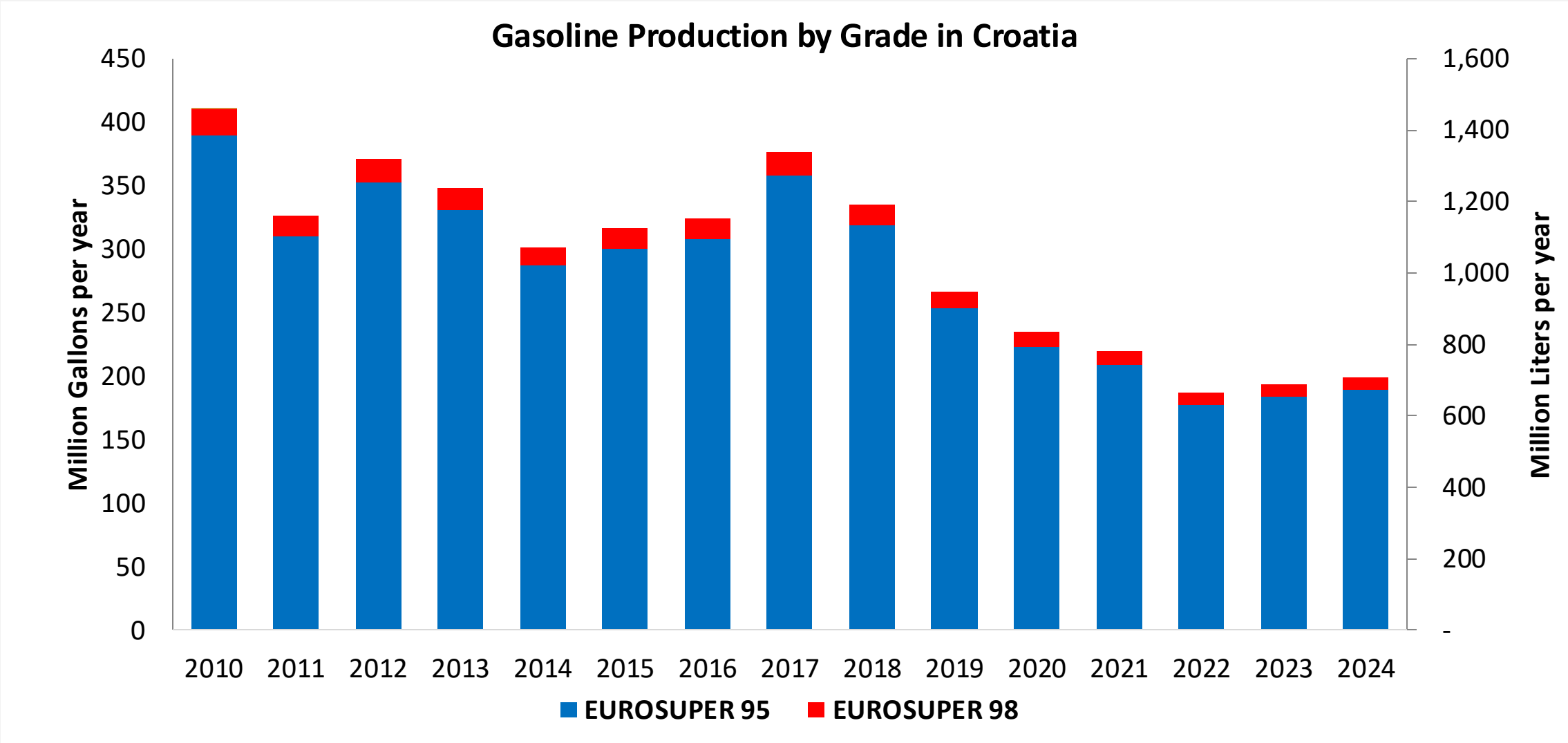
Source: Eurostat, European Commission

Gasoline Demand in Croatia



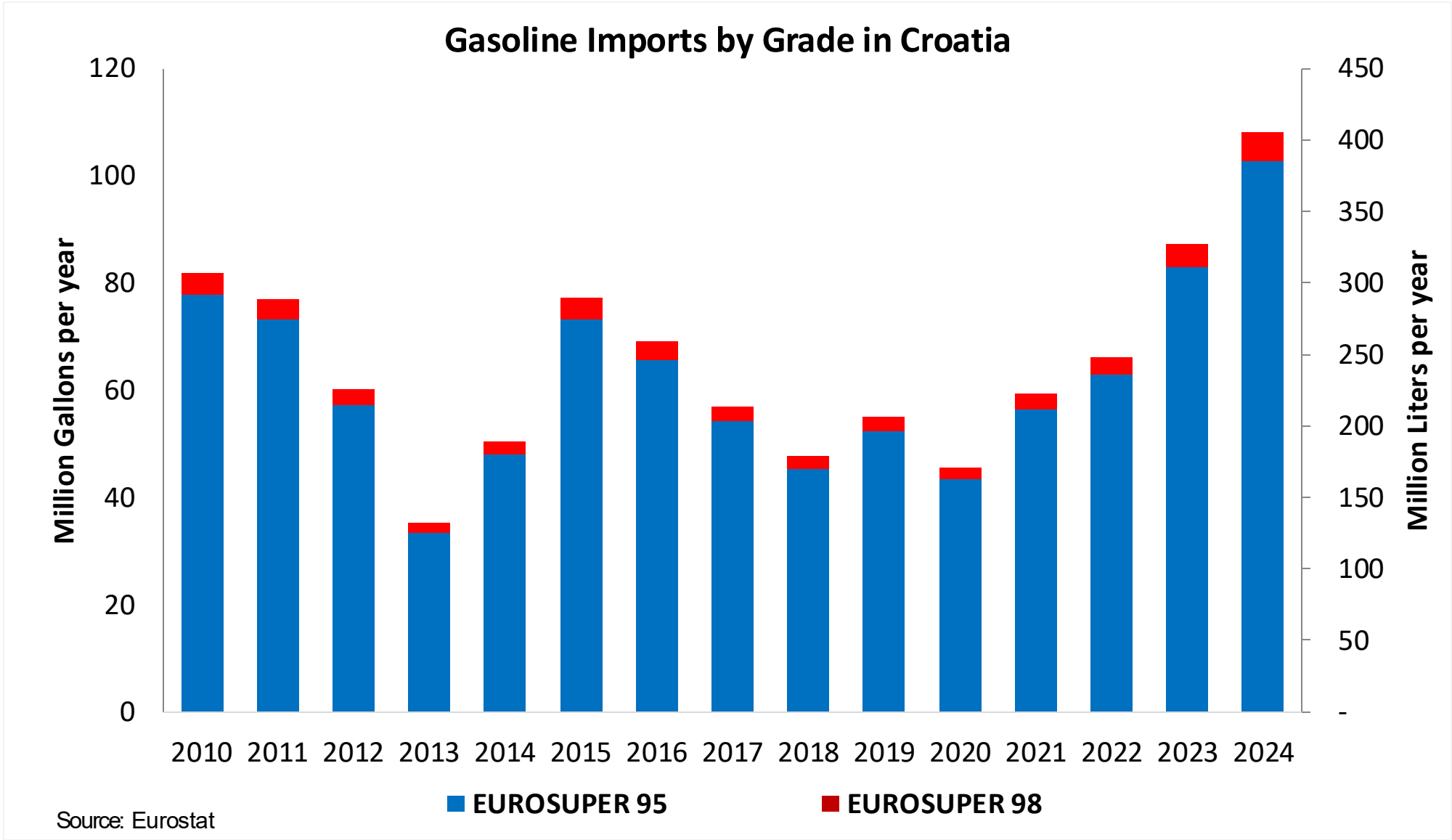
Source: Eurostat

Gasoline Production in Croatia

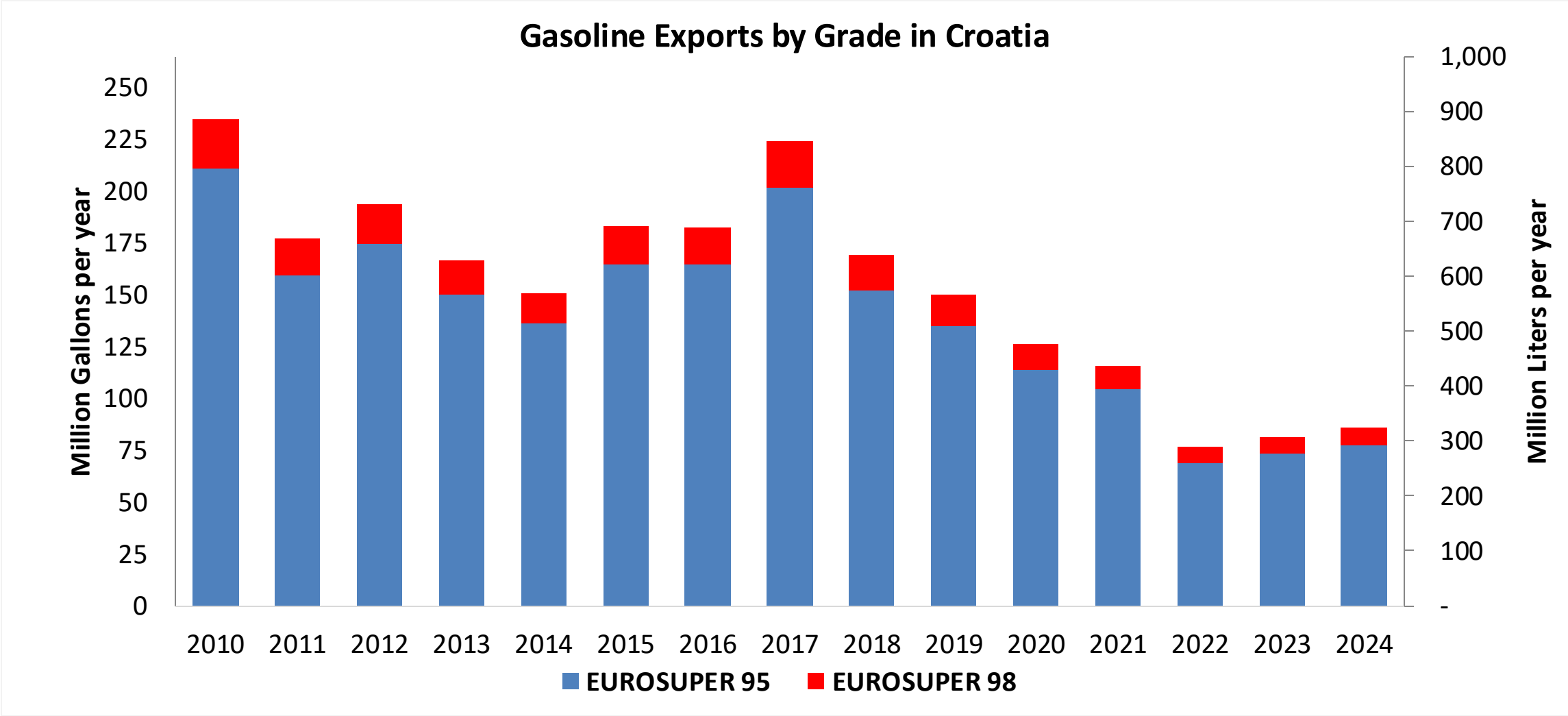


Source: Eurostat

Gasoline Imports in Croatia

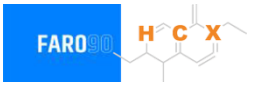


Gasoline Exports in Croatia



Source: Eurostat

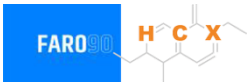
Ethanol Balance in Croatia



Croatia does not produce or consumes bioethanol in a commercial level. There is a small production of generic ethanol that is exported in minimal quantities to Italy, Slovenia, Austria and Bosnia.

Source: The Global Economy, WITS/Comtrade, Total Croatia News, Axens

Gasoline Quality in Croatia

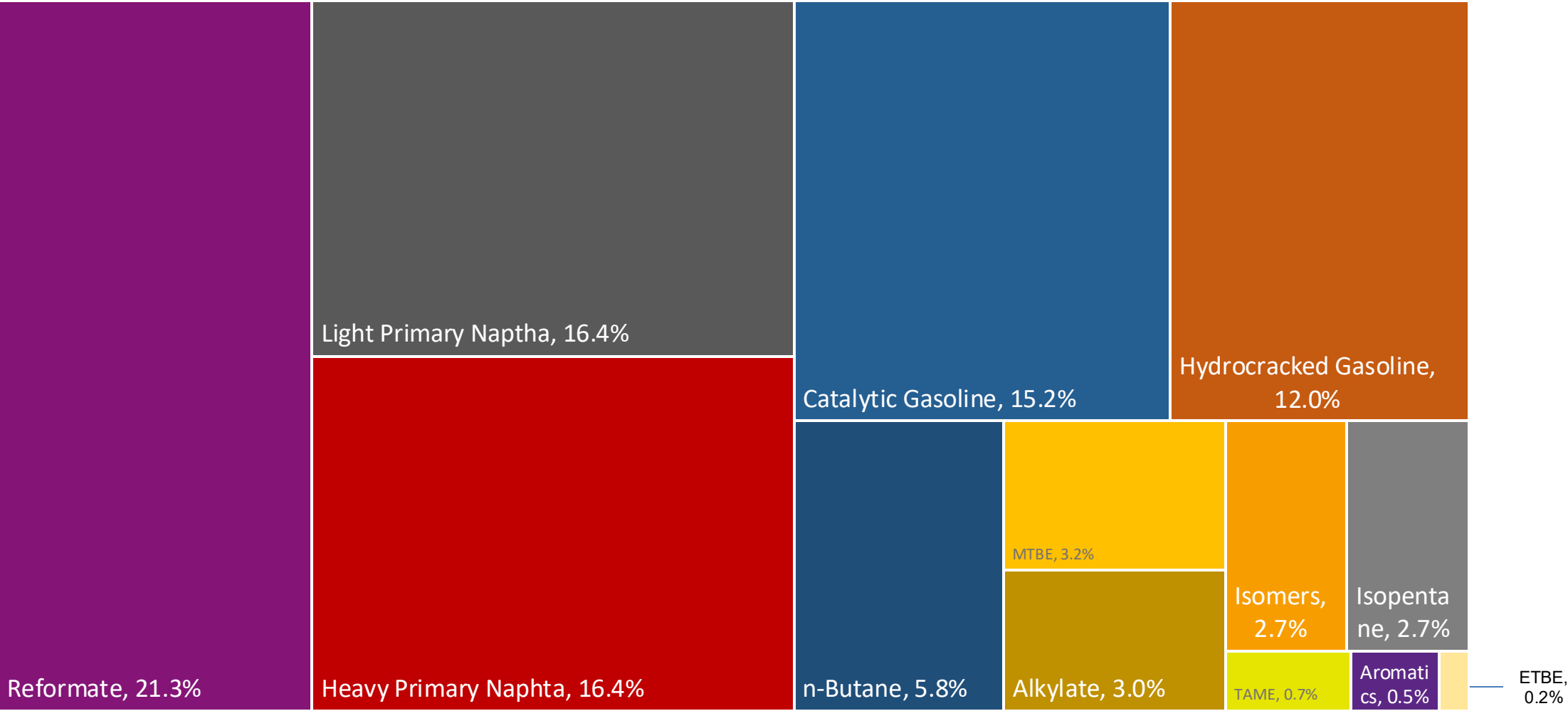


Name	EN 228:2025 (Euro 6); Country Level Fuel Ordinances							
Implementation Date	2025							
Applicability	All EU Contries + UK							
Grade	RON 95 E5		RON 95 E10		RON 98 E5		RON 98 E10	
	Min	Max	Min	Max	Min	Max	Min	Max
RON	95.0		95.0		98.0		98.0	
MON	> 85	> 88	> 85	> 88	> 85	> 88	> 85	> 88
AKI								
Benzene (%vol)	< 1							
Aromatics (% vol)	< 35							
Olefins (%vol)	< 18							
Lead (g/L)	< 0.005							
Manganese (mg/L)	< 2,0							
Sulfur (ppm)	10.0							
Oxygen (%w)*	< 2,7	< 3,7	< 2,7	< 3,7	< 2,7	< 3,7	< 2,7	< 3,7
Ethanol (%vol)	< 5	< 10	< 5	< 10	< 5	< 10	< 5	< 10
PVR 37.8°C (psi)	<> 45 - 100 kPa in 6 Volatility Classes							
Overall Biofuels (%cal)	8.8							
Advanced Biofuels (%cal)	-							
Bioethanol in Gasoline Sales (%cal)	1.0							
GHG Emission Reduction (%)	-							

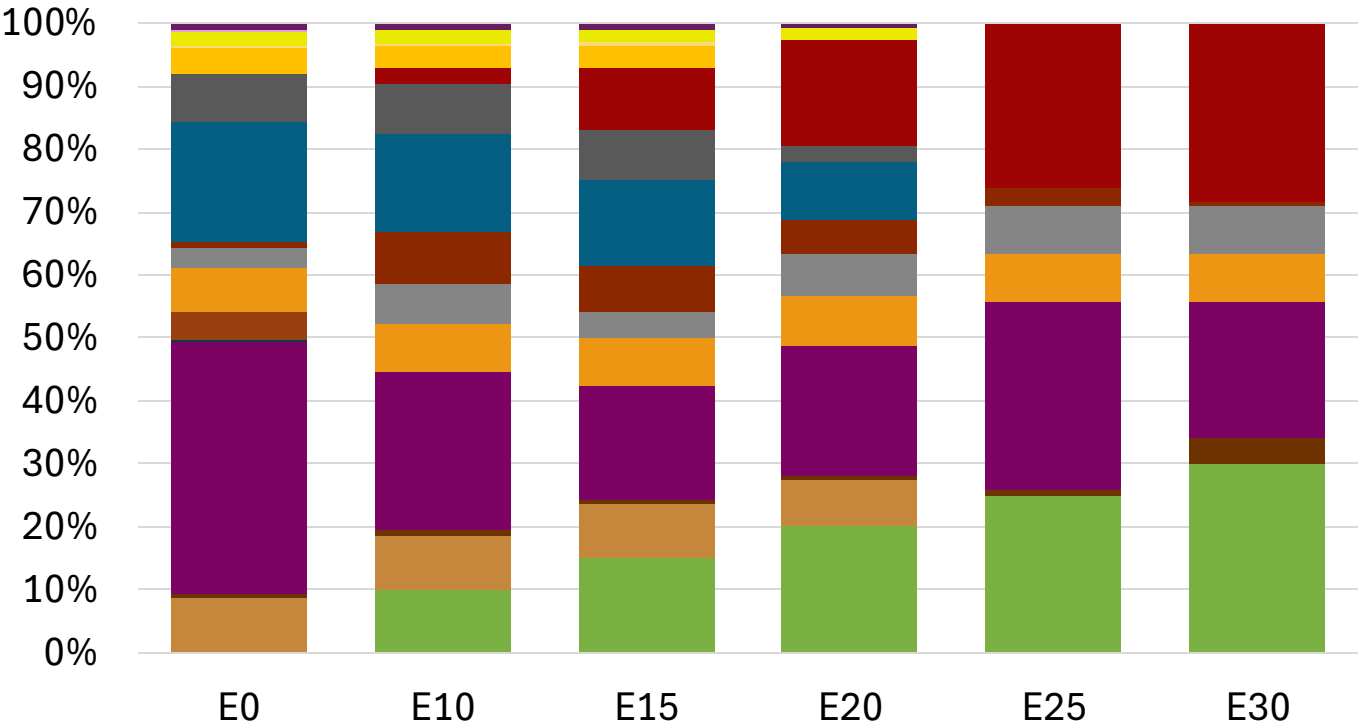
* Methanol 3%, Iso-propyl alcohol 12%, Ter-butyl alcohol 15%, Iso-butyl alcohol 15%, Ethers +5 Carbons 22%, Other Oxygenates 15%

Source: European Commision

Croatia Available Blending Components



Croatia – Gasoline 95 – Constant Octane

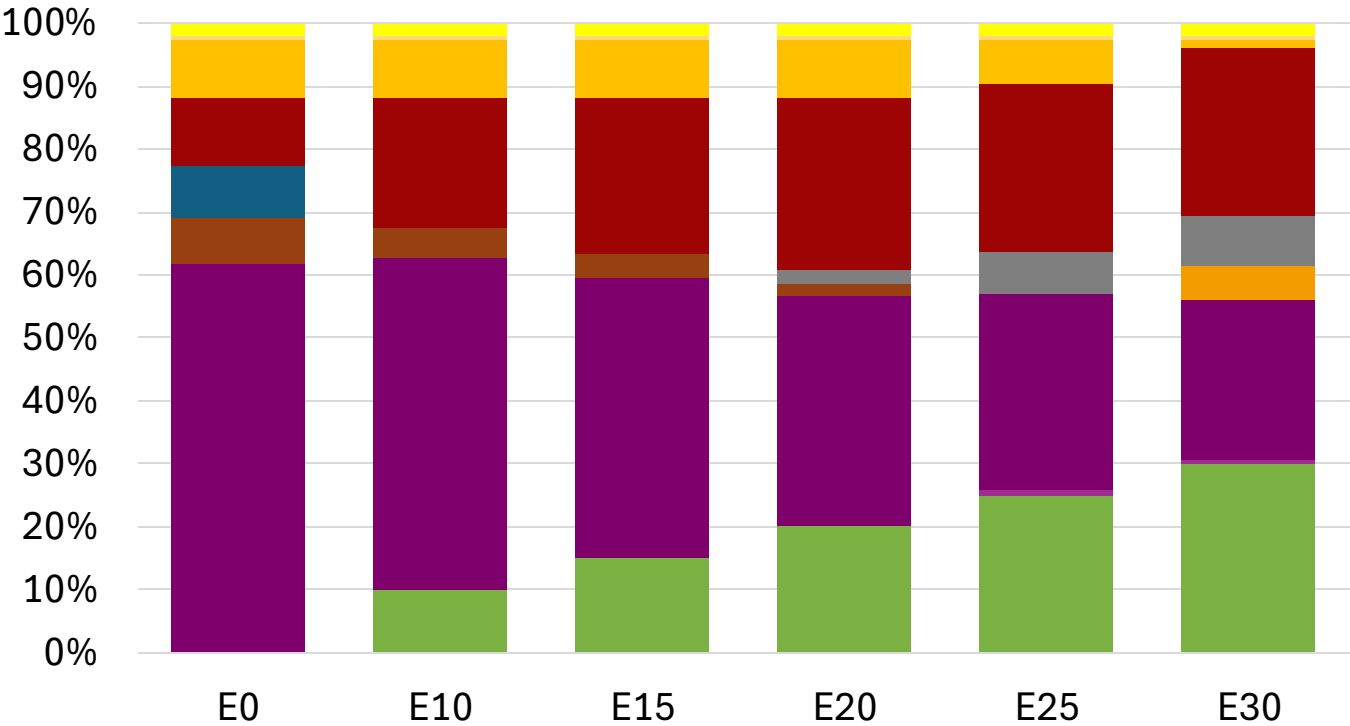
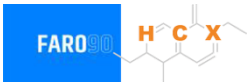


Average CIF 2024
Prices

Octane (RON)	95.0	95.0	95.0	95.0	95.4	95.8
Price (US\$/gal)	\$ 2.258	\$ 2.145	\$ 2.062	\$ 1.986	\$ 1.891	\$ 1.817

Source: Faro90

Croatia - Gasoline 98 - Constant Octane

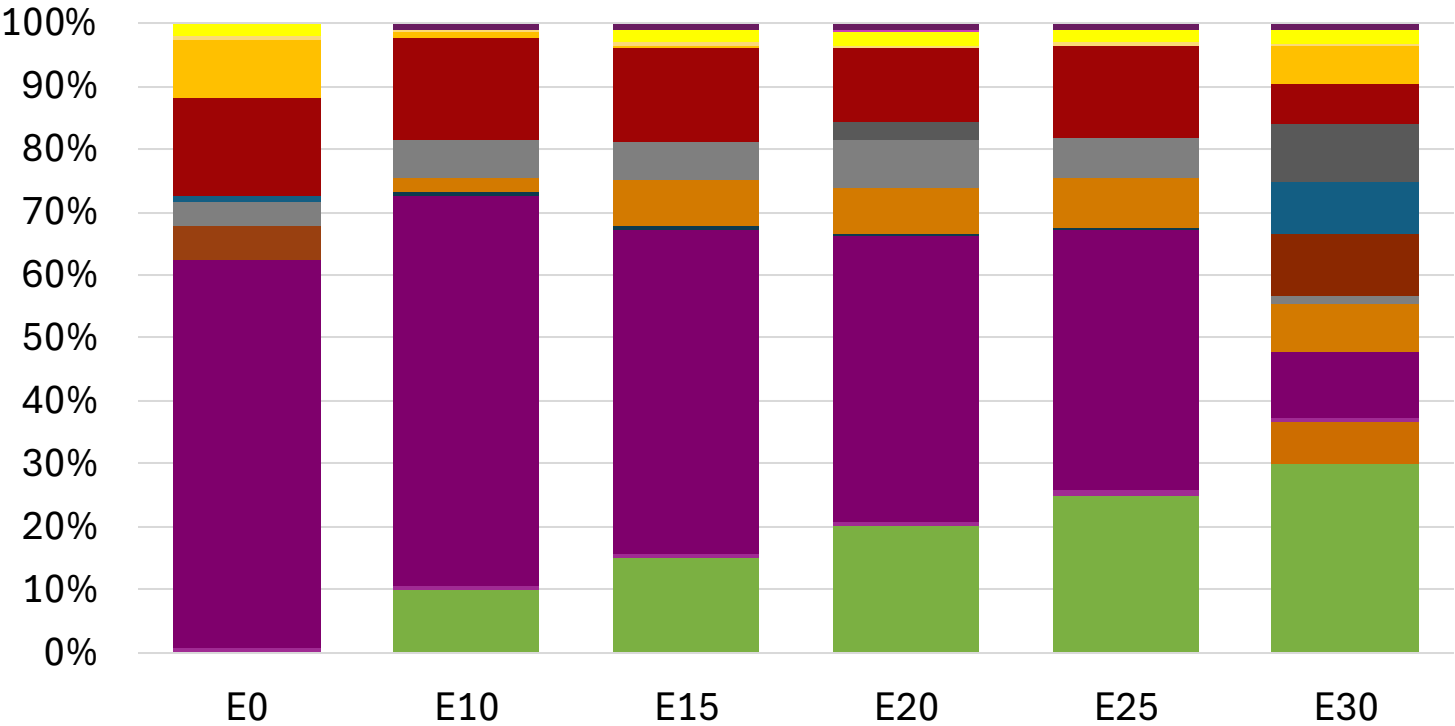


Average CIF 2024 Prices

Octane (RON)	98.0	98.0	98.0	98.0	98.0	98.0
Price (US\$/gal)	\$ 2.307	\$ 2.188	\$ 2.119	\$ 2.049	\$ 1.979	\$ 1.926

Source: Faro90

Croatia – Gasoline 95 – Increased Octane

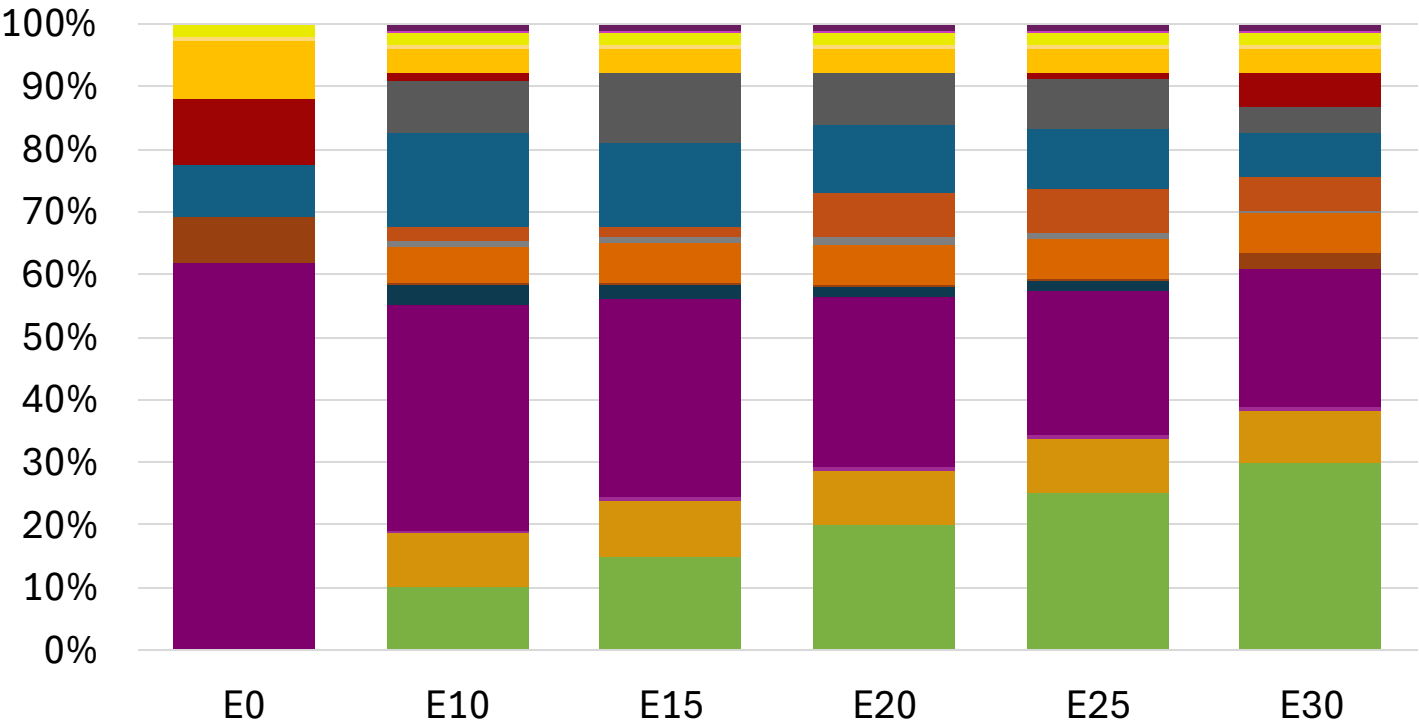
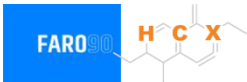


Average CIF 2024
Prices

Octane (RON)	95.0	95.9	97.5	99.0	100.5	101.8
Price (US\$/gal)	\$ 2.177	\$ 2.177	\$ 2.177	\$ 2.177	\$ 2.177	\$ 2.177

Source: Faro90

Croatia – Gasoline 98 – Increased Octane

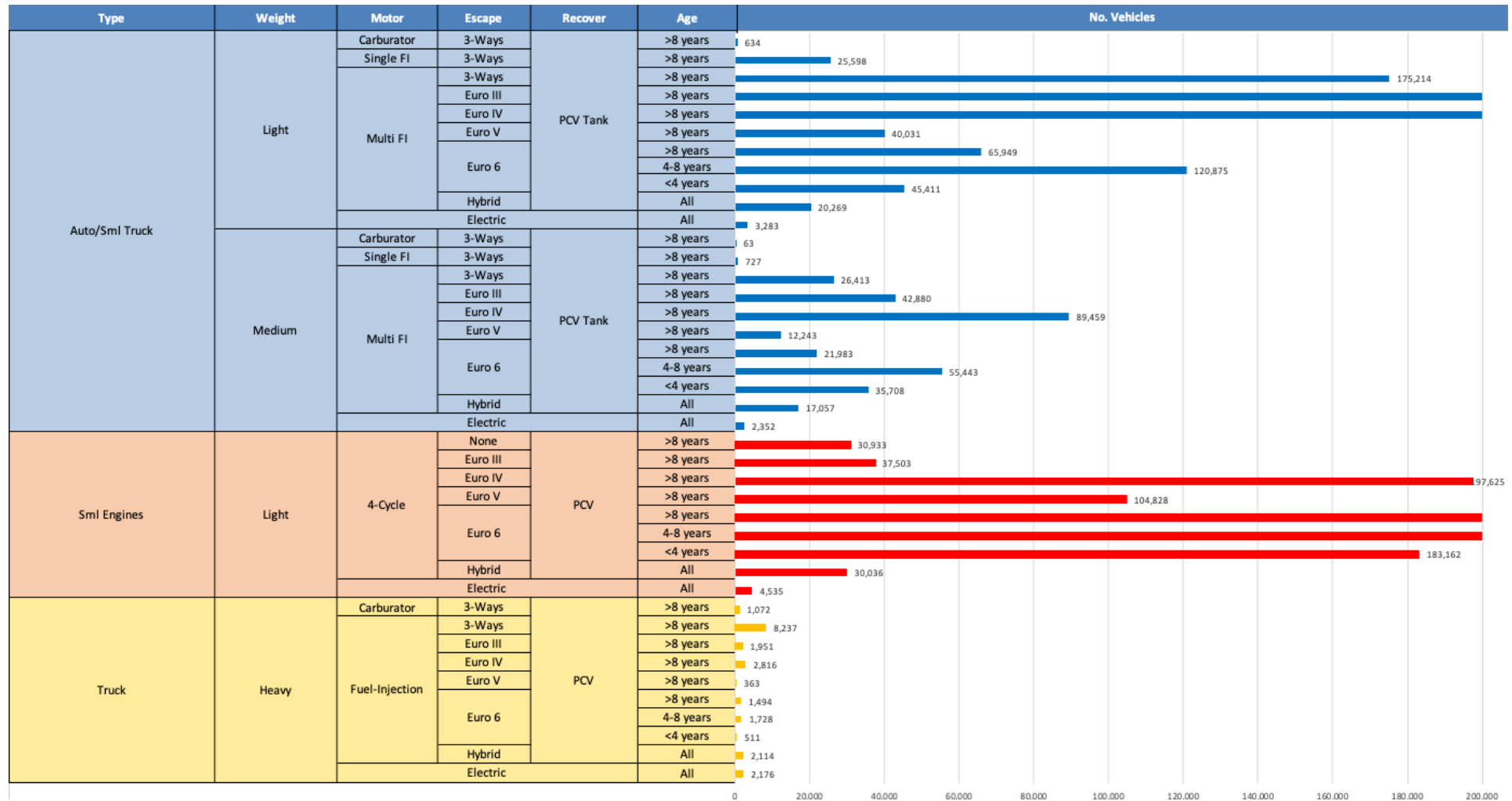


Average CIF 2024 Prices

Octane (RON)	98.0	98.8	100.1	101.3	102.6	104.1
Price (US\$/gal)	\$ 2.307	\$ 2.307	\$ 2.307	\$ 2.307	\$ 2.307	\$ 2.307

Source: Faro90

Gasoline Vehicle Fleet - Croatia

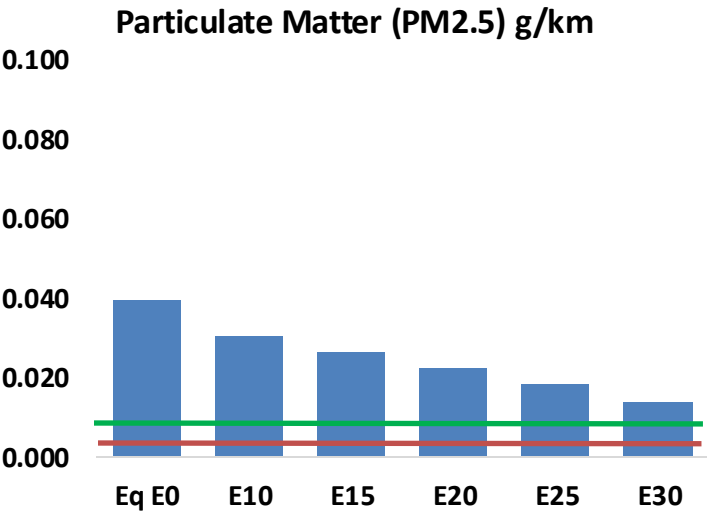
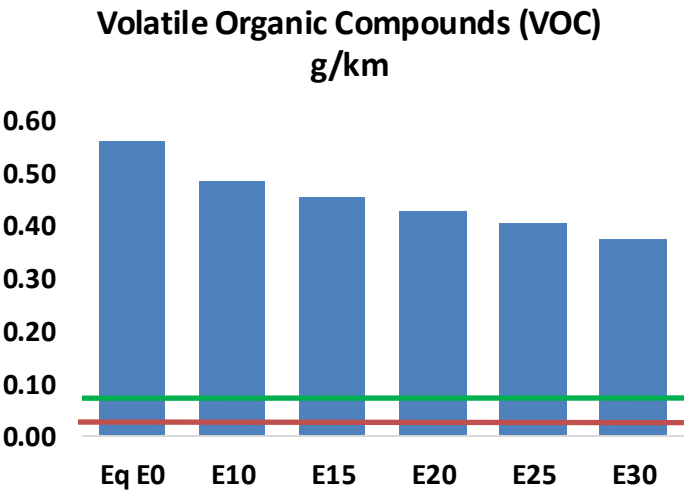
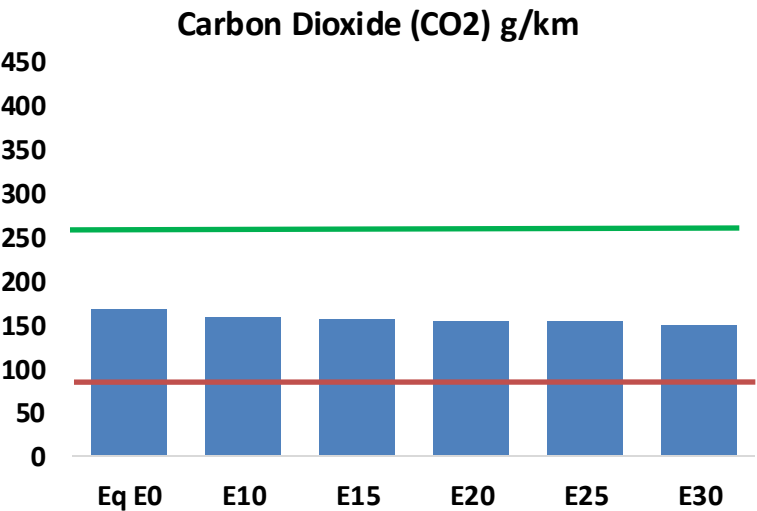
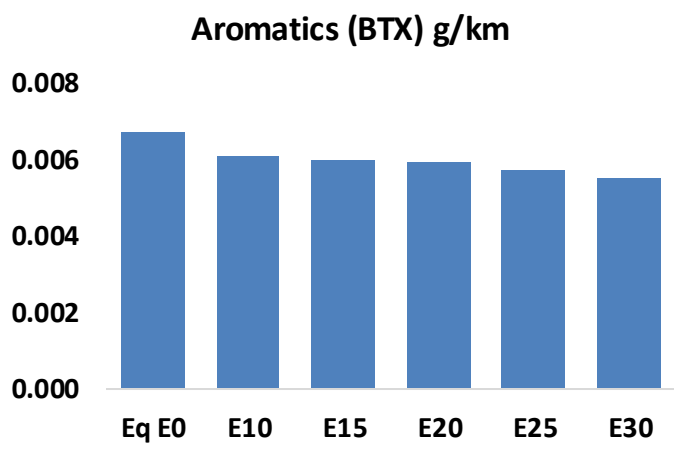
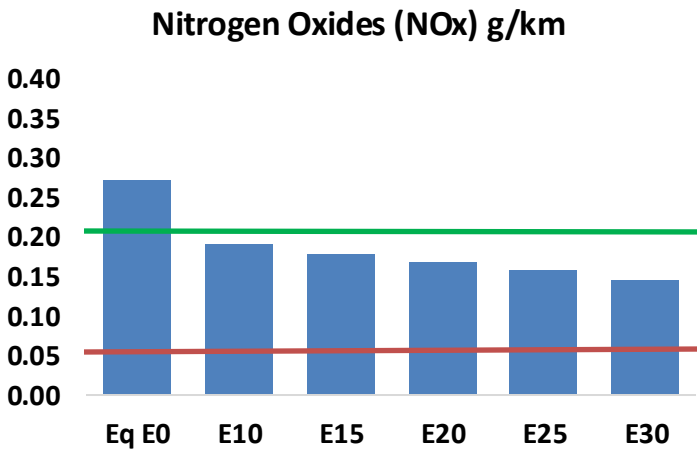
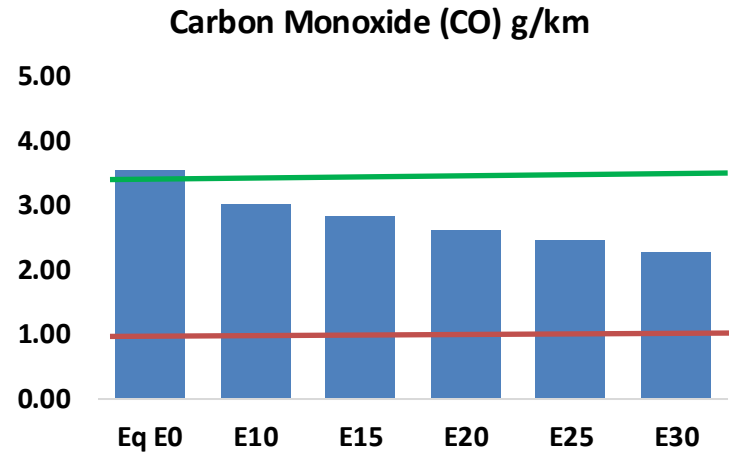
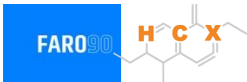


Vehicle Fleet: 2,358,574
Gasoline Vehicle Fleet: 923,614

Source: UNECE, Eurostat, ACEA

Croatia – Vehicular Emissions

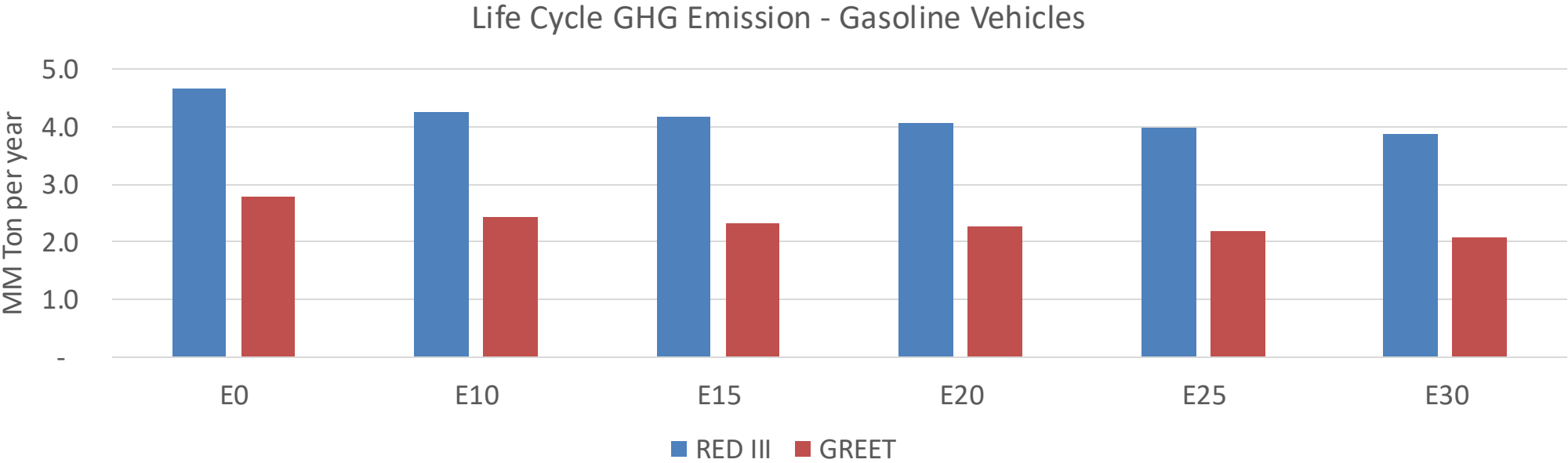
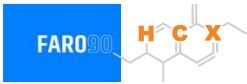
TIER III BIN 125 United States
Euro 6 Standard



Croatia – Vehicular Emissions

Vehicular Fleet Emissions (kg/day)	Eq E0	E5	E10	E15	E20	E25	E30	Reduction (%)	
								E10	E20
CO	91,726	84,984	78,241	72,777	67,305	63,106	58,353	-15%	-27%
CO2	4,317,386	4,206,367	4,101,516	4,019,054	3,978,249	3,955,342	3,882,433	-5%	-8%
VOC	14,448	13,483	12,518	11,755	11,001	10,425	9,643	-13%	-24%
NOx	7,031	5,273	4,922	4,639	4,375	4,084	3,776	-30%	-38%
SOx	48	45	41	38	35	32	28	-15%	-28%
NH3	1,642	1,626	1,610	1,617	1,635	1,648	1,659	-2%	0%
N2O	59	59	59	59	60	61	62	-1%	2%
CH4	3,492	3,492	3,492	3,545	3,622	3,688	3,751	0%	4%
Lead	-	-	-	-	-	-	-	0%	0%
Butadiene	101	96	92	89	86	85	82	-9%	-14%
Acetaldehyde	293	336	493	751	1,023	1,169	1,381	68%	249%
Formaldehyde	1,099	1,069	1,246	1,463	1,533	1,696	1,846	13%	39%
Benzene	173	167	158	155	154	147	143	-9%	-11%
PM2.5	293	260	228	198	168	136	103	-22%	-43%
PM10	722	641	560	487	413	335	254	-22%	-43%

Croatia – Gasoline Vehicle Fleet Life Cycle GHG Emissions



Country Emissions 2024 (MMT/año)	17.9
Target @ 2035 (MMT/year)	10.1
Delta	7.8

%Reduction vs E0	E0	E10	E15	E20	E25	E30
GREET 2024	0.0%	13.3%	16.6%	19.4%	22.1%	25.6%
RED II/III	0.0%	8.5%	10.8%	12.7%	14.5%	16.8%

% Target@2035	E0	E10	E15	E20	E25	E30
GREET 2024	0.0%	4.7%	5.9%	6.9%	7.9%	9.1%
RED II/III	0.0%	5.1%	6.4%	7.6%	8.6%	10.0%

Source: Greet, RED II/III, IPCC, climateactiontracker.org, F90

Ethanol Gasoline Blending in Cyprus



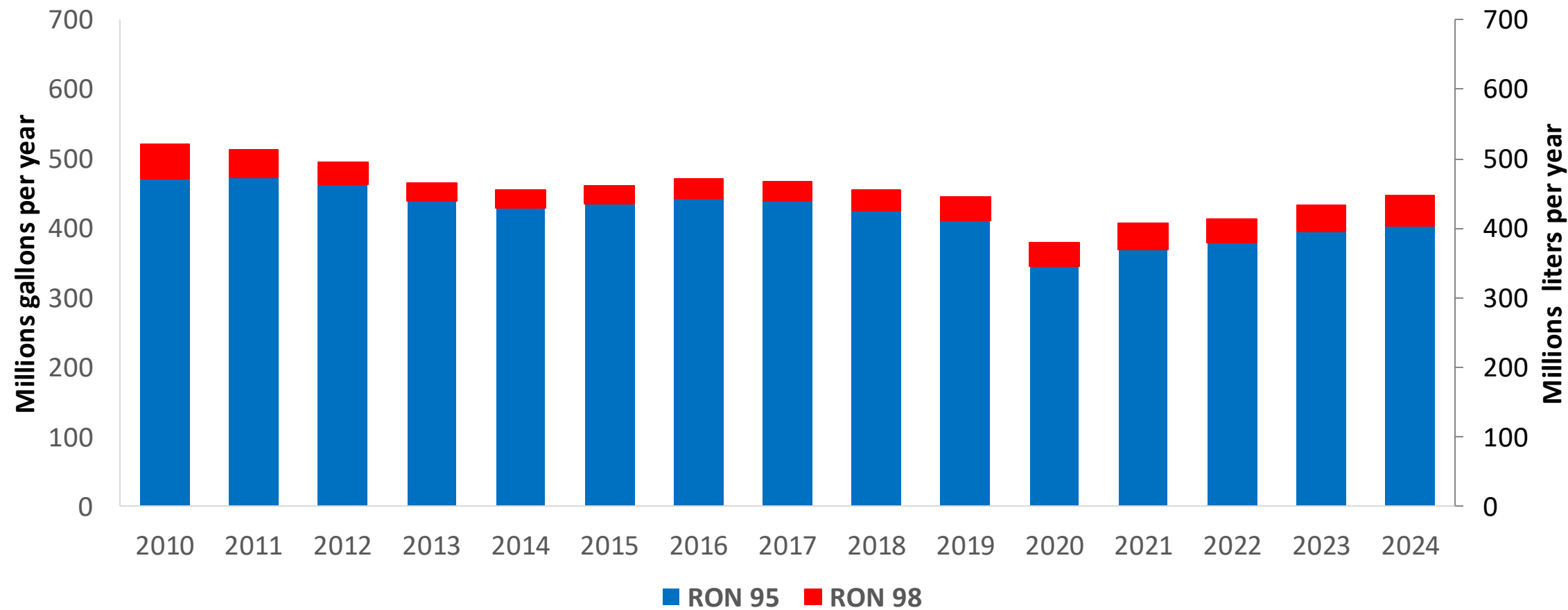
In 2024, gasoline consumption in Cyprus was 446 million liters (118 million gallons) and growth rate was 0.1%. Gasoline production and net imports were and 446 million liters (and 118 million gallons) correspondingly. Cyprus complies with the European gasoline standards and offers two main grades: RON 95 and RON 98) with an average octane of 95.3 RON.

Cyprus has no specific ethanol mandate and a target of 14% biofuels in the transport sector by 2030. Ethanol demand was 17.4 million liters in 2024 (4.6 million gallons) equivalent to 3.9% of gasoline demand. Ethanol is supplied by imports.

Source: Eurostat, European Commission

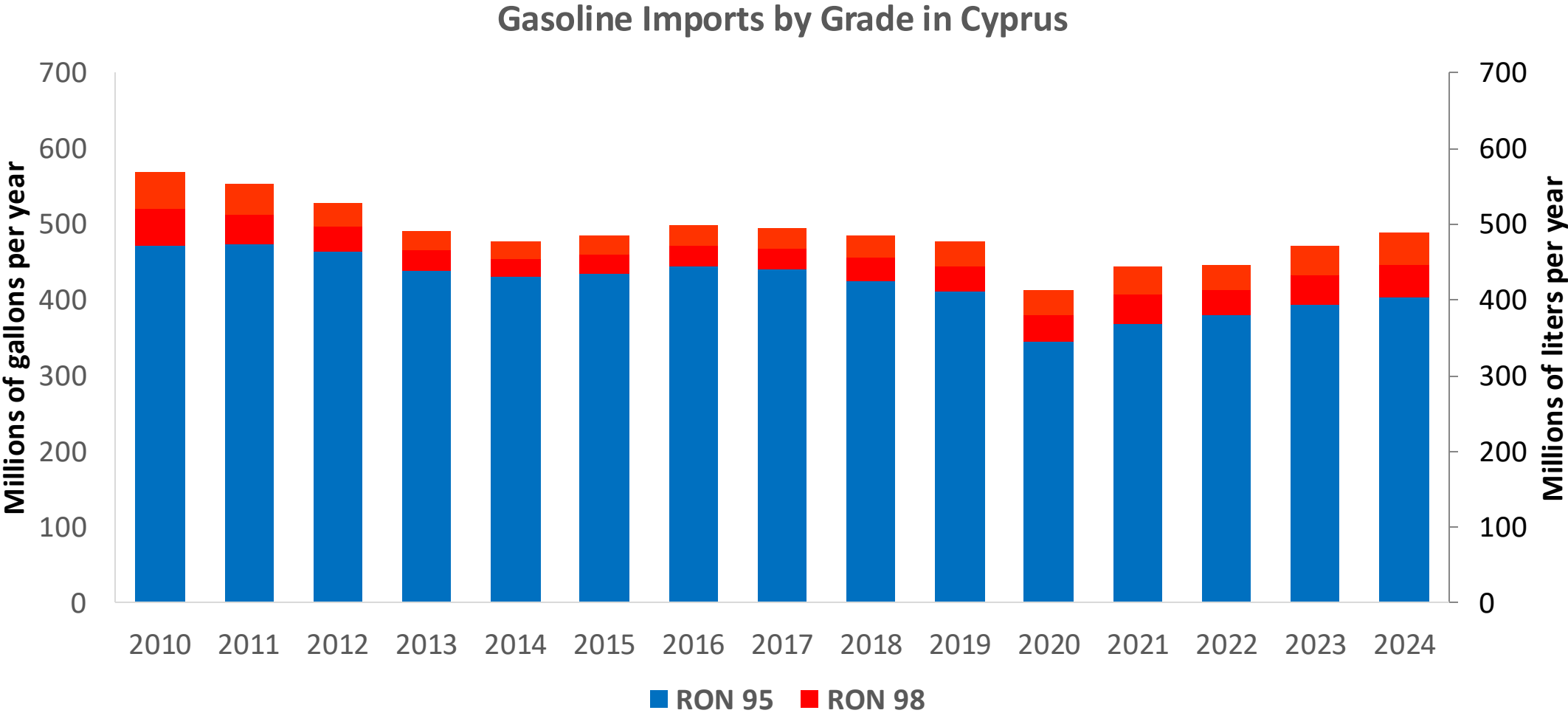
Gasoline Demand in Cyprus

Gasoline Demand by Grade in Cyprus



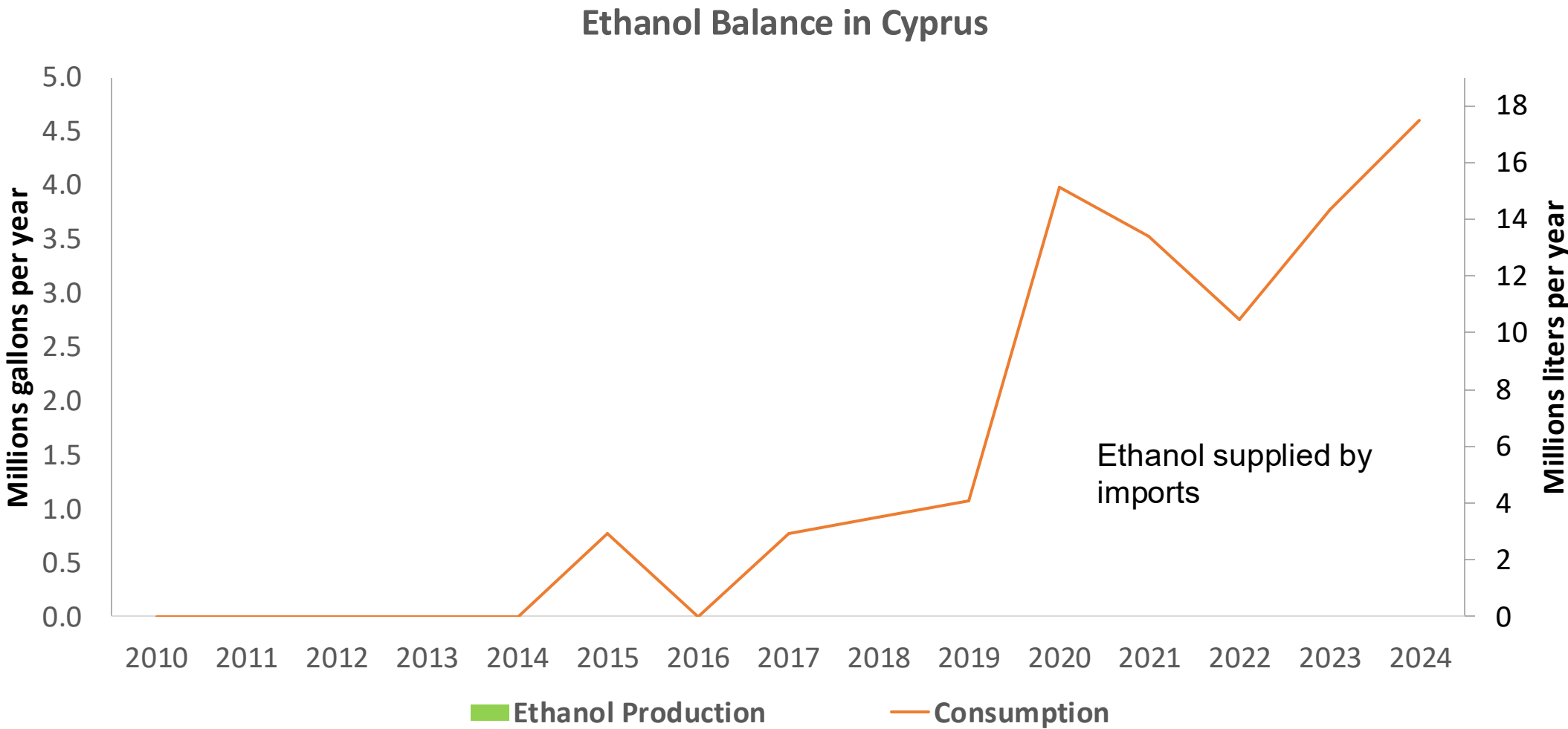
Source: Eurostat

Gasoline Imports in Cyprus



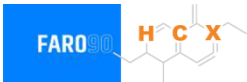
Source: Eurostat

Ethanol Balance in Cyprus



Source: Eurostat, Cyprus Statistics, Global Economy Report

Gasoline Quality in Cyprus

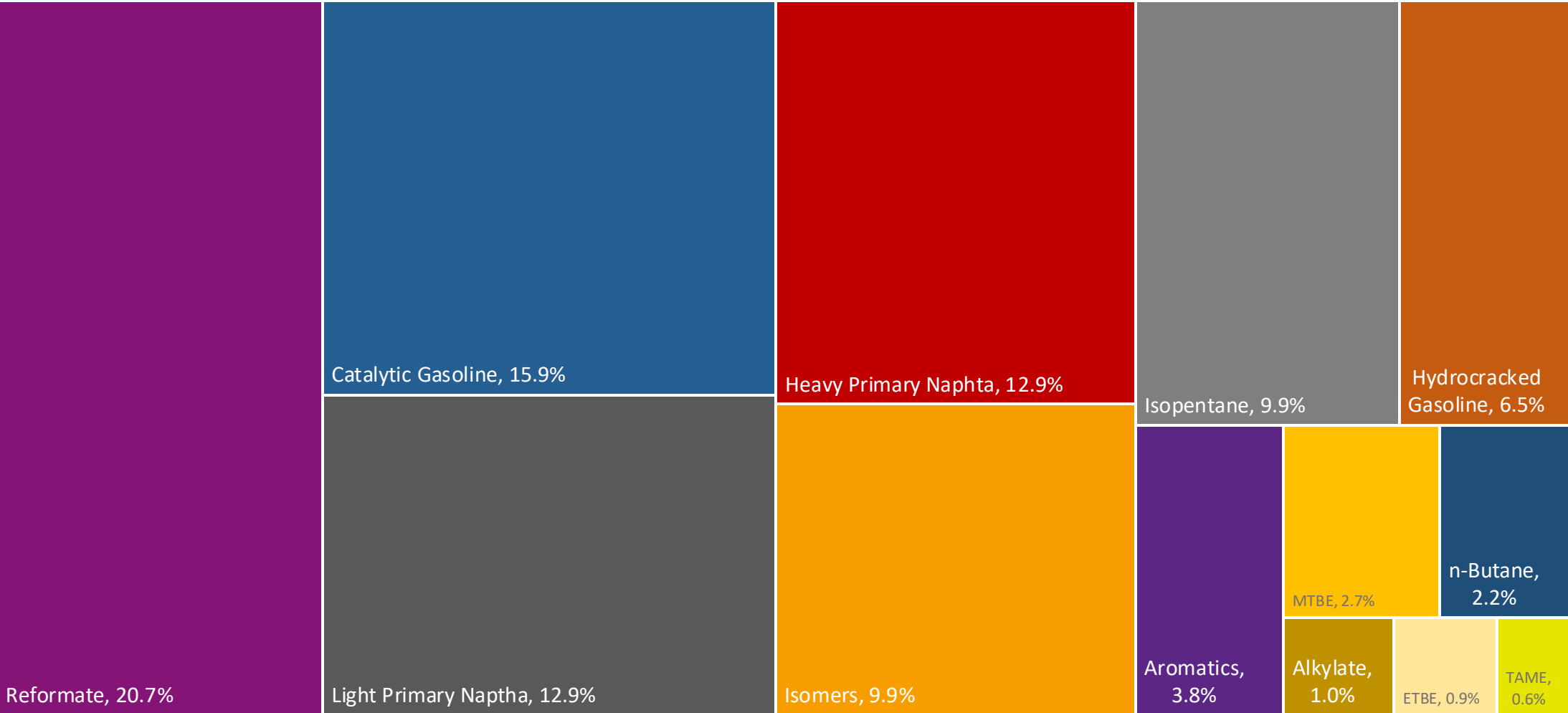
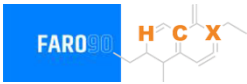


Name	EN 228:2025 (Euro 6); Country Level Fuel Ordinances							
Implementation Date	2025							
Applicability	All EU Countries + UK							
Grade	RON 95 E5		RON 95 E10		RON 98 E5		RON 98 E10	
	Min	Max	Min	Max	Min	Max	Min	Max
RON	95.0		95.0		98.0		98.0	
MON	> 85	> 88	> 85	> 88	> 85	> 88	> 85	> 88
AKI								
Benzene (%vol)	< 1							
Aromatics (% vol)	< 35							
Olefins (%vol)	< 18							
Lead (g/L)	< 0.005							
Manganese (mg/L)	< 2,0							
Sulfur (ppm)	10.0							
Oxygen (%w)*	< 2,7	< 3,7	< 2,7	< 3,7	< 2,7	< 3,7	< 2,7	< 3,7
Ethanol (%vol)	< 5	< 10	< 5	< 10	< 5	< 10	< 5	< 10
PVR 37.8°C (psi)	<> 45 - 100 kPa in 6 Volatility Classes							
Overall Biofuels (%cal)	-							
Advanced Biofuels (%cal)	0.2							
Bioethanol in Gasoline Sales (%cal)	-							
GHG Emission Reduction (%)	-							

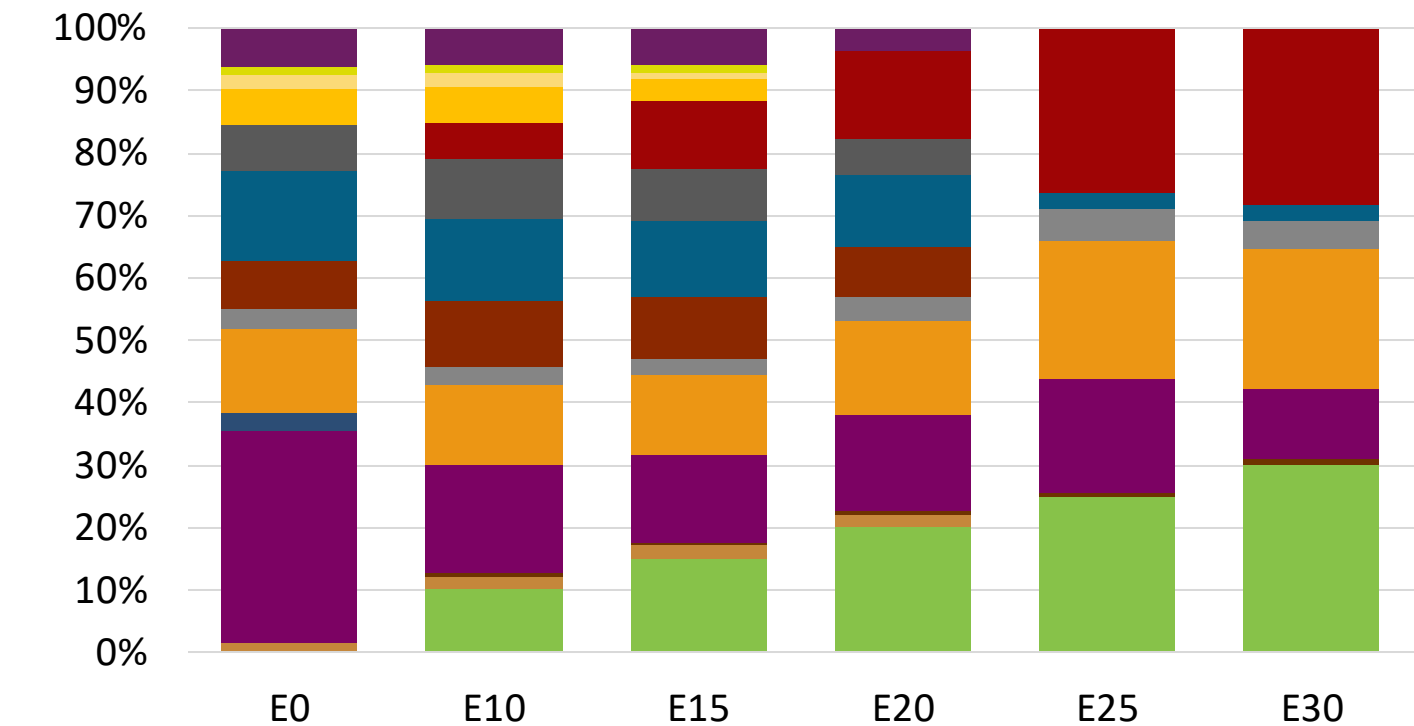
* Methanol 3%, Iso-propyl alcohol 12%, Ter-butyl alcohol 15%, Iso-butyl alcohol 15%, Ethers +5 Carbons 22%, Other Oxygenates 15%

Source: European Commission

Cyprus Available Blending Components



Cyprus – Gasoline 95 – Constant Octane

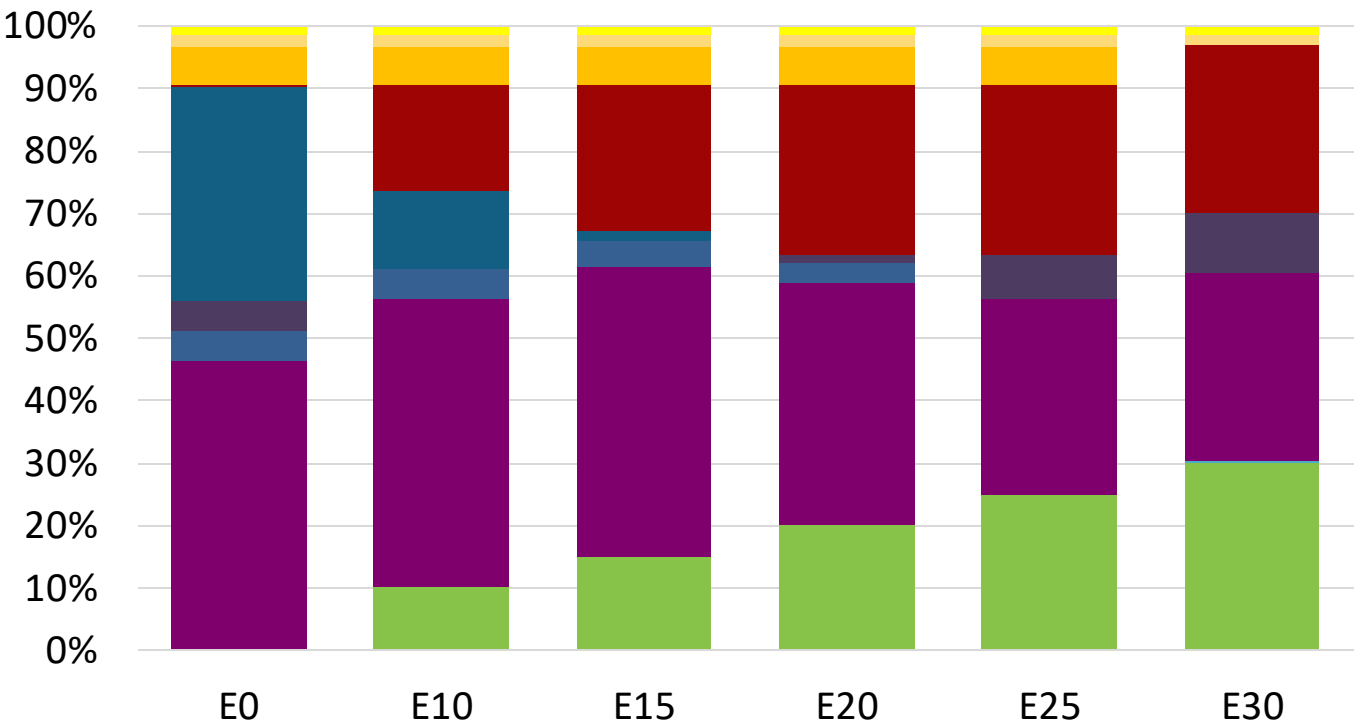
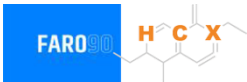


Average CIF 2024
Prices

Octane (RON)	95.0	95.0	95.0	95.0	95.0	96.0
Price (US\$/gal)	\$ 2.258	\$ 2.135	\$ 2.069	\$ 2.002	\$ 1.881	\$ 1.845

Source: Faro90

Cyprus - Gasoline 98 - Constant Octane

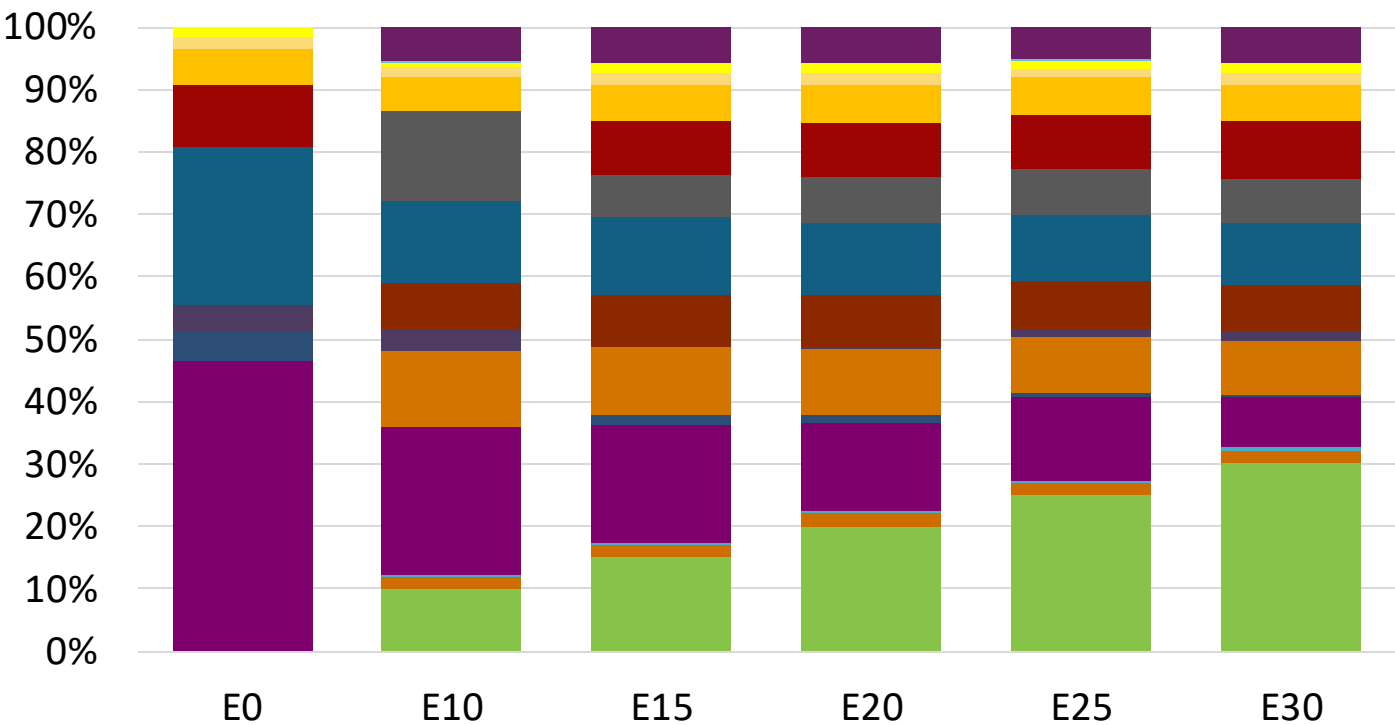
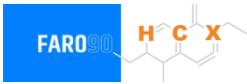


Average CIF 2024 Prices

Octane (RON)	98.0	98.0	98.0	98.0	98.0	98.0
Price (US\$/gal)	\$ 2.330	\$ 2.204	\$ 2.132	\$ 2.057	\$ 1.980	\$ 1.924

Source: Faro90

Cyprus – Gasoline 95 – Increased Octane

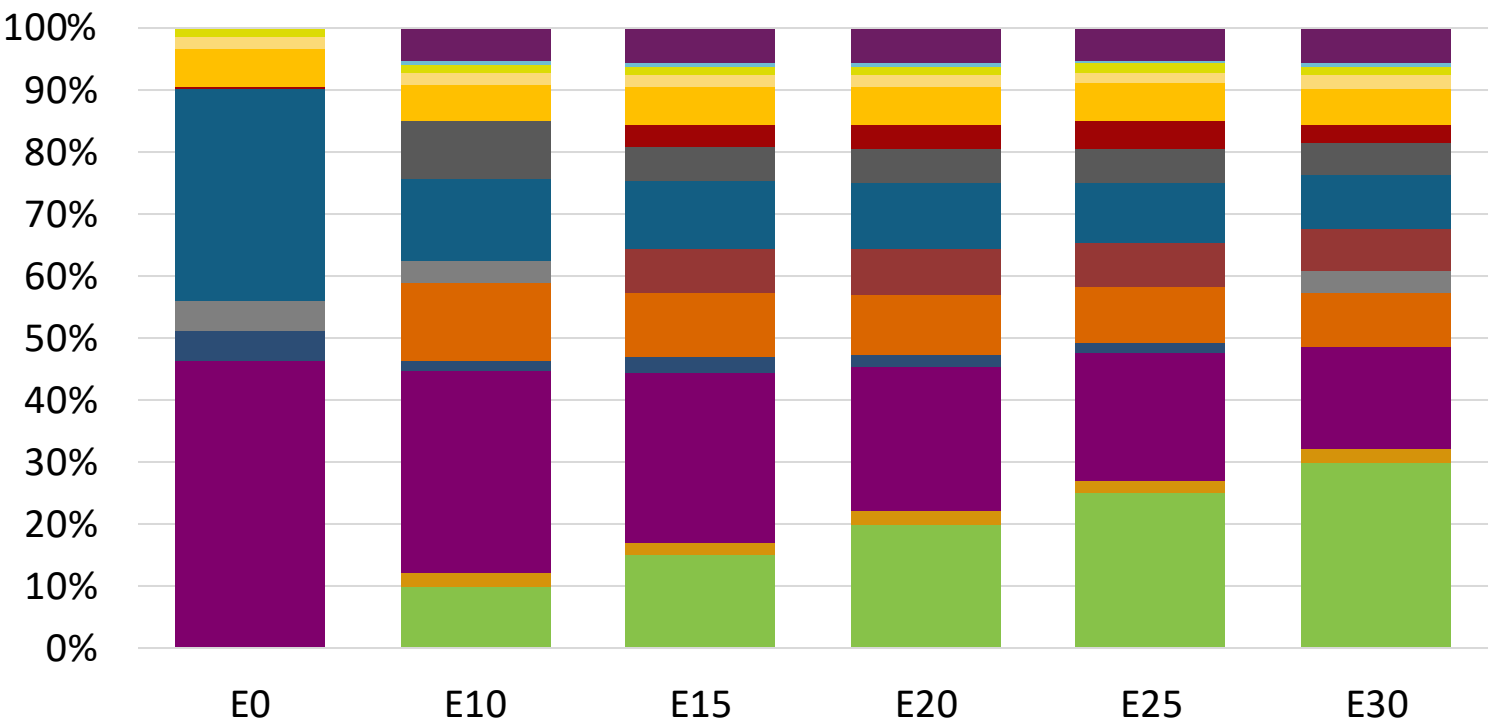
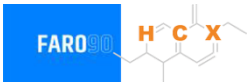


Average CIF 2024 Prices

Octane (RON)	95.0	96.2	97.9	99.4	100.8	102.2
Price (US\$/gal)	\$ 2.196	\$ 2.196	\$ 2.196	\$ 2.196	\$ 2.196	\$ 2.196

Source: Faro90

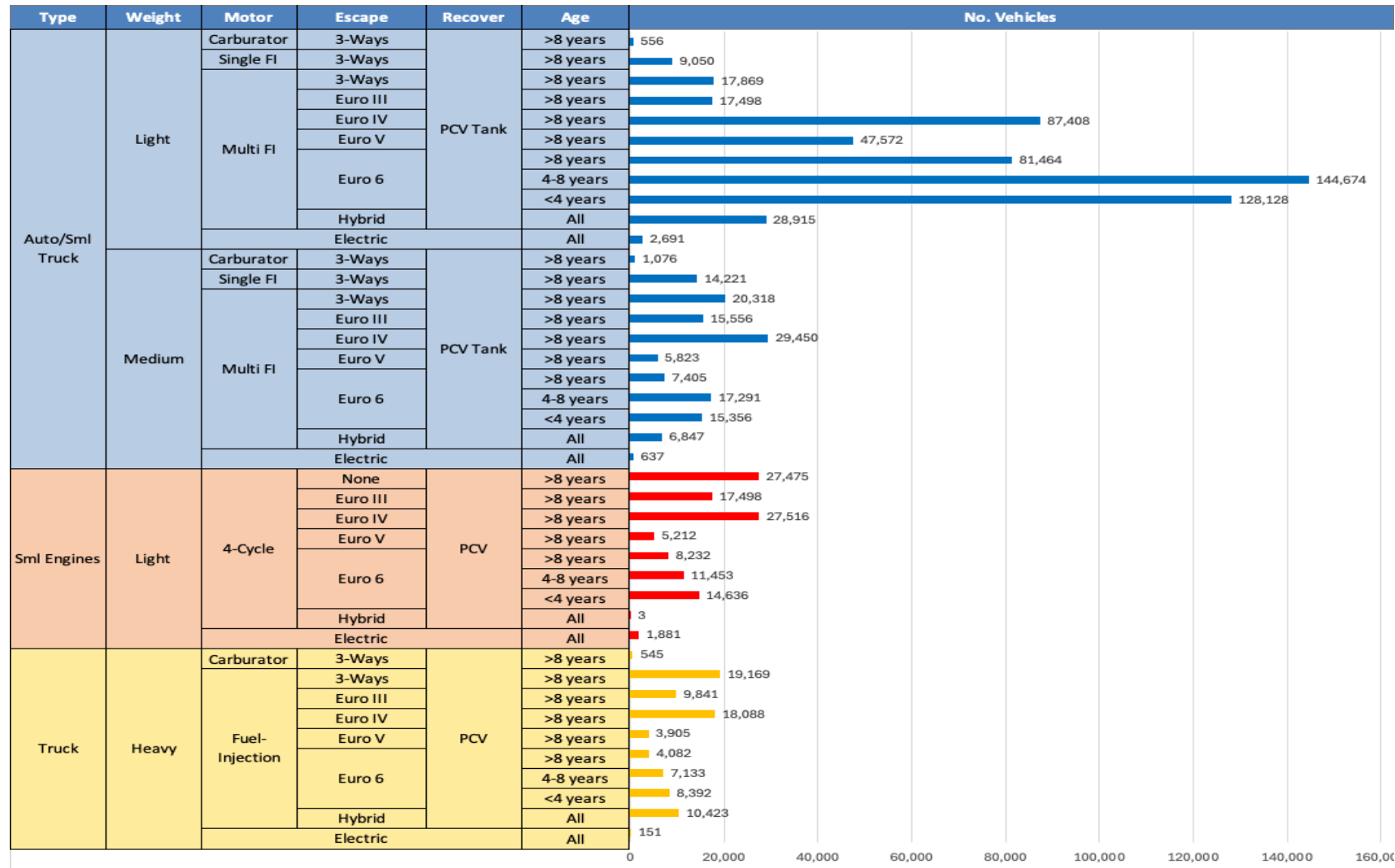
Cyprus – Gasoline 98 – Increased Octane



Average CIF 2024 Prices

Source: Faro90

Gasoline Vehicle Fleet - Cyprus

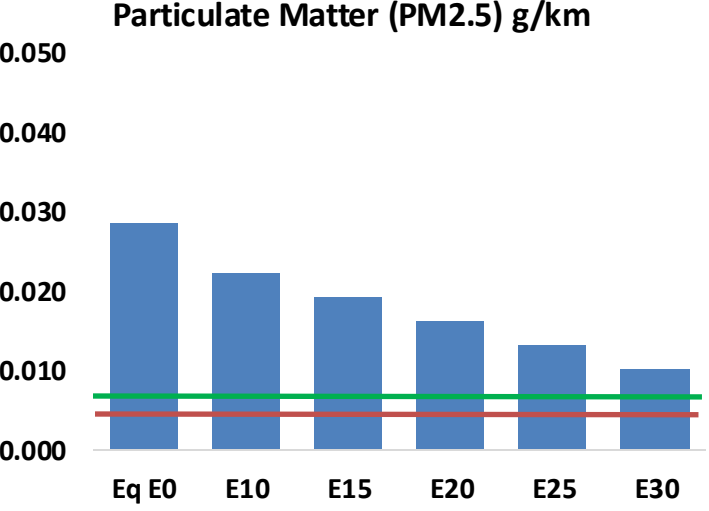
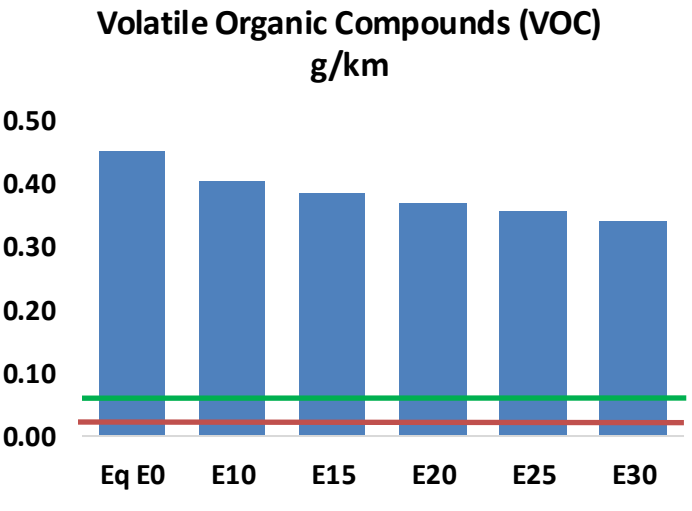
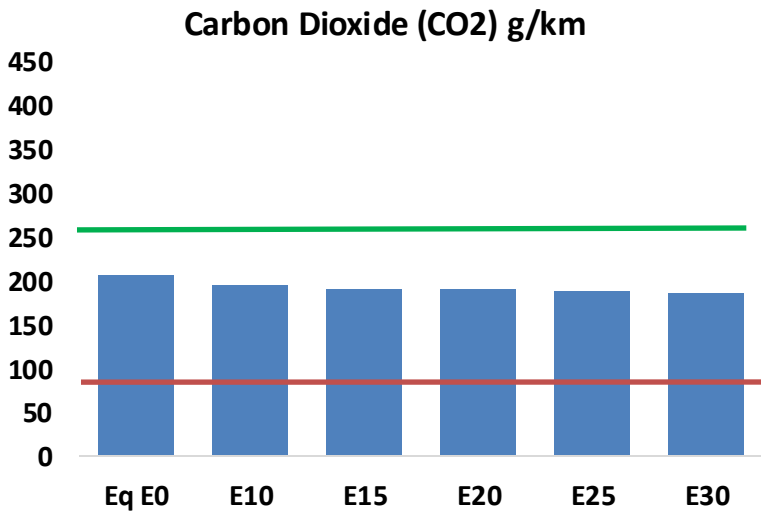
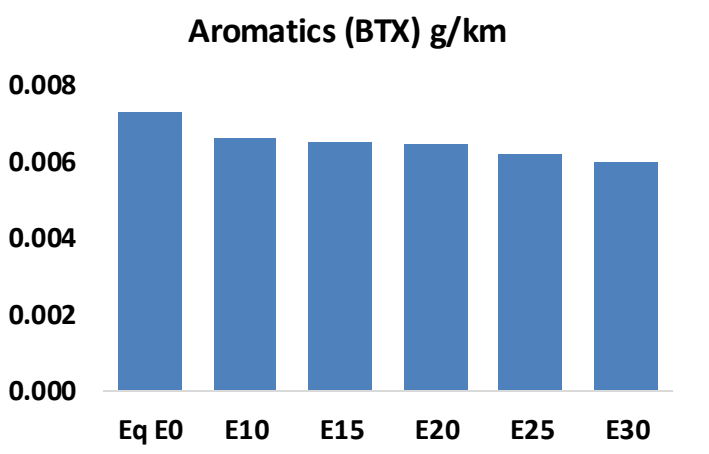
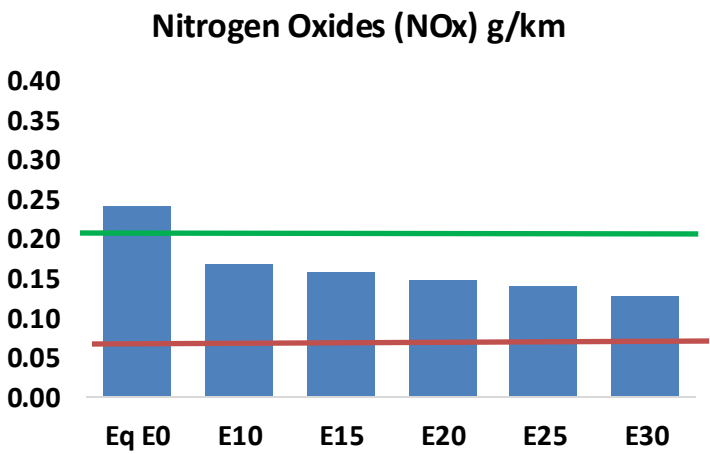
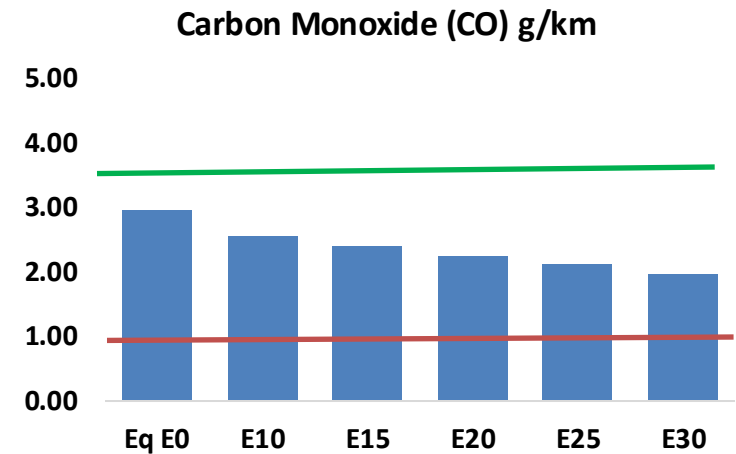
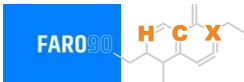


Vehicle Fleet: 895,440
Gasoline Vehicle Fleet: 6685,744

Source: Eurostat, ACEA

Cyprus – Vehicular Emissions

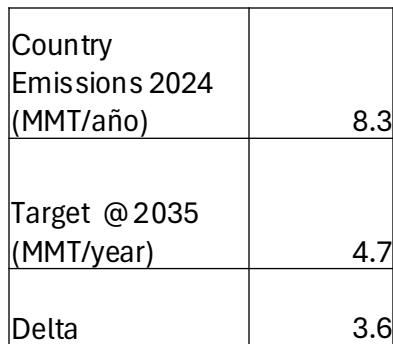
TIER III BIN 125 United States
Euro 6 Standard



Cyprus – Vehicular Emissions

Vehicular Fleet Emissions (kg/day)	Eq E0	E5	E10	E15	E20	E25	E30	Reduction (%)	
								E10	E20
CO	46,762	43,557	40,352	37,802	35,271	33,335	31,171	-14%	-25%
CO2	3,244,976	3,161,534	3,082,728	3,020,749	2,990,079	2,975,062	2,920,223	-5%	-8%
VOC	7,110	6,737	6,364	6,092	5,834	5,640	5,365	-10%	-18%
NOx	3,799	2,849	2,659	2,507	2,364	2,207	2,040	-30%	-38%
SOx	37	34	31	29	27	24	22	-15%	-28%
NH3	1,157	1,146	1,135	1,140	1,153	1,162	1,169	-2%	0%
N2O	43	43	43	43	44	44	45	-1%	2%
CH4	1,608	1,608	1,608	1,632	1,668	1,698	1,727	0%	4%
Lead	-	-	-	-	-	-	-	0%	0%
Butadiene	40	38	36	34	33	32	31	-10%	-17%
Acetaldehyde	103	118	173	263	359	410	485	68%	249%
Formaldehyde	372	362	422	496	519	574	625	13%	39%
Benzene	115	110	105	103	102	98	94	-9%	-11%
PM2.5	204	181	158	137	117	95	72	-22%	-43%
PM10	246	218	191	166	141	114	86	-22%	-43%

The FARO90 logo is shown in a blue box, and a chemical structure is displayed to its right. The chemical structure features a central carbon atom (C) bonded to a hydrogen atom (H) and a group X, with various other bonds extending from the ring system.



Source: Greet, RED II/III, IPCC, climateactiontracker.org, F90

Ethanol Gasoline Blending in Czechia

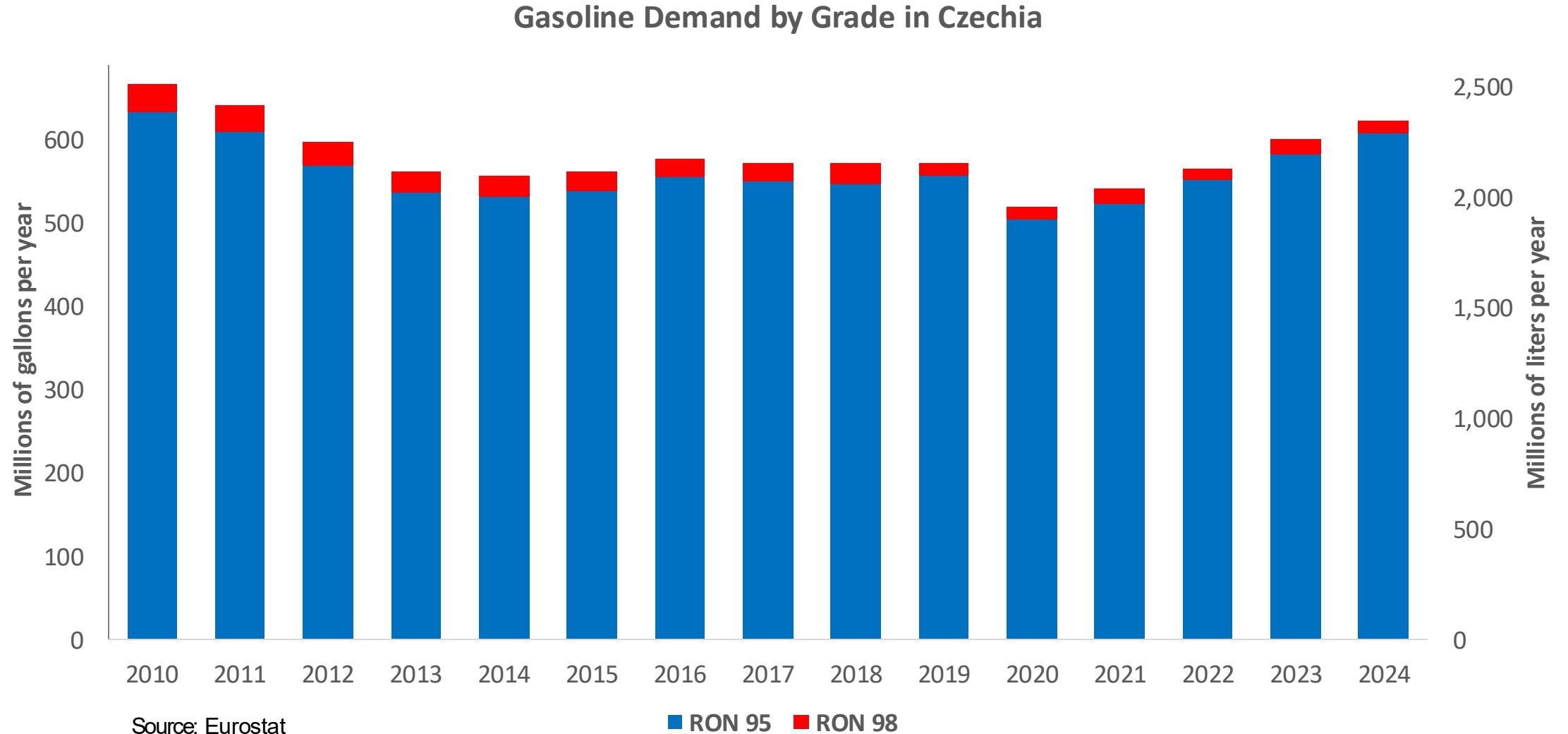


In 2024, gasoline consumption in Czechia was 2,347 million liters (620 million gallons) and growth rate was 1.7%. Gasoline production and net imports were 2,328 and -936 million liters (615 and -247 million gallons) correspondingly. Czechia complies with the European gasoline standards and offers two main grades: RON 95 and RON 98) with an average octane of 95.1 RON.

In 2024, ethanol consumption in Czechia was 96 million liters (25 million gallons) representing 4.1% of gasoline demand. Ethanol production and Net Imports were 76 and 21 million liters (25 and 5 million gallons) correspondingly. Ethanol sources are Corn 50% and Sugar Beet 50%.

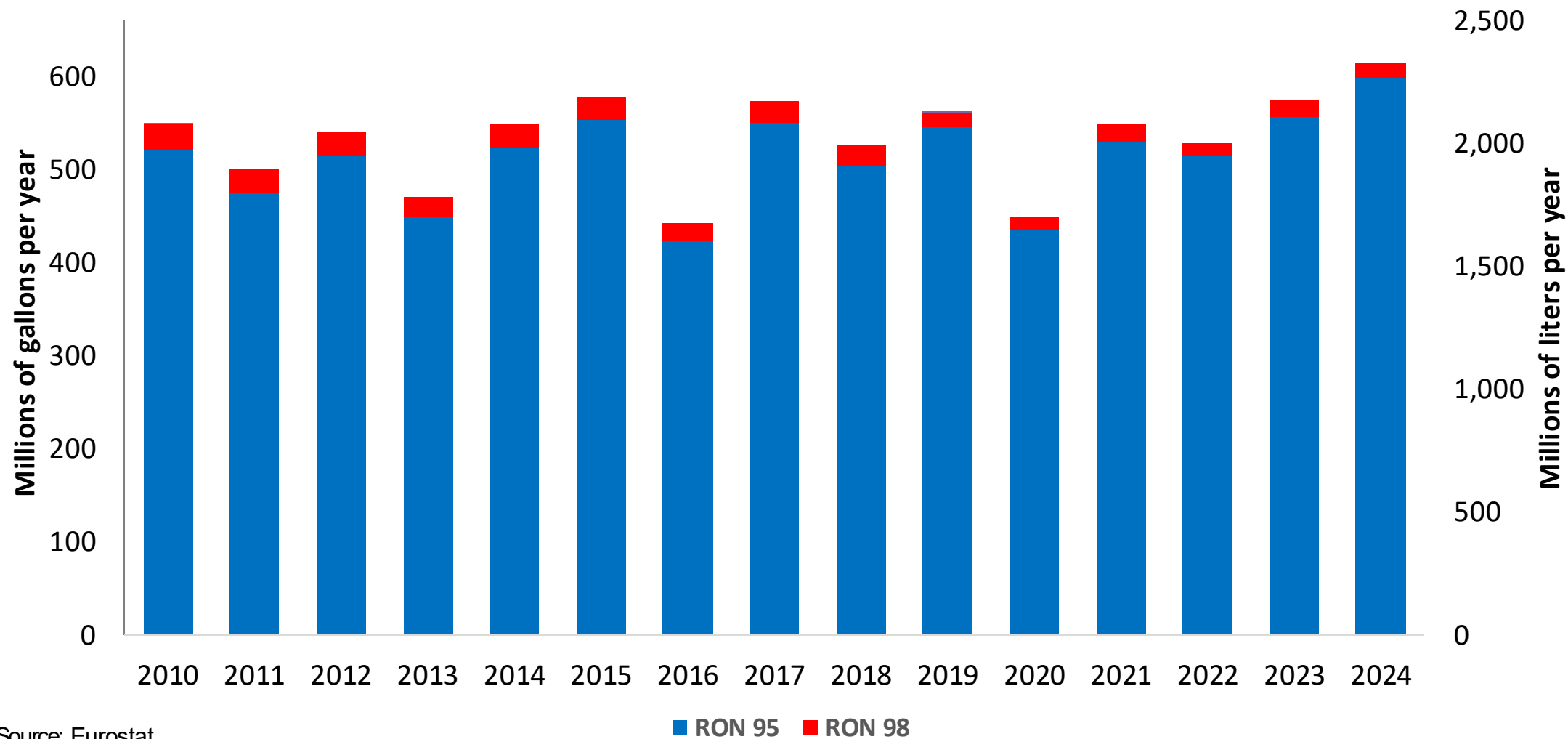
Source: Czechia Statistics, Eurostat

Gasoline Demand in Czechia



Gasoline Production in Czechia

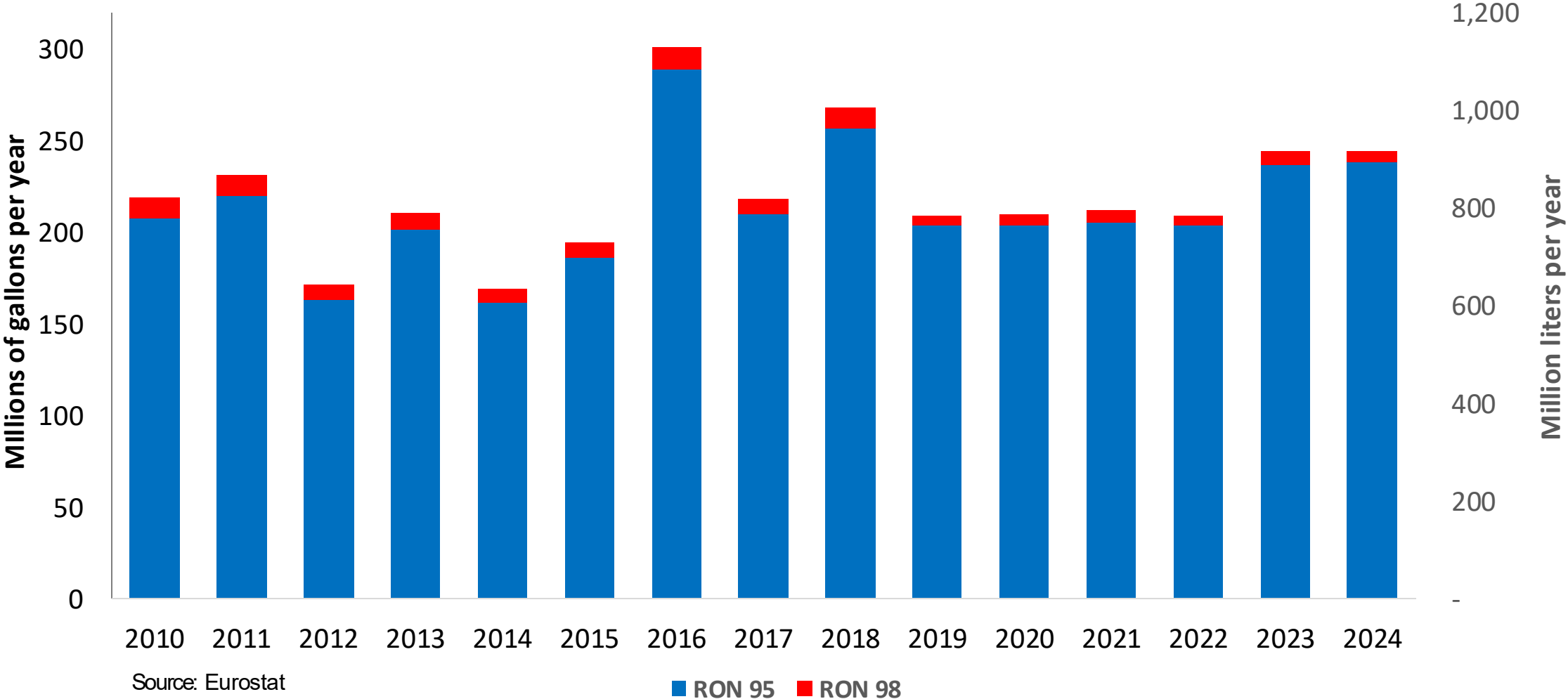
Gasoline Production by Grade in Czechia



Source: Eurostat

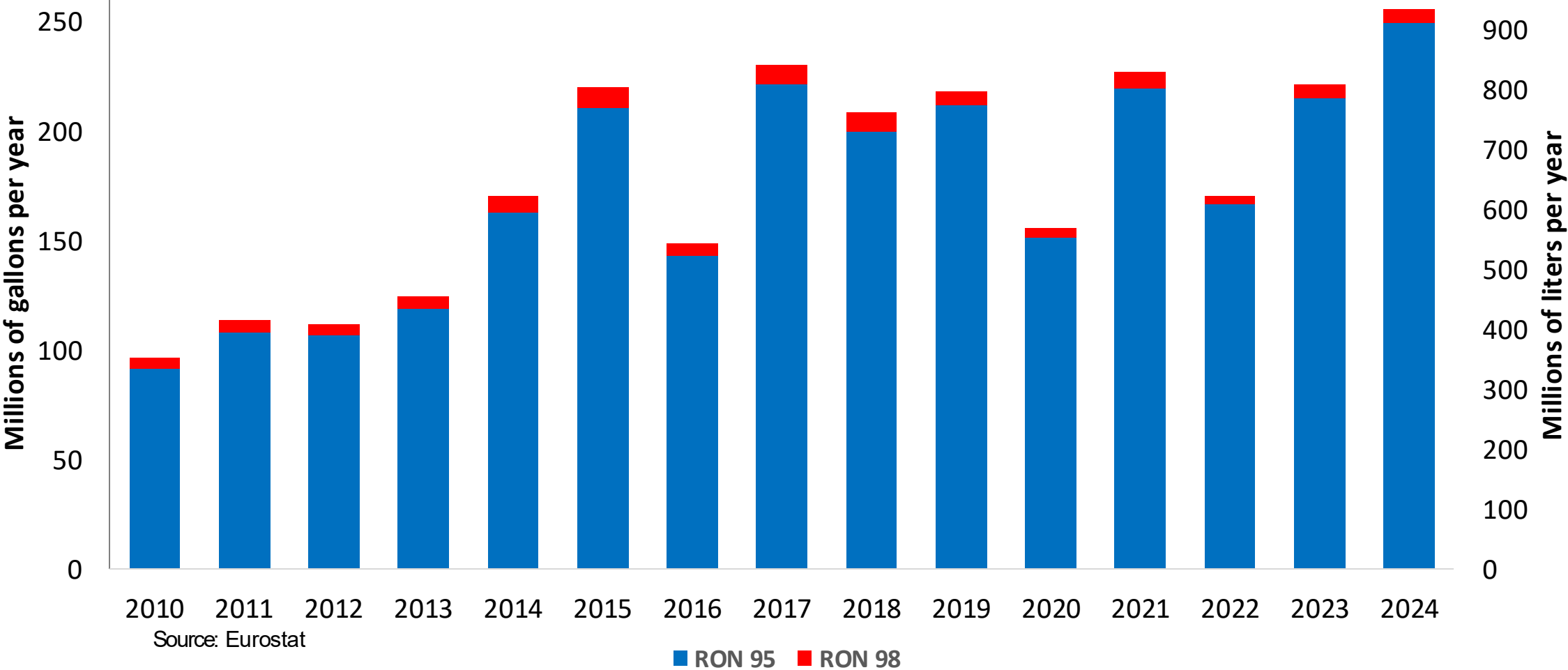
Gasoline Imports in Czechia

Gasoline Imports by Grade in Czechia

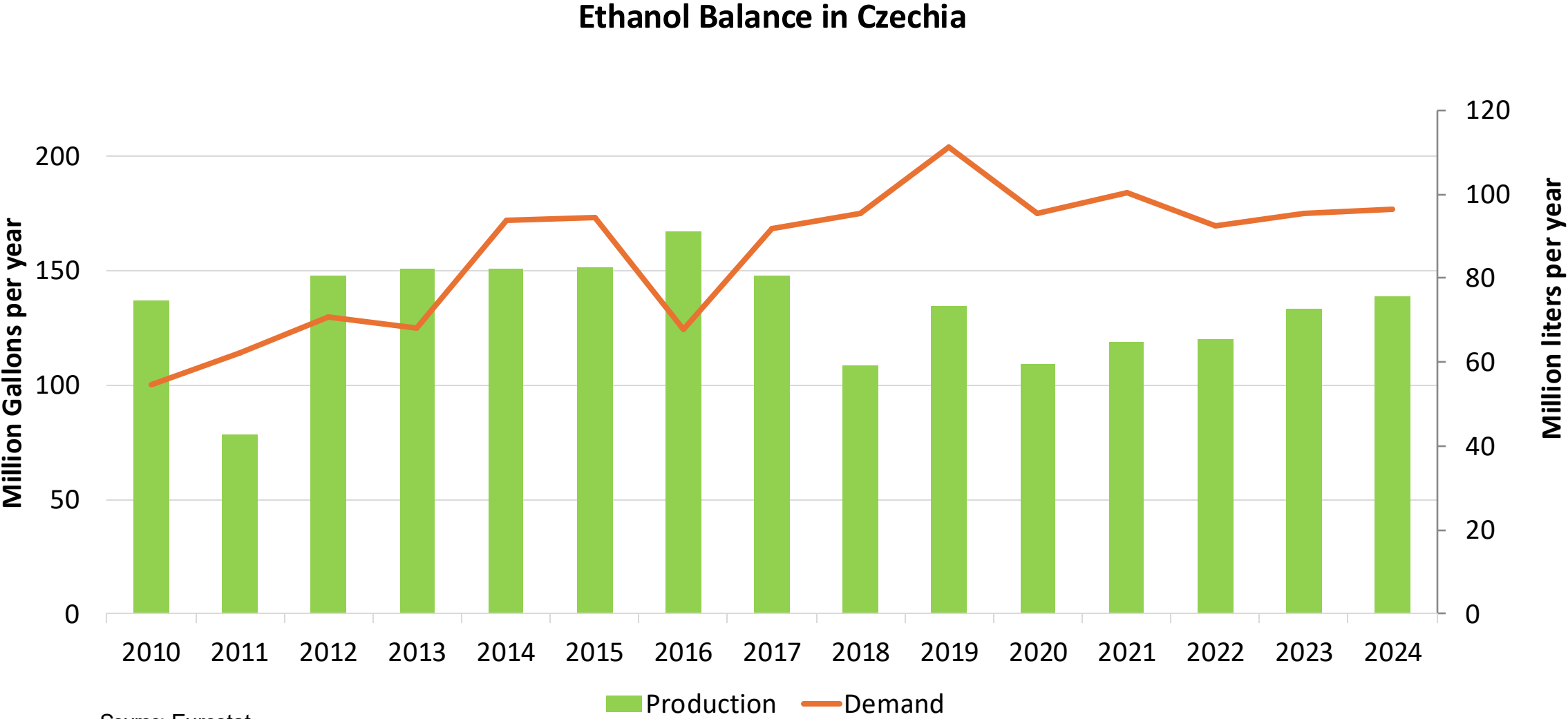


Gasoline Exports in Czechia

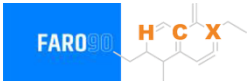
Gasoline Exports by Grade in Czechia



Ethanol Balance in Czechia



Gasoline Quality in Czechia

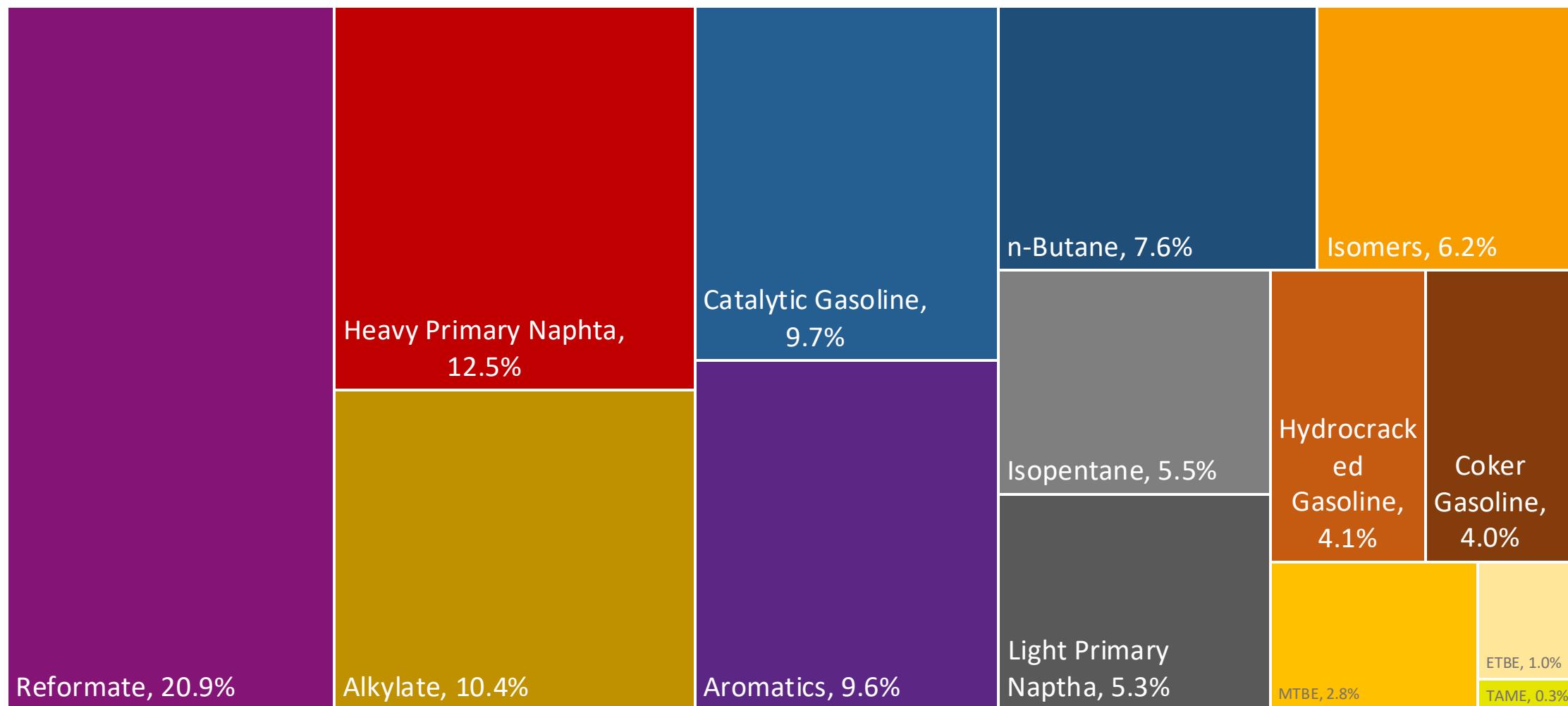


Name	EN 228:2025 (Euro 6); Country Level Fuel Ordinances							
Implementation Date	2025							
Applicability	All EU Contries + UK							
Grade	RON 95 E5		RON 95 E10		RON 98 E5		RON 98 E10	
	Min	Max	Min	Max	Min	Max	Min	Max
RON	95.0		95.0		98.0		98.0	
MON	> 85	> 88	> 85	> 88	> 85	> 88	> 85	> 88
AKI								
Benzene (%vol)	< 1							
Aromatics (% vol)	< 35							
Olefins (%vol)	< 18							
Lead (g/L)	< 0.005							
Manganese (mg/L)	< 2,0							
Sulfur (ppm)	10.0							
Oxygen (%w)*	< 2,7	< 3,7	< 2,7	< 3,7	< 2,7	< 3,7	< 2,7	< 3,7
Ethanol (%vol)	< 5	< 10	< 5	< 10	< 5	< 10	< 5	< 10
PVR 37.8°C (psi)	<> 45 - 100 kPa in 6 Volatility Classes							
Overall Biofuels (%cal)	-							
Advanced Biofuels (%cal)	1.1							
Bioethanol in Gasoline Sales (%cal)	-							
GHG Emission Reduction (%)	6.0							

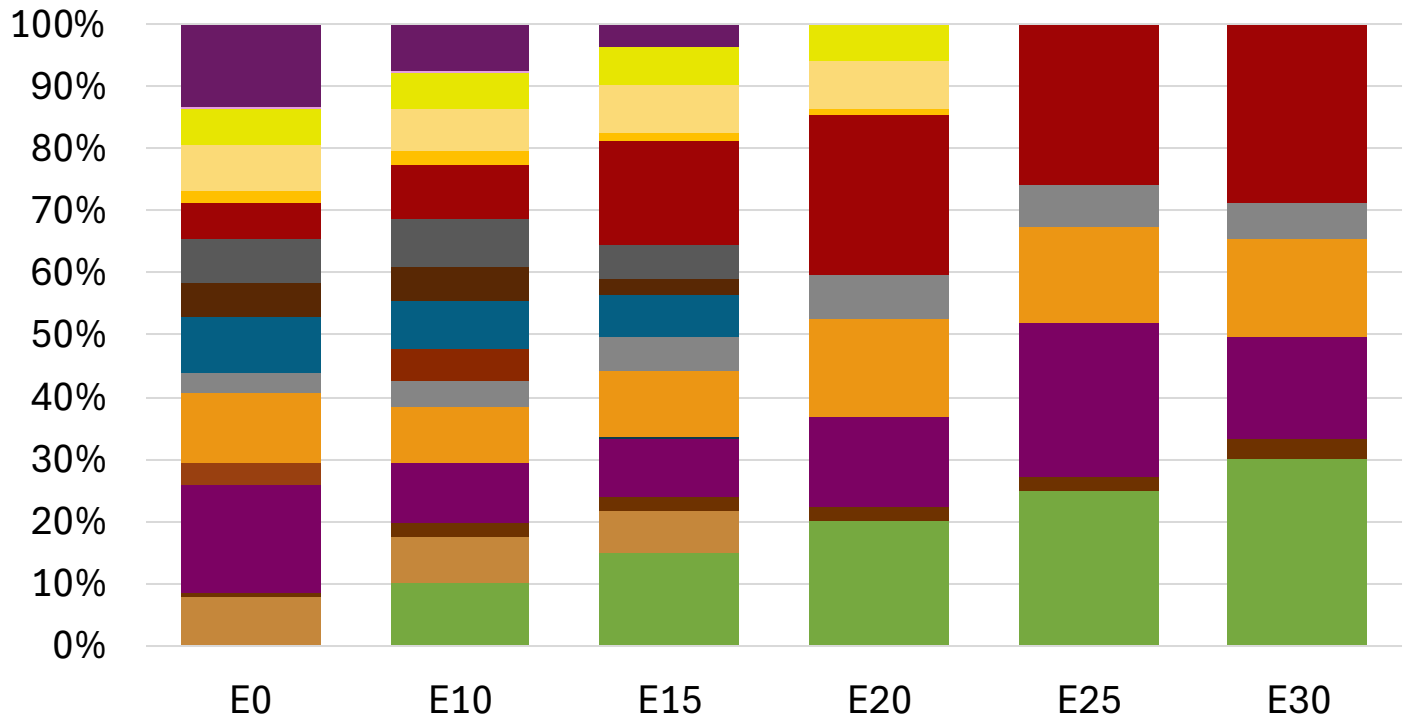
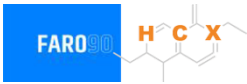
* Methanol 3%, Iso-propyl alcohol 12%, Ter-butyl alcohol 15%, Iso-butyl alcohol 15%, Ethers +5 Carbons 22%, Other Oxygenates 15%

Source: European Commision

Czechia Available Blending Components



Czechia – Gasoline 95 – Constant Octane

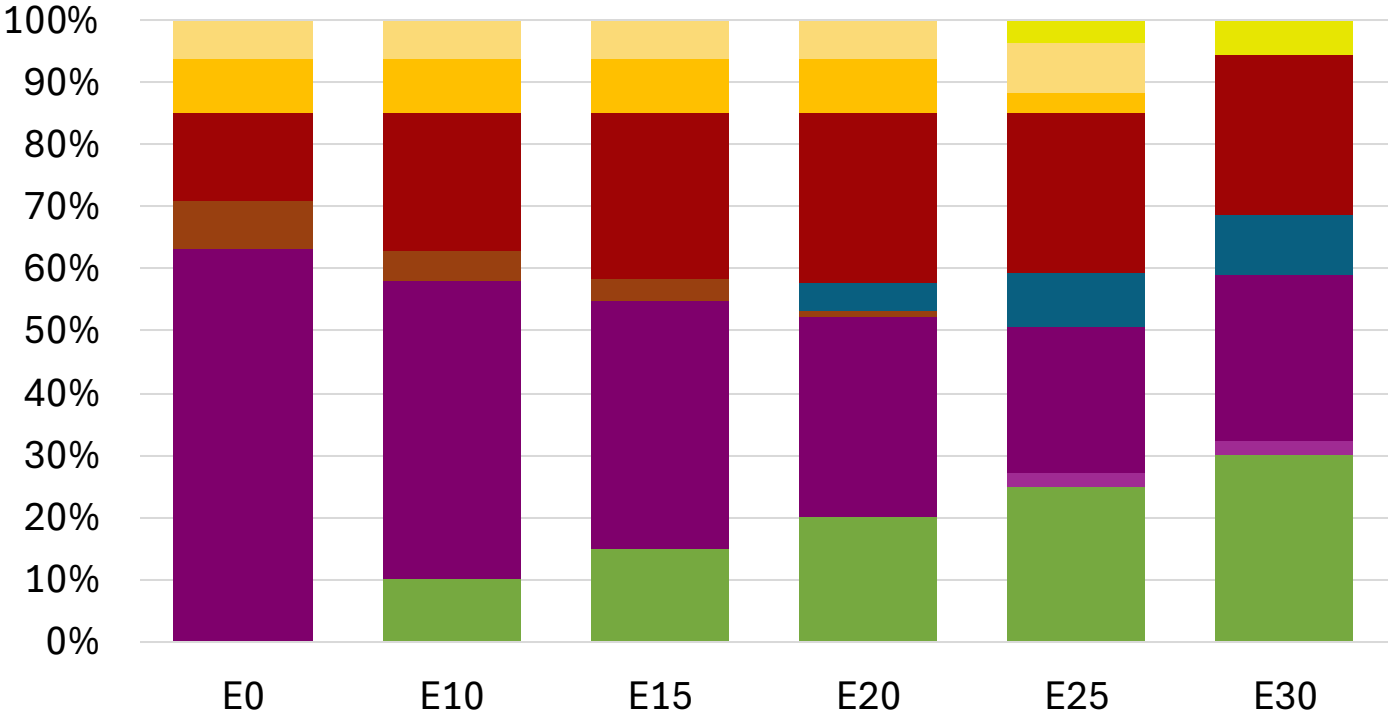
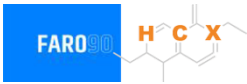


Average CIF 2024 Prices

Octane (RON)	95.0	95.0	95.0	95.0	95.0	95.8
Price (US\$/gal)	\$ 2.314	\$ 2.176	\$ 2.076	\$ 1.961	\$ 1.915	\$ 1.862

Source: Faro90

Czechia - Gasoline 98 - Constant Octane

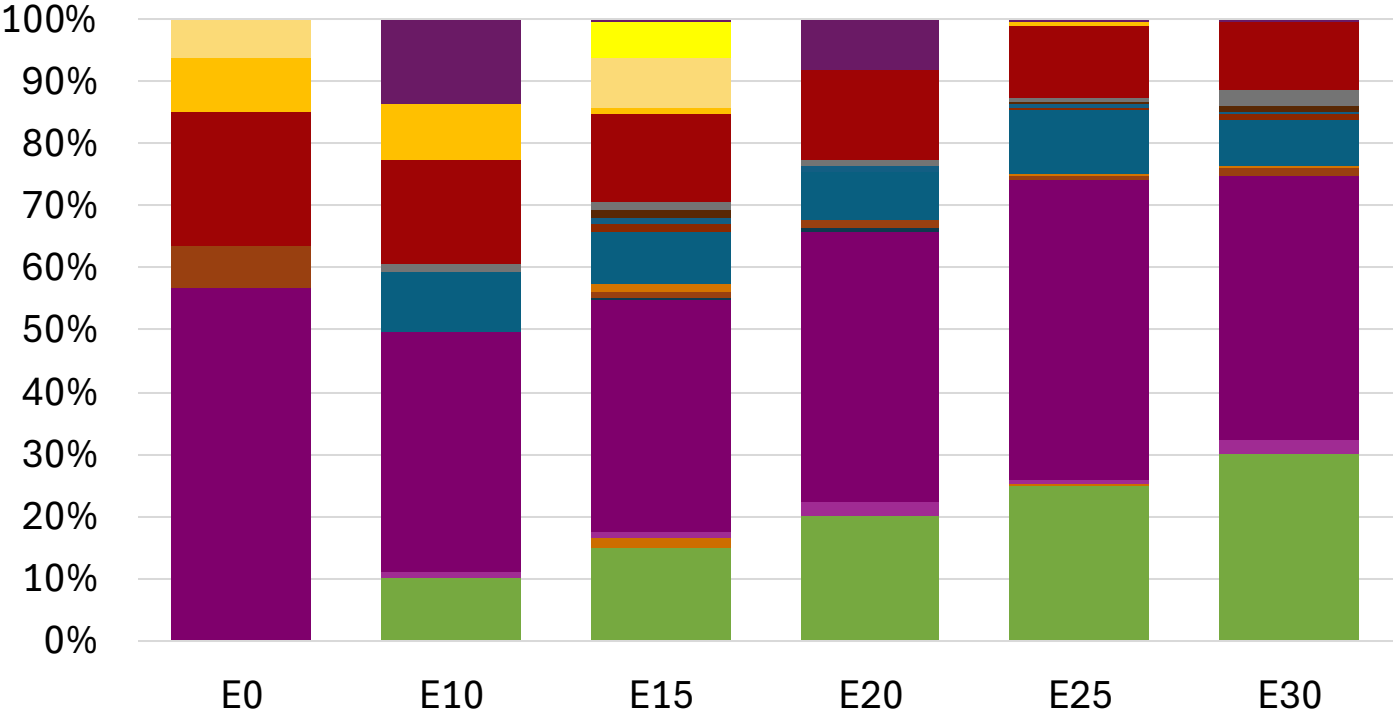


Average CIF 2024 Prices

Octane (RON)	98.0					
Price (US\$/gal)	\$ 2.334	\$ 2.217	\$ 2.148	\$ 2.076	\$ 2.011	\$ 1.961

Source: Faro90

Czechia – Gasoline 95 – Increased Octane

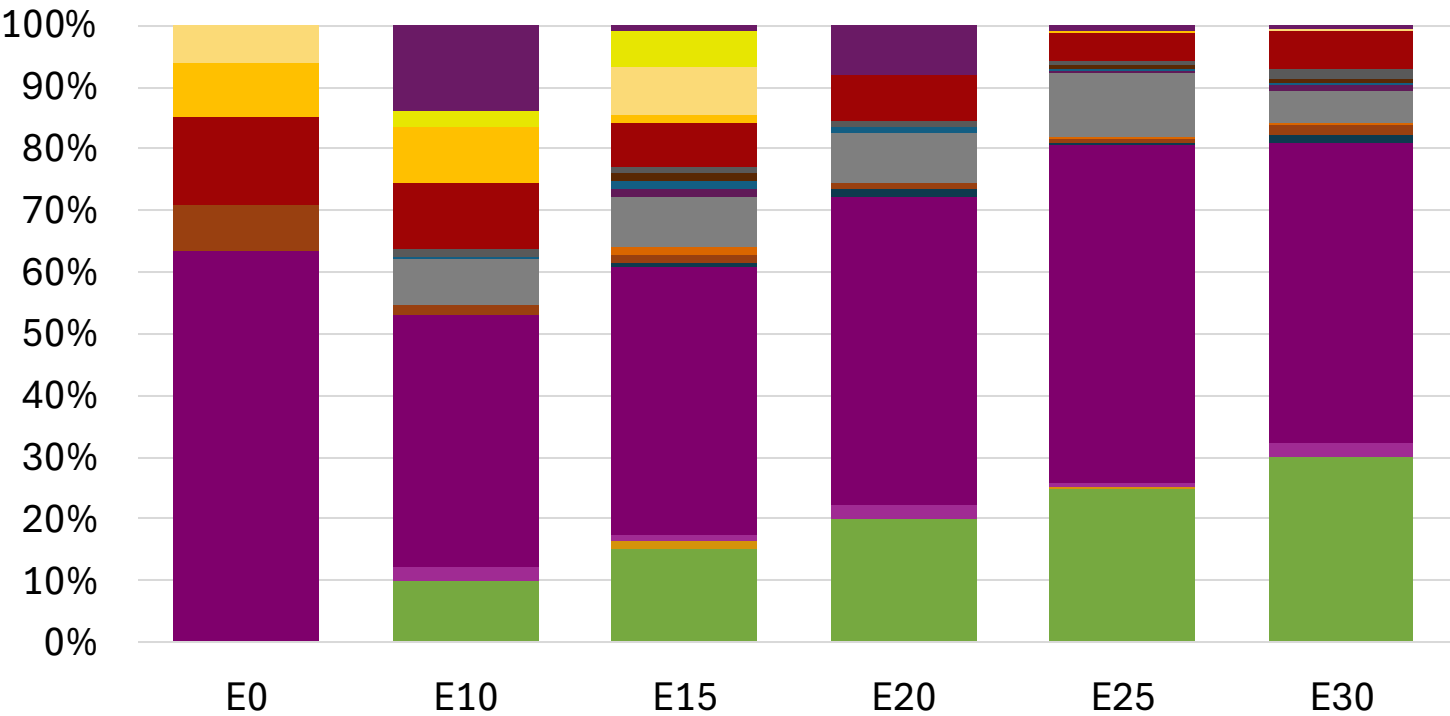


Average CIF 2024
Prices

Octane (RON)	95.0	96.2	98.3	98.7	100.5	101.8
Price (US\$/gal)	\$ 2.202	\$ 2.202	\$ 2.202	\$ 2.202	\$ 2.202	\$ 2.202

Source: Faro90

Czechia – Gasoline 98 – Increased Octane

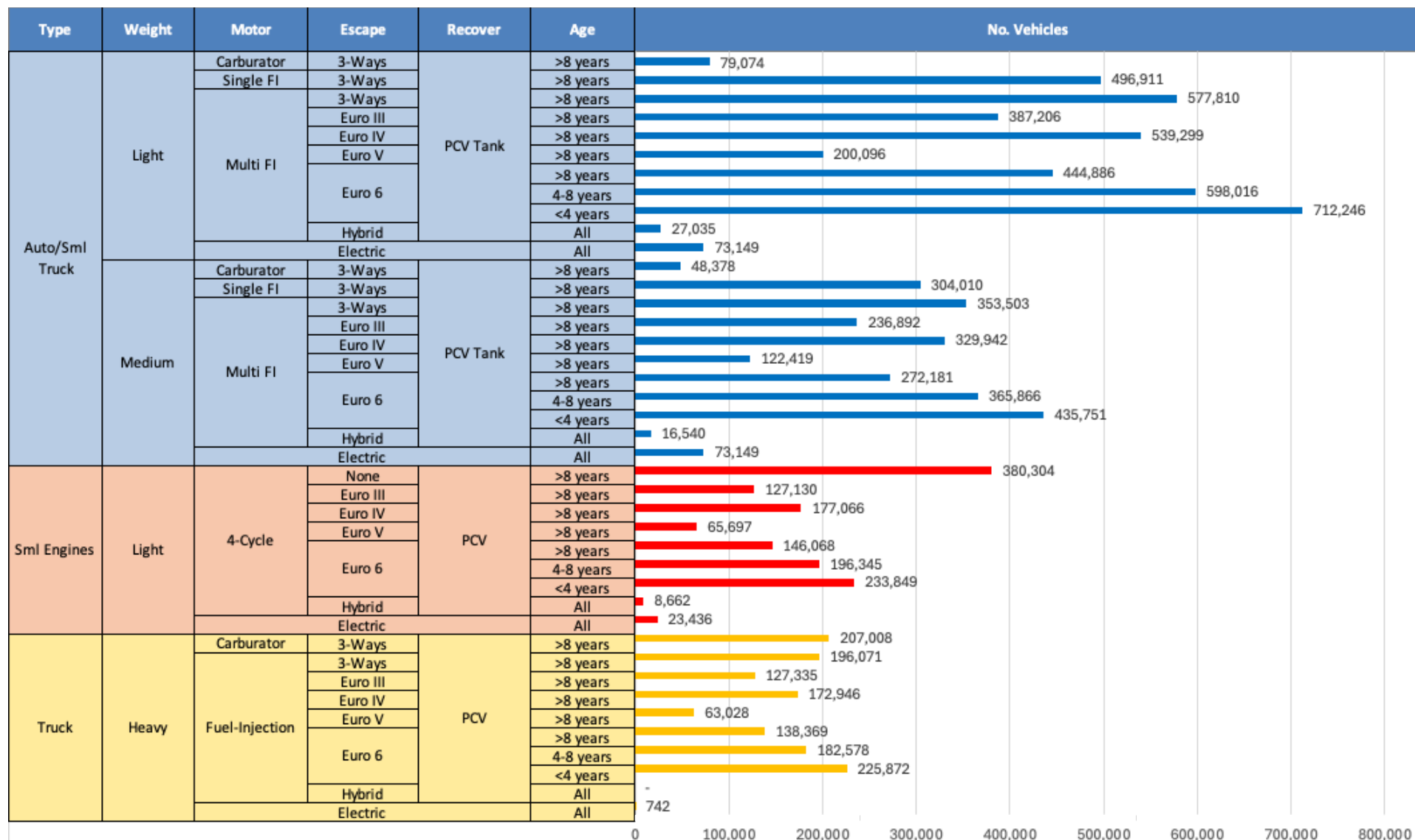


Average CIF 2024
Prices

Octane (RON)	98.0	99.0	100.9	101.2	102.8	104.1
Price (US\$/gal)	\$ 2.334	\$ 2.334	\$ 2.334	\$ 2.334	\$ 2.334	\$ 2.334

Source: Faro90

Gasoline Vehicle Fleet - Czechia

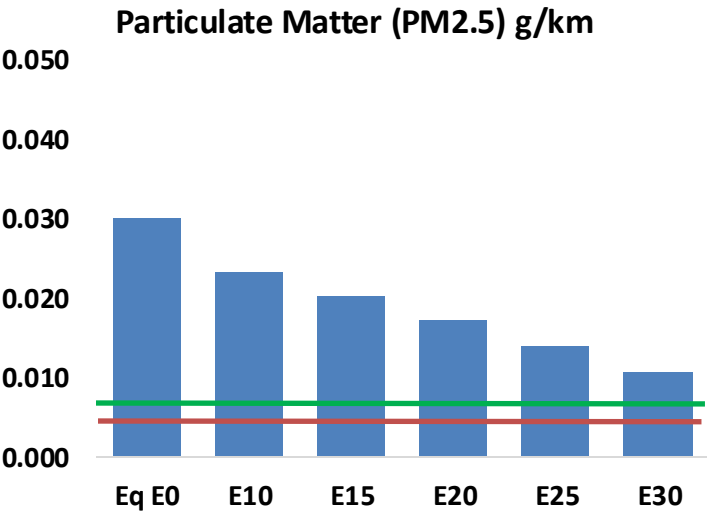
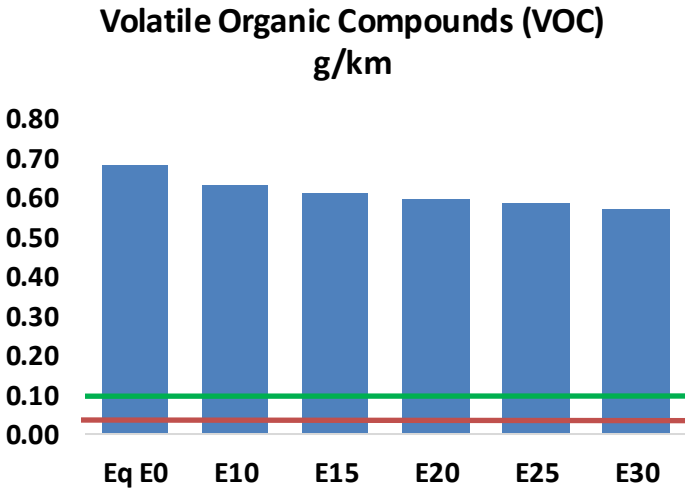
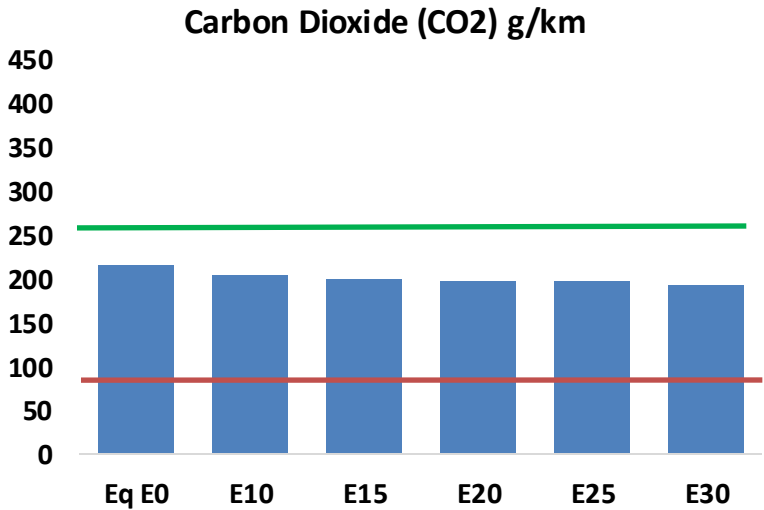
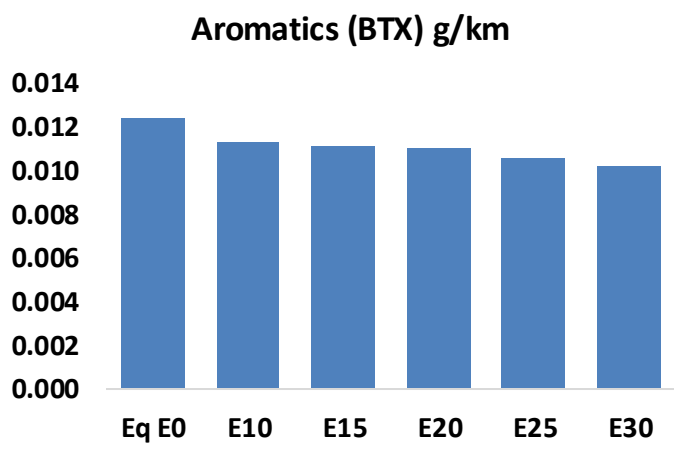
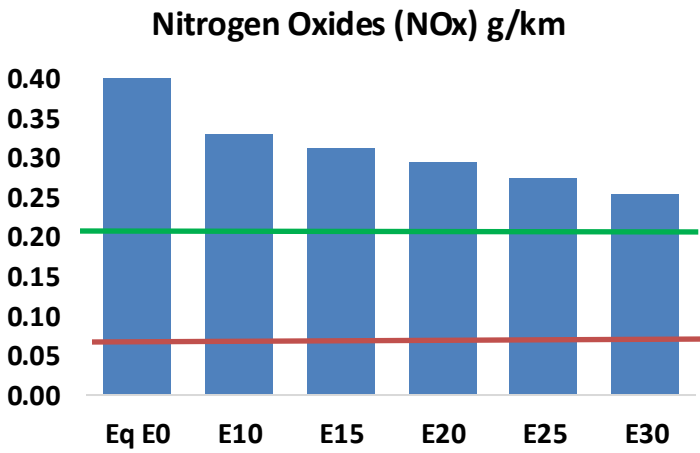
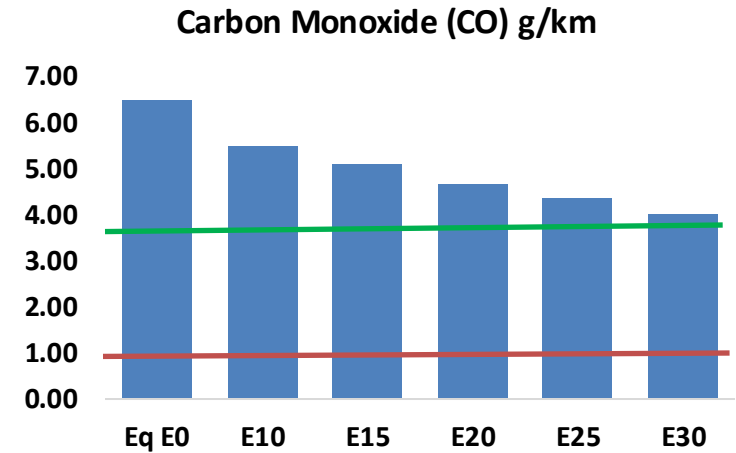
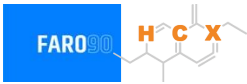


Vehicle Fleet: 9,366,865
Gasoline Vehicle Fleet: 3,468,160

Source: Eurostat, ACEA

Czechia – Vehicular Emissions

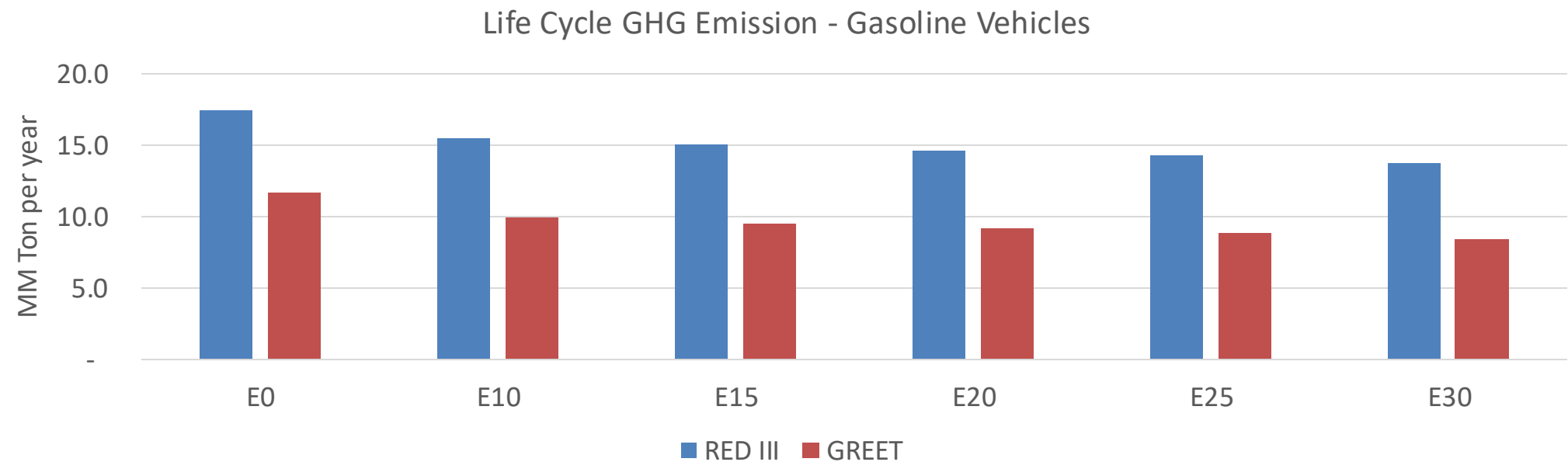
TIER III BIN 125 United States
Euro 6 Standard



Czechia – Vehicular Emissions

Vehicular Fleet Emissions (kg/day)	Eq E0	E5	E10	E15	E20	E25	E30	Reduction (%)	
								E10	E20
CO	514,816	475,239	435,661	403,229	370,591	345,495	316,625	-15%	-28%
CO2	17,030,957	16,593,018	16,179,409	15,854,118	15,693,151	15,618,037	15,330,151	-5%	-8%
VOC	54,073	51,964	49,856	48,528	47,375	46,544	45,240	-8%	-12%
NOx	37,565	28,173	26,295	24,783	23,373	21,817	20,172	-30%	-38%
SOx	196	181	166	154	142	129	115	-15%	-28%
NH3	5,868	5,810	5,752	5,779	5,845	5,890	5,928	-2%	0%
N2O	308	306	305	306	313	316	320	-1%	2%
CH4	11,875	11,875	11,875	12,053	12,314	12,540	12,754	0%	4%
Lead	-	-	-	-	-	-	-	0%	0%
Butadiene	361	343	324	310	297	288	274	-10%	-18%
Acetaldehyde	771	885	1,299	1,977	2,694	3,079	3,638	68%	249%
Formaldehyde	2,546	2,475	2,885	3,388	3,549	3,926	4,274	13%	39%
Benzene	987	948	898	884	877	839	812	-9%	-11%
PM2.5	1,023	909	794	689	585	475	360	-22%	-43%
PM10	1,353	1,202	1,050	912	774	628	476	-22%	-43%

Czechia – Gasoline Vehicle Fleet Life Cycle GHG Emissions



Country Emissions 2024 (MMT/año)	108.0
Target @ 2035 (MMT/year)	60.8
Delta	47.2

%Reduction vs E0	E0	E10	E15	E20	E25	E30
GREET 2024	0.0%	15.1%	18.6%	21.5%	24.3%	27.9%
RED II/III	0.0%	11.1%	13.8%	16.2%	18.4%	21.2%

% Target@2035	E0	E10	E15	E20	E25	E30
GREET 2024	0.0%	3.8%	4.6%	5.3%	6.0%	6.9%
RED II/III	0.0%	4.1%	5.1%	6.0%	6.8%	7.8%

Ethanol Gasoline Blending in Denmark

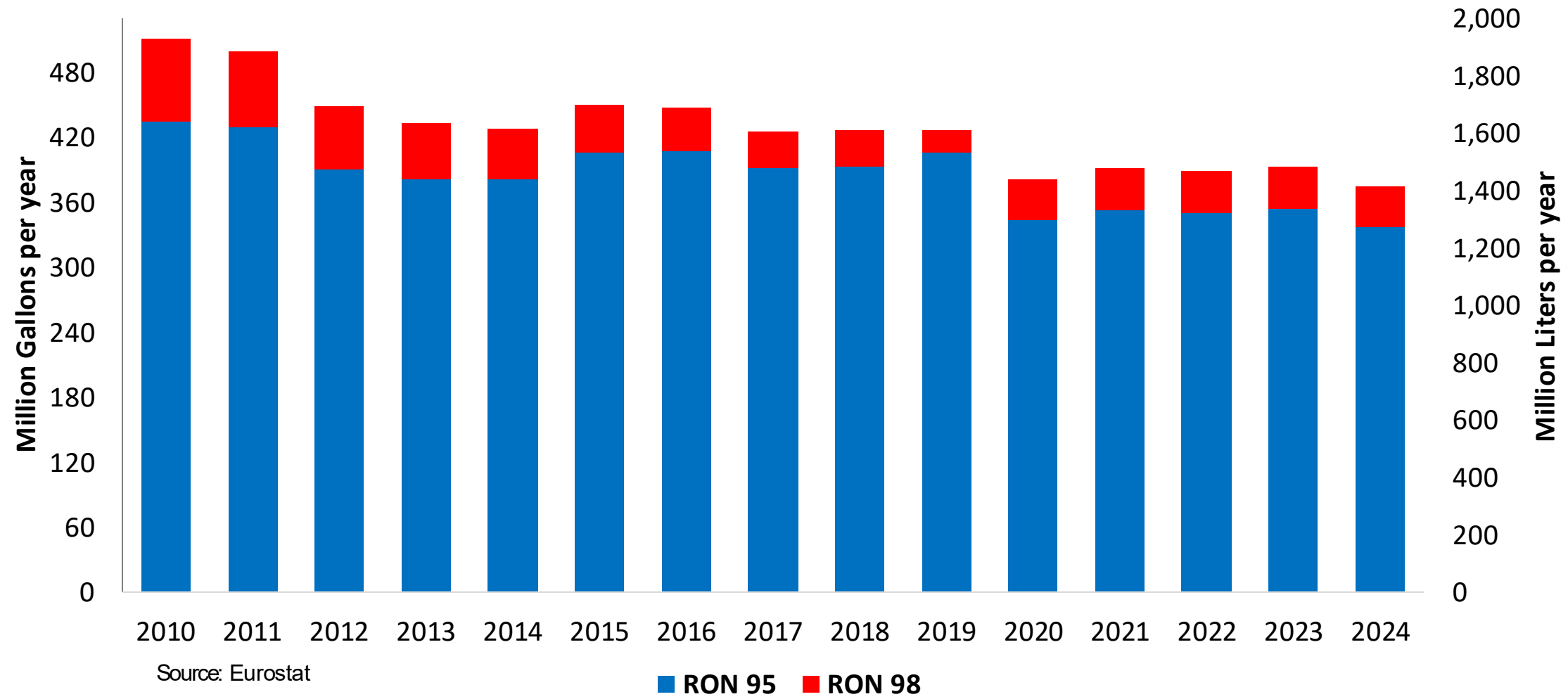


In 2024, gasoline consumption in Denmark was 1,415 million liters (374 million gallons) and growth rate was -2.6%. Gasoline production and net imports were 2,745 and -1,072 million liters (725 and -283 million gallons) correspondingly. Denmark complies with the European gasoline standards and offers two main grades: RON 95 and RON 98) with an average octane of 95.3 RON.

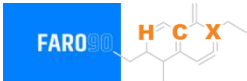
In 2024, ethanol consumption in Denmark was 168 million liters (44 million gallons) representing 11.9% of gasoline demand. Ethanol production and Net Imports were 160 and 8 million liters (44 and 2 million gallons) correspondingly.

Source: Eurostat, European Commission

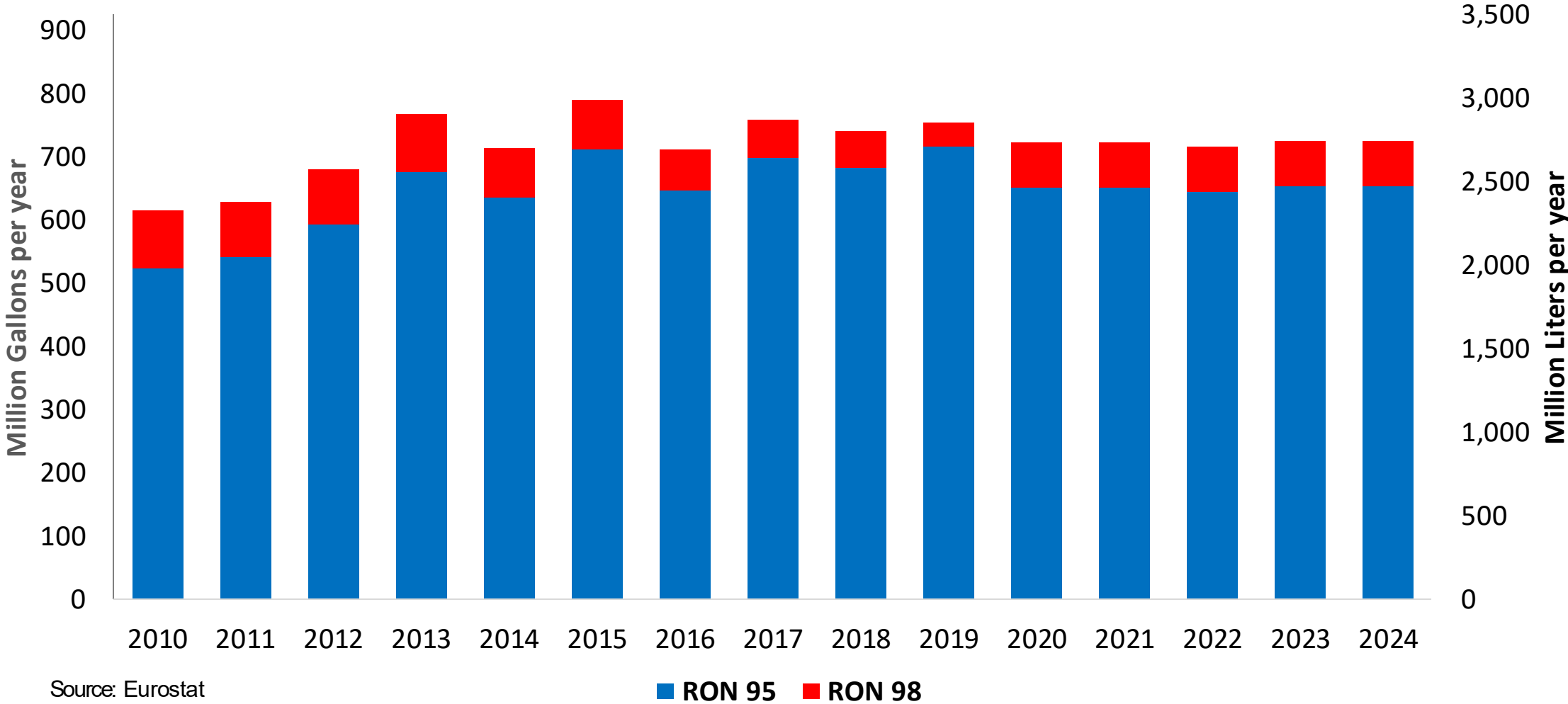
Gasoline Demand by Grade in Denmark



Gasoline Production in Denmark



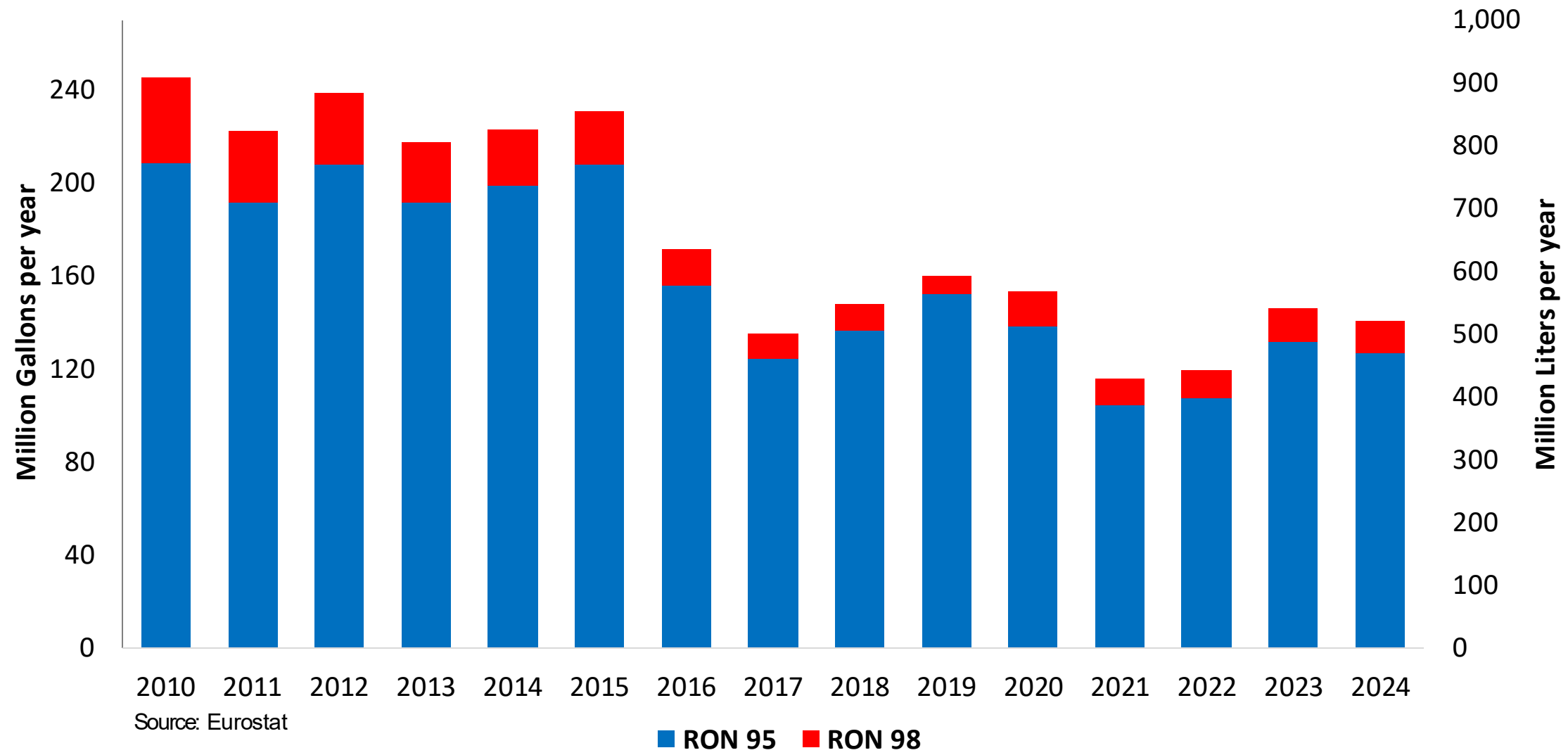
Gasoline Production by Grade in Denmark



Source: Eurostat

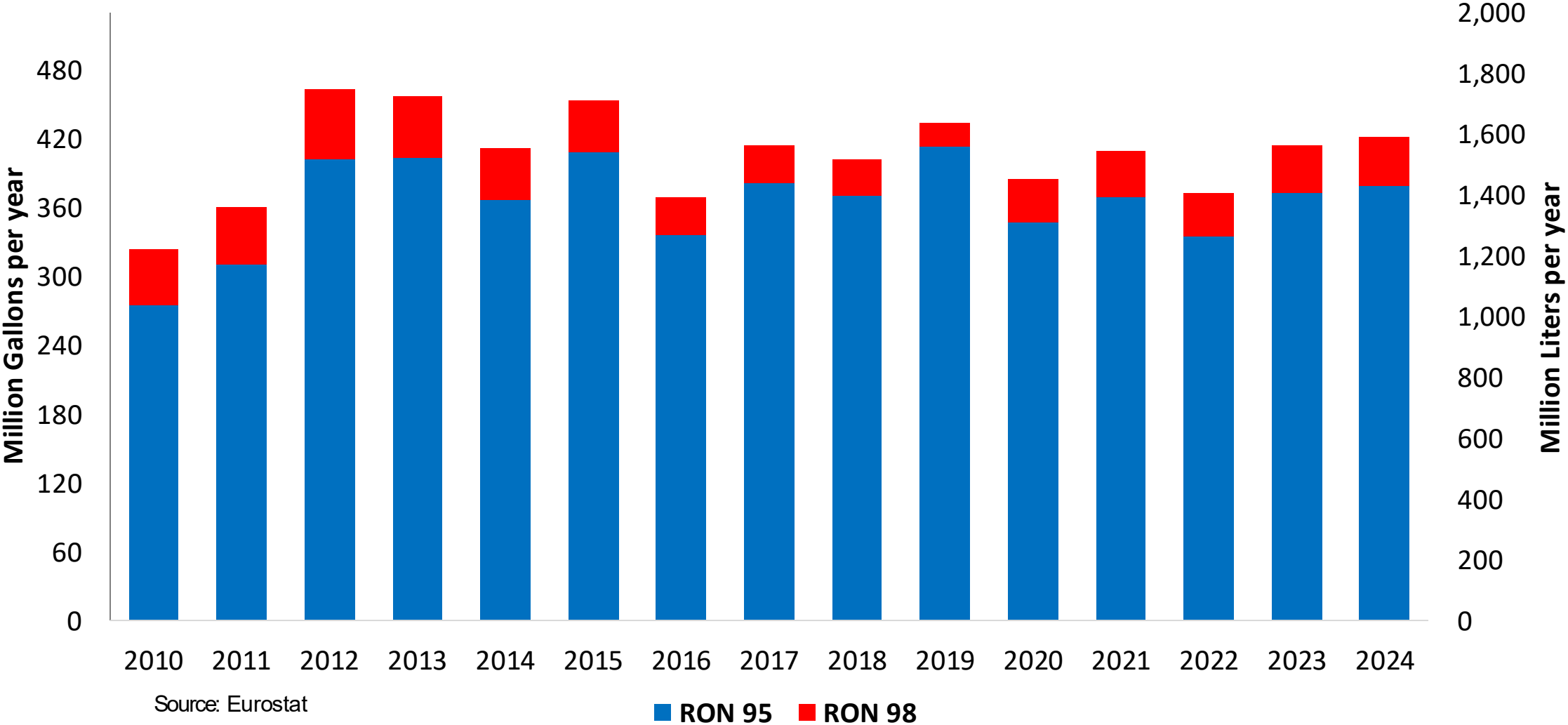
Gasoline Imports in Denmark

Gasoline Imports by Grade in Denmark

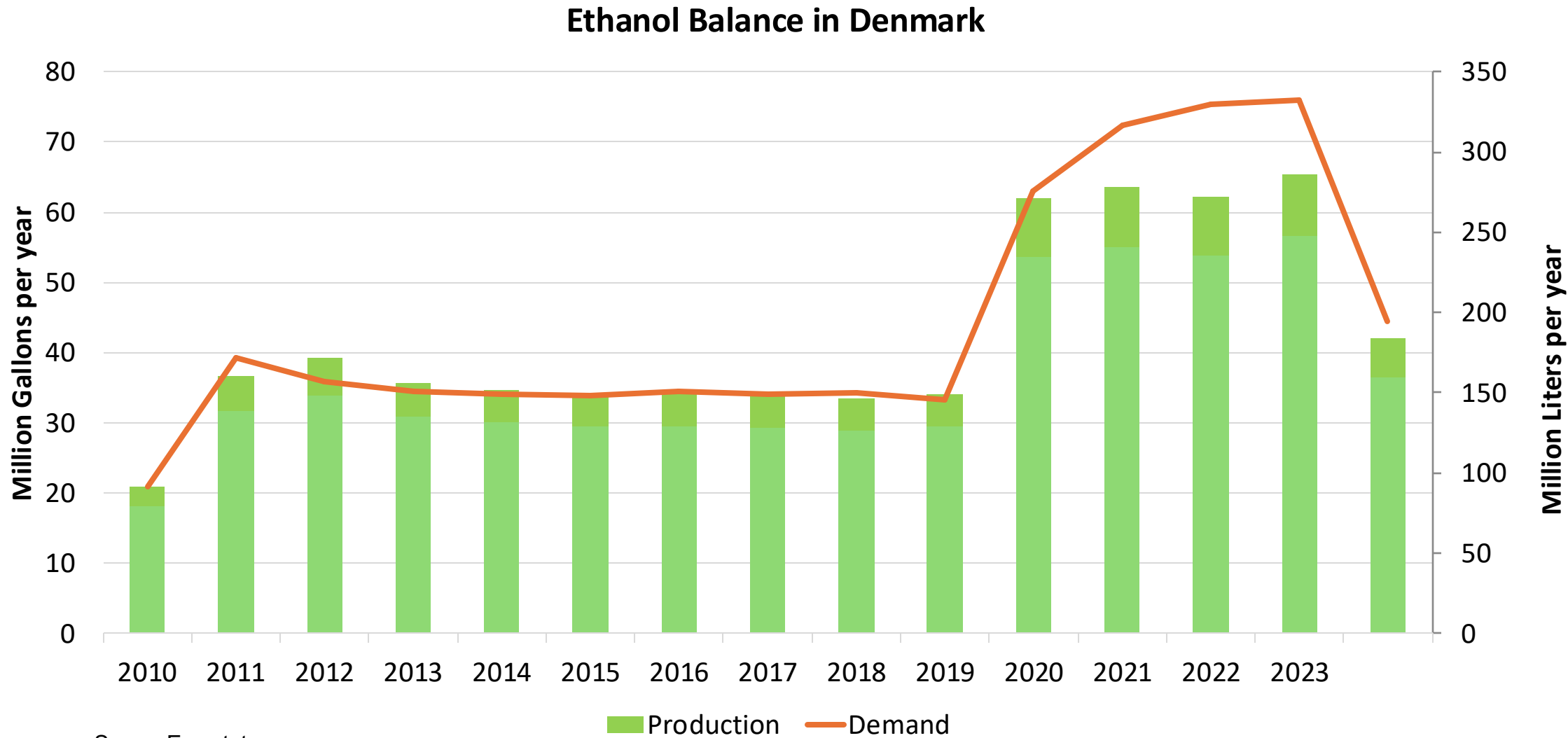


Gasoline Exports in Denmark

Gasoline Exports by Grade in Denmark



Ethanol Balance in Denmark



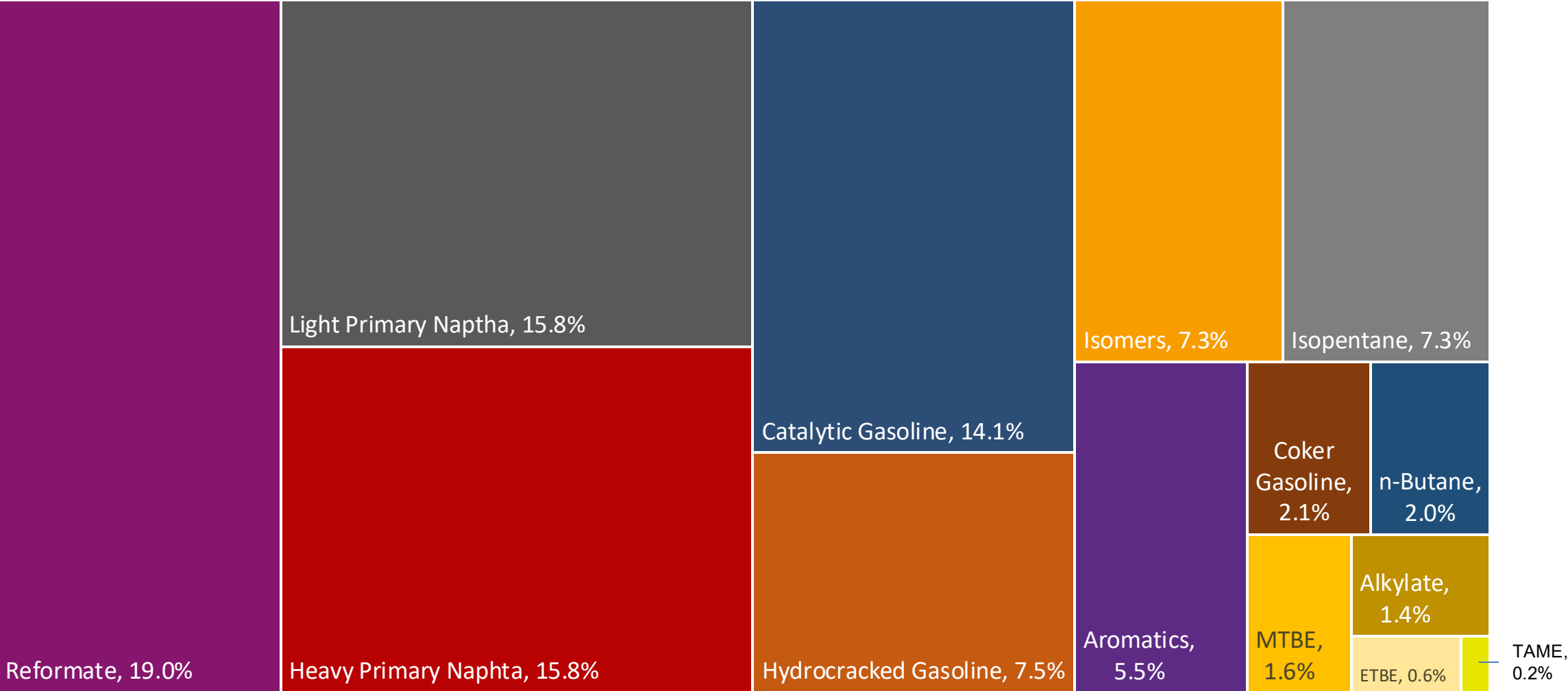
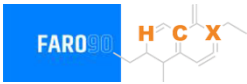
Gasoline Quality in Denmark

Name	EN 228:2025 (Euro 6); Country Level Fuel Ordinances							
Implementation Date	2025							
Applicability	All EU Contries + UK							
Grade	RON 95 E5		RON 95 E10		RON 98 E5		RON 98 E10	
	Min	Max	Min	Max	Min	Max	Min	Max
RON	95.0		95.0		98.0		98.0	
MON	> 85	> 88	> 85	> 88	> 85	> 88	> 85	> 88
AKI								
Benzene (%vol)	< 1							
Aromatics (% vol)	< 35							
Olefins (%vol)	< 18							
Lead (g/L)	< 0.005							
Manganese (mg/L)	< 2,0							
Sulfur (ppm)	10.0							
Oxygen (%w)*	< 2,7	< 3,7	< 2,7	< 3,7	< 2,7	< 3,7	< 2,7	< 3,7
Ethanol (%vol)	< 5	< 10	< 5	< 10	< 5	< 10	< 5	< 10
PVR 37.8°C (psi)	<> 45 - 100 kPa in 6 Volatility Classes							
Overall Biofuels (%cal)	-							
Advanced Biofuels (%cal)	1.0							
Bioethanol in Gasoline Sales (%cal)	-							
GHG Emission Reduction (%)	5.2							

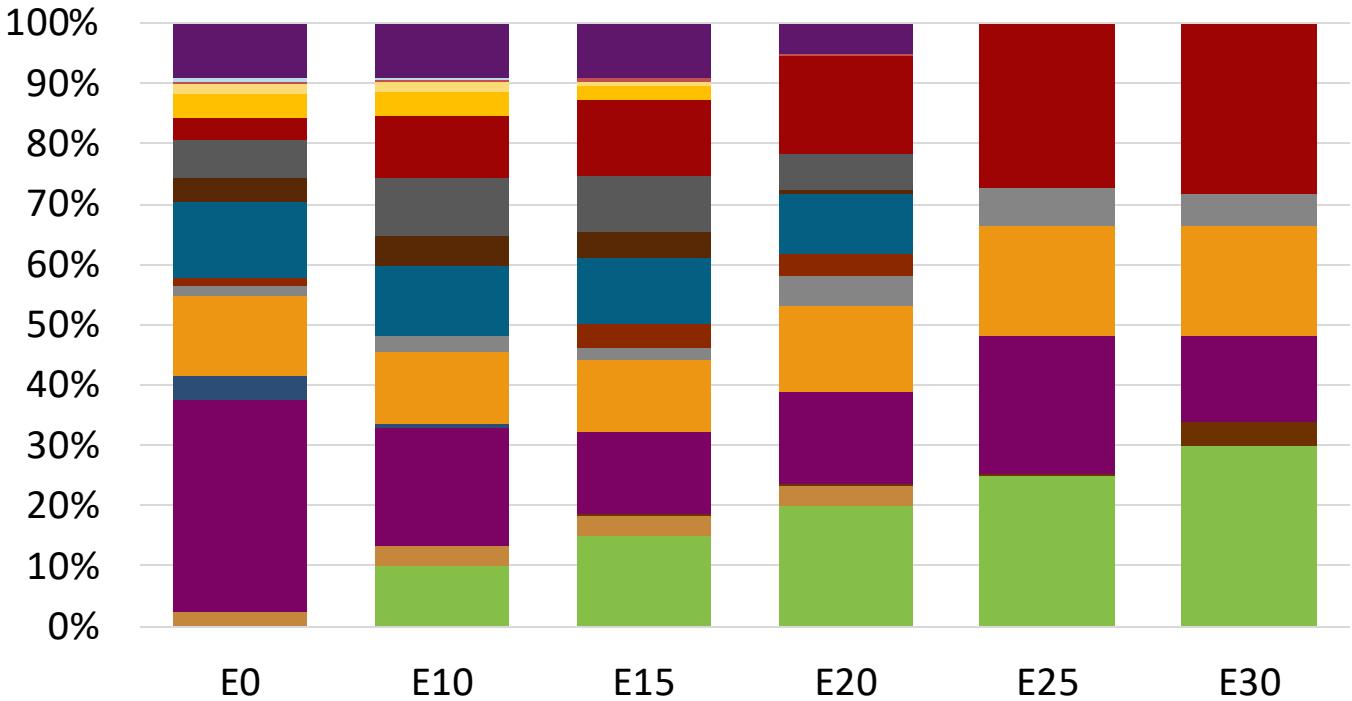
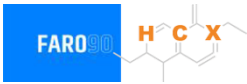
* Methanol 3%, Iso-propyl alcohol 12%, Ter-butyl alcohol 15%, Iso-butyl alcohol 15%, Ethers +5 Carbons 22%, Other Oxygenates 15%

Source: European Commision

Denmark Available Blending Components



Denmark – Gasoline 95 – Constant Octane

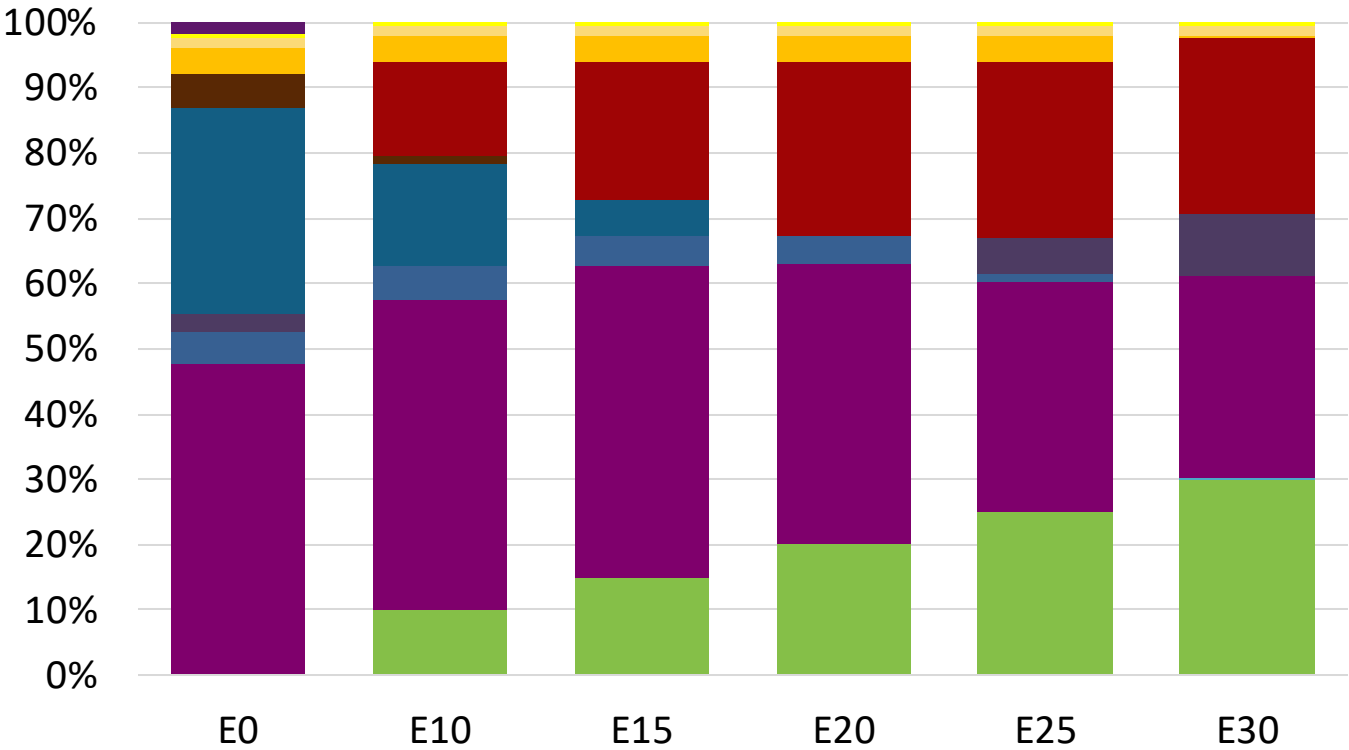
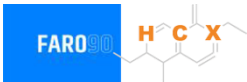


Average CIF 2024 Prices

Octane (RON)	95.0	95.0	95.0	95.0	95.1	95.6
Price (US\$/gal)	\$ 2.272	\$ 2.144	\$ 2.083	\$ 2.002	\$ 1.880	\$ 1.814

Source: Faro90

Denmark - Gasoline 98 - Constant Octane

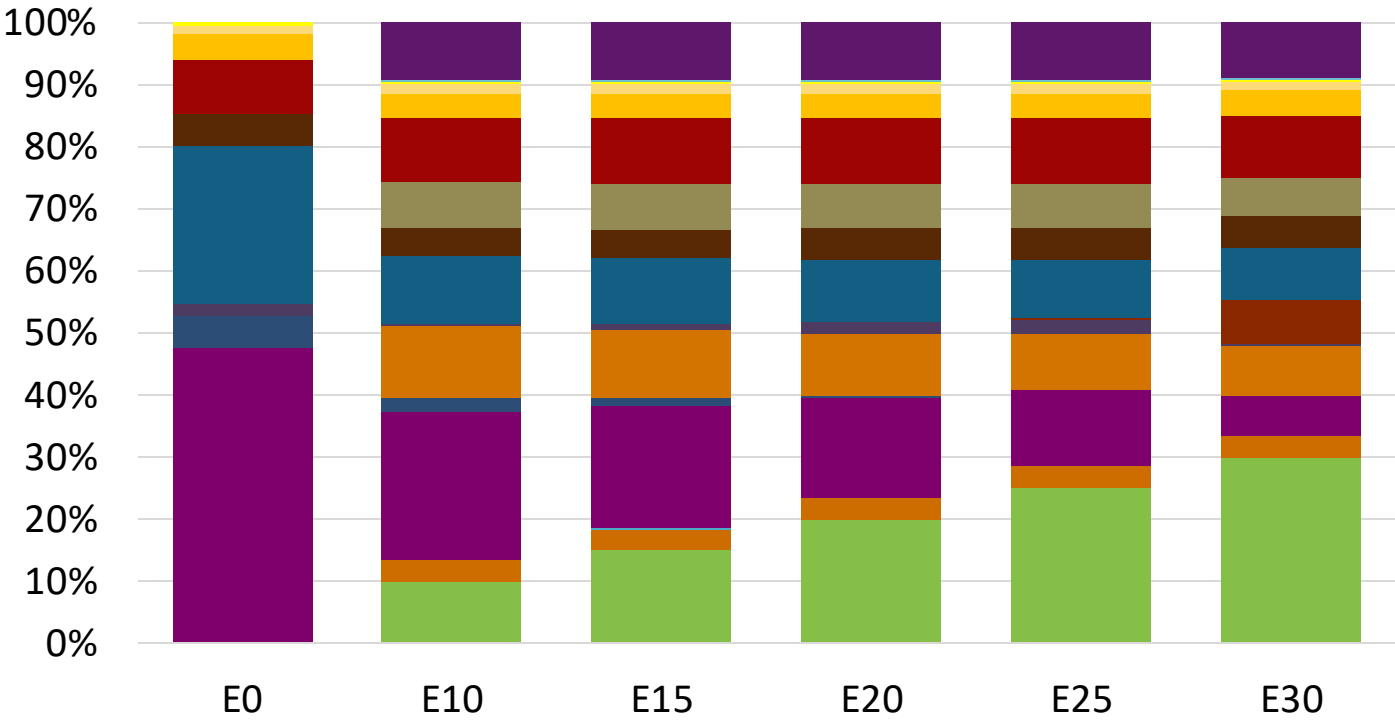
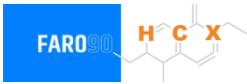


Average CIF 2024 Prices

Octane (RON)	98.0	98.0	98.0	98.0	98.0	98.0
Price (US\$/gal)	\$ 2.347	\$ 2.218	\$ 2.146	\$ 2.069	\$ 1.992	\$ 1.925

Source: Faro90

Denmark – Gasoline 95 – Increased Octane

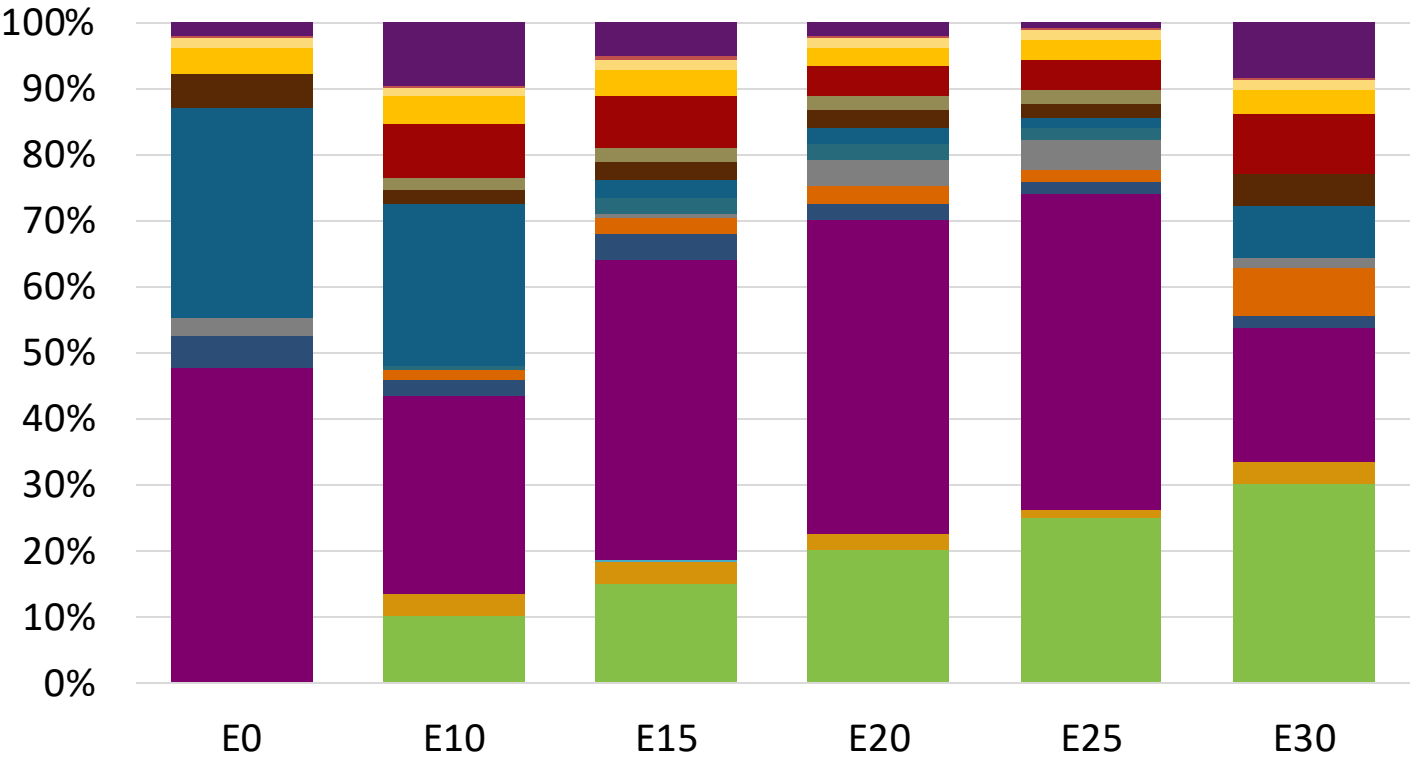


Average CIF 2024
Prices

Octane (RON)	95.0	96.5	97.9	99.4	100.8	102.0
Price (US\$/gal)	\$ 2.209	\$ 2.209	\$ 2.209	\$ 2.209	\$ 2.209	\$ 2.209

Source: Faro90

Denmark – Gasoline 98 – Increased Octane

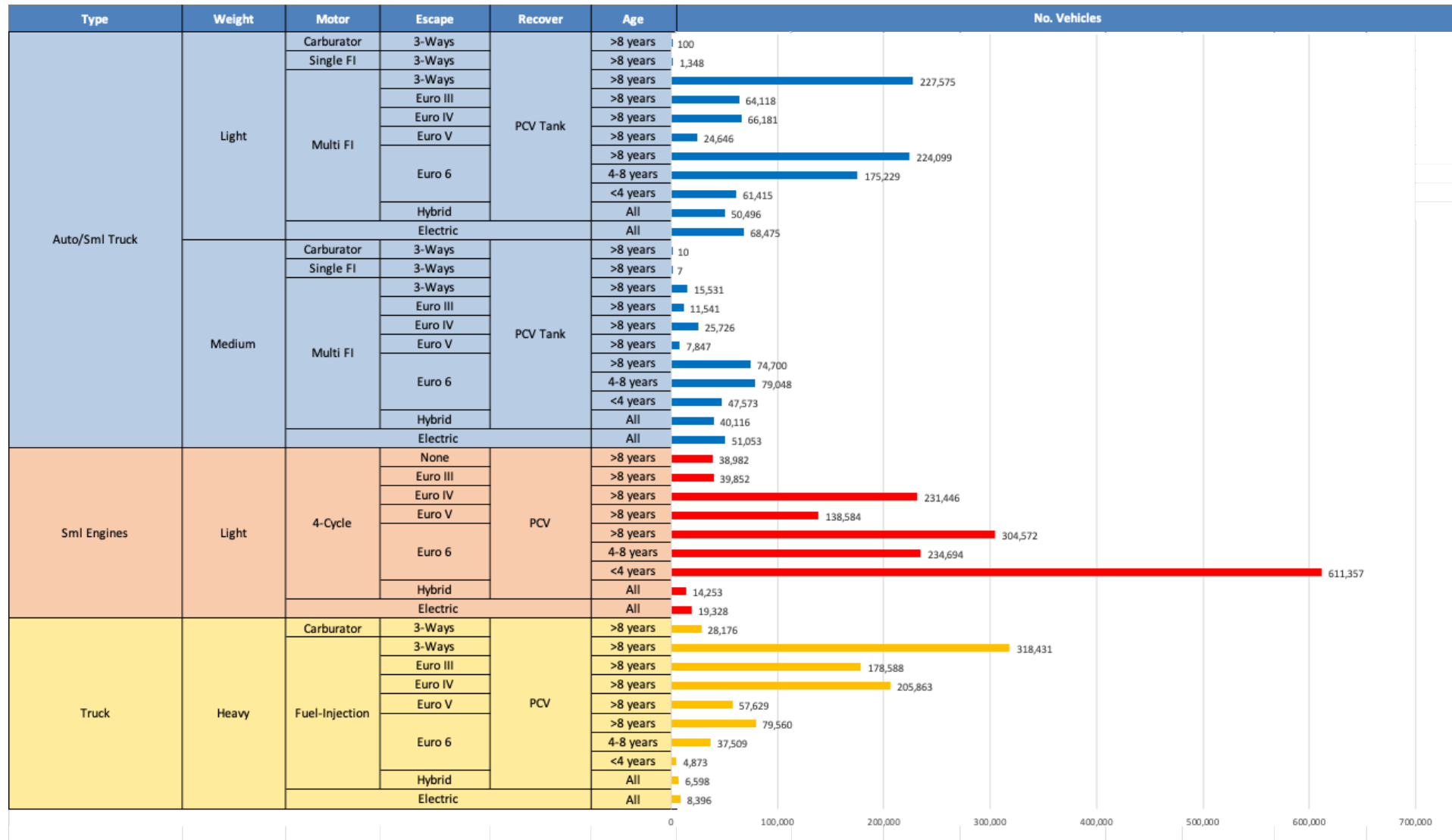


Average CIF 2024
Prices

Octane (RON)	98.0	99.2	101.3	102.6	104.0	104.7
Price (US\$/gal)	\$ 2.347	\$ 2.347	\$ 2.347	\$ 2.347	\$ 2.347	\$ 2.347

Source: Faro90

Gasoline Vehicle Fleet - Denmark

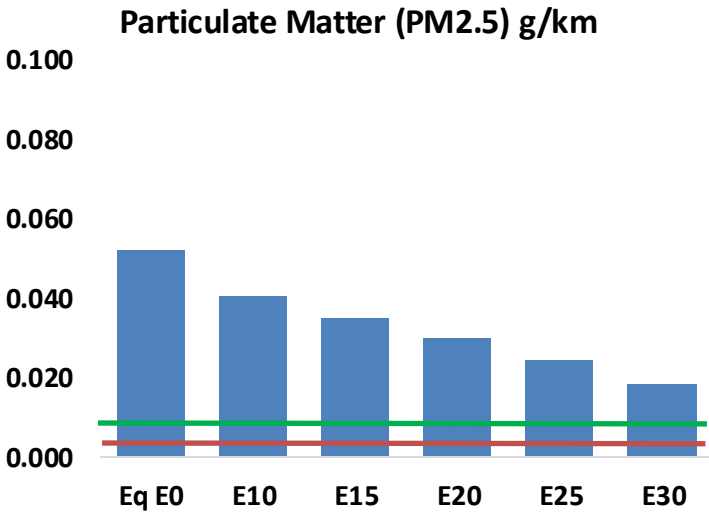
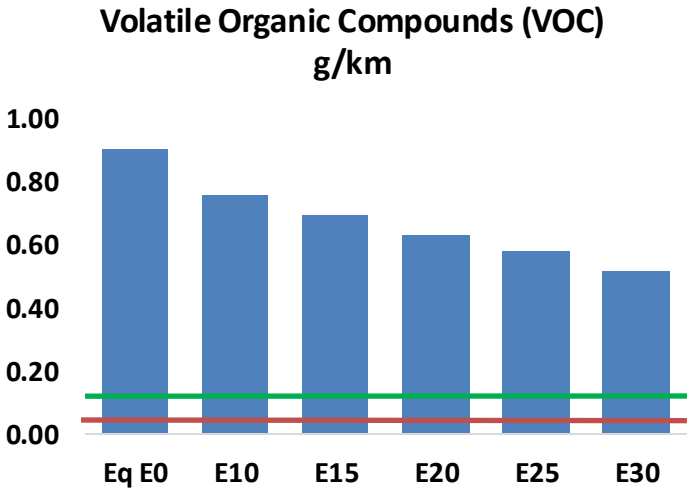
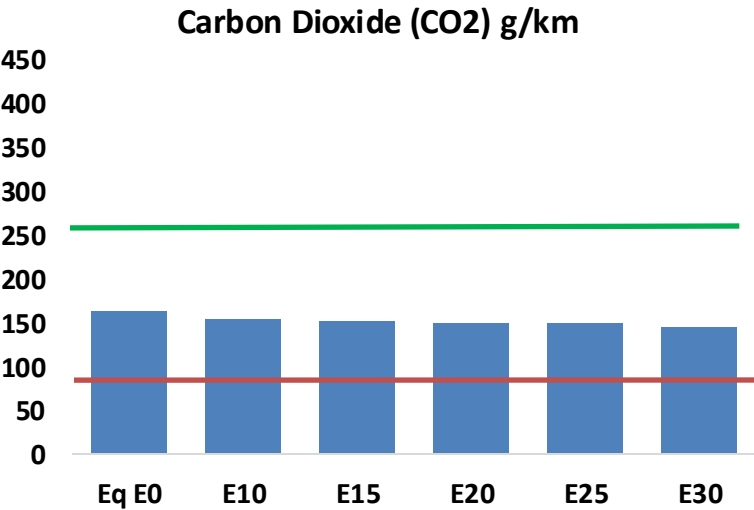
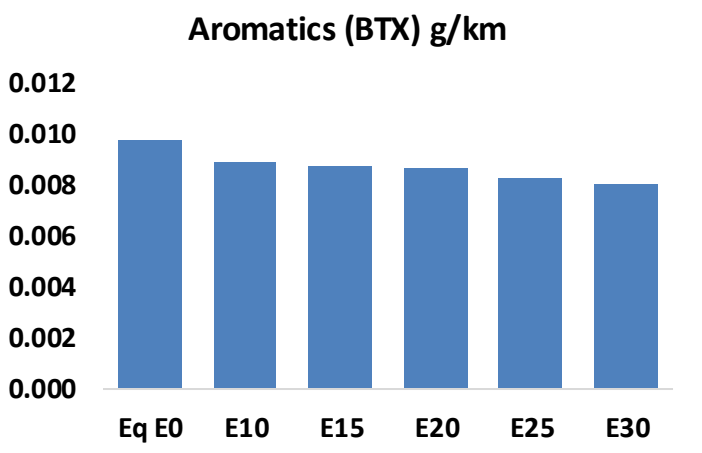
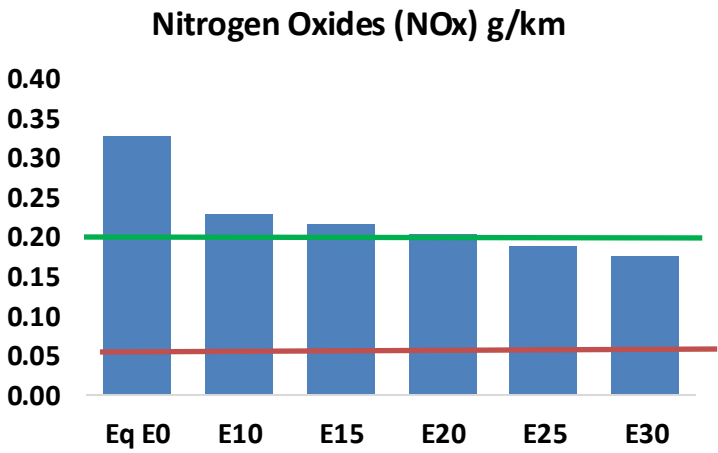
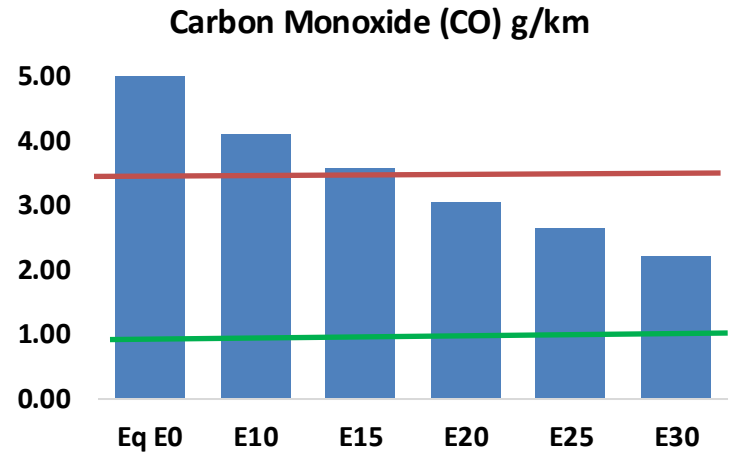
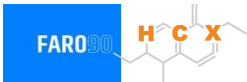


Vehicle Fleet: 3,875,526
Gasoline Vehicle Fleet: 2,055,566

Source: Eurostat, ACEA

Denmark – Vehicular Emissions

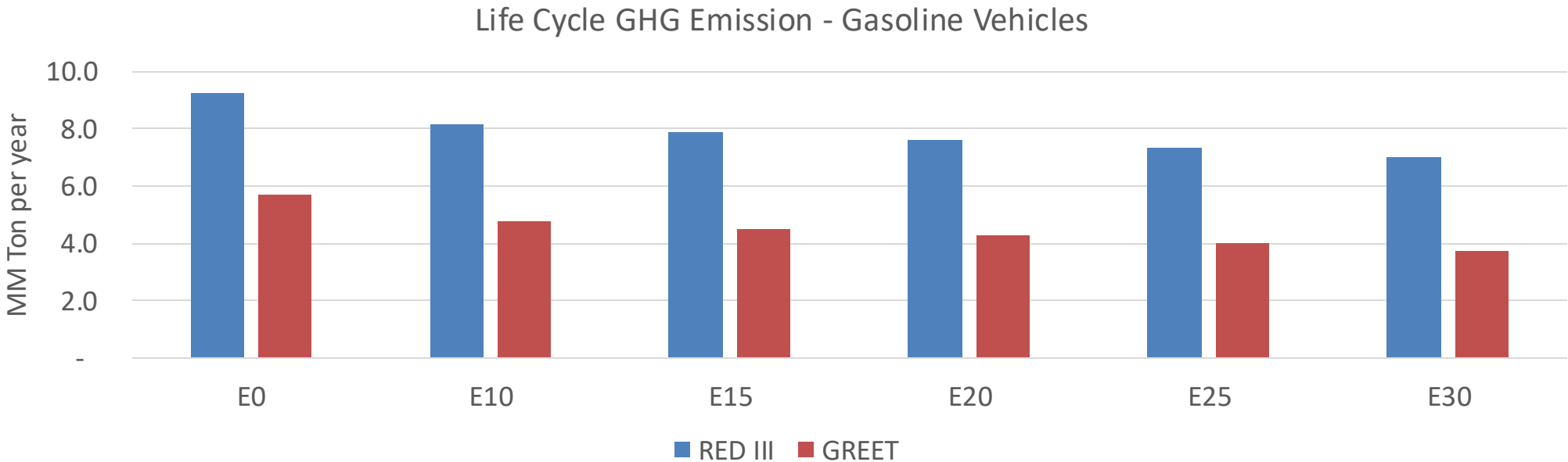
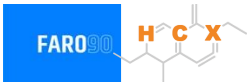
TIER III BIN 125 United States
Euro 6 Standard



Denmark – Vehicular Emissions

Vehicular Fleet Emissions (kg/day)	Eq E0	E5	E10	E15	E20	E25	E30	Reduction (%)	
								E10	E20
CO	268,045	238,243	208,441	182,226	155,019	133,849	112,318	-22%	-42%
CO2	8,251,751	8,039,563	7,839,163	7,681,555	7,603,564	7,562,067	7,422,676	-5%	-8%
VOC	46,038	42,219	38,400	35,216	31,986	29,495	26,198	-17%	-31%
NOx	16,648	12,486	11,654	10,984	10,359	9,669	8,940	-30%	-38%
SOx	97	90	82	76	70	64	57	-15%	-28%
NH3	3,339	3,306	3,272	3,288	3,325	3,351	3,373	-2%	0%
N2O	140	140	139	140	143	144	146	-1%	2%
CH4	10,288	10,288	10,288	10,442	10,669	10,864	11,049	0%	4%
Lead	-	-	-	-	-	-	-	0%	0%
Butadiene	269	252	236	223	210	200	187	-12%	-22%
Acetaldehyde	843	967	1,419	2,161	2,944	3,364	3,975	68%	249%
Formaldehyde	3,275	3,184	3,712	4,358	4,566	5,051	5,499	13%	39%
Benzene	498	478	453	446	442	423	409	-9%	-11%
PM2.5	557	494	432	375	318	258	196	-22%	-43%
PM10	2,084	1,850	1,617	1,404	1,192	967	733	-22%	-43%

Denmark – Gasoline Vehicle Fleet Life Cycle GHG Emissions



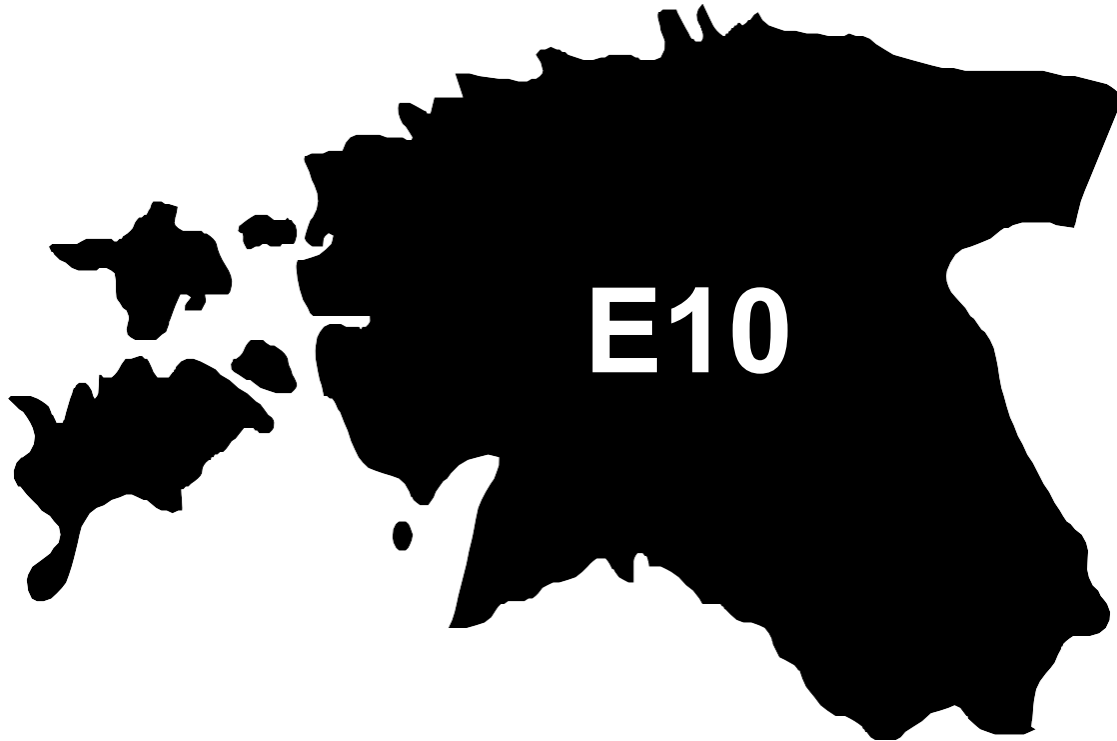
Country Emissions 2024 (MMT/año)	45.5
Target @ 2035 (MMT/year)	25.6
Delta	19.9

%Reduction vs E0	E0	E10	E15	E20	E25	E30
GREET 2024	0.0%	16.6%	21.2%	25.2%	29.3%	34.1%
RED II/III	0.0%	11.4%	14.7%	17.6%	20.5%	23.8%

% Target@2035	E0	E10	E15	E20	E25	E30
GREET 2024	0.0%	4.8%	6.1%	7.2%	8.4%	9.8%
RED II/III	0.0%	5.3%	6.8%	8.2%	9.5%	11.0%

Source: Greet, RED II/III, IPCC, climateactiontracker.org, F90

Ethanol Gasoline Blending in Estonia

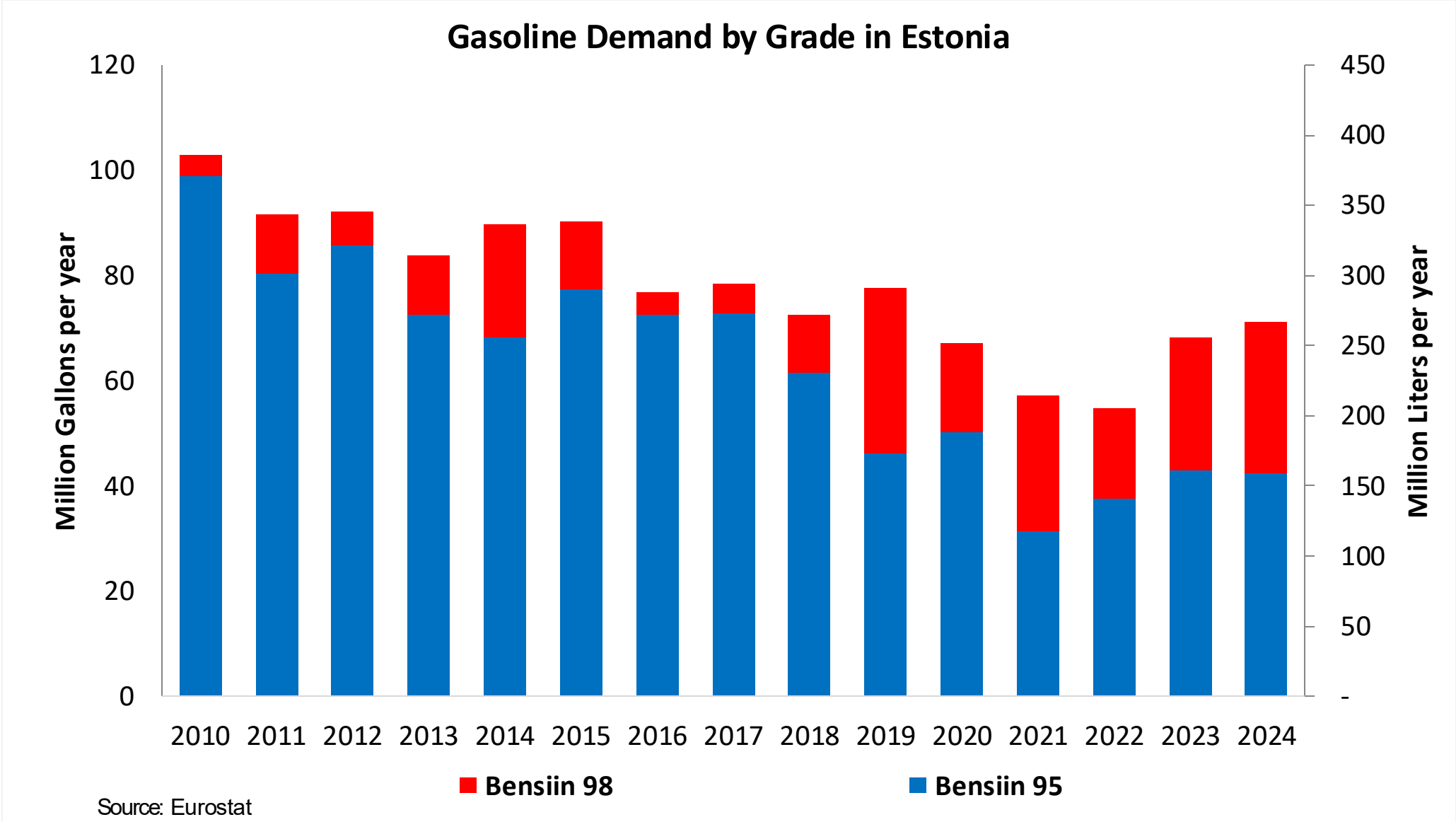


In 2024, gasoline consumption in Estonia was 267 million liters (71 million gallons) and growth rate was -1.8%. There is no gasoline production. Gasoline net imports were 267 million liters (70 million gallons). Estonia complies with the European gasoline standards and offers two main grades: RON 95 and RON 98) with an average octane of 96.2 RON.

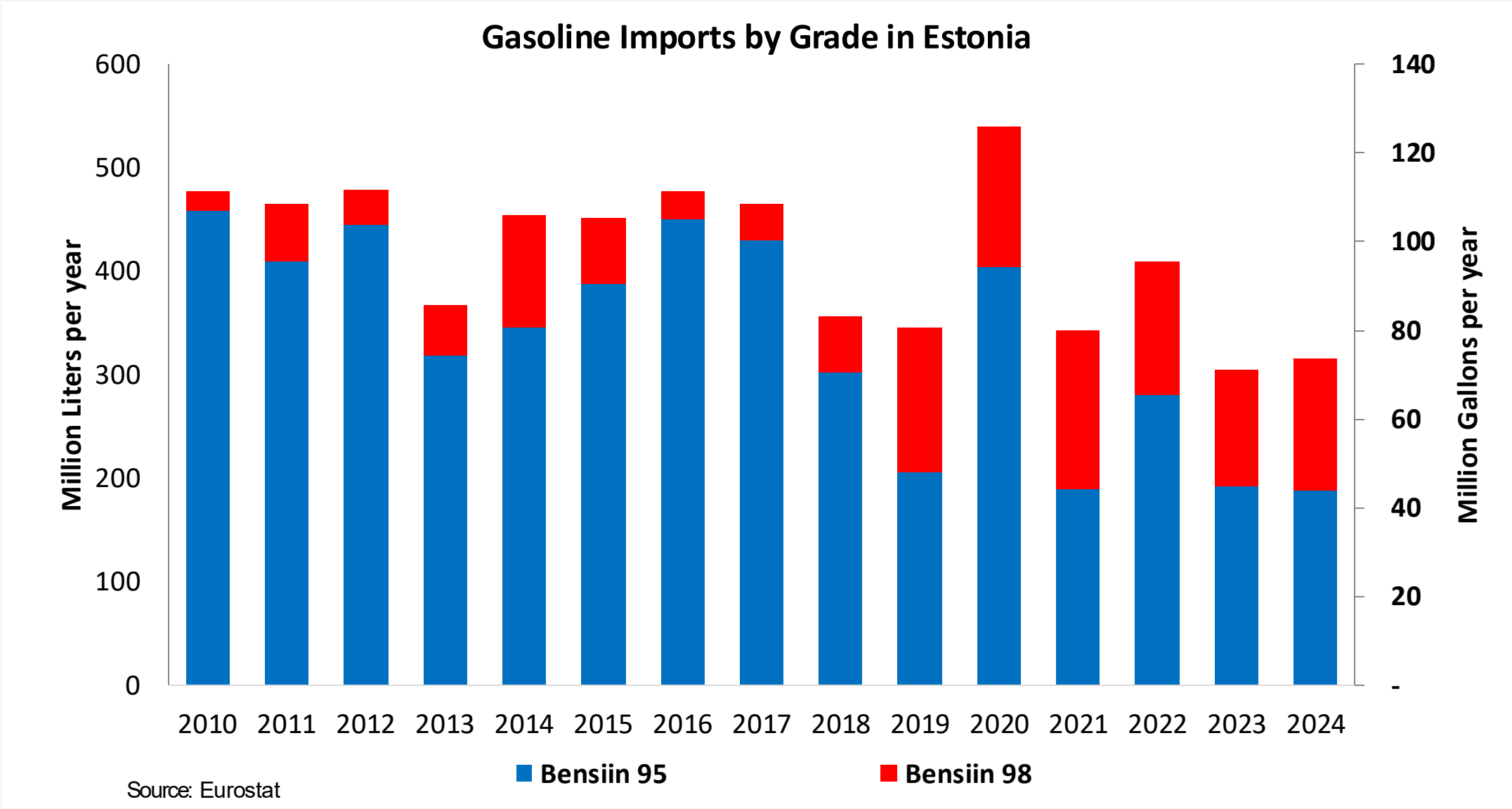
In 2024, ethanol consumption in Estonia was 5 million liters (1 million gallons) representing 1.8% of gasoline demand. Ethanol imports were 5 million liters (1 million gallons). Ethanol sources is Cereals 100%.

Source: Eurostat, European Commission

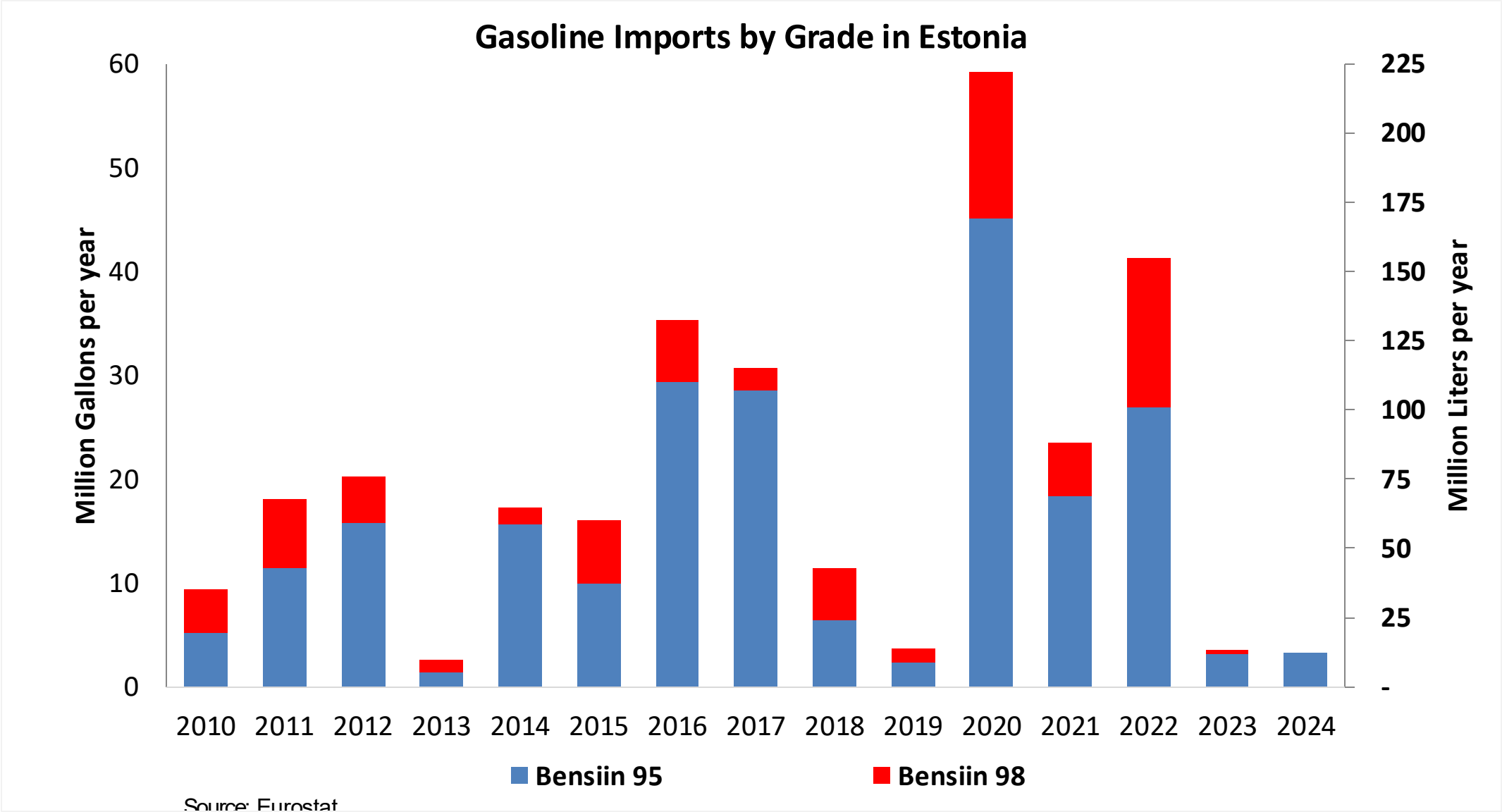
Gasoline Demand in Estonia



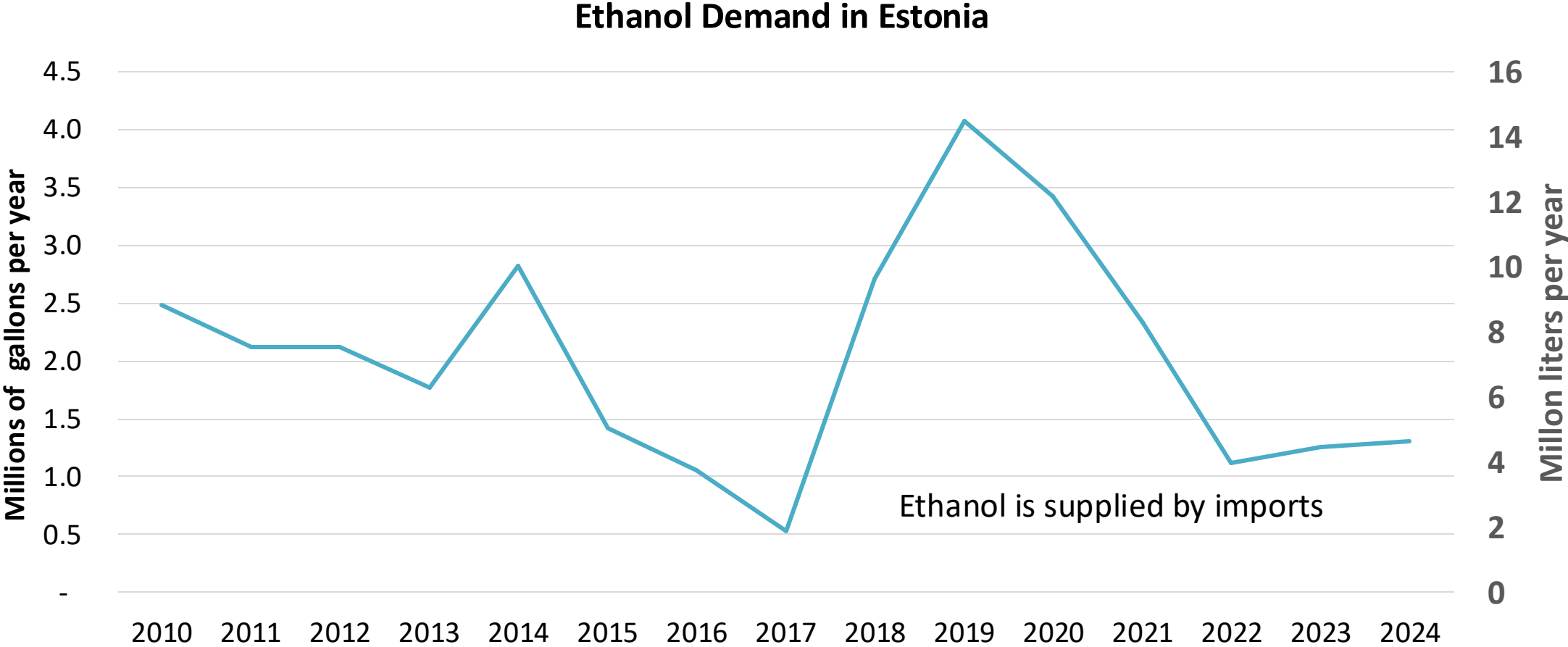
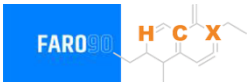
Gasoline Imports in Estonia



Gasoline Exports in Estonia

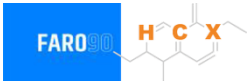


Ethanol Balance in Estonia



Source: Eurostat, Statistics Estonia

Gasoline Quality in Estonia

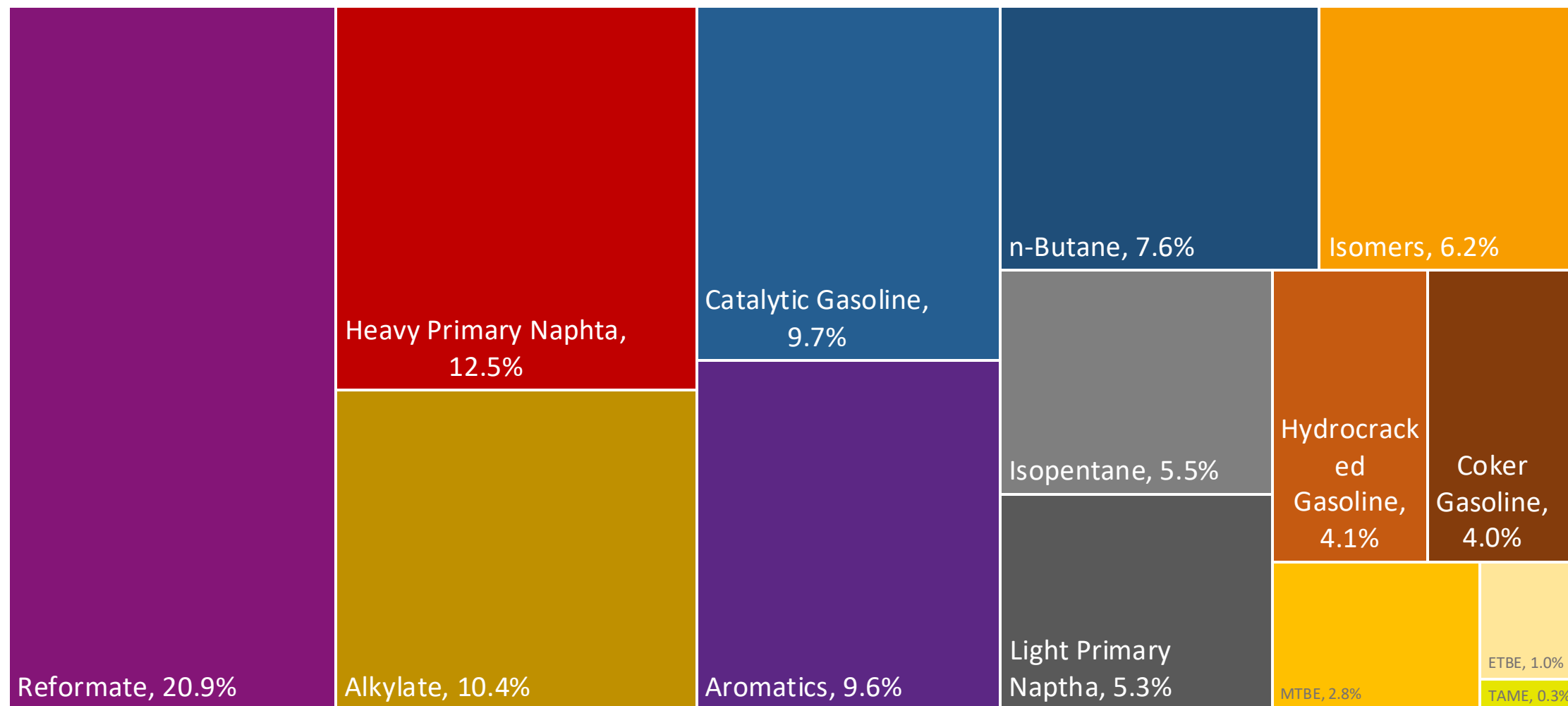


Name	EN 228:2025 (Euro 6); Country Level Fuel Ordinances							
Implementation Date	2025							
Applicability	All EU Contries + UK							
Grade	RON 95 E5		RON 95 E10		RON 98 E5		RON 98 E10	
	Min	Max	Min	Max	Min	Max	Min	Max
RON	95.0		95.0		98.0		98.0	
MON	> 85	> 88	> 85	> 88	> 85	> 88	> 85	> 88
AKI								
Benzene (%vol)	< 1							
Aromatics (% vol)	< 35							
Olefins (%vol)	< 18							
Lead (g/L)	< 0.005							
Manganese (mg/L)	< 2,0							
Sulfur (ppm)	10.0							
Oxygen (%w)*	< 2,7	< 3,7	< 2,7	< 3,7	< 2,7	< 3,7	< 2,7	< 3,7
Ethanol (%vol)	< 5	< 10	< 5	< 10	< 5	< 10	< 5	< 10
PVR 37.8°C (psi)	<> 45 - 100 kPa in 6 Volatility Classes							
Overall Biofuels (%cal)	7.5							
Advanced Biofuels (%cal)	0.5							
Bioethanol in Gasoline Sales (%cal)	-							
GHG Emission Reduction (%)	-							

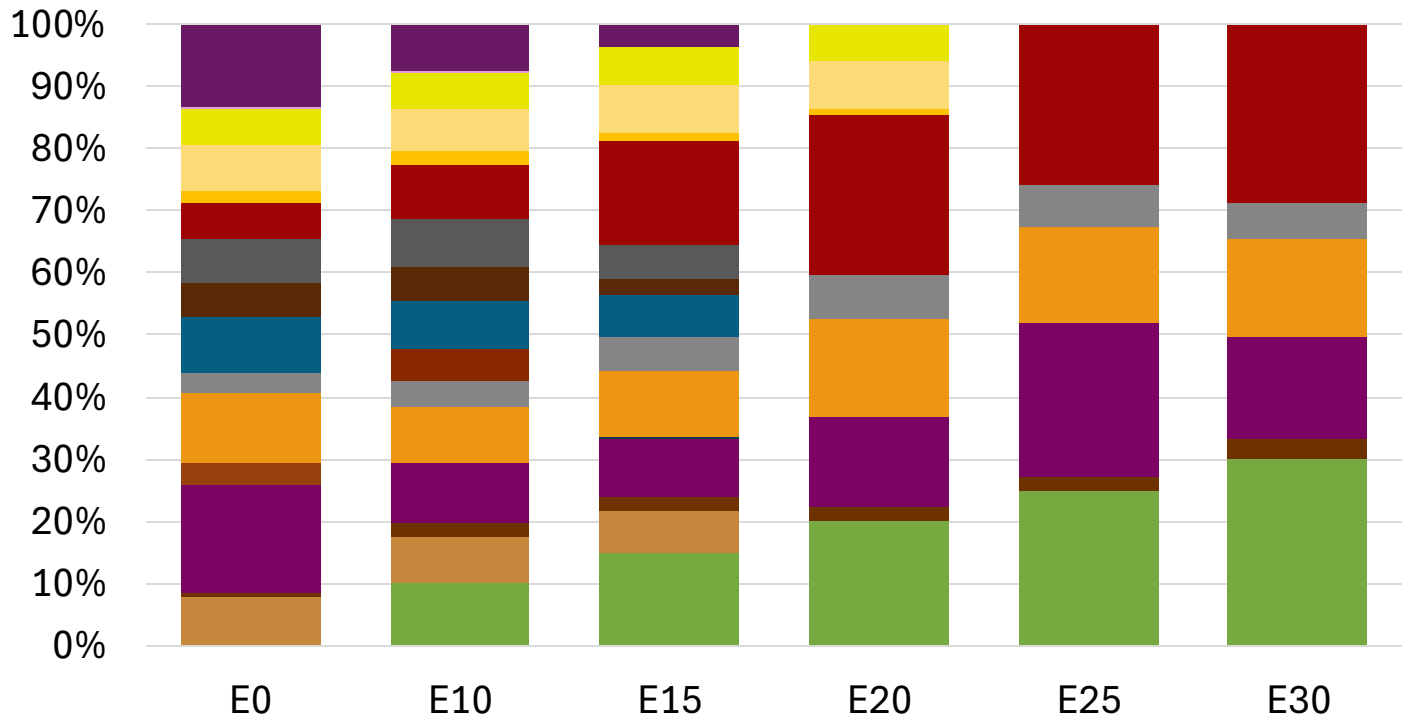
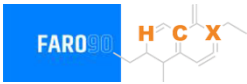
* Methanol 3%, Iso-propyl alcohol 12%, Ter-butyl alcohol 15%, Iso-butyl alcohol 15%, Ethers +5 Carbons 22%, Other Oxygenates 15%

Source: European Commision

Estonia Available Blending Components



Estonia – Gasoline 95 – Constant Octane

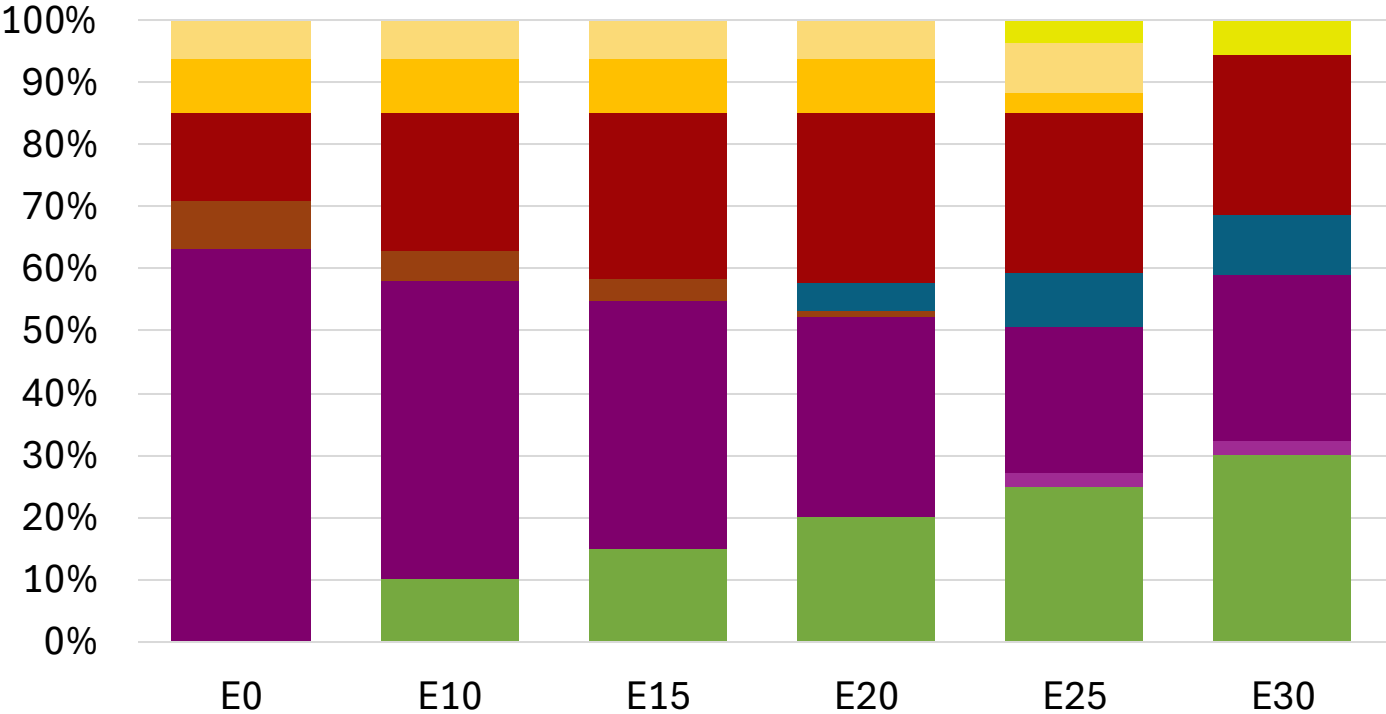
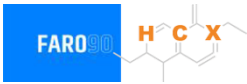


Average CIF 2024 Prices

Octane (RON)	95.0	95.0	95.0	95.0	95.0	95.8
Price (US\$/gal)	\$ 2.274	\$ 2.136	\$ 2.036	\$ 1.921	\$ 1.875	\$ 1.822

Source: Faro90

Estonia - Gasoline 98 - Constant Octane



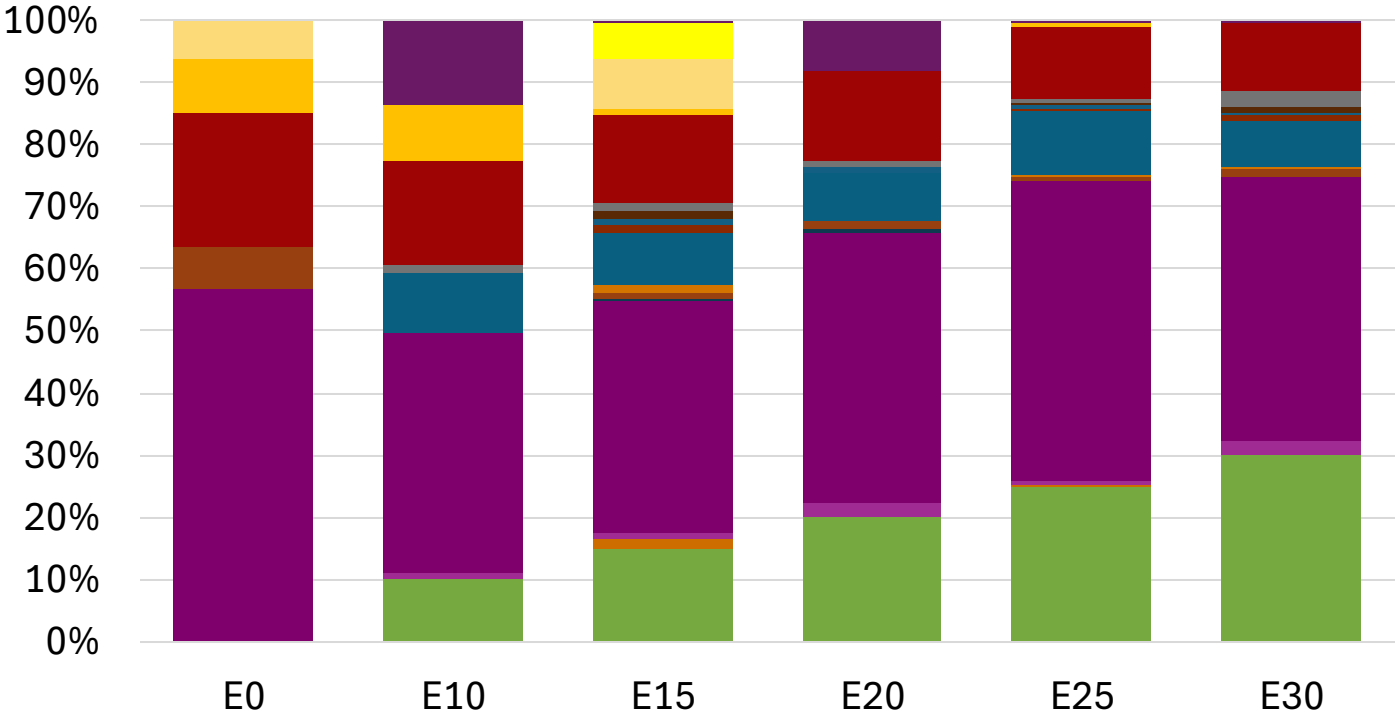
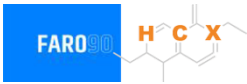
Ethanol
Alkylate
Raffinate
Reformate
n-Butane
Isomerate
Isopentane
Hydrocracked Gasoline
Catalytic Gasoline
Coker Naphtha
Light Primary Naphtha
Heavy Primary Naphtha
MTBE
ETBE
TAME
Aromatics

Average CIF 2024 Prices

Octane (RON)	98.0	98.0	98.0	98.0	98.0	98.0
Price (US\$/gal)	\$ 2.294	\$ 2.177	\$ 2.108	\$ 2.036	\$ 1.971	\$ 1.921

Source: Faro90

Estonia – Gasoline 95 – Increased Octane

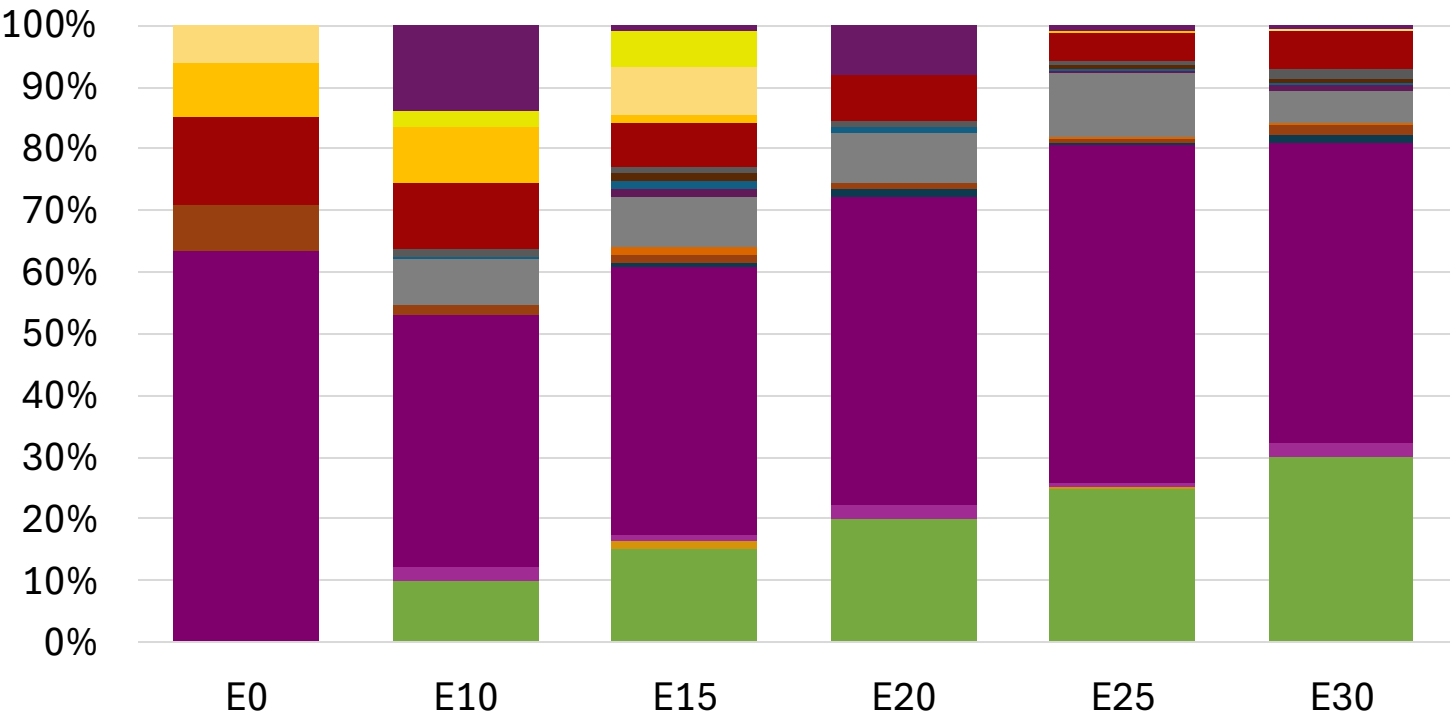
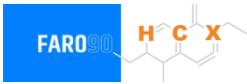


Average CIF 2024
Prices

Octane (RON)	95.0	96.2	98.3	98.7	100.5	101.8
Price (US\$/gal)	\$ 2.162	\$ 2.162	\$ 2.162	\$ 2.162	\$ 2.162	\$ 2.162

Source: Faro90

Estonia – Gasoline 98 – Increased Octane

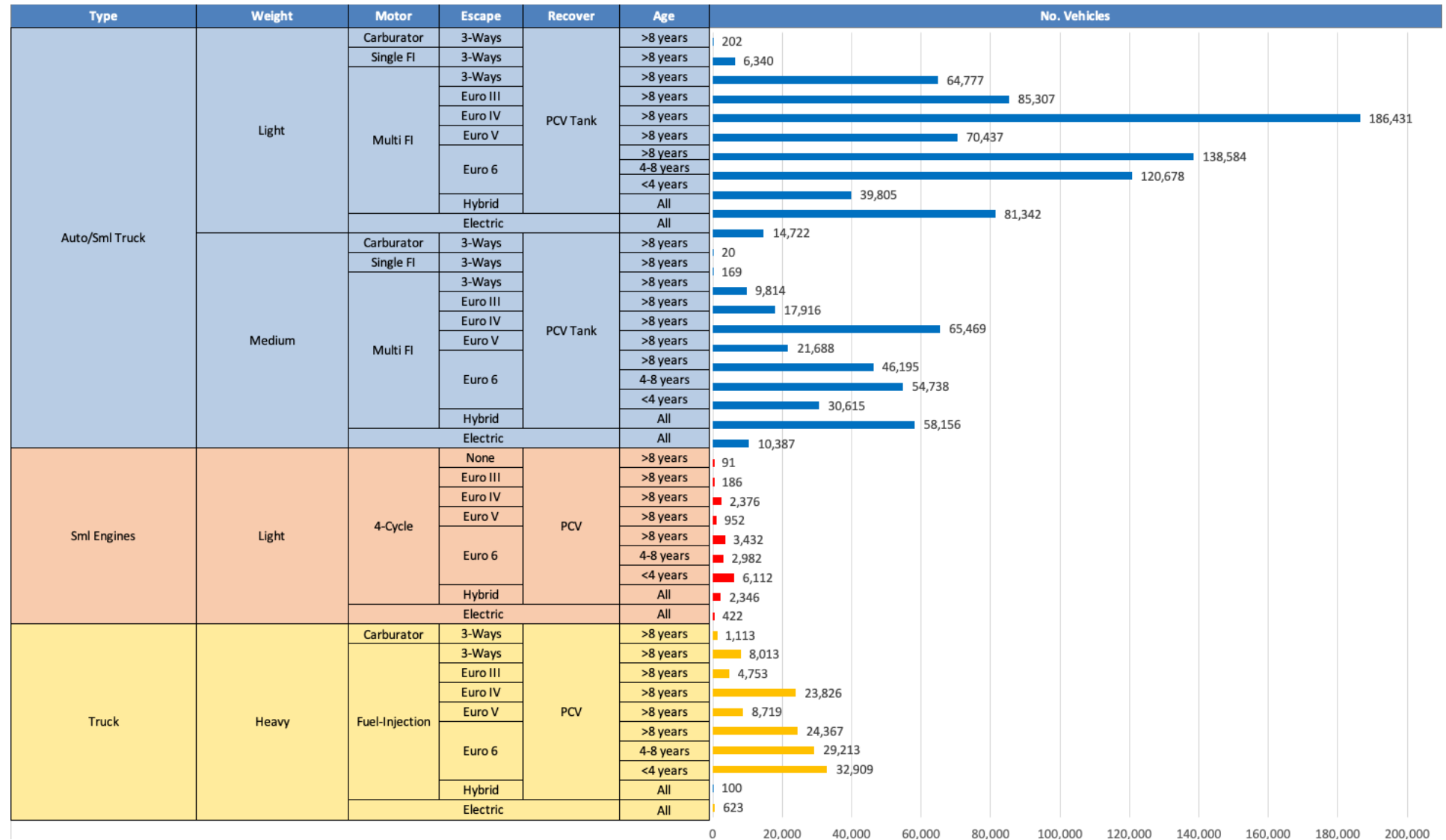


Average CIF 2024
Prices

Octane (RON)	98.0	99.0	100.9	101.2	102.8	104.1
Price (US\$/gal)	\$ 2.294	\$ 2.294	\$ 2.294	\$ 2.294	\$ 2.294	\$ 2.294

Source: Faro90

Gasoline Vehicle Fleet - Estonia

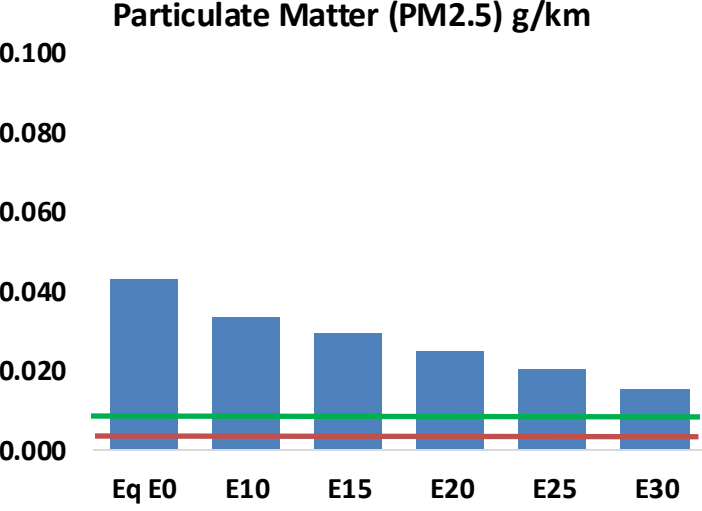
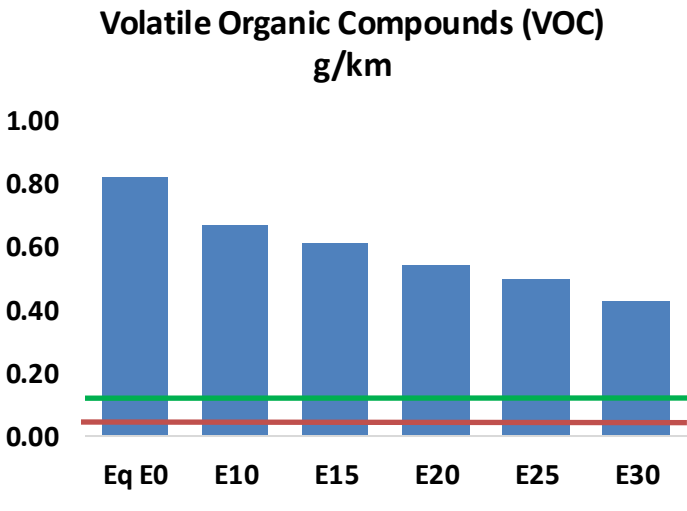
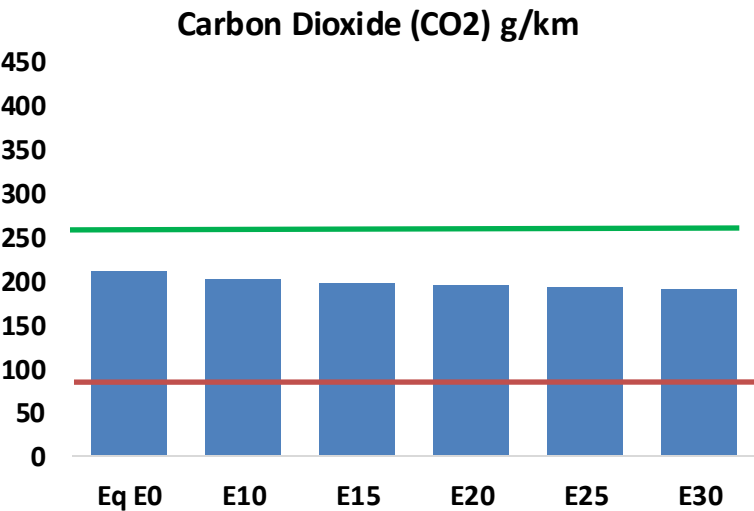
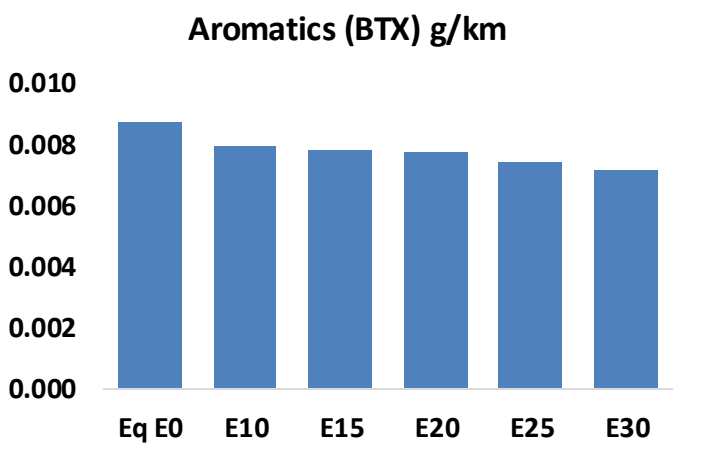
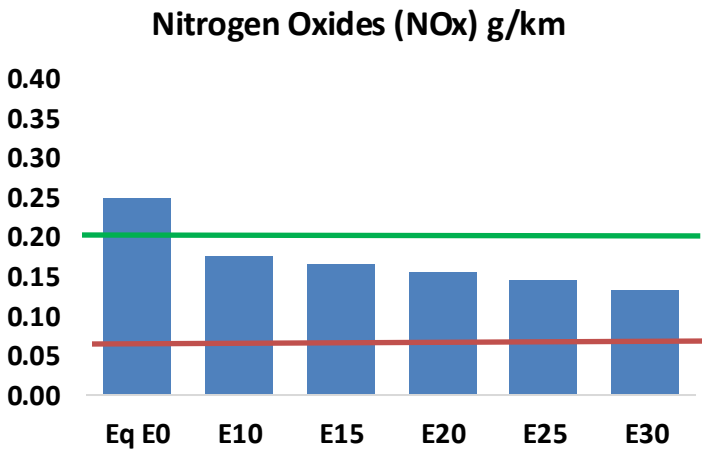
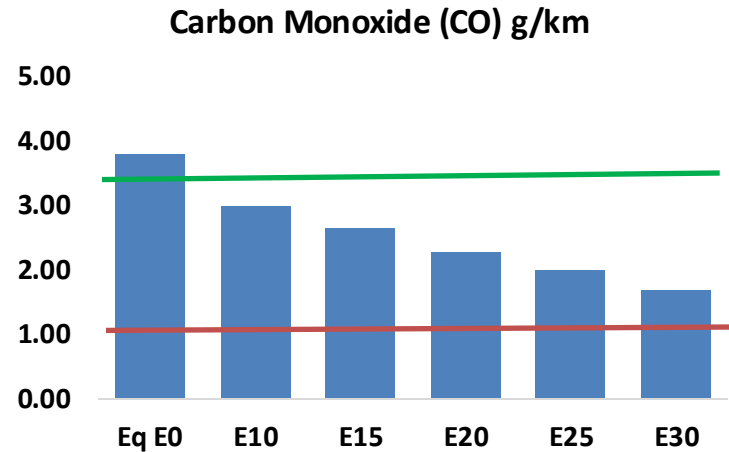
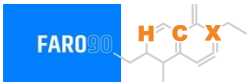


Vehicle Fleet: 1,276,323
Gasoline Vehicle Fleet: 616,792

Source: Eurostat, ACEA

Estonia – Vehicular Emissions

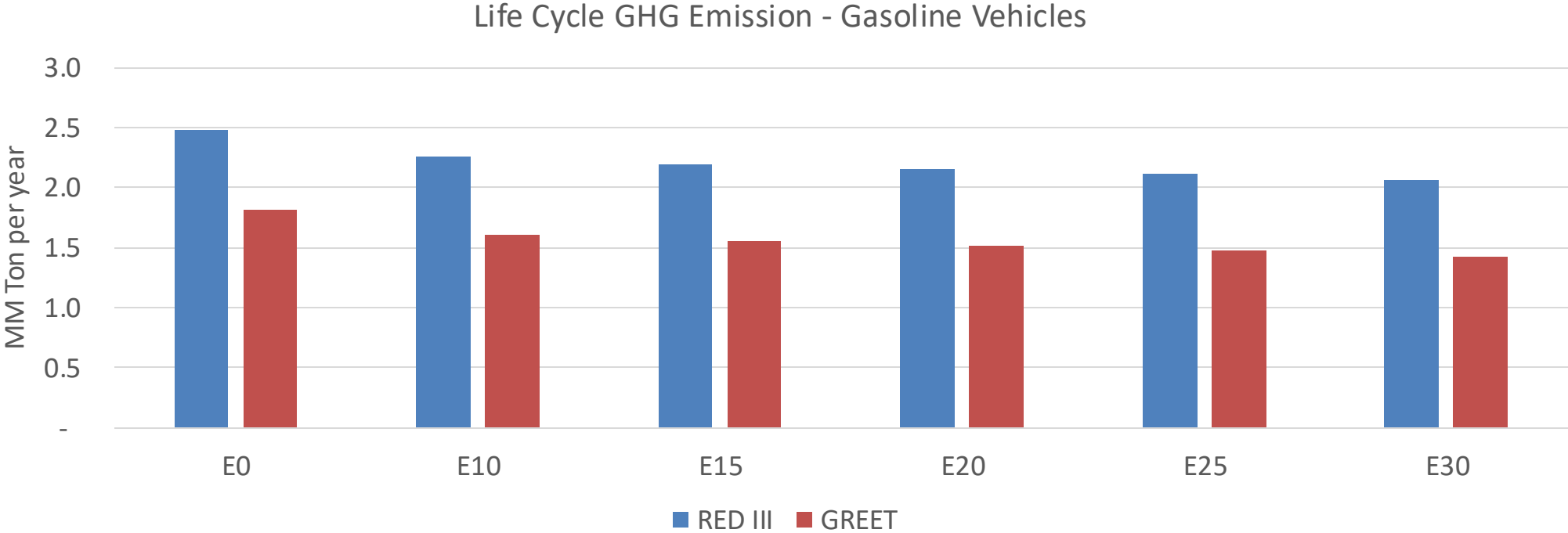
TIER III BIN 125 United States
Euro 6 Standard



Estonia – Vehicular Emissions

Vehicular Fleet Emissions (kg/day)	Eq E0	E5	E10	E15	E20	E25	E30	Reduction (%)	
								E10	E20
CO	59,676	53,345	47,014	41,486	35,767	31,322	26,777	-21%	-40%
CO2	3,336,894	3,251,088	3,170,049	3,106,315	3,074,776	3,056,519	3,000,178	-5%	-8%
VOC	12,895	11,734	10,573	9,589	8,583	7,805	6,784	-18%	-33%
NOx	3,952	2,964	2,766	2,607	2,459	2,295	2,122	-30%	-38%
SOx	40	37	34	31	29	26	24	-15%	-28%
NH3	1,269	1,257	1,244	1,250	1,264	1,274	1,282	-2%	0%
N2O	53	53	53	53	54	55	55	-1%	2%
CH4	2,578	2,578	2,578	2,616	2,673	2,722	2,769	0%	4%
Lead	-	-	-	-	-	-	-	0%	0%
Butadiene	57	54	51	49	47	45	43	-10%	-18%
Acetaldehyde	199	228	335	510	695	794	938	68%	249%
Formaldehyde	758	737	859	1,009	1,057	1,169	1,273	13%	39%
Benzene	137	132	125	123	122	117	113	-9%	-11%
PM2.5	207	184	161	139	118	96	73	-22%	-43%
PM10	474	421	368	319	271	220	167	-22%	-43%

Estonia – Gasoline Vehicle Fleet Life Cycle GHG Emissions



Country Emissions 2024 (MMT/año)	15.6
Target @ 2035 (MMT/year)	8.8
Delta	6.8

%Reduction vs E0	E0	E10	E15	E20	E25	E30
GREET 2024	0.0%	11.7%	14.5%	16.7%	18.7%	21.5%
RED II/III	0.0%	8.9%	11.1%	12.8%	14.4%	16.6%

% Target@2035	E0	E10	E15	E20	E25	E30
GREET 2024	0.0%	3.1%	3.9%	4.4%	5.0%	5.7%
RED II/III	0.0%	3.2%	4.0%	4.7%	5.2%	6.0%

Source: Greet, RED II/III, IPCC, climateactiontracker.org, F90

Ethanol Gasoline Blending in Finland

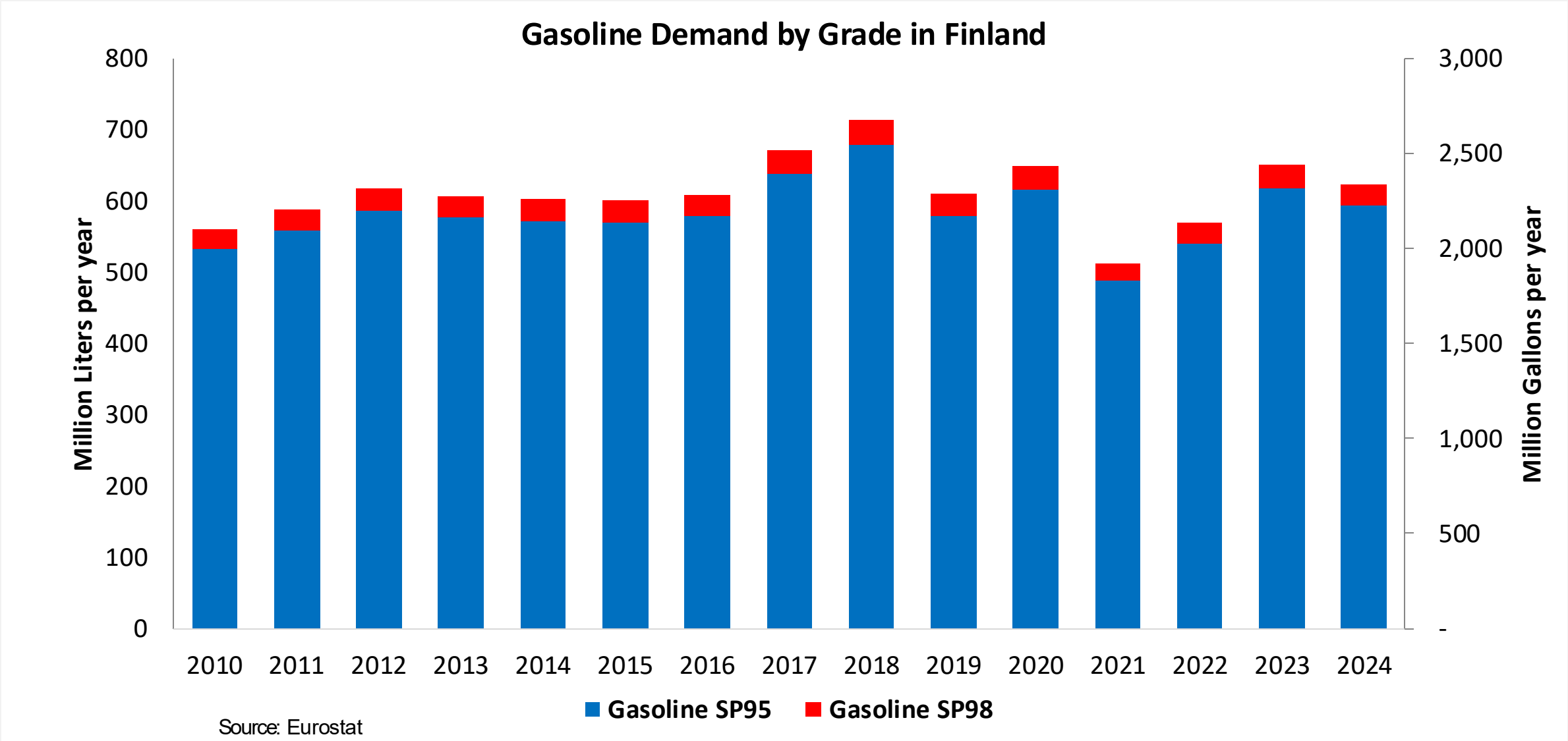


In 2024, gasoline consumption in Finland was 2,341 million liters (618 million gallons) and growth rate was 0.5%. Gasoline production and net imports were 5,450 and -3,109 million liters (1,440 and -821 million gallons) correspondingly. Finland complies with the European gasoline standards and offers two main grades: RON 95 and RON 98) with an average octane of 95.2 RON.

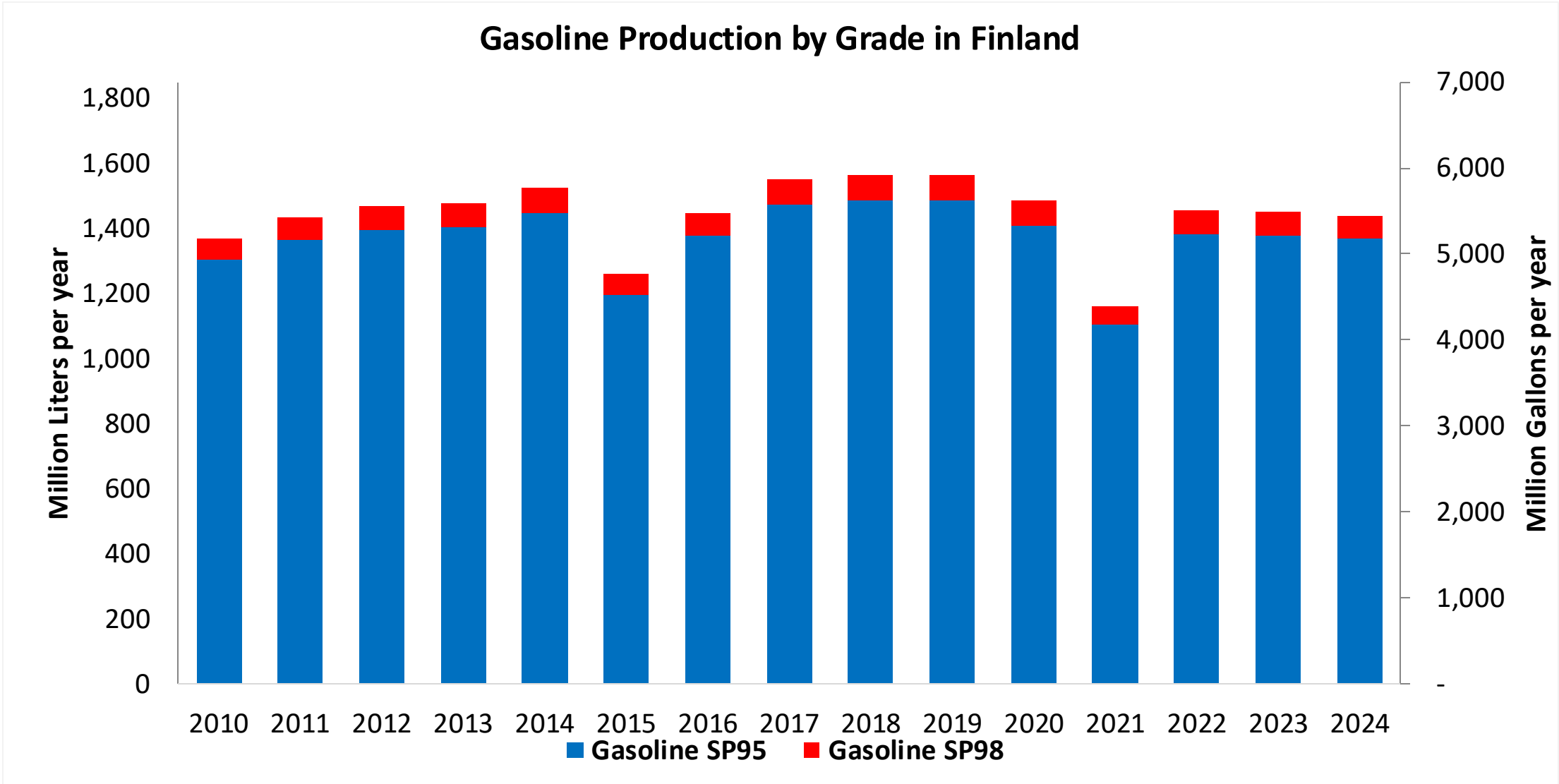
In 2024, ethanol consumption in Finland was 299 million liters (79 million gallons) representing 12.8% of gasoline demand. Ethanol production and Net Imports were 67 and 233 million liters (79 and 61 million gallons) correspondingly. Ethanol sources are Corn 5%, Cereals 85% and 2G lignocelulosic Ethanol 10%.

Source: Eurostat, European Commission

Gasoline Demand in Finland

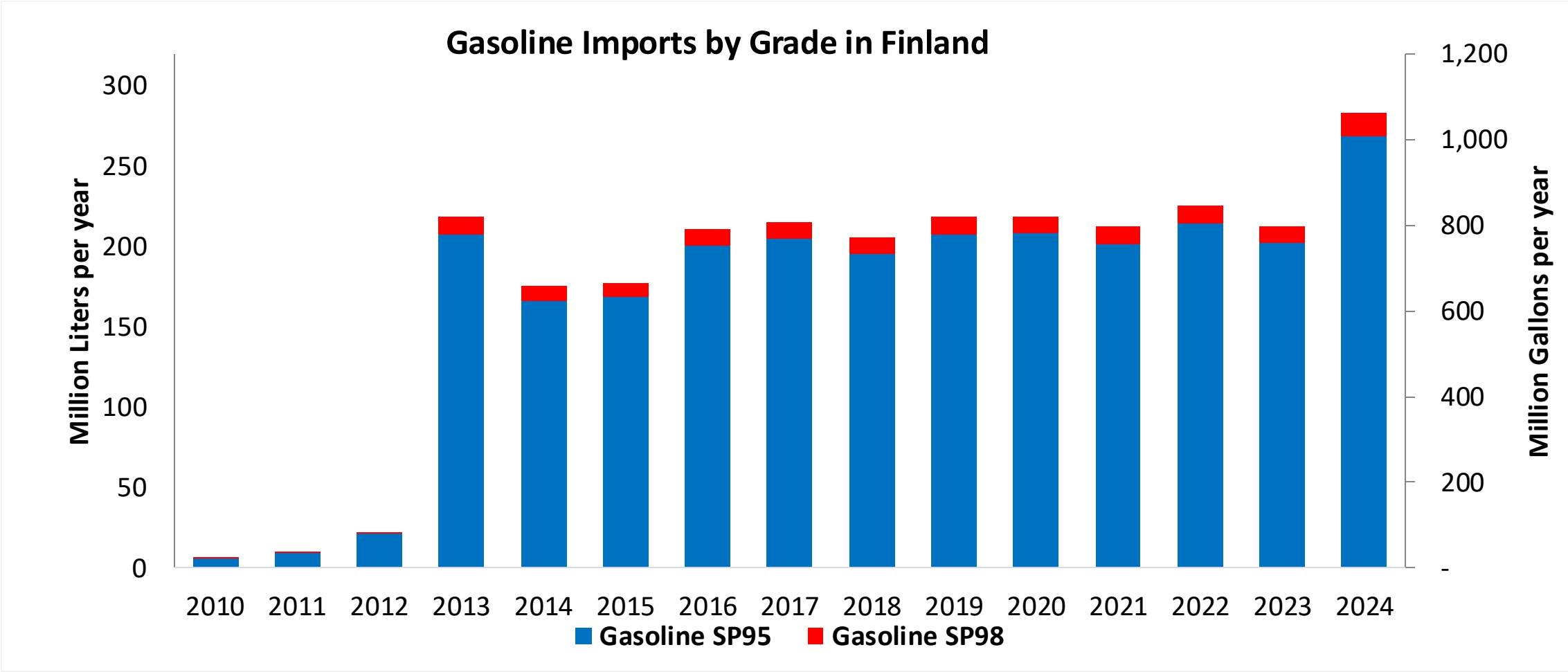


Gasoline Production in Finland



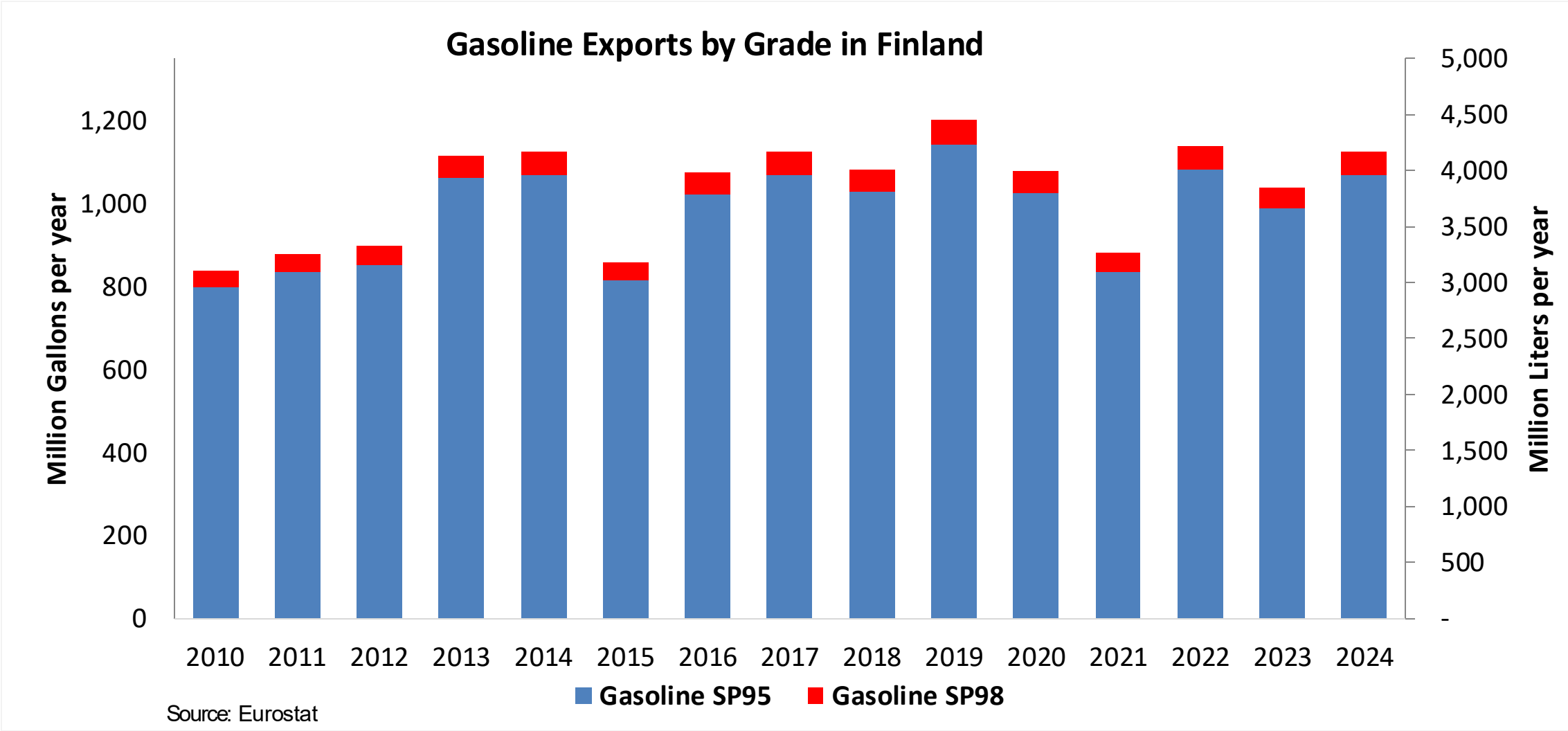
Source: Eurostat

Gasoline Imports in Finland

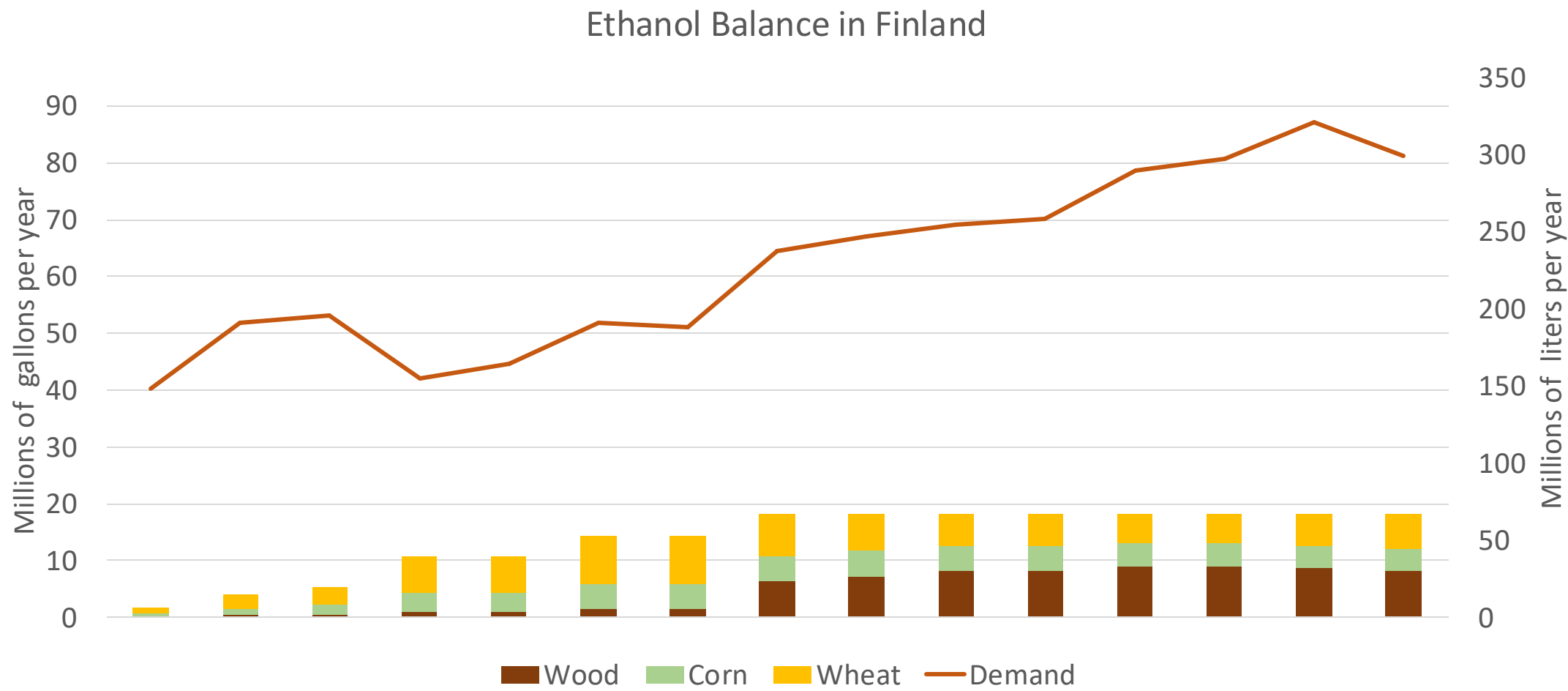
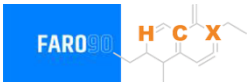


Source: Eurostat

Gasoline Exports in Finland

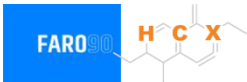


Ethanol Balance in Finland



Source: Eurostat

Gasoline Quality in Finland

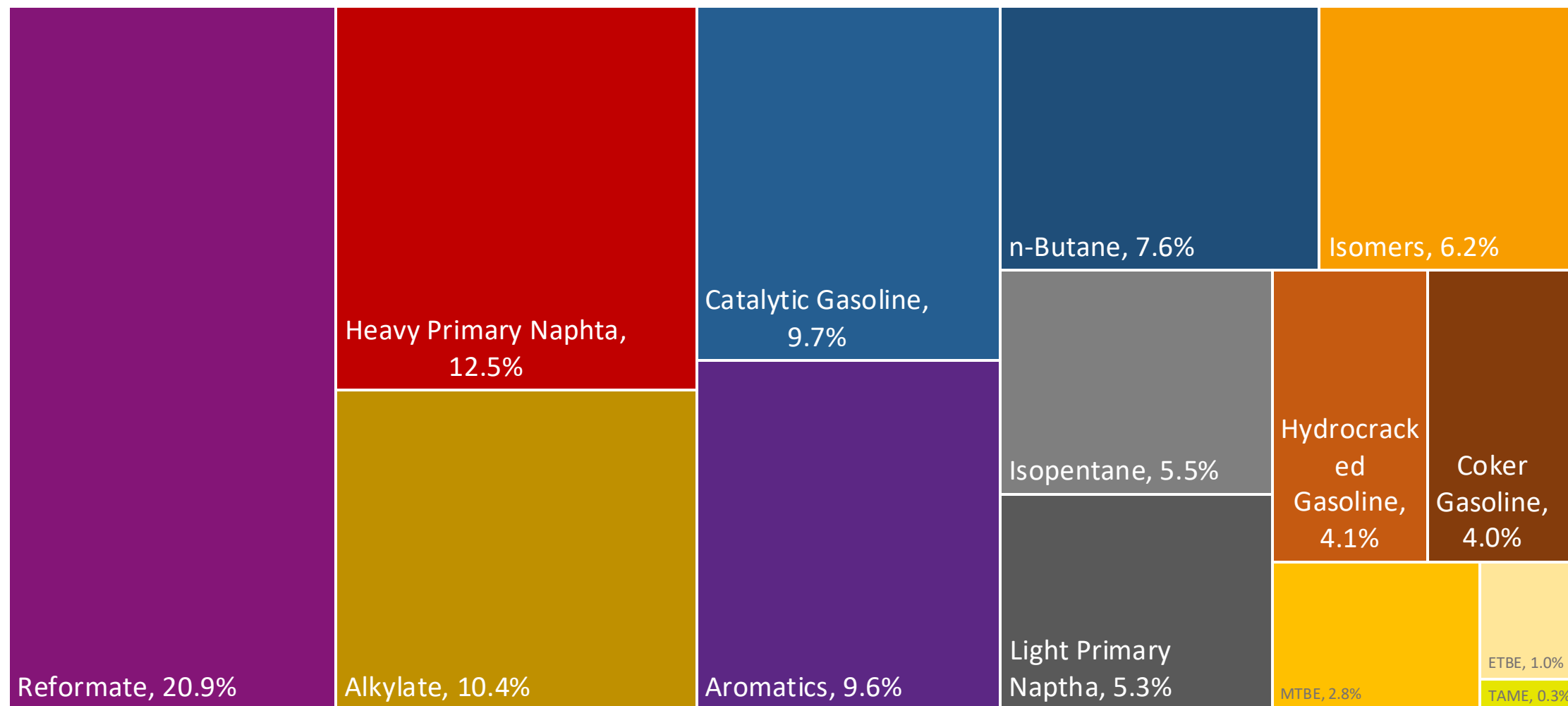


Name	EN 228:2025 (Euro 6); Country Level Fuel Ordinances							
Implementation Date	2025							
Applicability	All EU Contries + UK							
Grade	RON 95 E5		RON 95 E10		RON 98 E5		RON 98 E10	
	Min	Max	Min	Max	Min	Max	Min	Max
RON	95.0		95.0		98.0		98.0	
MON	> 85	> 88	> 85	> 88	> 85	> 88	> 85	> 88
AKI								
Benzene (%vol)	< 1							
Aromatics (% vol)	< 35							
Olefins (%vol)	< 18							
Lead (g/L)	< 0.005							
Manganese (mg/L)	< 2,0							
Sulfur (ppm)	10.0							
Oxygen (%w)*	< 2,7	< 3,7	< 2,7	< 3,7	< 2,7	< 3,7	< 2,7	< 3,7
Ethanol (%vol)	< 5	< 10	< 5	< 10	< 5	< 10	< 5	< 10
PVR 37.8°C (psi)	<> 45 - 100 kPa in 6 Volatility Classes							
Overall Biofuels (%cal)	16.5							
Advanced Biofuels (%cal)	4.0							
Bioethanol in Gasoline Sales (%cal)	-							
GHG Emission Reduction (%)	-							

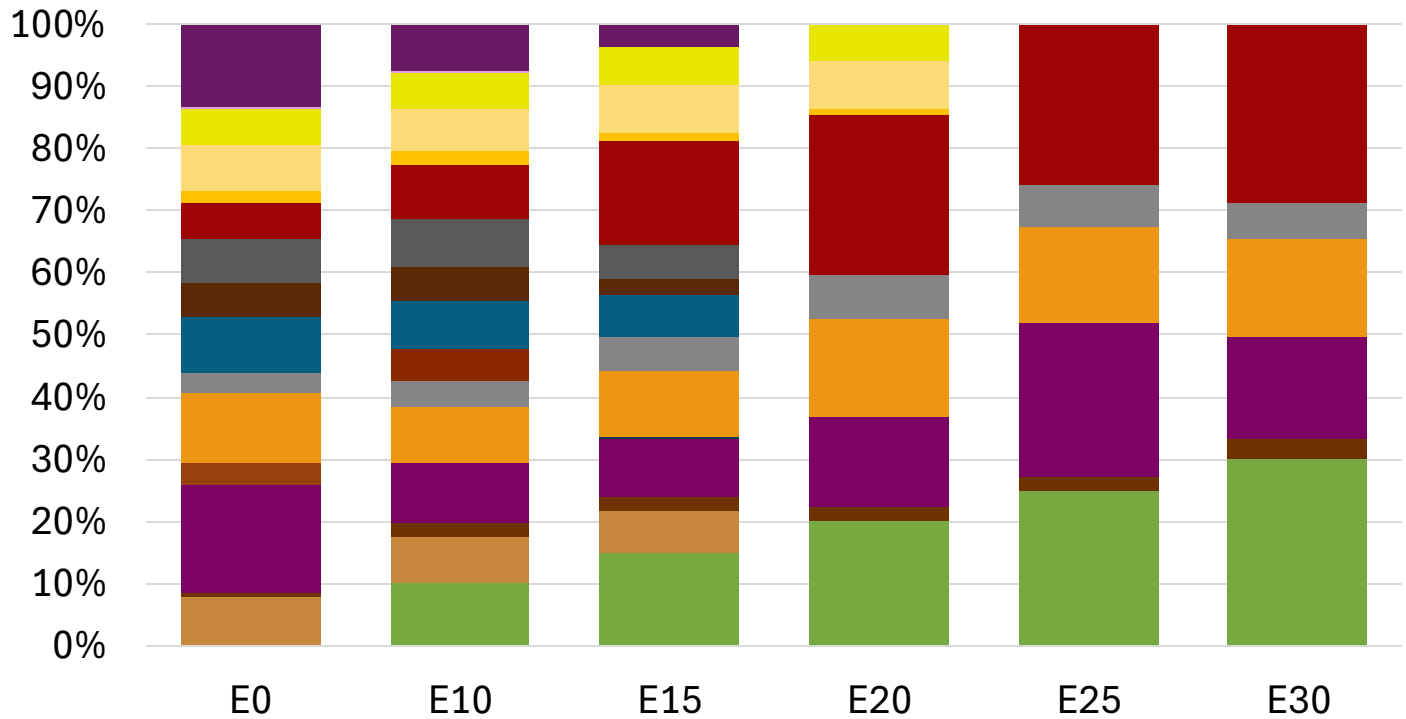
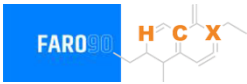
* Methanol 3%, Iso-propyl alcohol 12%, Ter-butyl alcohol 15%, Iso-butyl alcohol 15%, Ethers +5 Carbons 22%, Other Oxygenates 15%

Source: European Commision

Finland Available Blending Components



Finland – Gasoline 95 – Constant Octane

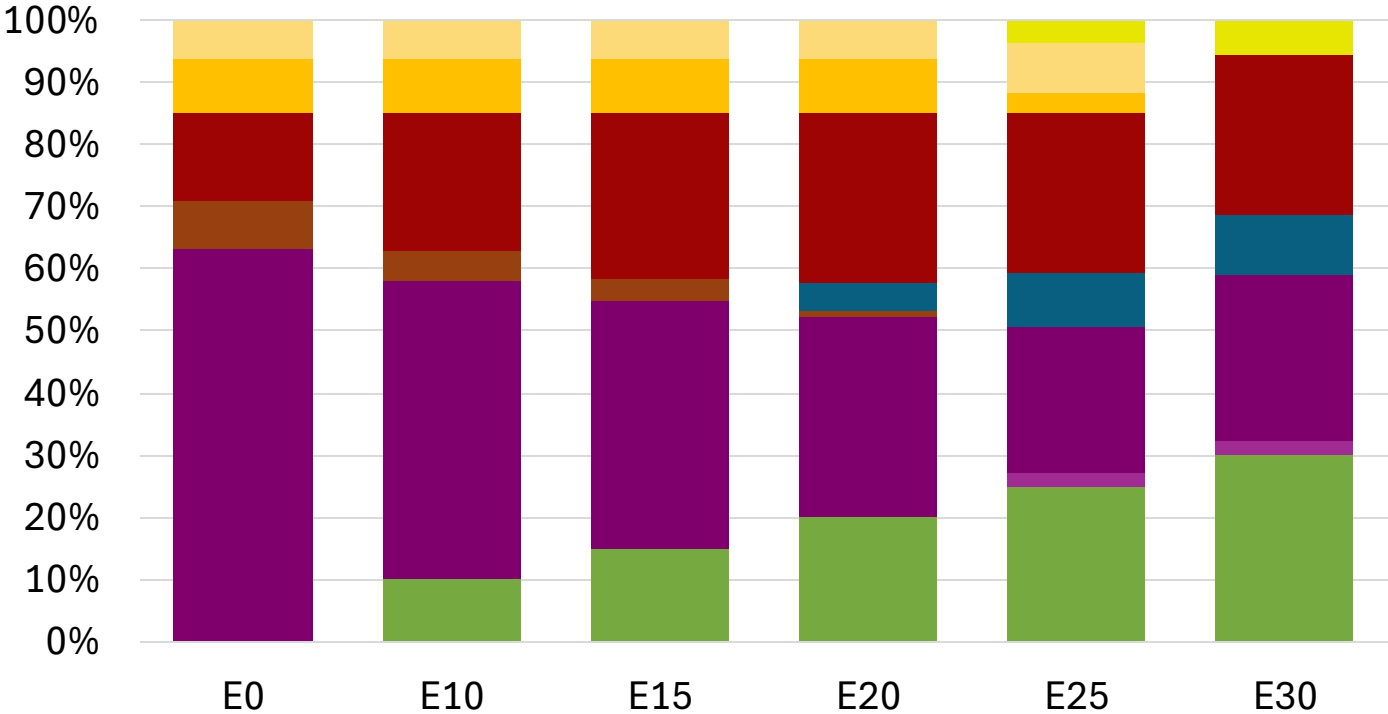
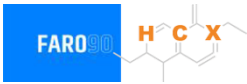


Average CIF 2024
Prices

Octane (RON)	95.0	95.0	95.0	95.0	95.0	95.8
Price (US\$/gal)	\$ 2.294	\$ 2.156	\$ 2.056	\$ 1.941	\$ 1.895	\$ 1.842

Source: Faro90

Finland - Gasoline 98 - Constant Octane

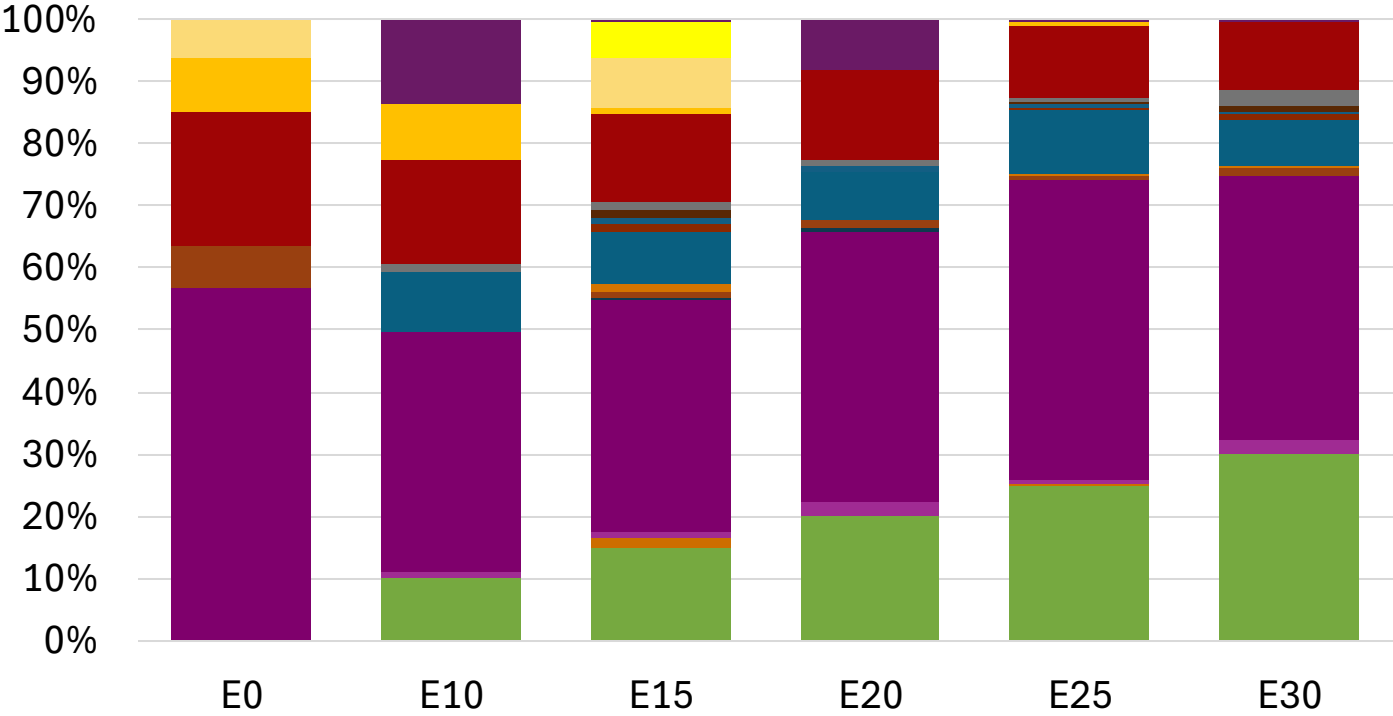


- Ethanol
- Alkylate
- Raffinate
- Reformate
- n-Butane
- Isomerate
- Isopentane
- Hydrocracked Gasoline
- Catalytic Gasoline
- Coker Naphtha
- Light Primary Naphtha
- Heavy Primary Naphtha
- MTBE
- ETBE
- TAME
- Aromatics

Average CIF 2024
Prices

Source: Faro90

Finland – Gasoline 95 – Increased Octane

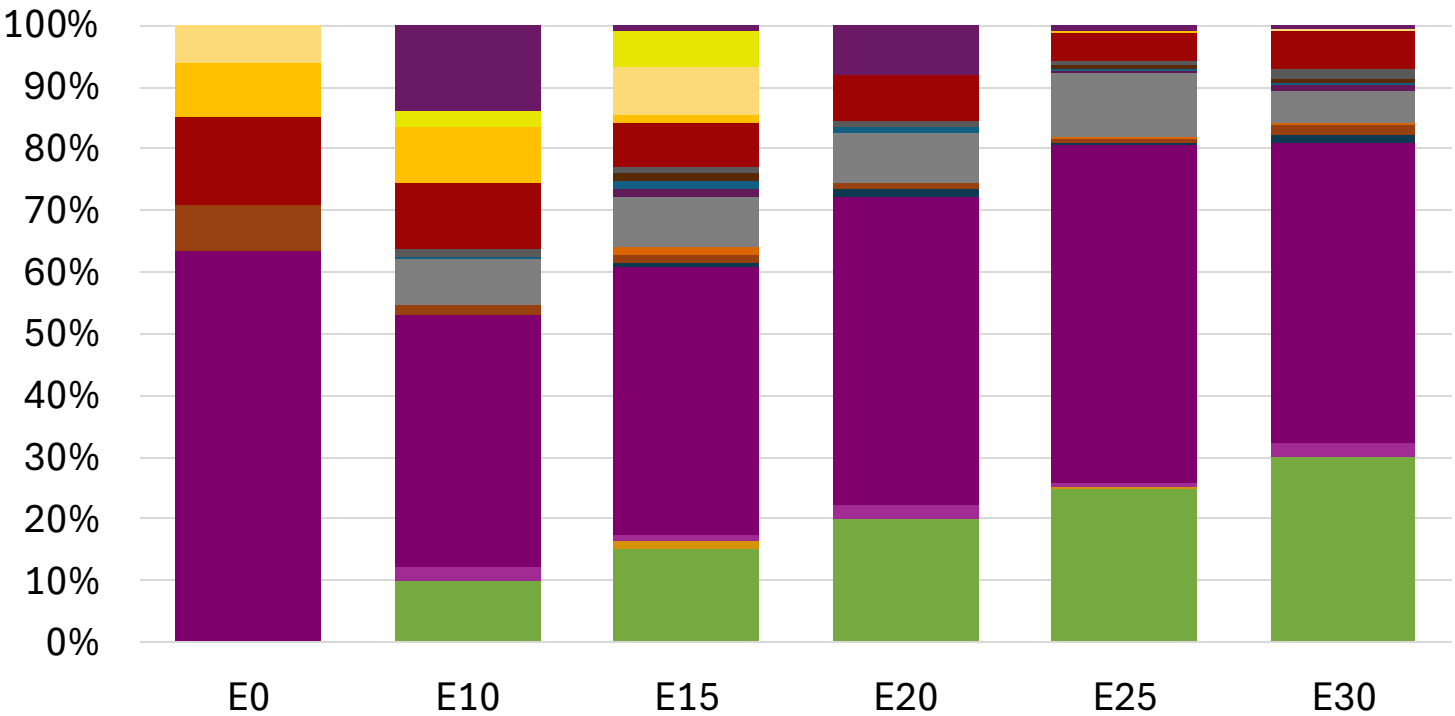
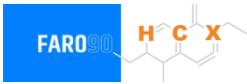


Average CIF 2024
Prices

Octane (RON)	95.0	96.2	98.3	98.7	100.5	101.8
Price (US\$/gal)	\$ 2.182	\$ 2.182	\$ 2.182	\$ 2.182	\$ 2.182	\$ 2.182

Source: Faro90

Finland – Gasoline 98 – Increased Octane

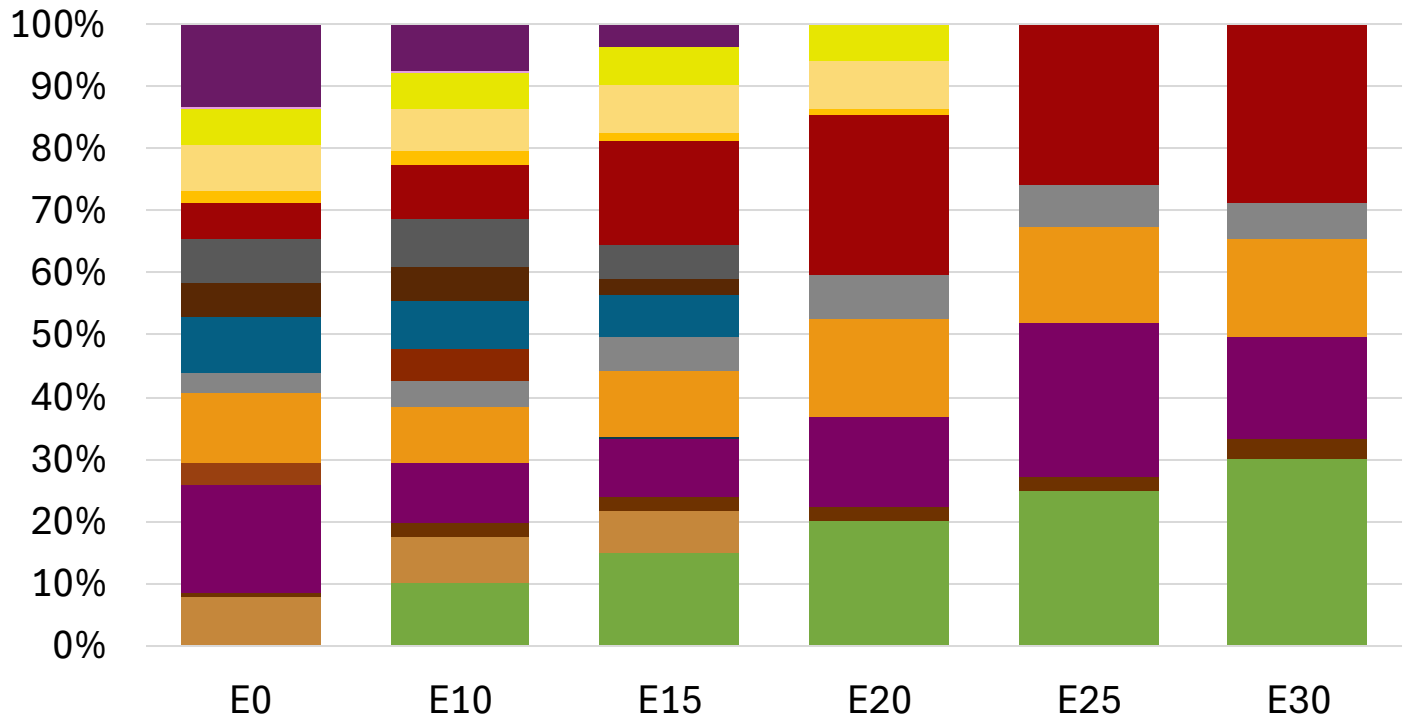
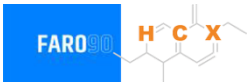


Average CIF 2024 Prices

Octane (RON)		98.0		99.0		100.9		101.2		102.8		104.1
Price (US\$/gal)	\$	2.314	\$	2.314	\$	2.314	\$	2.314	\$	2.314	\$	2.314

Source: Faro90

Finland – Gasoline 95 – Constant Octane

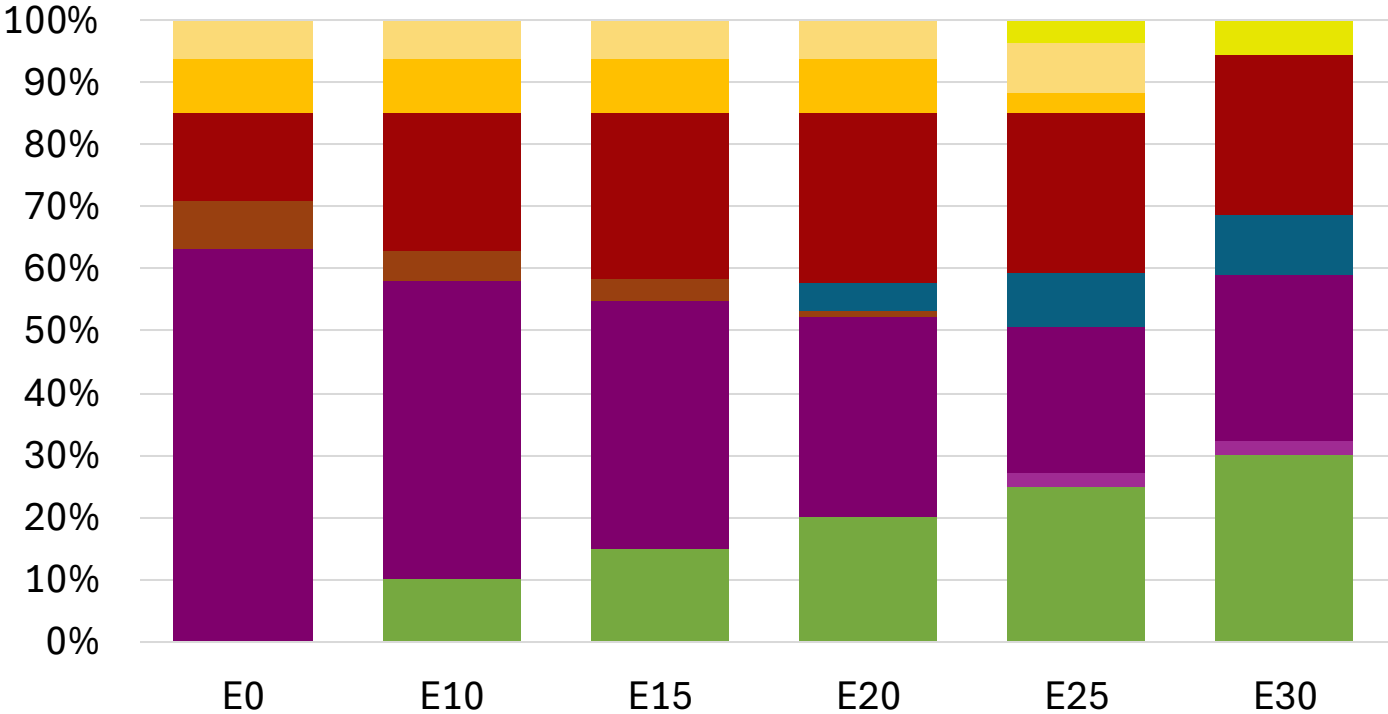
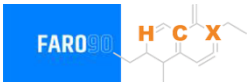


Average CIF 2024 Prices

Octane (RON)	95.0	95.0	95.0	95.0	95.0	95.8
Price (US\$/gal)	\$ 2.274	\$ 2.136	\$ 2.036	\$ 1.921	\$ 1.875	\$ 1.822

Source: Faro90

Finland - Gasoline 98 - Constant Octane



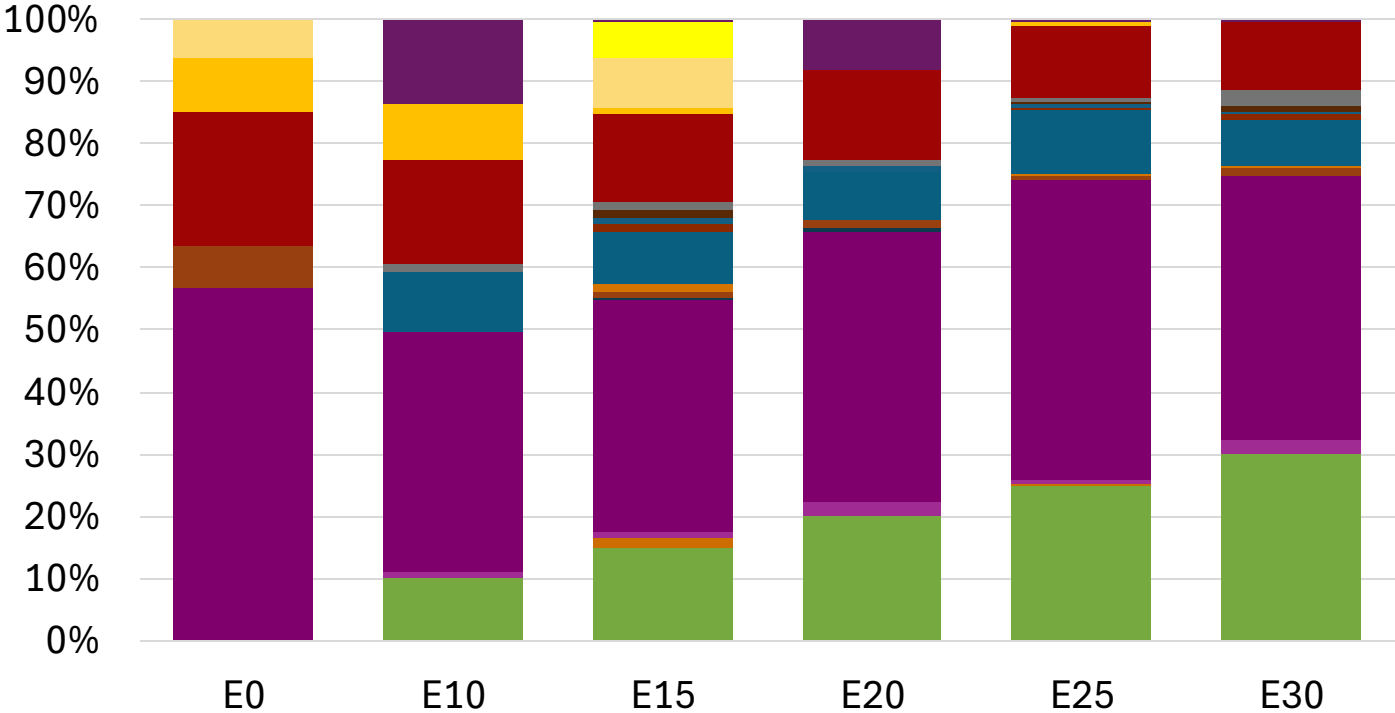
- Ethanol
- Alkylate
- Raffinate
- Reformate
- n-Butane
- Isomerate
- Isopentane
- Hydrocracked Gasoline
- Catalytic Gasoline
- Coker Naphtha
- Light Primary Naphtha
- Heavy Primary Naphtha
- MTBE
- ETBE
- TAME
- Aromatics

Average CIF 2024 Prices

Octane (RON)	98.0	98.0	98.0	98.0	98.0	98.0
Price (US\$/gal)	\$ 2.294	\$ 2.177	\$ 2.108	\$ 2.036	\$ 1.971	\$ 1.921

Source: Faro90

Finland – Gasoline 95 – Increased Octane

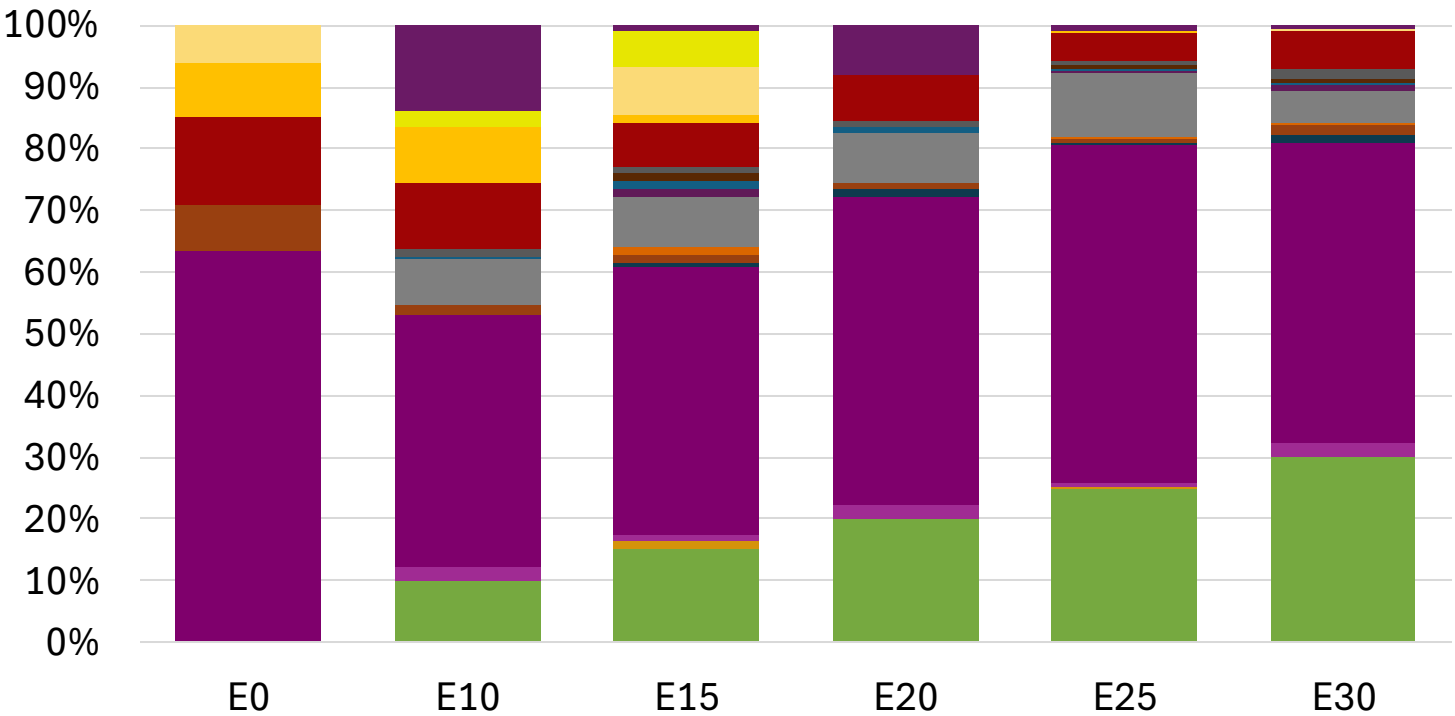
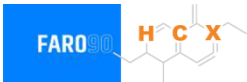


Average CIF 2024
Prices

Octane (RON)	95.0	96.2	98.3	98.7	100.5	101.8
Price (US\$/gal)	\$ 2.162	\$ 2.162	\$ 2.162	\$ 2.162	\$ 2.162	\$ 2.162

Source: Faro90

Finland – Gasoline 98 – Increased Octane

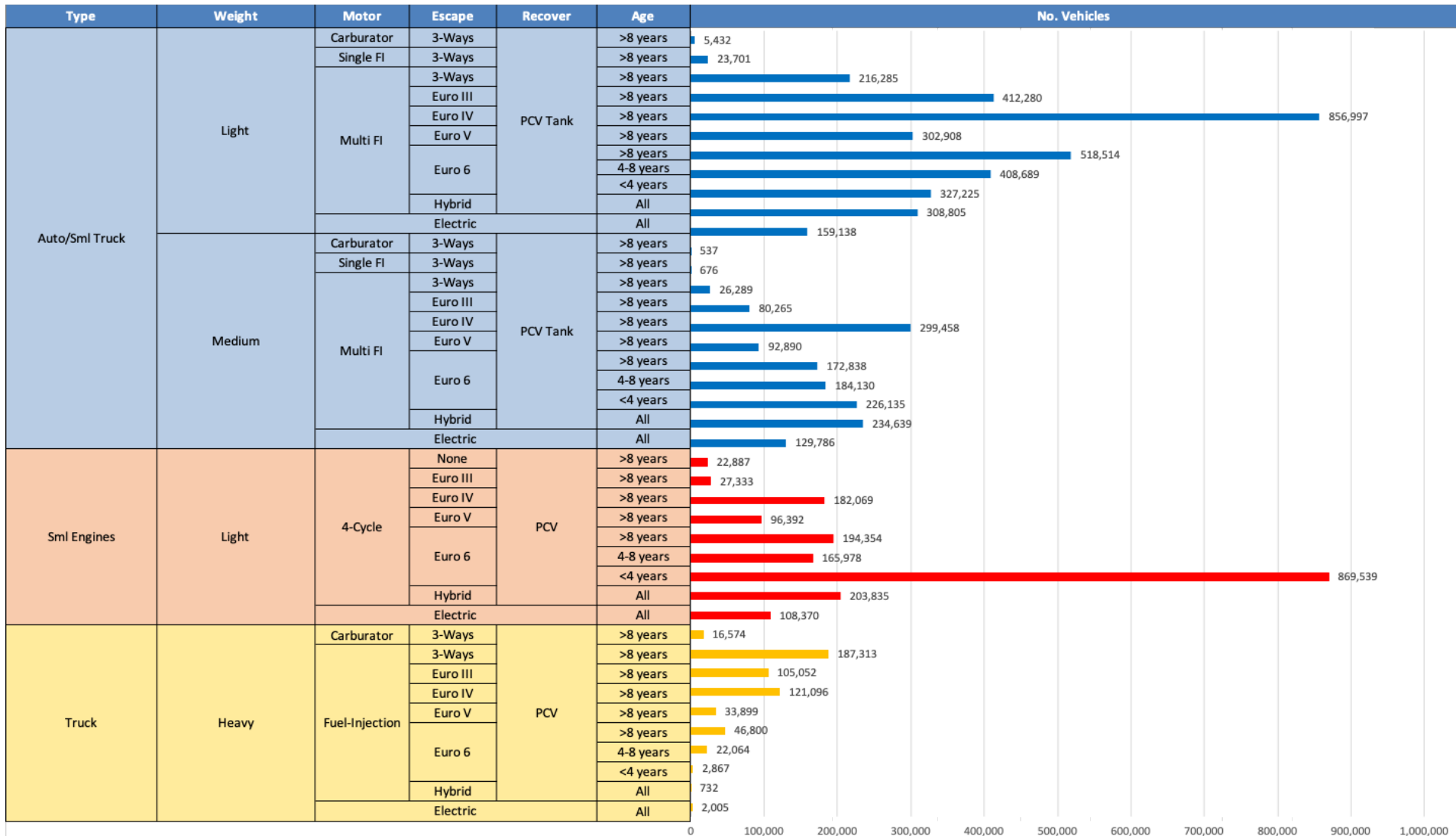


Average CIF 2024 Prices

Octane (RON)	98.0	99.0	100.9	101.2	102.8	104.1
Price (US\$/gal)	\$ 2.294	\$ 2.294	\$ 2.294	\$ 2.294	\$ 2.294	\$ 2.294

Source: Faro90

Gasoline Vehicle Fleet - Finland

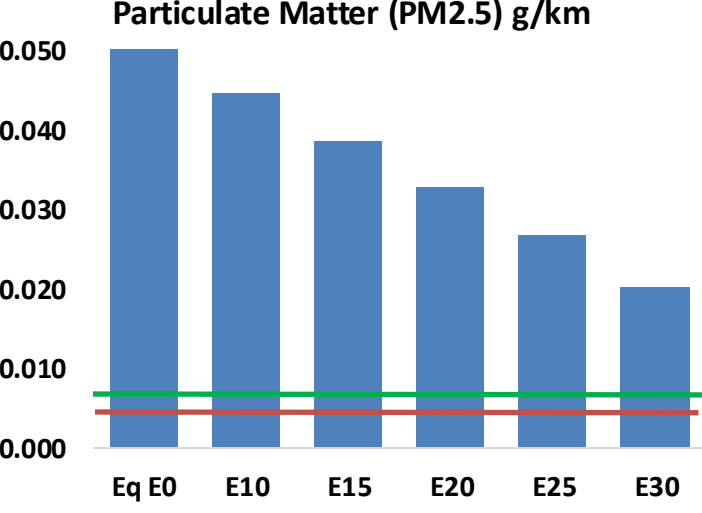
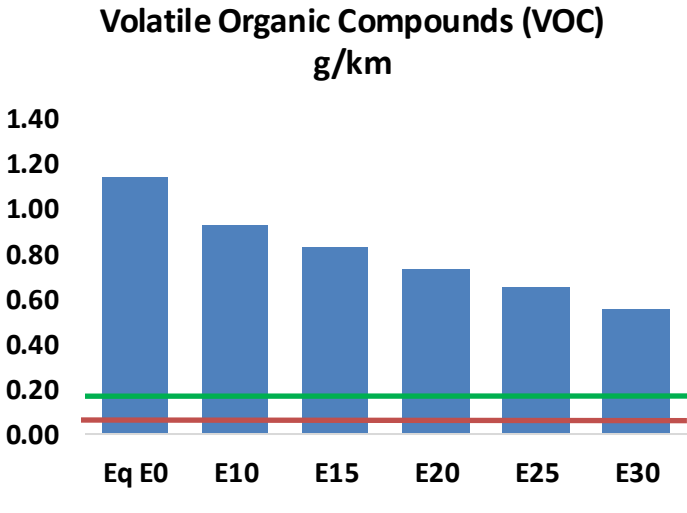
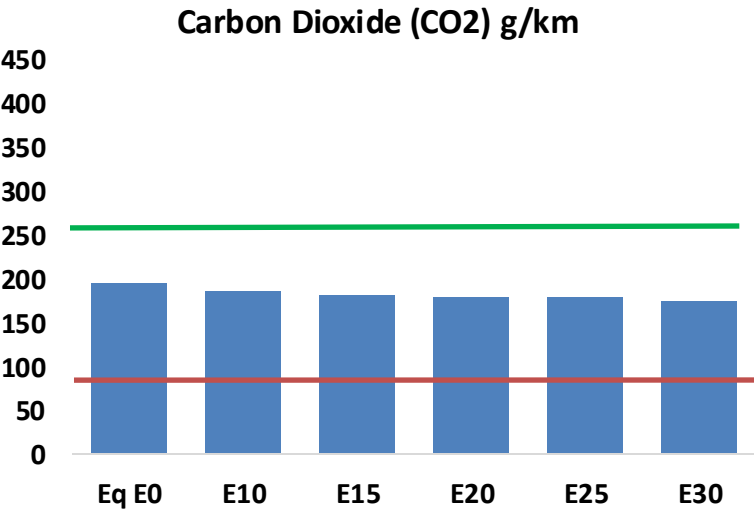
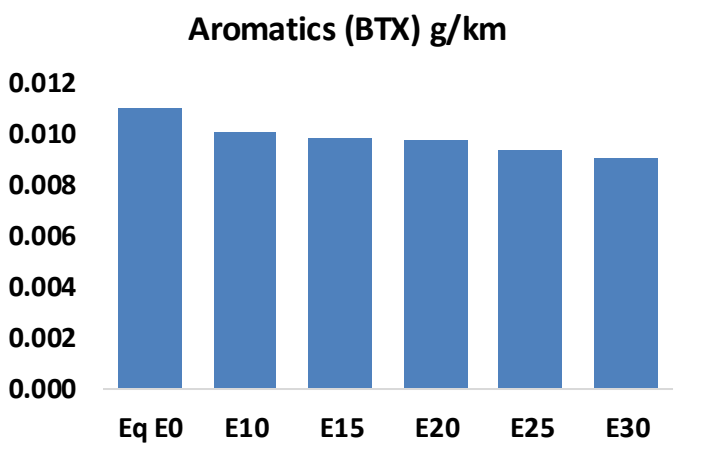
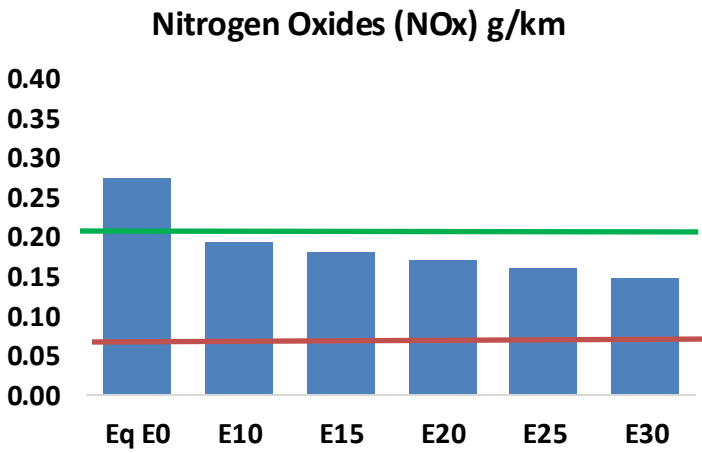
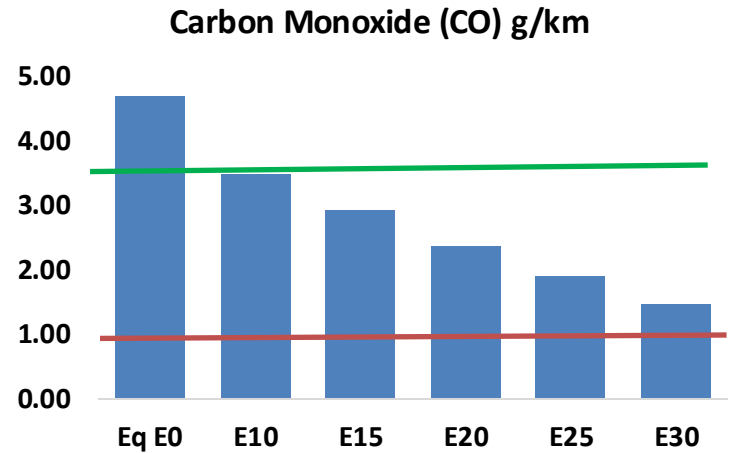
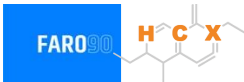


Vehicle Fleet: 7,396,775
Gasoline Vehicle Fleet: 3,837,401

Source: Eurostat, ACEA

Finland – Vehicular Emissions

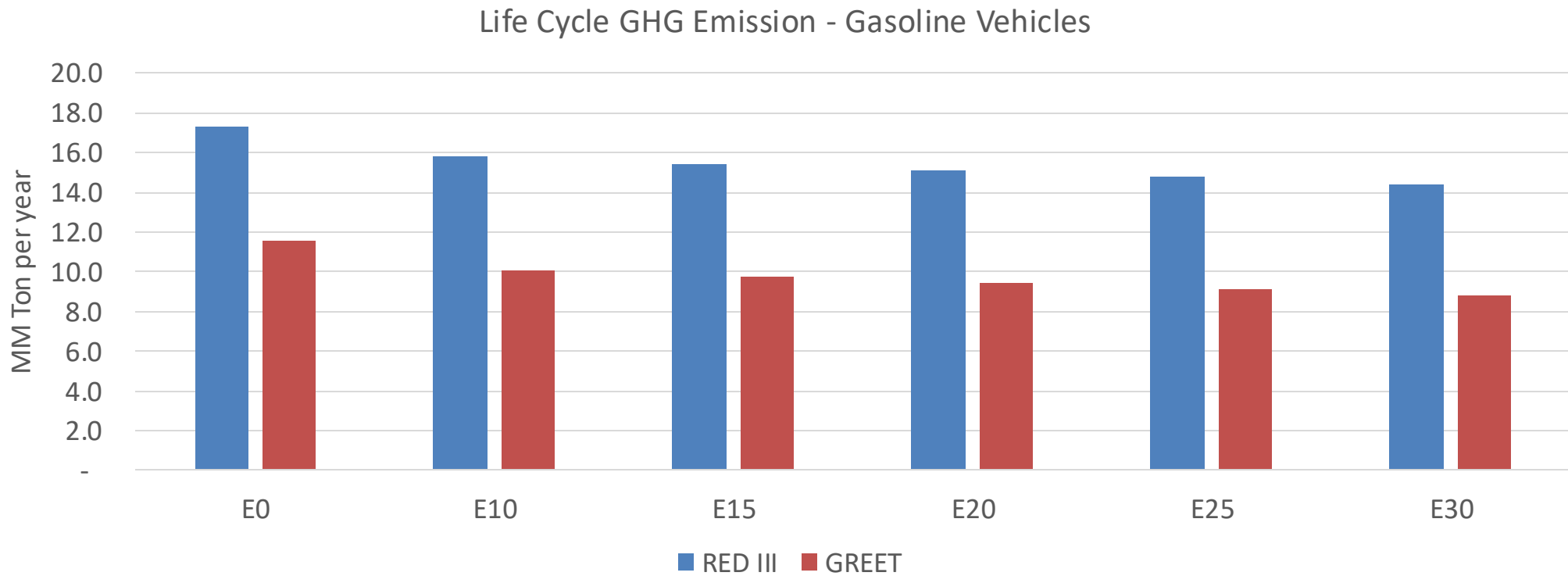
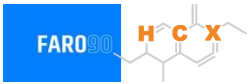
TIER III BIN 125 United States
Euro 6 Standard



Finland – Vehicular Emissions

Vehicular Fleet Emissions (kg/day)	Eq E0	E5	E10	E15	E20	E25	E30	Reduction (%)	
								E10	E20
CO	469,869	408,719	347,570	292,577	234,989	190,028	145,320	-26%	-50%
CO2	19,452,404	18,952,199	18,479,784	18,108,243	17,924,390	17,818,561	17,490,113	-5%	-8%
VOC	114,075	103,064	92,053	82,593	72,871	65,335	55,493	-19%	-36%
NOx	27,517	20,638	19,262	18,154	17,121	15,982	14,777	-30%	-38%
SOx	221	204	188	174	160	145	130	-15%	-28%
NH3	7,866	7,788	7,710	7,747	7,834	7,895	7,946	-2%	0%
N2O	324	322	321	323	329	333	337	-1%	2%
CH4	22,977	22,977	22,977	23,322	23,828	24,264	24,678	0%	4%
Lead	-	-	-	-	-	-	-	0%	0%
Butadiene	467	434	400	373	347	326	299	-14%	-26%
Acetaldehyde	1,791	2,056	3,017	4,594	6,259	7,152	8,451	68%	249%
Formaldehyde	7,151	6,952	8,104	9,516	9,969	11,029	12,006	13%	39%
Benzene	1,102	1,059	1,003	987	979	936	906	-9%	-11%
PM2.5	1,230	1,092	954	828	703	571	433	-22%	-43%
PM10	4,504	3,999	3,494	3,035	2,577	2,091	1,585	-22%	-43%

Finland – Gasoline Vehicle Fleet Life Cycle GHG Emissions



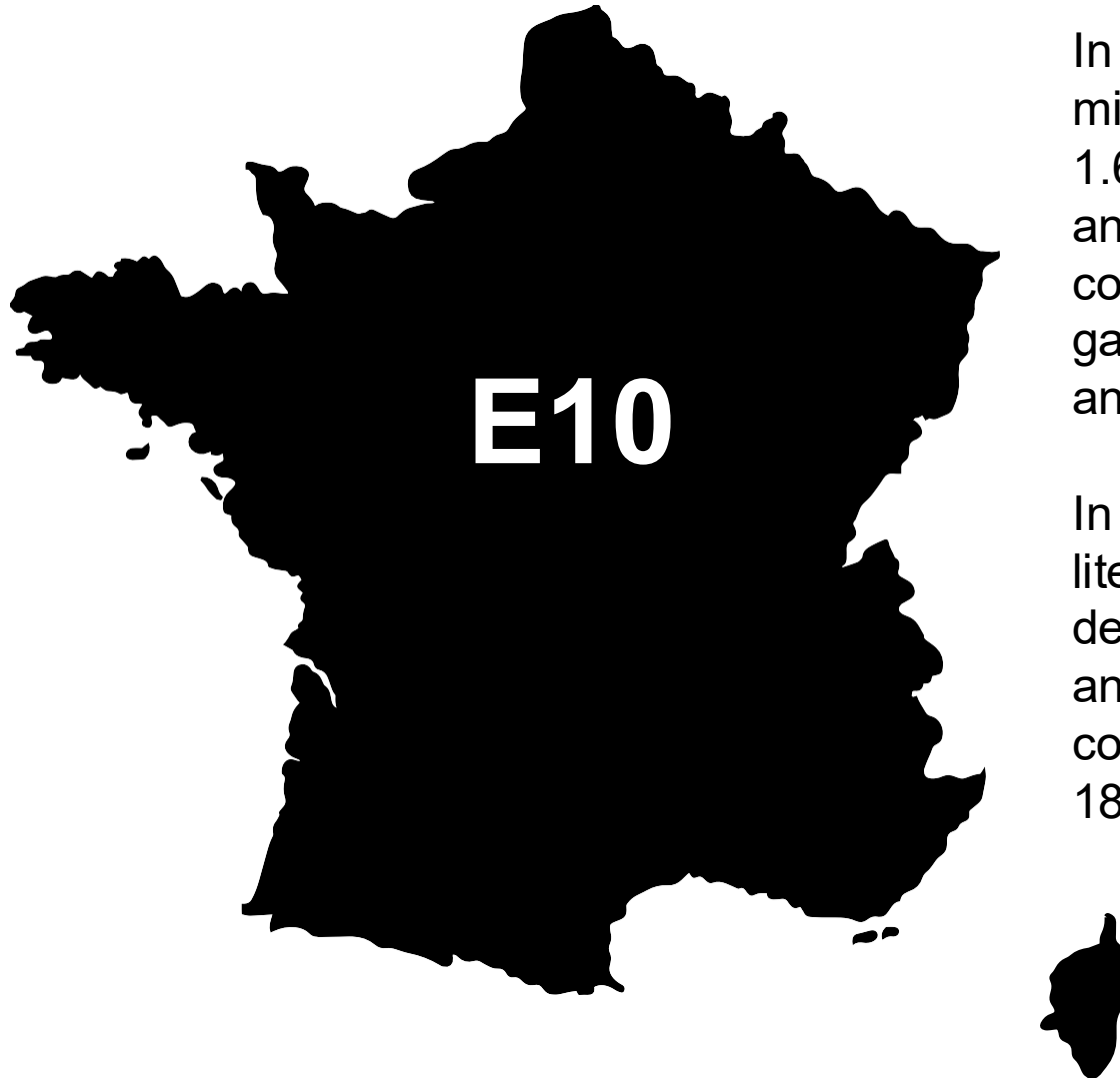
Country Emissions 2024 (MMT/año)	64.9
Target @ 2035 (MMT/year)	36.6
Delta	28.4

%Reduction vs E0	E0	E10	E15	E20	E25	E30
GREET 2024	0.0%	12.7%	15.9%	18.5%	21.0%	24.2%
RED II/III	0.0%	8.9%	11.3%	13.1%	14.9%	17.2%

% Target@2035	E0	E10	E15	E20	E25	E30
GREET 2024	0.0%	5.2%	6.5%	7.6%	8.6%	9.9%
RED II/III	0.0%	5.5%	6.9%	8.0%	9.1%	10.5%

Source: Greet, RED II/III, IPCC, climateactiontracker.org, F90

Ethanol Gasoline Blending in France

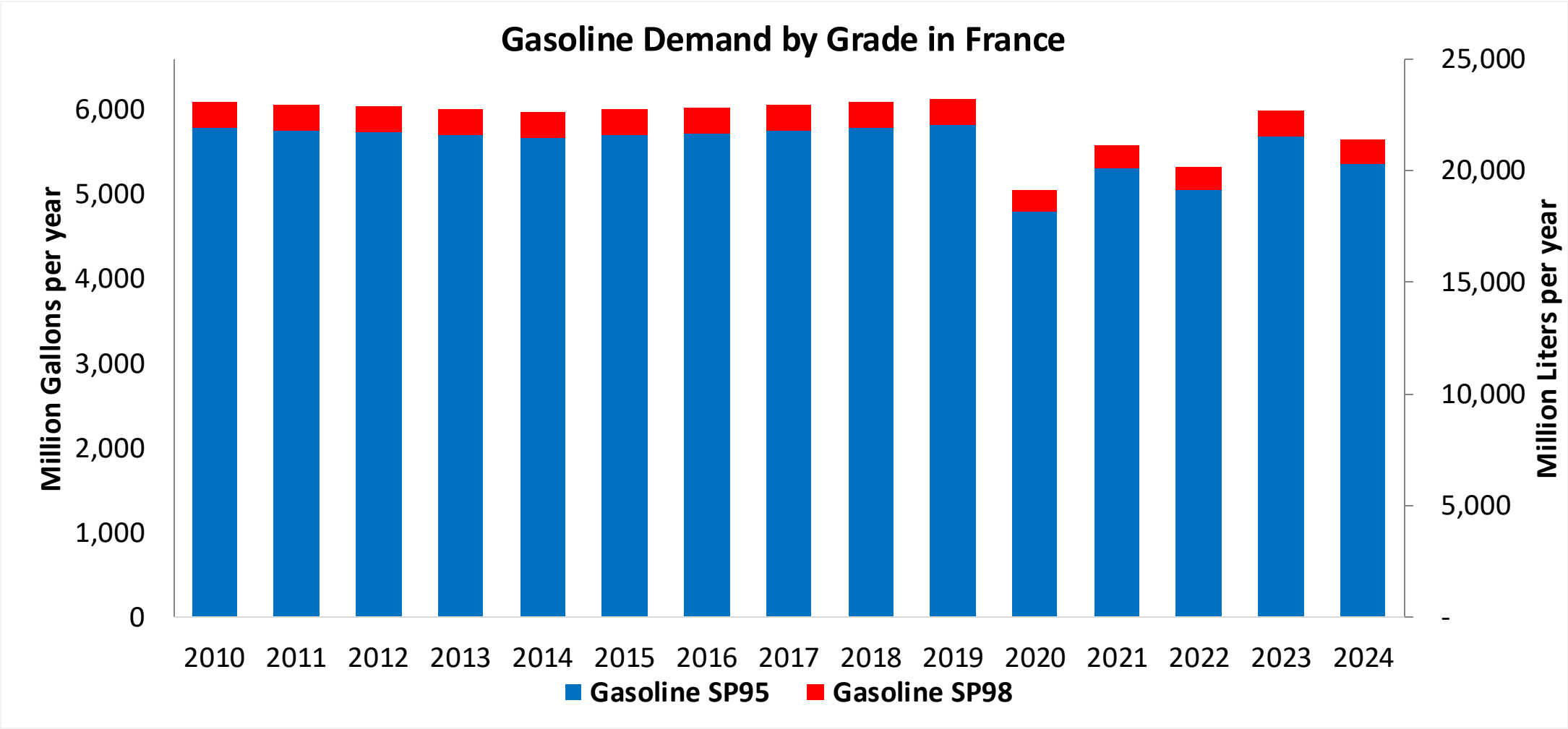


In 2024, gasoline consumption in France was 21,379 million liters (5,648 million gallons) and growth rate was - 1.6%. Gasoline production and net imports were 10,333 and 11,046 million liters (2,730 and 2,918 million gallons) correspondingly. France complies with the European gasoline standards and offers two main grades: RON 95 and RON 98) with an average octane of 95.2 RON.

In 2024, ethanol consumption in France was 2,622 million liters (693 million gallons) representing 12.3% of gasoline demand. Ethanol production and Net Imports were 2,154 and 468 million liters (693 and 124 million gallons) correspondingly. Ethanol sources are Corn 27%, Cereals 18% and Sugar Beet 37%.

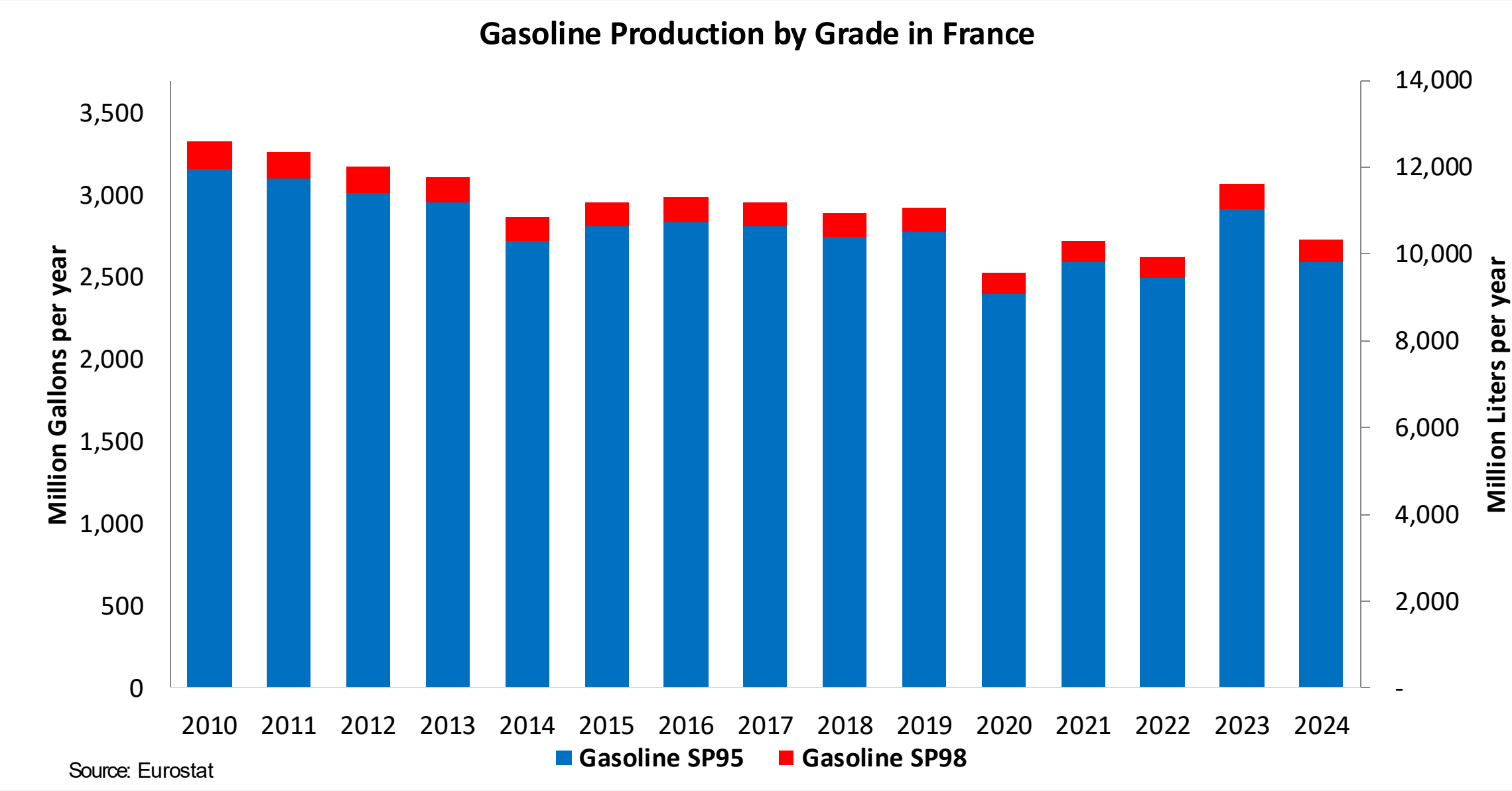
Source: Eurostat, European Commission

Gasoline Demand in France

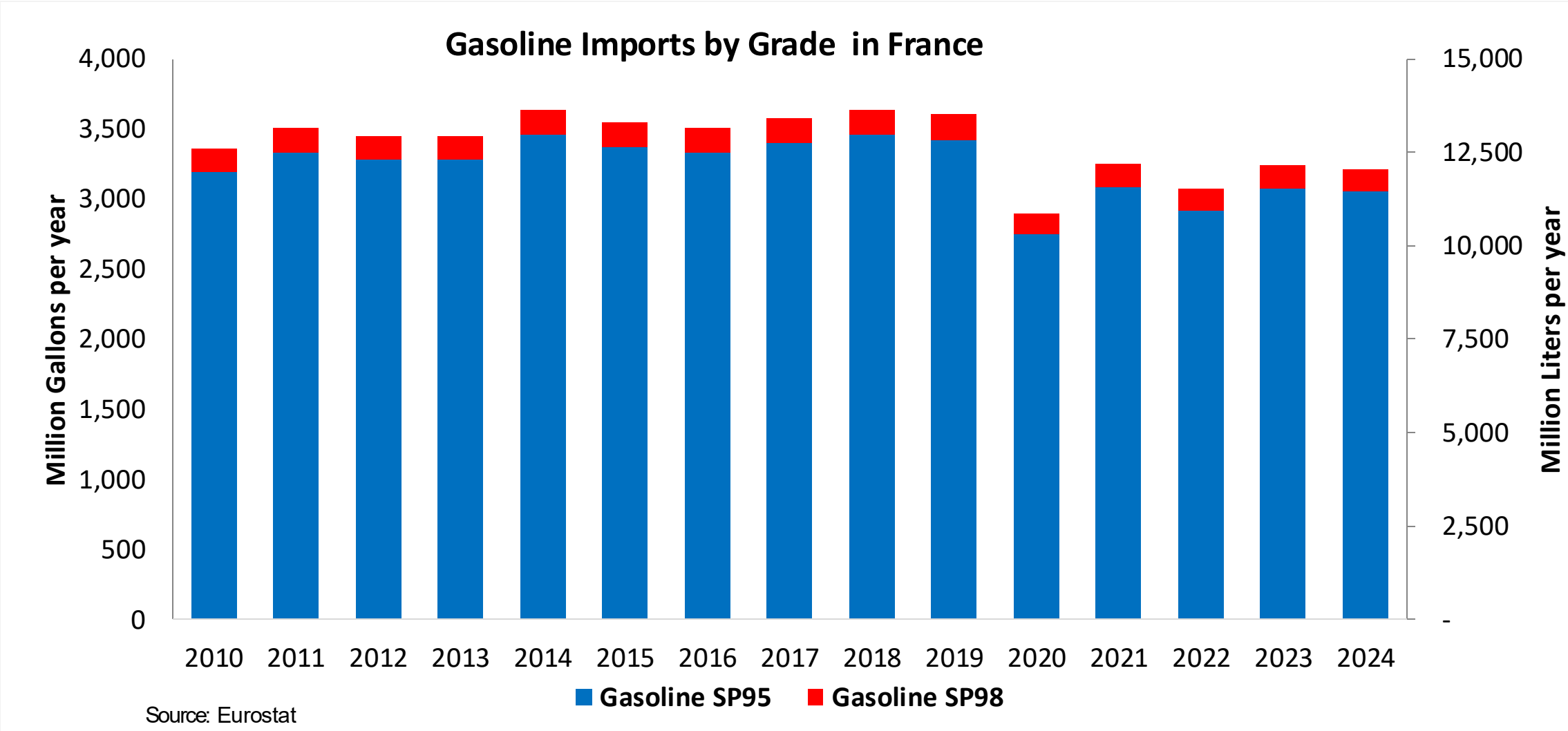


Source: Eurostat

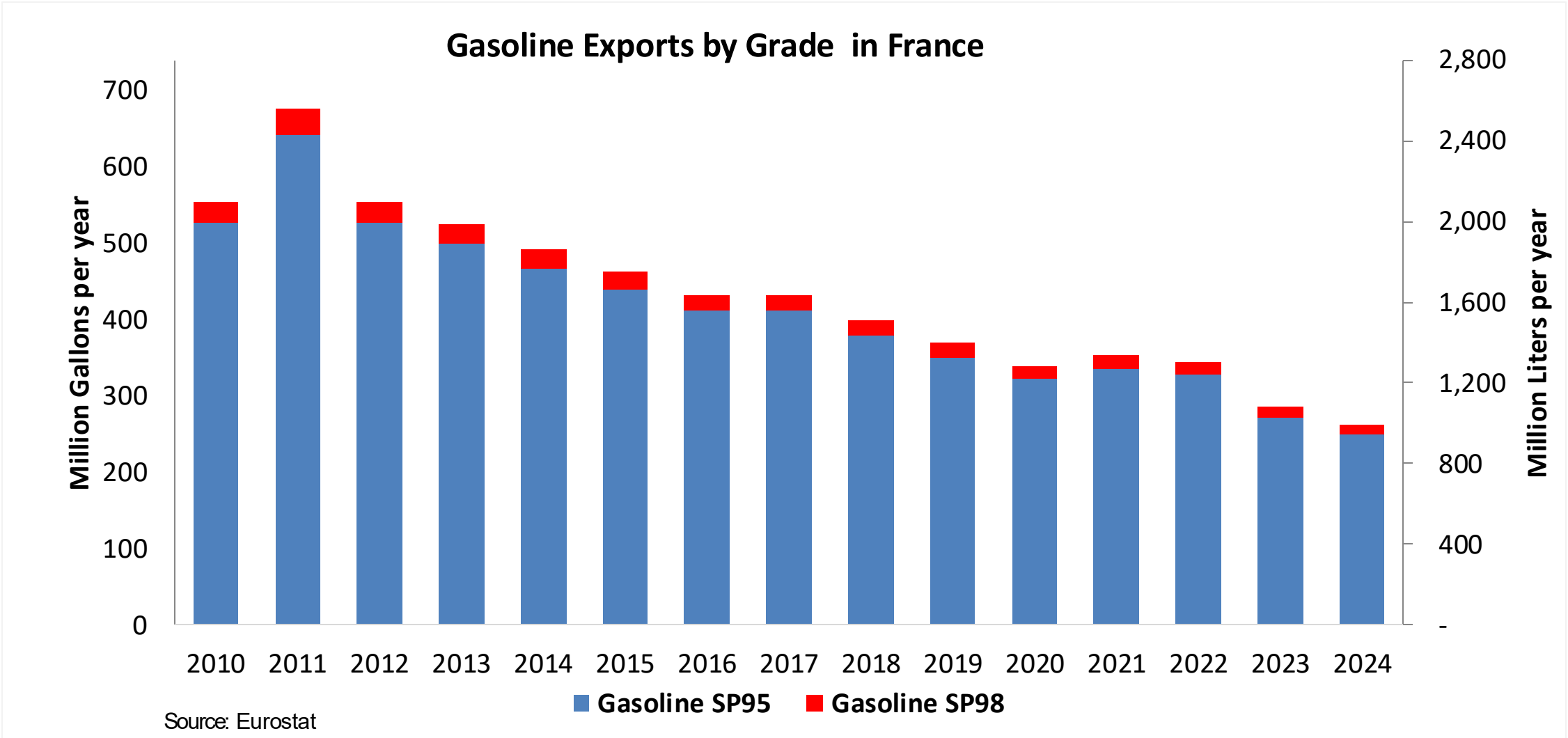
Gasoline Production in France



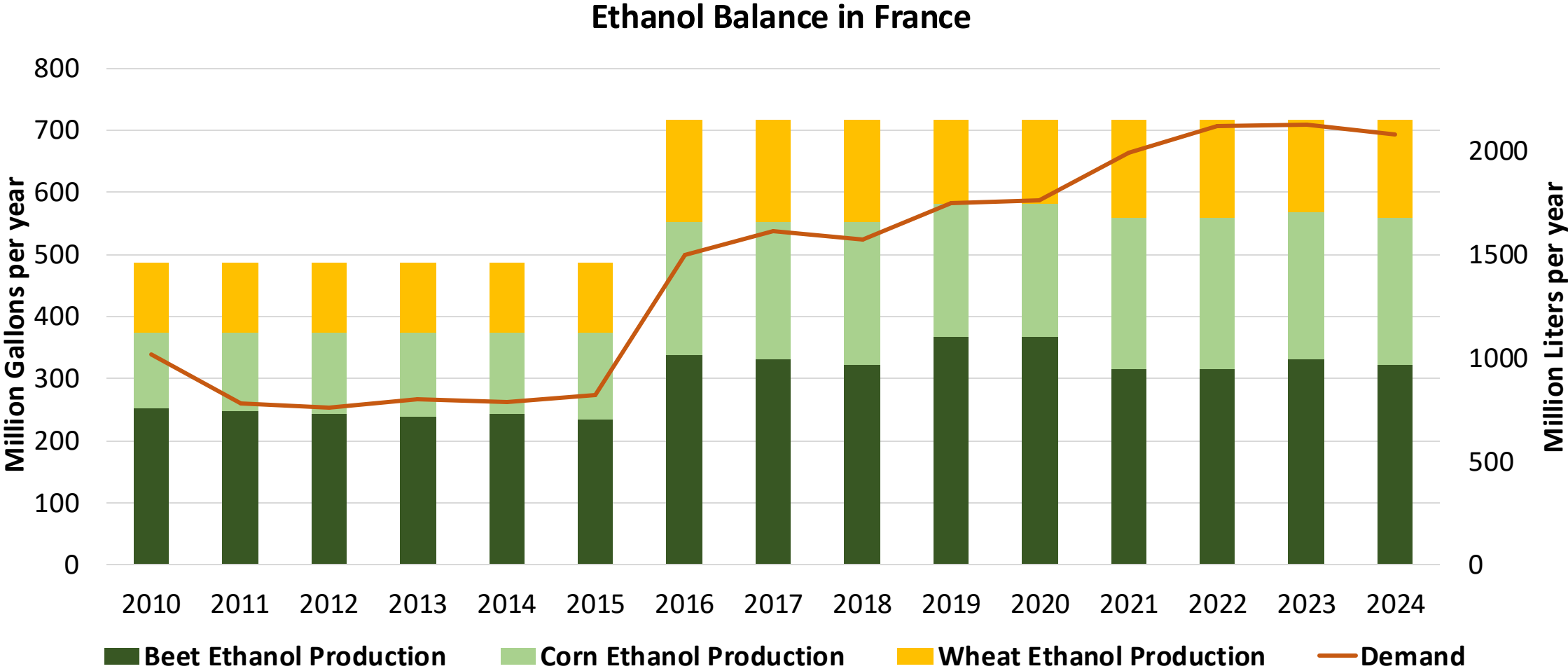
Gasoline Imports in France



Gasoline Exports in France

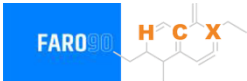


Ethanol Balance in France



Source: Eurostat

Gasoline Quality in France

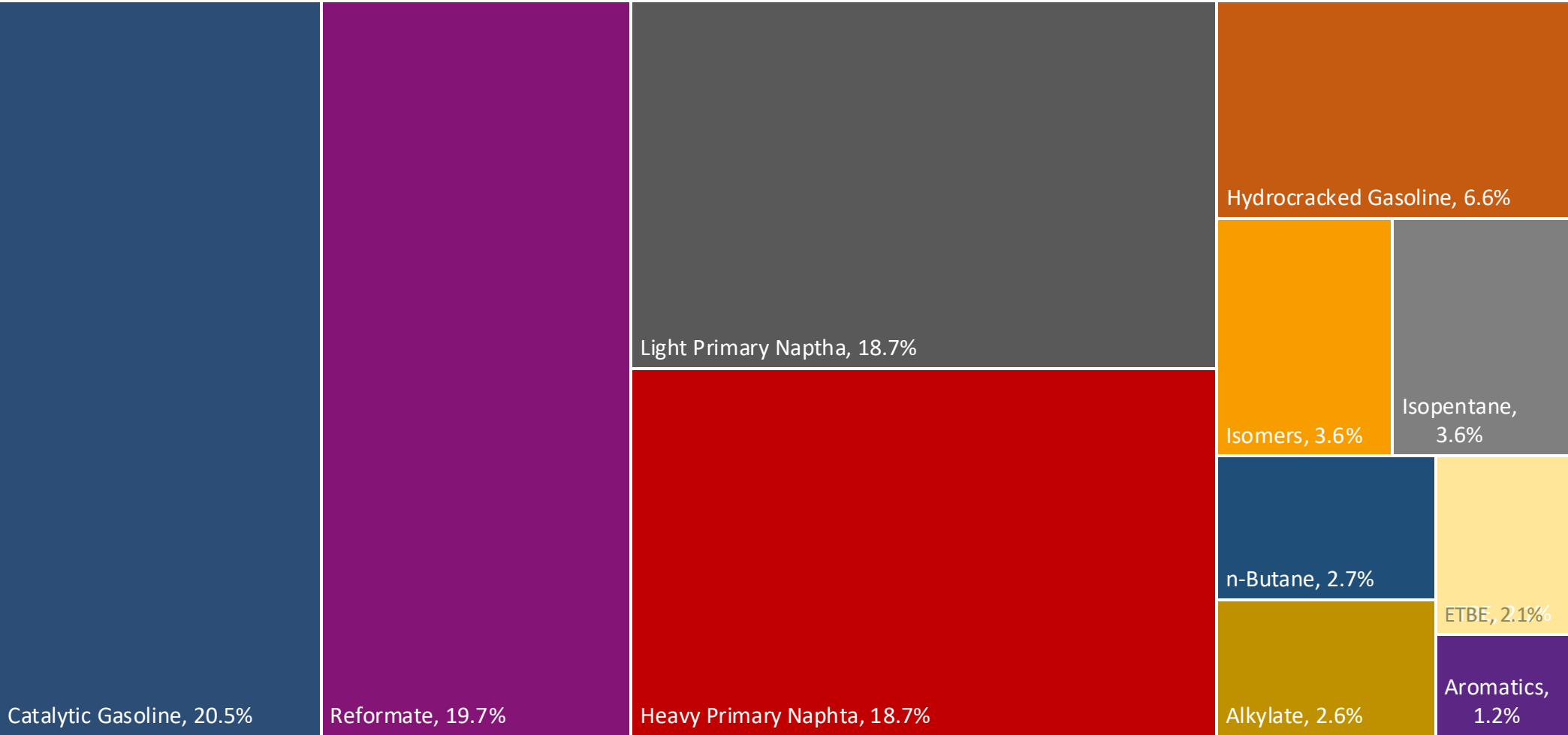
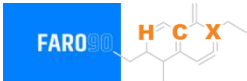


Name	EN 228:2025 (Euro 6); Country Level Fuel Ordinances							
Implementation Date	2025							
Applicability	All EU Contries + UK							
Grade	RON 95 E5		RON 95 E10		RON 98 E5		RON 98 E10	
	Min	Max	Min	Max	Min	Max	Min	Max
RON	95.0		95.0		98.0		98.0	
MON	> 85	> 88	> 85	> 88	> 85	> 88	> 85	> 88
AKI								
Benzene (%vol)	< 1							
Aromatics (% vol)	< 35							
Olefins (%vol)	< 18							
Lead (g/L)	< 0.005							
Manganese (mg/L)	< 2,0							
Sulfur (ppm)	10.0							
Oxygen (%w)*	< 2,7	< 3,7	< 2,7	< 3,7	< 2,7	< 3,7	< 2,7	< 3,7
Ethanol (%vol)	< 5	< 10	< 5	< 10	< 5	< 10	< 5	< 10
PVR 37.8°C (psi)	<> 45 - 100 kPa in 6 Volatility Classes							
Overall Biofuels (%cal)	-							
Advanced Biofuels (%cal)	1.2							
Bioethanol in Gasoline Sales (%cal)	8.6							
GHG Emission Reduction (%)	-							

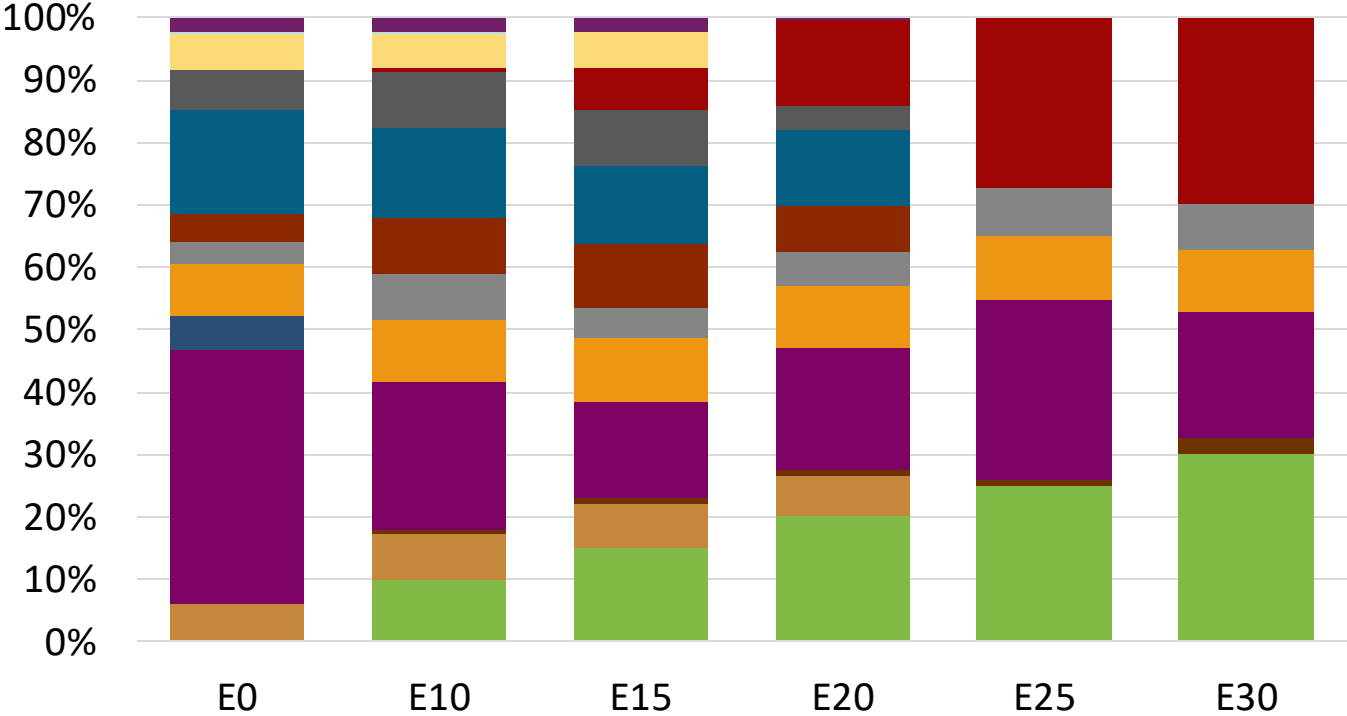
* Methanol 3%, Iso-propyl alcohol 12%, Ter-butyl alcohol 15%, Iso-butyl alcohol 15%, Ethers +5 Carbons 22%, Other Oxygenates 15%

Source: European Commision

France Available Blending Components



France – Gasoline 95 – Constant Octane

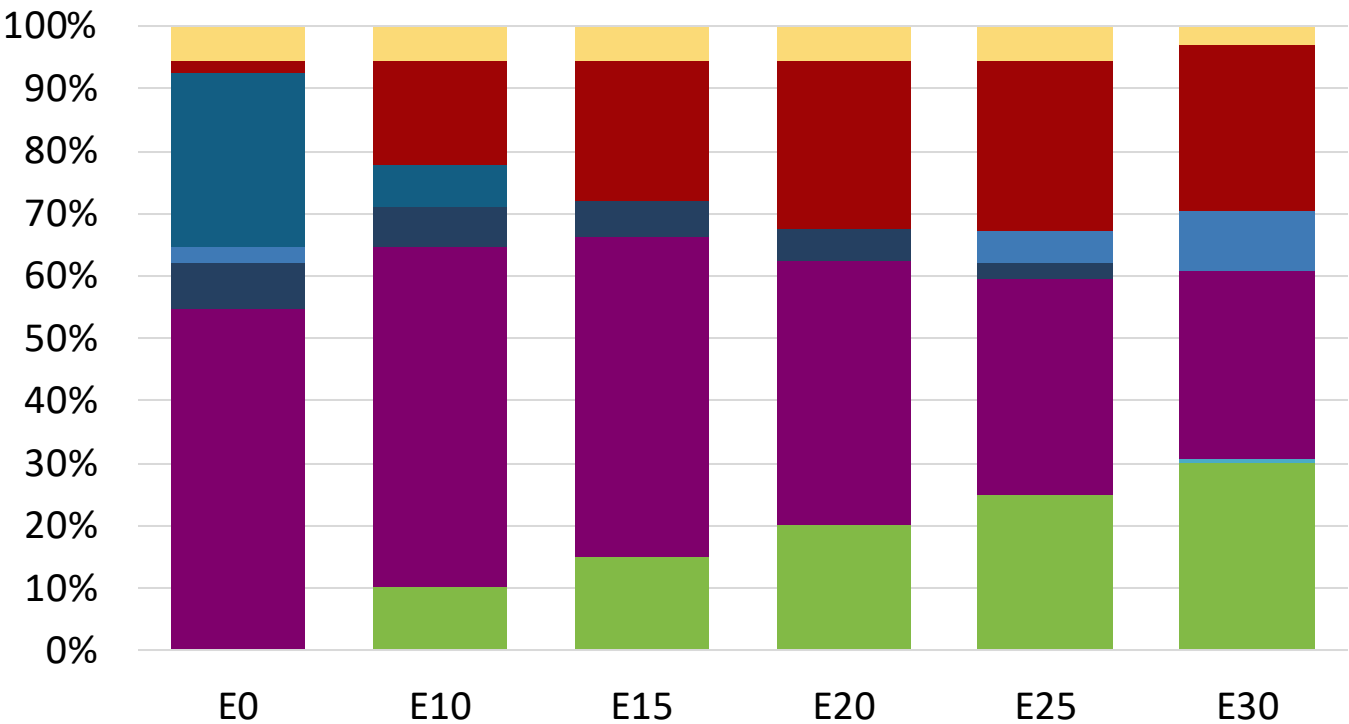
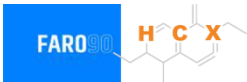


Average CIF 2024
Prices

Octane (RON)	95.0	95.0	95.0	95.0	95.2	95.7
Price (US\$/gal)	\$ 2.268	\$ 2.158	\$ 2.079	\$ 1.996	\$ 1.882	\$ 1.810

Source: Faro90

France - Gasoline 98 - Constant Octane

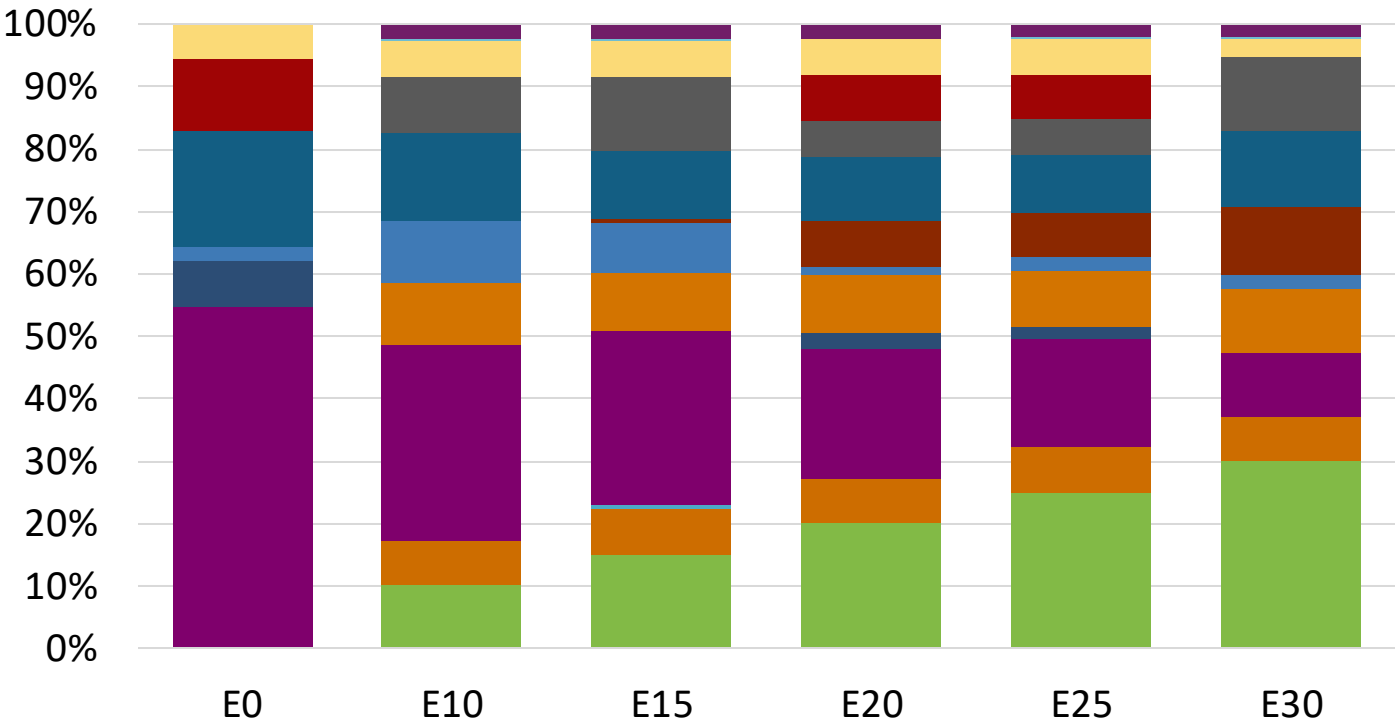


Average CIF 2024 Prices

Octane (RON)	98.0					
Price (US\$/gal)	\$ 2.346	\$ 2.222	\$ 2.151	\$ 2.076	\$ 1.998	\$ 1.923

Source: Faro90

France – Gasoline 95 – Increased Octane

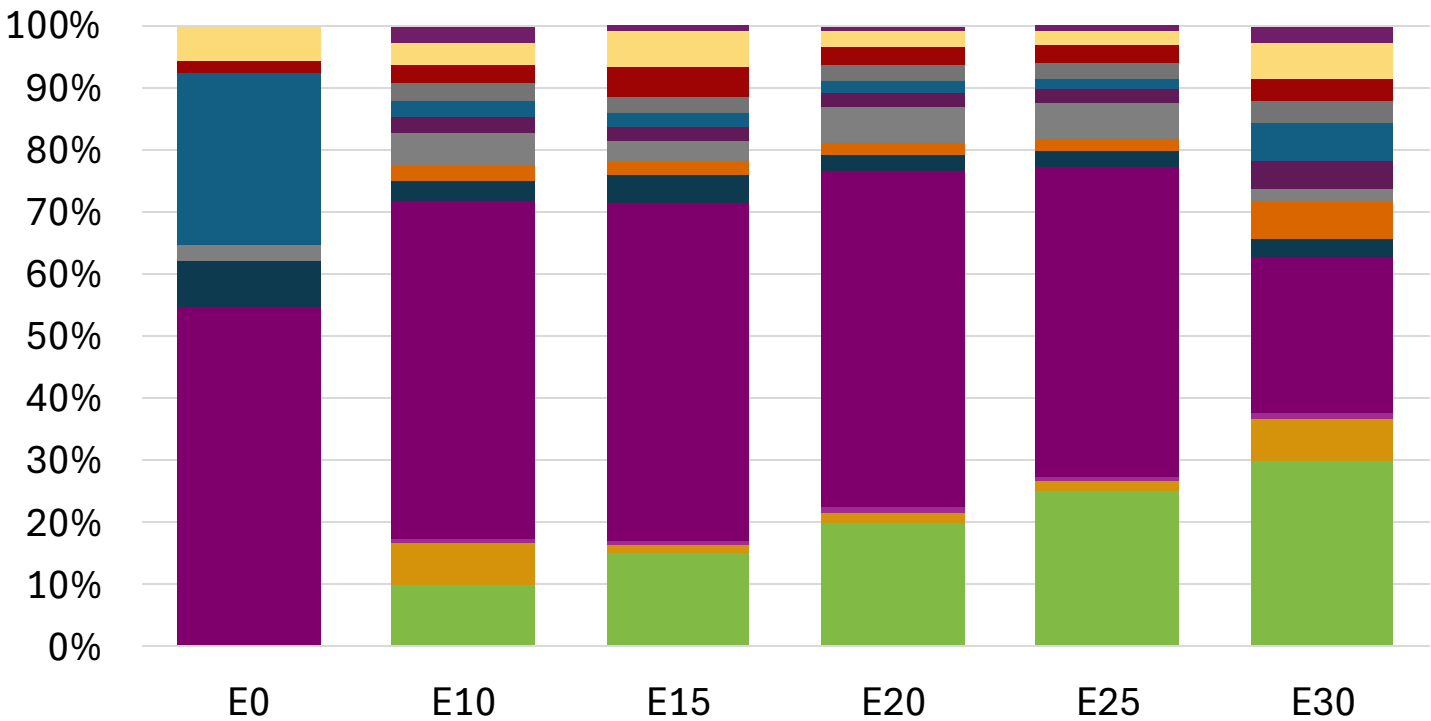
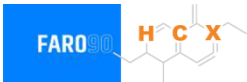


Average CIF 2024
Prices

Octane (RON)	95.0	96.5	97.9	99.4	100.8	101.7
Price (US\$/gal)	\$ 2.212	\$ 2.212	\$ 2.212	\$ 2.212	\$ 2.212	\$ 2.212

Source: Faro90

France – Gasoline 98 – Increased Octane

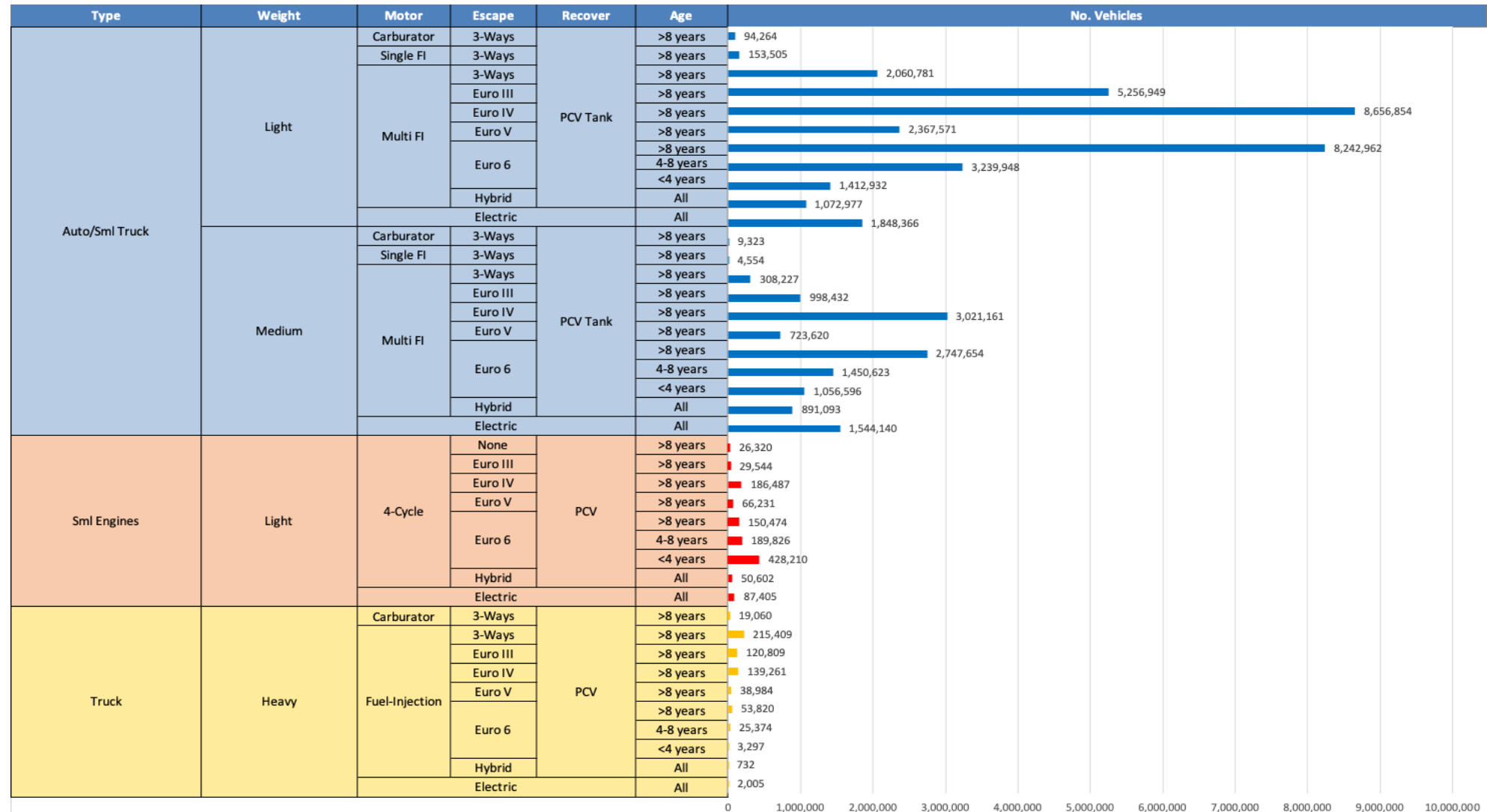


Average CIF 2024
Prices

Octane (RON)	98.0	99.3	101.4	102.4	103.7	104.6
Price (US\$/gal)	\$ 2.346	\$ 2.346	\$ 2.346	\$ 2.346	\$ 2.346	\$ 2.346

Source: Faro90

Gasoline Vehicle Fleet - France

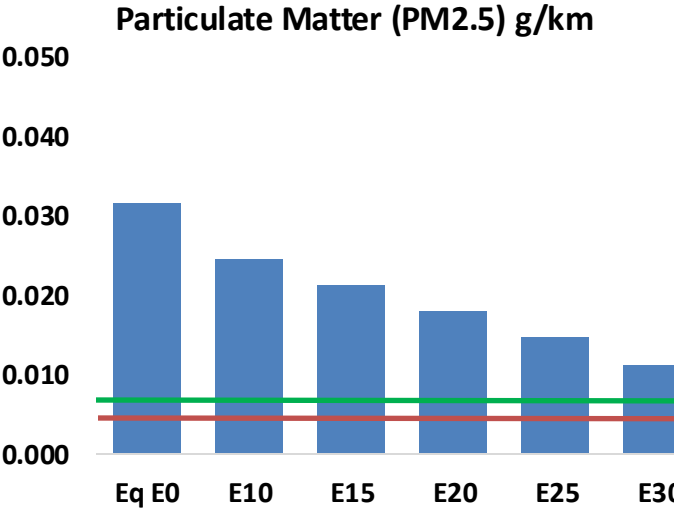
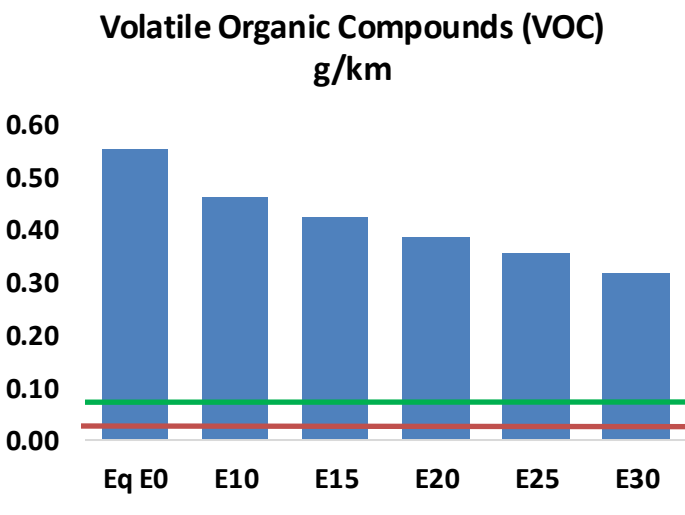
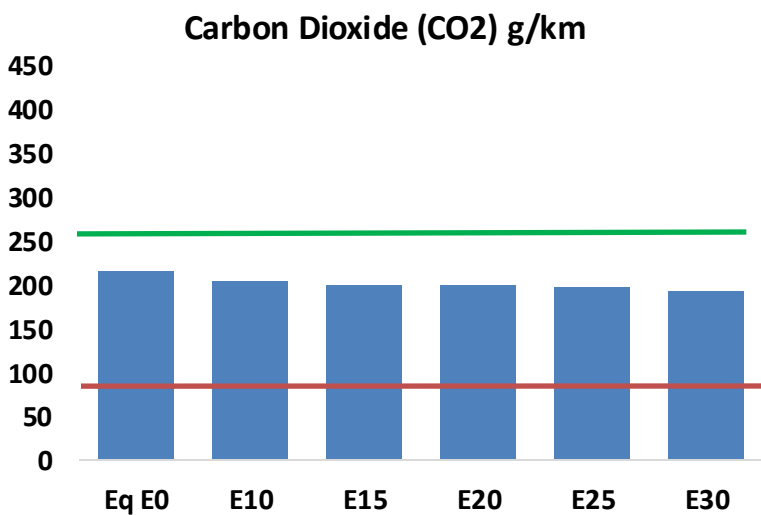
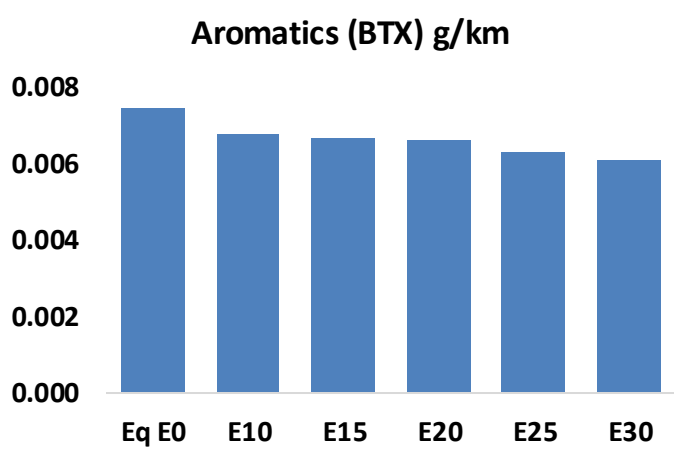
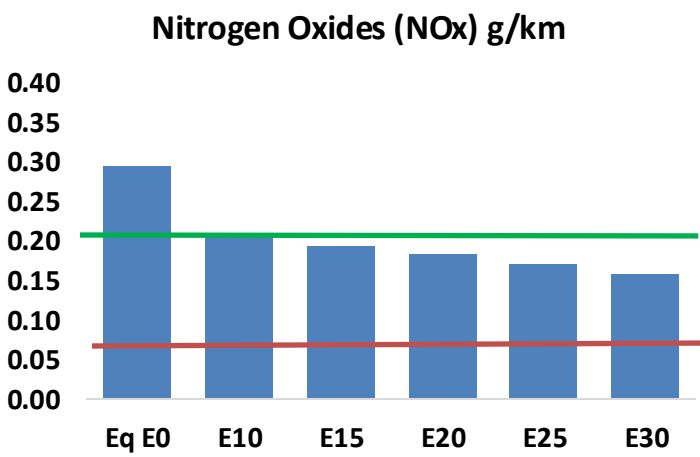
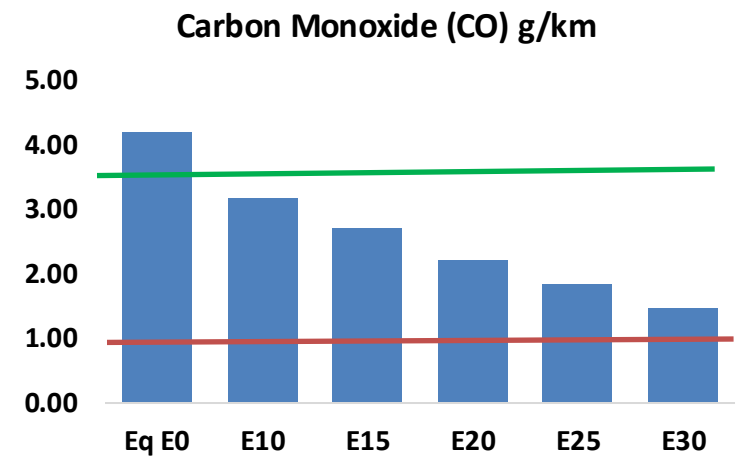
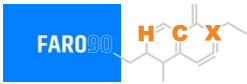


Vehicle Fleet: 48,964,974
Gasoline Vehicle Fleet: 24,096,952

Source: Eurostat, ACEA

France – Vehicular Emissions

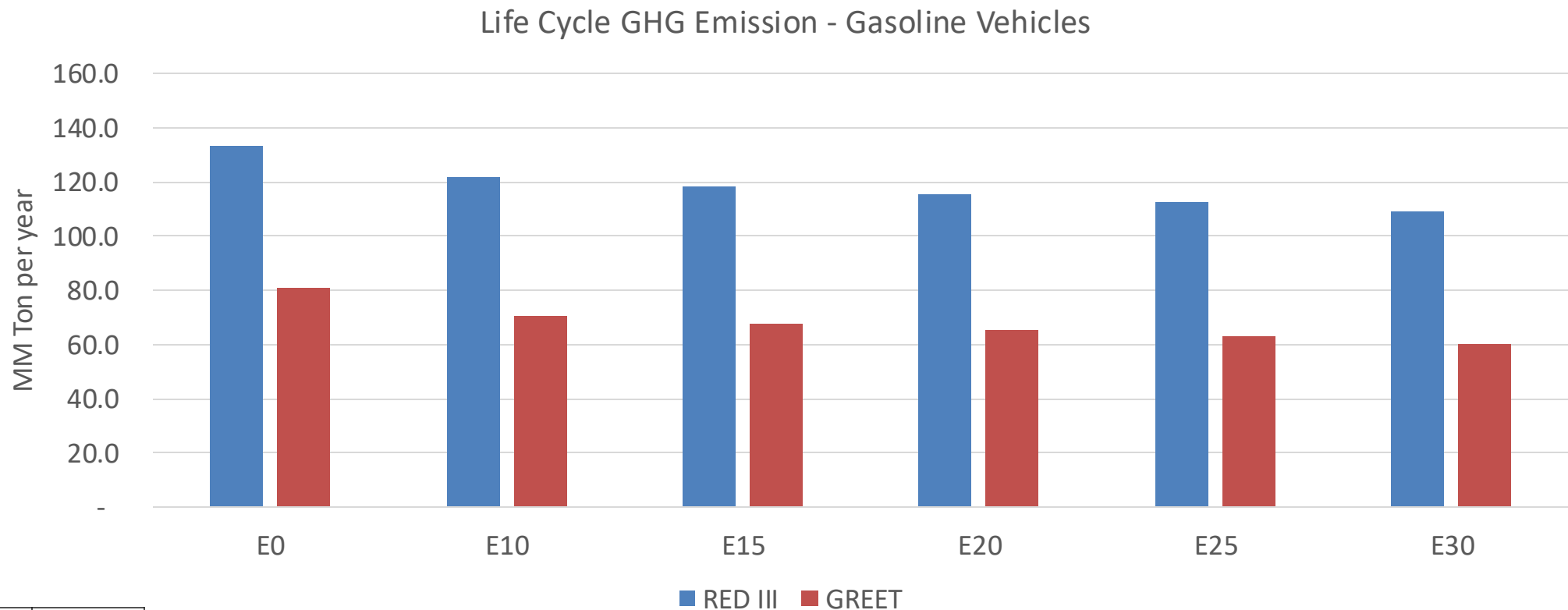
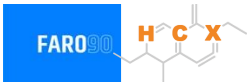
TIER III BIN 125 United States
Euro 6 Standard



France – Vehicular Emissions

Vehicular Fleet Emissions (kg/day)	Eq E0	E5	E10	E15	E20	E25	E30	Reduction (%)	
								E10	E20
CO	2,605,439	2,284,795	1,964,150	1,677,912	1,379,054	1,145,988	912,581	-25%	-47%
CO2	133,470,121	130,038,032	126,796,615	124,247,335	122,985,852	122,244,413	119,991,097	-5%	-8%
VOC	343,295	314,878	286,460	262,778	238,764	220,244	195,723	-17%	-30%
NOx	182,003	136,502	127,402	120,076	113,242	105,707	97,736	-30%	-38%
SOx	1,536	1,420	1,304	1,206	1,112	1,010	902	-15%	-28%
NH3	47,099	46,633	46,166	46,385	46,907	47,274	47,579	-2%	0%
N2O	1,983	1,972	1,964	1,974	2,014	2,037	2,060	-1%	2%
CH4	70,260	70,260	70,260	71,314	72,859	74,194	75,459	0%	4%
Lead	-	-	-	-	-	-	-	0%	0%
Butadiene	2,334	2,199	2,065	1,963	1,864	1,790	1,686	-12%	-20%
Acetaldehyde	5,934	6,810	9,993	15,218	20,733	23,691	27,994	68%	249%
Formaldehyde	20,829	20,251	23,606	27,719	29,040	32,126	34,973	13%	39%
Benzene	4,604	4,424	4,192	4,126	4,092	3,913	3,787	-9%	-11%
PM2.5	8,142	7,229	6,317	5,486	4,658	3,780	2,864	-22%	-43%
PM10	11,339	10,069	8,798	7,640	6,488	5,265	3,989	-22%	-43%

France – Gasoline Vehicle Fleet Life Cycle GHG Emissions



Country Emissions 2024 (MMT/año)	331.1
Target @ 2035 (MMT/year)	186.4
Delta	144.7

%Reduction vs E0	E0	E10	E15	E20	E25	E30
GREET 2024	0.0%	12.9%	16.4%	19.2%	22.0%	25.6%
RED II/III	0.0%	8.8%	11.4%	13.6%	15.6%	18.2%

% Target@2035	E0	E10	E15	E20	E25	E30
GREET 2024	0.0%	7.2%	9.2%	10.8%	12.3%	14.3%
RED II/III	0.0%	8.1%	10.5%	12.5%	14.4%	16.8%

Source: Greet, RED II/III, IPCC, climateactiontracker.org, F90

Ethanol Gasoline Blending in Germany



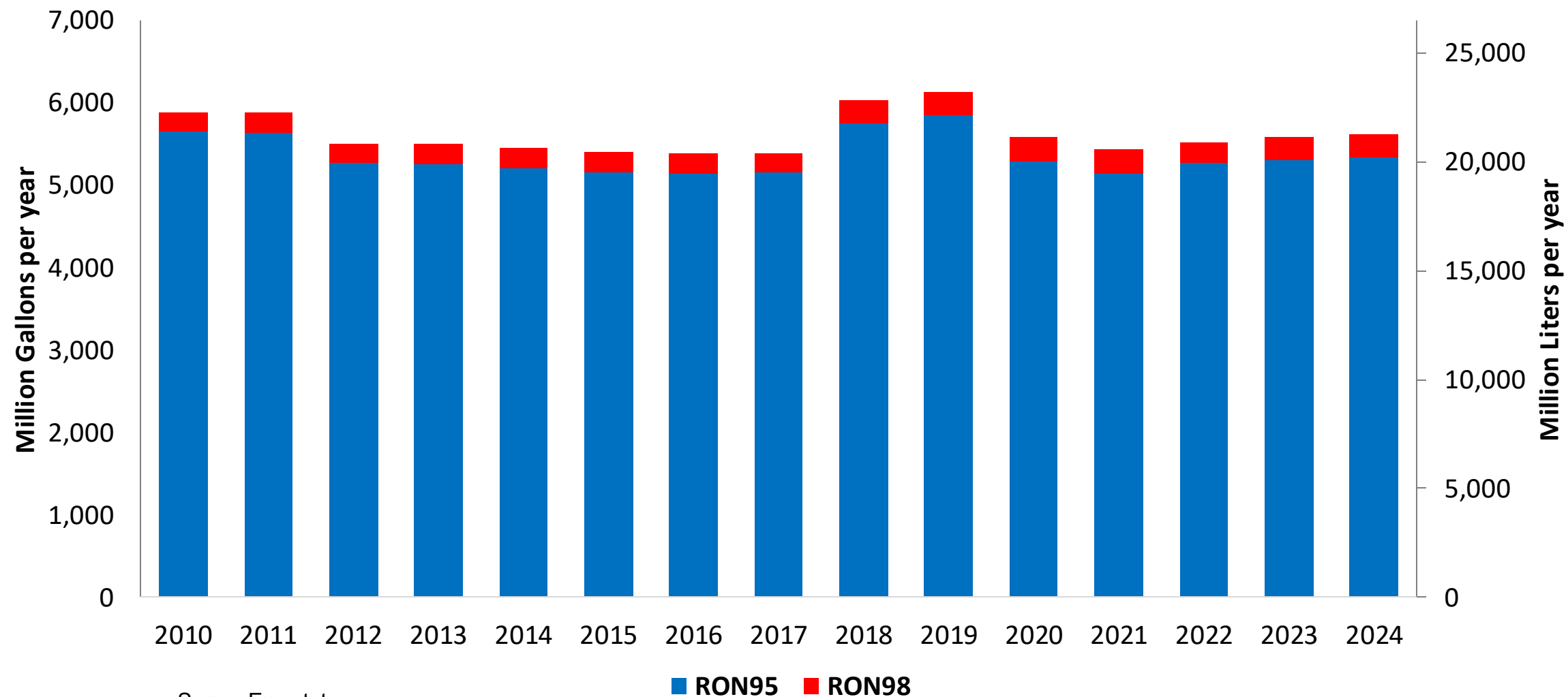
In 2024, gasoline consumption in Germany was 21,292 million liters (5,625 million gallons) and growth rate was -1.7%. Gasoline production and net imports were 22,284 and -3,611 million liters (5,887 and -954 million gallons) correspondingly. Germany complies with the European gasoline standards and offers two main grades: RON 95 and RON 98) with an average octane of 95.2 RON.

In 2024, ethanol consumption in Germany was 2,262 million liters (598 million gallons) representing 10.6% of gasoline demand. Ethanol production and Net Imports were 1,333 and 929 million liters (598 and 245 million gallons) correspondingly. Ethanol sources are Cereals 64% and Sugar Beet 33%.

Source: Eurostat, European Commission

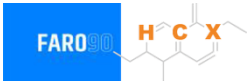
Gasoline Demand in Germany

Gasoline Demand by Grade in Germany

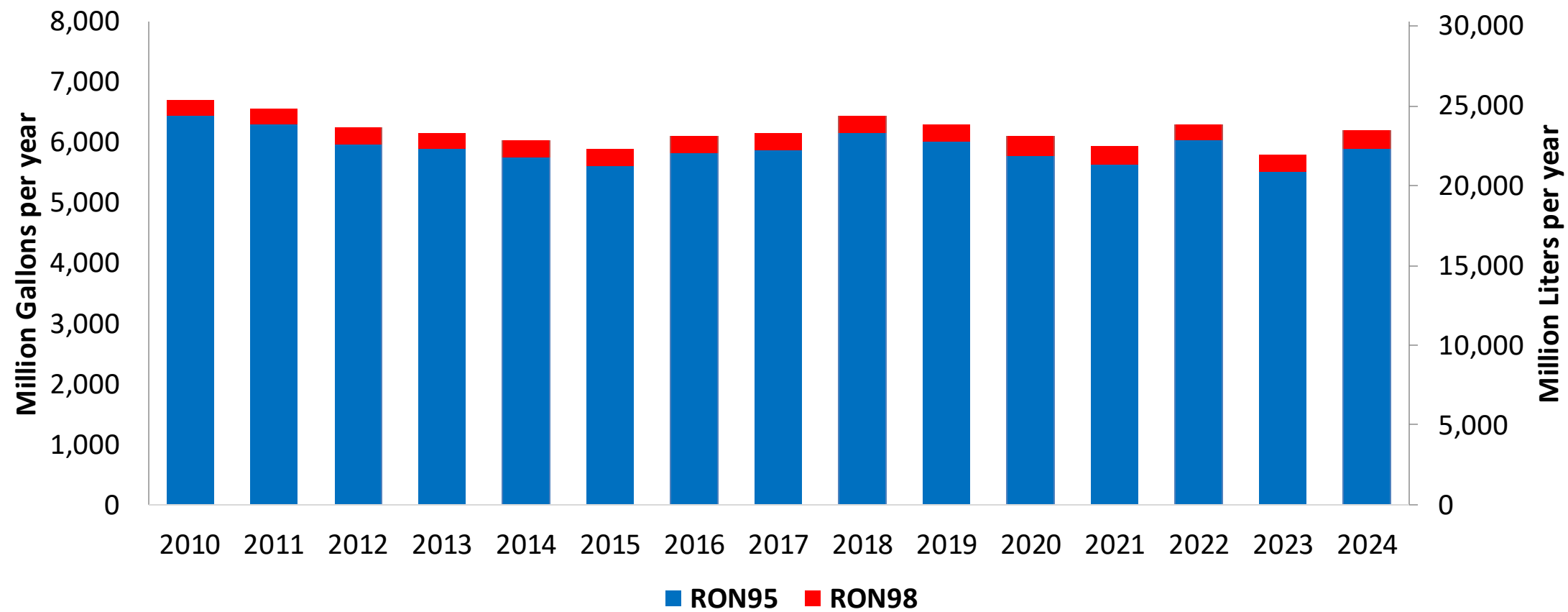


Source: Eurostat

Gasoline Production in Germany



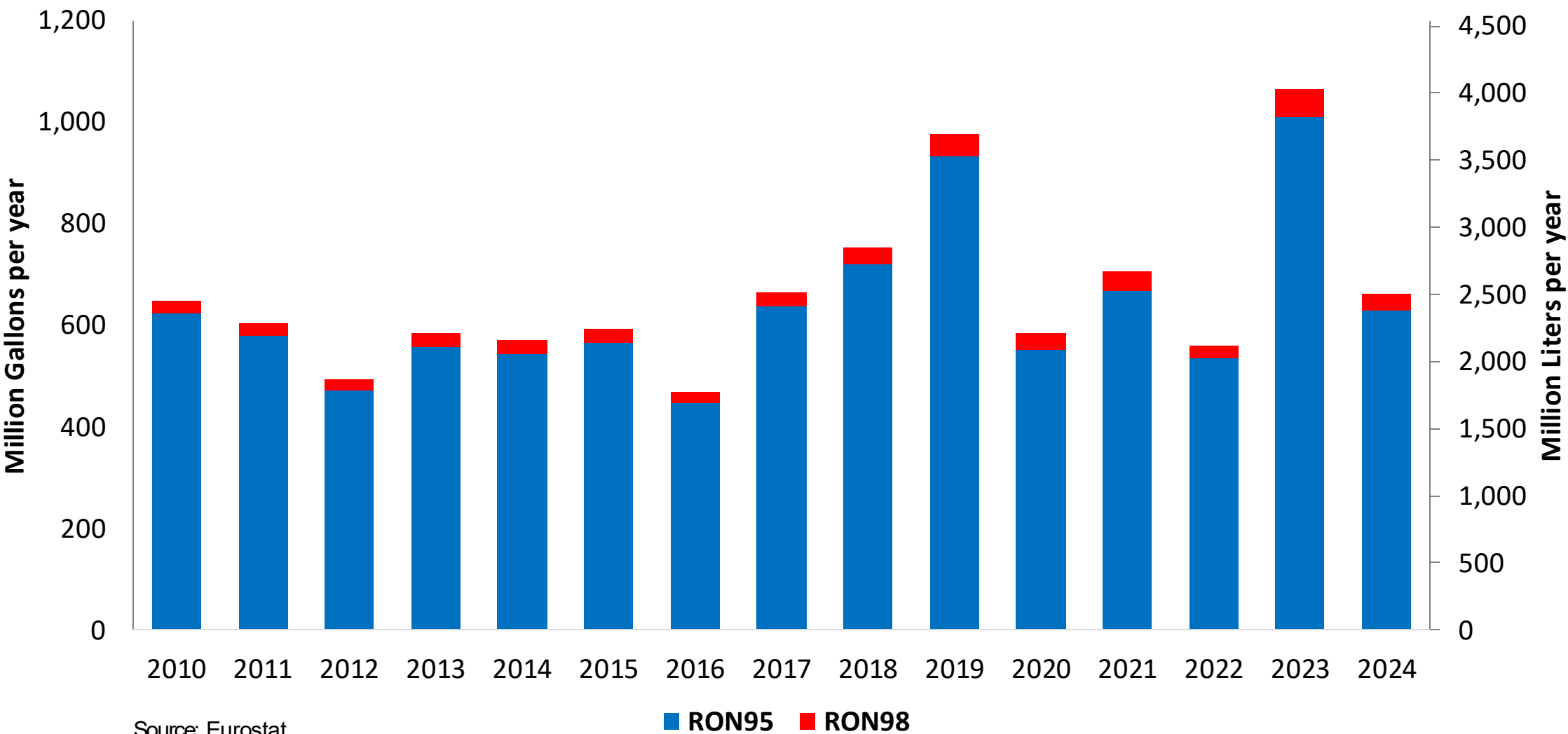
Gasoline Production by Grade in Germany



Source: Eurostat

Gasoline Imports in Germany

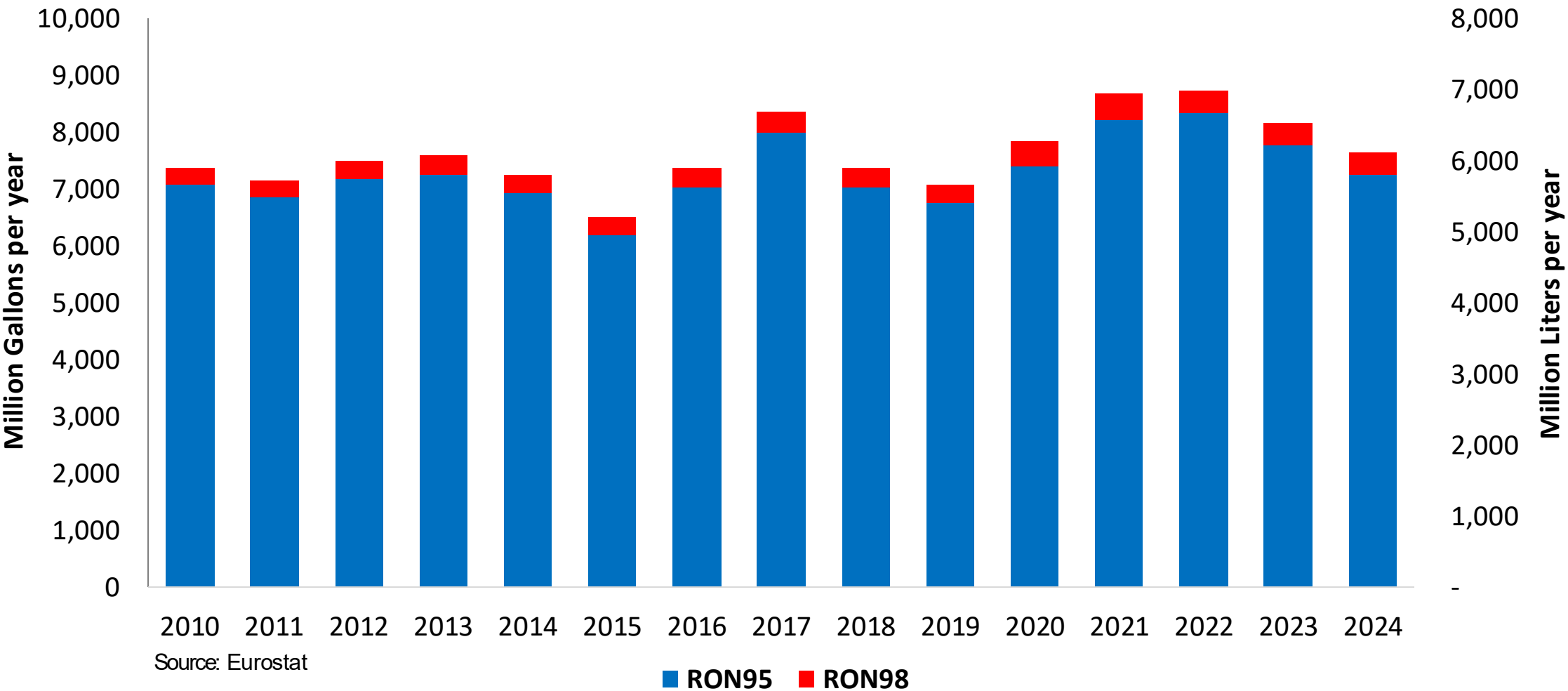
Gasoline Imports by Grade in Germany



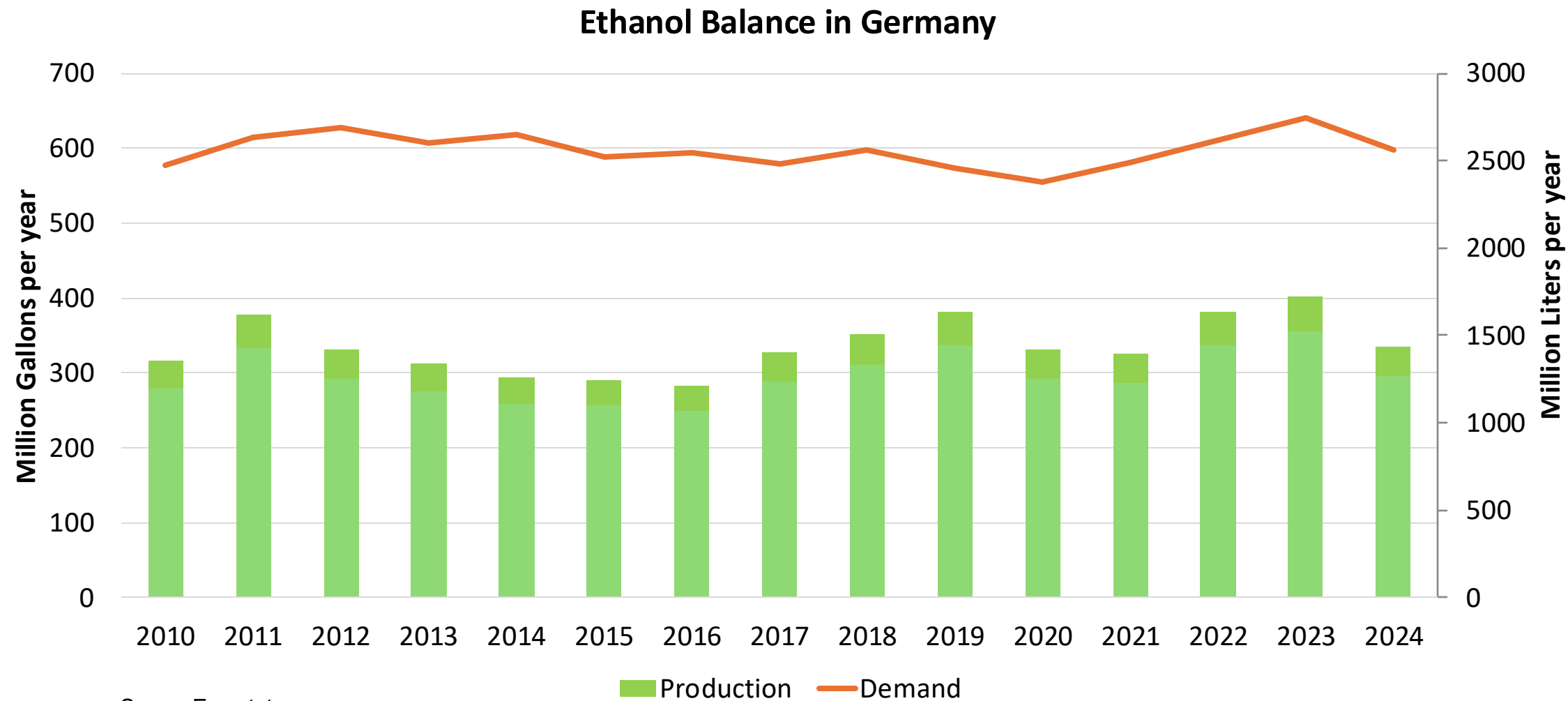
Source: Eurostat

Gasoline Exports in Germany

Gasoline Exports by Grade in Germany

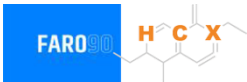


Ethanol Balance in Germany



Source: Eurostat

Gasoline Quality in Germany

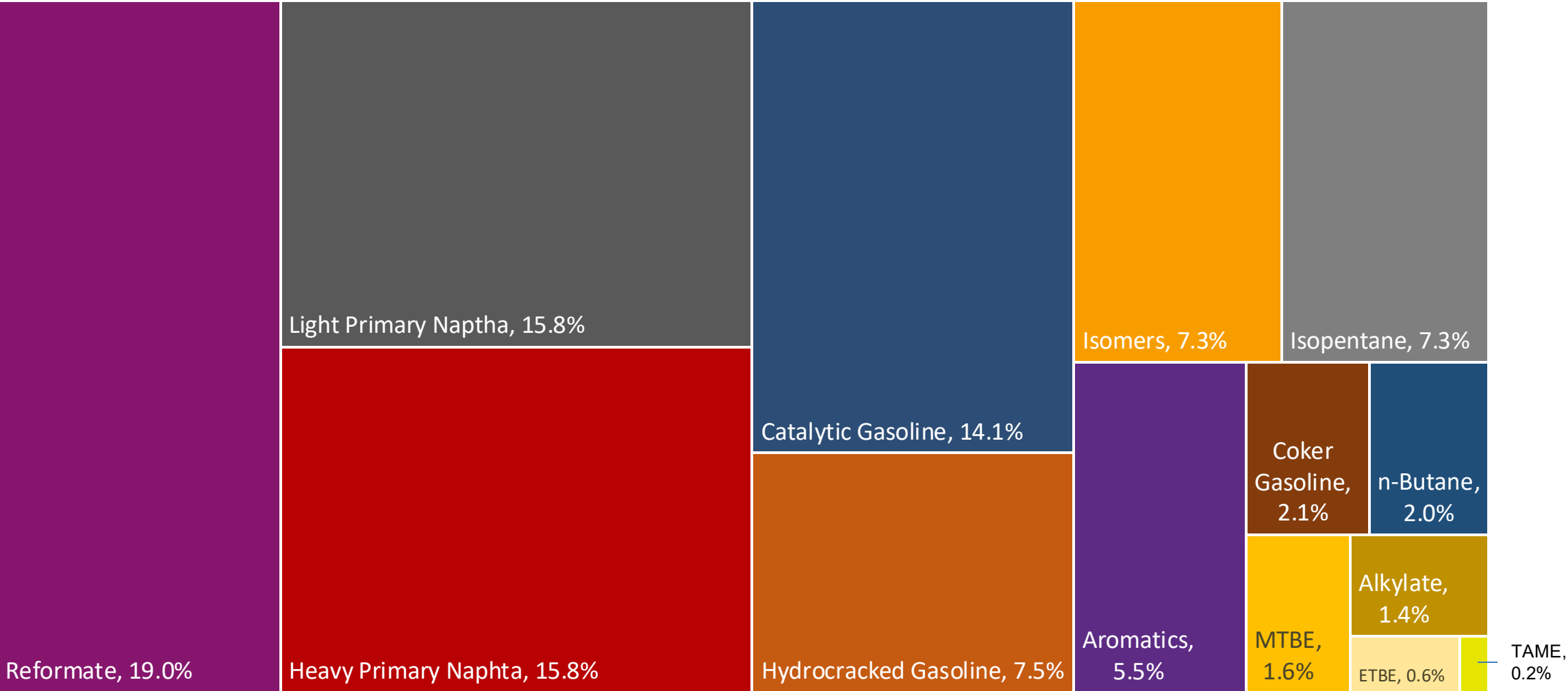
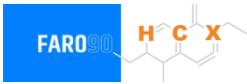


Name	EN 228:2025 (Euro 6); Country Level Fuel Ordinances							
Implementation Date	2025							
Applicability	All EU Contries + UK							
Grade	RON 95 E5		RON 95 E10		RON 98 E5		RON 98 E10	
	Min	Max	Min	Max	Min	Max	Min	Max
RON	95.0		95.0		98.0		98.0	
MON	> 85	> 88	> 85	> 88	> 85	> 88	> 85	> 88
AKI								
Benzene (%vol)	< 1							
Aromatics (% vol)	< 35							
Olefins (%vol)	< 18							
Lead (g/L)	< 0.005							
Manganese (mg/L)	< 2,0							
Sulfur (ppm)	10.0							
Oxygen (%w)*	< 2,7	< 3,7	< 2,7	< 3,7	< 2,7	< 3,7	< 2,7	< 3,7
Ethanol (%vol)	< 5	< 10	< 5	< 10	< 5	< 10	< 5	< 10
PVR 37.8°C (psi)	<> 45 - 100 kPa in 6 Volatility Classes							
Overall Biofuels (%cal)	-							
Advanced Biofuels (%cal)	0.7							
Bioethanol in Gasoline Sales (%cal)	-							
GHG Emission Reduction (%)	10.5							

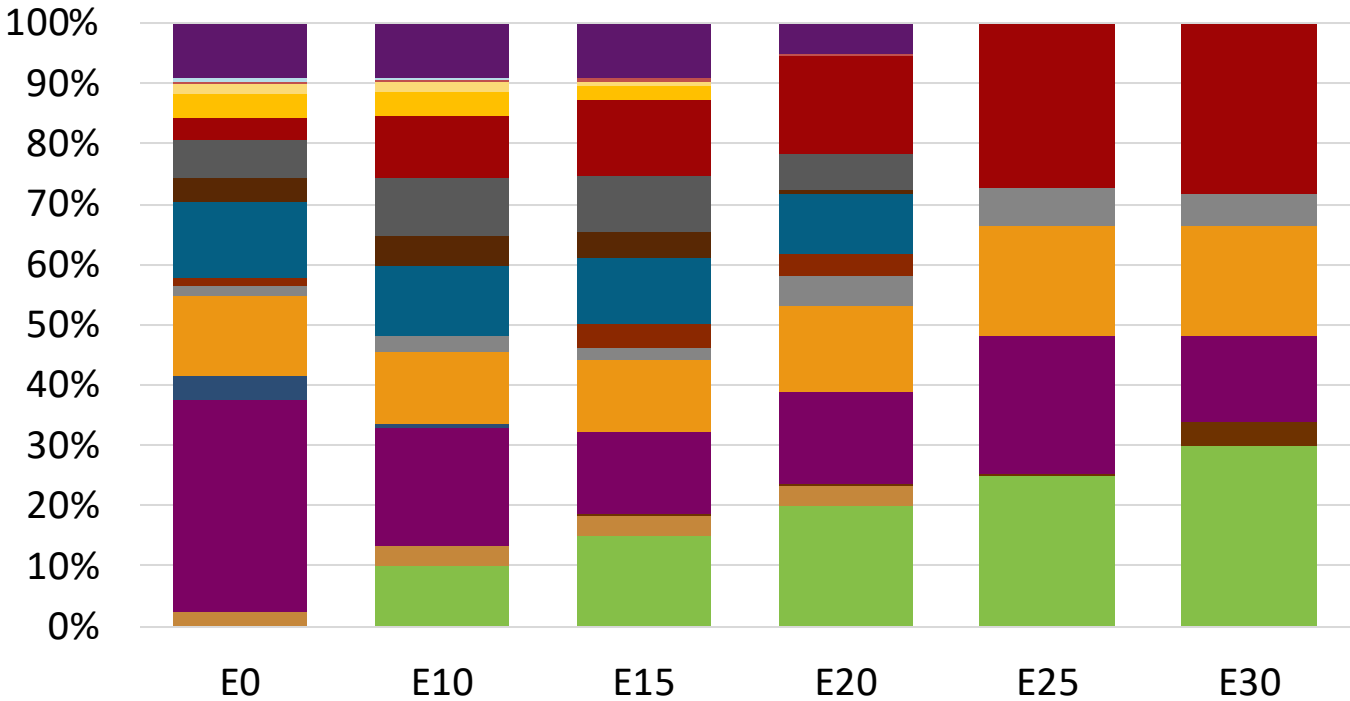
* Methanol 3%, Iso-propyl alcohol 12%, Ter-butyl alcohol 15%, Iso-butyl alcohol 15%, Ethers +5 Carbons 22%, Other Oxygenates 15%

Source: European Commision

Germany Available Blending Components



Germany – Gasoline 95 – Constant Octane

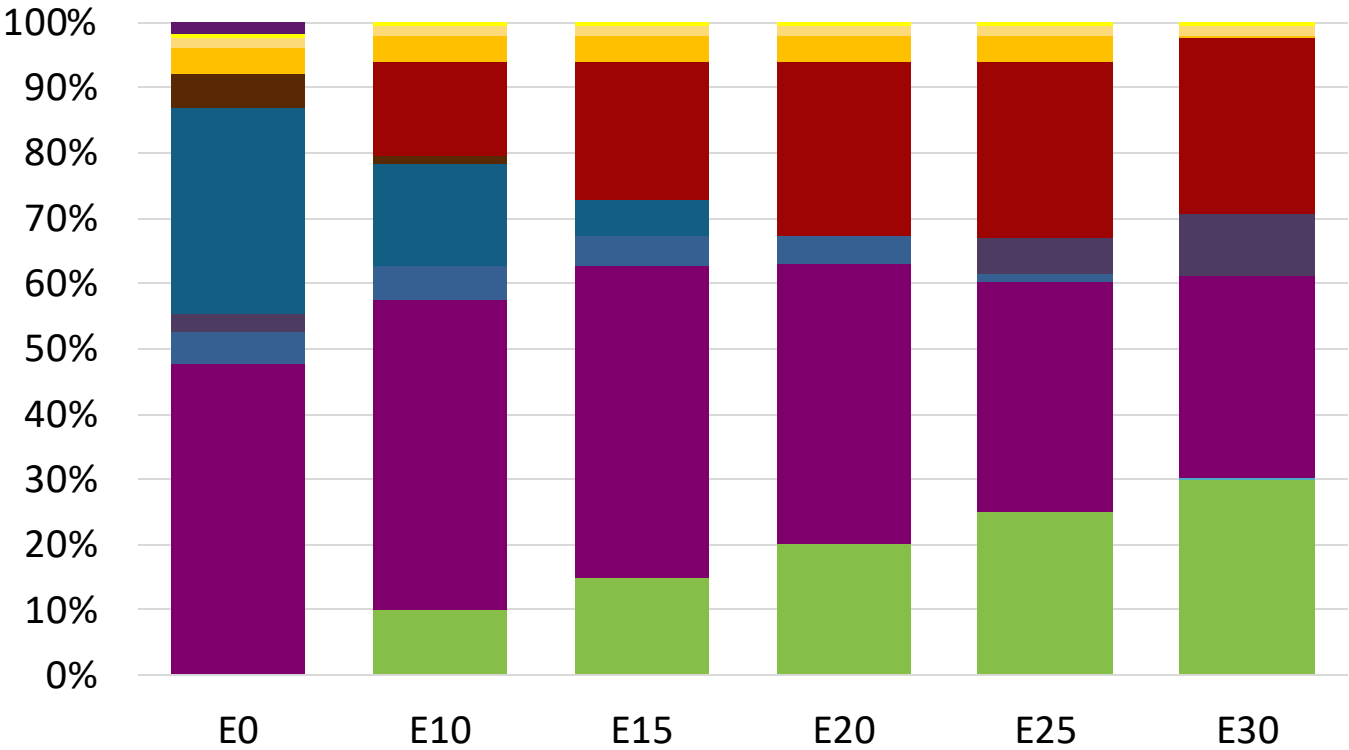
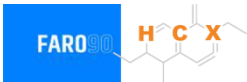


Average CIF 2024
Prices

Octane (RON)	95.0	95.0	95.0	95.0	95.1	95.6
Price (US\$/gal)	\$ 2.272	\$ 2.144	\$ 2.083	\$ 2.002	\$ 1.880	\$ 1.814

Source: Faro90

Germany - Gasoline 98 - Constant Octane

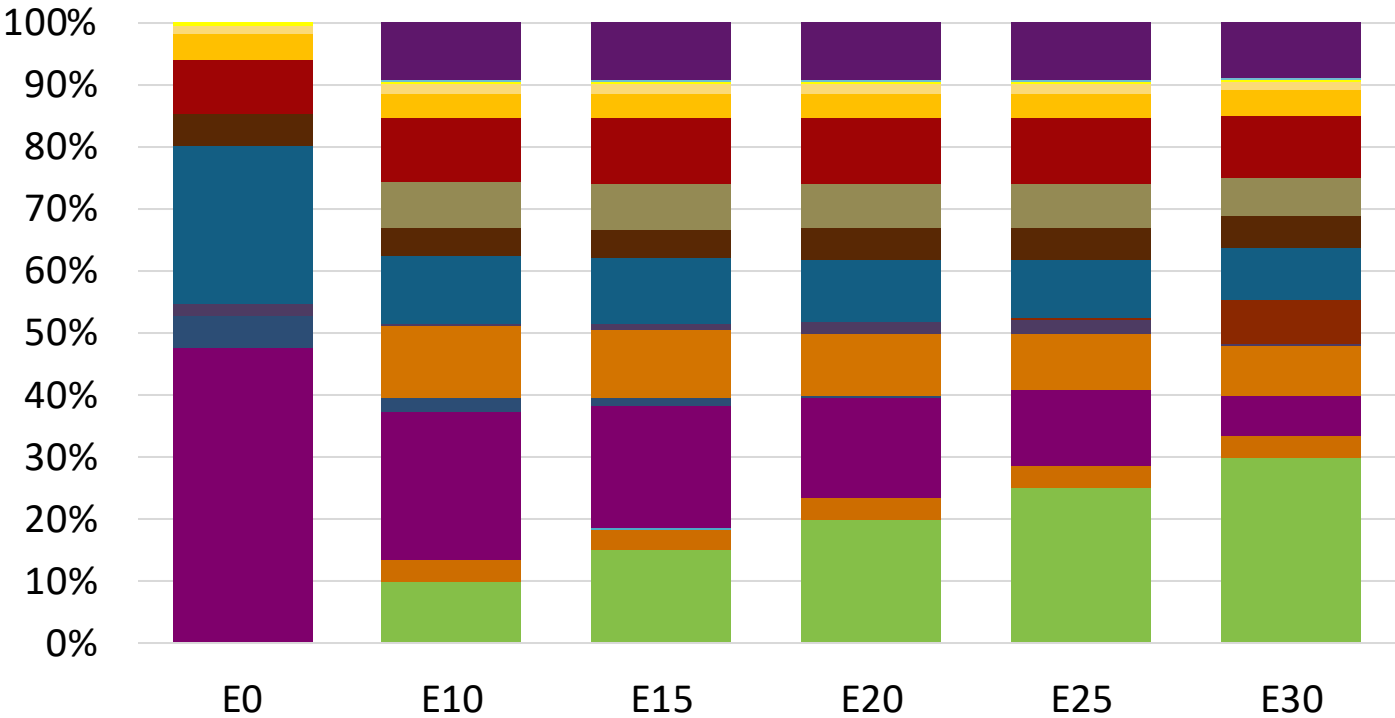
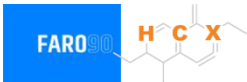


Average CIF 2024
Prices

Octane (RON)	98.0	98.0	98.0	98.0	98.0	98.0
Price (US\$/gal)	\$ 2.347	\$ 2.218	\$ 2.146	\$ 2.069	\$ 1.992	\$ 1.925

Source: Faro90

Germany – Gasoline 95 – Increased Octane

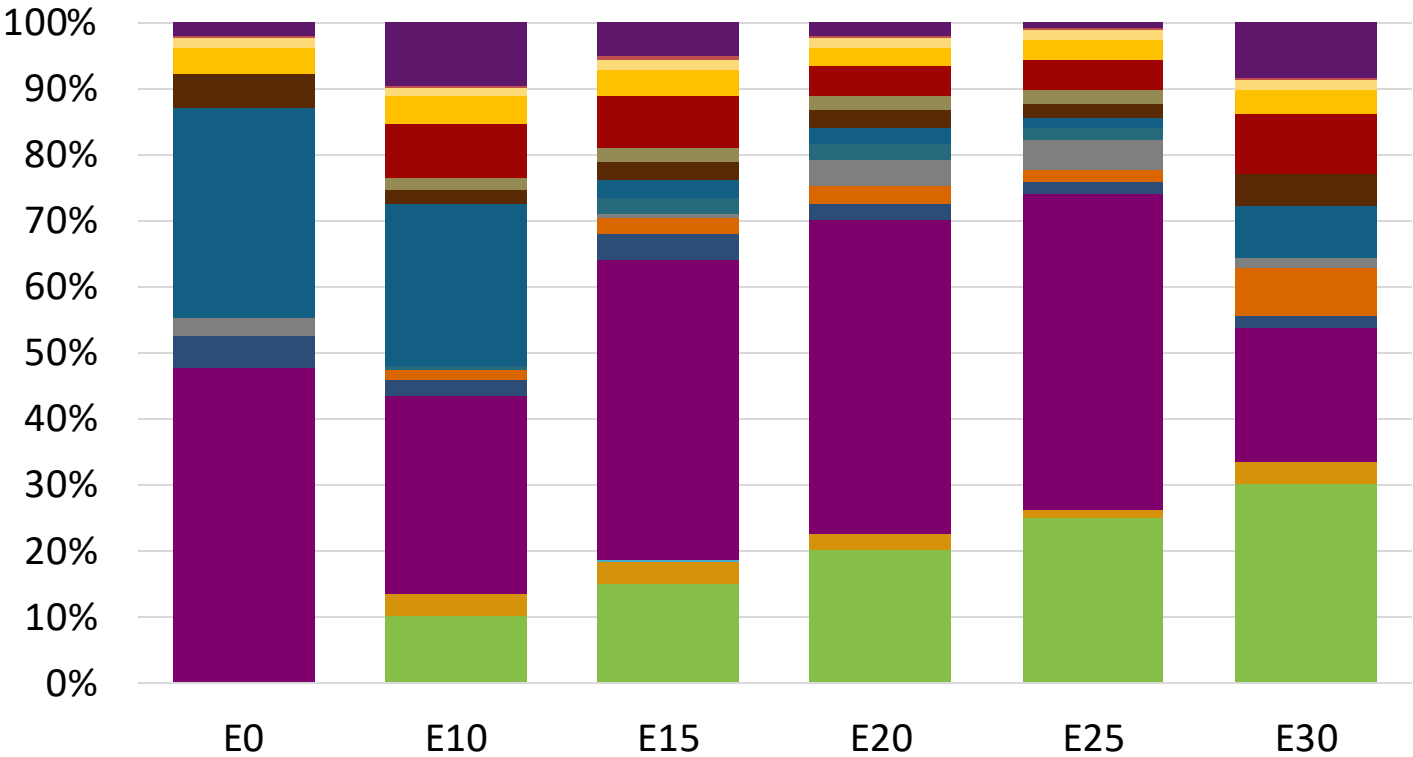
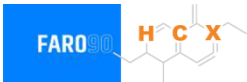


Average CIF 2024
Prices

Octane (RON)	95.0	96.5	97.9	99.4	100.8	102.0
Price (US\$/gal)	\$ 2.209	\$ 2.209	\$ 2.209	\$ 2.209	\$ 2.209	\$ 2.209

Source: Faro90

Germany – Gasoline 98 – Increased Octane

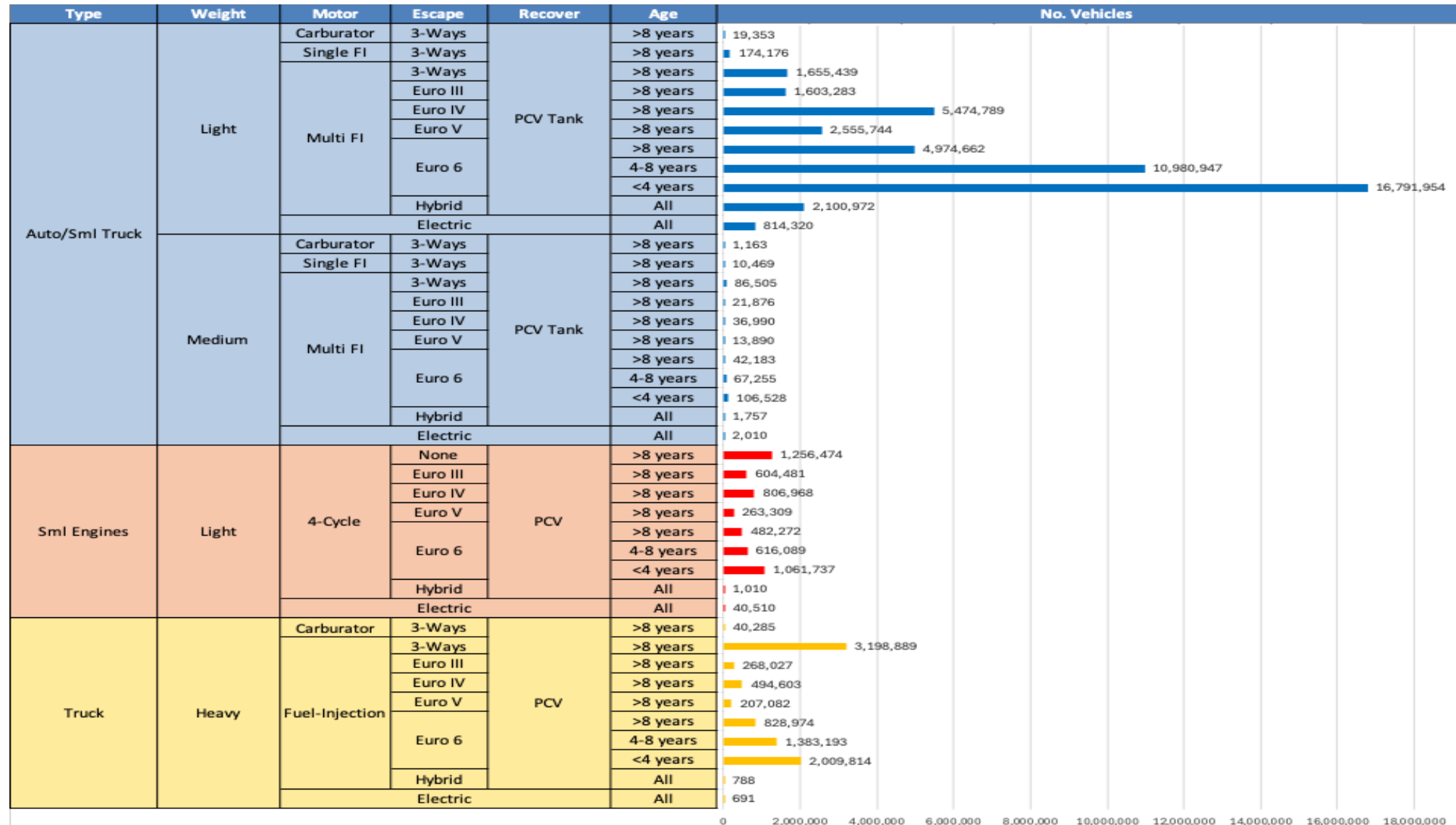


Average CIF 2024
Prices

Octane (RON)	98.0	99.2	101.3	102.6	104.0	104.7
Price (US\$/gal)	\$ 2.347	\$ 2.347	\$ 2.347	\$ 2.347	\$ 2.347	\$ 2.347

Source: Faro90

Gasoline Vehicle Fleet - Germany

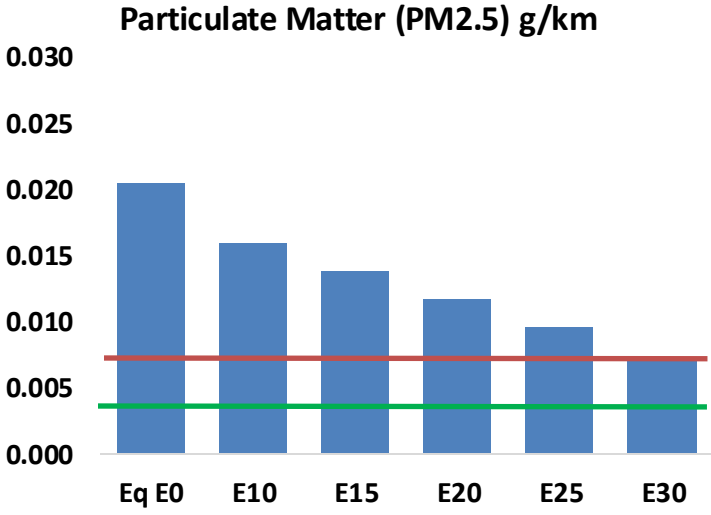
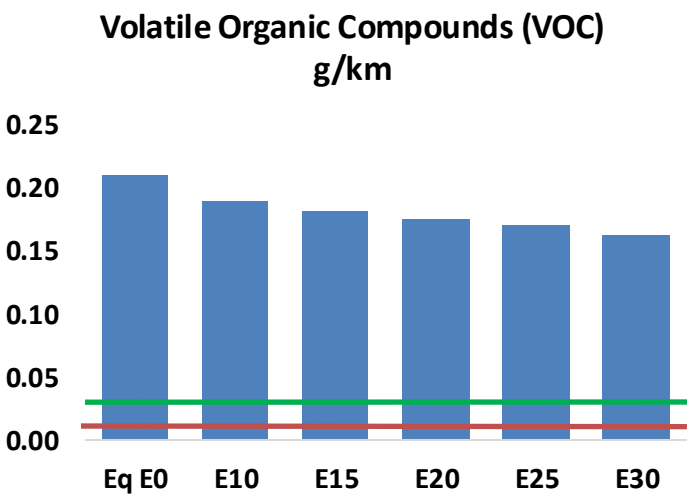
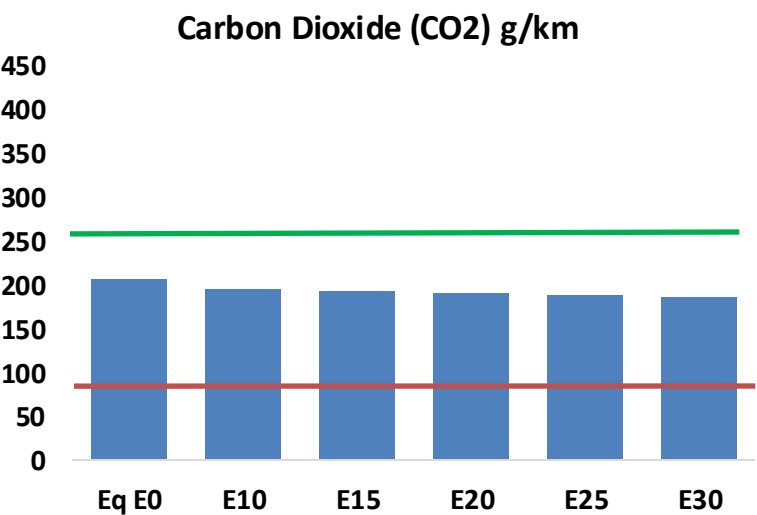
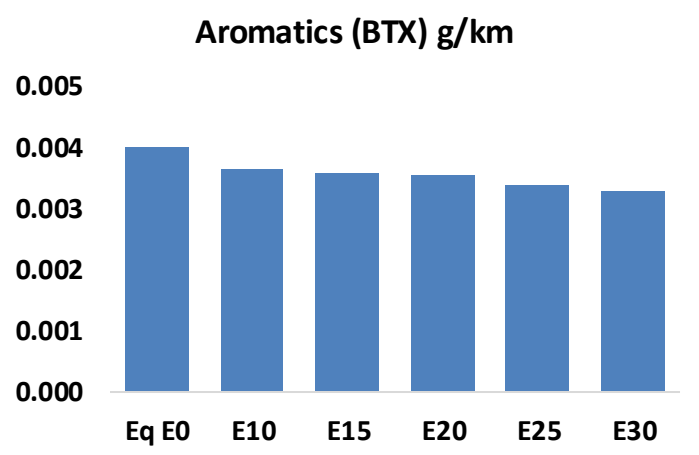
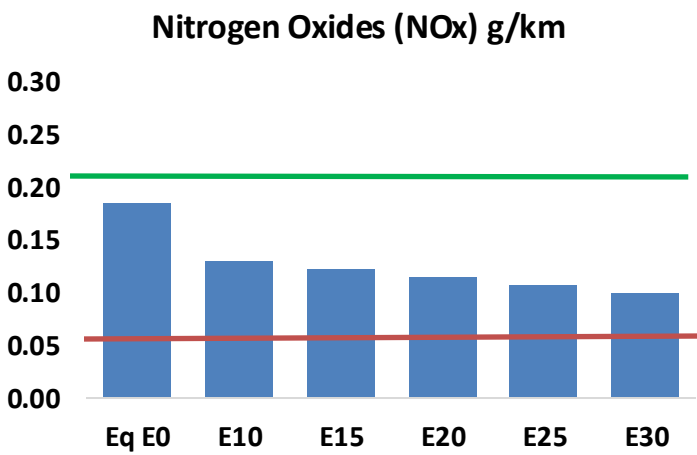
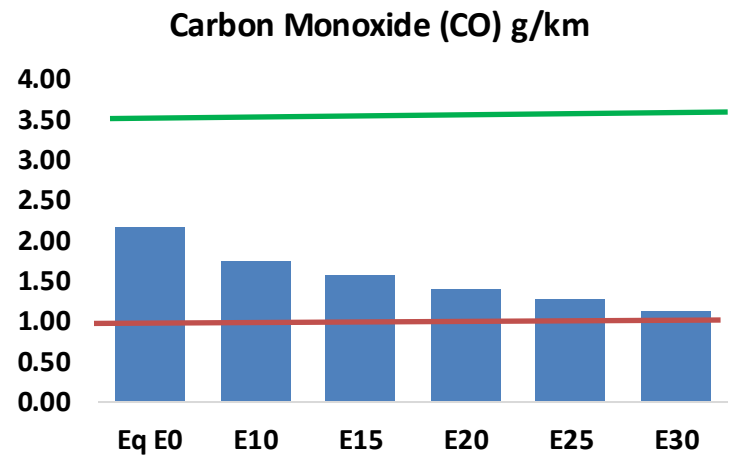
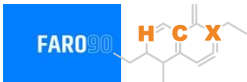


Vehicle Fleet: 61,101,463
Gasoline Vehicle Fleet: 32,331,793

Source: Eurostat, ACEA

Germany – Vehicular Emissions

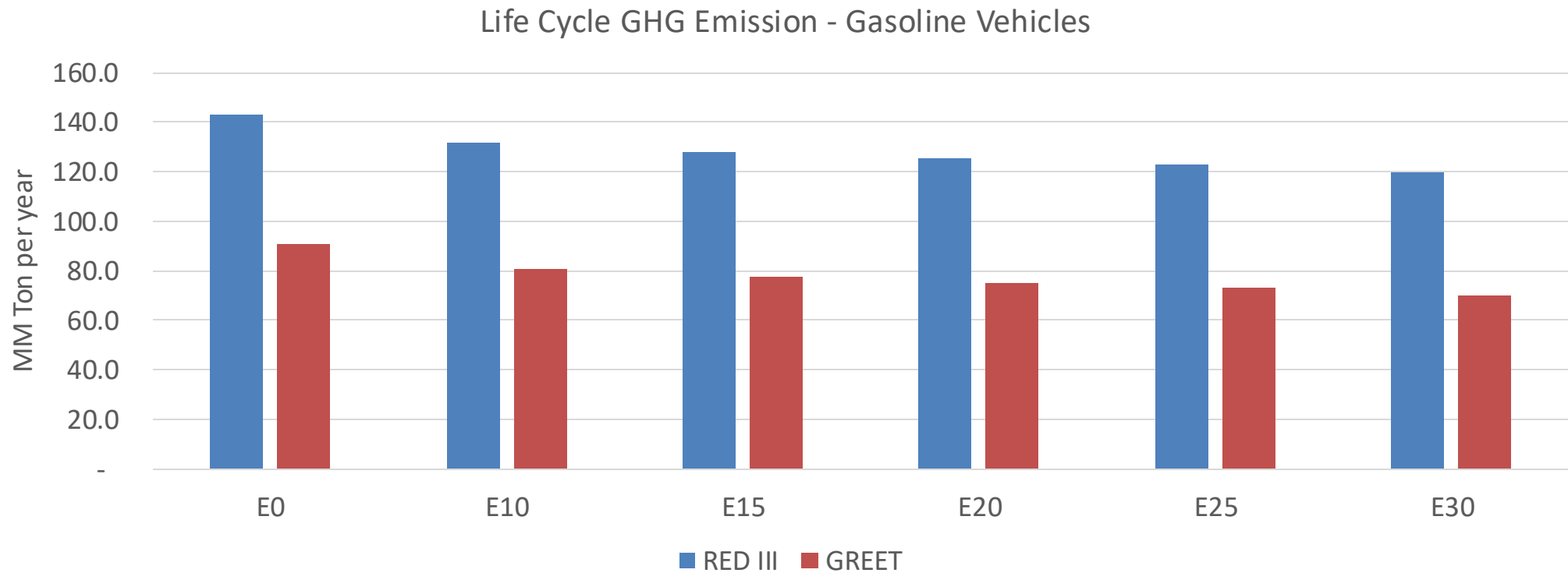
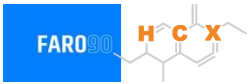
TIER III BIN 125 United States
Euro 6 Standard



Germany – Vehicular Emissions

Vehicular Fleet Emissions (kg/day)	Eq E0	E5	E10	E15	E20	E25	E30	Reduction (%)	
								E10	E20
CO	1,772,395	1,605,976	1,439,558	1,297,299	1,151,430	1,038,448	921,561	-19%	-35%
CO2	169,687,313	165,323,925	161,202,948	157,961,920	156,358,133	155,431,613	152,566,562	-5%	-8%
VOC	172,340	163,809	155,277	149,207	143,521	139,277	133,163	-10%	-17%
NOx	152,188	114,141	106,531	100,406	94,691	88,391	81,725	-30%	-38%
SOx	1,977	1,827	1,678	1,551	1,430	1,299	1,161	-15%	-28%
NH3	58,379	57,801	57,223	57,495	58,141	58,596	58,974	-2%	0%
N2O	1,820	1,810	1,803	1,811	1,849	1,869	1,891	-1%	2%
CH4	41,275	41,275	41,275	41,894	42,802	43,586	44,329	0%	4%
Lead	-	-	-	-	-	-	-	0%	0%
Butadiene	1,662	1,583	1,505	1,451	1,401	1,363	1,309	-9%	-16%
Acetaldehyde	3,372	3,870	5,679	8,648	11,783	13,464	15,909	68%	249%
Formaldehyde	10,885	10,583	12,337	14,486	15,176	16,789	18,277	13%	39%
Benzene	3,279	3,151	2,986	2,939	2,914	2,787	2,697	-9%	-11%
PM2.5	10,785	9,576	8,368	7,267	6,171	5,007	3,795	-22%	-43%
PM10	6,036	5,360	4,683	4,067	3,453	2,802	2,124	-22%	-43%

Germany – Gasoline Vehicle Fleet Life Cycle GHG Emissions



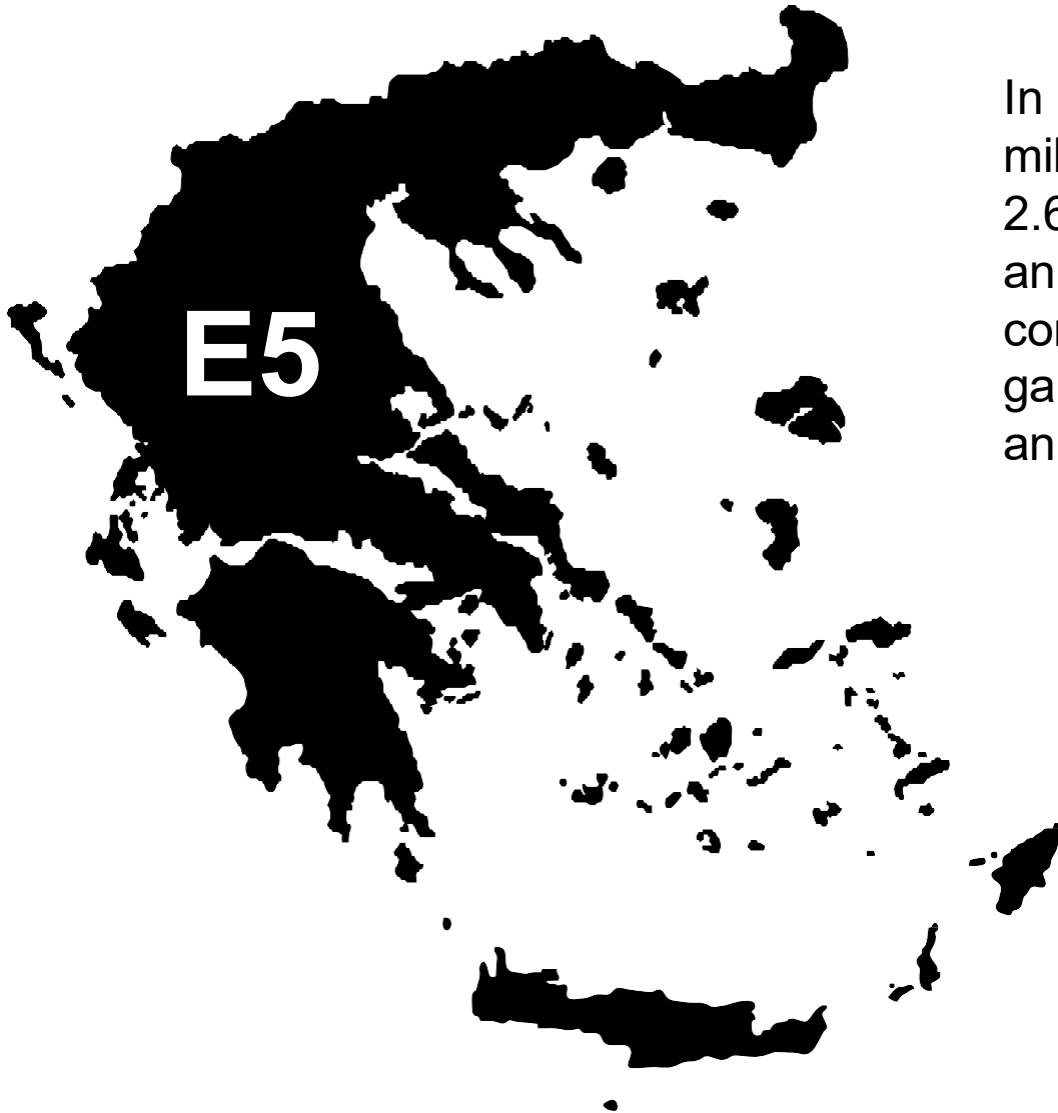
Country Emissions 2024 (MMT/año)	717.1
Target @ 2035 (MMT/year)	403.6
Delta	313.4

%Reduction vs E0	E0	E10	E15	E20	E25	E30
GREET 2024	0.0%	11.2%	14.4%	16.9%	19.3%	22.6%
RED II/III	0.0%	7.9%	10.3%	12.2%	14.1%	16.4%

% Target@2035	E0	E10	E15	E20	E25	E30
GREET 2024	0.0%	3.2%	4.2%	4.9%	5.6%	6.5%
RED II/III	0.0%	3.6%	4.7%	5.6%	6.4%	7.5%

Source: Greet, RED II/III, IPCC, climateactiontracker.org, F90

Ethanol Gasoline Blending in Greece

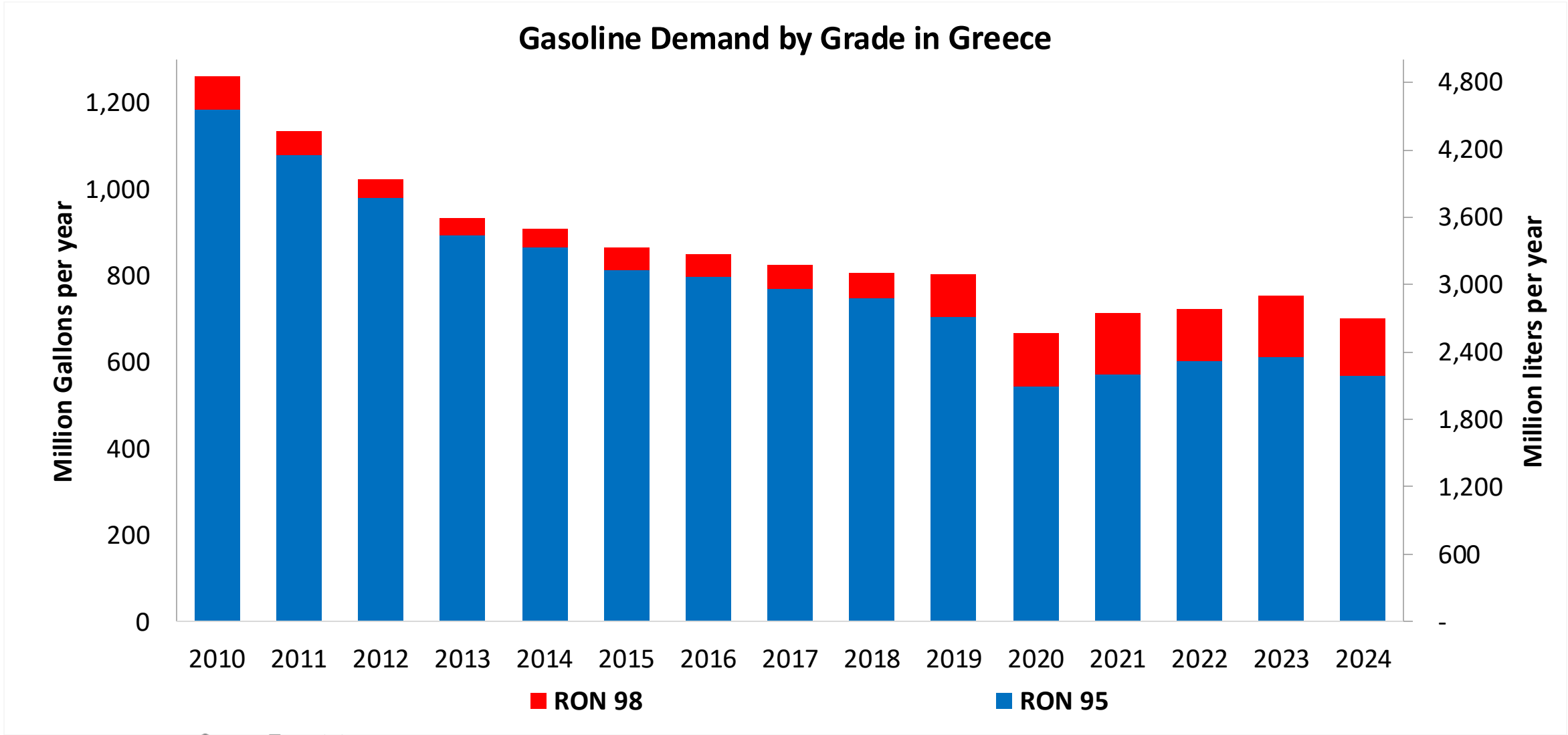


In 2024, gasoline consumption in Greece was 2,660 million liters (703 million gallons) and growth rate was -2.6%. Gasoline production and net imports were 2,811 and -151 million liters (743 and -40 million gallons) correspondingly. Greece complies with the European gasoline standards and offers two main grades: RON 95 and RON 98) with an average octane of 95.5 RON.

Ethanol consumption in Greece is determined by 3.3% bioethanol content mandate (by energy). Demand was 134 million liters (35 million gallons) representing 5.0% of gasoline demand. Ethanol is supplied by imports.

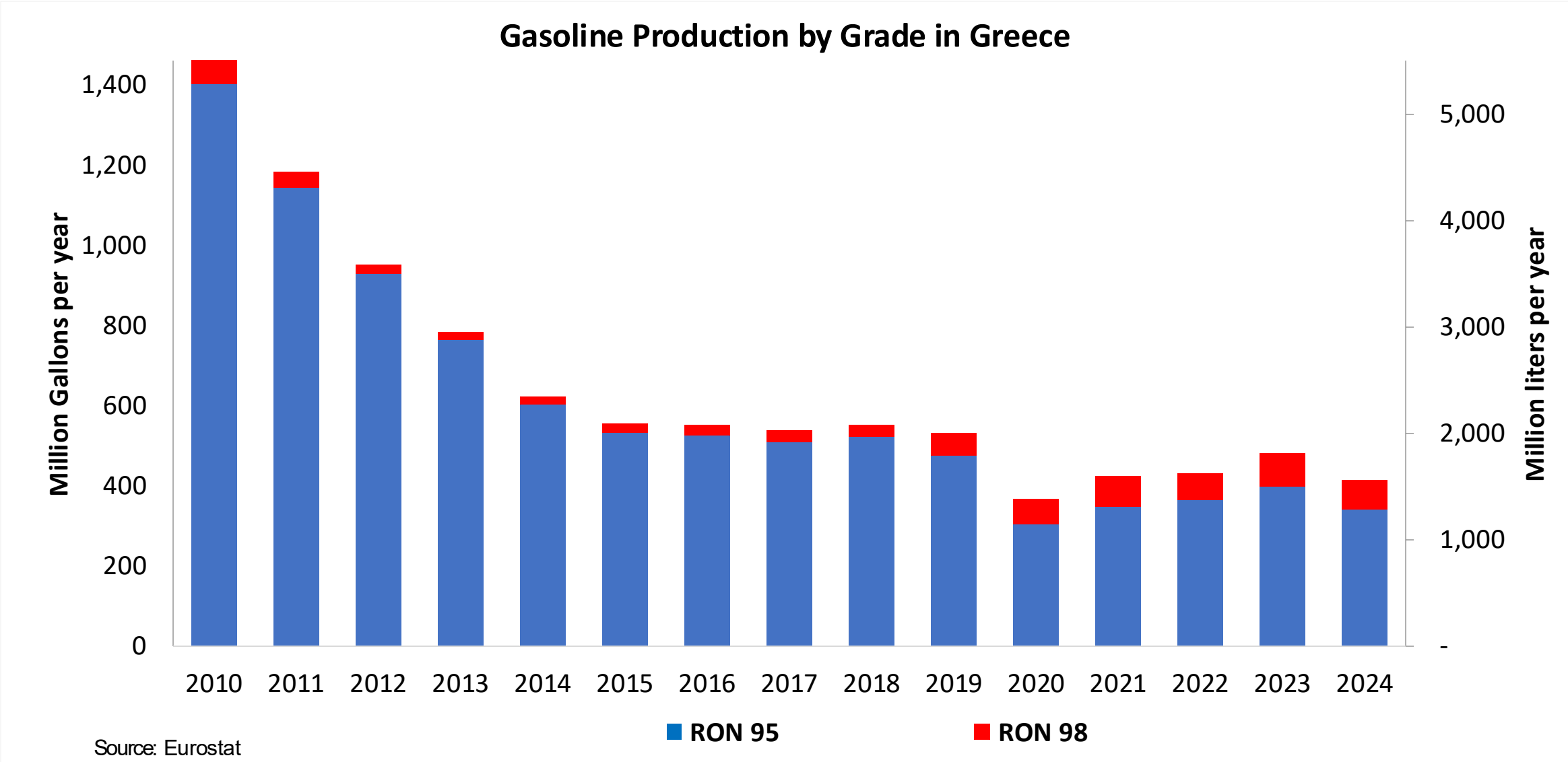
Source: Eurostat, European Commission, USDA

Gasoline Demand in Greece

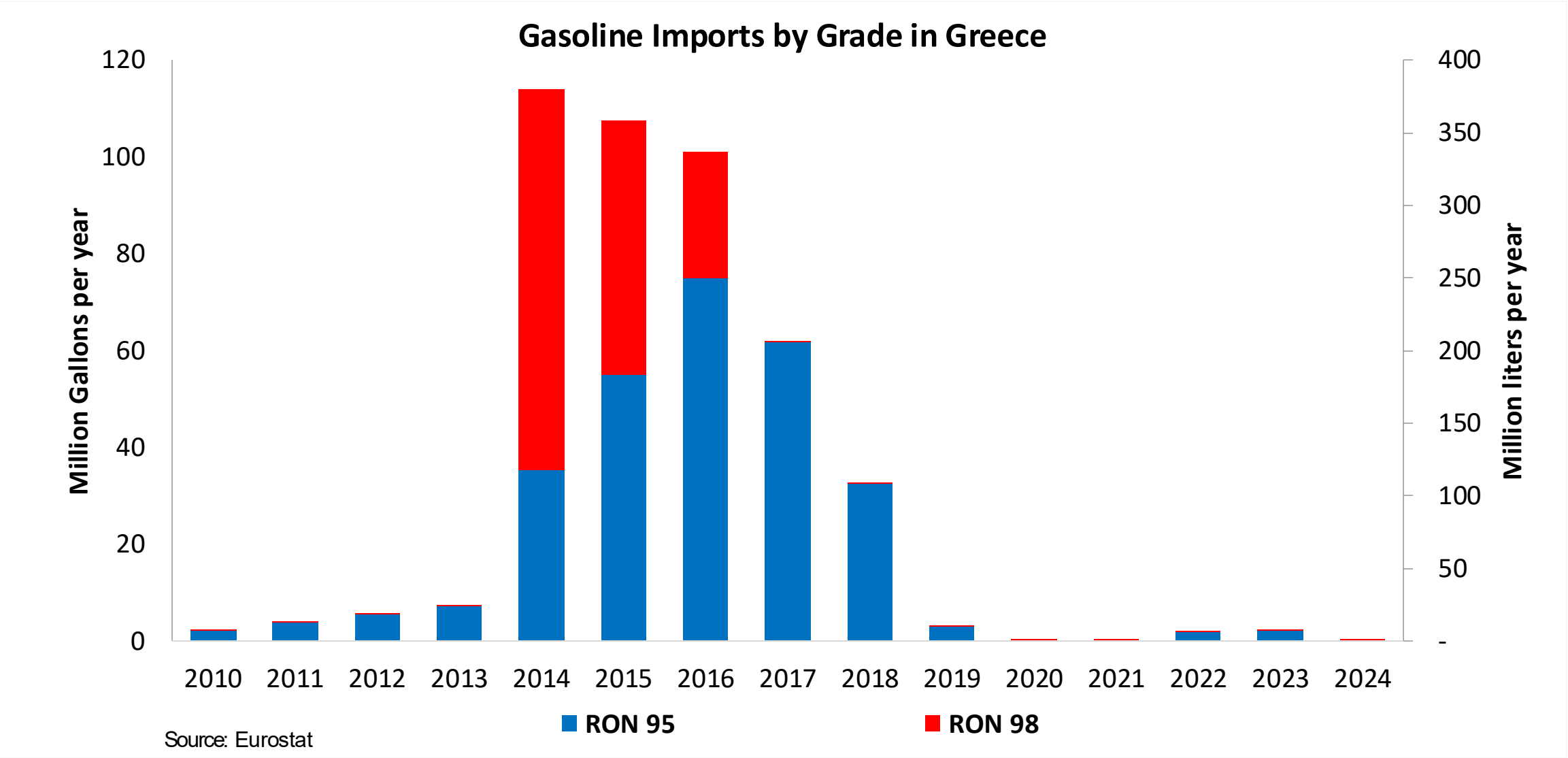


Source: Eurostat

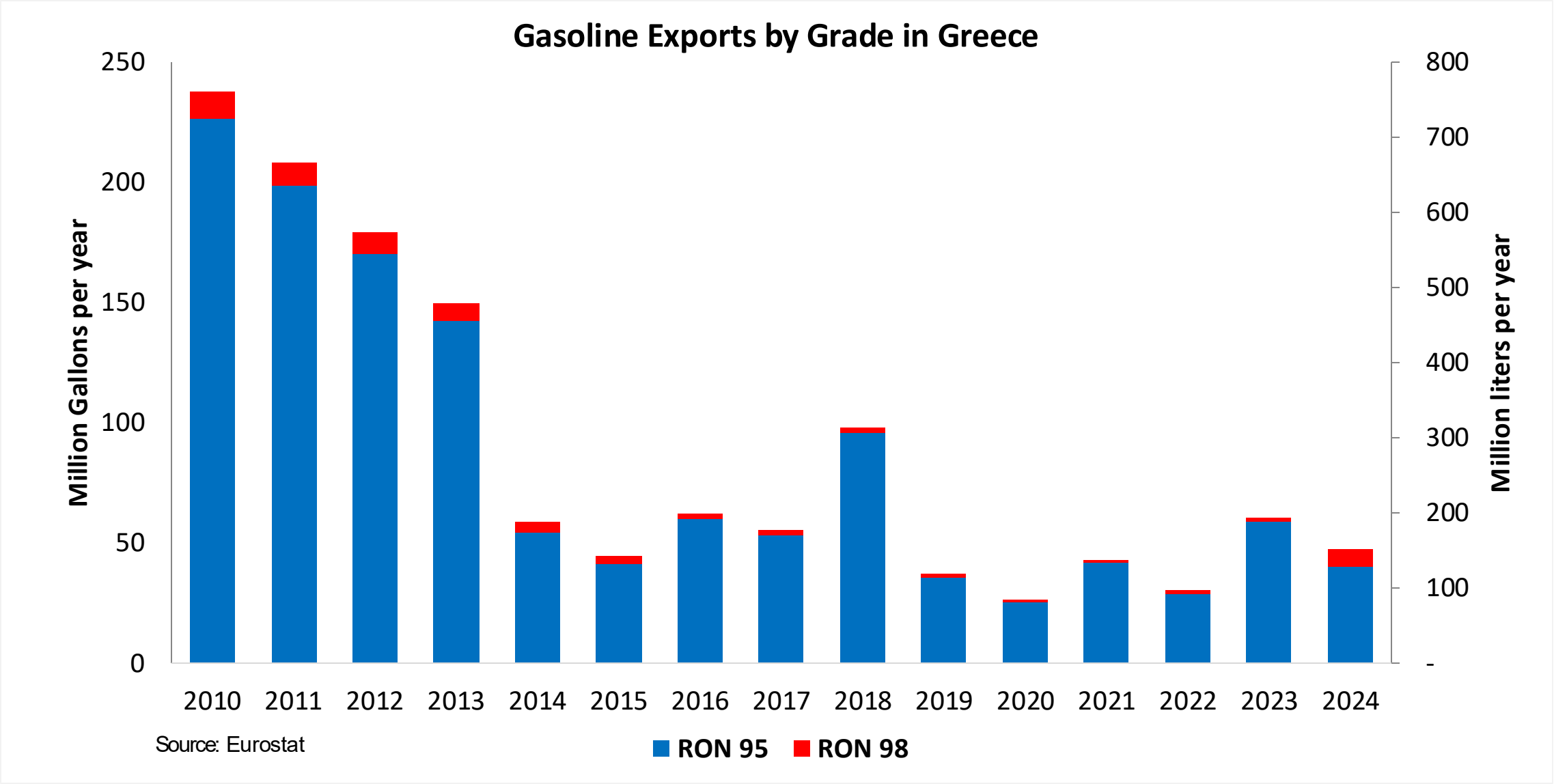
Gasoline Production in Greece



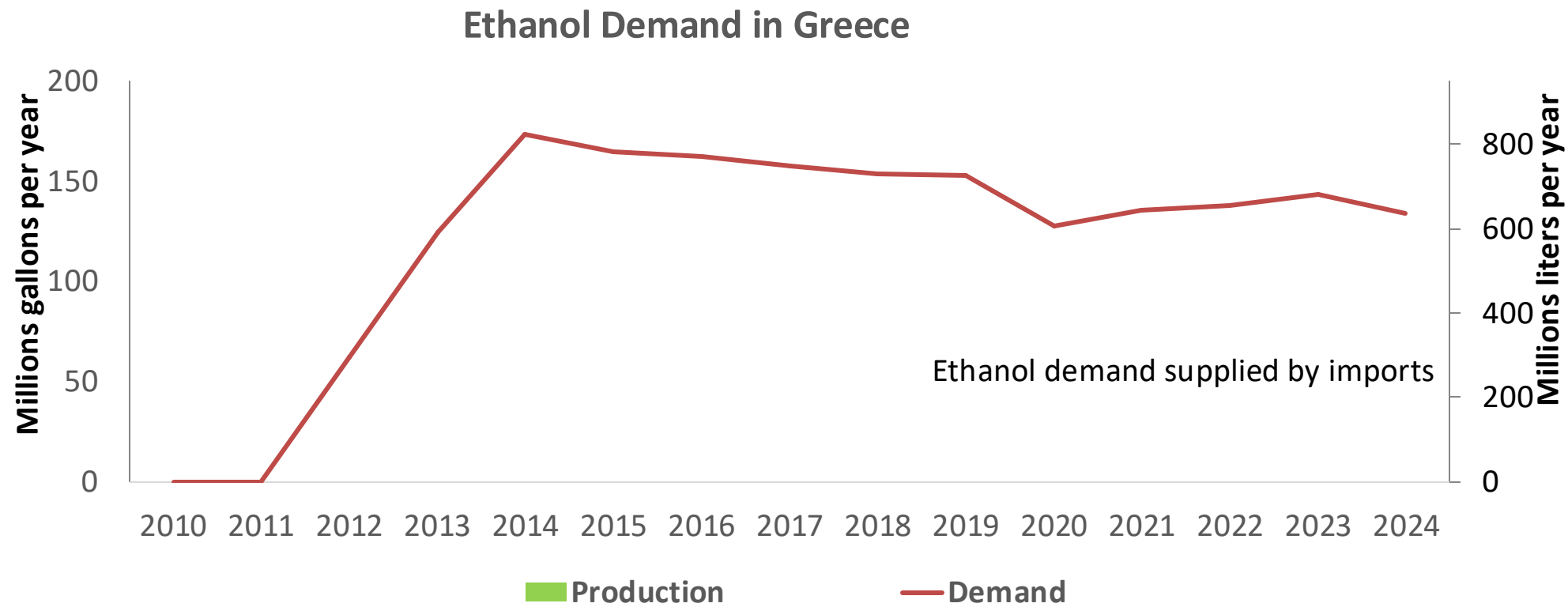
Gasoline Imports in Greece



Gasoline Exports in Greece



Ethanol Balance in Greece



Source: Eurostat, USDA – Fuel Mandates in EU

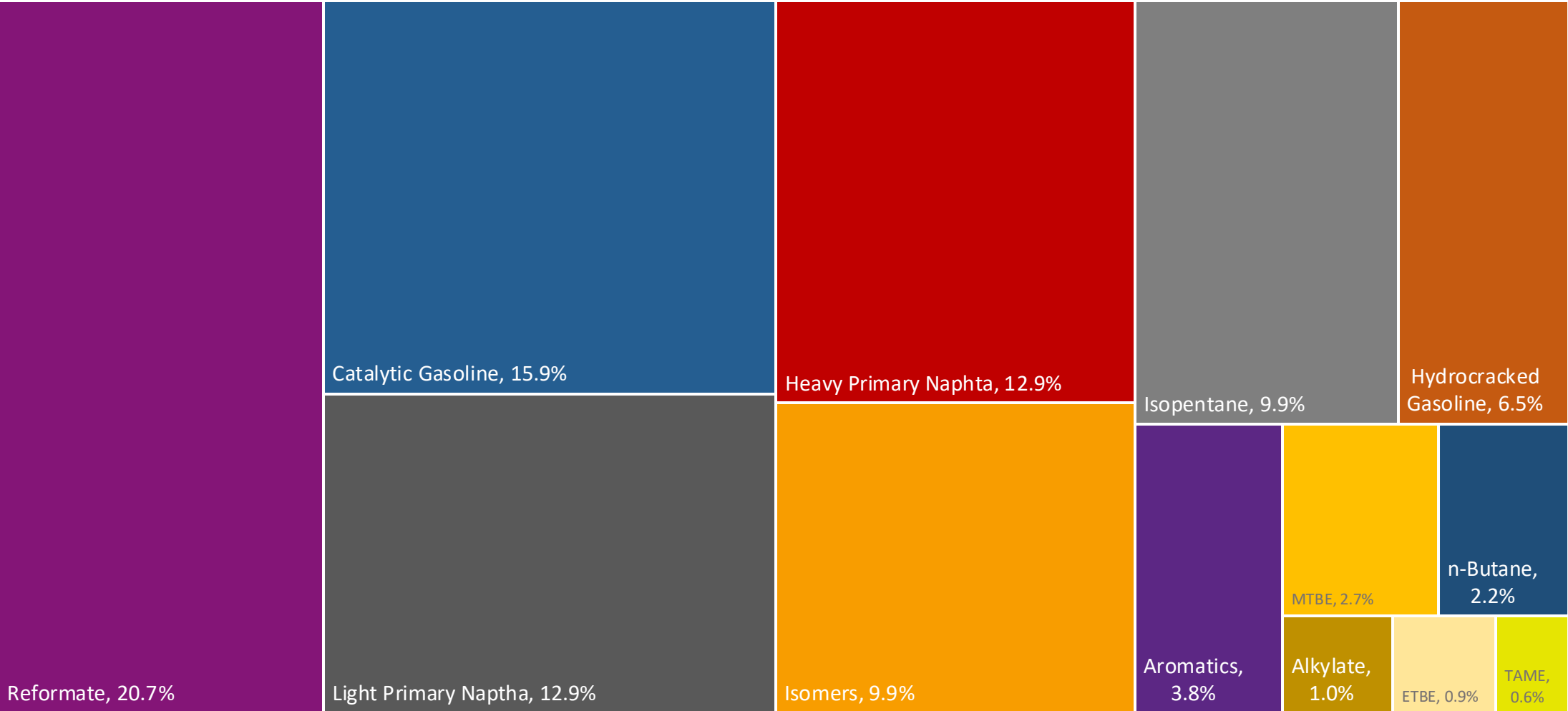
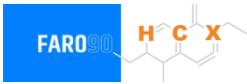
Gasoline Quality in Greece

Name	EN 228:2025 (Euro 6); Country Level Fuel Ordinances							
Implementation Date	2025							
Applicability	All EU Contries + UK							
Grade	RON 95 E5		RON 95 E10		RON 98 E5		RON 98 E10	
	Min	Max	Min	Max	Min	Max	Min	Max
RON	95.0		95.0		98.0		98.0	
MON	> 85	> 88	> 85	> 88	> 85	> 88	> 85	> 88
AKI								
Benzene (%vol)	< 1							
Aromatics (% vol)	< 35							
Olefins (%vol)	< 18							
Lead (g/L)	< 0.005							
Manganese (mg/L)	< 2,0							
Sulfur (ppm)	10.0							
Oxygen (%w)*	< 2,7	< 3,7	< 2,7	< 3,7	< 2,7	< 3,7	< 2,7	< 3,7
Ethanol (%vol)	< 5	< 10	< 5	< 10	< 5	< 10	< 5	< 10
PVR 37.8°C (psi)	<> 45 - 100 kPa in 6 Volatility Classes							
Overall Biofuels (%cal)	10.0							
Advanced Biofuels (%cal)	-							
Bioethanol in Gasoline Sales (%cal)	3.3							
GHG Emission Reduction (%)	-							

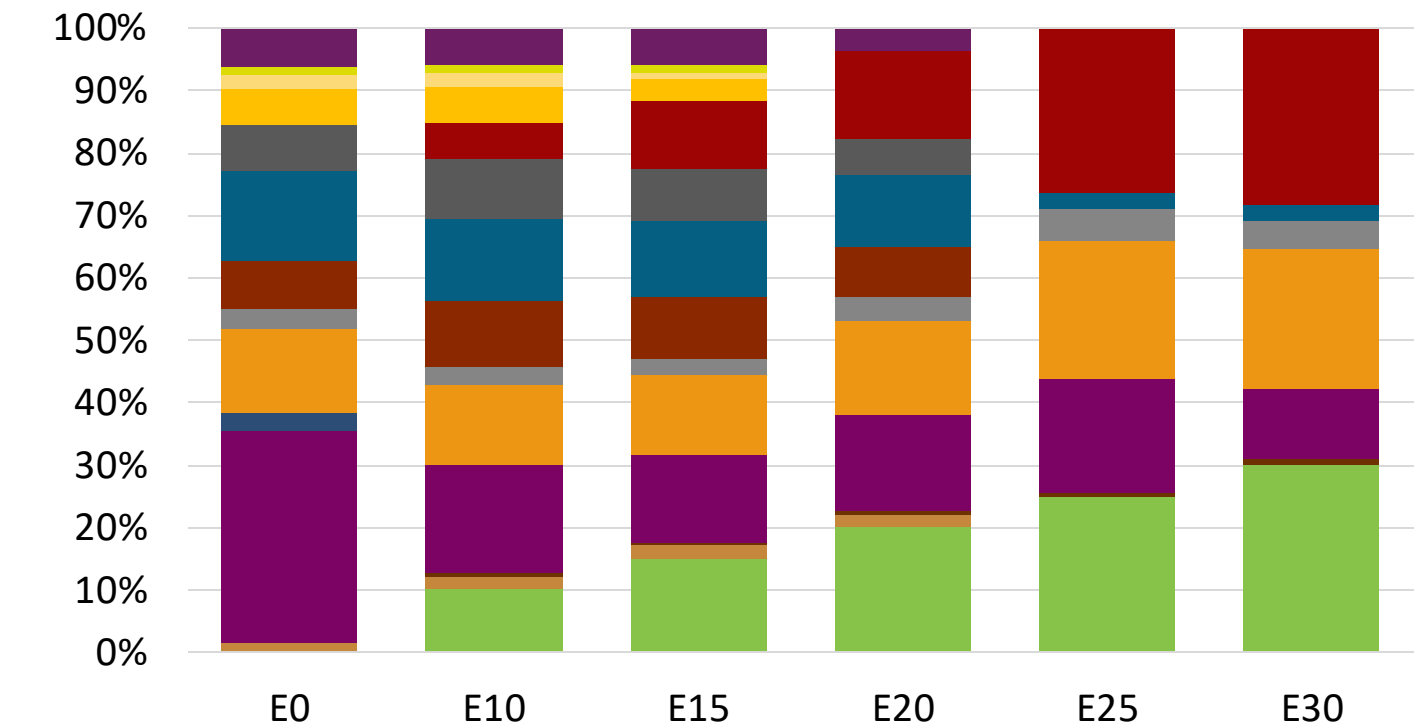
* Methanol 3%, Iso-propyl alcohol 12%, Ter-butyl alcohol 15%, Iso-butyl alcohol 15%, Ethers +5 Carbons 22%, Other Oxygenates 15%

Source: European Commision

Greece Available Blending Components



Greece – Gasoline 95 – Constant Octane

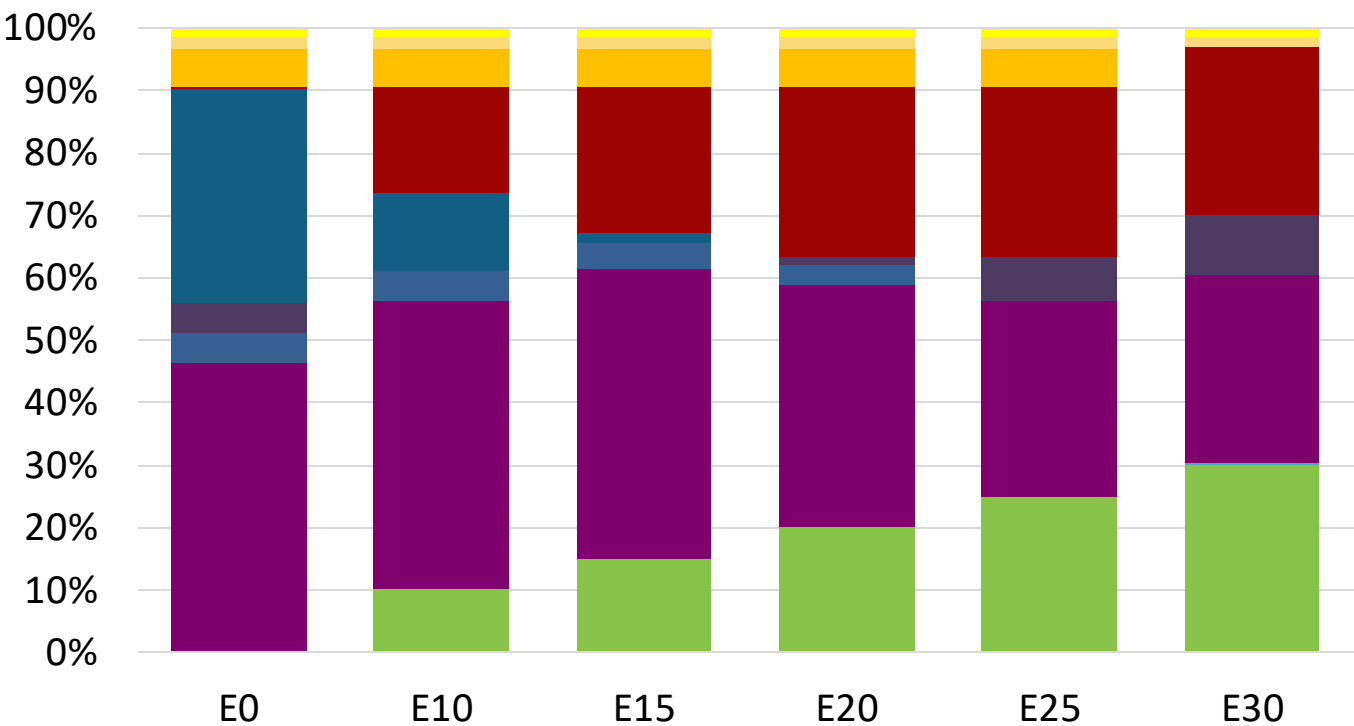


Average CIF 2024
Prices

Octane (RON)	95.0	95.0	95.0	95.0	95.0	96.0
Price (US\$/gal)	\$ 2.258	\$ 2.135	\$ 2.069	\$ 2.002	\$ 1.881	\$ 1.845

Source: Faro90

Greece - Gasoline 98 - Constant Octane

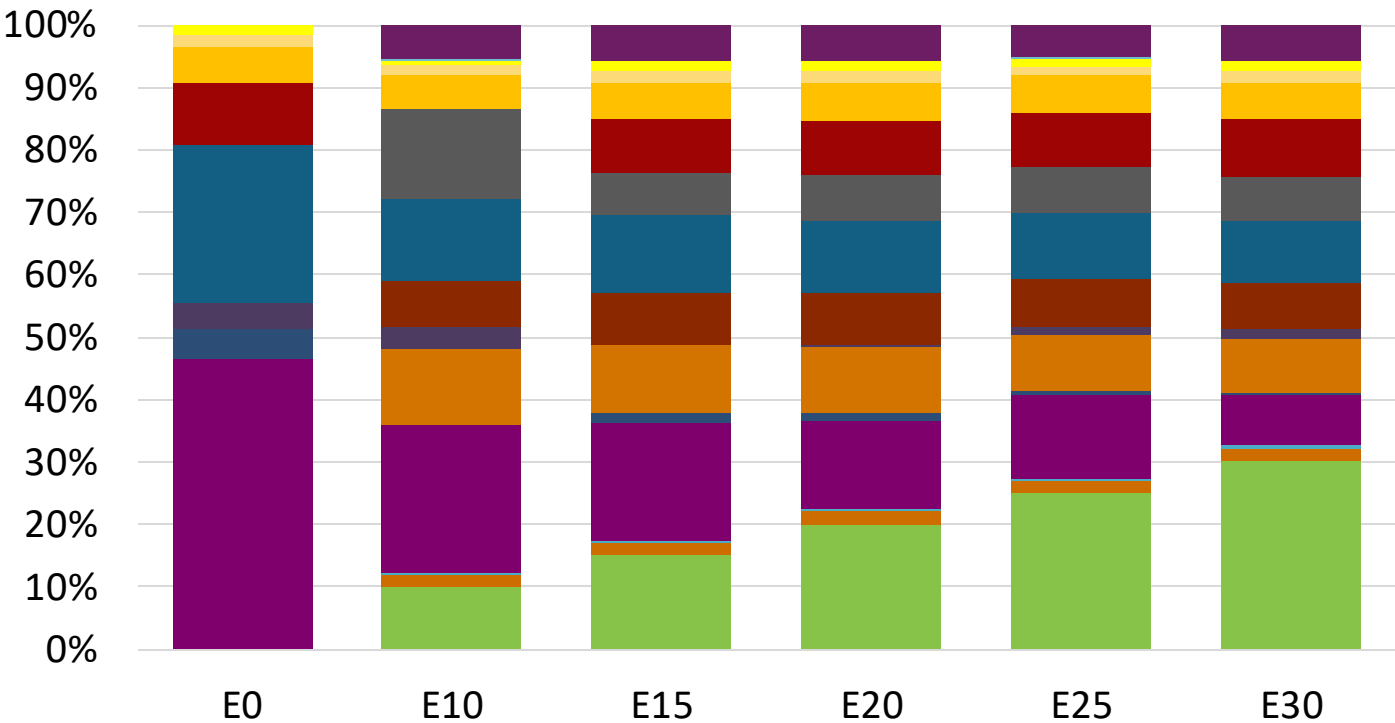
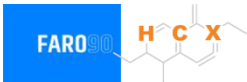


Average CIF 2024
Prices

Octane (RON)	98.0	98.0	98.0	98.0	98.0	98.0
Price (US\$/gal)	\$ 2.330	\$ 2.204	\$ 2.132	\$ 2.057	\$ 1.980	\$ 1.924

Source: Faro90

Greece – Gasoline 95 – Increased Octane

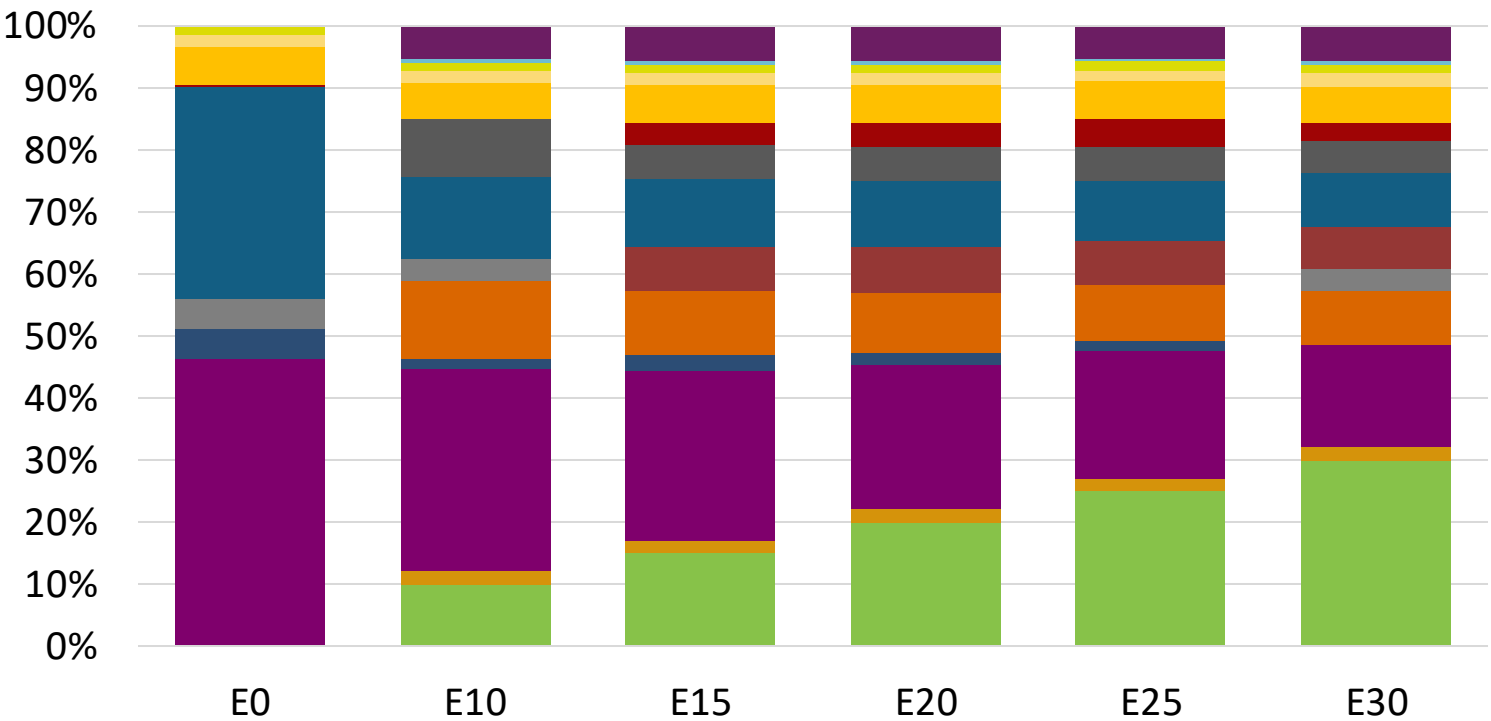
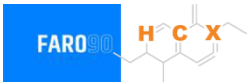


Average CIF 2024
Prices

Octane (RON)	95.0	96.2	97.9	99.4	100.8	102.2
Price (US\$/gal)	\$ 2.196	\$ 2.196	\$ 2.196	\$ 2.196	\$ 2.196	\$ 2.196

Source: Faro90

Greece – Gasoline 98 – Increased Octane

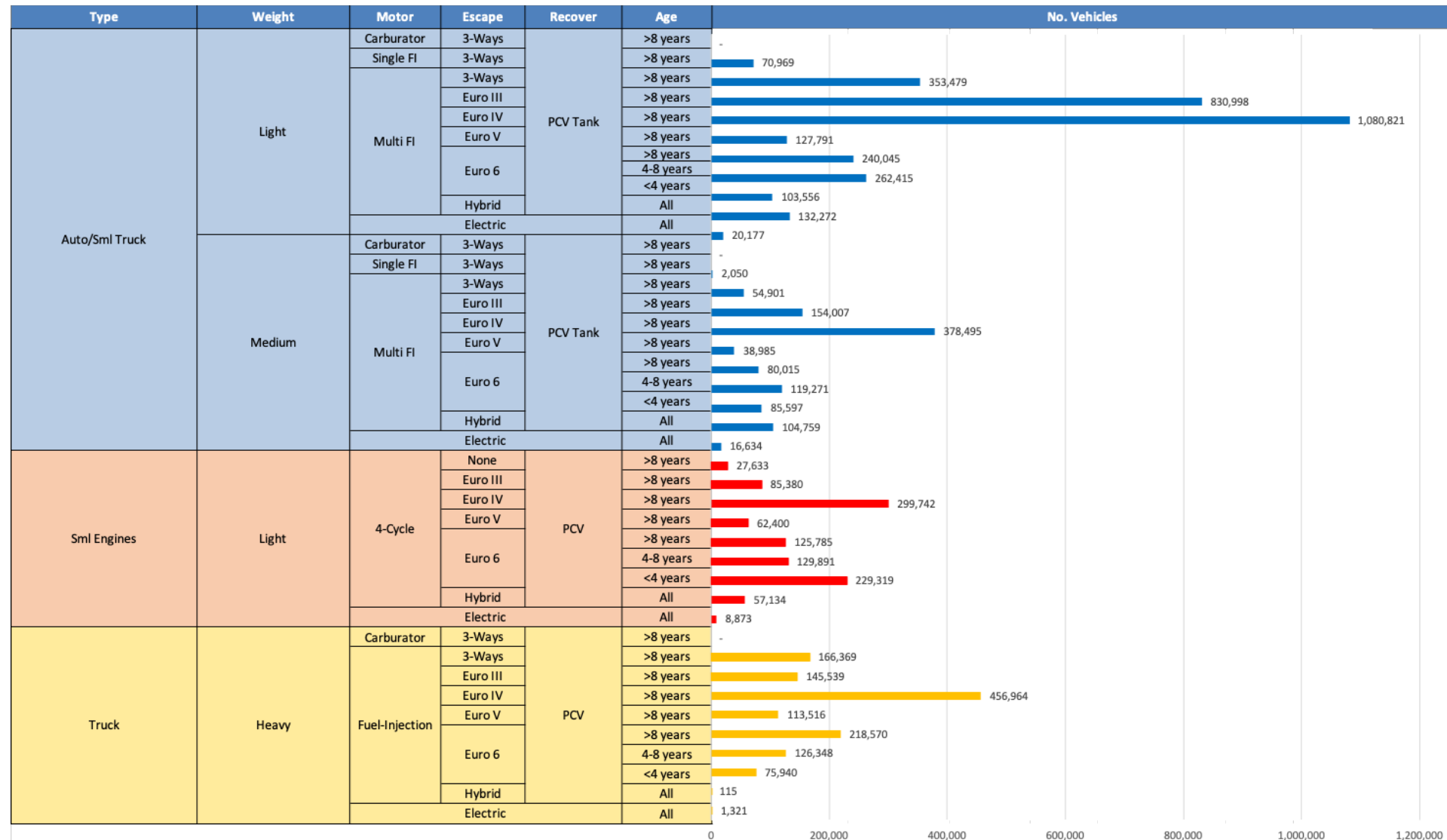


Average CIF 2024 Prices

Octane (RON)		98.0	99.4	100.7	102.0	103.3	104.5
Price (US\$/gal)	\$	2.330	\$ 2.330	\$ 2.330	\$ 2.330	\$ 2.330	\$ 2.330

Source: Faro90

Gasoline Vehicle Fleet - Greece

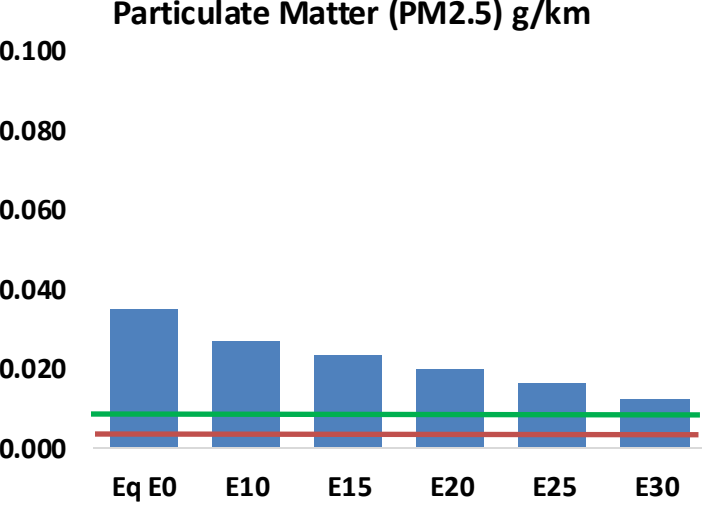
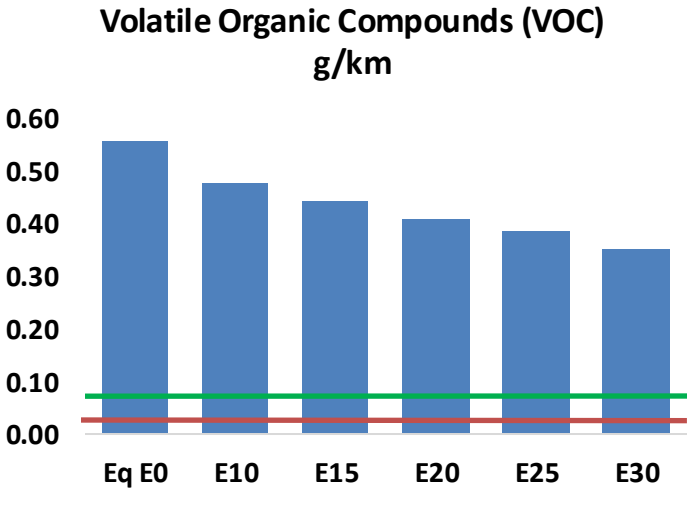
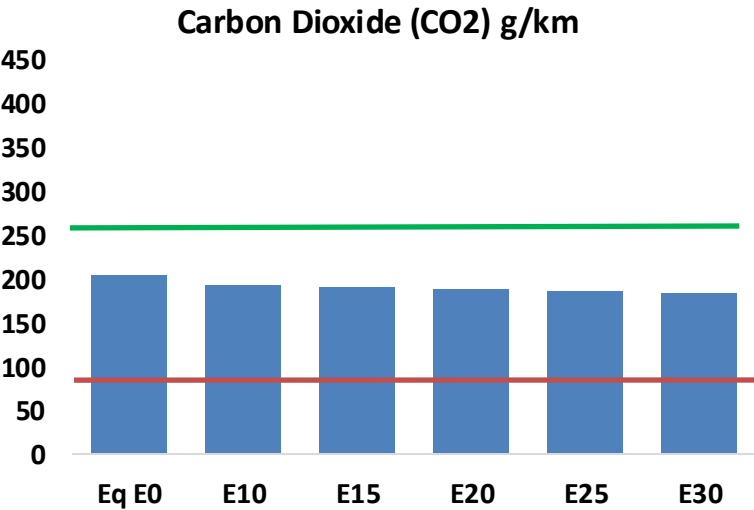
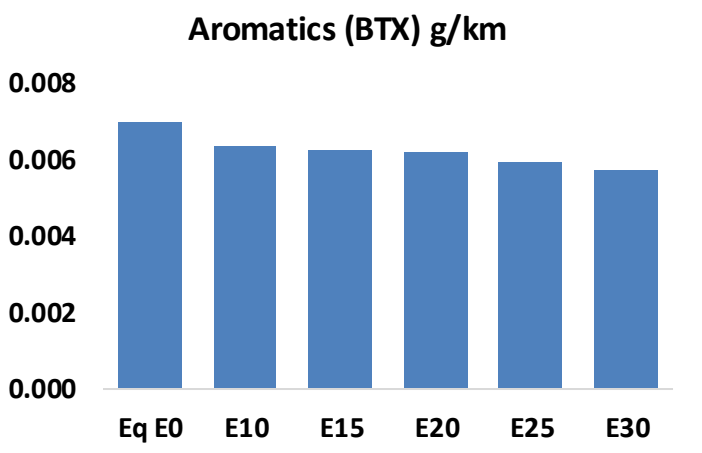
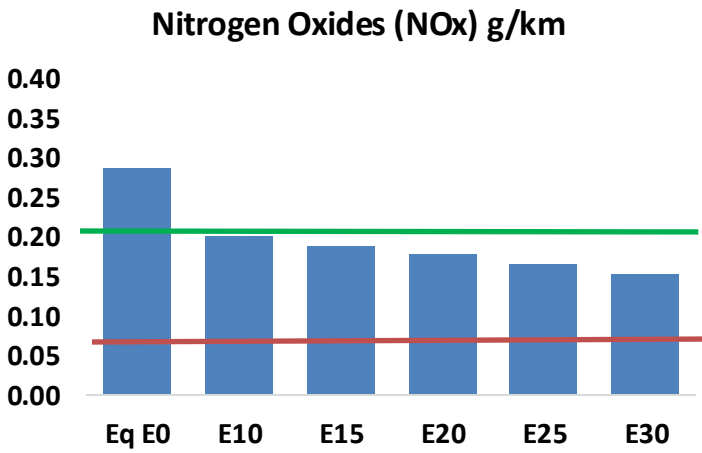
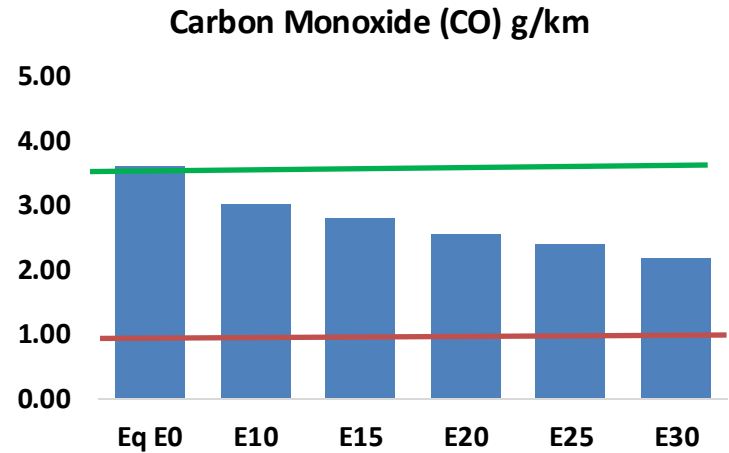
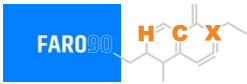


Vehicle Fleet: 6,588,079
Gasoline Vehicle Fleet: 4,489,264

Source: Eurostat, ACEA

Greece – Vehicular Emissions

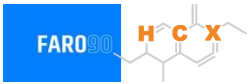
TIER III BIN 125 United States
Euro 6 Standard



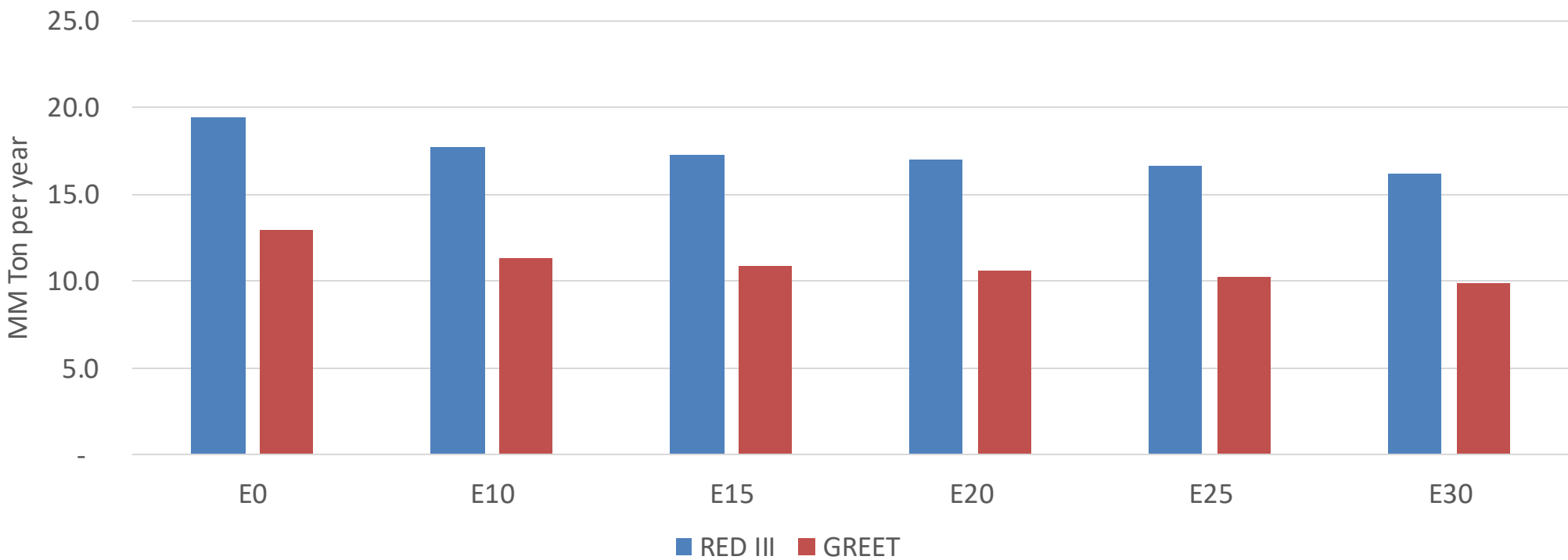
Greece – Vehicular Emissions

Vehicular Fleet Emissions (kg/day)	Eq E0	E5	E10	E15	E20	E25	E30	Reduction (%)	
								E10	E20
CO	385,776	355,555	325,335	300,460	275,378	256,076	234,621	-16%	-29%
CO2	21,880,668	21,318,022	20,786,634	20,368,713	20,161,910	20,045,323	19,675,830	-5%	-8%
VOC	59,679	55,319	50,959	47,431	43,901	41,193	37,557	-15%	-26%
NOx	30,711	23,034	21,498	20,262	19,109	17,837	16,492	-30%	-38%
SOx	260	240	221	204	188	171	153	-15%	-28%
NH3	7,990	7,911	7,832	7,869	7,958	8,020	8,072	-2%	0%
N2O	301	299	298	299	305	309	312	-1%	2%
CH4	12,854	12,854	12,854	13,047	13,330	13,574	13,806	0%	4%
Lead	-	-	-	-	-	-	-	0%	0%
Butadiene	391	376	360	351	342	336	326	-8%	-13%
Acetaldehyde	1,097	1,259	1,847	2,812	3,831	4,378	5,173	68%	249%
Formaldehyde	3,922	3,813	4,445	5,220	5,469	6,050	6,586	13%	39%
Benzene	749	719	682	671	665	636	616	-9%	-11%
PM2.5	1,379	1,225	1,070	929	789	640	485	-22%	-43%
PM10	2,338	2,076	1,814	1,575	1,337	1,085	822	-22%	-43%

Greece – Gasoline Vehicle Fleet Life Cycle GHG Emissions



Life Cycle GHG Emission - Gasoline Vehicles



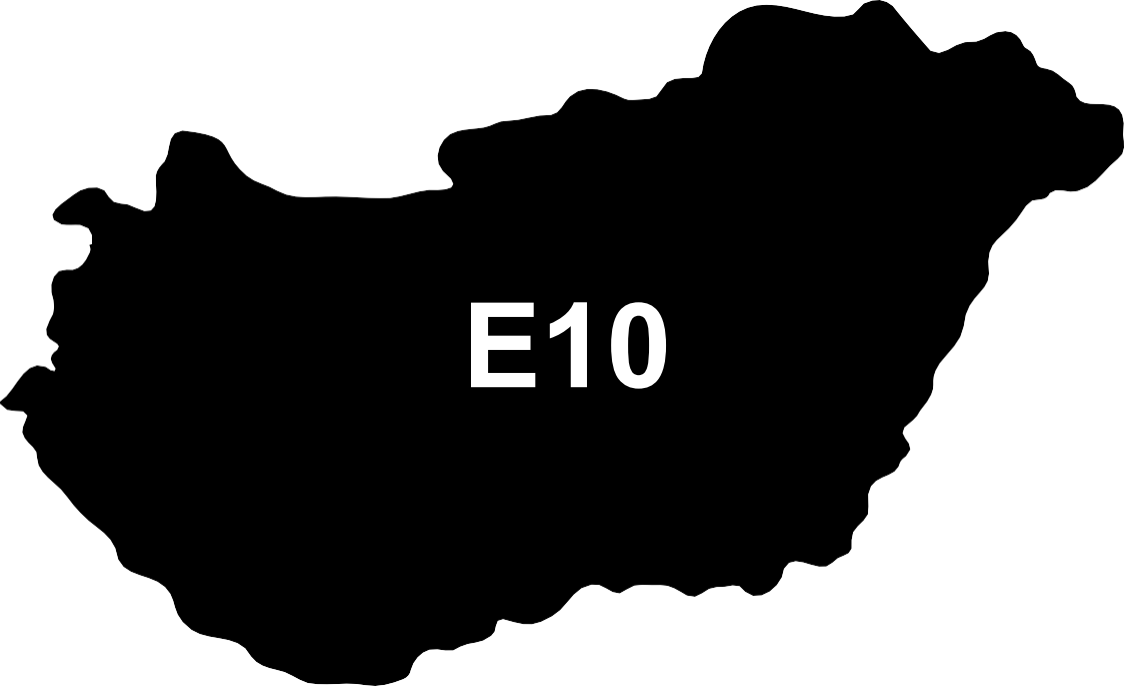
Country Emissions 2024 (MMT/año)	74.3
Target @ 2035 (MMT/year)	41.8
Delta	32.5

%Reduction vs E0	E0	E10	E15	E20	E25	E30
GREET 2024	0.0%	12.6%	15.7%	18.1%	20.5%	23.7%
RED II/III	0.0%	8.8%	11.0%	12.8%	14.5%	16.7%

% Target@2035	E0	E10	E15	E20	E25	E30
GREET 2024	0.0%	5.0%	6.2%	7.2%	8.2%	9.4%
RED II/III	0.0%	5.2%	6.6%	7.7%	8.7%	10.0%

Source: Greet, RED II/III, IPCC, climateactiontracker.org, F90

Ethanol Gasoline Blending in Hungary



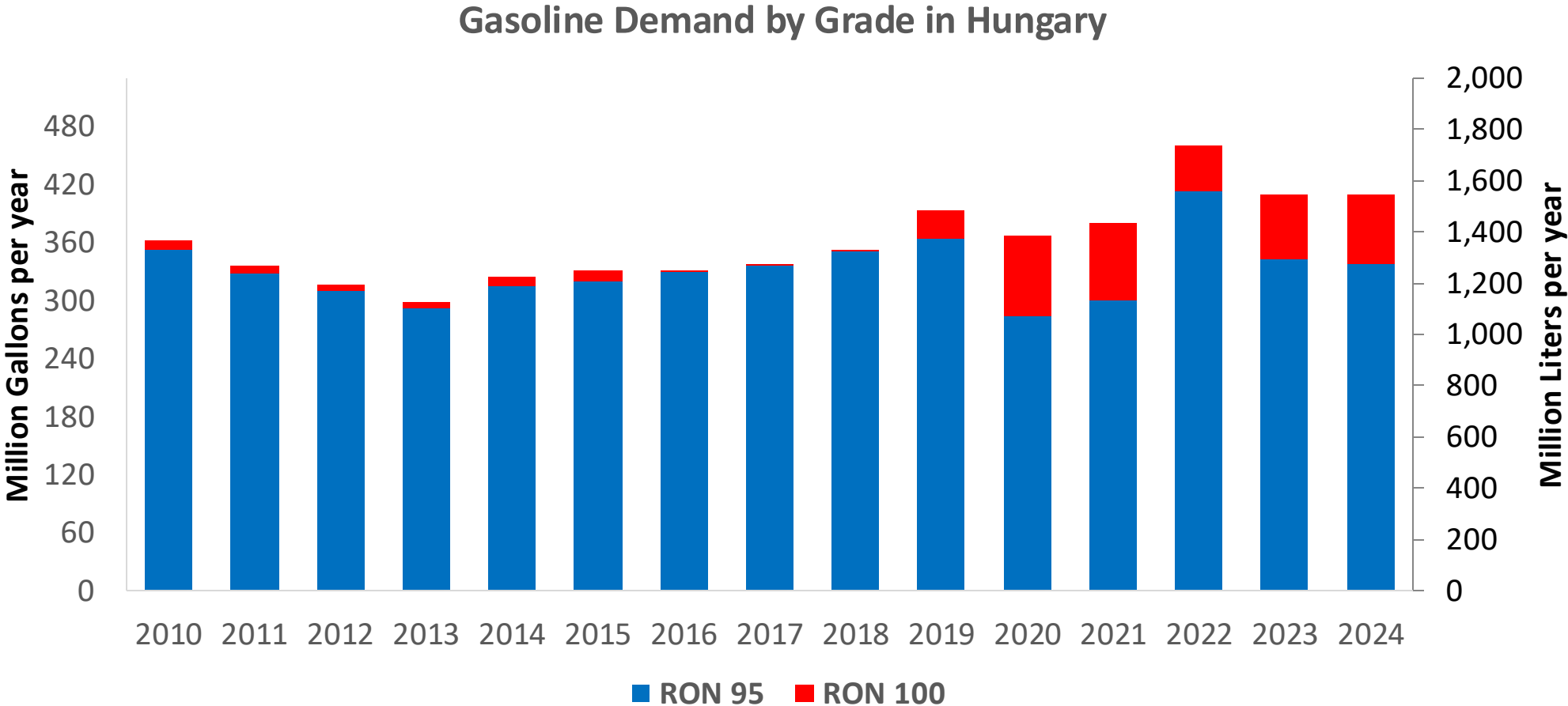
E10

In 2024, gasoline consumption in Hungary was 1,544 million liters (408 million gallons) and growth rate was 0.8%. Gasoline production and net imports were 1,701 and 79 million liters (449 and 21 million gallons) correspondingly. Hungary complies with the European gasoline standards and offers two main grades: RON 95 and RON 98) with an average octane of 95.9 RON.

In 2024, ethanol consumption in Hungary was 152 million liters (40 million gallons) representing 9.8% of gasoline demand. Ethanol production and Net Imports were 609 and -457 million liters (40 and -121 million gallons) correspondingly. Ethanol source is Corn 100%.

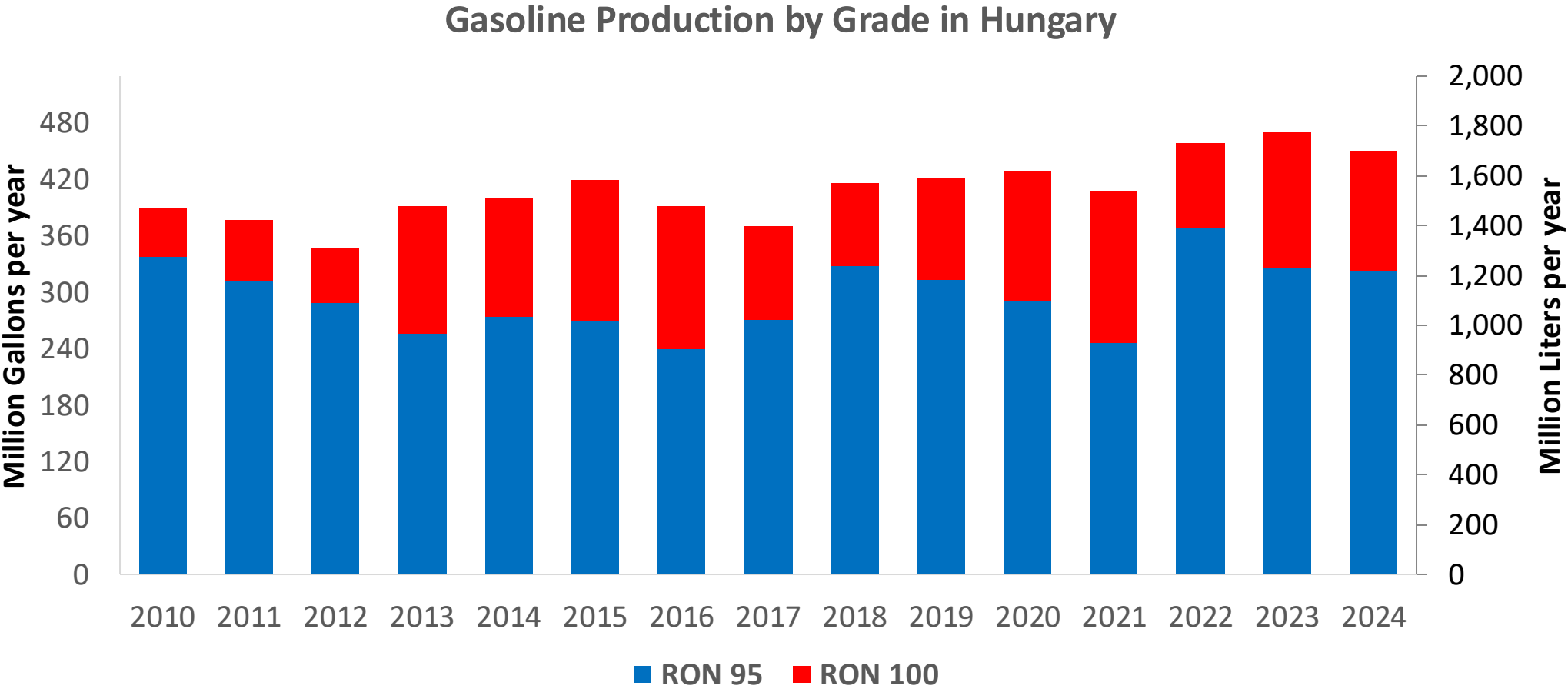
Source: Eurostat, European Commission

Gasoline Demand in Hungary

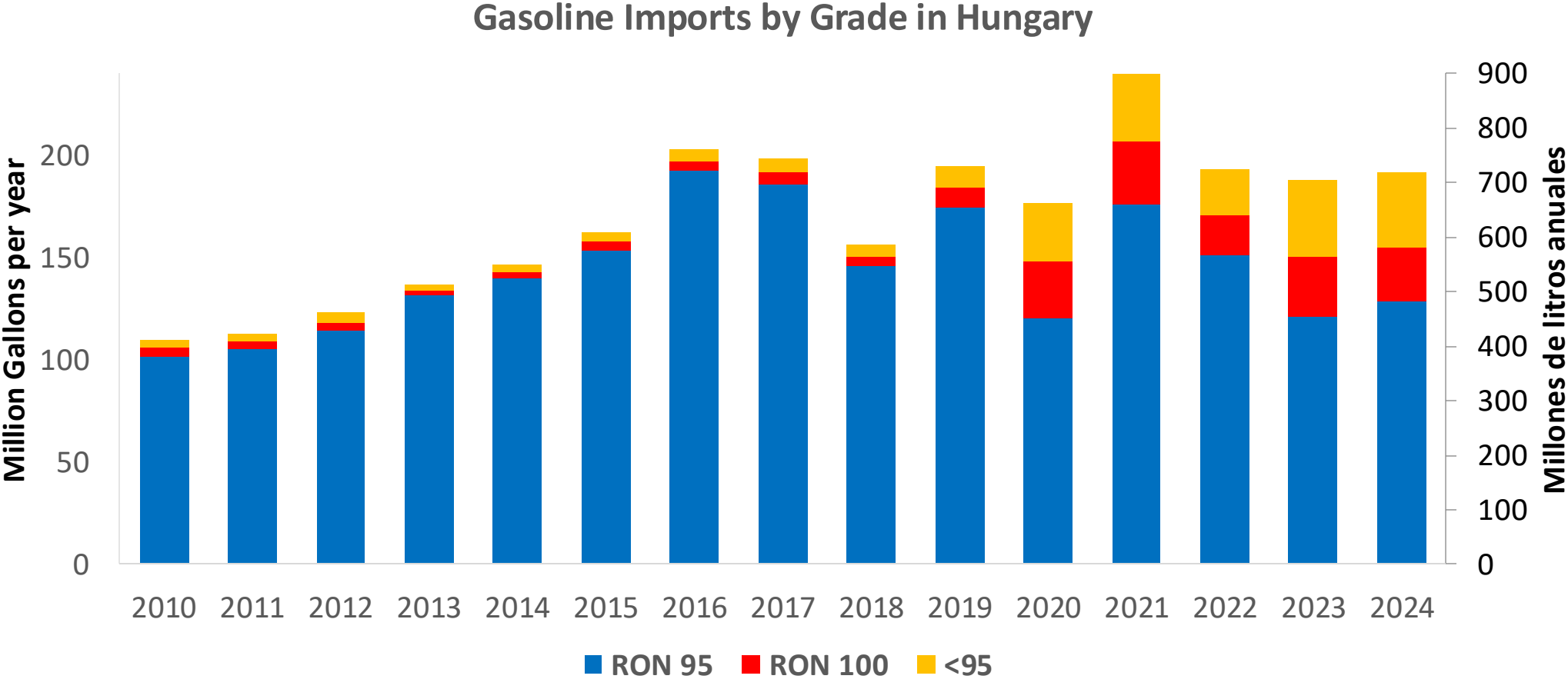


Source: Eurostat

Gasoline Production in Hungary

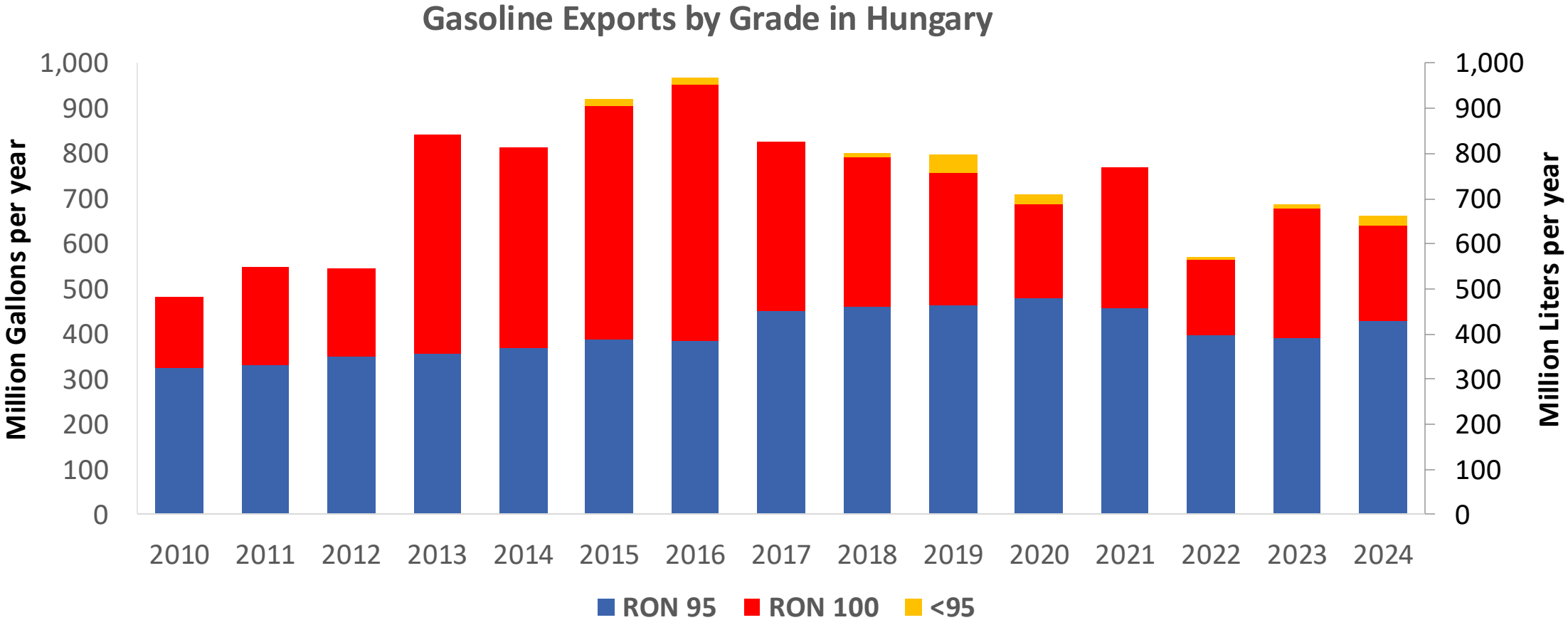


Source: Eurostat



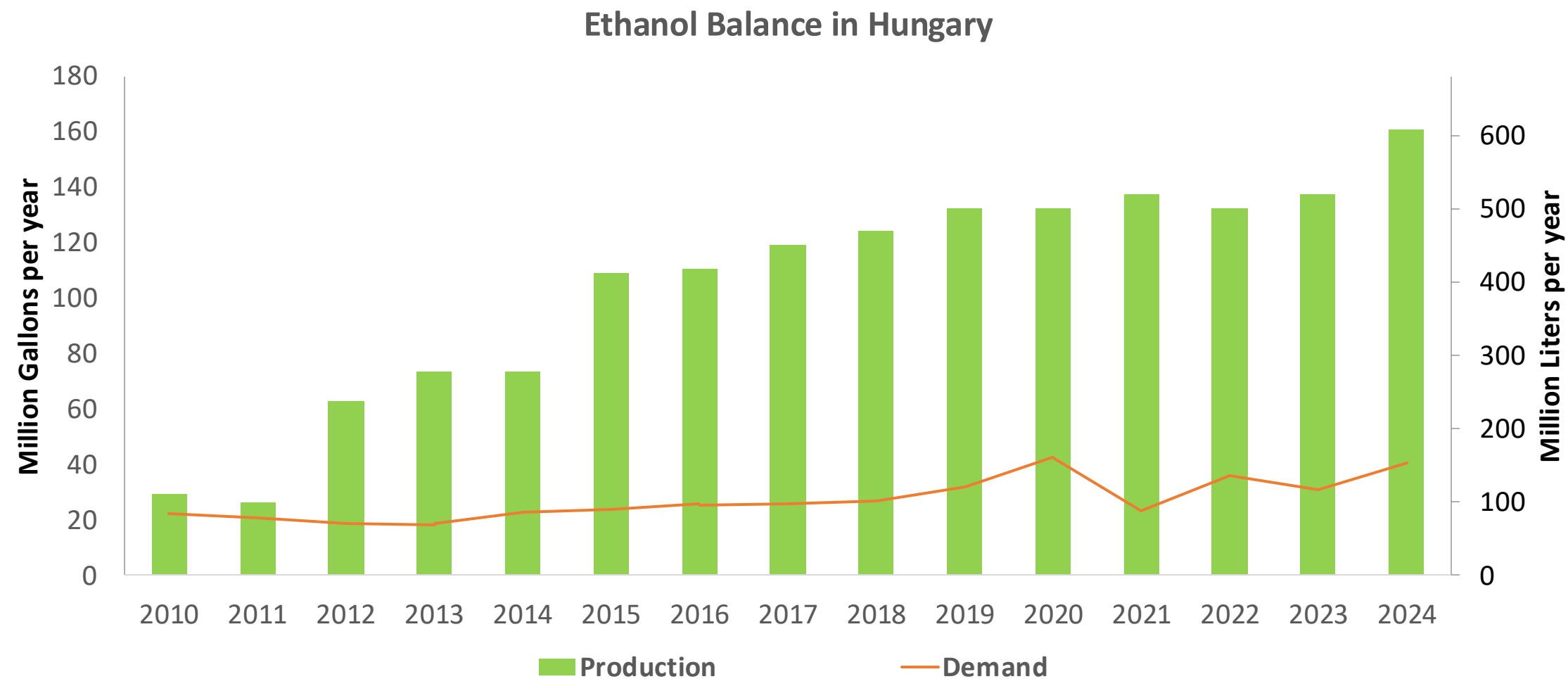
Source: Eurostat

Gasoline Exports in Hungary



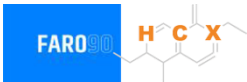
Source: Eurostat

Ethanol Balance in Hungary



Source: Eurostat

Gasoline Quality in Hungary

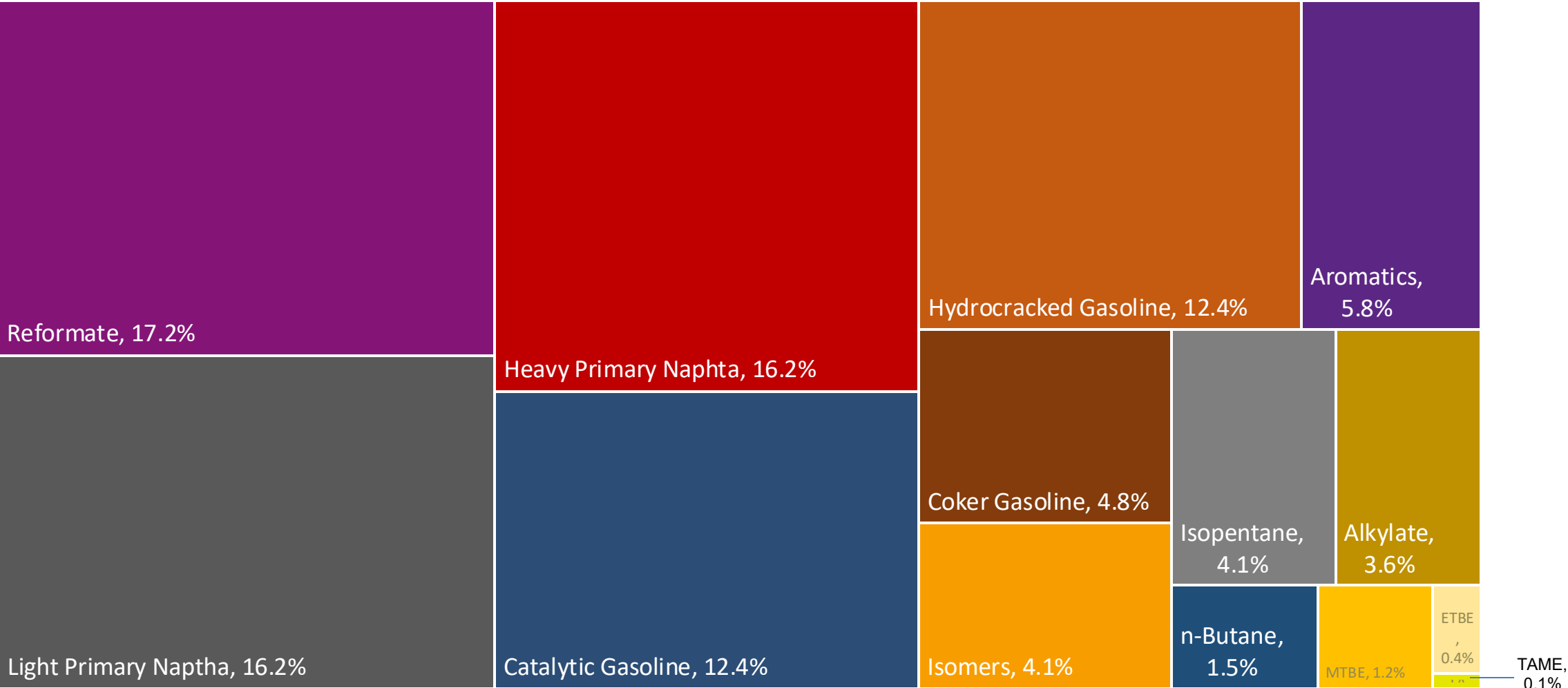
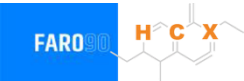


Name	EN 228:2025 (Euro 6); Country Level Fuel Ordinances							
Implementation Date	2025							
Applicability	All EU Contries + UK							
Grade	RON 95 E5		RON 95 E10		RON 98 E5		RON 98 E10	
	Min	Max	Min	Max	Min	Max	Min	Max
RON	95.0		95.0		98.0		98.0	
MON	> 85	> 88	> 85	> 88	> 85	> 88	> 85	> 88
AKI								
Benzene (%vol)	< 1							
Aromatics (% vol)	< 35							
Olefins (%vol)	< 18							
Lead (g/L)	< 0.005							
Manganese (mg/L)	< 2,0							
Sulfur (ppm)	10.0							
Oxygen (%w)*	< 2,7	< 3,7	< 2,7	< 3,7	< 2,7	< 3,7	< 2,7	< 3,7
Ethanol (%vol)	< 5	< 10	< 5	< 10	< 5	< 10	< 5	< 10
PVR 37.8°C (psi)	<> 45 - 100 kPa in 6 Volatility Classes							
Overall Biofuels (%cal)	-							
Advanced Biofuels (%cal)	0.5							
Bioethanol in Gasoline Sales (%cal)	6.1							
GHG Emission Reduction (%)	-							

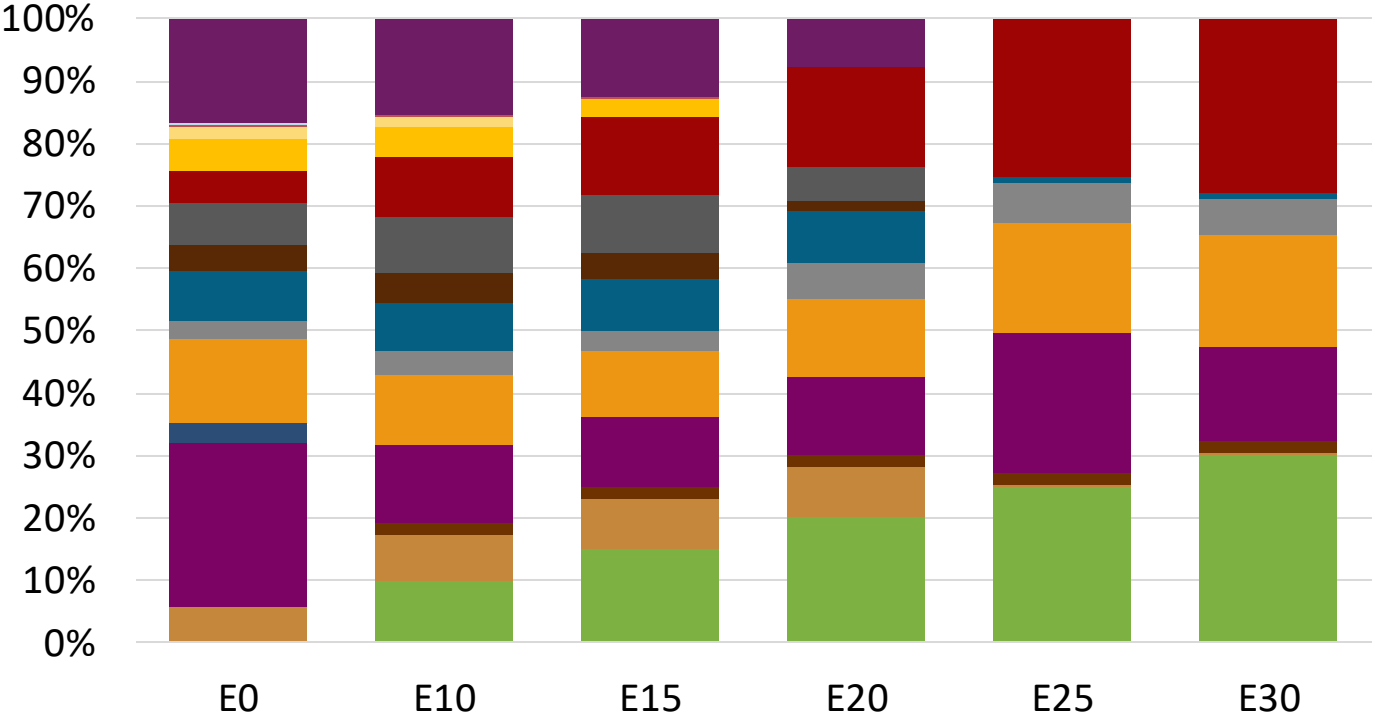
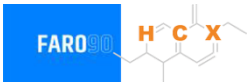
* Methanol 3%, Iso-propyl alcohol 12%, Ter-butyl alcohol 15%, Iso-butyl alcohol 15%, Ethers +5 Carbons 22%, Other Oxygenates 15%

Source: European Commision

Hungary Available Blending Components



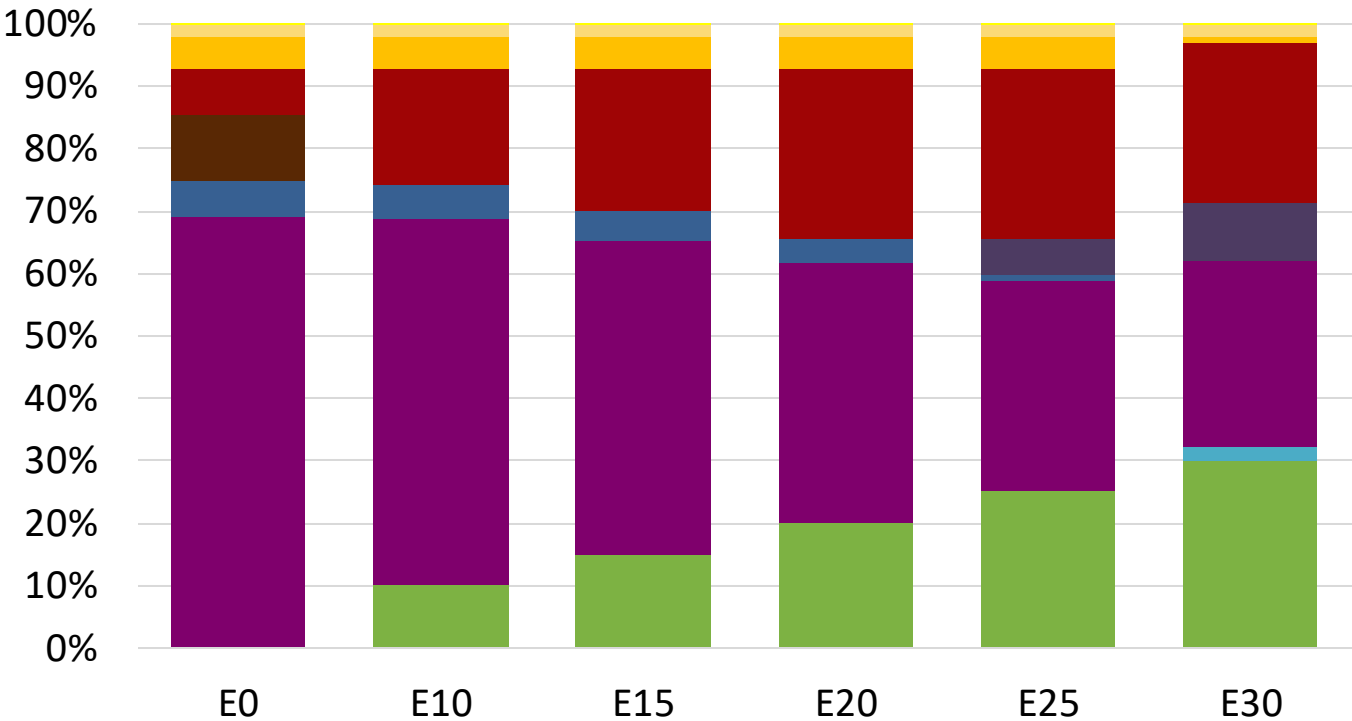
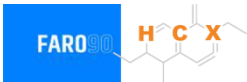
Hungary – Gasoline 95 – Constant Octane



Average CIF 2024
Prices

Source: Faro90

Hungary - Gasoline 98 - Constant Octane

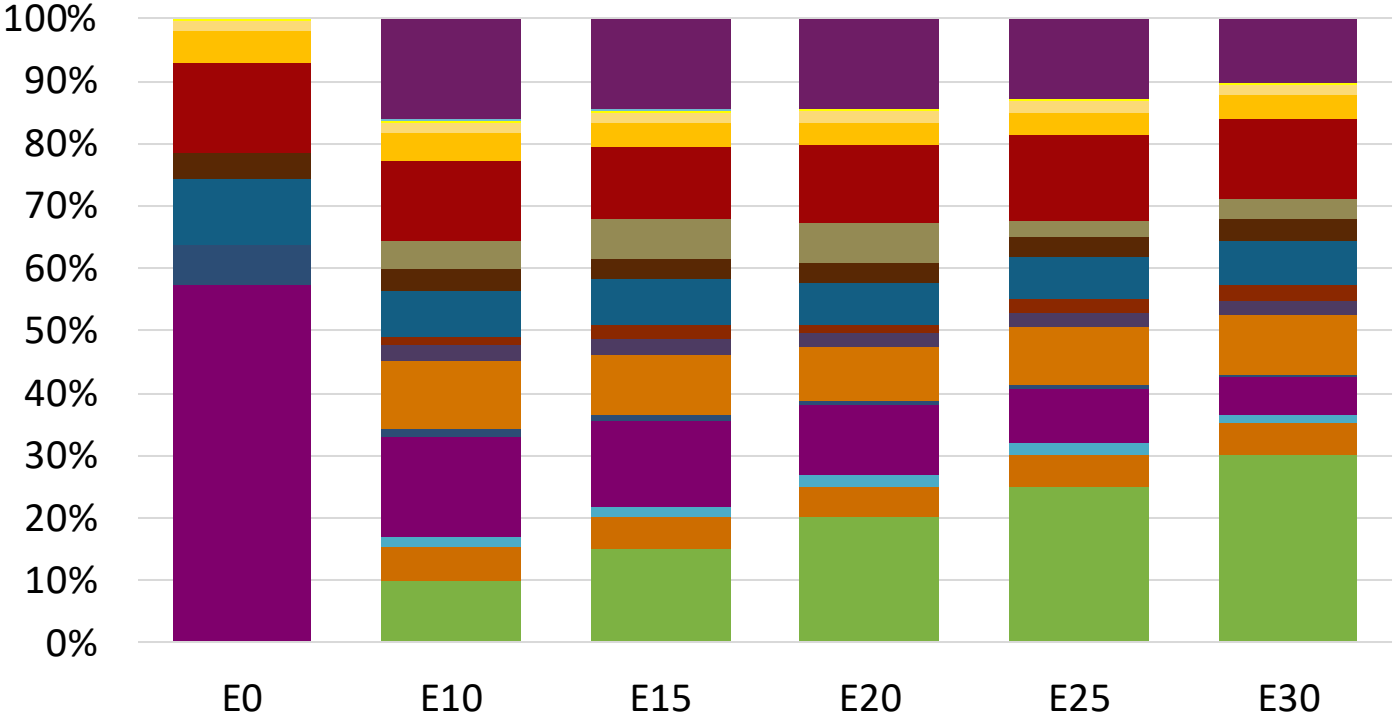
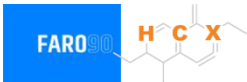


Average CIF 2024 Prices

Octane (RON)	98.0	98.0	98.0	98.0	98.0	98.0
Price (US\$/gal)	\$ 2.329	\$ 2.208	\$ 2.139	\$ 2.064	\$ 1.987	\$ 1.923

Source: Faro90

Hungary – Gasoline 95 – Increased Octane

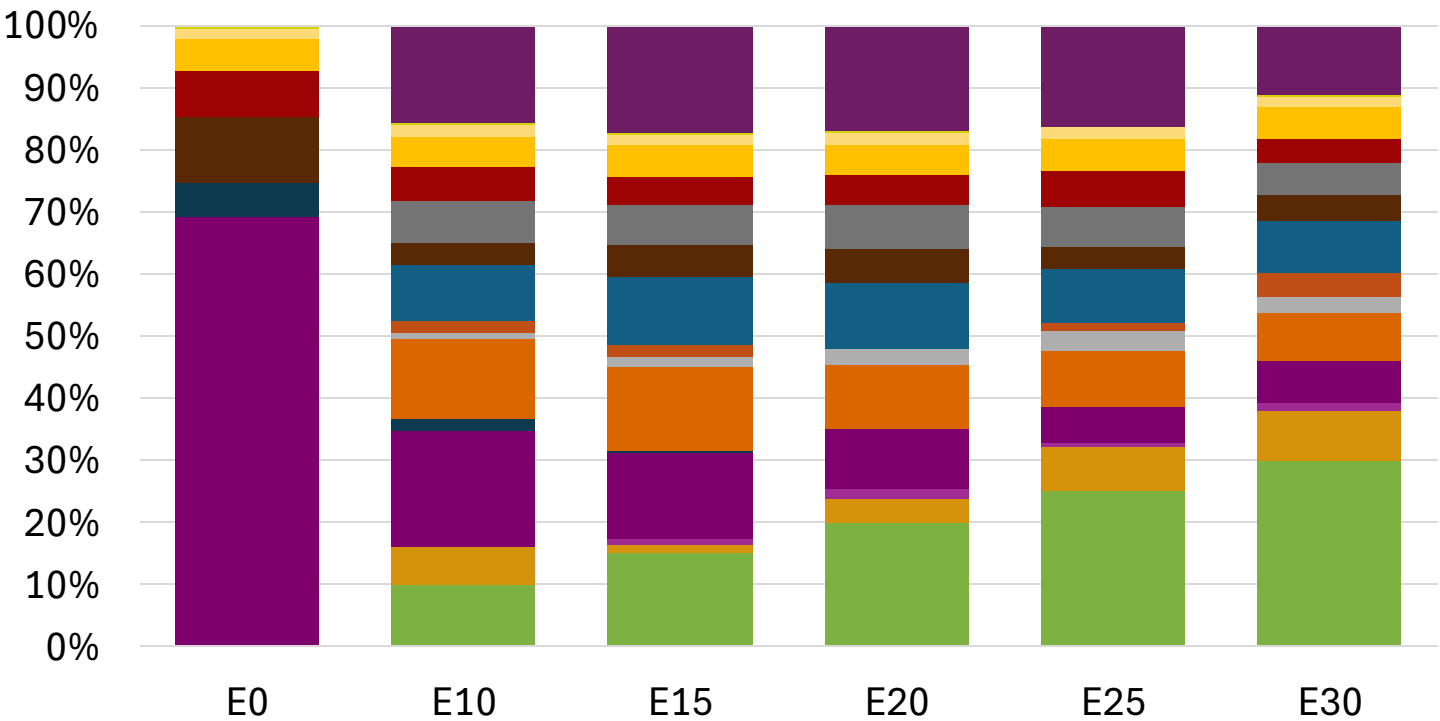
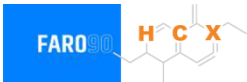


Average CIF 2024 Prices

Octane (RON)	95.0	95.7	97.2	98.8	100.3	101.8
Price (US\$/gal)	\$ 2.198	\$ 2.198	\$ 2.198	\$ 2.198	\$ 2.198	\$ 2.198

Source: Faro90

Hungary – Gasoline 98 – Increased Octane

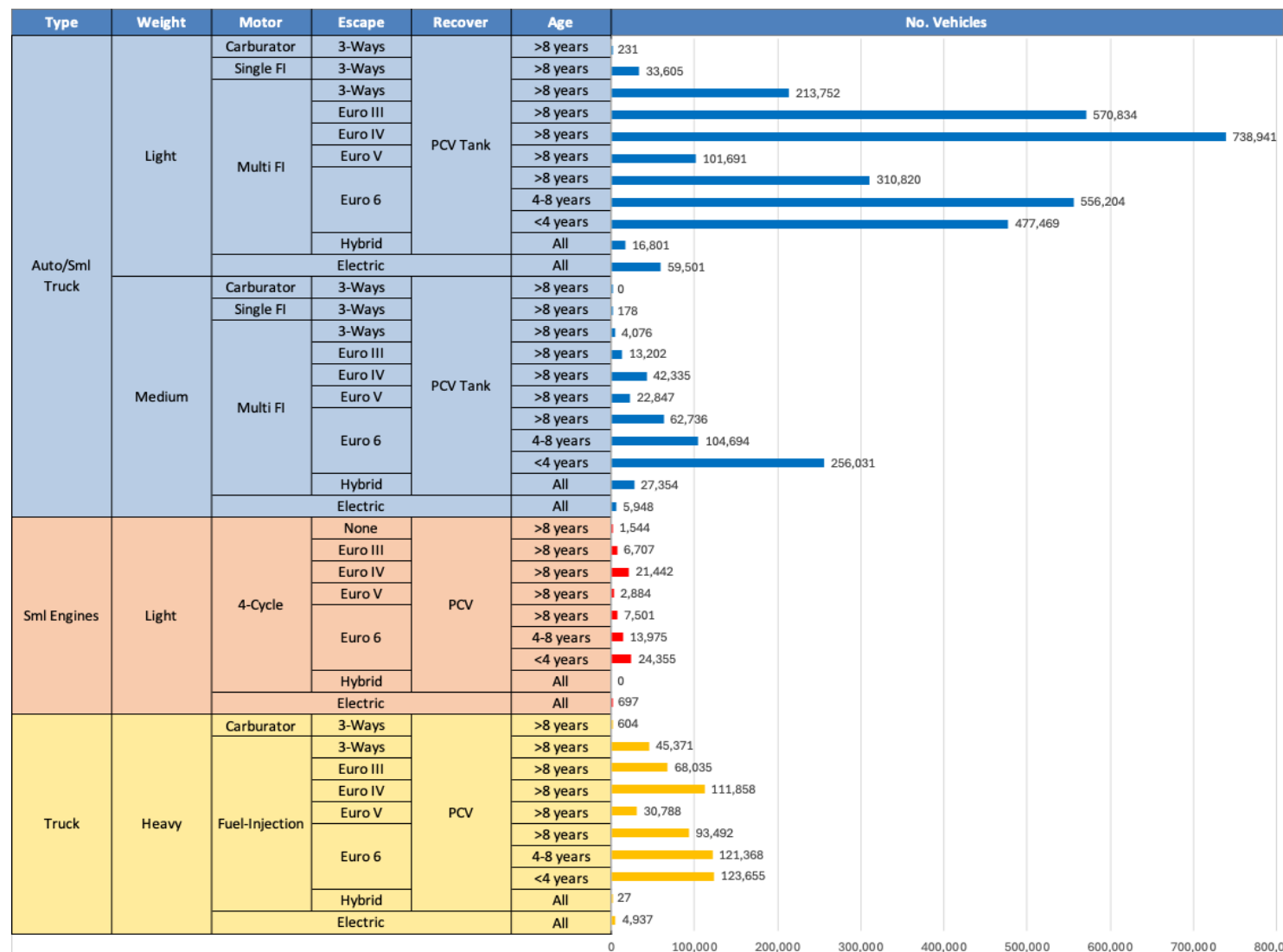


Average CIF 2024 Prices

Octane (RON)	98.0	98.5	99.9	101.2	102.5	104.0
Price (US\$/gal)	\$ 2.329	\$ 2.329	\$ 2.329	\$ 2.329	\$ 2.329	\$ 2.329

Source: Faro90

Gasoline Vehicle Fleet - Hungary

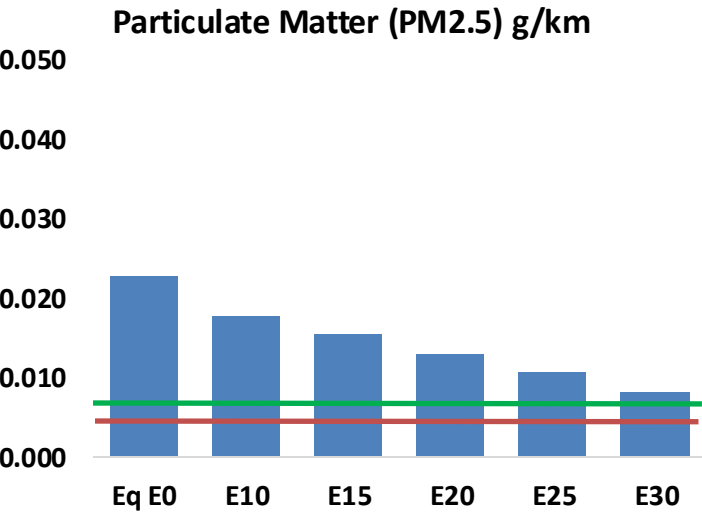
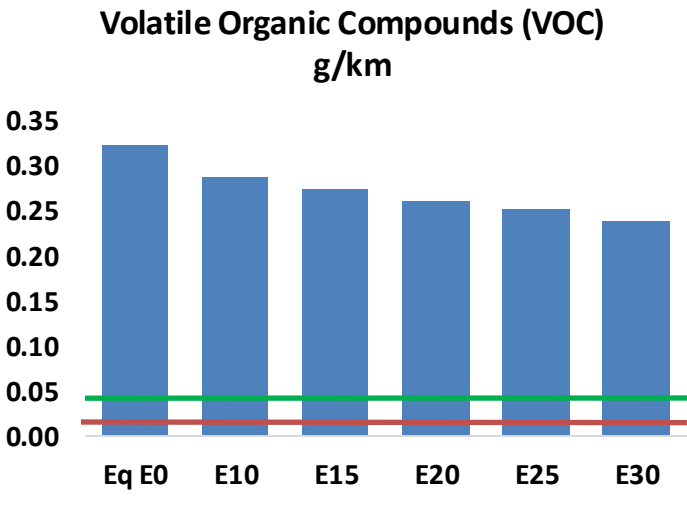
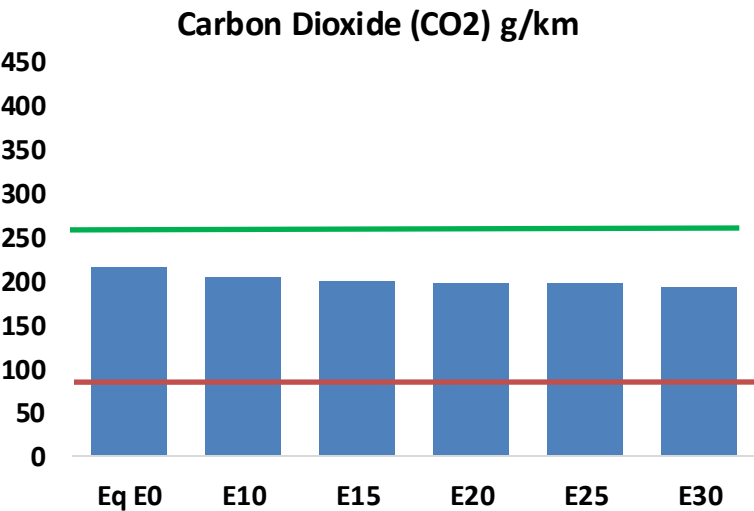
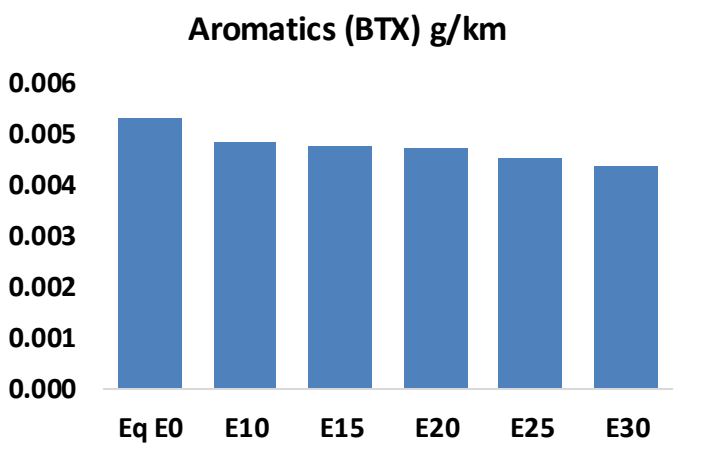
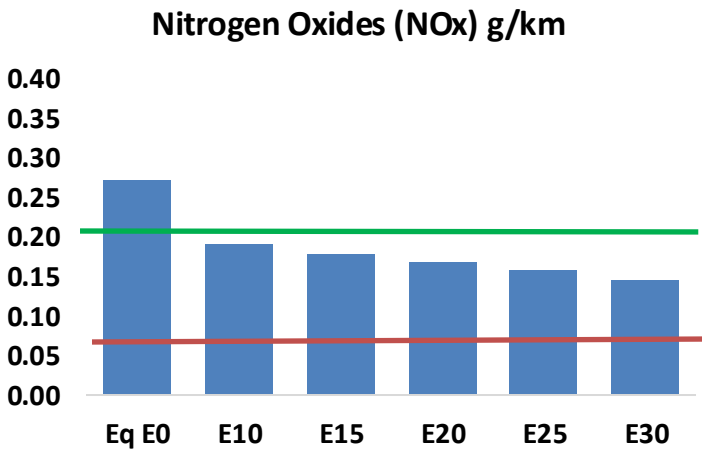
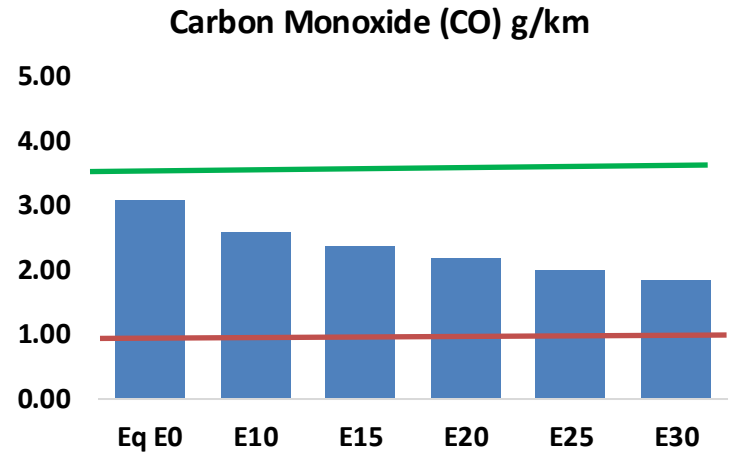
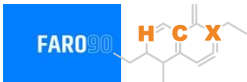


Vehicle Fleet: 4,298,491
Gasoline Vehicle Fleet: 2,317,855

Source: Eurostat, ACEA

Hungary – Vehicular Emissions

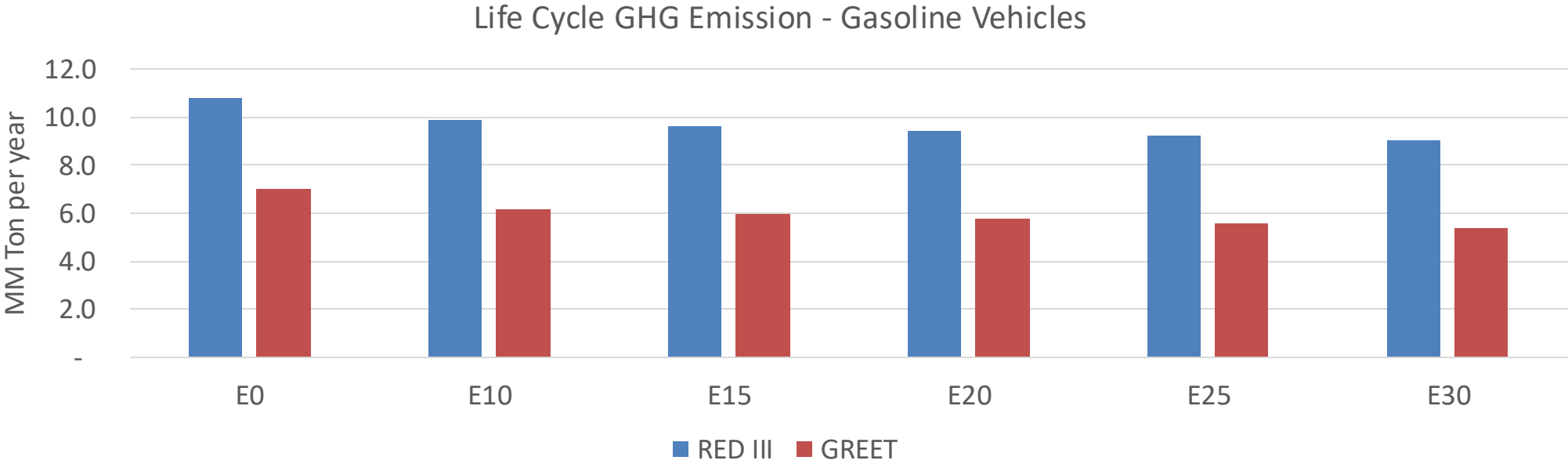
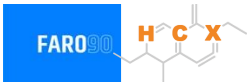
TIER III BIN 125 United States
Euro 6 Standard



Hungary – Vehicular Emissions

Vehicular Fleet Emissions (kg/day)	Eq E0	E5	E10	E15	E20	E25	E30	Reduction (%)	
								E10	E20
CO	174,918	160,652	146,385	134,534	122,535	113,286	103,266	-16%	-30%
CO2	12,154,422	11,841,879	11,546,701	11,314,551	11,199,675	11,138,457	10,933,143	-5%	-8%
VOC	18,282	17,261	16,241	15,479	14,748	14,196	13,422	-11%	-19%
NOx	15,402	11,552	10,782	10,162	9,583	8,946	8,271	-30%	-38%
SOx	138	127	117	108	100	90	81	-15%	-28%
NH3	4,122	4,081	4,041	4,060	4,105	4,138	4,164	-2%	0%
N2O	188	187	186	187	191	193	195	-1%	2%
CH4	3,983	3,983	3,983	4,043	4,131	4,206	4,278	0%	4%
Lead	-	-	-	-	-	-	-	0%	0%
Butadiene	159	153	148	145	142	141	138	-7%	-10%
Acetaldehyde	342	393	576	877	1,195	1,366	1,614	68%	249%
Formaldehyde	1,074	1,044	1,217	1,429	1,497	1,657	1,803	13%	39%
Benzene	302	290	275	270	268	256	248	-9%	-11%
PM2.5	750	666	582	506	429	348	264	-22%	-43%
PM10	537	477	417	362	307	249	189	-22%	-43%

Hungary – Gasoline Vehicle Fleet Life Cycle GHG Emissions



Country Emissions 2024 (MMT/año)	64.6
Target @ 2035 (MMT/year)	36.4
Delta	28.2

%Reduction vs E0	E0	E10	E15	E20	E25	E30
GREET 2024	0.0%	12.2%	15.3%	17.8%	20.1%	23.3%
RED II/III	0.0%	8.4%	10.7%	12.5%	14.2%	16.4%

% Target@2035	E0	E10	E15	E20	E25	E30
GREET 2024	0.0%	3.0%	3.8%	4.4%	5.0%	5.8%
RED II/III	0.0%	3.2%	4.1%	4.8%	5.4%	6.3%

Source: Greet, RED II/III, IPCC, climateactiontracker.org, F90

Ethanol Gasoline Blending in Ireland



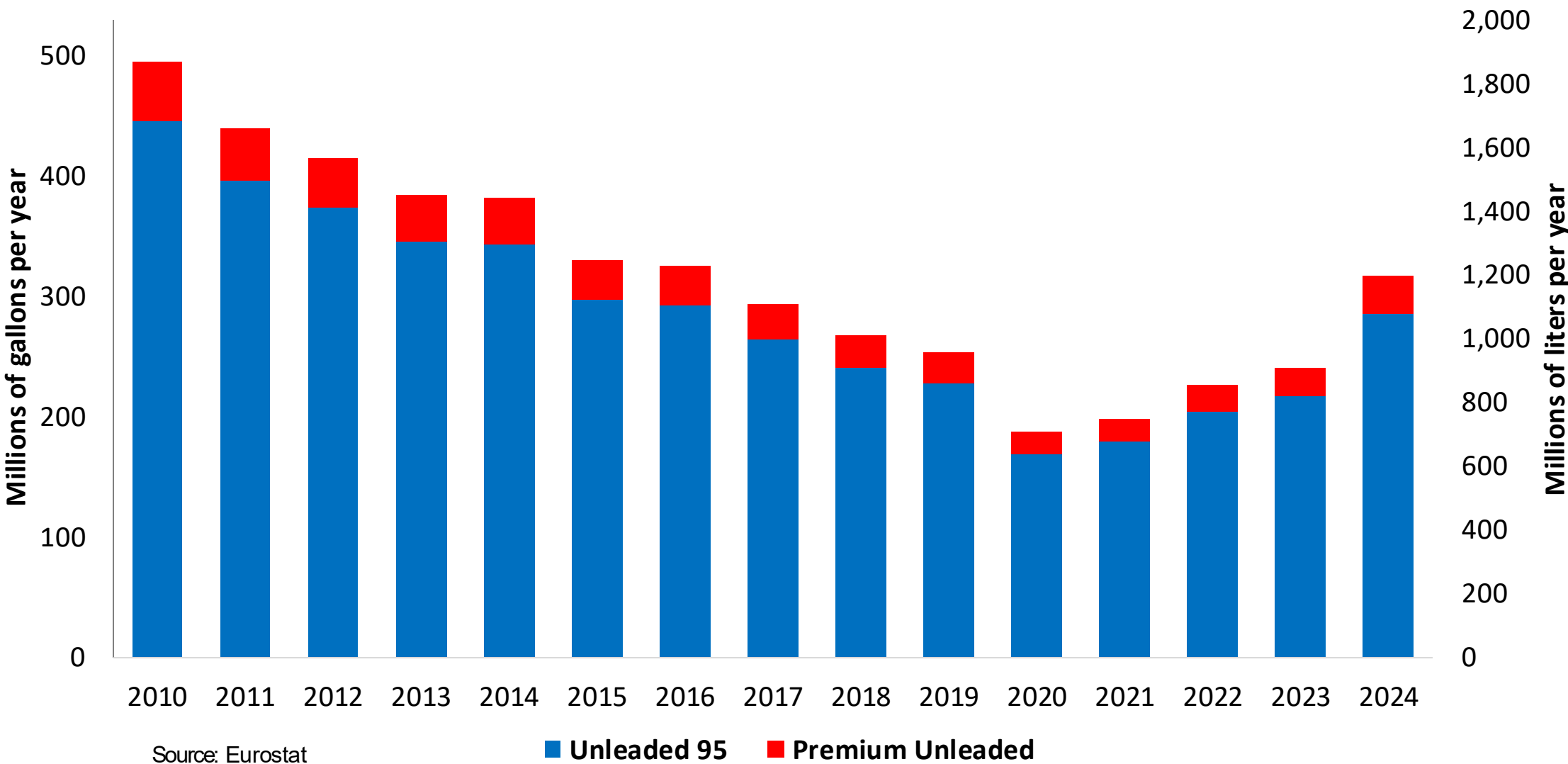
In 2024, gasoline consumption in Ireland was 1,197 million liters (316 million gallons) and growth rate was 4.6%. Gasoline production and net imports were 676 and 521 million liters (179 and 138 million gallons) correspondingly. Ireland complies with the European gasoline standards and offers two main grades: RON 95 and RON 98) with an average octane of 95.3 RON.

In 2024, ethanol consumption in Ireland was 60 million liters (16 million gallons) representing 5.0% of gasoline demand. Ethanol production and Net Imports were 22 and 39 million liters (16 and 10 million gallons) correspondingly. Ethanol source is Cereals 100%.

Source: Eurostat, European Commission

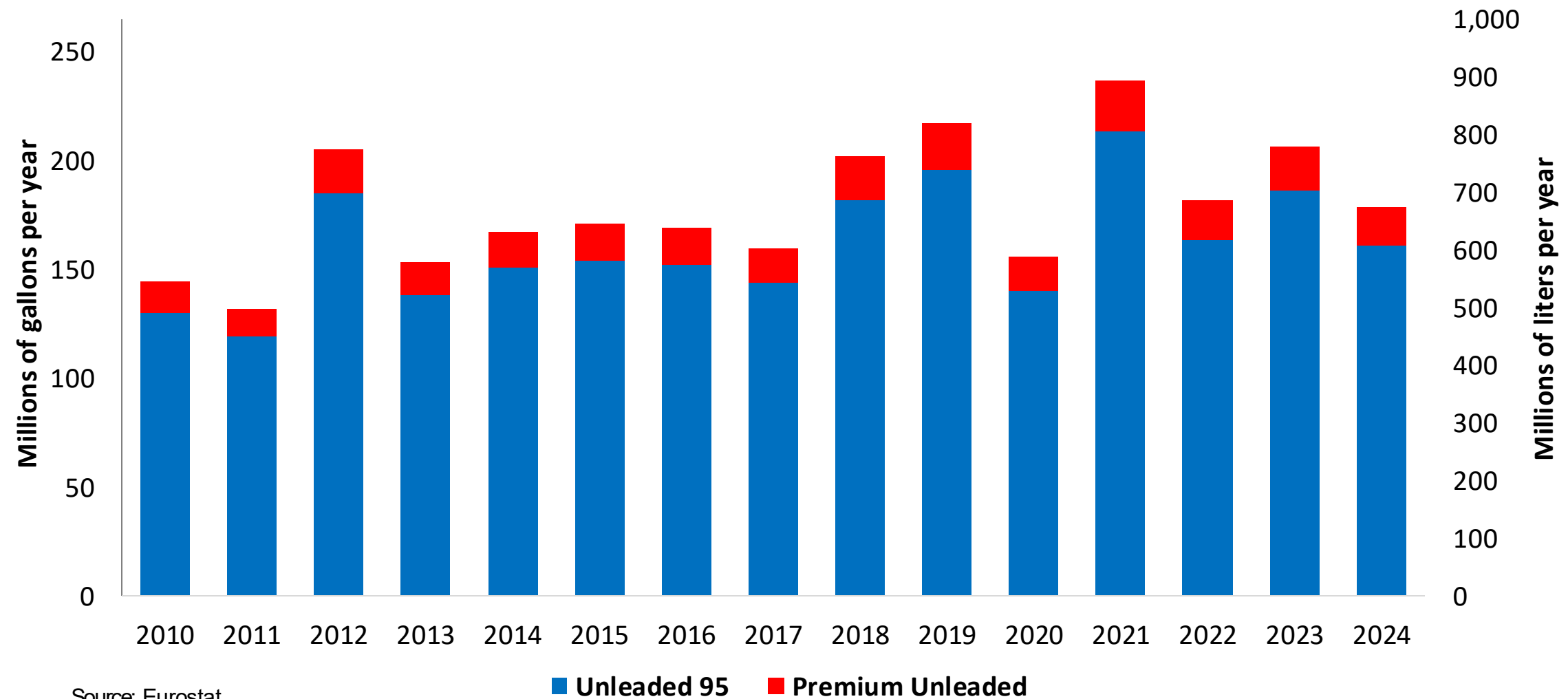
Gasoline Demand in Ireland

Gasoline Demand by Grade in Ireland



Gasoline Production in Ireland

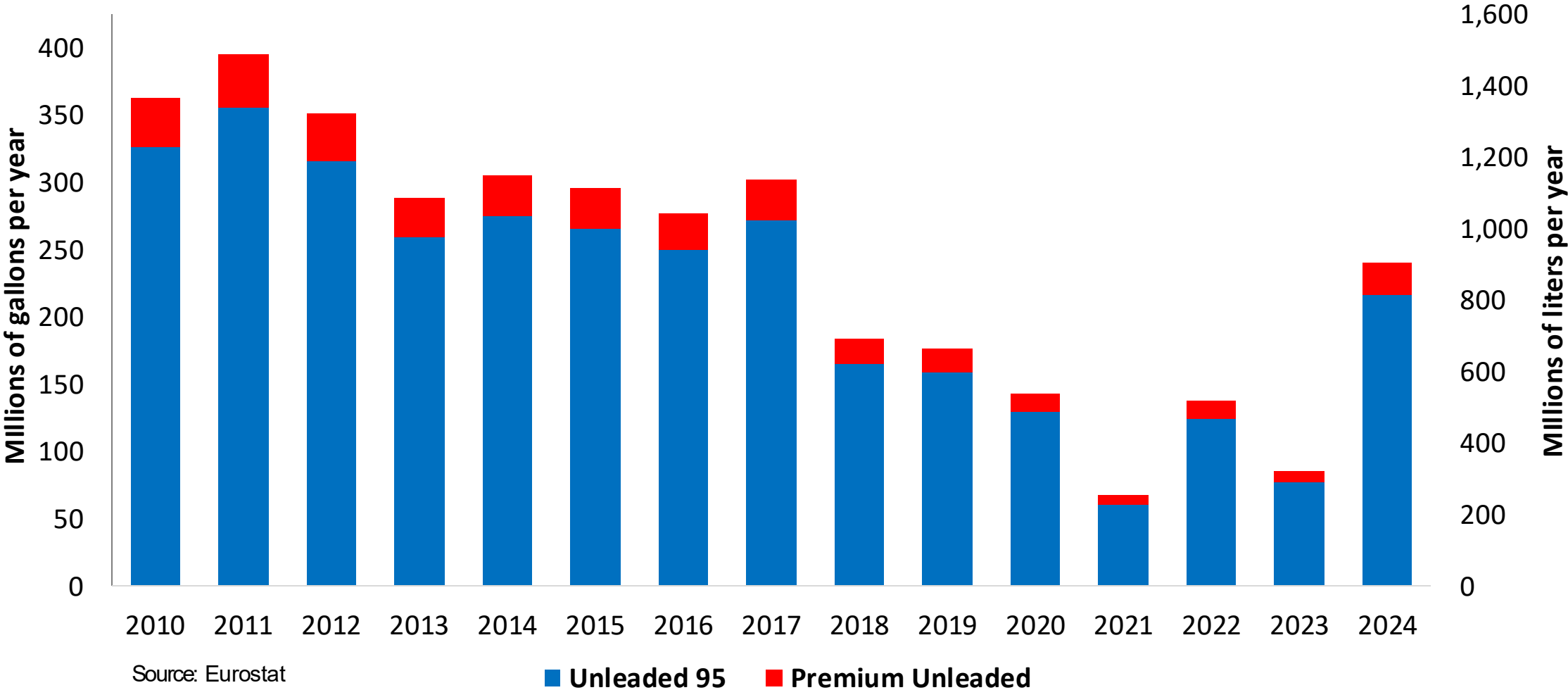
Gasoline Production by Grade in Ireland



Source: Eurostat

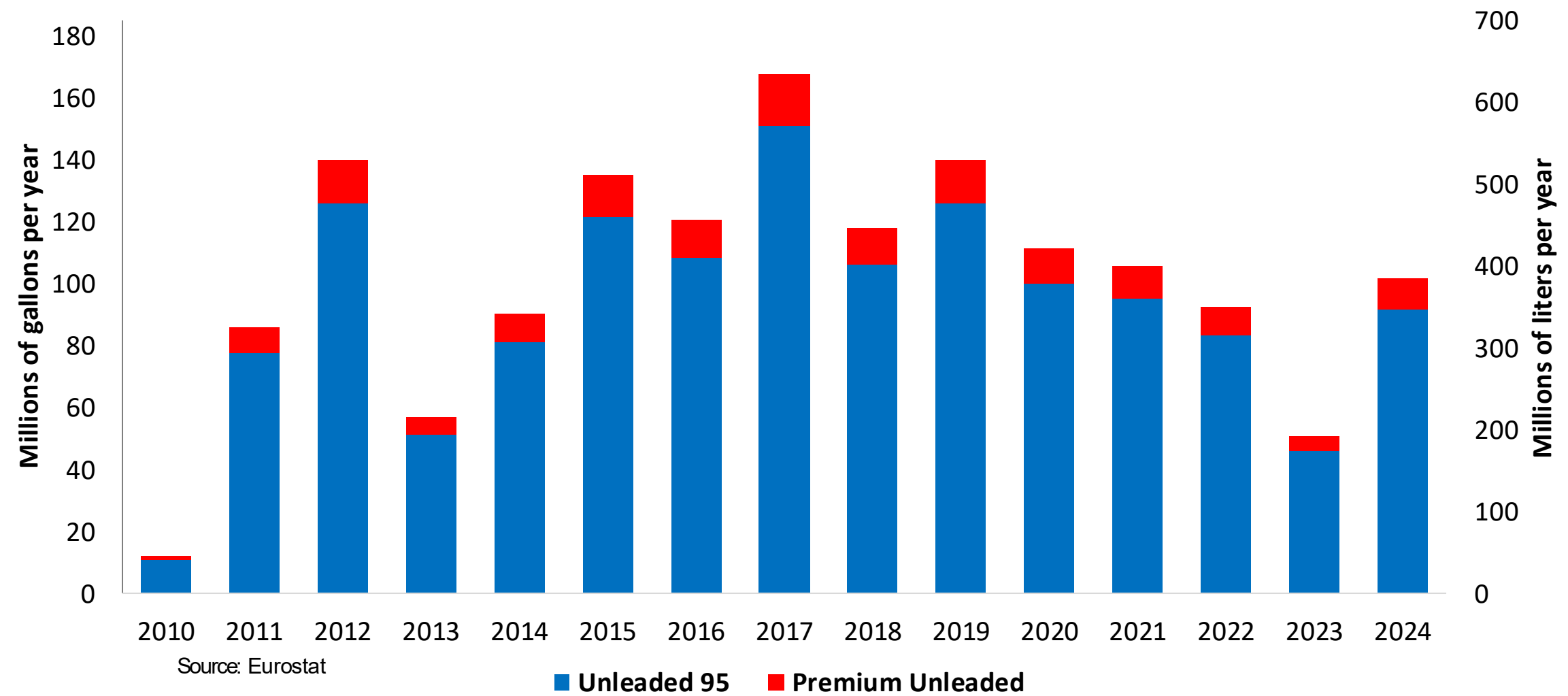
Gasoline Imports in Ireland

Gasoline Imports by Grade in Ireland

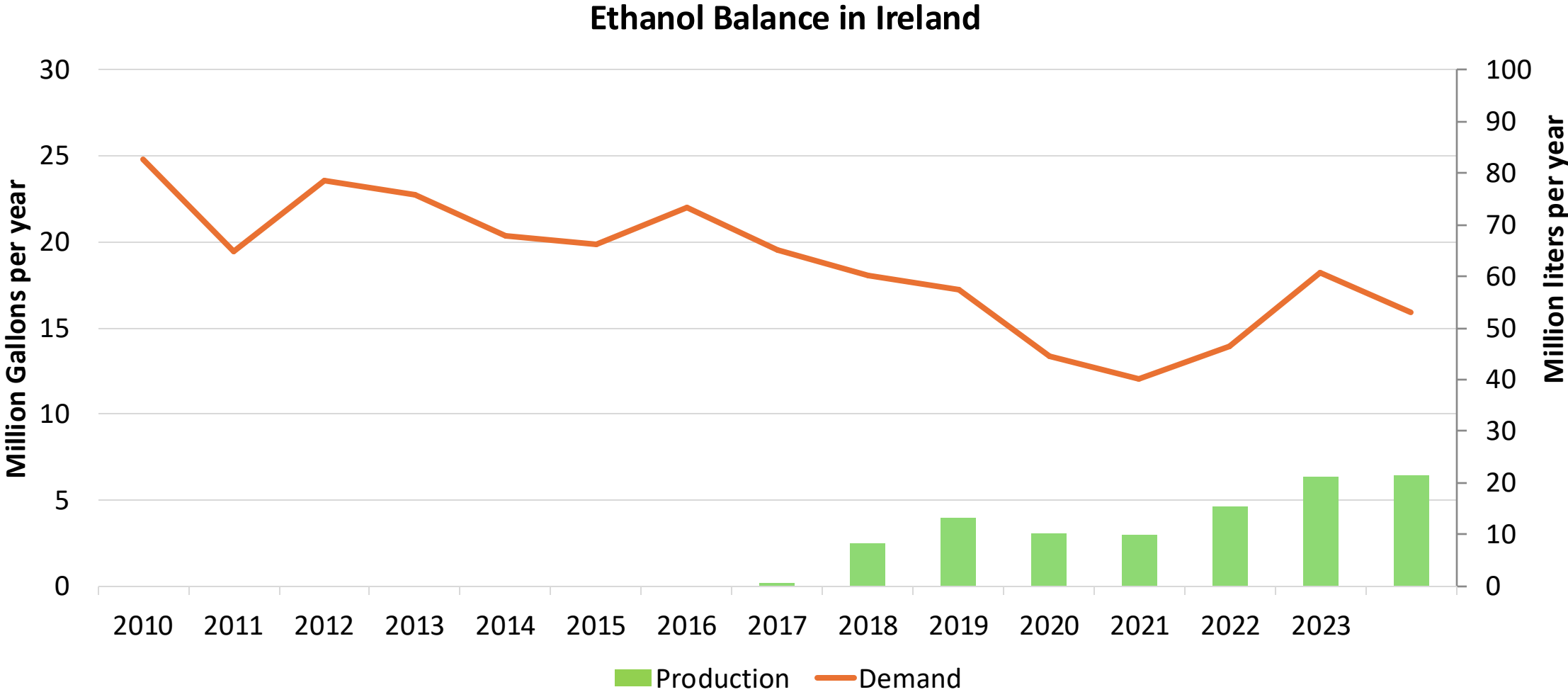


Gasoline Exports in Ireland

Gasoline Exports by Grade in Ireland

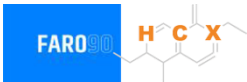


Ethanol Balance in Ireland



Source: Eurostat

Gasoline Quality in Ireland

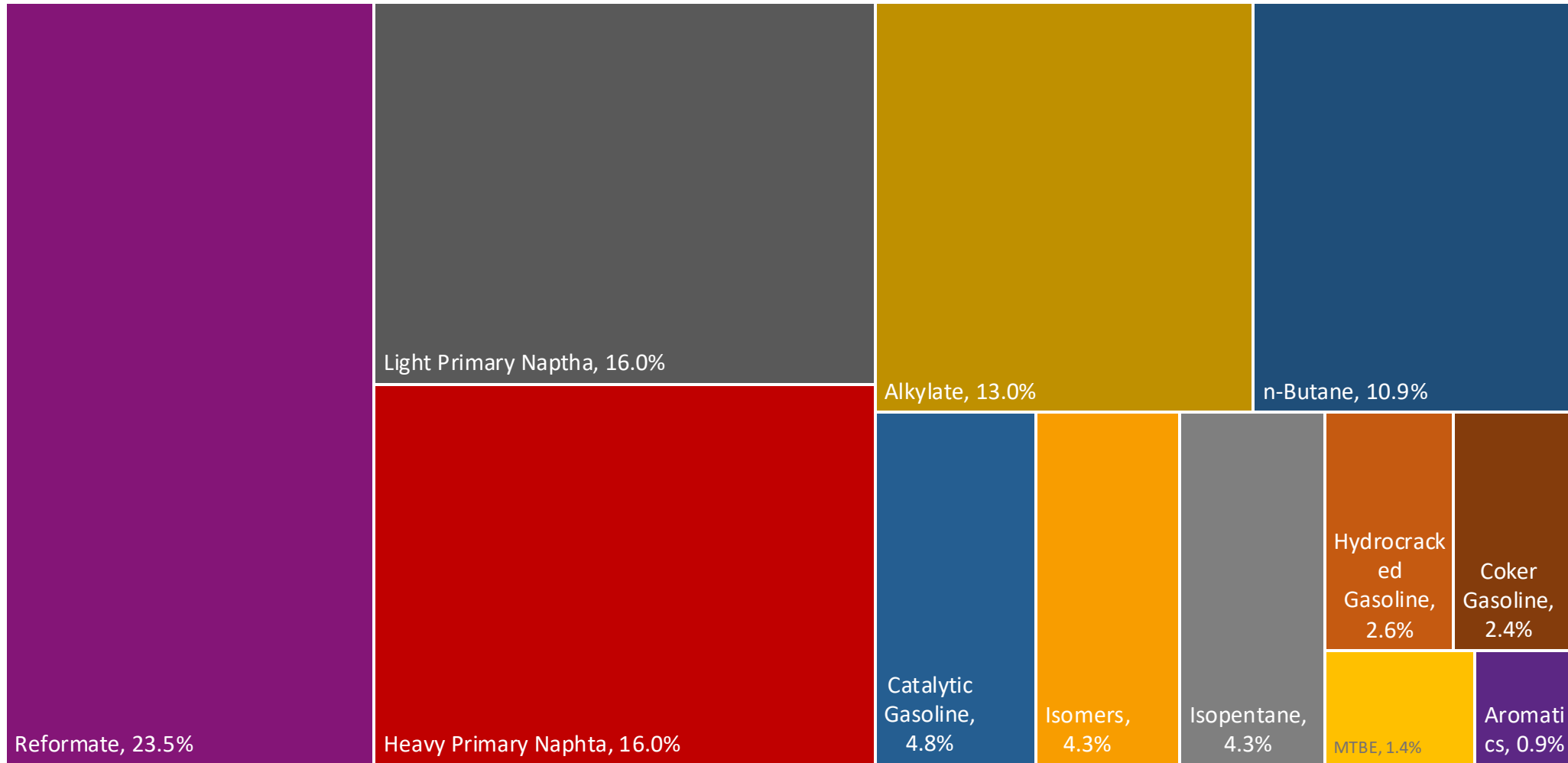


Name	EN 228:2025 (Euro 6); Country Level Fuel Ordinances							
Implementation Date	2025							
Applicability	All EU Contries + UK							
Grade	RON 95 E5		RON 95 E10		RON 98 E5		RON 98 E10	
	Min	Max	Min	Max	Min	Max	Min	Max
RON	95.0		95.0		98.0		98.0	
MON	> 85	> 88	> 85	> 88	> 85	> 88	> 85	> 88
AKI								
Benzene (%vol)	< 1							
Aromatics (% vol)	< 35							
Olefins (%vol)	< 18							
Lead (g/L)	< 0.005							
Manganese (mg/L)	< 2,0							
Sulfur (ppm)	10.0							
Oxygen (%w)*	< 2,7	< 3,7	< 2,7	< 3,7	< 2,7	< 3,7	< 2,7	< 3,7
Ethanol (%vol)	< 5	< 10	< 5	< 10	< 5	< 10	< 5	< 10
PVR 37.8°C (psi)	<> 45 - 100 kPa in 6 Volatility Classes							
Overall Biofuels (%cal)	21.0							
Advanced Biofuels (%cal)	1.0							
Bioethanol in Gasoline Sales (%cal)	-							
GHG Emission Reduction (%)	-							

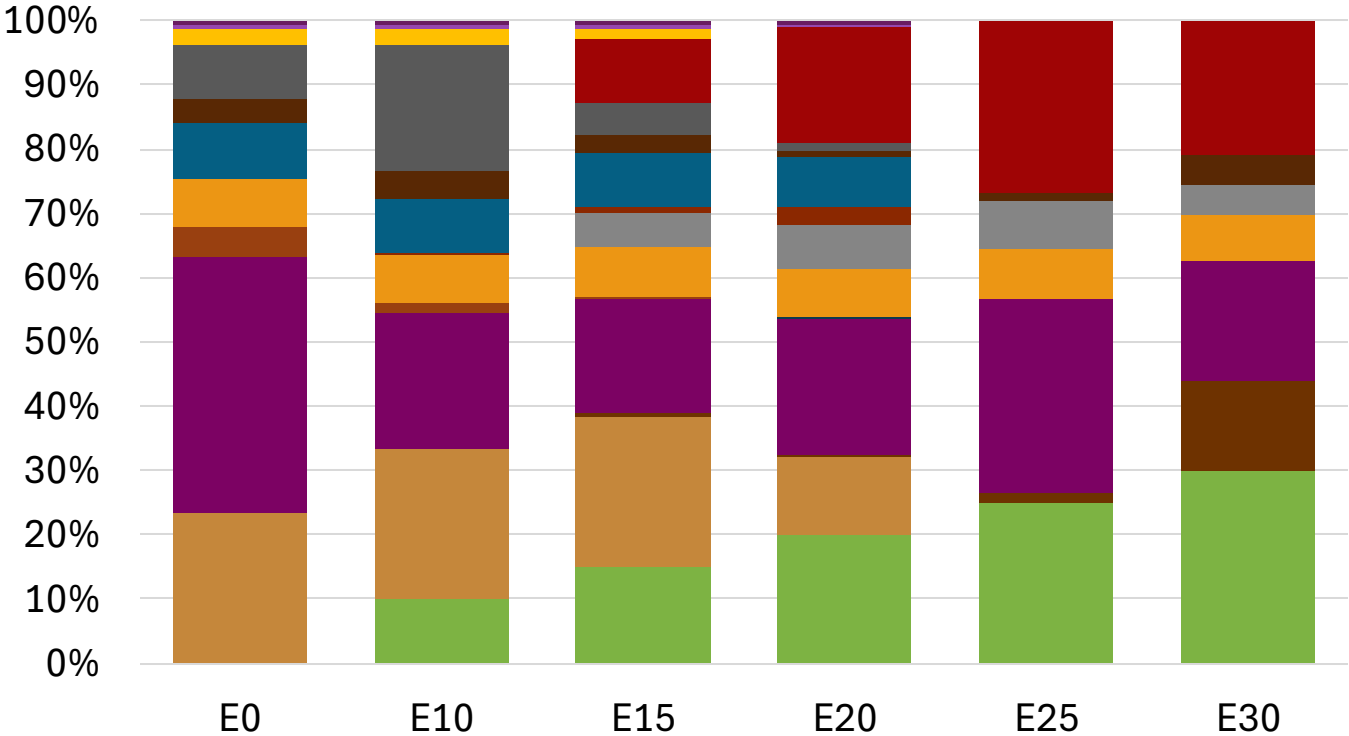
* Methanol 3%, Iso-propyl alcohol 12%, Ter-butyl alcohol 15%, Iso-butyl alcohol 15%, Ethers +5 Carbons 22%, Other Oxygenates 15%

Source: European Commision

Ireland Available Blending Components



Ireland – Gasoline 95 – Constant Octane

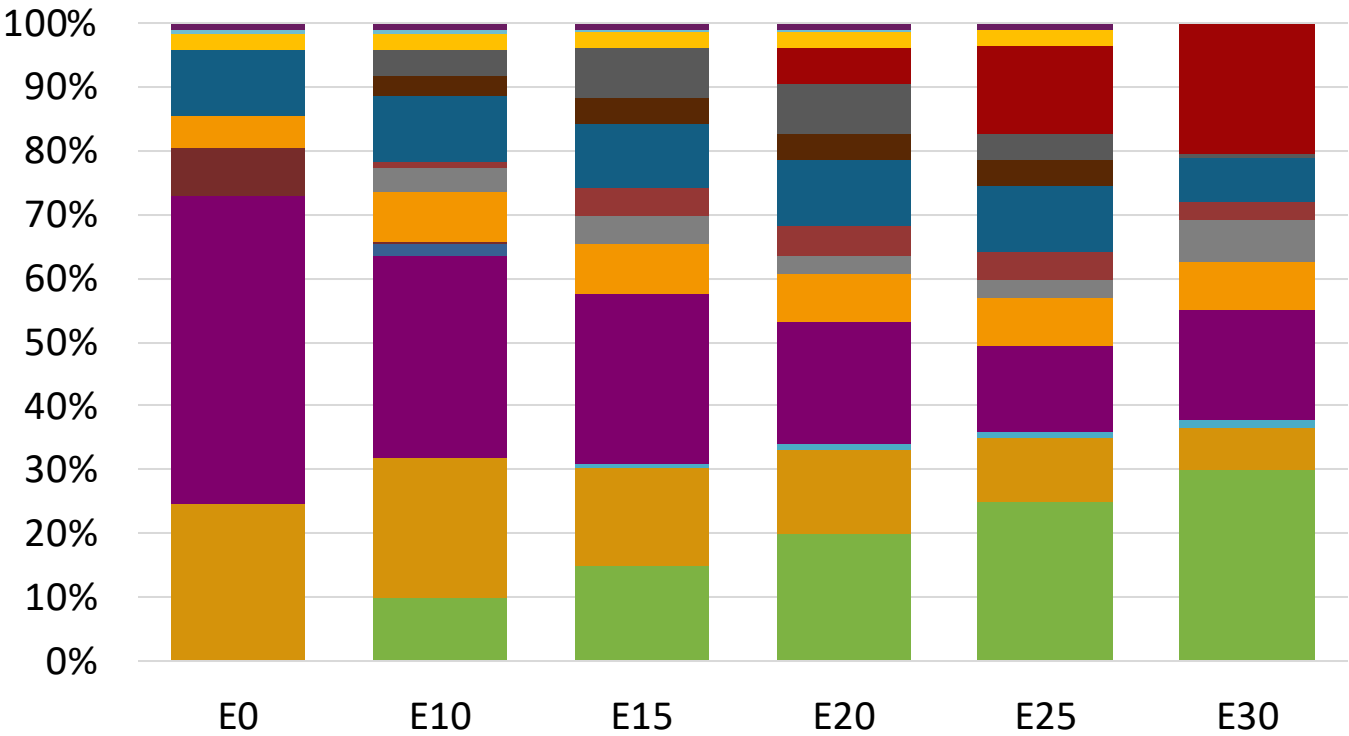
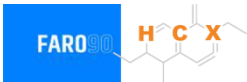


Average CIF 2024
Prices

Octane (RON)	95.0	95.0	95.0	95.0	95.4	95.8
Price (US\$/gal)	\$ 2.317	\$ 2.187	\$ 2.105	\$ 1.996	\$ 1.886	\$ 1.821

Source: Faro90

Ireland - Gasoline 98 - Constant Octane

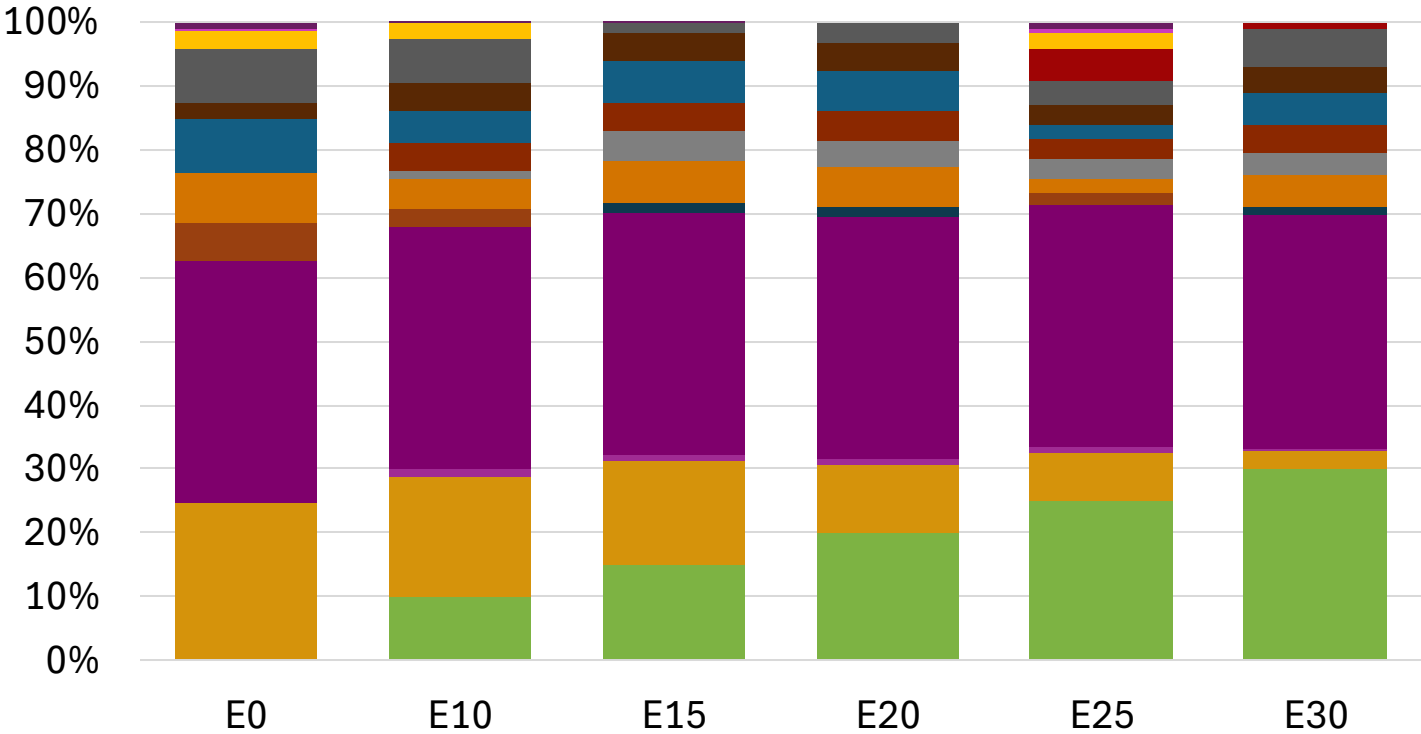
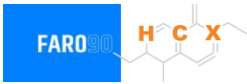


Average CIF 2024 Prices

Octane (RON)	98.0	98.0	98.0	98.0	98.0	98.0
Price (US\$/gal)	\$ 2.440	\$ 2.317	\$ 2.240	\$ 2.157	\$ 2.062	\$ 1.967

Source: Faro90

Ireland – Gasoline 95 – Increased Octane

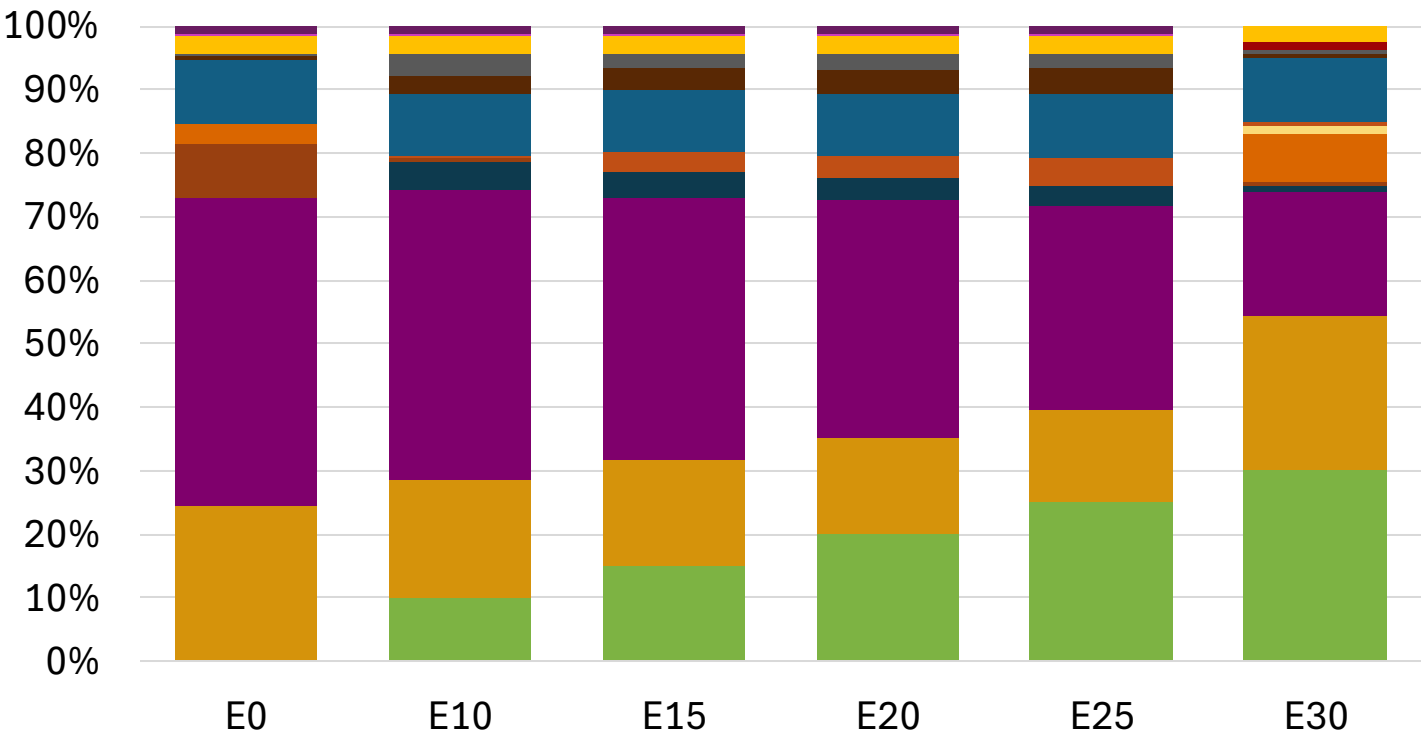
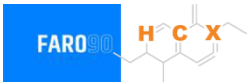


Average CIF 2024
Prices

Octane (RON)		95.0	97.9	99.3	100.9	102.6	103.9
Price (US\$/gal)	\$	2.301	\$ 2.301	\$ 2.301	\$ 2.301	\$ 2.301	\$ 2.301

Source: Faro90

Ireland – Gasoline 98 – Increased Octane

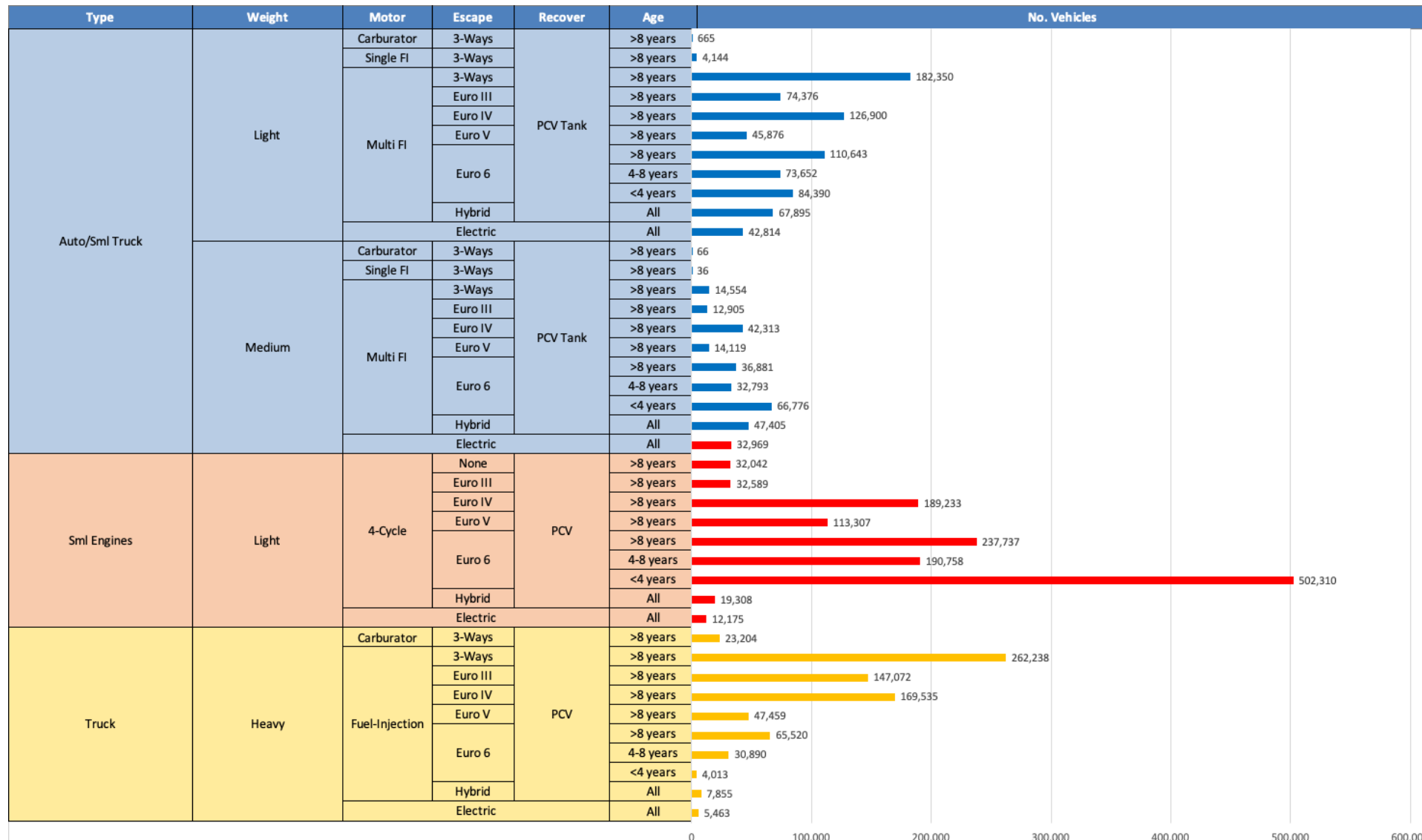


Average CIF 2024 Prices

Octane (RON)	98.0	100.7	102.0	103.3	104.5	105.0
Price (US\$/gal)	\$ 2.425	\$ 2.425	\$ 2.425	\$ 2.425	\$ 2.425	\$ 2.425

Source: Faro90

Gasoline Vehicle Fleet - Ireland

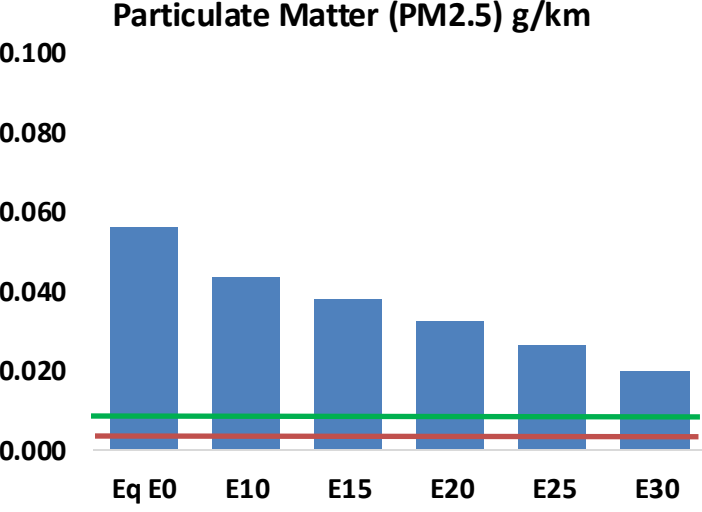
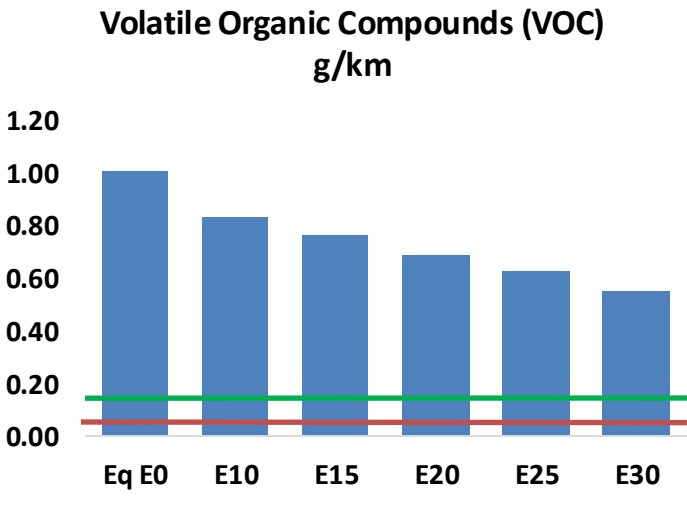
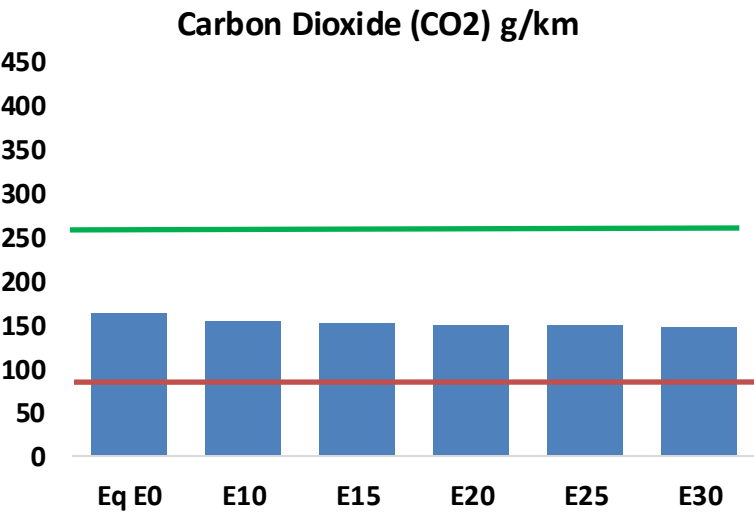
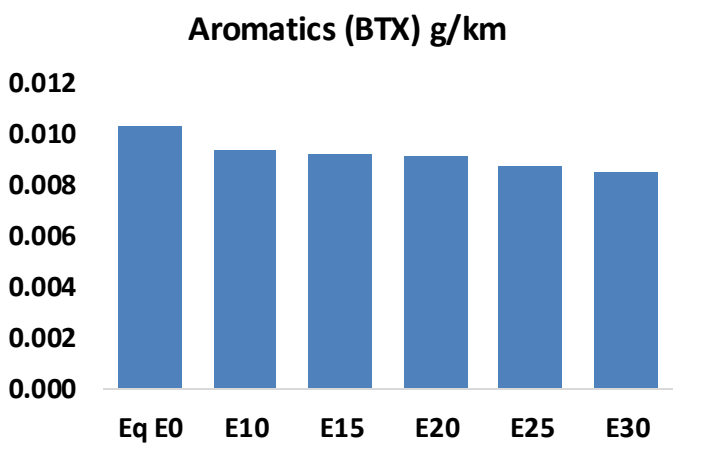
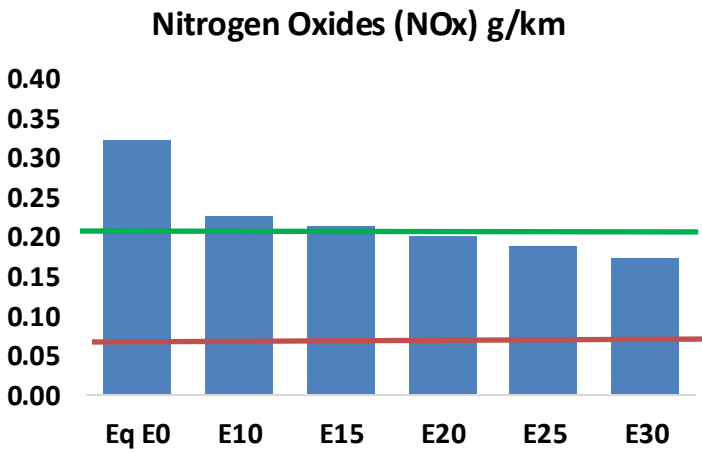
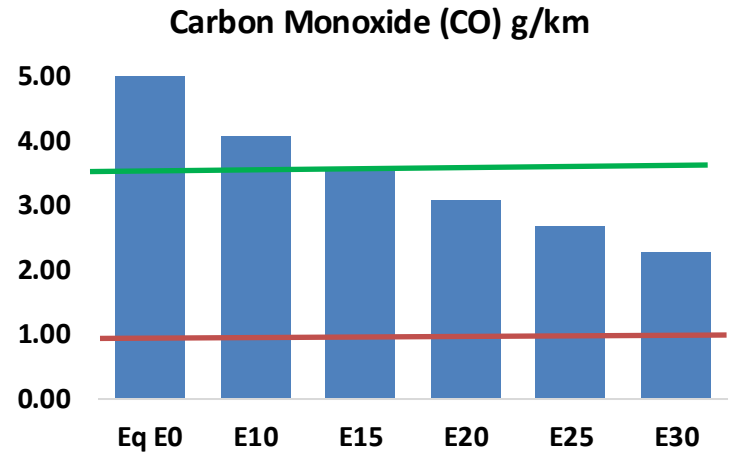
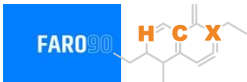


Vehicle Fleet: 3,207,228
Gasoline Vehicle Fleet: 1,641,194

Source: Eurostat, ACEA

Ireland – Vehicular Emissions

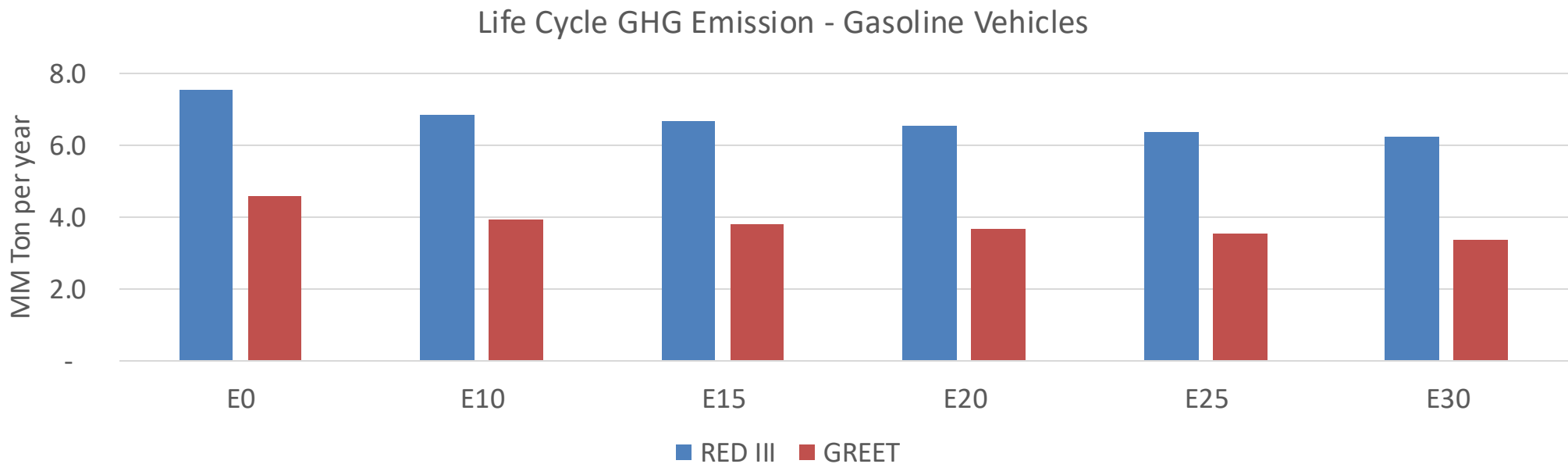
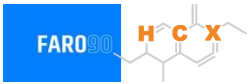
TIER III BIN 125 United States
Euro 6 Standard



Ireland – Vehicular Emissions

Vehicular Fleet Emissions (kg/day)	Eq E0	E5	E10	E15	E20	E25	E30	Reduction (%)	
								E10	E20
CO	211,328	188,438	165,548	145,494	124,716	108,559	92,049	-22%	-41%
CO2	6,629,738	6,459,259	6,298,251	6,171,623	6,108,963	6,075,101	5,963,120	-5%	-8%
VOC	40,883	37,376	33,869	30,924	27,928	25,614	22,561	-17%	-32%
NOx	13,148	9,861	9,204	8,675	8,181	7,637	7,061	-30%	-38%
SOx	74	68	63	58	53	49	43	-15%	-28%
NH3	2,749	2,722	2,695	2,708	2,738	2,760	2,777	-2%	0%
N2O	118	117	116	117	119	121	122	-1%	2%
CH4	8,968	8,968	8,968	9,103	9,300	9,471	9,632	0%	4%
Lead	-	-	-	-	-	-	-	0%	0%
Butadiene	216	203	190	180	170	162	152	-12%	-22%
Acetaldehyde	725	832	1,221	1,860	2,534	2,895	3,421	68%	249%
Formaldehyde	2,850	2,771	3,230	3,793	3,973	4,396	4,785	13%	39%
Benzene	420	404	383	377	374	357	346	-9%	-11%
PM2.5	447	397	347	301	256	208	157	-22%	-43%
PM10	1,837	1,631	1,425	1,238	1,051	853	646	-22%	-43%

Ireland – Gasoline Vehicle Fleet Life Cycle GHG Emissions



Country Emissions 2024 (MMT/año)	70.0
Target @ 2035 (MMT/year)	39.4
Delta	30.6

%Reduction vs E0	E0	E10	E15	E20	E25	E30
GREET 2024	0.0%	14.1%	17.5%	20.3%	23.1%	26.6%
RED II/III	0.0%	9.1%	11.4%	13.3%	15.1%	17.4%

% Target@2035	E0	E10	E15	E20	E25	E30
GREET 2024	0.0%	2.1%	2.6%	3.1%	3.5%	4.0%
RED II/III	0.0%	2.2%	2.8%	3.3%	3.7%	4.3%

Source: Greet, RED II/III, IPCC, climateactiontracker.org, F90

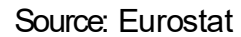
Ethanol Gasoline Blending in Italy



In 2024, gasoline consumption in Italy was 11,426 million liters (3,018 million gallons) and growth rate was 3.2%. Gasoline production and net imports were 17,947 and -6,521 million liters (4,741 and -1,723 million gallons) correspondingly. Italy complies with the European gasoline standards and offers two main grades: RON 95 and RON 98) with an average octane of 95.2 RON.

In 2024, ethanol consumption in Italy was 640 million liters (169 million gallons) representing 5.6% of gasoline demand. Ethanol production and Net Imports were 700 and -60 million liters (169 and -16 million gallons) correspondingly. Ethanol sources are Wine and Second Generation Lignocelulosic ethanol.

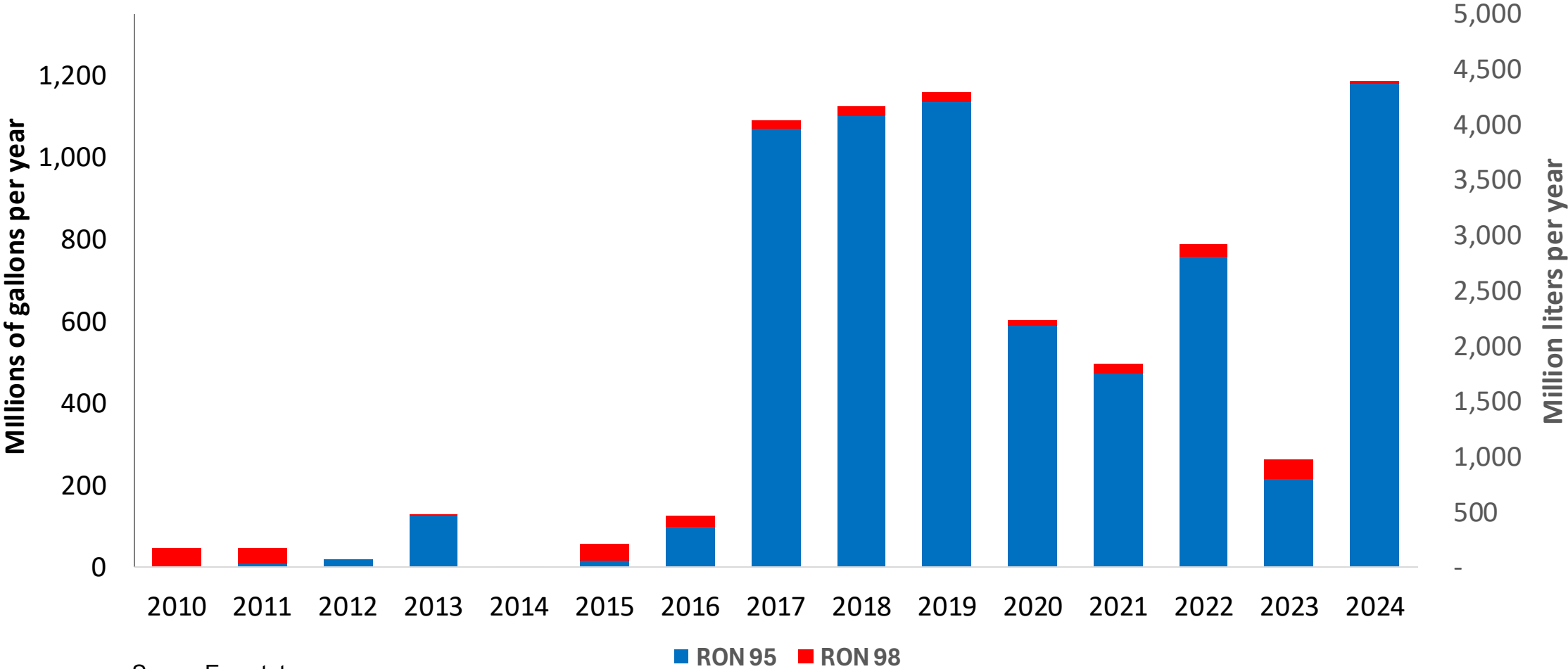
The FARO90 logo is shown in a blue box, followed by a chemical structure of a substituted cyclohexadiene. The structure features a six-membered ring with two double bonds. Substituents include a methyl group, a propyl group, and a group labeled 'X'. The letters 'H', 'C', and 'X' are highlighted in orange in the original image.



The FARO90 logo is shown in a blue box, followed by a chemical structure of a substituted cyclohexadiene. The structure features a six-membered ring with two double bonds. Substituents include a methyl group, a propyl group, and a group labeled 'X'. The letters 'H', 'C', and 'X' are highlighted in orange in the original image.

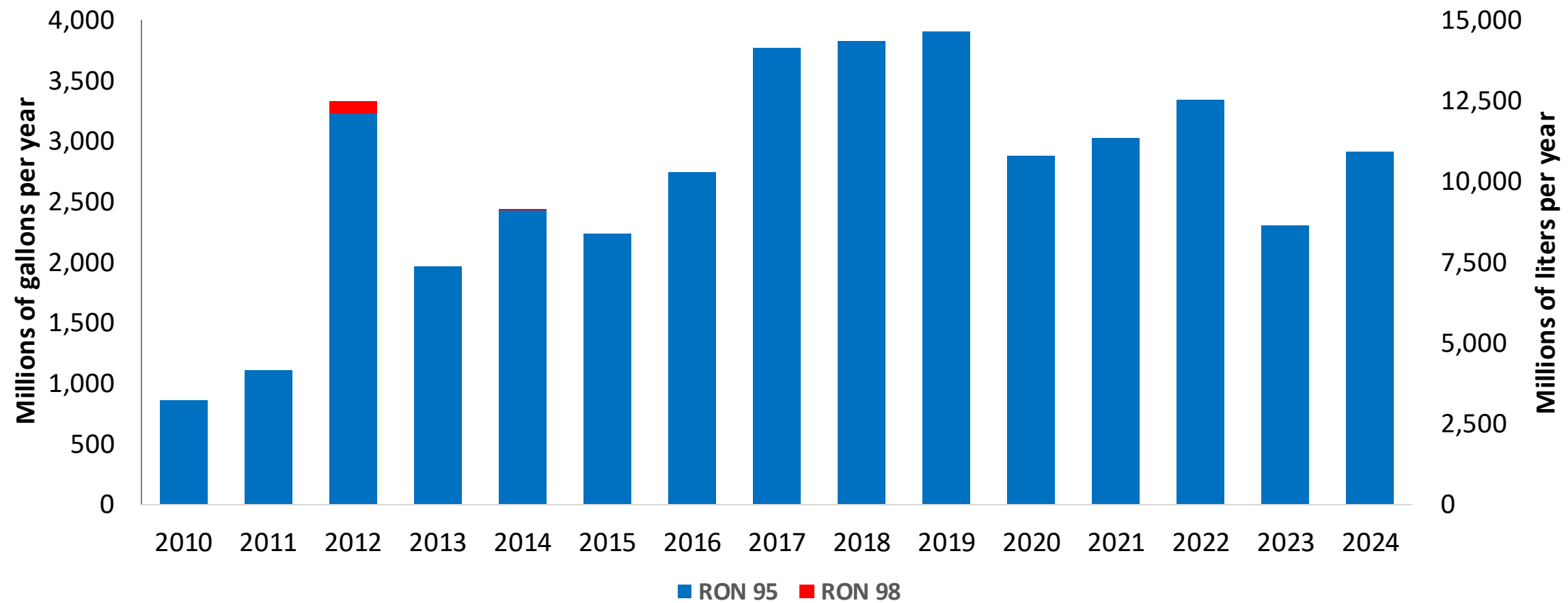


Gasoline Imports by Grade in Italy



Source: Eurostat

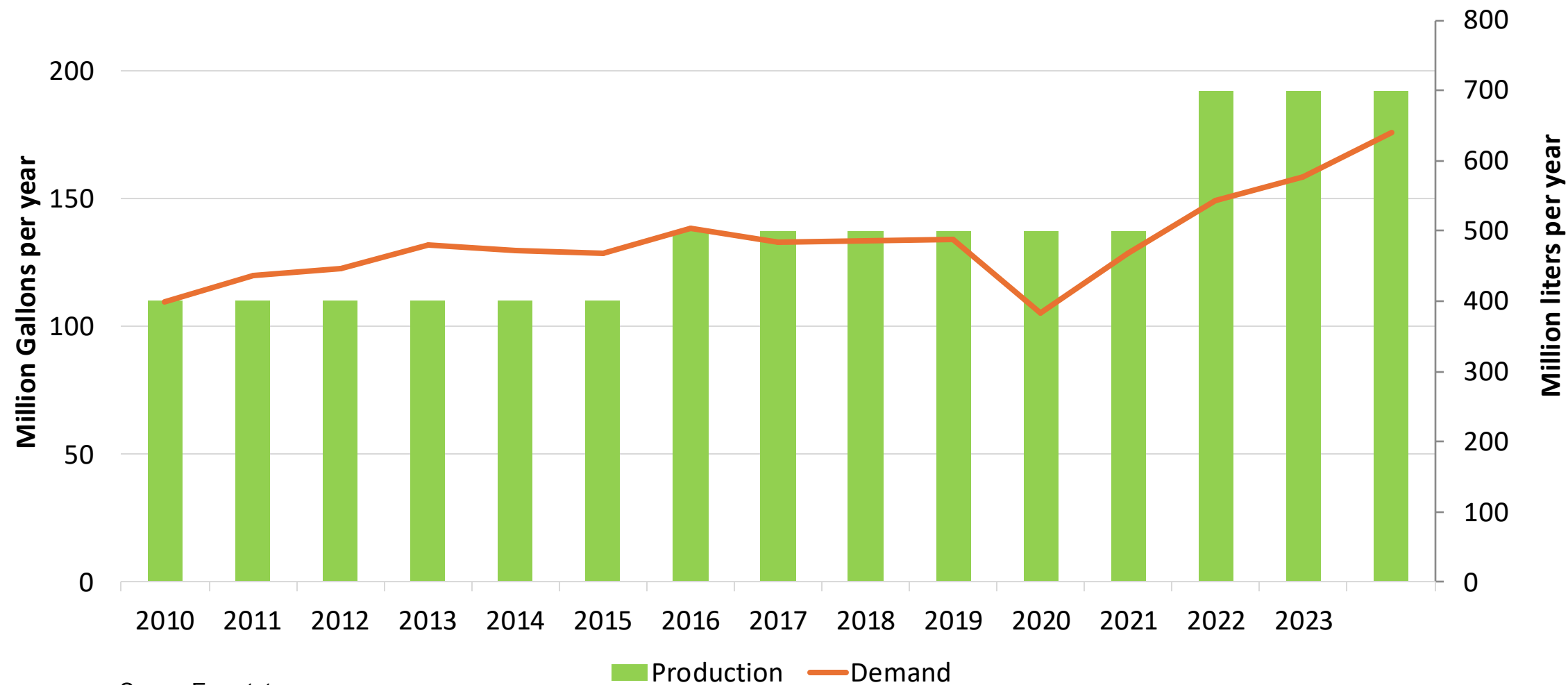
Gasoline Exports by Grade in Italy



Source: Eurostat

Ethanol Balance in Italy

Ethanol Balance in Italy



Source: Eurostat

Gasoline Quality in Italy

Name	EN 228:2025 (Euro 6); Country Level Fuel Ordinances							
Implementation Date	2025							
Applicability	All EU Contries + UK							
Grade	RON 95 E5		RON 95 E10		RON 98 E5		RON 98 E10	
	Min	Max	Min	Max	Min	Max	Min	Max
RON	95.0		95.0		98.0		98.0	
MON	> 85	> 88	> 85	> 88	> 85	> 88	> 85	> 88
AKI								
Benzene (%vol)	< 1							
Aromatics (% vol)	< 35							
Olefins (%vol)	< 18							
Lead (g/L)	< 0.005							
Manganese (mg/L)	< 2,0							
Sulfur (ppm)	10.0							
Oxygen (%w)*	< 2,7	< 3,7	< 2,7	< 3,7	< 2,7	< 3,7	< 2,7	< 3,7
Ethanol (%vol)	< 5	< 10	< 5	< 10	< 5	< 10	< 5	< 10
PVR 37.8°C (psi)	<> 45 - 100 kPa in 6 Volatility Classes							
Overall Biofuels (%cal)	11.7							
Advanced Biofuels (%cal)	4.9							
Bioethanol in Gasoline Sales (%cal)	-							
GHG Emission Reduction (%)	-							

* Methanol 3%, Iso-propyl alcohol 12%, Ter-butyl alcohol 15%, Iso-butyl alcohol 15%, Ethers +5 Carbons 22%, Other Oxygenates 15%

Source: European Commision

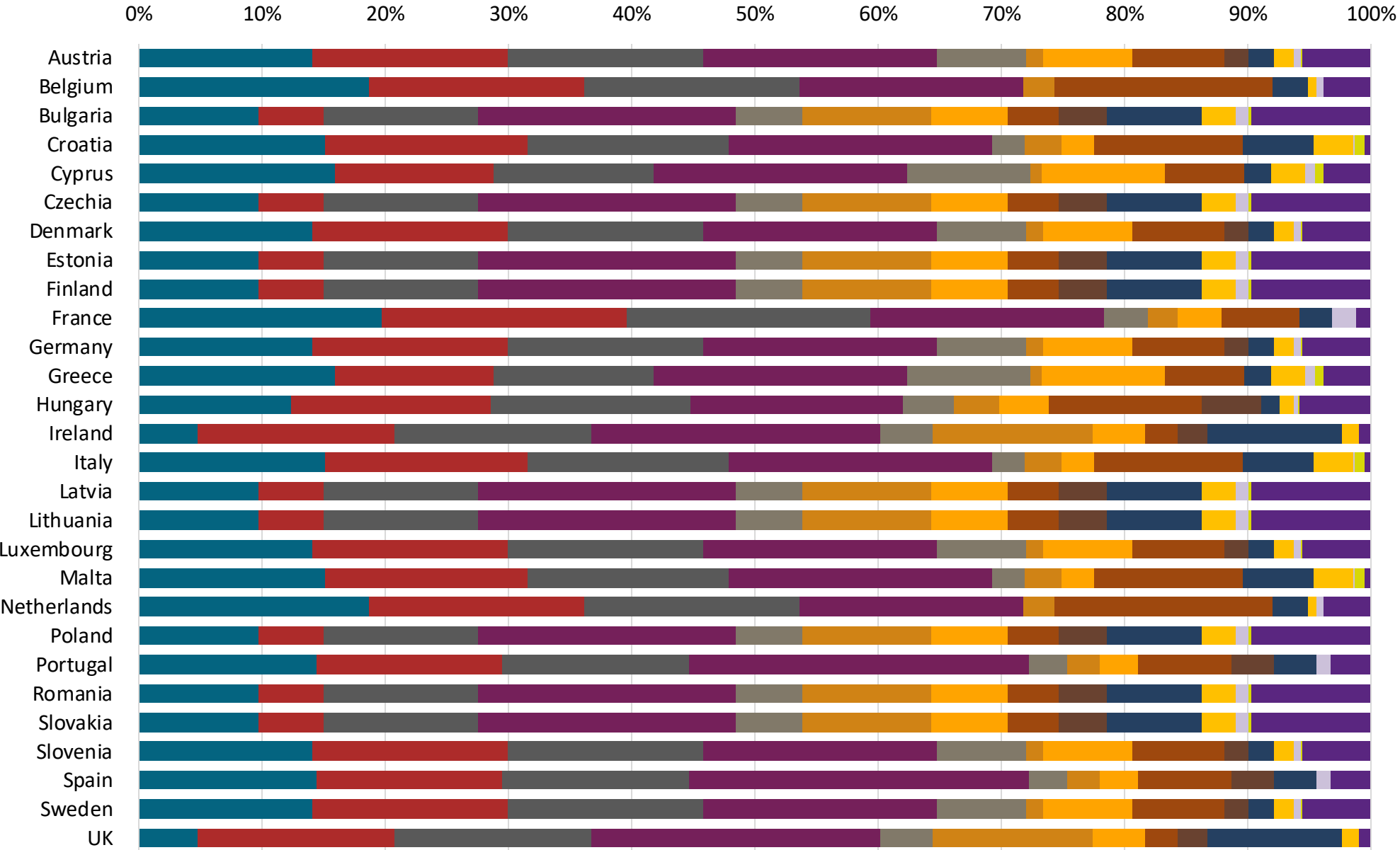
Gasoline Component Blending in Europe



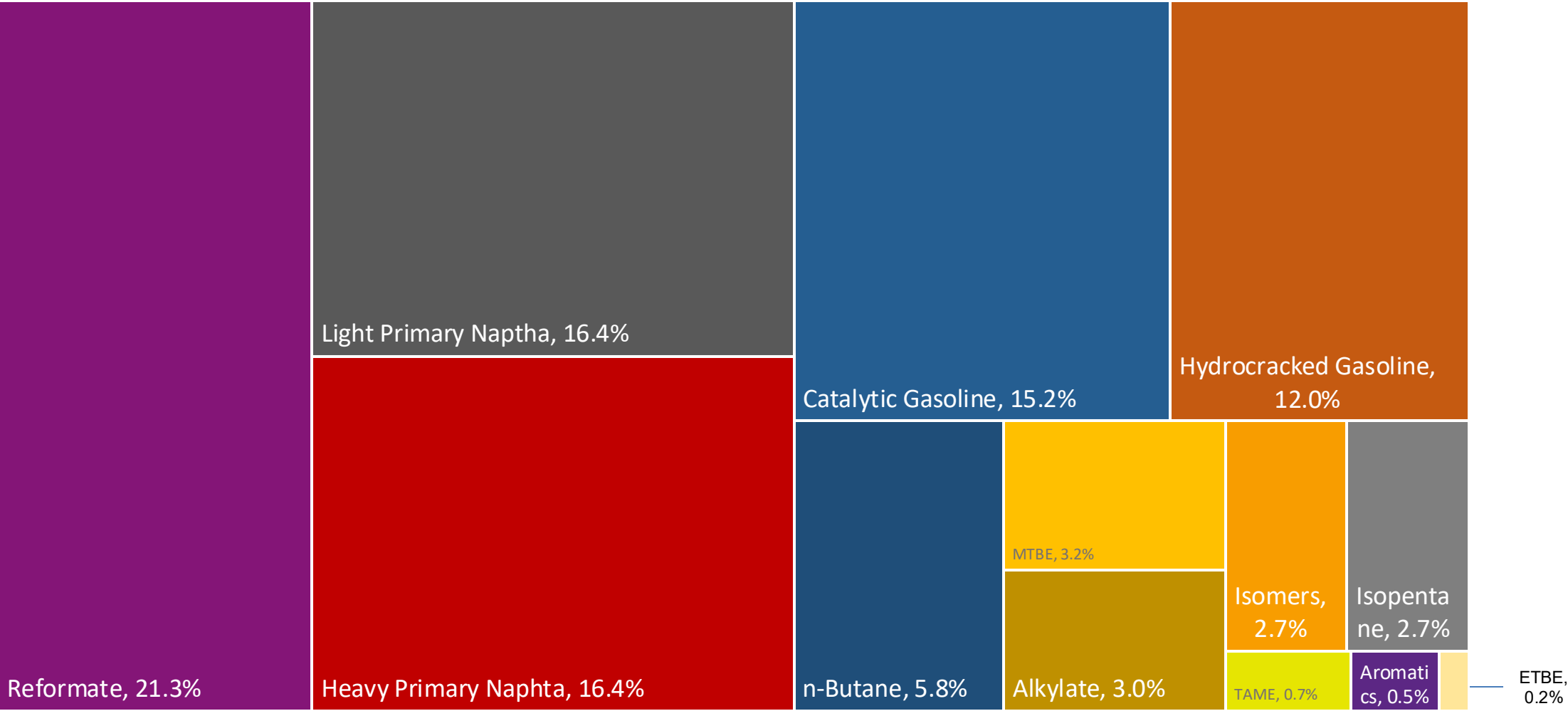
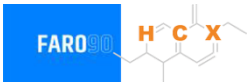
Gasoline is a blend of a base gasoline and other components. Each component provides different properties to the final blend, for example, isomerates, alkylates and butanes increase the octane. The components commonly used in Europe are:

- Catalytic Gasoline
- Light Primary Naphta
- Heavy Primary Naphta
- Reformate
- Isopentane
- Alkylate
- Isomerate
- Hydrocracked Gasoline
- Coker Naphta
- n-Butane
- MTBE
- ETBE
- TAME
- Aromatics

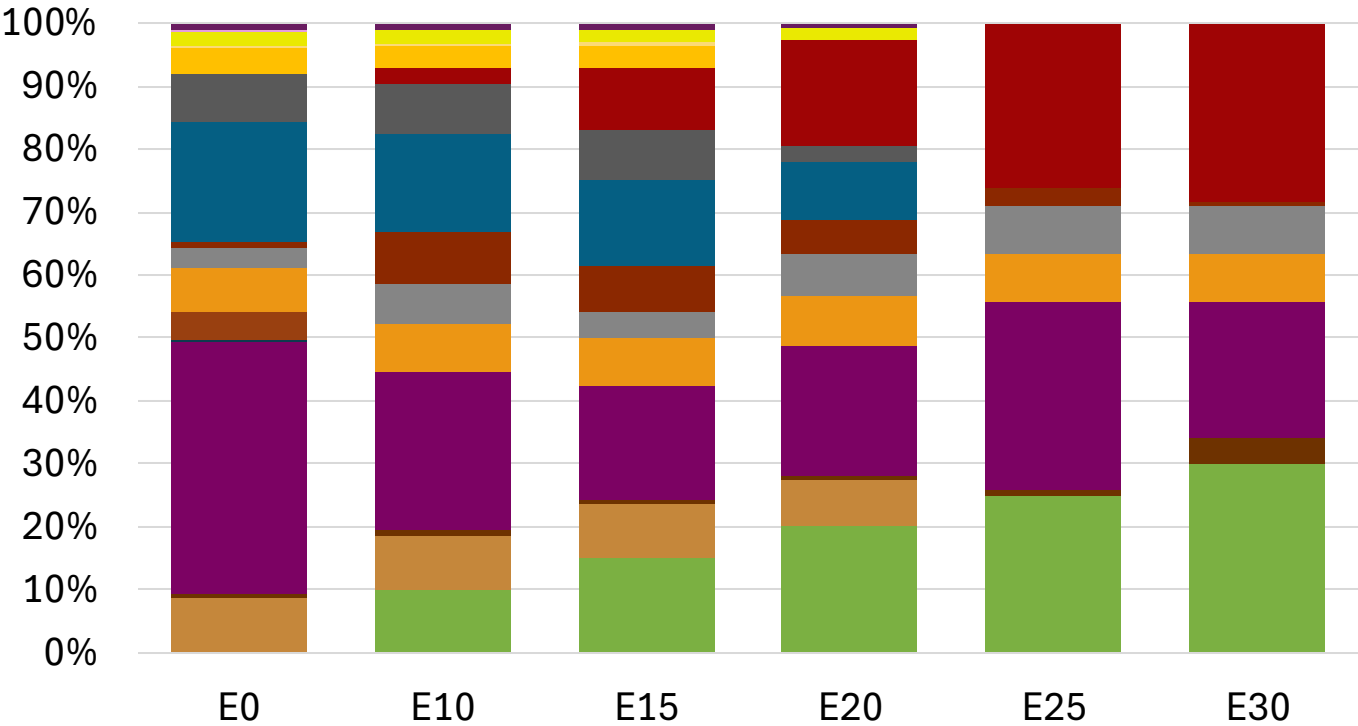
Source: Faro90



Italy Available Blending Components



Italy – Gasoline 95 – Constant Octane

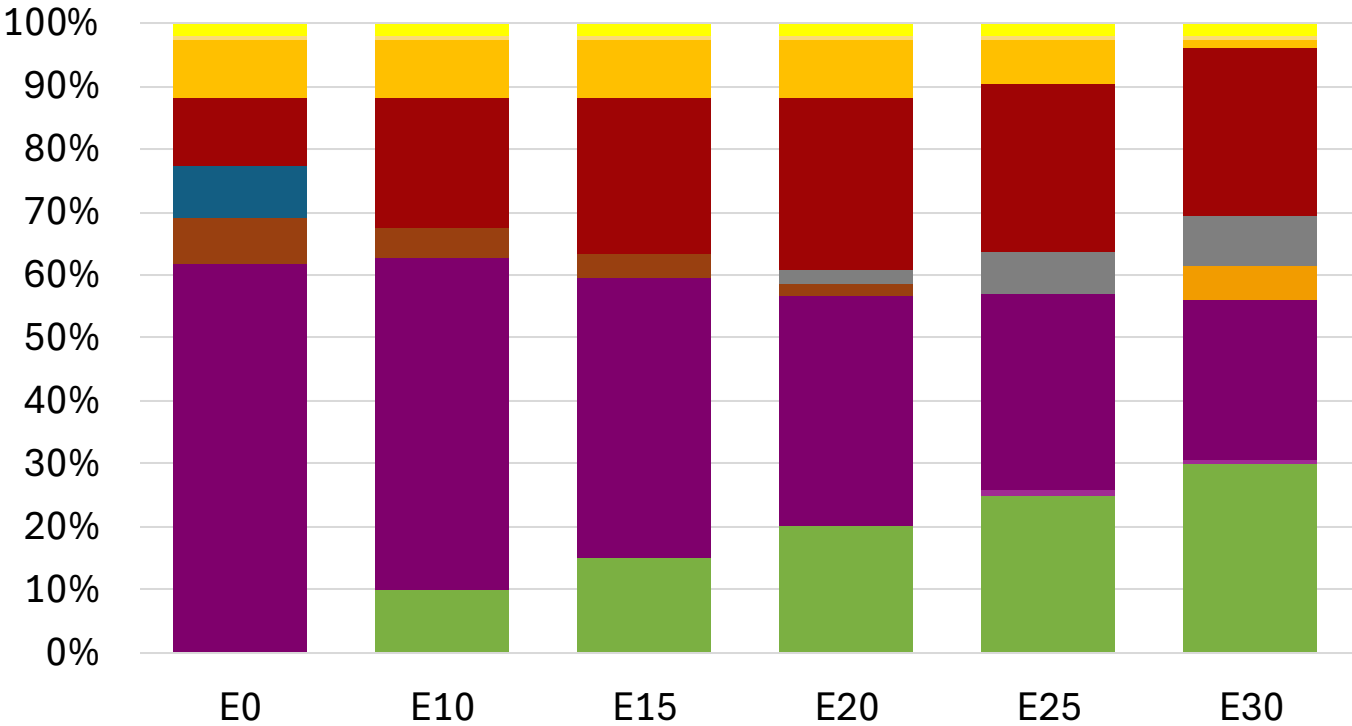
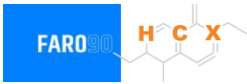


Average CIF 2024
Prices

Octane (RON)	95.0	95.0	95.0	95.0	95.4	95.8
Price (US\$/gal)	\$ 2.258	\$ 2.145	\$ 2.062	\$ 1.986	\$ 1.891	\$ 1.817

Source: Faro90

Italy - Gasoline 98 - Constant Octane

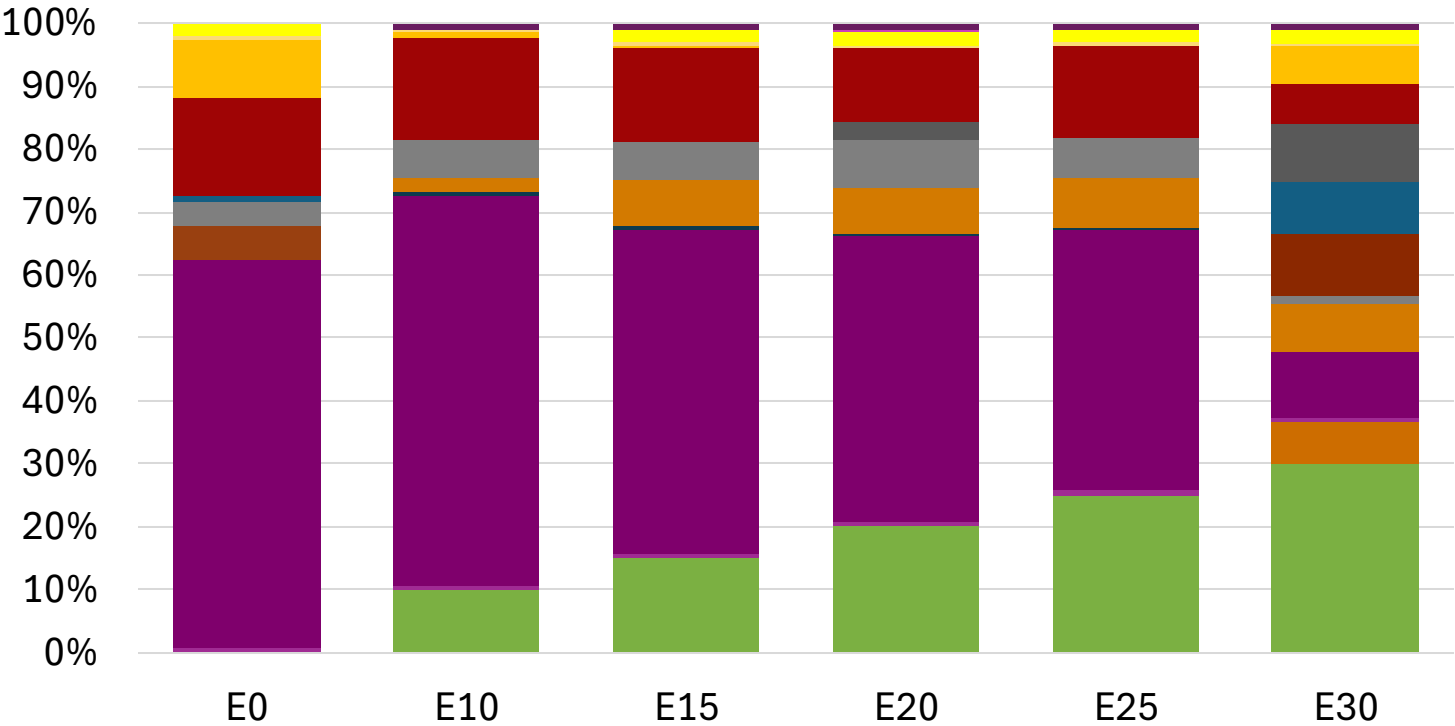
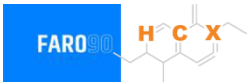


Average CIF 2024 Prices

Octane (RON)	98.0	98.0	98.0	98.0	98.0	98.0
Price (US\$/gal)	\$ 2.307	\$ 2.188	\$ 2.119	\$ 2.049	\$ 1.979	\$ 1.926

Source: Faro90

Italy – Gasoline 95 – Increased Octane

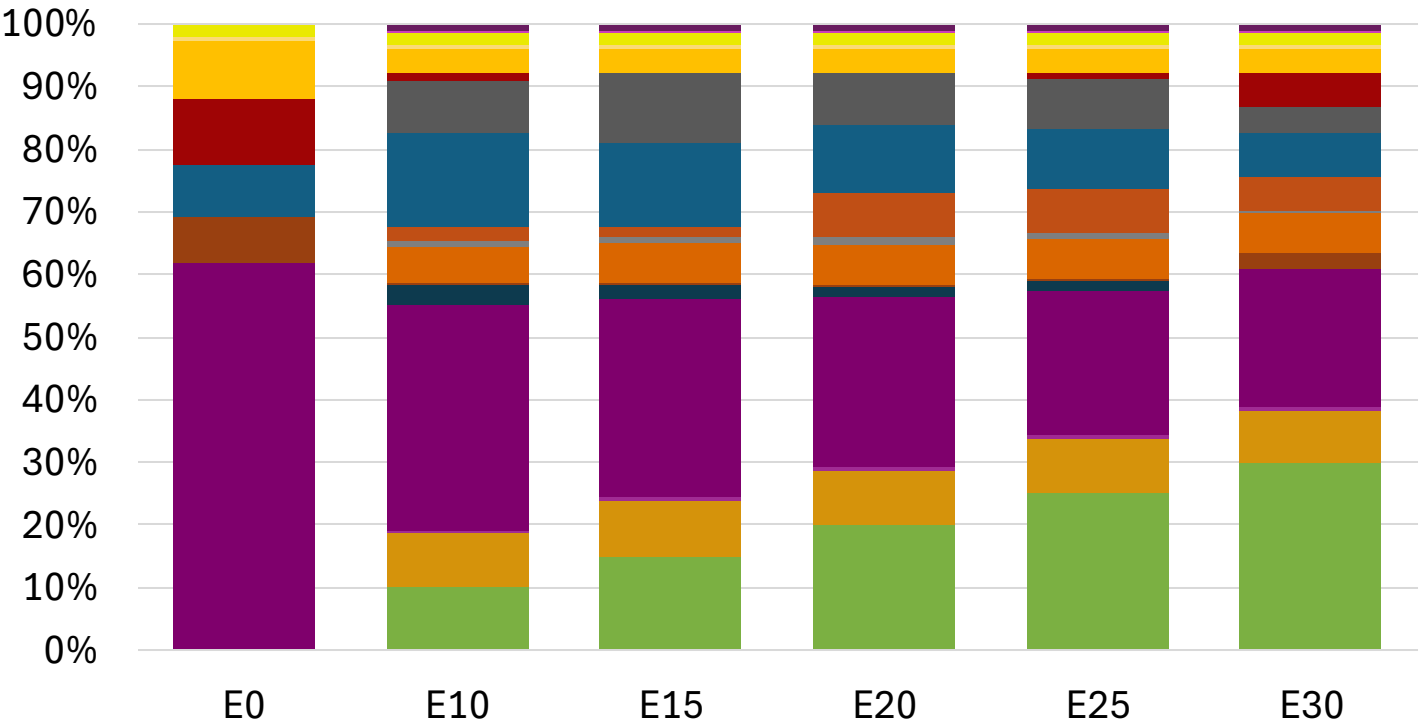


Average CIF 2024 Prices

Octane (RON)	95.0	95.9	97.5	99.0	100.5	101.8
Price (US\$/gal)	\$ 2.177	\$ 2.177	\$ 2.177	\$ 2.177	\$ 2.177	\$ 2.177

Source: Faro90

Italy – Gasoline 98 – Increased Octane

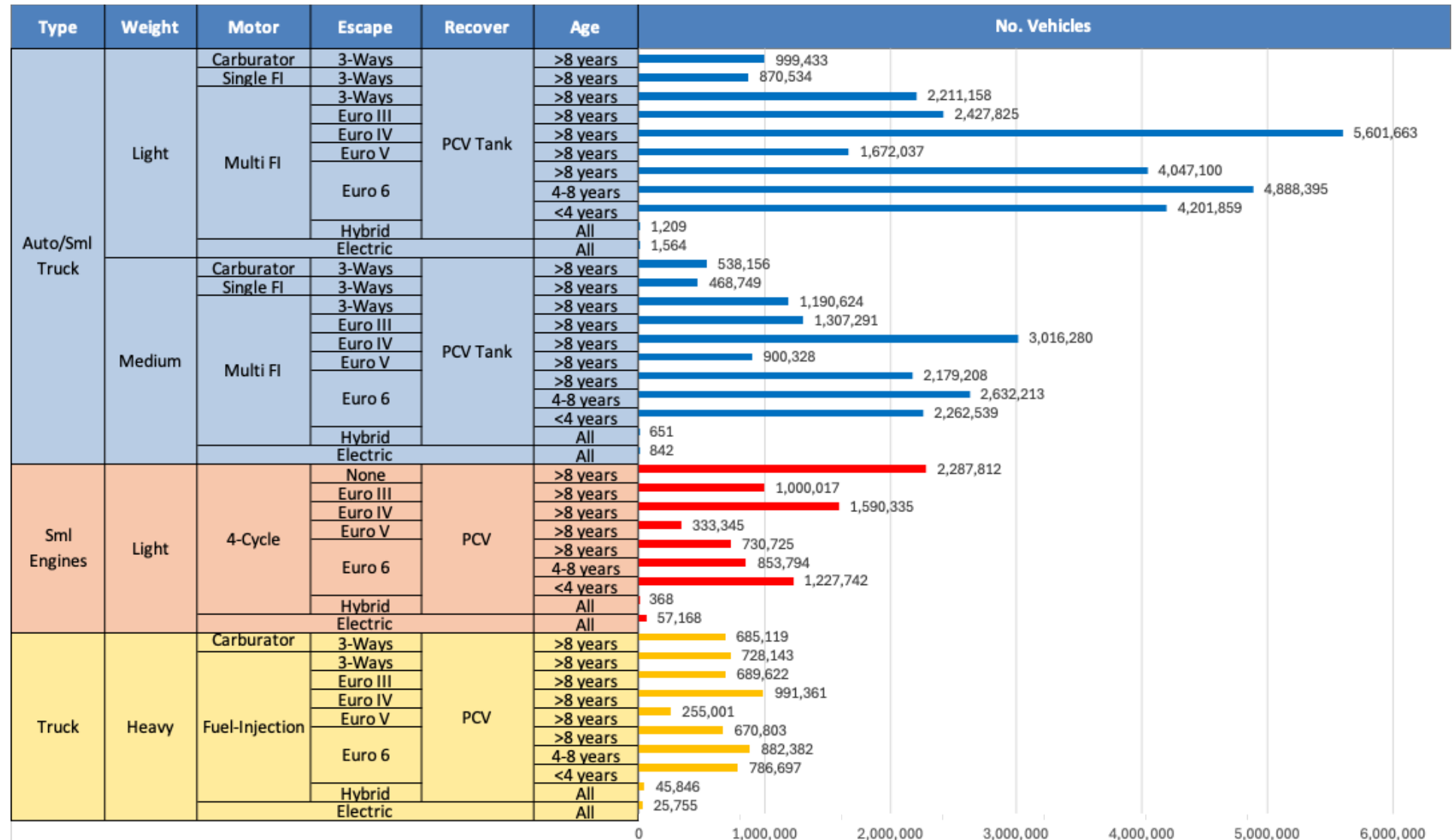


Average CIF 2024
Prices

Octane (RON)	98.0	98.8	100.1	101.3	102.6	104.1
Price (US\$/gal)	\$ 2.307	\$ 2.307	\$ 2.307	\$ 2.307	\$ 2.307	\$ 2.307

Source: Faro90

Gasoline Vehicle Fleet - Italy

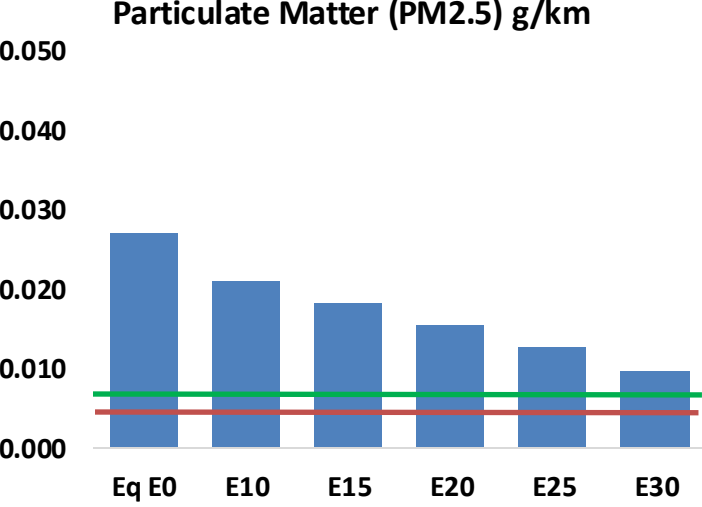
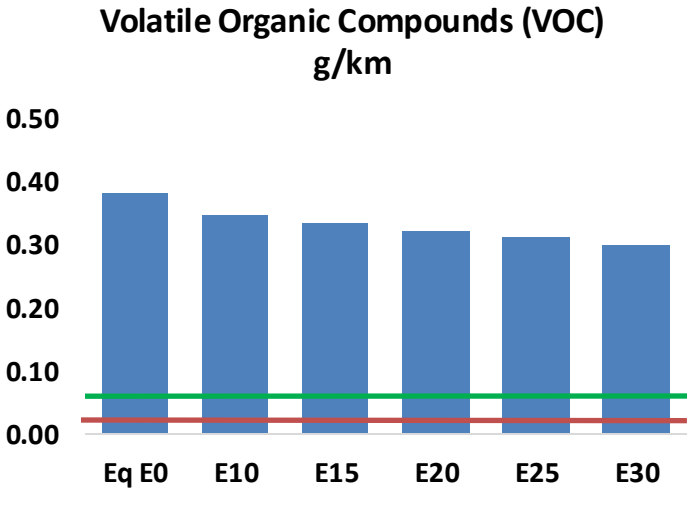
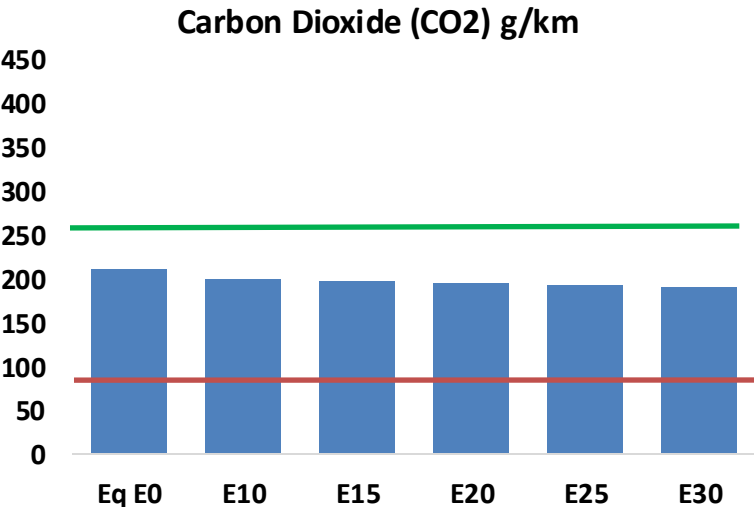
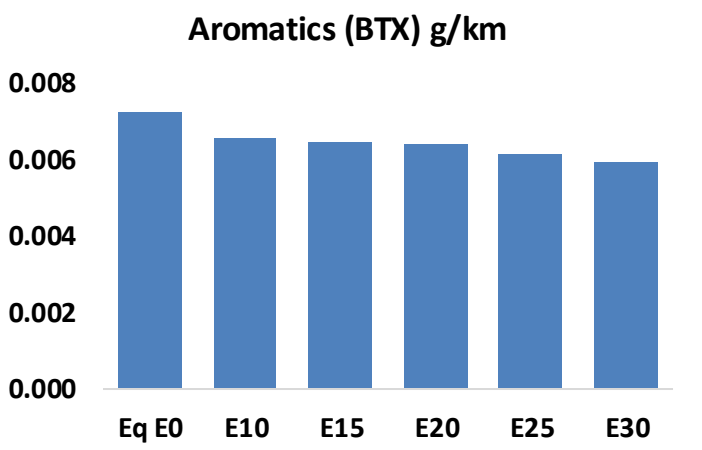
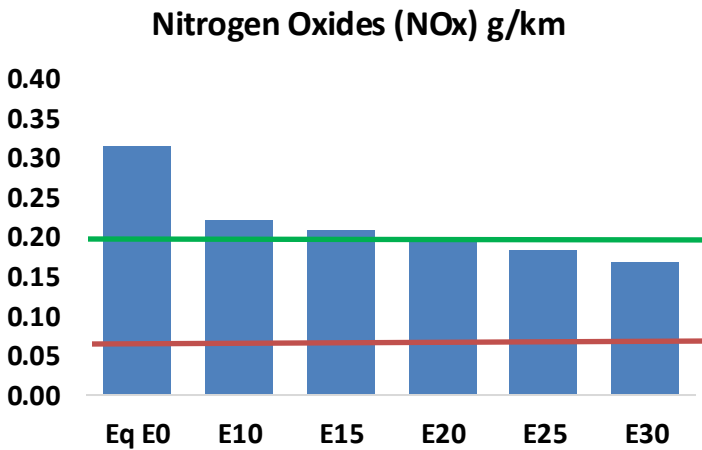
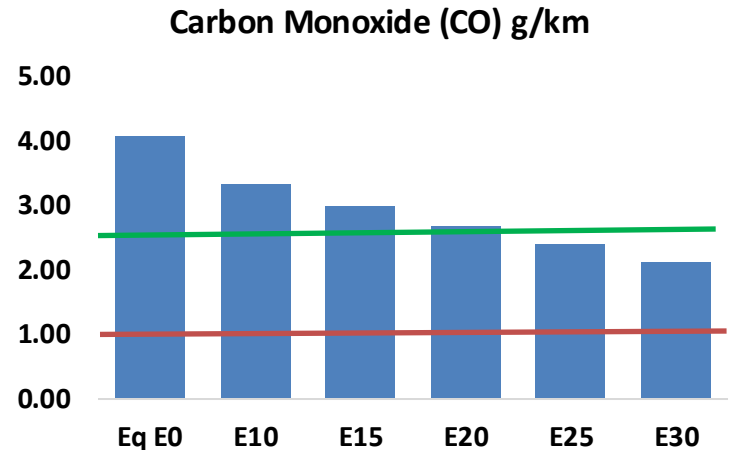
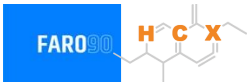


Vehicle Fleet: 55,261,692
Gasoline Vehicle Fleet: 19,966,496

Source: Eurostat, ACEA

Italy – Vehicular Emissions

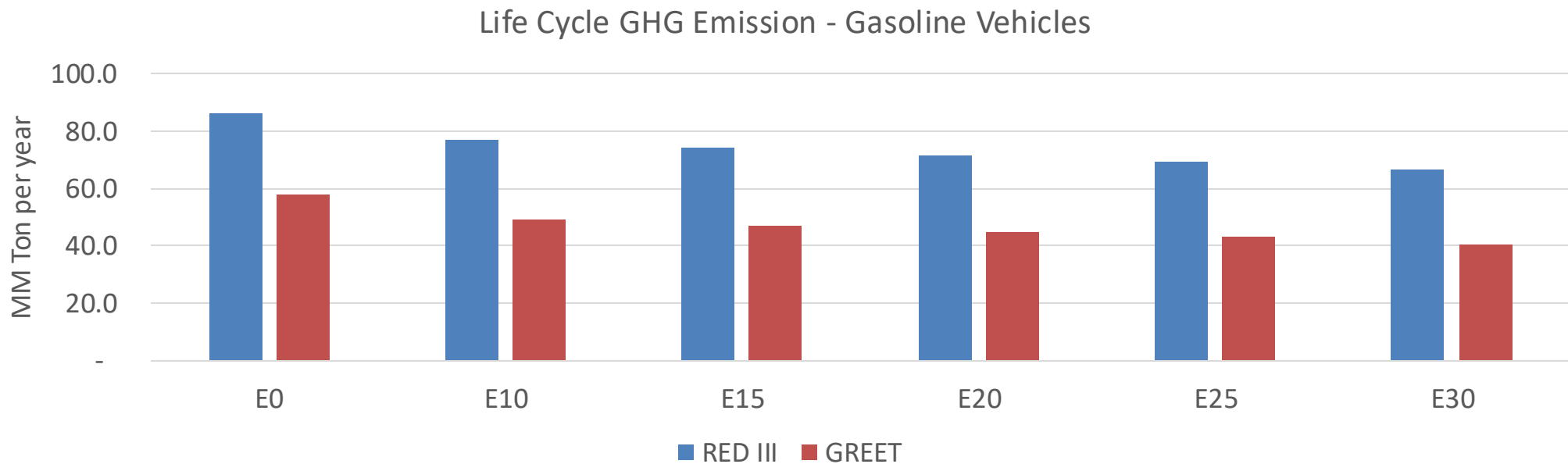
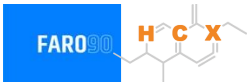
TIER III BIN 125 United States
Euro 6 Standard



Italy – Vehicular Emissions

Vehicular Fleet Emissions (kg/day)	Eq E0	E5	E10	E15	E20	E25	E30	Reduction (%)	
								E10	E20
CO	1,868,067	1,694,065	1,520,062	1,371,545	1,219,358	1,101,511	973,684	-19%	-35%
CO2	96,686,516	94,200,291	91,852,190	90,005,478	89,091,652	88,644,245	87,010,277	-5%	-8%
VOC	175,091	166,801	158,512	152,728	147,372	143,394	137,590	-9%	-16%
NOx	144,409	108,307	101,087	95,274	89,852	83,873	77,548	-30%	-38%
SOx	1,112	1,028	944	873	805	731	653	-15%	-28%
NH3	33,122	32,794	32,466	32,620	32,987	33,245	33,460	-2%	0%
N2O	1,578	1,570	1,563	1,571	1,603	1,621	1,640	-1%	2%
CH4	40,511	40,511	40,511	41,118	42,010	42,779	43,509	0%	4%
Lead	-	-	-	-	-	-	-	0%	0%
Butadiene	1,540	1,462	1,384	1,328	1,275	1,236	1,179	-10%	-17%
Acetaldehyde	3,291	3,777	5,542	8,440	11,498	13,139	15,525	68%	249%
Formaldehyde	10,962	10,658	12,424	14,588	15,283	16,907	18,406	13%	39%
Benzene	3,310	3,181	3,014	2,967	2,942	2,814	2,723	-9%	-11%
PM2.5	5,909	5,246	4,584	3,981	3,381	2,743	2,079	-22%	-43%
PM10	6,477	5,751	5,025	4,364	3,706	3,007	2,279	-22%	-43%

Italy – Gasoline Vehicle Fleet Life Cycle GHG Emissions



Country Emissions 2024 (MMT/año)	357.1
Target @ 2035 (MMT/year)	201.0
Delta	156.1

%Reduction vs E0	E0	E10	E15	E20	E25	E30
GREET 2024	0.0%	14.7%	18.8%	22.3%	25.7%	30.0%
RED II/III	0.0%	10.8%	14.0%	16.7%	19.3%	22.4%

% Target@2035	E0	E10	E15	E20	E25	E30
GREET 2024	0.0%	5.5%	7.0%	8.3%	9.5%	11.1%
RED II/III	0.0%	6.0%	7.7%	9.2%	10.6%	12.4%

Source: Greet, RED II/III, IPCC, climateactiontracker.org, F90

Ethanol Gasoline Blending in Latvia

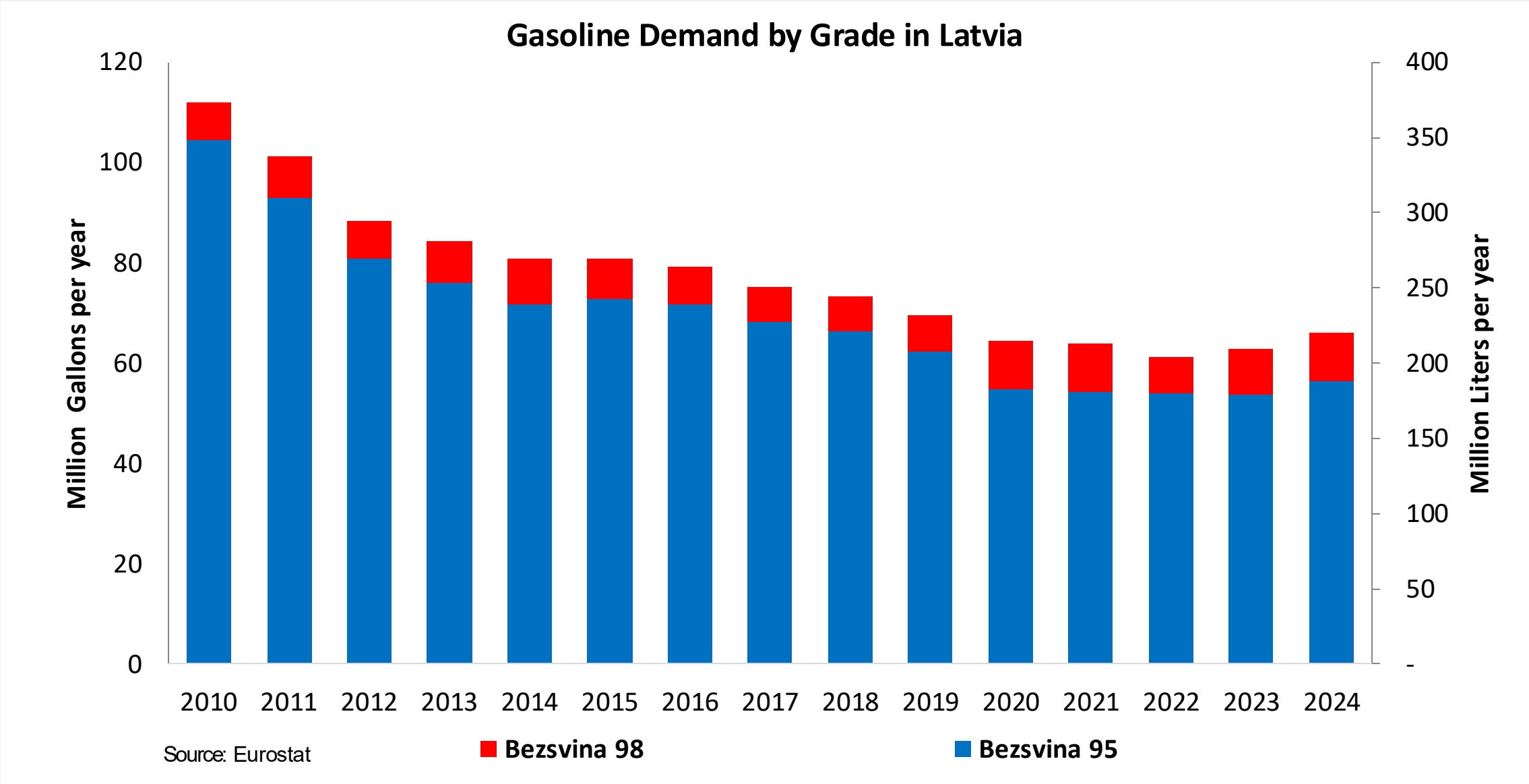


In 2024, gasoline consumption in Latvia was 220 million liters (58 million gallons) and growth rate was -1.1%. There is no gasoline production. Gasoline net imports were 200 million liters (53 million gallons). Latvia complies with the European gasoline standards and offers two main grades: RON 95 and RON 98) with an average octane of 95.4 RON.

In 2024, ethanol consumption in Latvia was 23 million liters (6 million gallons) representing 10.4% of gasoline demand. Ethanol production and Net Imports were 25 and -3 million liters (6 and -1 million gallons) correspondingly. Ethanol source Cereals 100%.

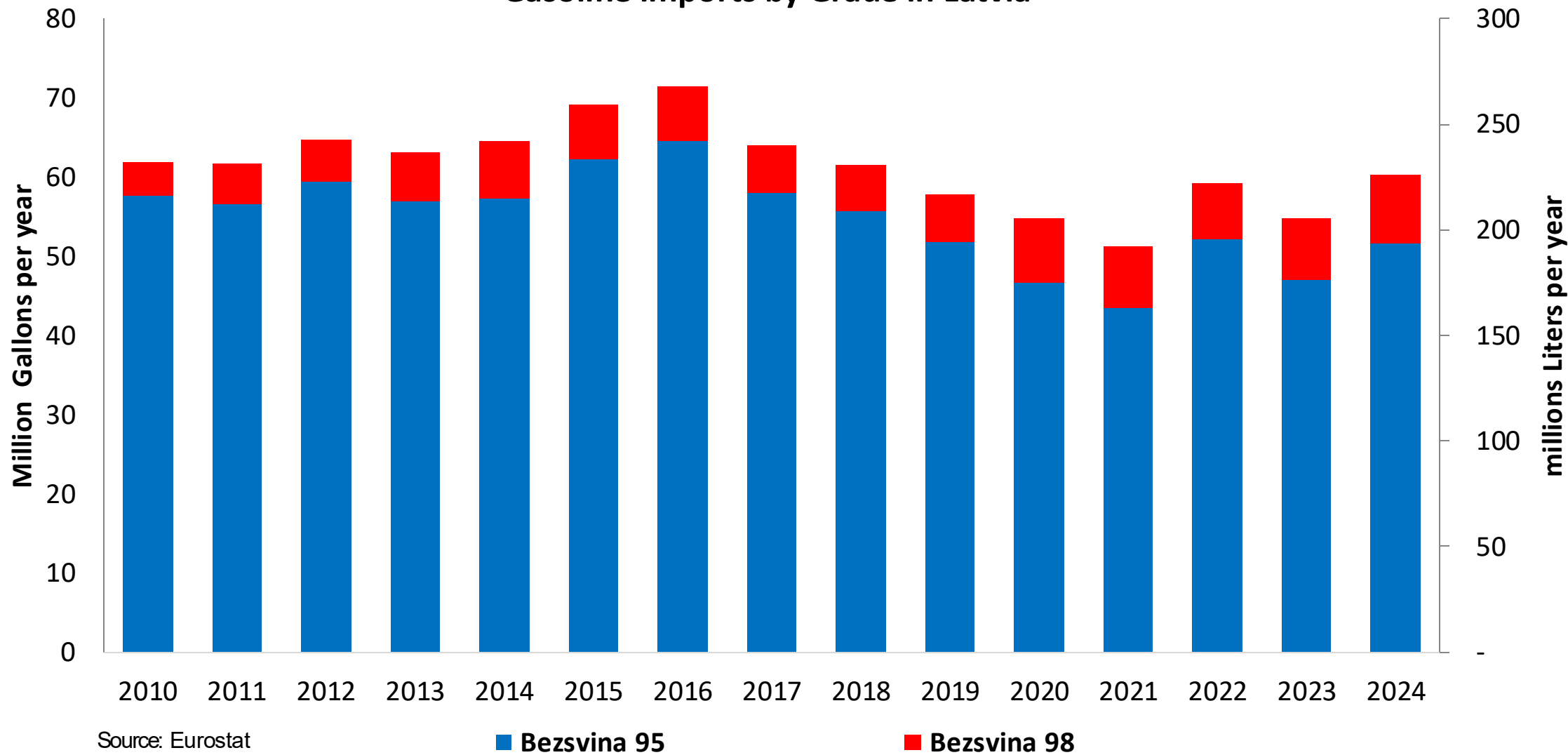
Source: Eurostat, European Commission

Gasoline Demand in Latvia

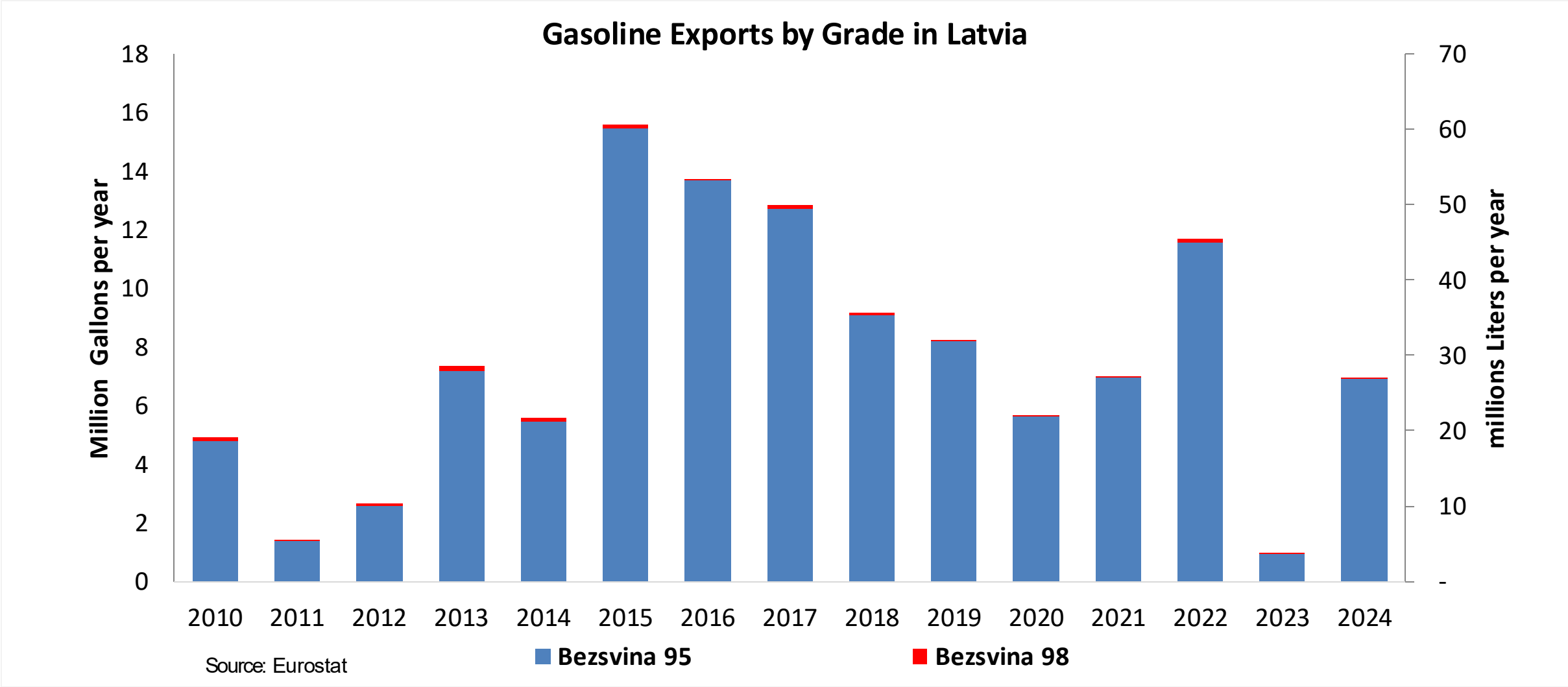


Gasoline Imports in Latvia

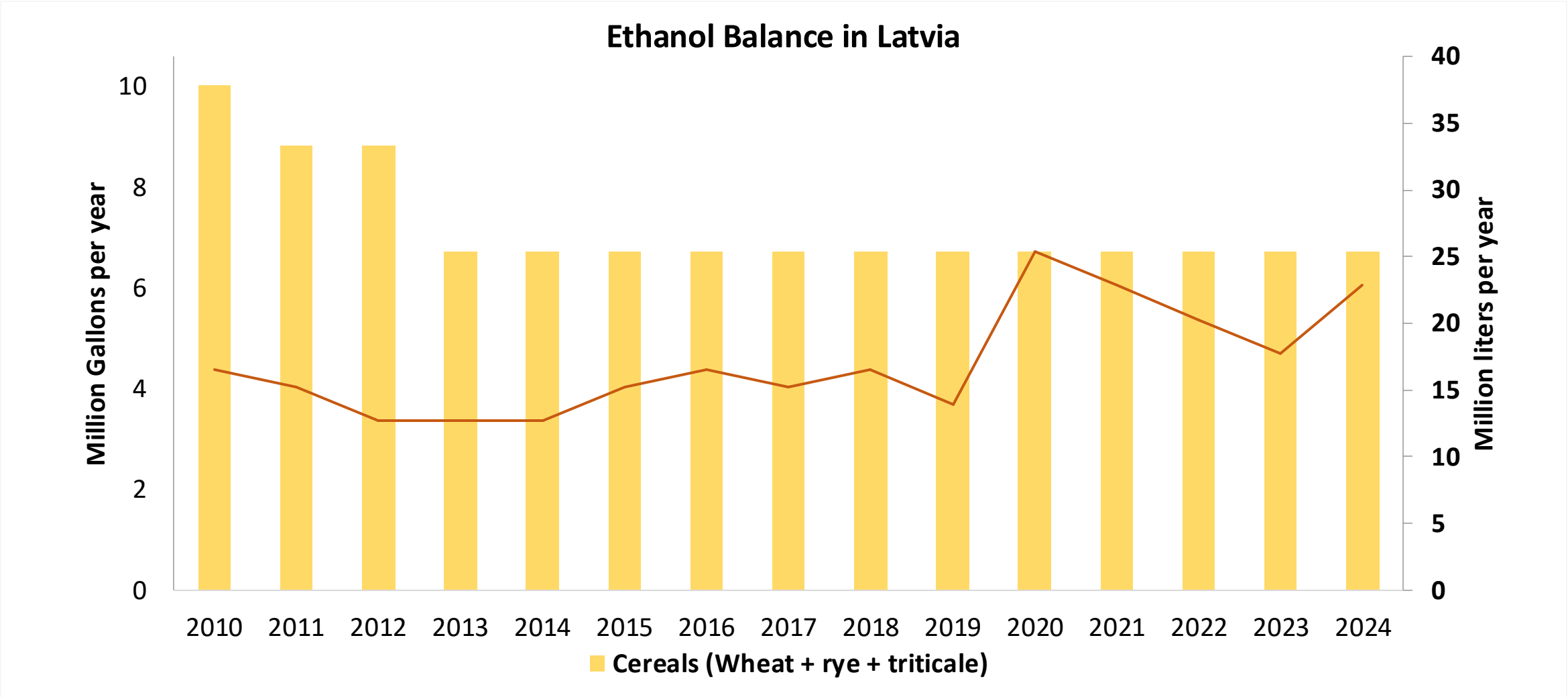
Gasoline imports by Grade in Latvia



Gasoline Exports in Latvia

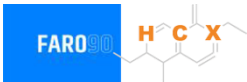


Ethanol Balance in Latvia



Source: Eurostat, Latvia Statistics

Gasoline Quality in Latvia

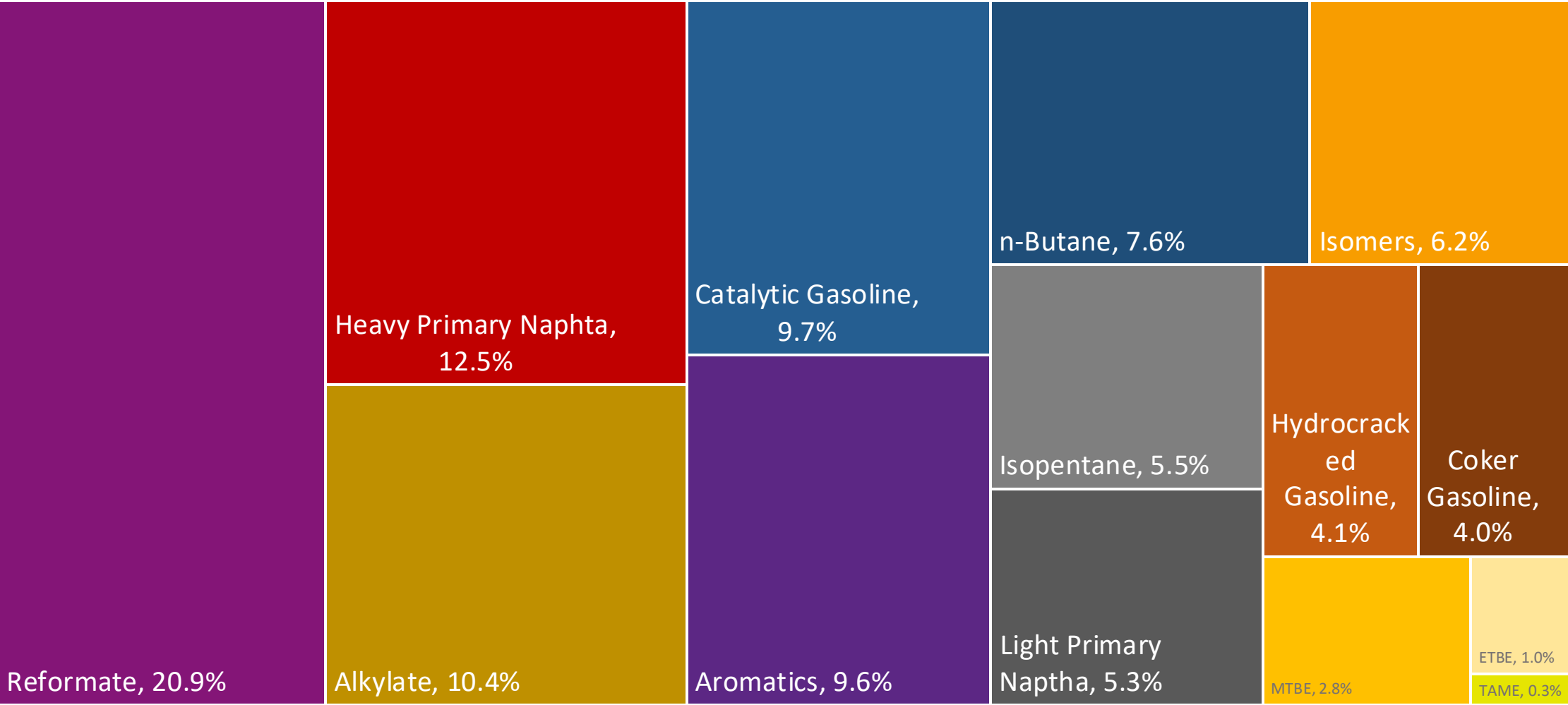
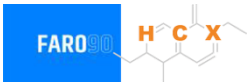


Name	EN 228:2025 (Euro 6); Country Level Fuel Ordinances							
Implementation Date	2025							
Applicability	All EU Contries + UK							
Grade	RON 95 E5		RON 95 E10		RON 98 E5		RON 98 E10	
	Min	Max	Min	Max	Min	Max	Min	Max
RON	95.0		95.0		98.0		98.0	
MON	> 85	> 88	> 85	> 88	> 85	> 88	> 85	> 88
AKI								
Benzene (%vol)	< 1							
Aromatics (% vol)	< 35							
Olefins (%vol)	< 18							
Lead (g/L)	< 0.005							
Manganese (mg/L)	< 2,0							
Sulfur (ppm)	10.0							
Oxygen (%w)*	< 2,7	< 3,7	< 2,7	< 3,7	< 2,7	< 3,7	< 2,7	< 3,7
Ethanol (%vol)	< 5	< 10	< 5	< 10	< 5	< 10	< 5	< 10
PVR 37.8°C (psi)	<> 45 - 100 kPa in 6 Volatility Classes							
Overall Biofuels (%cal)	-							
Advanced Biofuels (%cal)	0.2							
Bioethanol in Gasoline Sales (%cal)	-							
GHG Emission Reduction (%)	-							

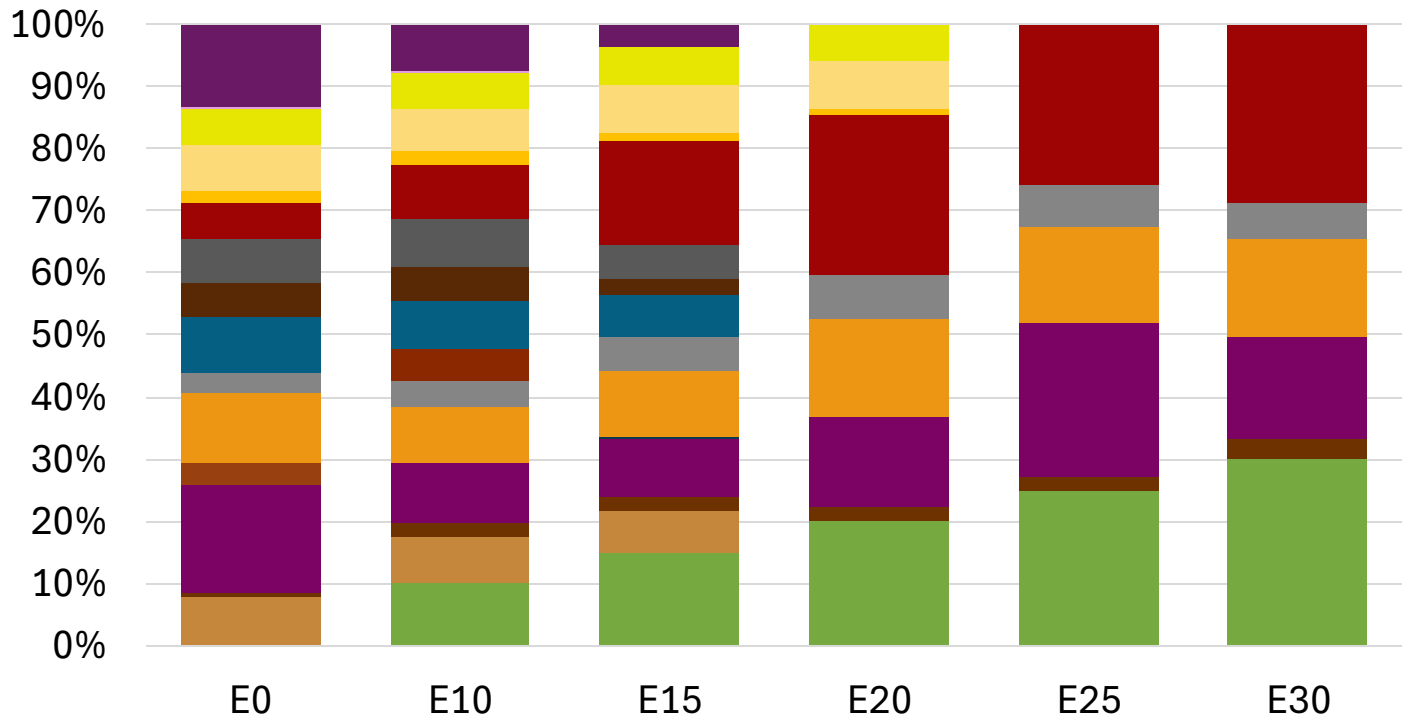
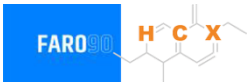
* Methanol 3%, Iso-propyl alcohol 12%, Ter-butyl alcohol 15%, Iso-butyl alcohol 15%, Ethers +5 Carbons 22%, Other Oxygenates 15%

Source: European Commision

Latvia Available Blending Components



Latvia – Gasoline 95 – Constant Octane

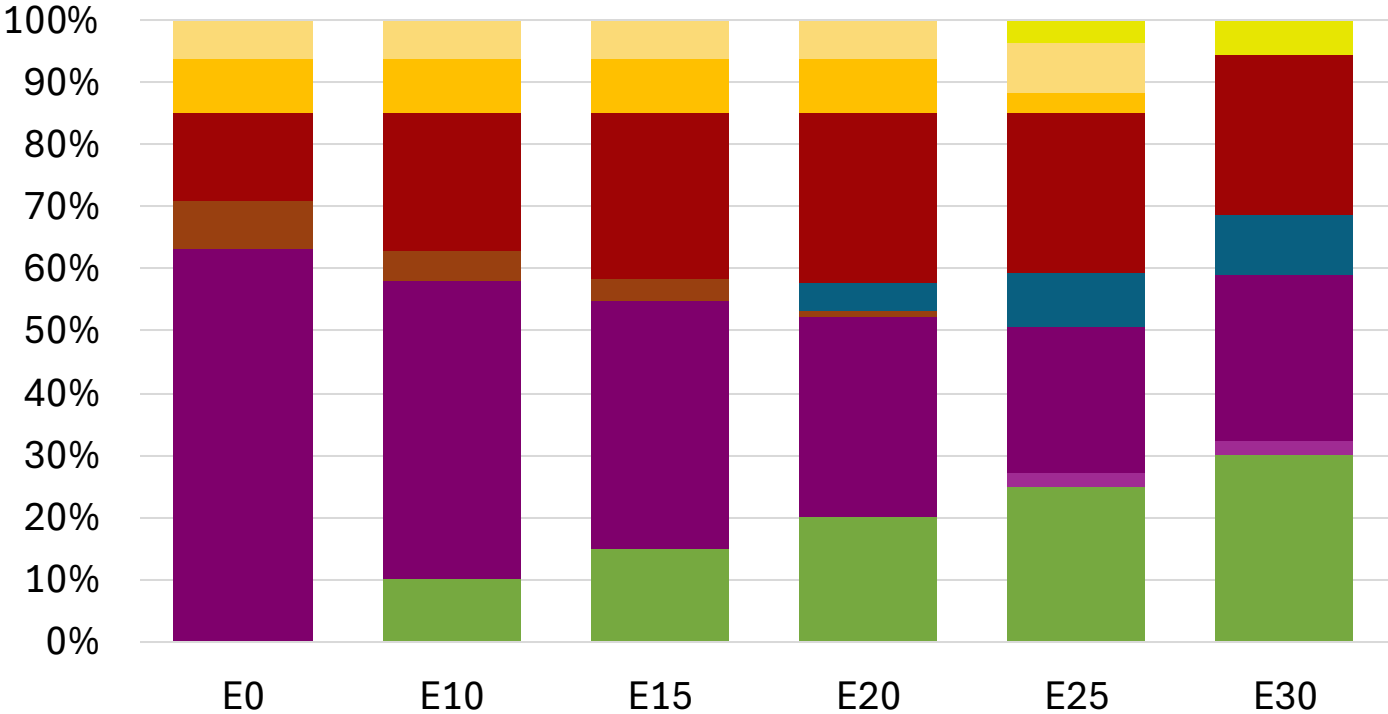
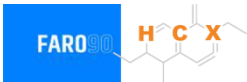


Average CIF 2024 Prices

Octane (RON)	95.0	95.0	95.0	95.0	95.0	95.8
Price (US\$/gal)	\$ 2.274	\$ 2.136	\$ 2.036	\$ 1.921	\$ 1.875	\$ 1.822

Source: Faro90

Latvia - Gasoline 98 - Constant Octane



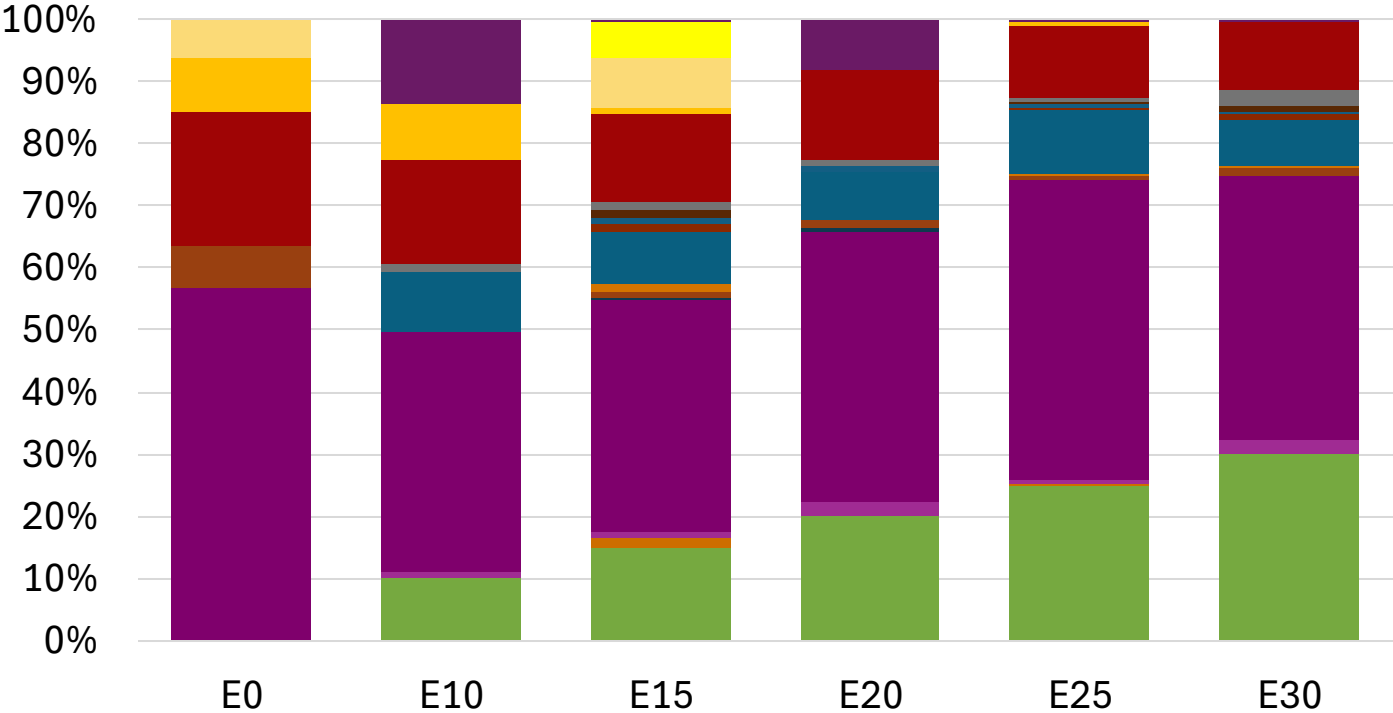
Ethanol
Alkylate
Raffinate
Reformate
n-Butane
Isomerate
Isopentane
Hydrocracked Gasoline
Catalytic Gasoline
Coker Naphtha
Light Primary Naphtha
Heavy Primary Naphtha
MTBE
ETBE
TAME
Aromatics

Average CIF 2024 Prices

Octane (RON)	98.0	98.0	98.0	98.0	98.0	98.0
Price (US\$/gal)	\$ 2.294	\$ 2.177	\$ 2.108	\$ 2.036	\$ 1.971	\$ 1.921

Source: Faro90

Latvia – Gasoline 95 – Increased Octane



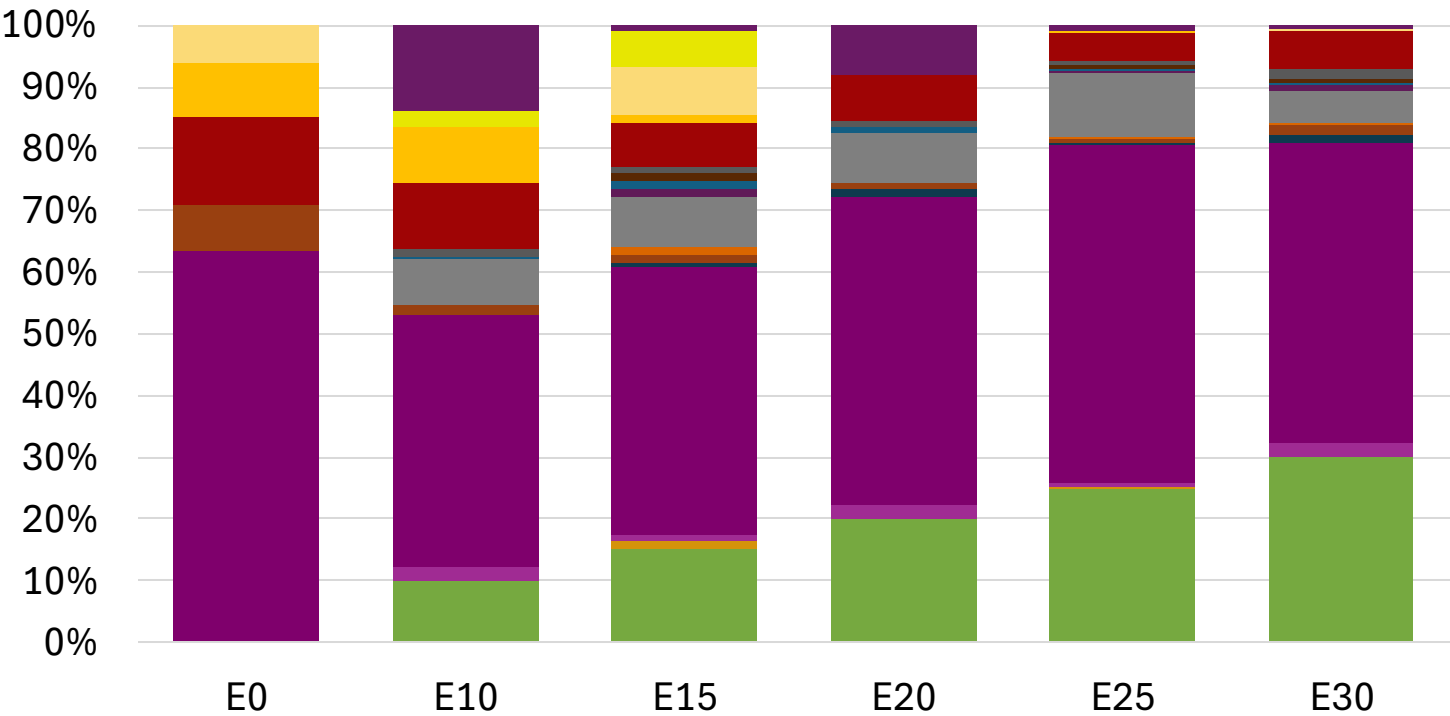
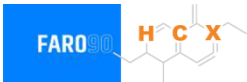
Ethanol
Alkylate
Raffinate
Reformate
n-Butane
Isomerate
Isopentane
Hydrocracked Gasoline
Catalytic Gasoline
Coker Naphtha
Light Primary Naphtha
Heavy Primary Naphtha
MTBE
ETBE
TAME
Aromatics

Average CIF 2024
Prices

Octane (RON)	95.0	96.2	98.3	98.7	100.5	101.8
Price (US\$/gal)	\$ 2.162	\$ 2.162	\$ 2.162	\$ 2.162	\$ 2.162	\$ 2.162

Source: Faro90

Latvia – Gasoline 98 – Increased Octane

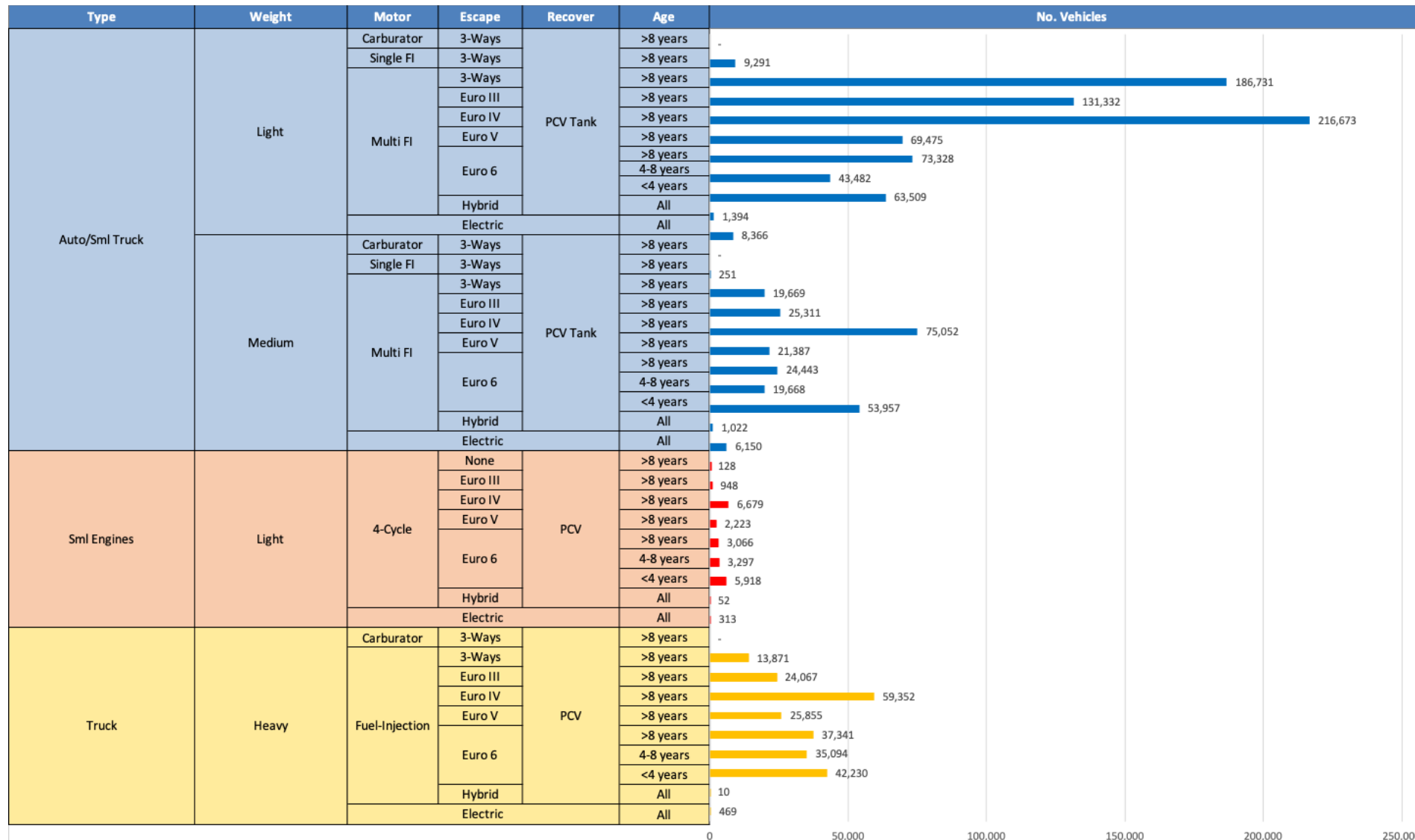


Average CIF 2024 Prices

Octane (RON)	98.0	99.0	100.9	101.2	102.8	104.1
Price (US\$/gal)	\$ 2.294	\$ 2.294	\$ 2.294	\$ 2.294	\$ 2.294	\$ 2.294

Source: Faro90

Gasoline Vehicle Fleet - Latvia

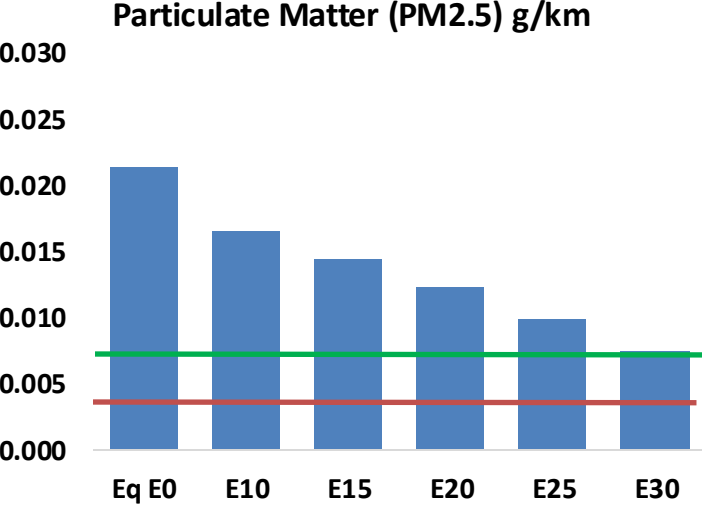
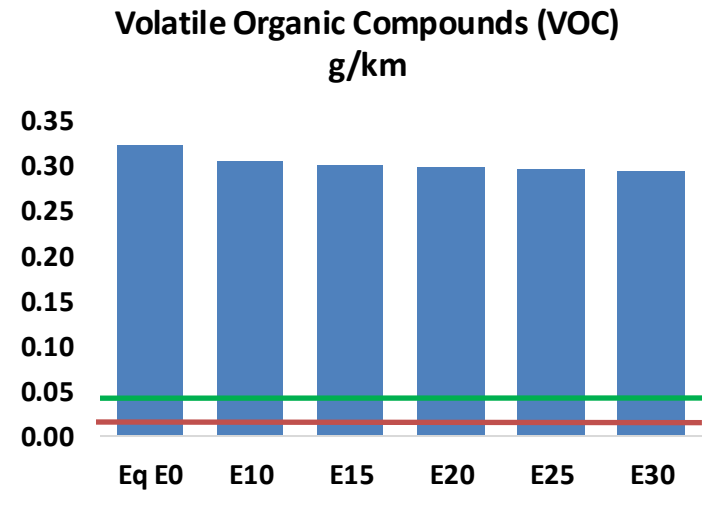
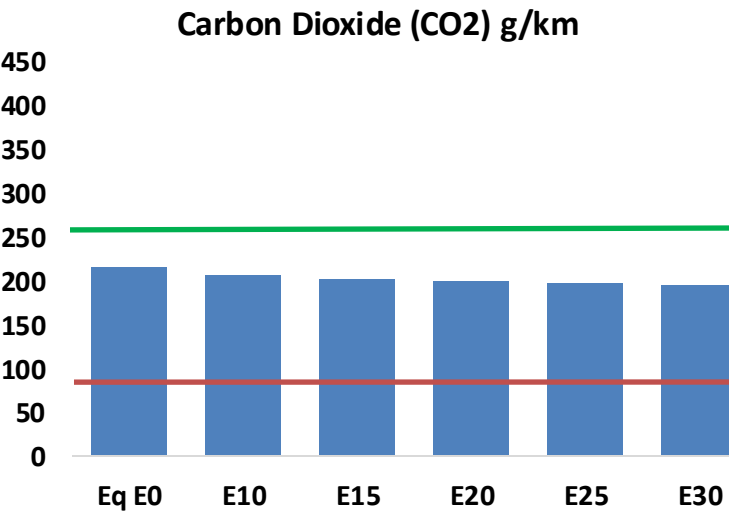
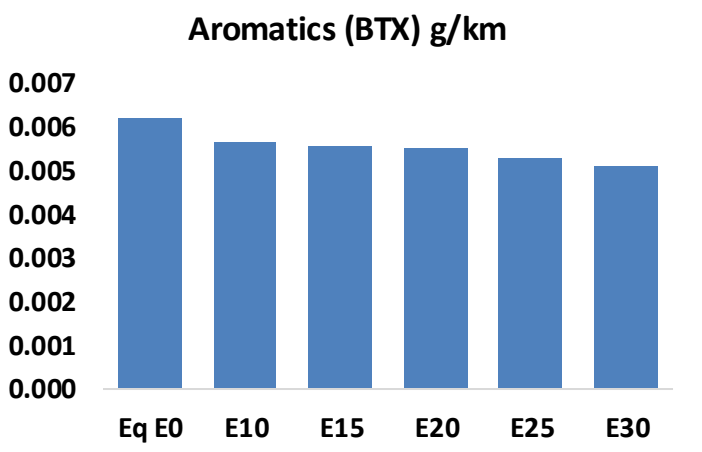
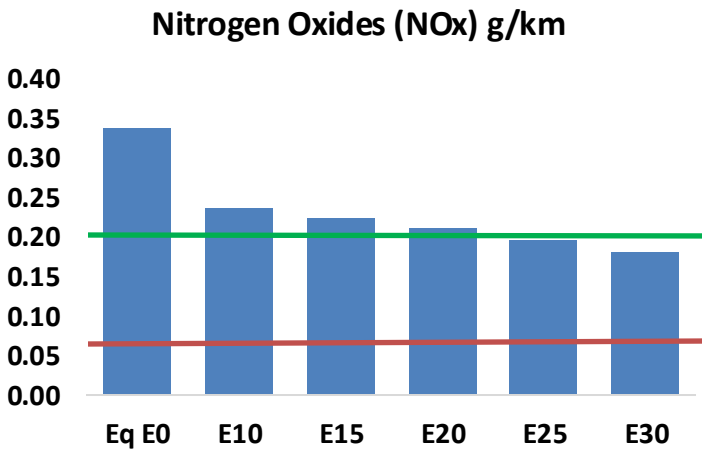
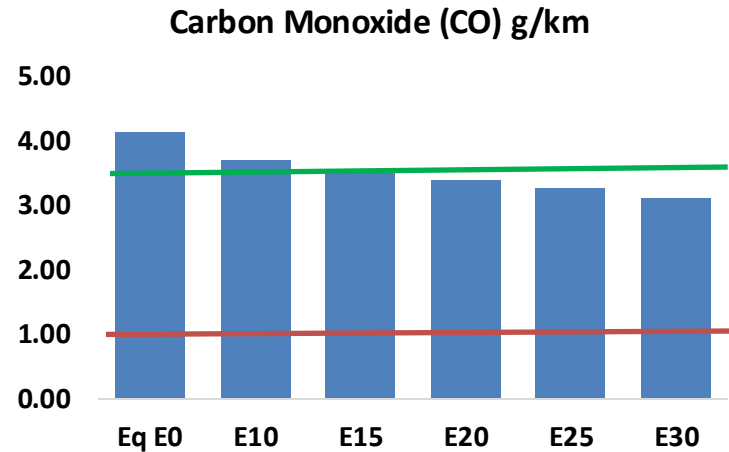
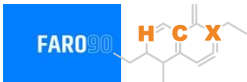


Vehicle Fleet: 1,311,403
Gasoline Vehicle Fleet: 333,918

Source: Eurostat, ACEA

Latvia – Vehicular Emissions

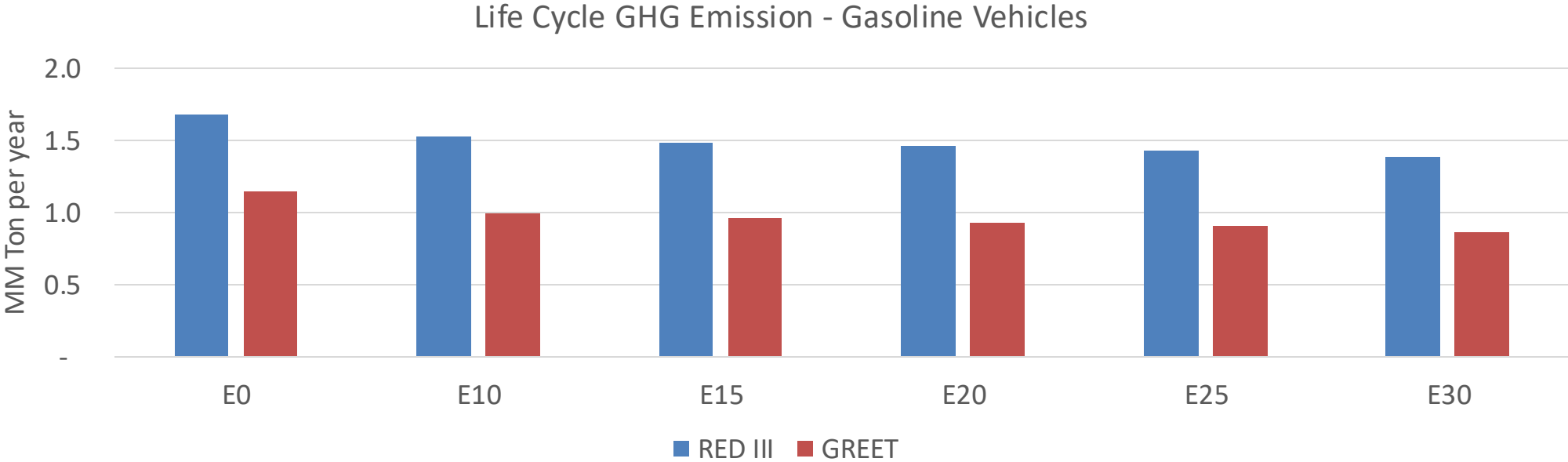
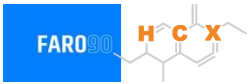
TIER III BIN 125 United States
Euro 6 Standard



Latvia – Vehicular Emissions

Vehicular Fleet Emissions (kg/day)	Eq E0	E5	E10	E15	E20	E25	E30	Reduction (%)	
								E10	E20
CO	36,253	34,335	32,417	31,014	29,681	28,680	27,419	-11%	-18%
CO2	1,902,185	1,853,272	1,807,076	1,770,744	1,752,766	1,741,942	1,709,833	-5%	-8%
VOC	2,835	2,759	2,682	2,646	2,624	2,610	2,578	-5%	-7%
NOx	2,975	2,232	2,083	1,963	1,851	1,728	1,598	-30%	-38%
SOx	23	21	19	18	16	15	13	-15%	-28%
NH3	639	633	626	629	636	641	646	-2%	0%
N2O	29	29	29	29	30	30	30	-1%	2%
CH4	629	629	629	638	652	664	676	0%	4%
Lead	-	-	-	-	-	-	-	0%	0%
Butadiene	35	34	34	33	33	33	33	-4%	-4%
Acetaldehyde	68	78	114	174	237	270	319	68%	249%
Formaldehyde	197	191	223	262	274	303	330	13%	39%
Benzene	55	53	50	49	49	46	45	-9%	-11%
PM2.5	117	104	91	79	67	54	41	-22%	-43%
PM10	71	63	55	48	41	33	25	-22%	-43%

Latvia – Gasoline Vehicle Fleet Life Cycle GHG Emissions



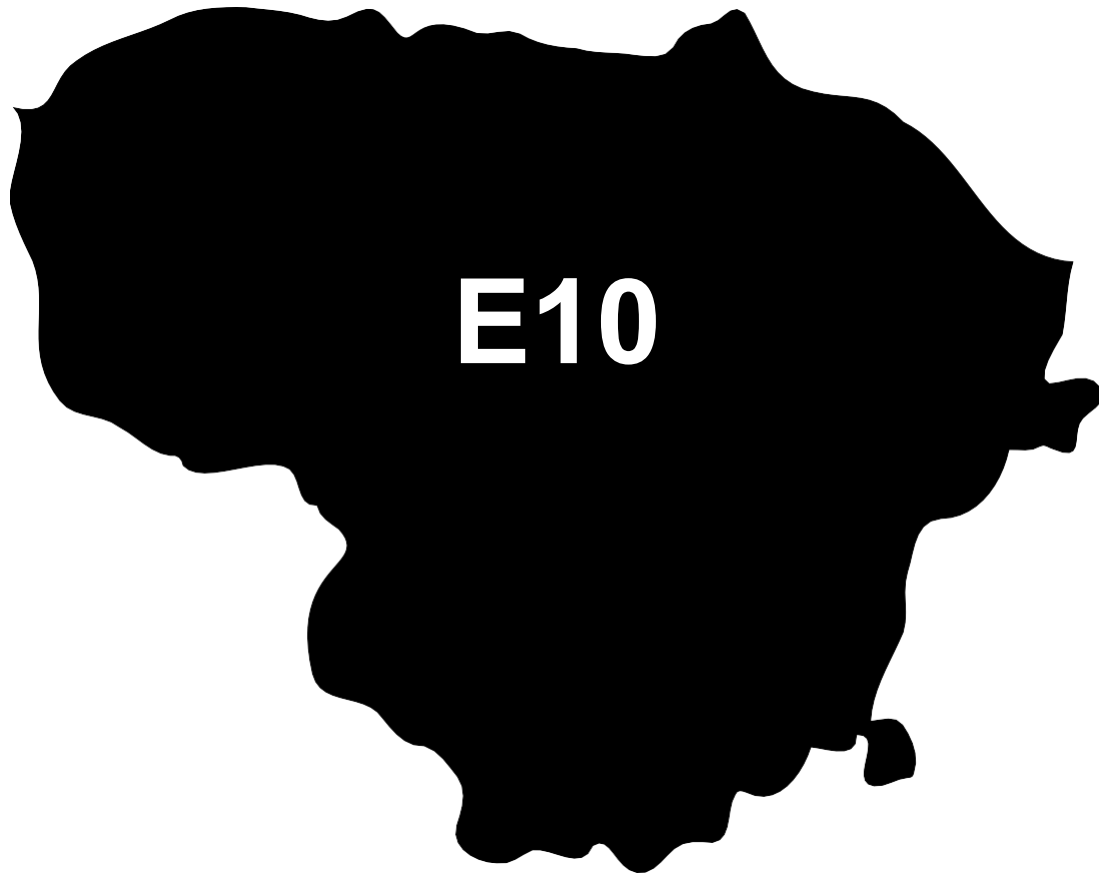
Country Emissions 2024 (MMT/año)	12.1
Target @ 2035 (MMT/year)	6.8
Delta	5.3

%Reduction vs E0	E0	E10	E15	E20	E25	E30
GREET 2024	0.0%	13.1%	16.2%	18.7%	21.1%	24.3%
RED II/III	0.0%	9.3%	11.6%	13.4%	15.2%	17.5%

% Target@2035	E0	E10	E15	E20	E25	E30
GREET 2024	0.0%	2.8%	3.5%	4.0%	4.6%	5.3%
RED II/III	0.0%	3.0%	3.7%	4.3%	4.8%	5.6%

Source: Greet, RED II/III, IPCC, climateactiontracker.org, F90

Ethanol Gasoline Blending in Lithuania

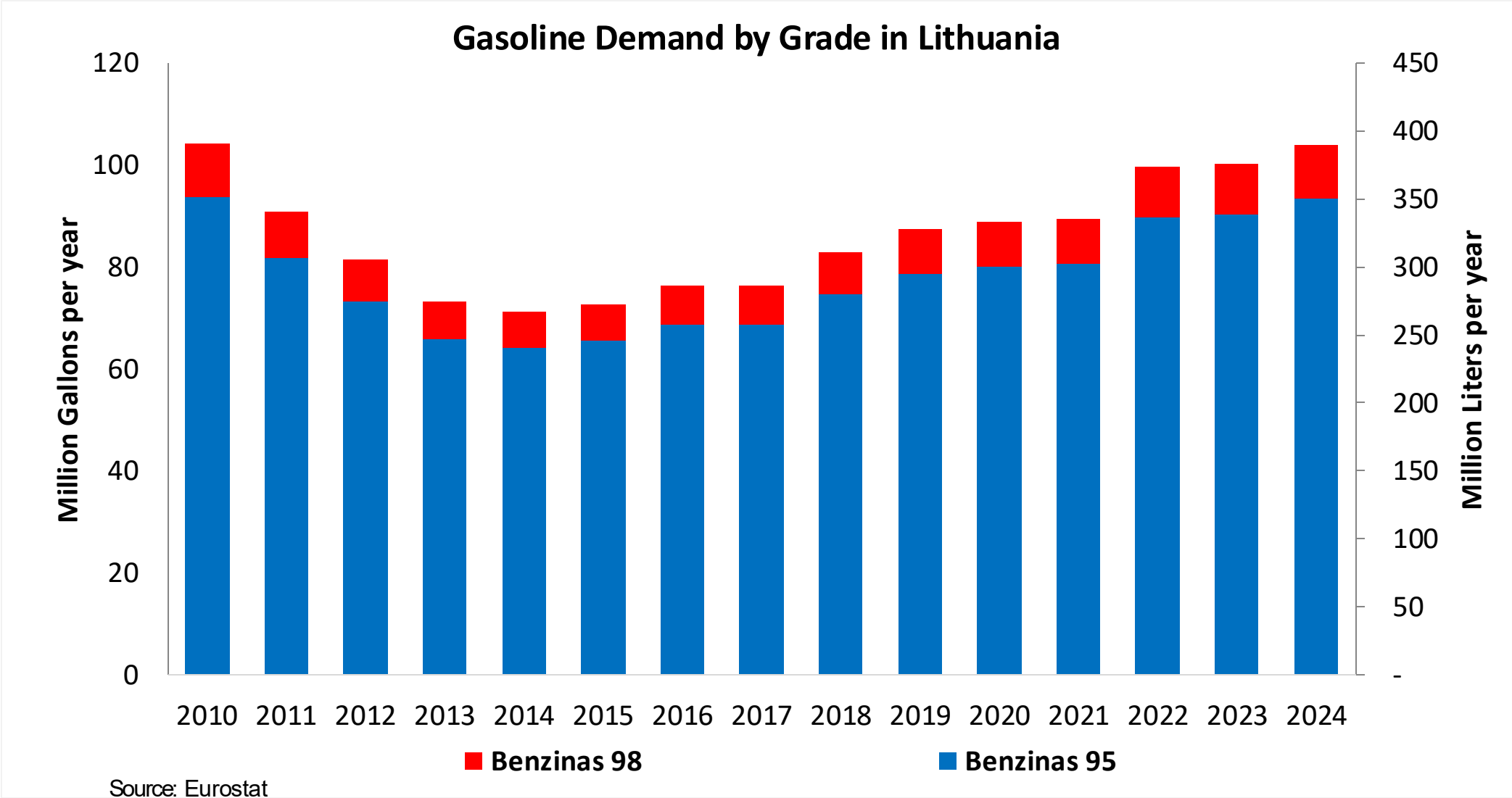


In 2024, gasoline consumption in Lithuania was 390 million liters (103 million gallons) and growth rate was 3.5%. Gasoline production and net imports were 3,762 and -3,361 million liters (994 and -888 million gallons) correspondingly. Lithuania complies with the European gasoline standards and offers two main grades: RON 95 and RON 98) with an average octane of 95.3 RON.

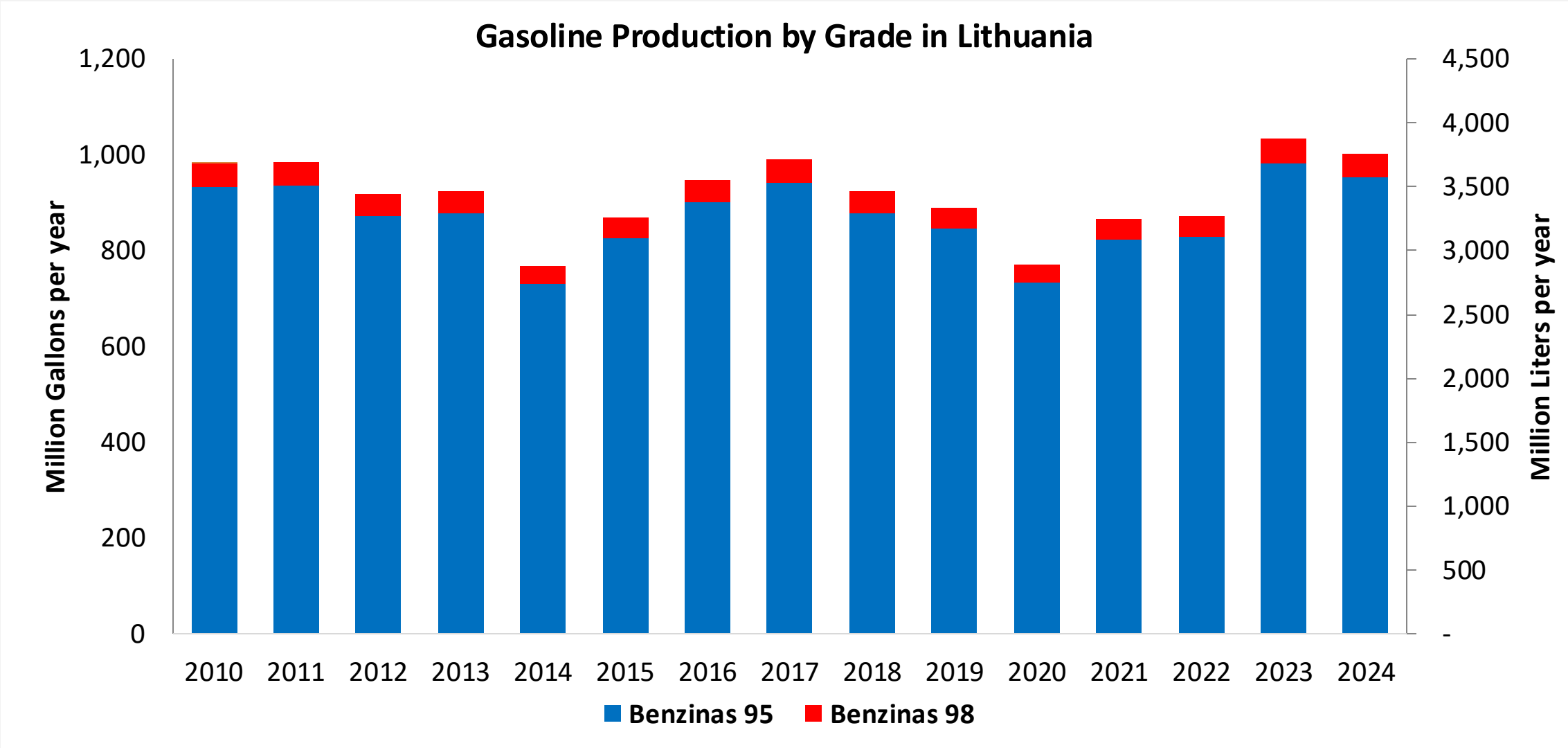
In 2024, ethanol consumption in Lithuania was 45 million liters (12 million gallons) representing 11.4% of gasoline demand. Ethanol production and Net Imports were 26 and 19 million liters (12 and 5 million gallons) correspondingly. Ethanol sources are Corn 15% and Cereals 85%.

Source: Eurostat, European Commission

Gasoline Demand in Lithuania

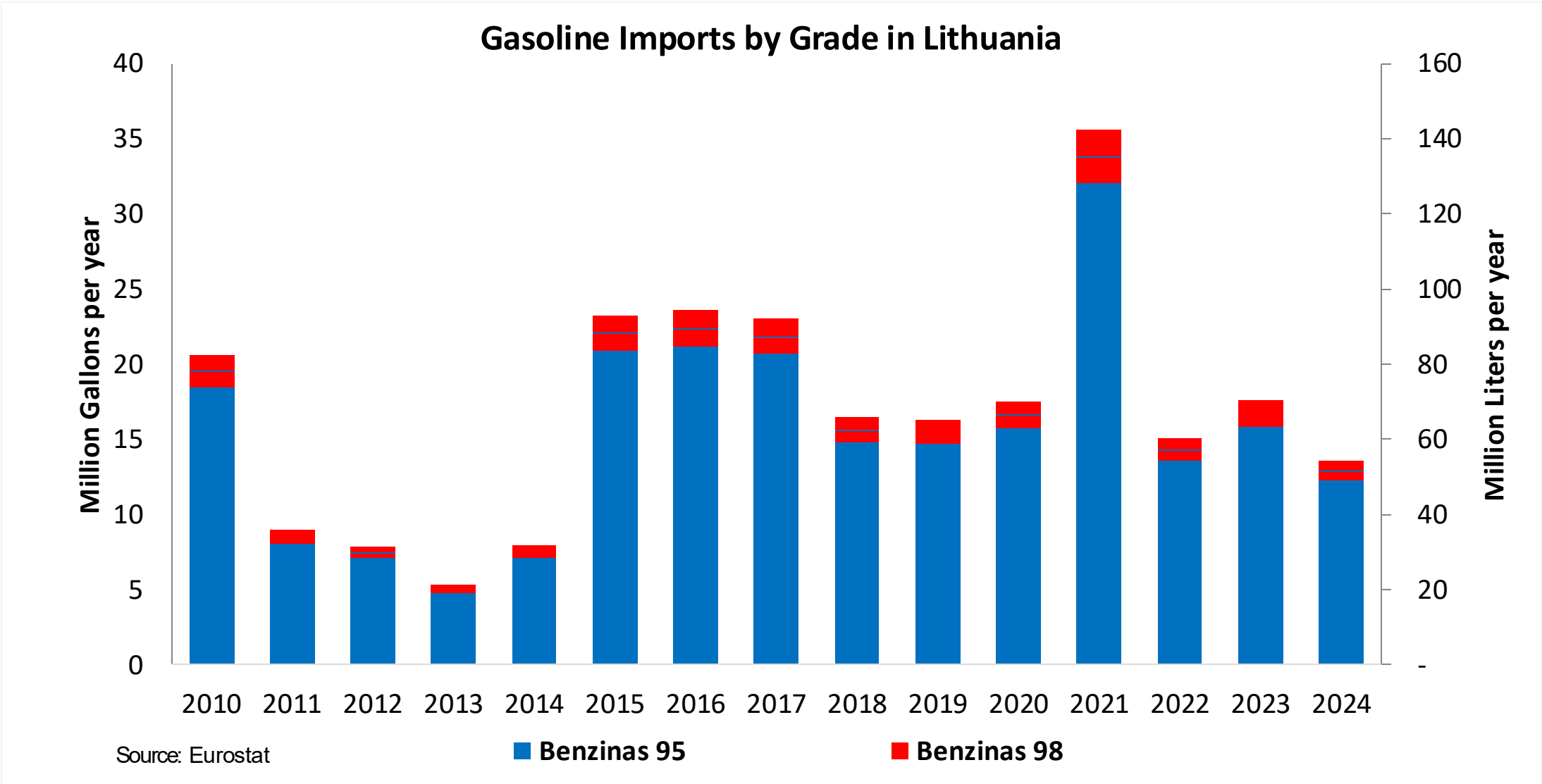


Gasoline Production in Lithuania



Source: Eurostat

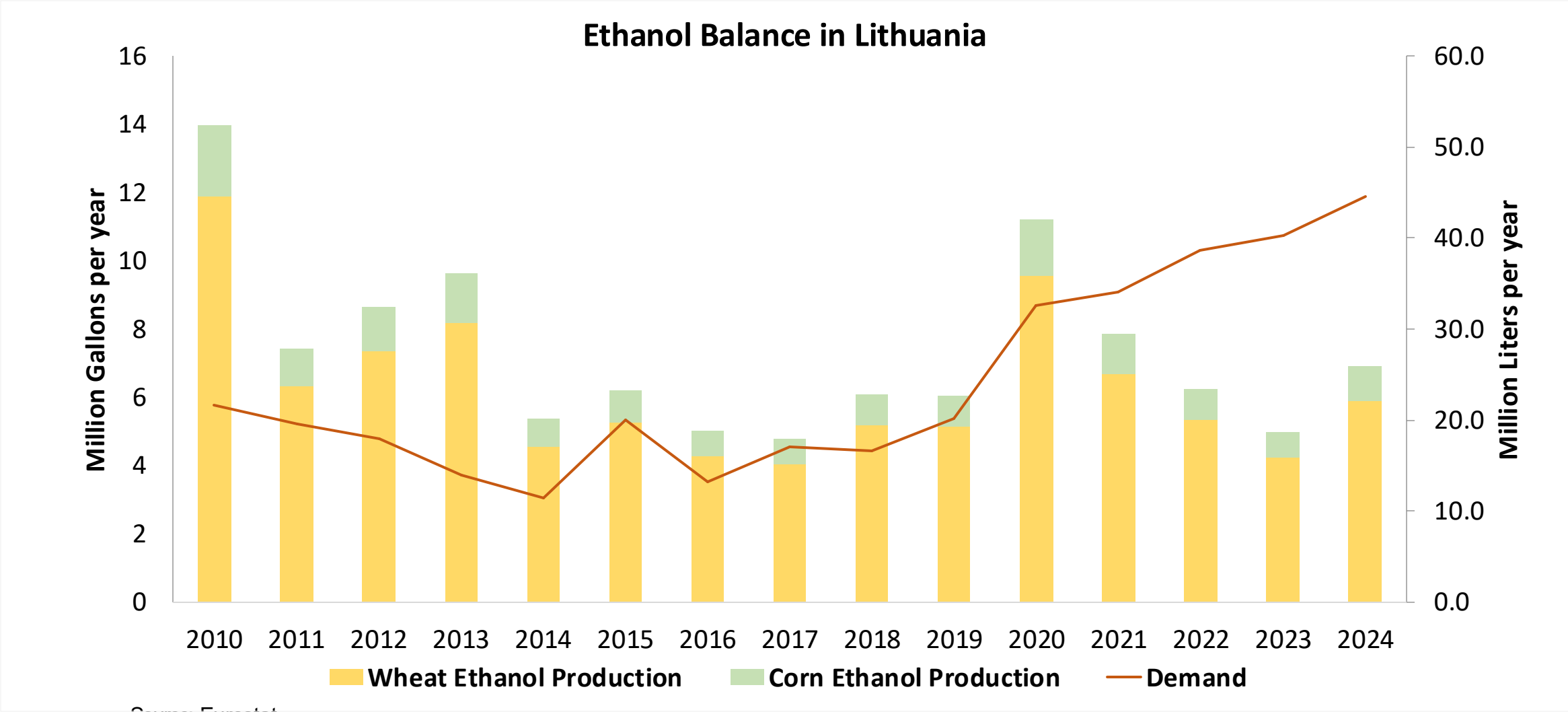
Gasoline Imports in Lithuania



The FARO90 logo is shown in a blue box, followed by a chemical structure of a substituted cyclohexadiene. The structure features a six-membered ring with two double bonds. Substituents include a methyl group, a propyl group, and a group labeled 'X'. The letters 'H', 'C', and 'X' are highlighted in orange in the original image.

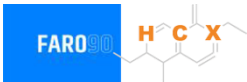


Ethanol Balance in Lithuania



Source: Eurostat

Gasoline Quality in Lithuania

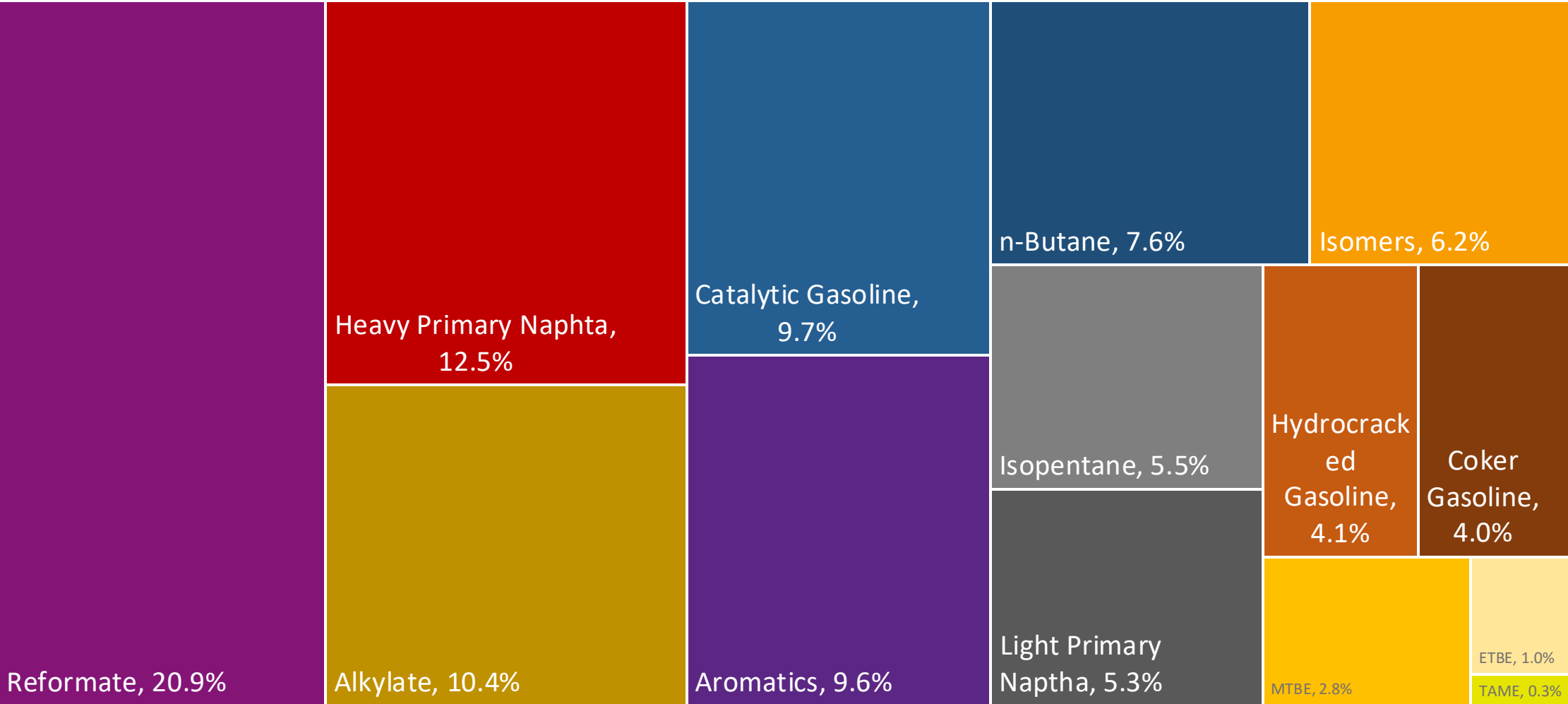
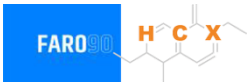


Name	EN 228:2025 (Euro 6); Country Level Fuel Ordinances							
Implementation Date	2025							
Applicability	All EU Contries + UK							
Grade	RON 95 E5		RON 95 E10		RON 98 E5		RON 98 E10	
	Min	Max	Min	Max	Min	Max	Min	Max
RON	95.0		95.0		98.0		98.0	
MON	> 85	> 88	> 85	> 88	> 85	> 88	> 85	> 88
AKI								
Benzene (%vol)	< 1							
Aromatics (% vol)	< 35							
Olefins (%vol)	< 18							
Lead (g/L)	< 0.005							
Manganese (mg/L)	< 2,0							
Sulfur (ppm)	10.0							
Oxygen (%w)*	< 2,7	< 3,7	< 2,7	< 3,7	< 2,7	< 3,7	< 2,7	< 3,7
Ethanol (%vol)	< 5	< 10	< 5	< 10	< 5	< 10	< 5	< 10
PVR 37.8°C (psi)	<> 45 - 100 kPa in 6 Volatility Classes							
Overall Biofuels (%cal)	8.6							
Advanced Biofuels (%cal)	1.0							
Bioethanol in Gasoline Sales (%cal)	-							
GHG Emission Reduction (%)	-							

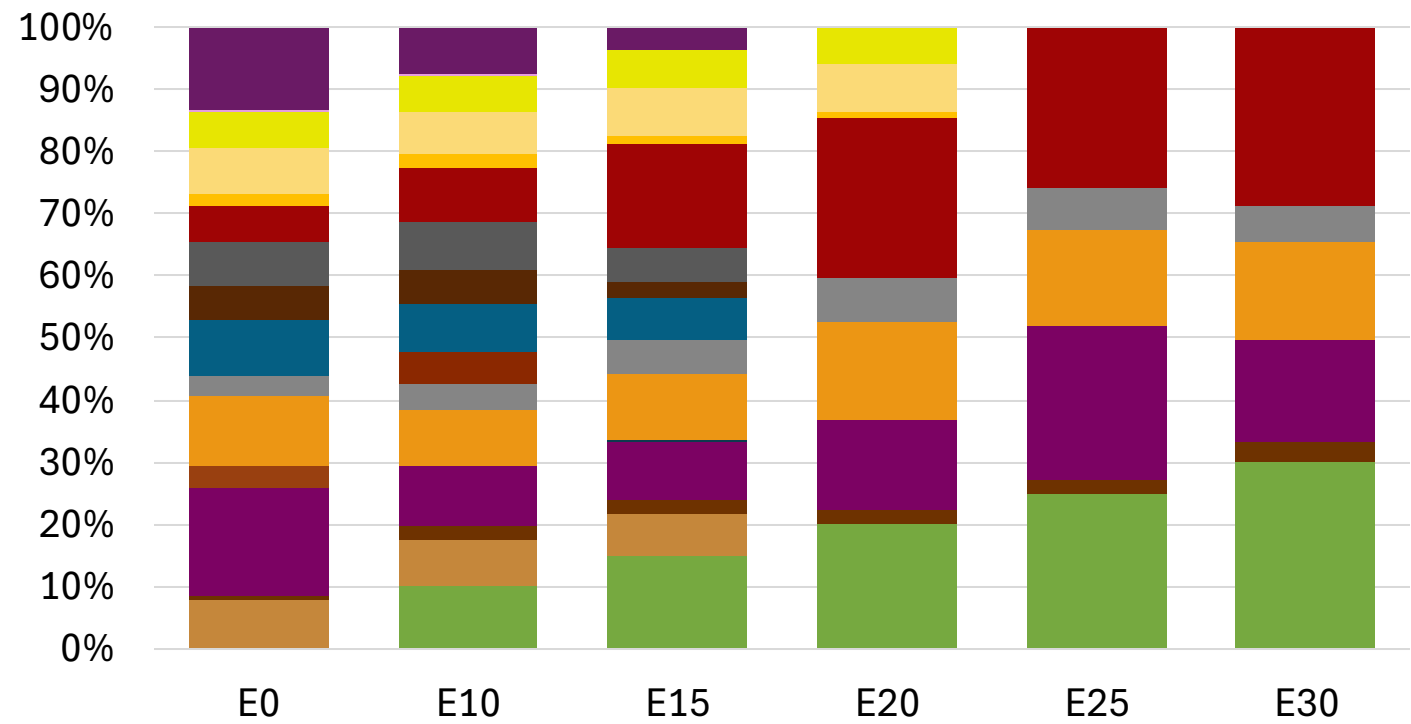
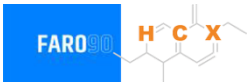
* Methanol 3%, Iso-propyl alcohol 12%, Ter-butyl alcohol 15%, Iso-butyl alcohol 15%, Ethers +5 Carbons 22%, Other Oxygenates 15%

Source: European Commision

Lithuania Available Blending Components



Lithuania – Gasoline 95 – Constant Octane

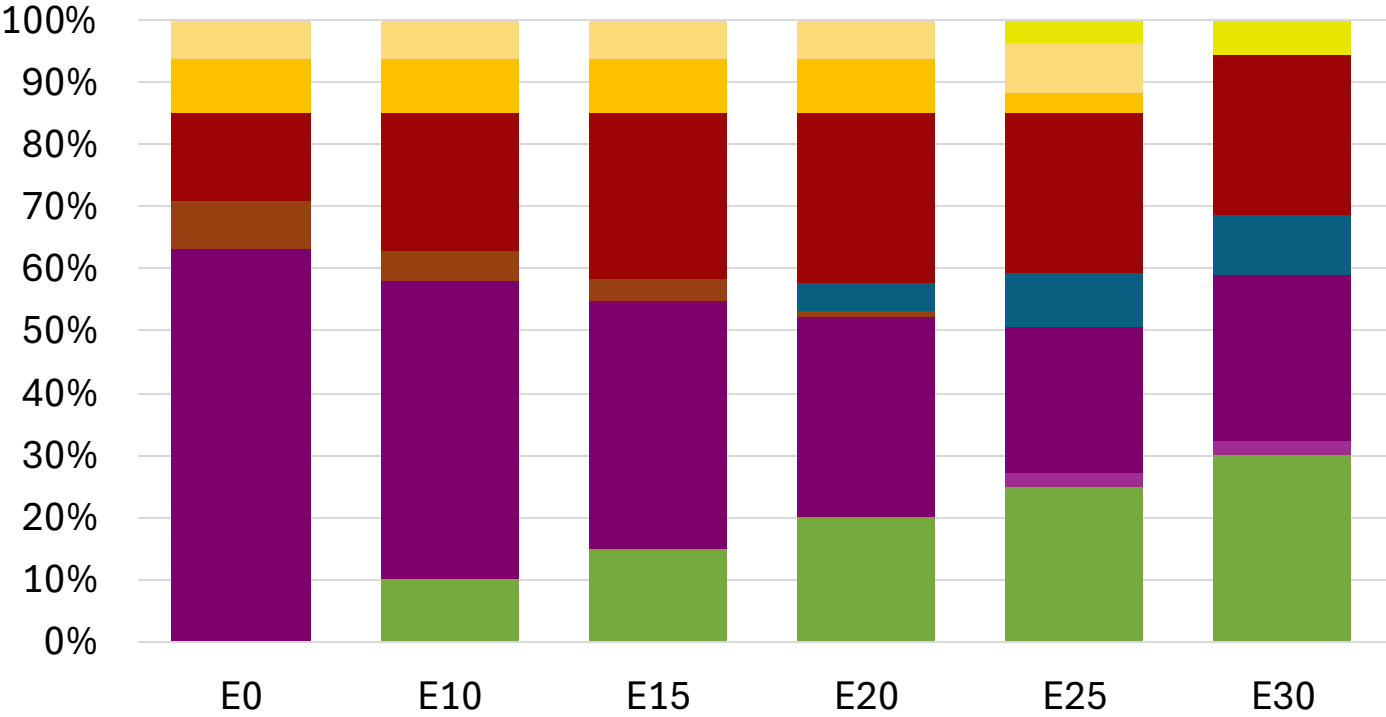
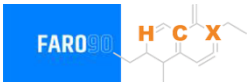


Average CIF 2024 Prices

Octane (RON)	95.0	95.0	95.0	95.0	95.0	95.8
Price (US\$/gal)	\$ 2.274	\$ 2.136	\$ 2.036	\$ 1.921	\$ 1.875	\$ 1.822

Source: Faro90

Lithuania - Gasoline 98 - Constant Octane

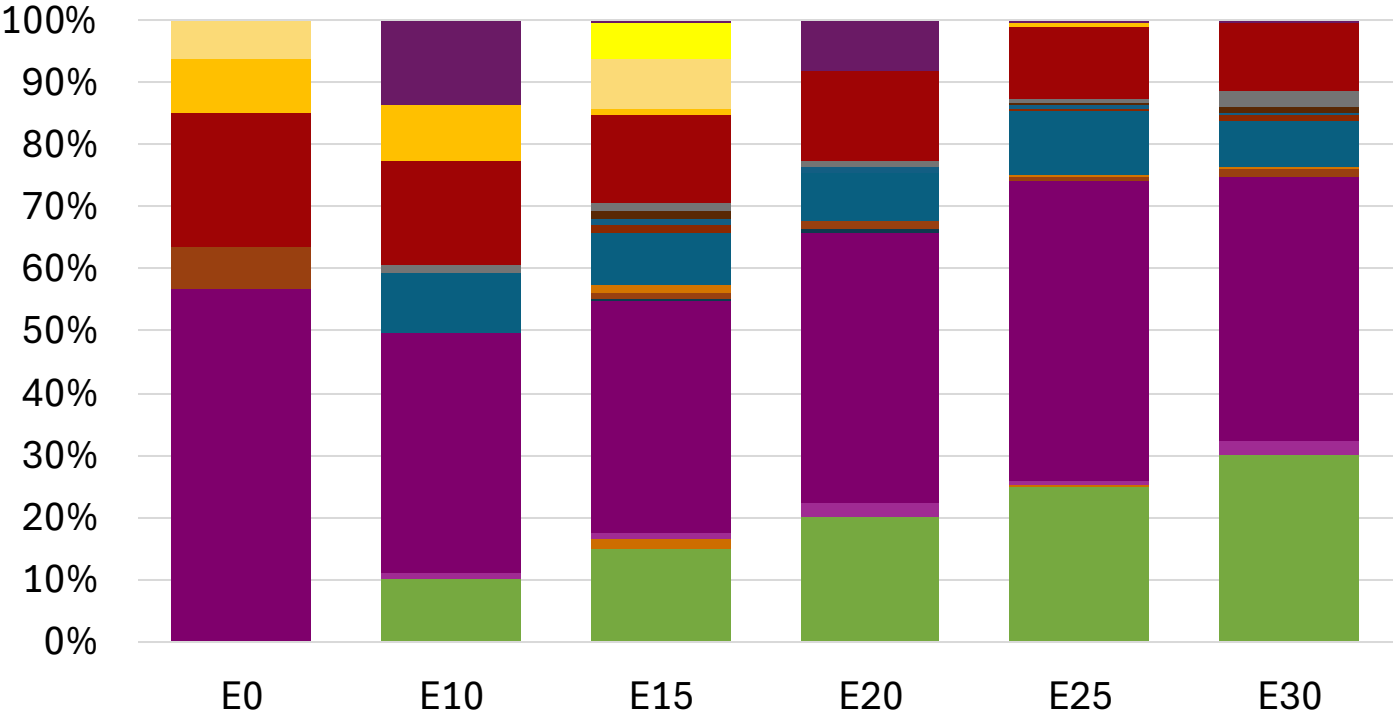
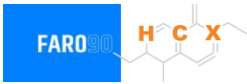


Average CIF 2024 Prices

Octane (RON)	98.0	98.0	98.0	98.0	98.0	98.0
Price (US\$/gal)	\$ 2.294	\$ 2.177	\$ 2.108	\$ 2.036	\$ 1.971	\$ 1.921

Source: Faro90

Lithuania – Gasoline 95 – Increased Octane



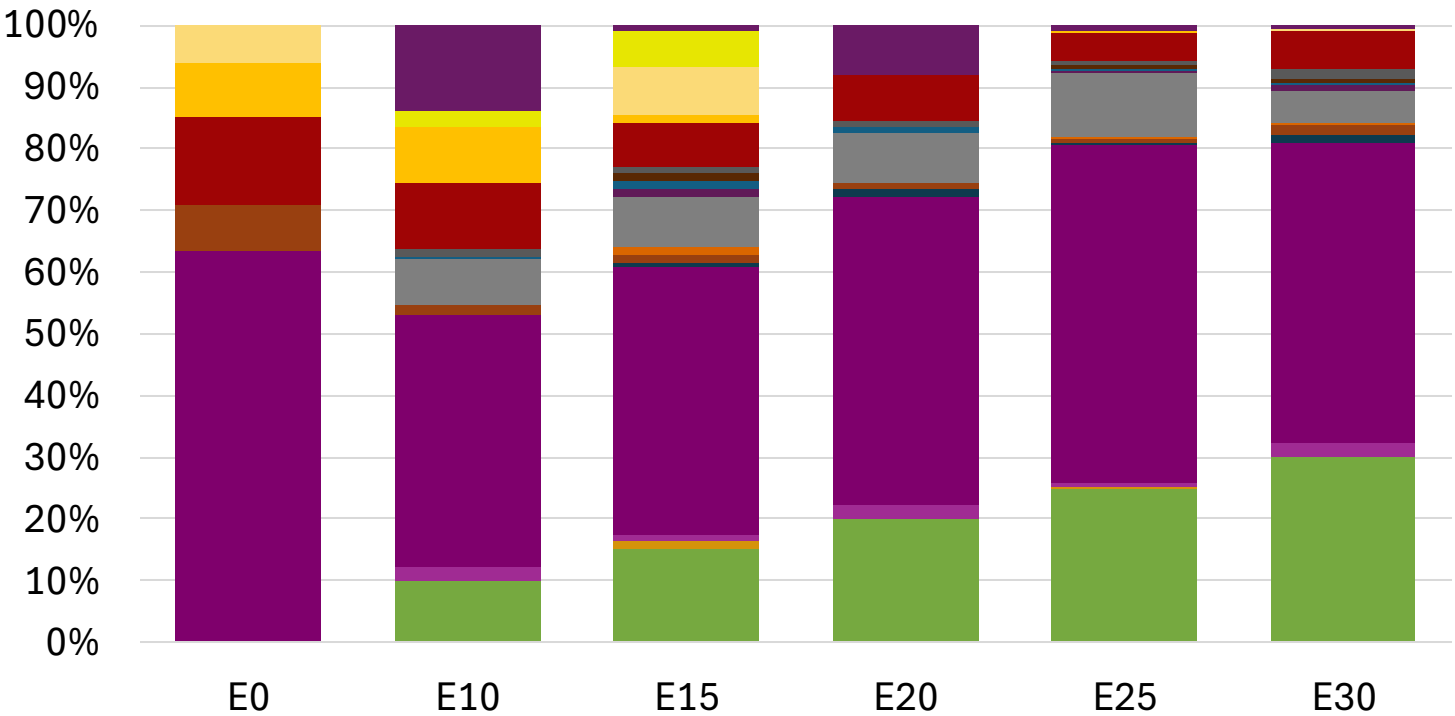
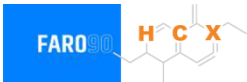
Ethanol
Alkylate
Raffinate
Reformate
n-Butane
Isomerate
Isopentane
Hydrocracked Gasoline
Catalytic Gasoline
Coker Naphtha
Light Primary Naphtha
Heavy Primary Naphtha
MTBE
ETBE
TAME
Aromatics

Average CIF 2024 Prices

Octane (RON)	95.0	96.2	98.3	98.7	100.5	101.8
Price (US\$/gal)	\$ 2.162	\$ 2.162	\$ 2.162	\$ 2.162	\$ 2.162	\$ 2.162

Source: Faro90

Lithuania – Gasoline 98 – Increased Octane

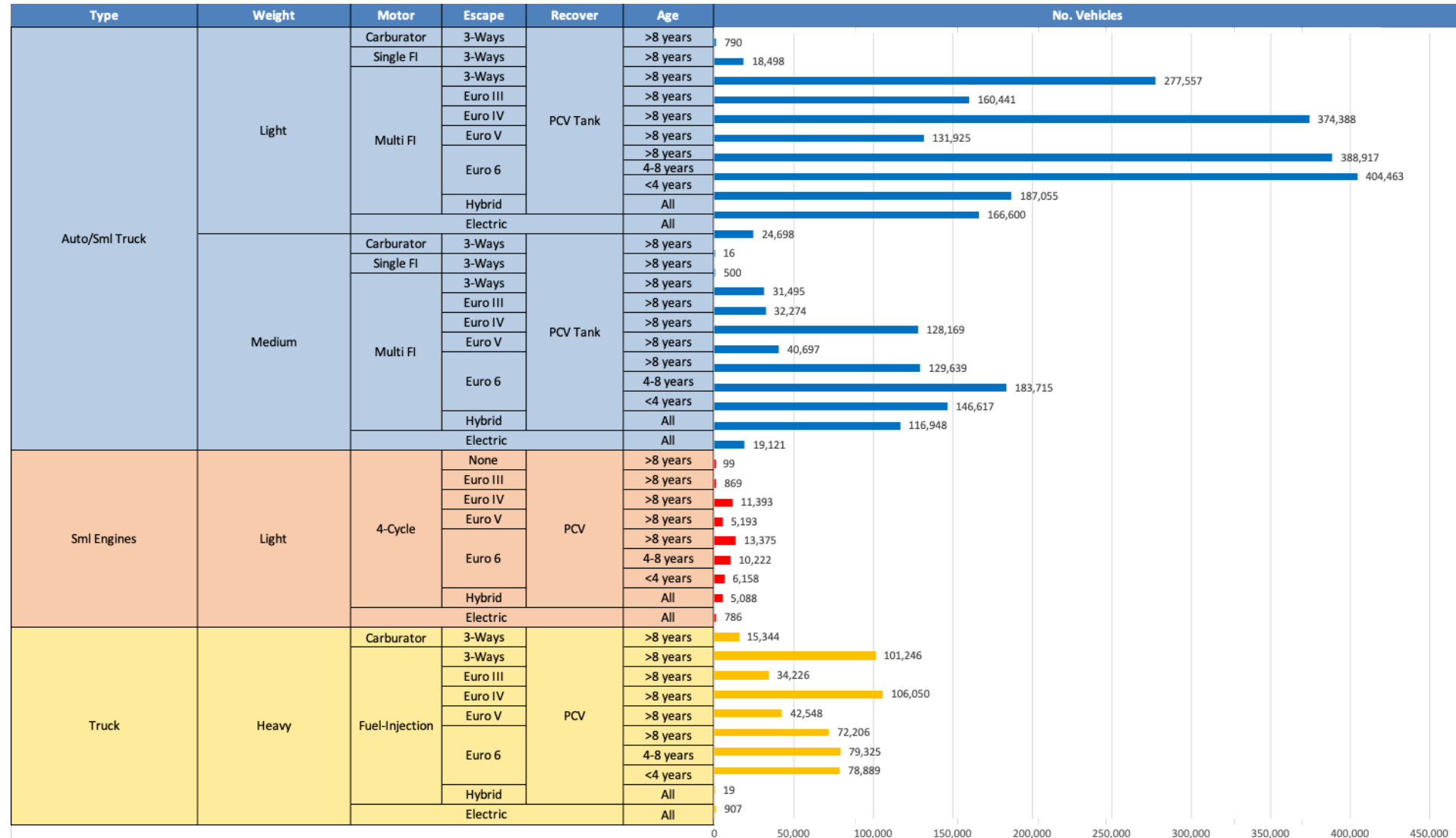


Average CIF 2024 Prices

Octane (RON)	98.0	99.0	100.9	101.2	102.8	104.1
Price (US\$/gal)	\$ 2.294	\$ 2.294	\$ 2.294	\$ 2.294	\$ 2.294	\$ 2.294

Source: Faro90

Gasoline Vehicle Fleet - Lithuania

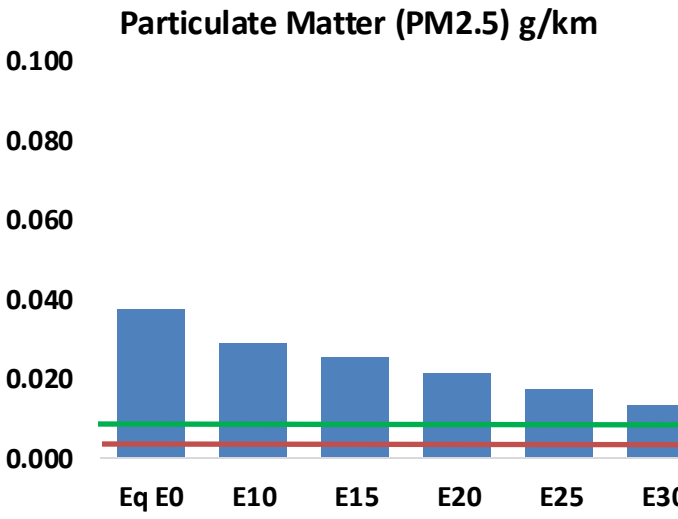
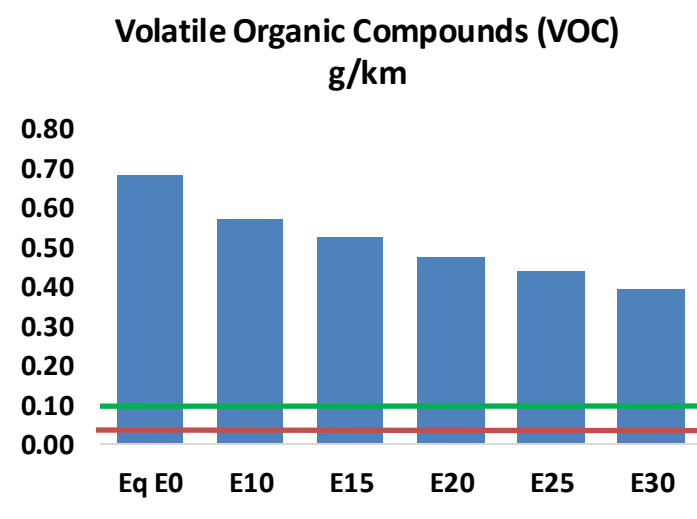
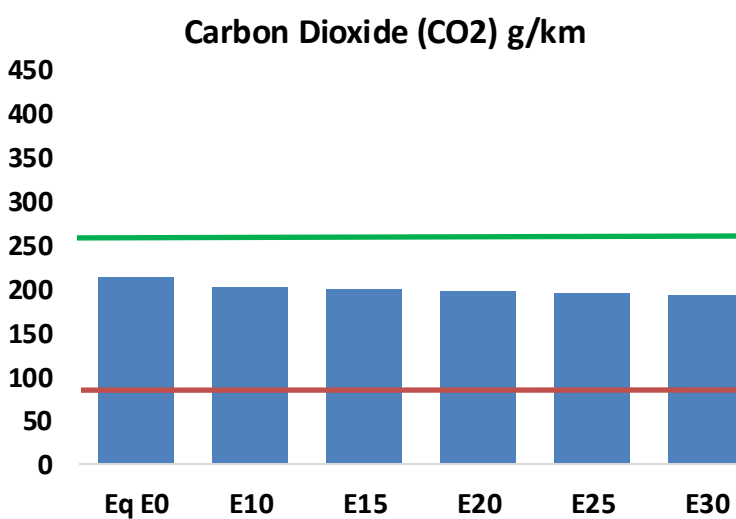
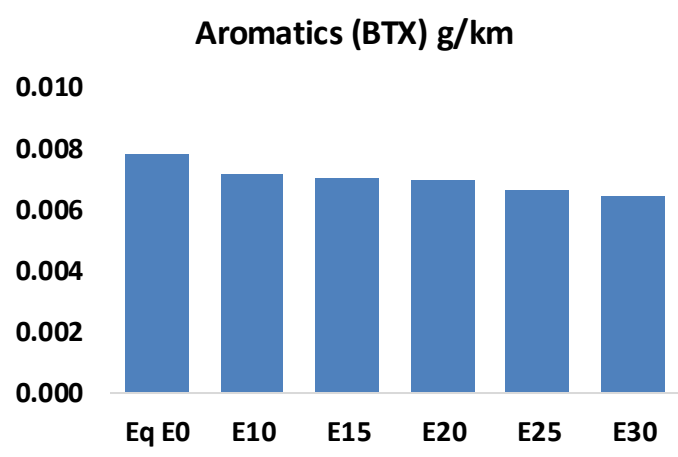
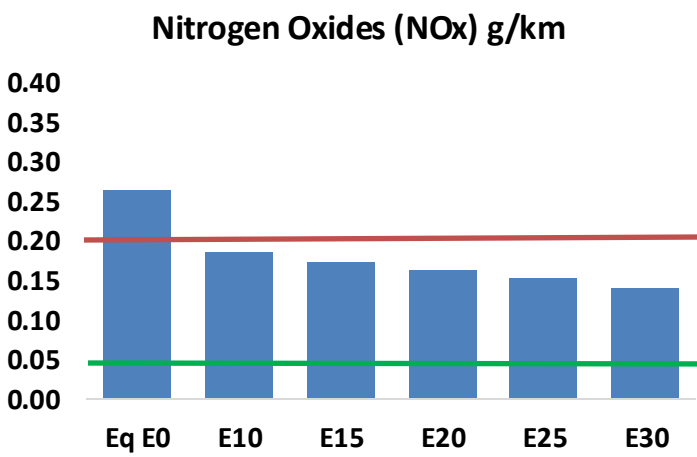
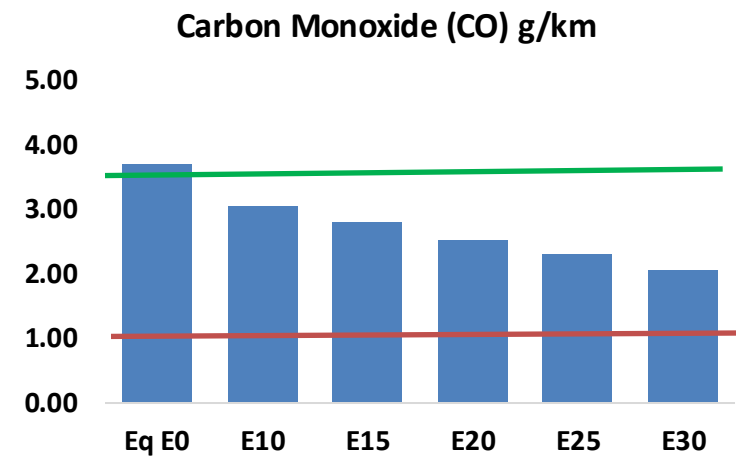
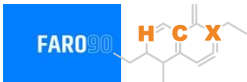


Vehicle Fleet: 3,548,467
Gasoline Vehicle Fleet 745,407

Source: Eurostat, ACEA

Lithuania – Vehicular Emissions

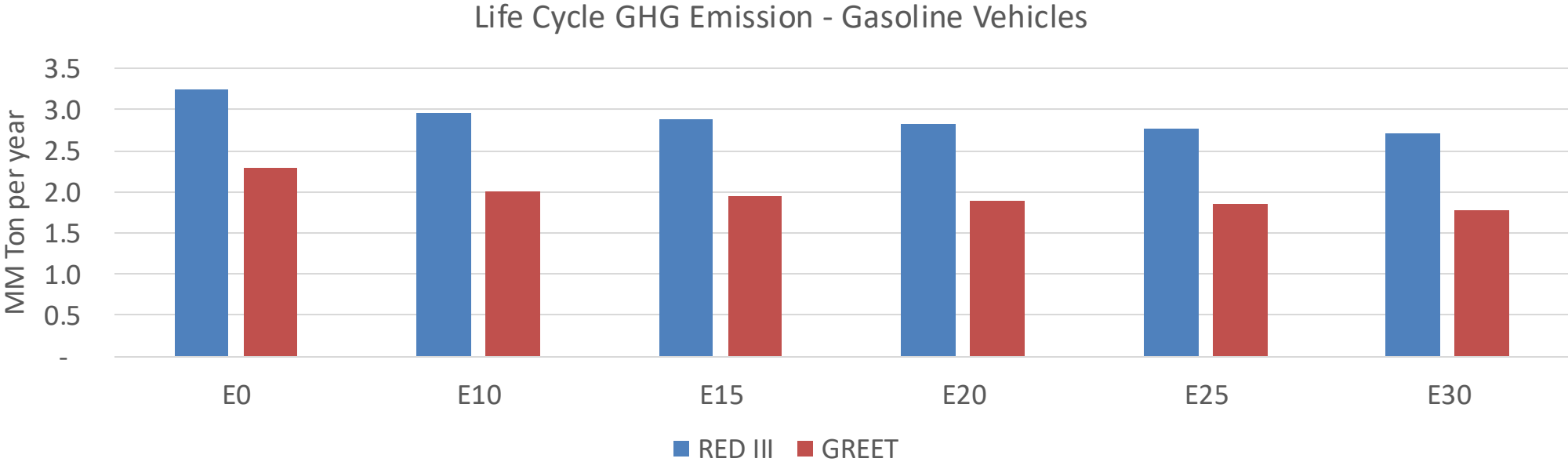
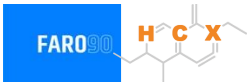
TIER III BIN 125 United States
Euro 6 Standard



Lithuania – Vehicular Emissions

Vehicular Fleet Emissions (kg/day)	Eq E0	E5	E10	E15	E20	E25	E30	Reduction (%)	
								E10	E20
CO	71,126	64,920	58,715	53,486	48,158	44,041	39,682	-17%	-32%
CO2	4,091,103	3,985,903	3,886,547	3,808,407	3,769,741	3,747,441	3,678,365	-5%	-8%
VOC	13,034	11,965	10,896	10,007	9,106	8,411	7,490	-16%	-30%
NOx	5,080	3,810	3,556	3,352	3,161	2,950	2,728	-30%	-38%
SOx	49	45	42	39	35	32	29	-15%	-28%
NH3	1,503	1,488	1,473	1,480	1,497	1,508	1,518	-2%	0%
N2O	69	68	68	68	70	70	71	-1%	2%
CH4	2,644	2,644	2,644	2,684	2,742	2,792	2,840	0%	4%
Lead	-	-	-	-	-	-	-	0%	0%
Butadiene	67	64	62	60	58	57	55	-8%	-14%
Acetaldehyde	210	242	354	540	735	840	993	68%	249%
Formaldehyde	773	752	876	1,029	1,078	1,193	1,298	13%	39%
Benzene	151	145	137	135	134	128	124	-9%	-11%
PM2.5	252	224	196	170	144	117	89	-22%	-43%
PM10	463	411	359	312	265	215	163	-22%	-43%

Lithuania – Gasoline Vehicle Fleet Life Cycle GHG Emissions



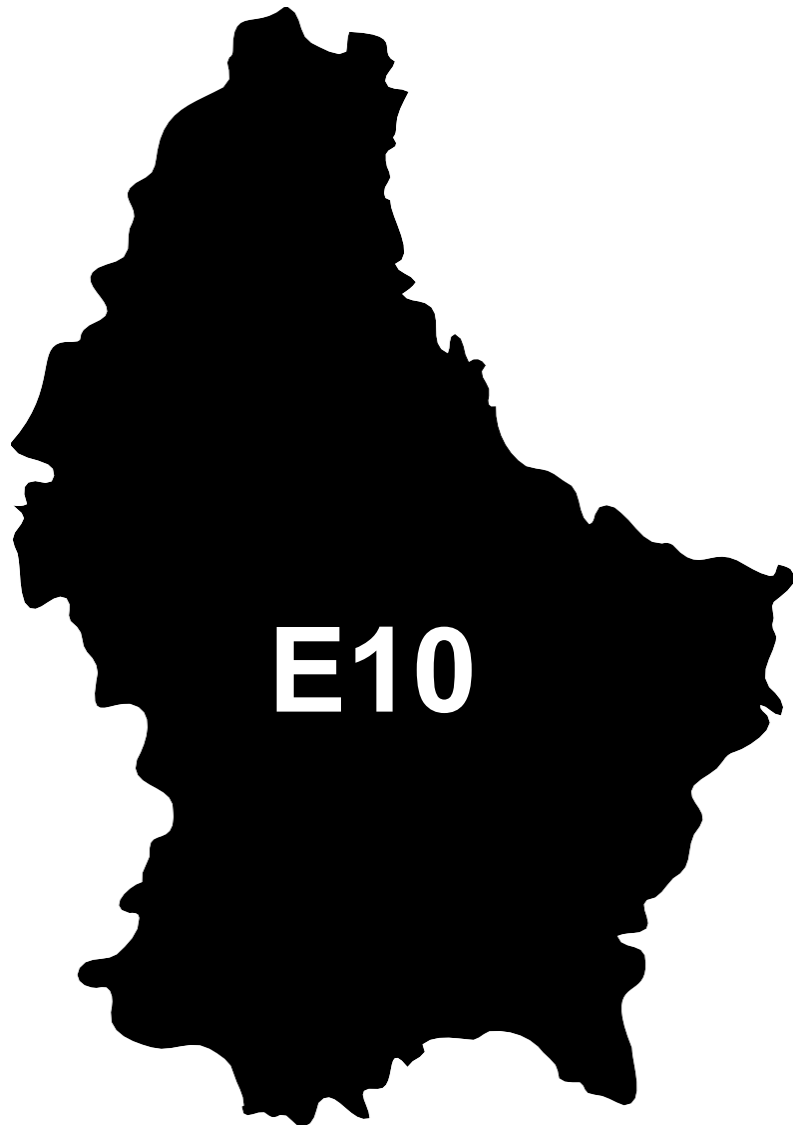
Country Emissions 2024 (MMT/año)	25.4
Target @ 2035 (MMT/year)	14.3
Delta	11.1

%Reduction vs E0	E0	E10	E15	E20	E25	E30
GREET 2024	0.0%	11.9%	14.9%	17.2%	19.3%	22.3%
RED II/III	0.0%	8.8%	11.0%	12.8%	14.4%	16.6%

% Target@2035	E0	E10	E15	E20	E25	E30
GREET 2024	0.0%	2.5%	3.1%	3.5%	4.0%	4.6%
RED II/III	0.0%	2.6%	3.2%	3.7%	4.2%	4.8%

Source: Greet, RED II/III, IPCC, climateactiontracker.org, F90

Ethanol Gasoline Blending in Luxembourg

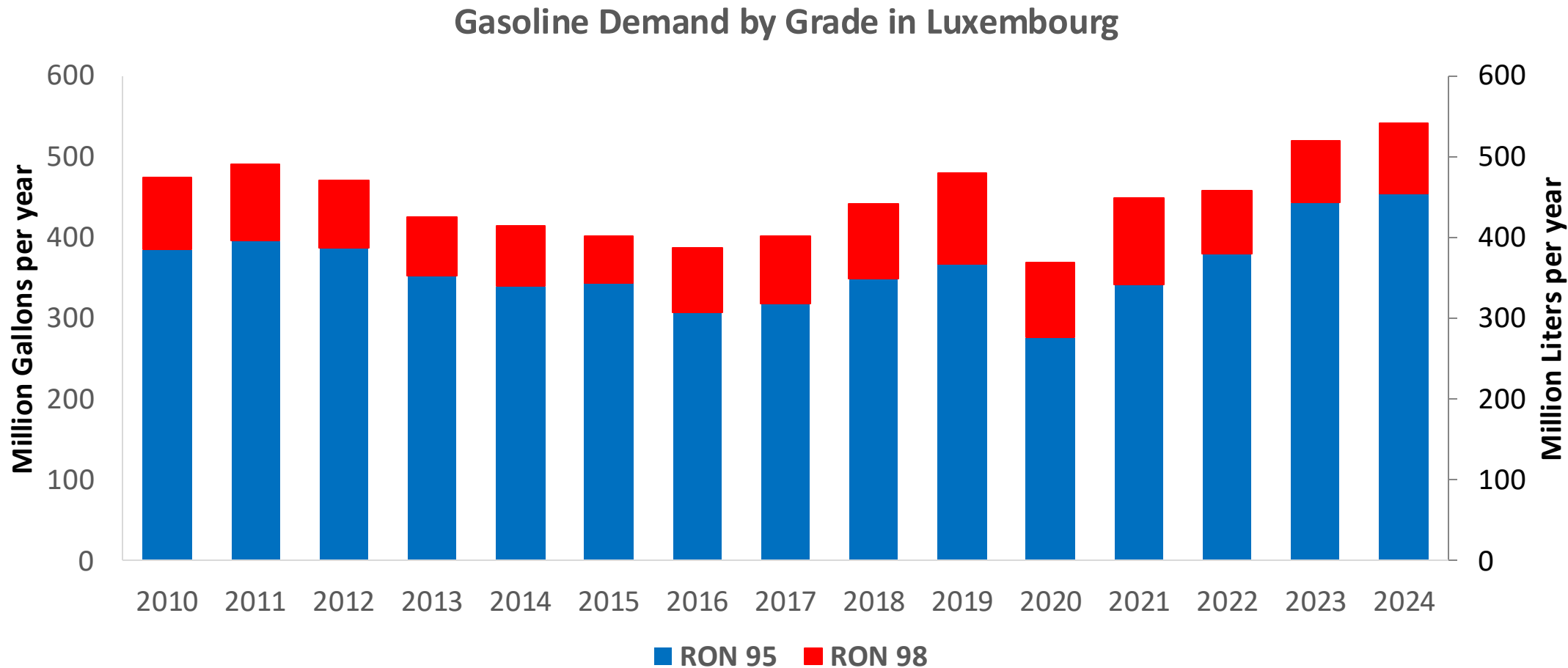


In 2024, gasoline consumption in Luxembourg was 540 million liters (143 million gallons) and growth rate was 2.4%. There is no gasoline production. Gasoline net imports were 540 million liters (143 million gallons) correspondingly. Luxembourg complies with the European gasoline standards and offers two main grades: RON 95 and RON 98) with an average octane of 95.5 RON.

In 2024, ethanol consumption in Luxembourg was 29 million liters (8 million gallons) representing 5.3% of gasoline demand. Net Imports were 29 million liters (8 million gallons). Ethanol sources are Sugar Cane 80% and Corn 20%.

Source: Eurostat, European Commission

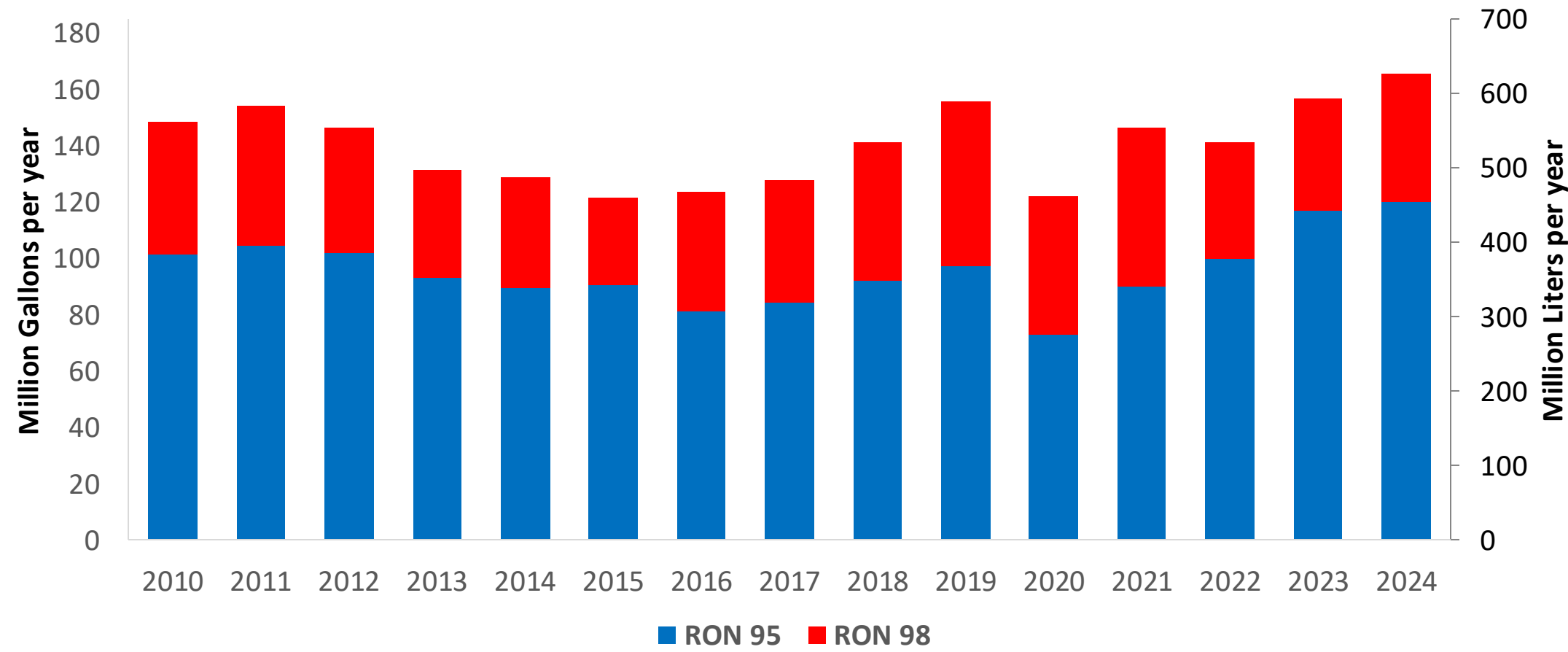
Gasoline Demand in Luxembourg



Source: Eurostat

Gasoline Imports in Luxembourg

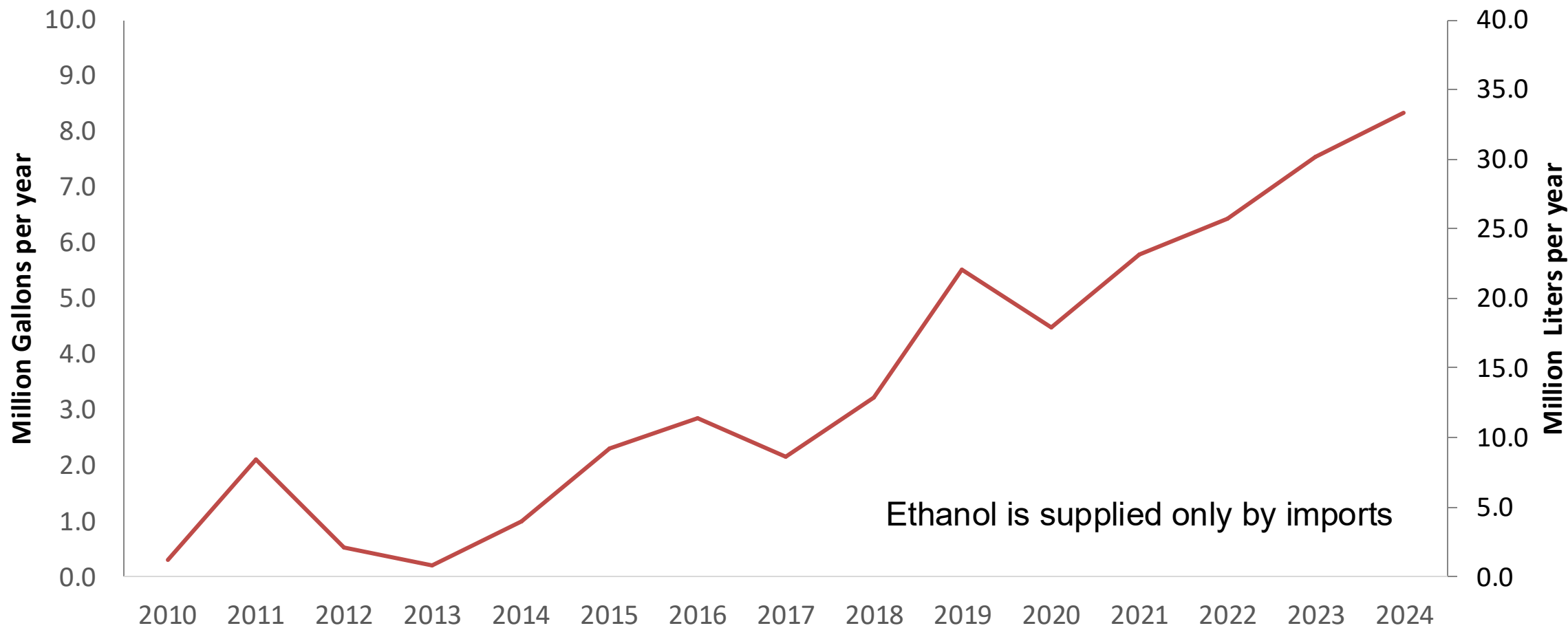
Gasoline Imports by Grade in Luxembourg



Source: Eurostat

Ethanol Balance in Luxembourg

Ethanol Demand in Luxembourg



Source: Eurostat

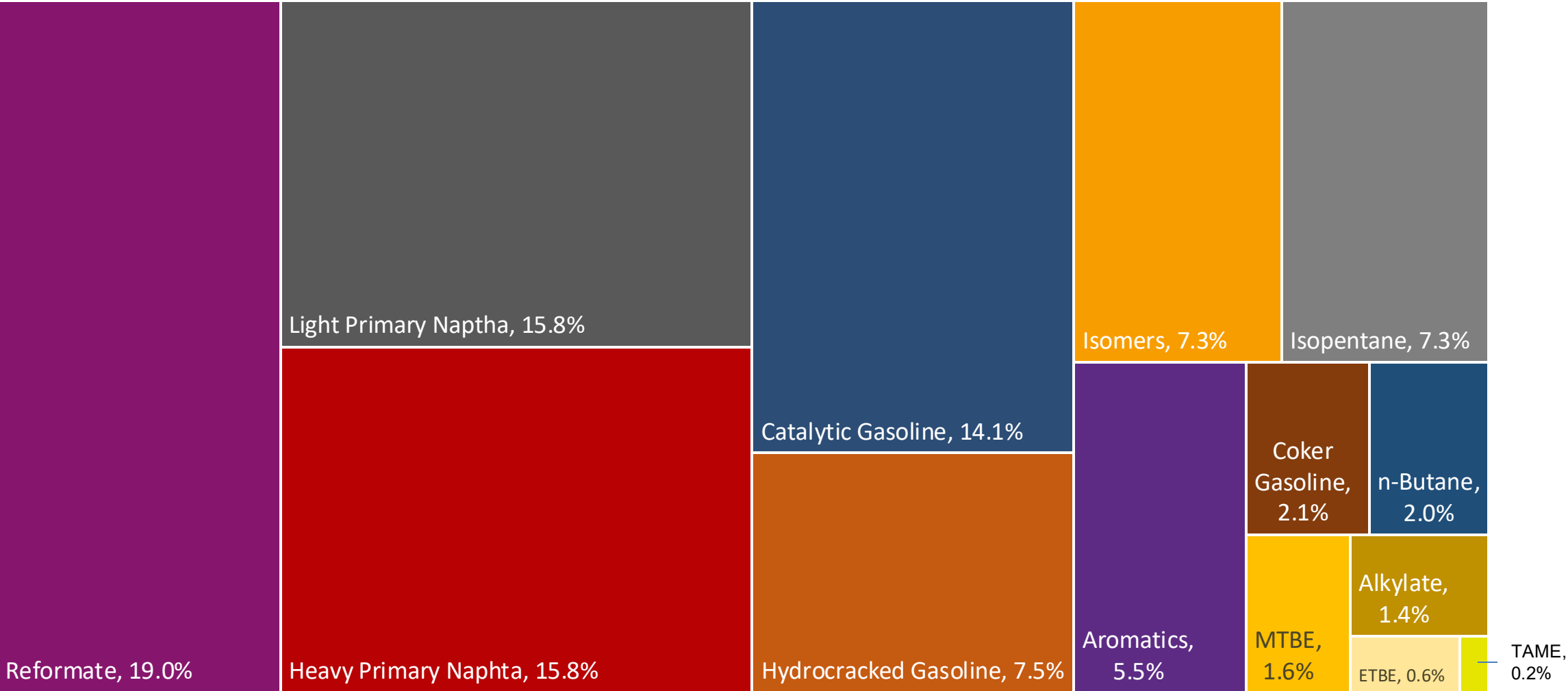
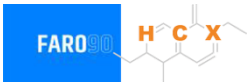
Gasoline Quality in Luxembourg

Name	EN 228:2025 (Euro 6); Country Level Fuel Ordinances							
Implementation Date	2025							
Applicability	All EU Contries + UK							
Grade	RON 95 E5		RON 95 E10		RON 98 E5		RON 98 E10	
	Min	Max	Min	Max	Min	Max	Min	Max
RON	95.0		95.0		98.0		98.0	
MON	> 85	> 88	> 85	> 88	> 85	> 88	> 85	> 88
AKI								
Benzene (%vol)	< 1							
Aromatics (% vol)	< 35							
Olefins (%vol)	< 18							
Lead (g/L)	< 0.005							
Manganese (mg/L)	< 2,0							
Sulfur (ppm)	10.0							
Oxygen (%w)*	< 2,7	< 3,7	< 2,7	< 3,7	< 2,7	< 3,7	< 2,7	< 3,7
Ethanol (%vol)	< 5	< 10	< 5	< 10	< 5	< 10	< 5	< 10
PVR 37.8°C (psi)	<> 45 - 100 kPa in 6 Volatility Classes							
Overall Biofuels (%cal)	7.7							
Advanced Biofuels (%cal)	-							
Bioethanol in Gasoline Sales (%cal)	-							
GHG Emission Reduction (%)	6.0							

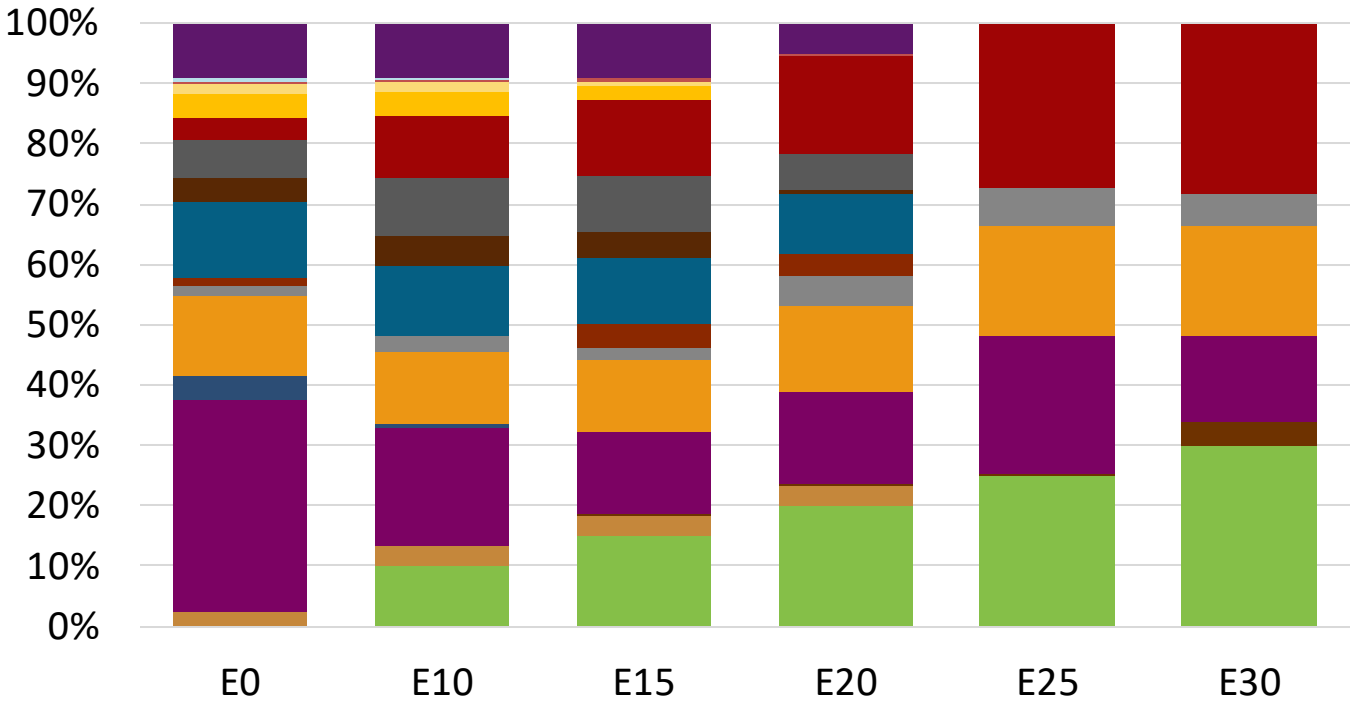
* Methanol 3%, Iso-propyl alcohol 12%, Ter-butyl alcohol 15%, Iso-butyl alcohol 15%, Ethers +5 Carbons 22%, Other Oxygenates 15%

Source: European Commision

Luxembourg Available Blending Components



Luxembourg – Gasoline 95 – Constant Octane

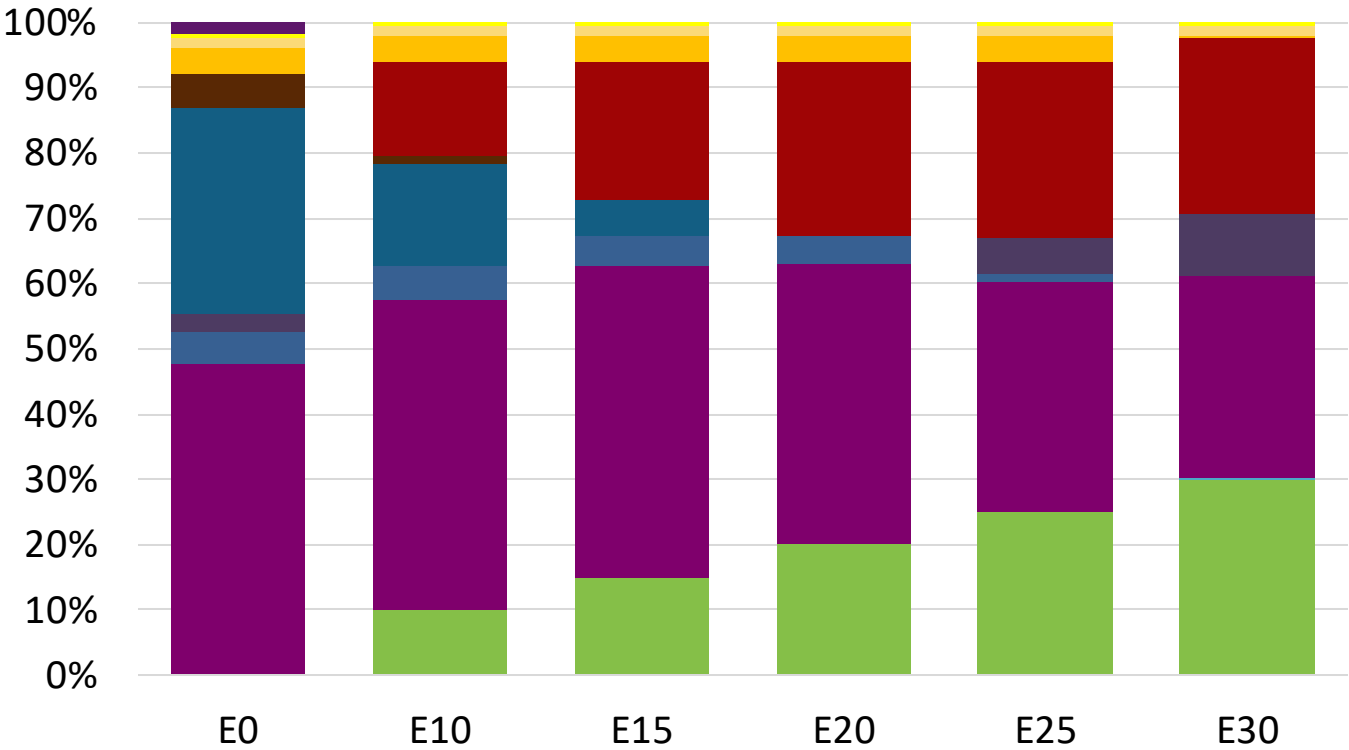


Average CIF 2024
Prices

Octane (RON)	95.0	95.0	95.0	95.0	95.1	95.6
Price (US\$/gal)	\$ 2.272	\$ 2.144	\$ 2.083	\$ 2.002	\$ 1.880	\$ 1.814

Source: Faro90

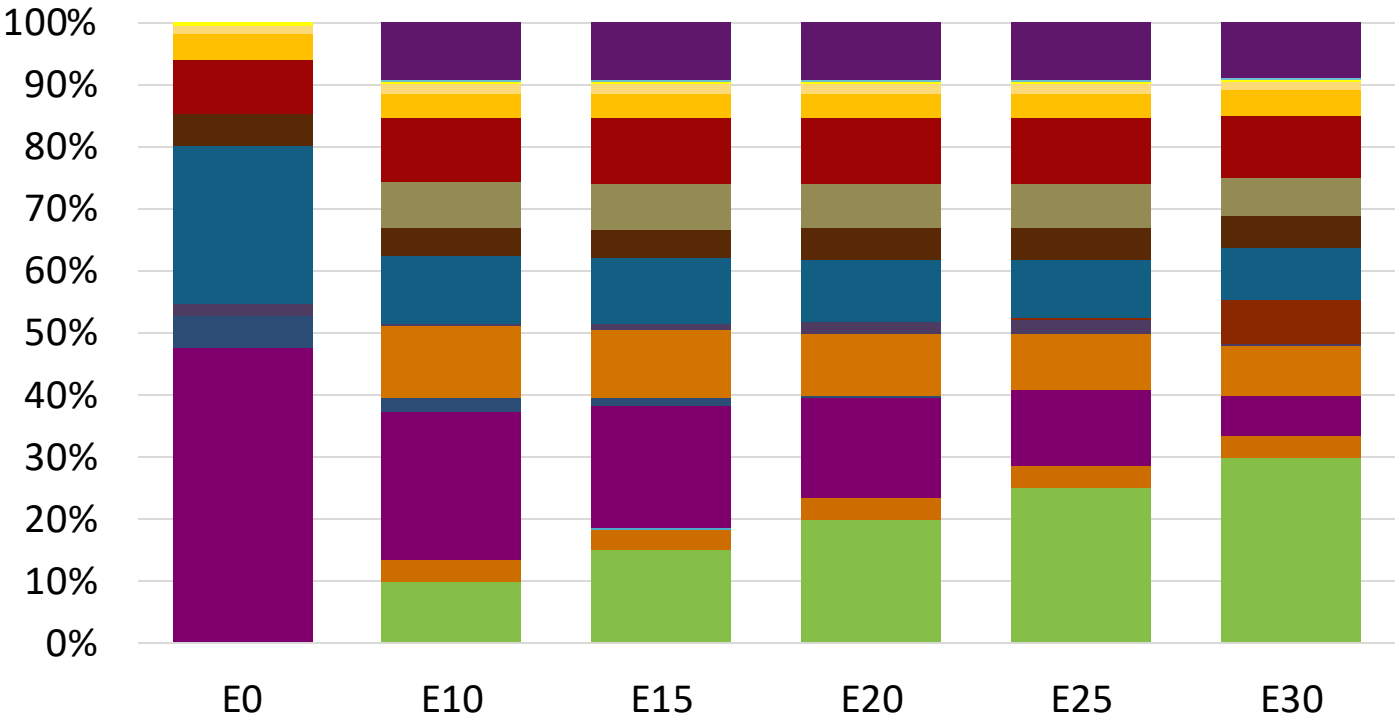
Luxembourg - Gasoline 98 - Constant Octane



Average CIF 2024
Prices

Source: Faro90

Luxembourg – Gasoline 95 – Increased Octane

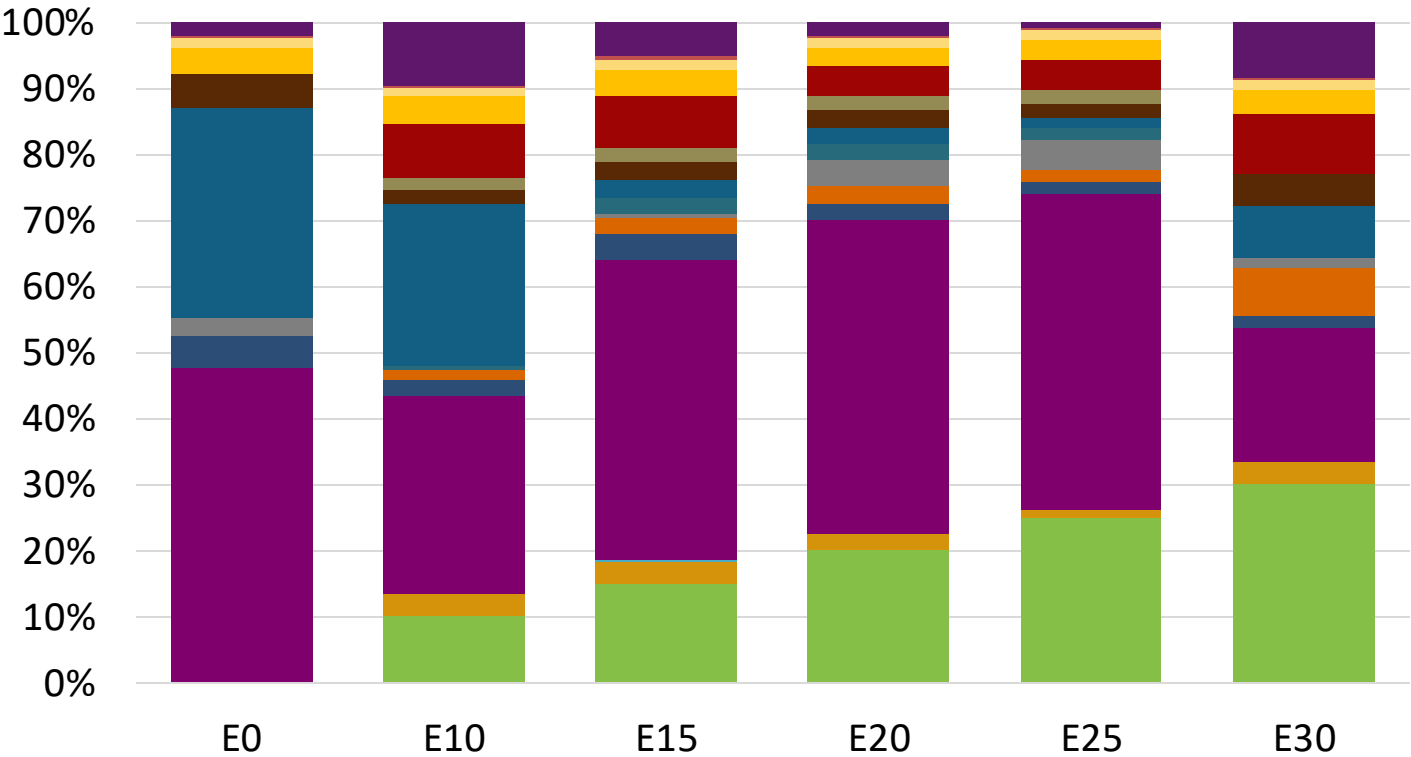
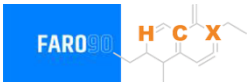


Average CIF 2024
Prices

Octane (RON)	95.0	96.5	97.9	99.4	100.8	102.0
Price (US\$/gal)	\$ 2.209	\$ 2.209	\$ 2.209	\$ 2.209	\$ 2.209	\$ 2.209

Source: Faro90

Luxembourg – Gasoline 98 – Increased Octane

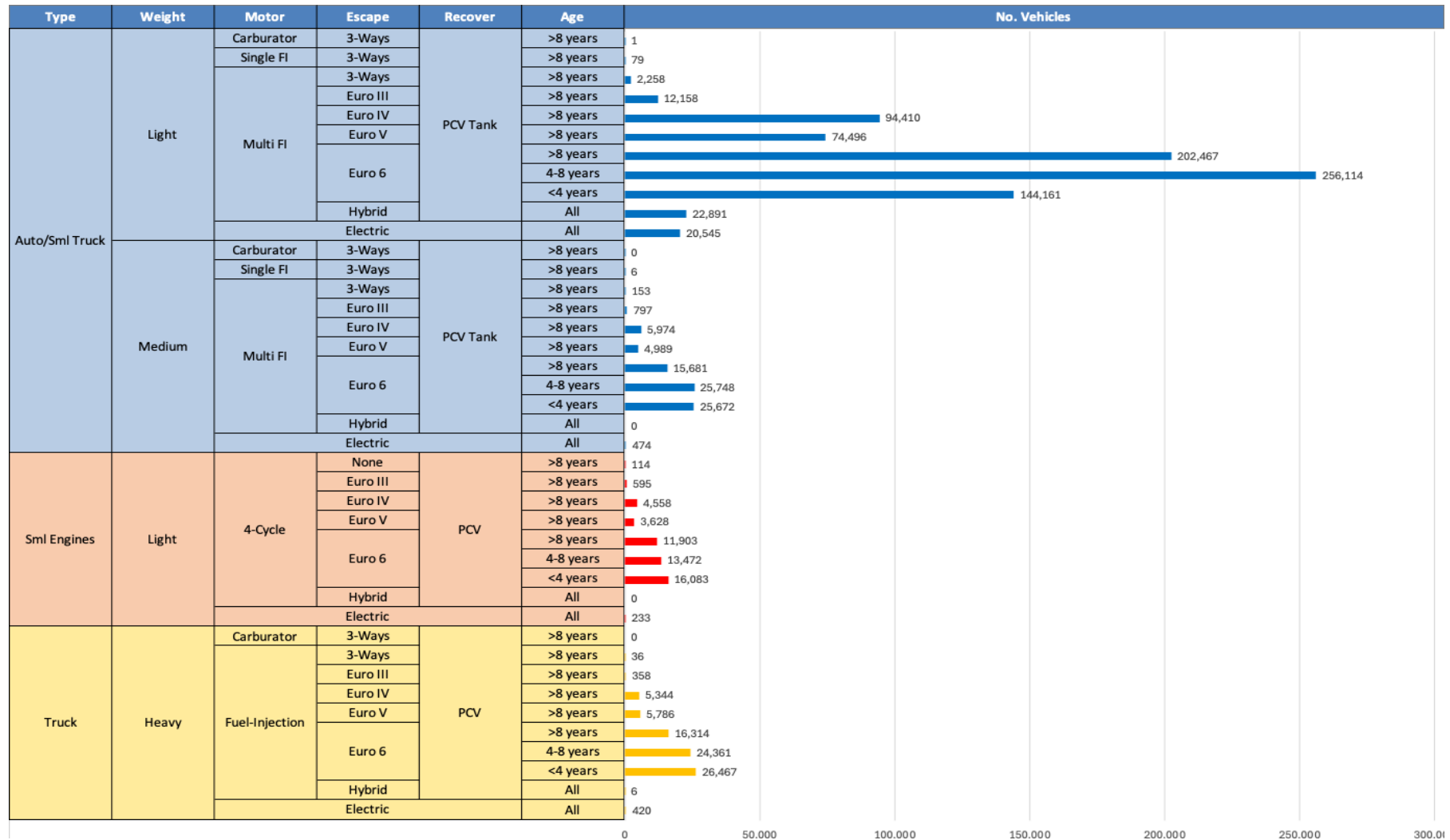


Average CIF 2024
Prices

Octane (RON)	98.0	99.2	101.3	102.6	104.0	104.7
Price (US\$/gal)	\$ 2.347	\$ 2.347	\$ 2.347	\$ 2.347	\$ 2.347	\$ 2.347

Source: Faro90

Gasoline Vehicle Fleet - Luxembourg

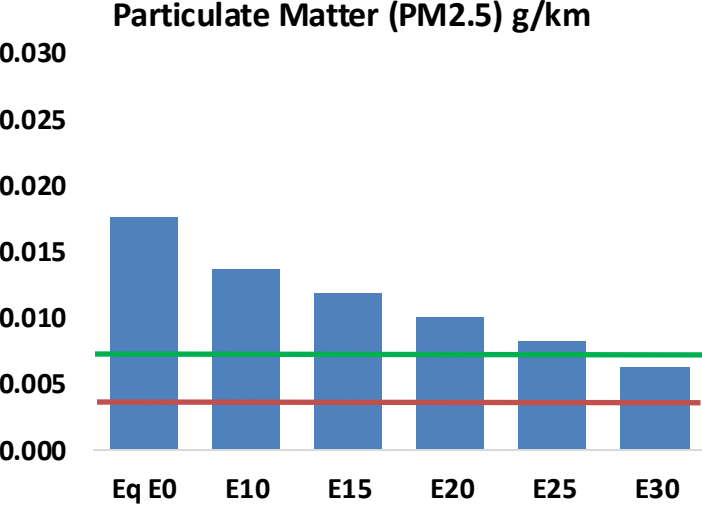
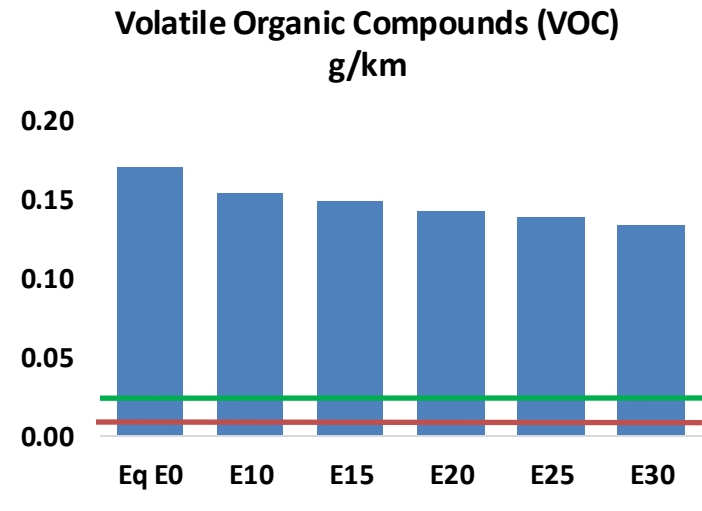
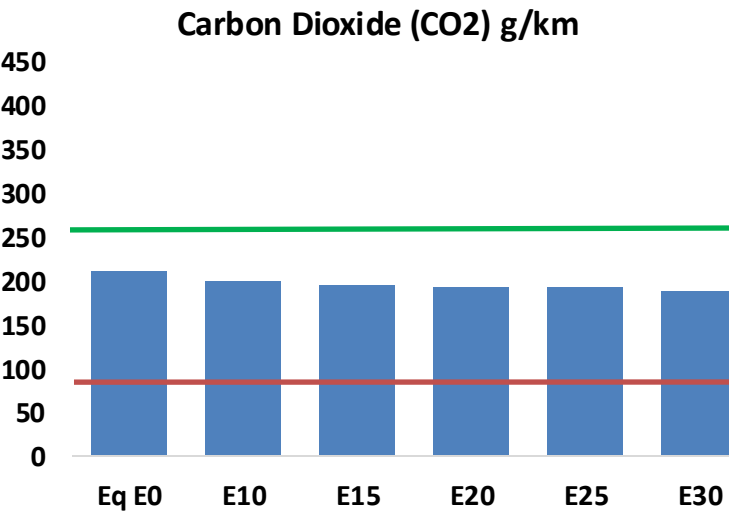
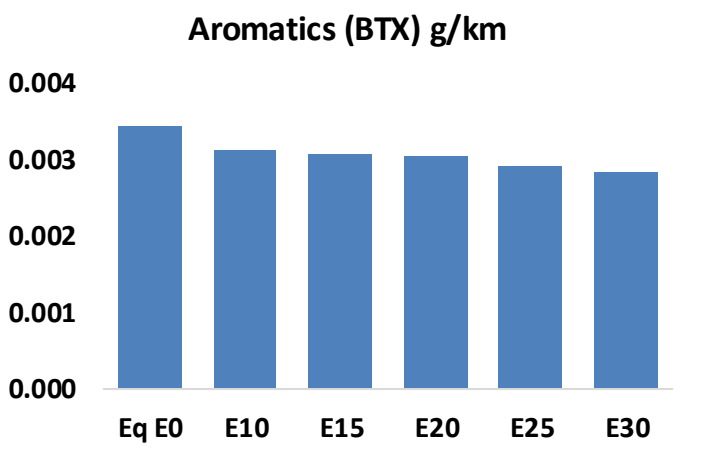
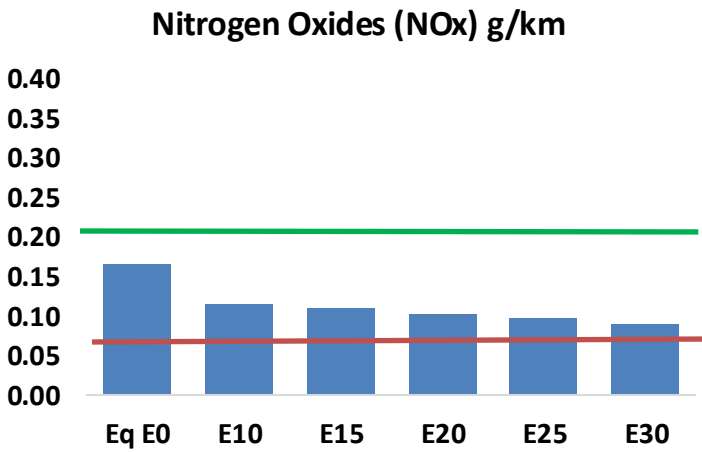
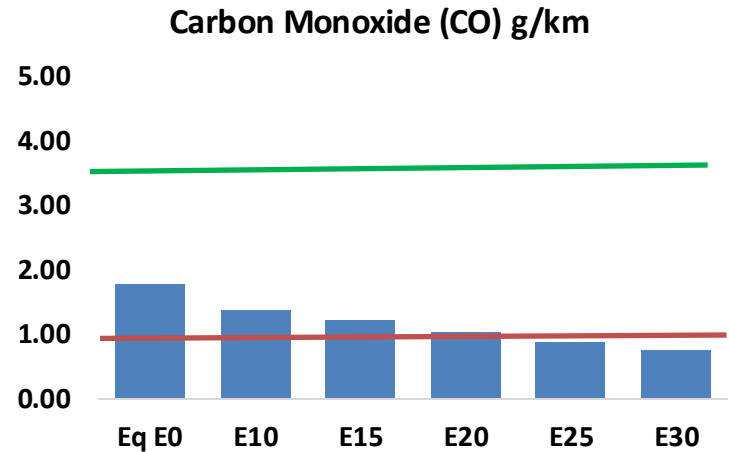
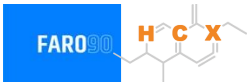


Vehicle Fleet: 1,038,755
Gasoline Vehicle Fleet: 694,397

Source: Eurostat, ACEA

Luxembourg – Vehicular Emissions

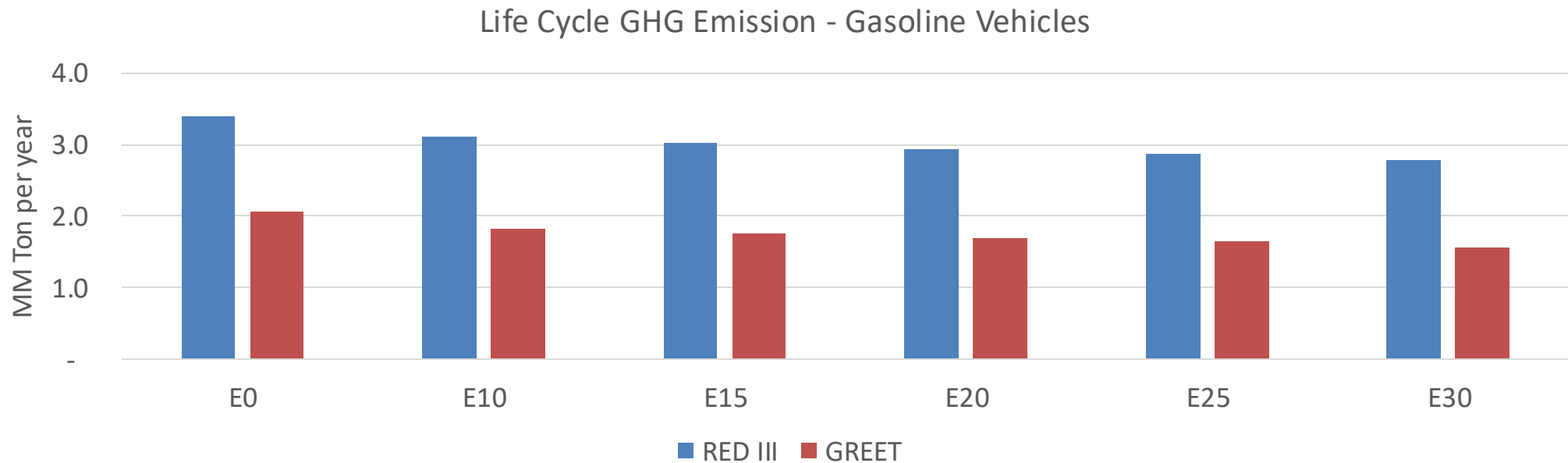
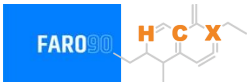
TIER III BIN 125 United States
Euro 6 Standard



Luxembourg – Vehicular Emissions

Vehicular Fleet Emissions (kg/day)	Eq E0	E5	E10	E15	E20	E25	E30	Reduction (%)	
								E10	E20
CO	32,931	29,249	25,567	22,326	18,961	16,342	13,726	-22%	-42%
CO2	3,882,918	3,783,071	3,688,772	3,614,608	3,577,909	3,556,182	3,490,631	-5%	-8%
VOC	3,142	2,992	2,843	2,738	2,642	2,570	2,465	-10%	-16%
NOx	3,081	2,311	2,157	2,033	1,917	1,790	1,655	-30%	-38%
SOx	44	41	37	34	32	29	26	-15%	-28%
NH3	1,305	1,292	1,279	1,285	1,299	1,309	1,318	-2%	0%
N2O	48	47	47	47	48	49	49	-1%	2%
CH4	747	747	747	758	775	789	802	0%	4%
Lead	-	-	-	-	-	-	-	0%	0%
Butadiene	31	29	28	27	27	26	25	-8%	-14%
Acetaldehyde	55	64	93	142	194	221	261	68%	249%
Formaldehyde	162	157	184	216	226	250	272	13%	39%
Benzene	63	61	58	57	56	54	52	-9%	-11%
PM2.5	244	216	189	164	139	113	86	-22%	-43%
PM10	81	72	62	54	46	37	28	-22%	-43%

Luxembourg – Gasoline Vehicle Fleet Life Cycle GHG Emissions



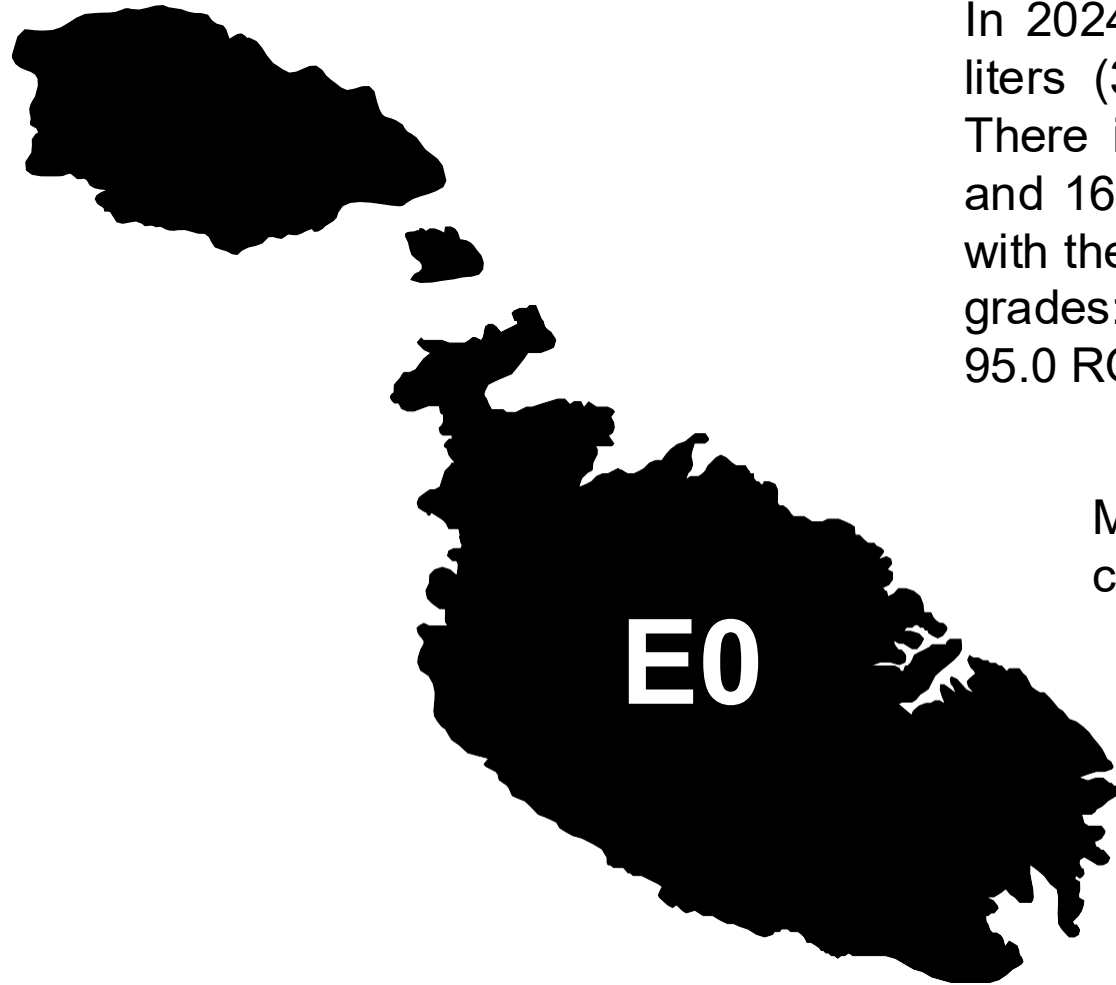
Country Emissions 2024 (MMT/año)	9.0
Target @ 2035 (MMT/year)	5.1
Delta	3.9

%Reduction vs E0	E0	E10	E15	E20	E25	E30
GREET 2024	0.0%	11.2%	14.7%	17.5%	20.2%	23.8%
RED II/III	0.0%	8.1%	10.8%	13.0%	15.2%	17.8%

% Target@2035	E0	E10	E15	E20	E25	E30
GREET 2024	0.0%	5.9%	7.7%	9.2%	10.6%	12.5%
RED II/III	0.0%	7.0%	9.3%	11.2%	13.1%	15.4%

Source: Greet, RED II/III, IPCC, climateactiontracker.org, F90

Ethanol Gasoline Blending in Malta

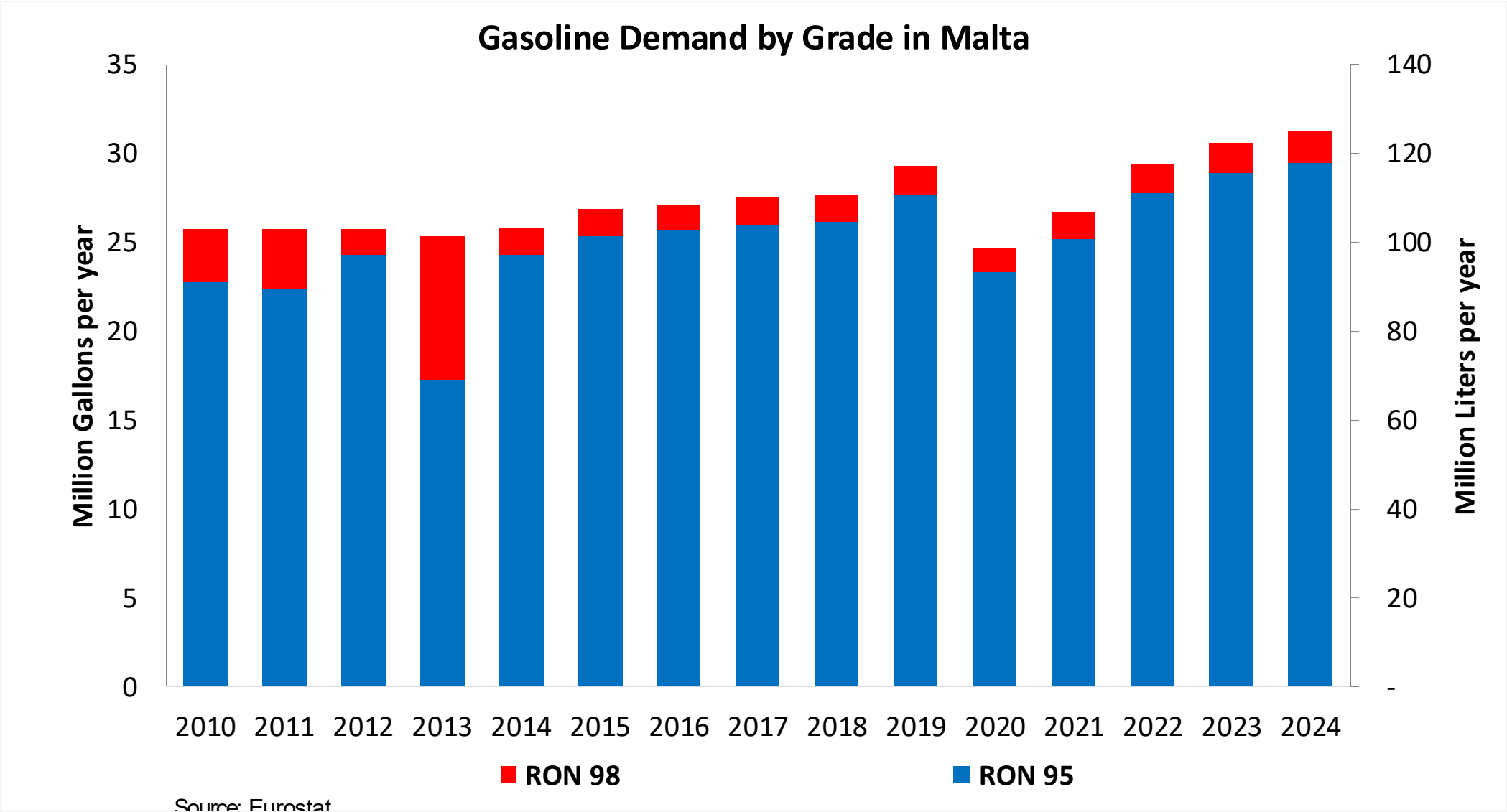


In 2024, gasoline consumption in Malta was 118 million liters (31 million gallons) and growth rate was 1.2%. There is no gasoline production. Gasoline imports were 163 million liters (43 million gallons). Malta complies with the European gasoline standards and offers two main grades: RON 95 and RON 98) with an average octane of 95.0 RON.

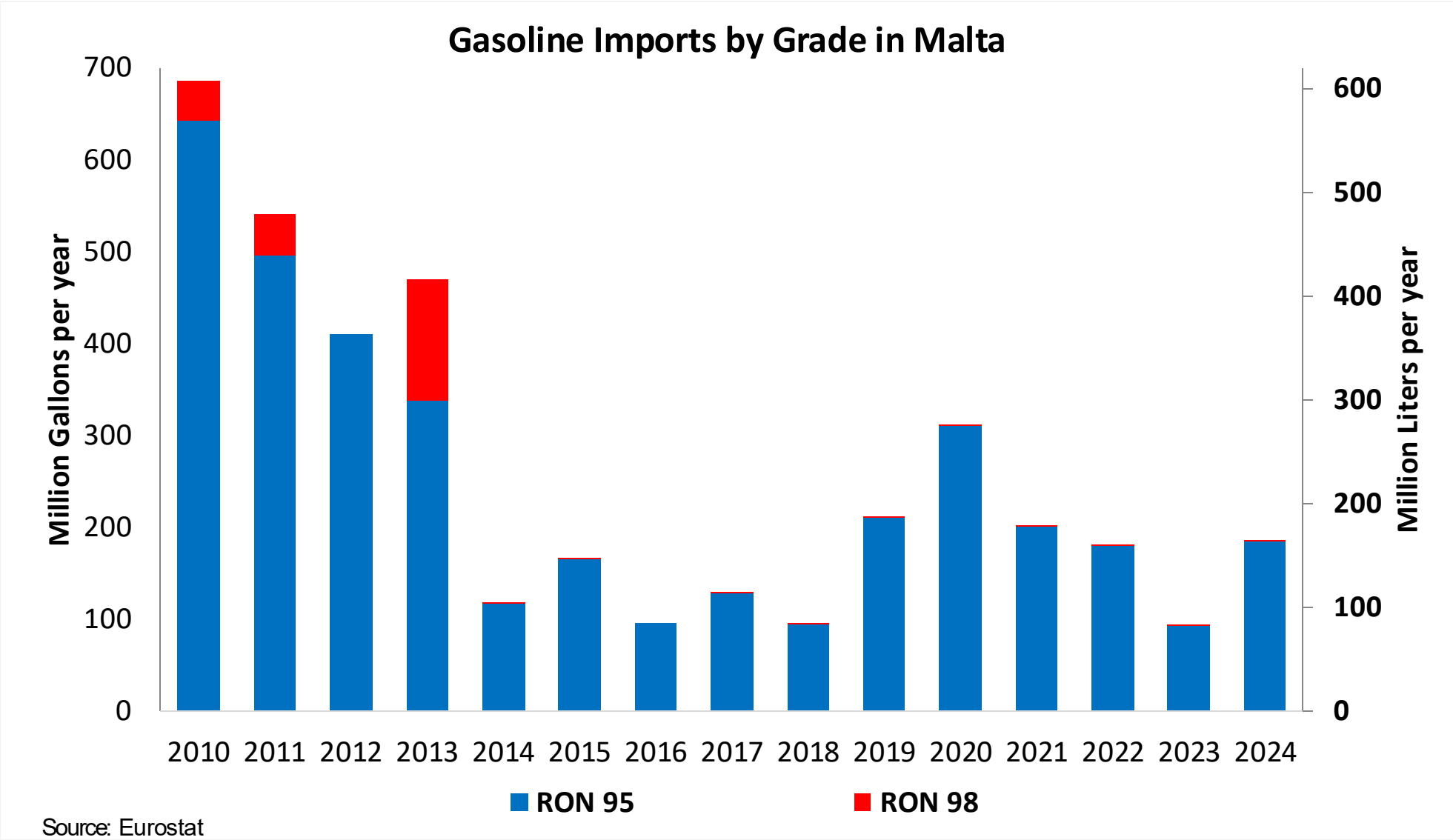
Malta does not produce or consume bioethanol at a commercial level

Source: European Commission, Eurostat

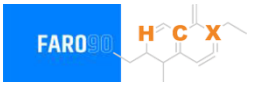
Gasoline Demand in Malta



Gasoline Imports in Malta



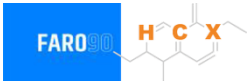
Ethanol Balance in Malta



Malta does not produce or consume bioethanol at a commercial level

Source: Eurostat

Gasoline Quality in Malta

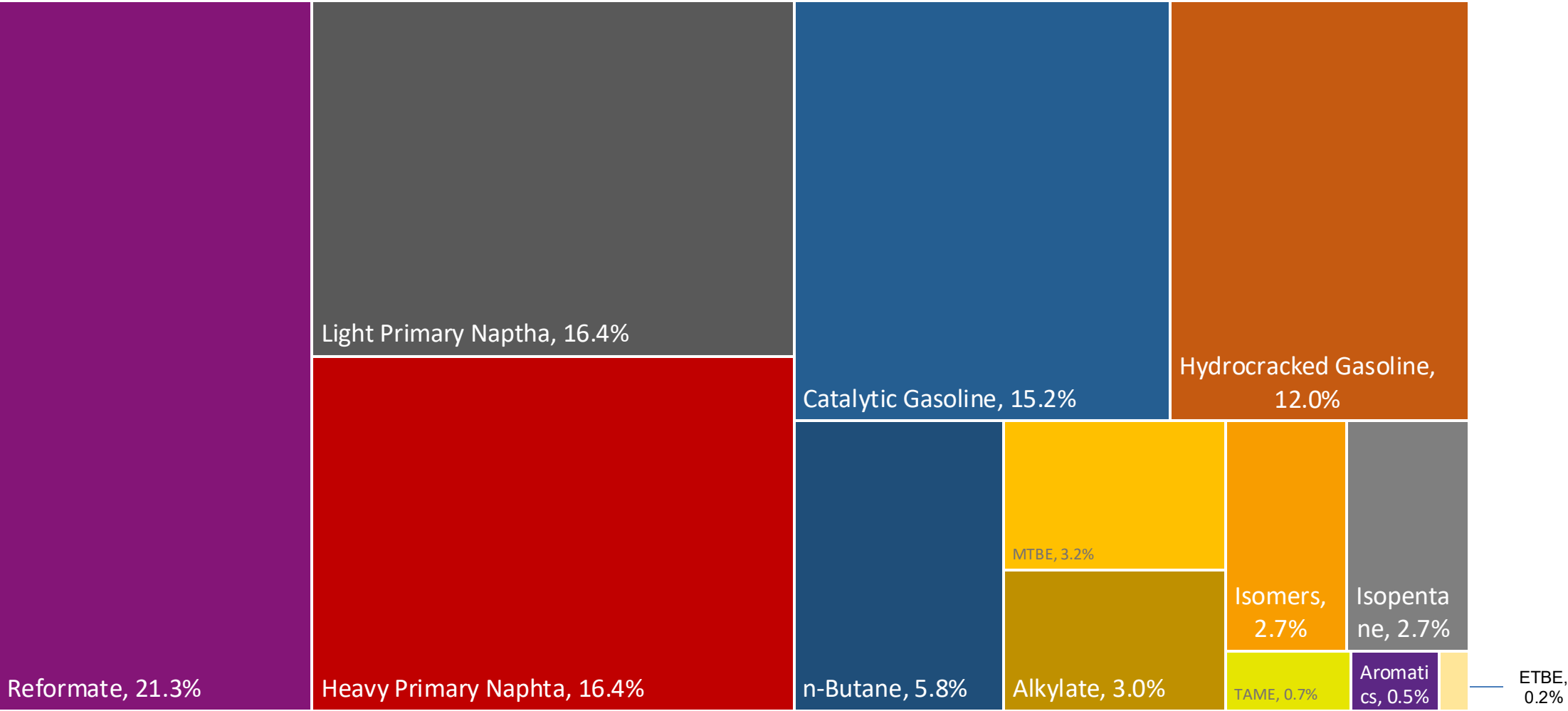
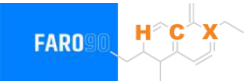


Name	EN 228:2025 (Euro 6); Country Level Fuel Ordinances							
Implementation Date	2025							
Applicability	All EU Countries + UK							
Grade	RON 95 E5		RON 95 E10		RON 98 E5		RON 98 E10	
	Min	Max	Min	Max	Min	Max	Min	Max
RON	95.0		95.0		98.0		98.0	
MON	> 85	> 88	> 85	> 88	> 85	> 88	> 85	> 88
AKI								
Benzene (%vol)	< 1							
Aromatics (% vol)	< 35							
Olefins (%vol)	< 18							
Lead (g/L)	< 0.005							
Manganese (mg/L)	< 2,0							
Sulfur (ppm)	10.0							
Oxygen (%w)*	< 2,7	< 3,7	< 2,7	< 3,7	< 2,7	< 3,7	< 2,7	< 3,7
Ethanol (%vol)	< 5	< 10	< 5	< 10	< 5	< 10	< 5	< 10
PVR 37.8°C (psi)	<> 45 - 100 kPa in 6 Volatility Classes							
Overall Biofuels (%cal)	-							
Advanced Biofuels (%cal)	0.2							
Bioethanol in Gasoline Sales (%cal)	-							
GHG Emission Reduction (%)	-							

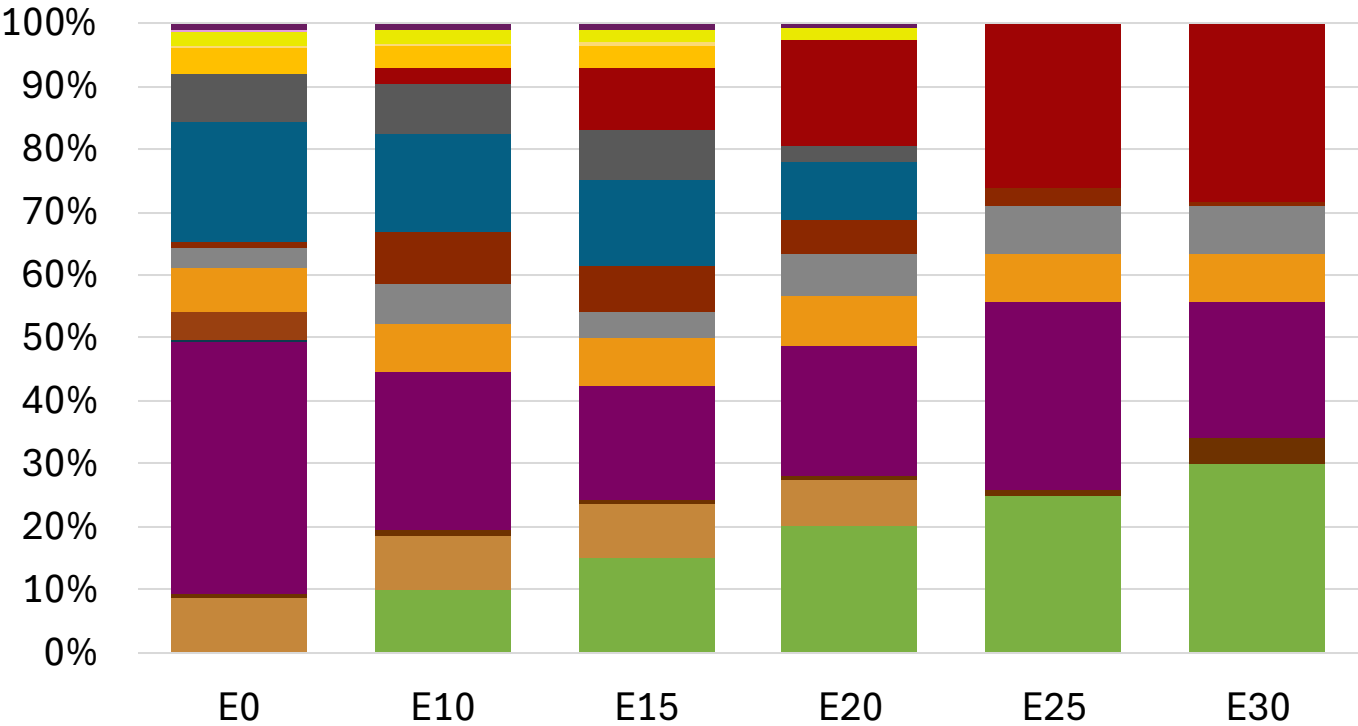
* Methanol 3%, Iso-propyl alcohol 12%, Ter-butyl alcohol 15%, Iso-butyl alcohol 15%, Ethers +5 Carbons 22%, Other Oxygenates 15%

Source: European Commission

Malta Available Blending Components



Malta – Gasoline 95 – Constant Octane

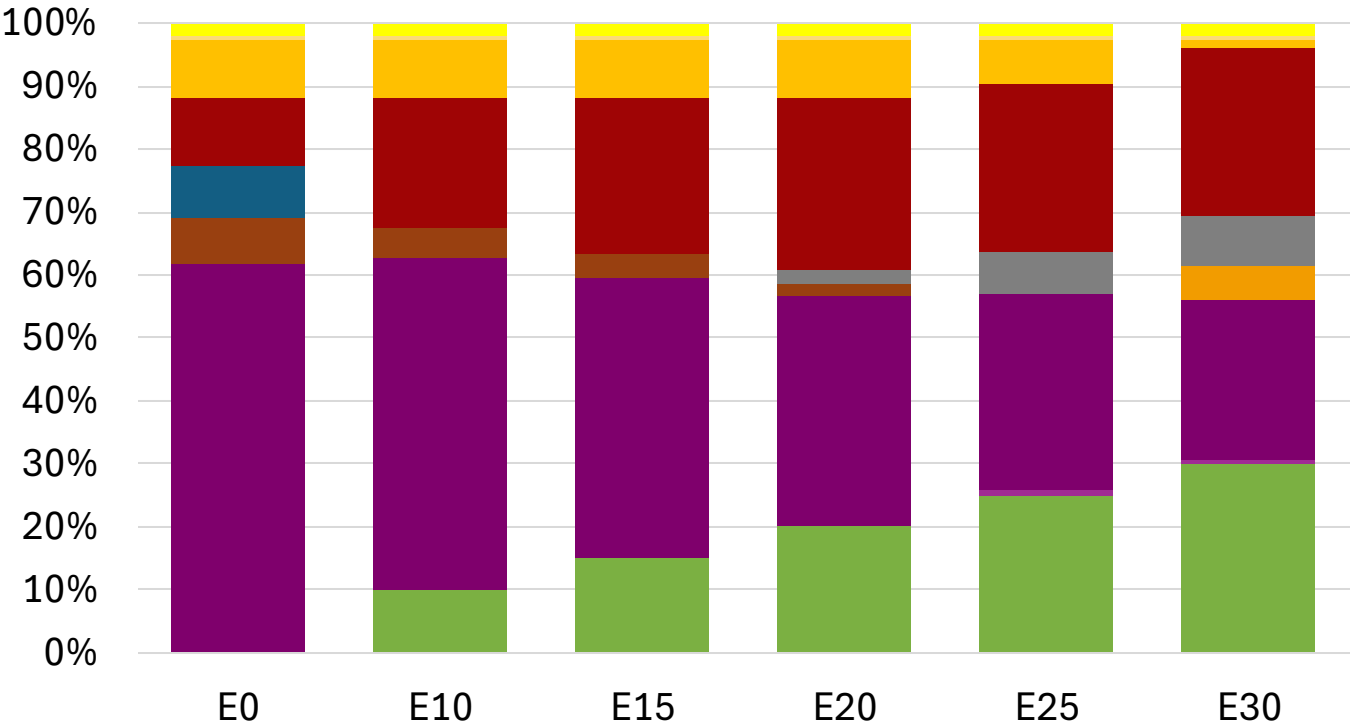


Average CIF 2024
Prices

Octane (RON)	95.0	95.0	95.0	95.0	95.4	95.8
Price (US\$/gal)	\$ 2.258	\$ 2.145	\$ 2.062	\$ 1.986	\$ 1.891	\$ 1.817

Source: Faro90

Malta - Gasoline 98 - Constant Octane

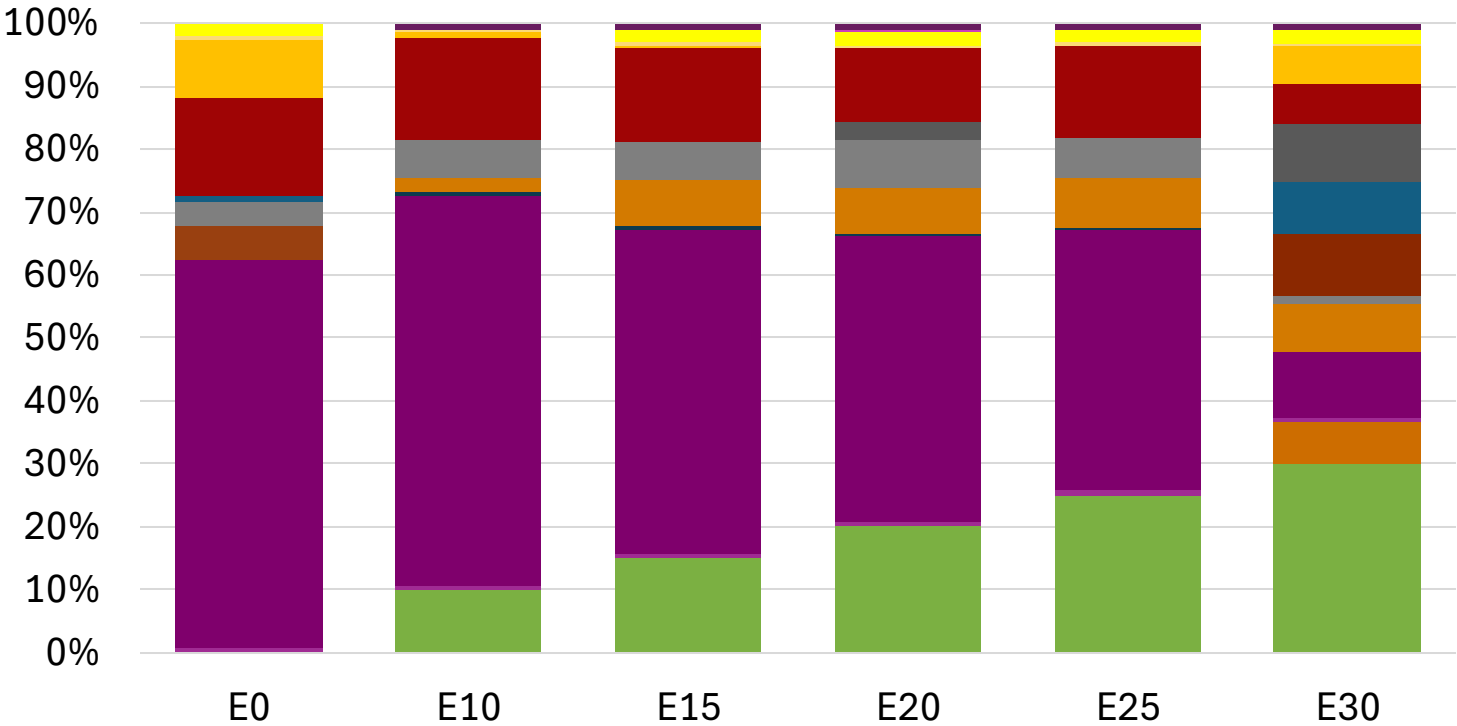
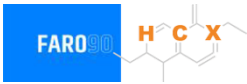


Average CIF 2024
Prices

Octane (RON)	98.0	98.0	98.0	98.0	98.0	98.0
Price (US\$/gal)	\$ 2.307	\$ 2.188	\$ 2.119	\$ 2.049	\$ 1.979	\$ 1.926

Source: Faro90

Malta – Gasoline 95 – Increased Octane

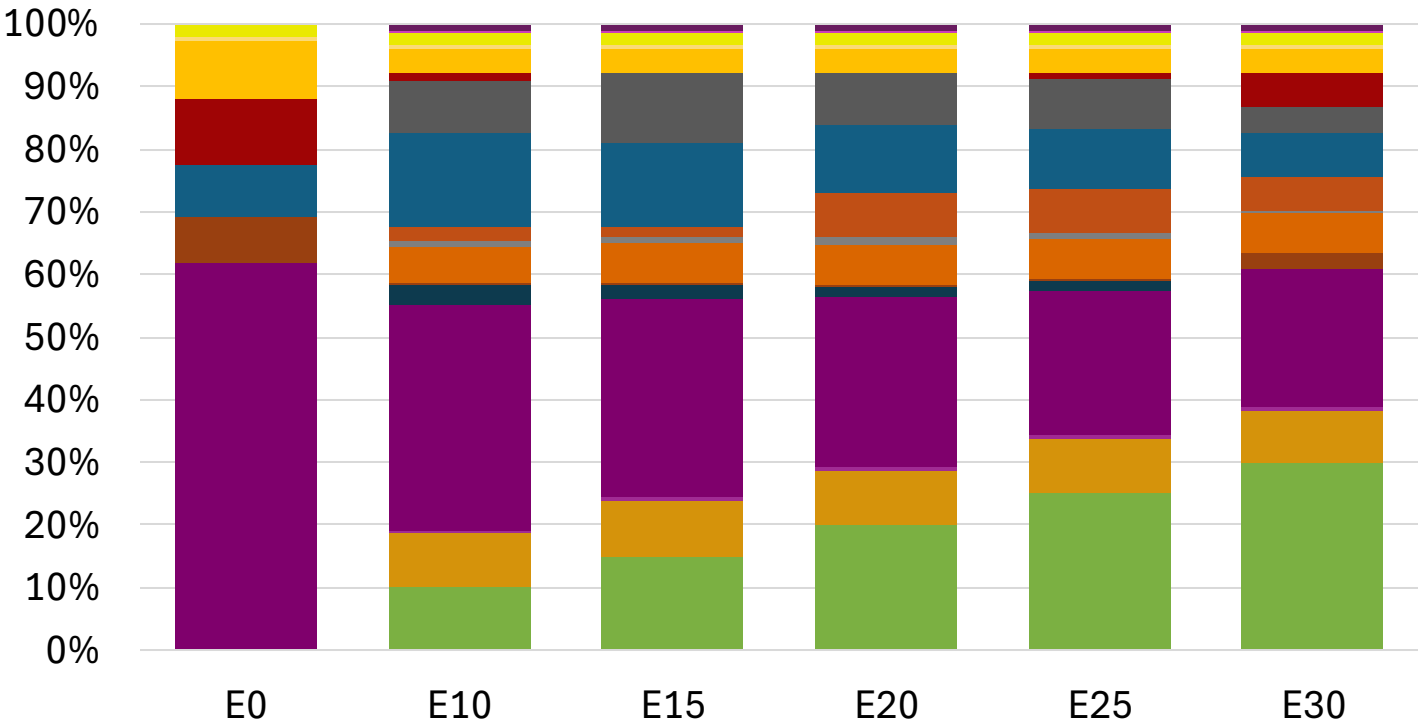
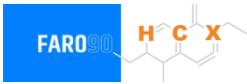


Average CIF 2024 Prices

Octane (RON)	95.0	95.9	97.5	99.0	100.5	101.8
Price (US\$/gal)	\$ 2.177	\$ 2.177	\$ 2.177	\$ 2.177	\$ 2.177	\$ 2.177

Source: Faro90

Malta – Gasoline 98 – Increased Octane

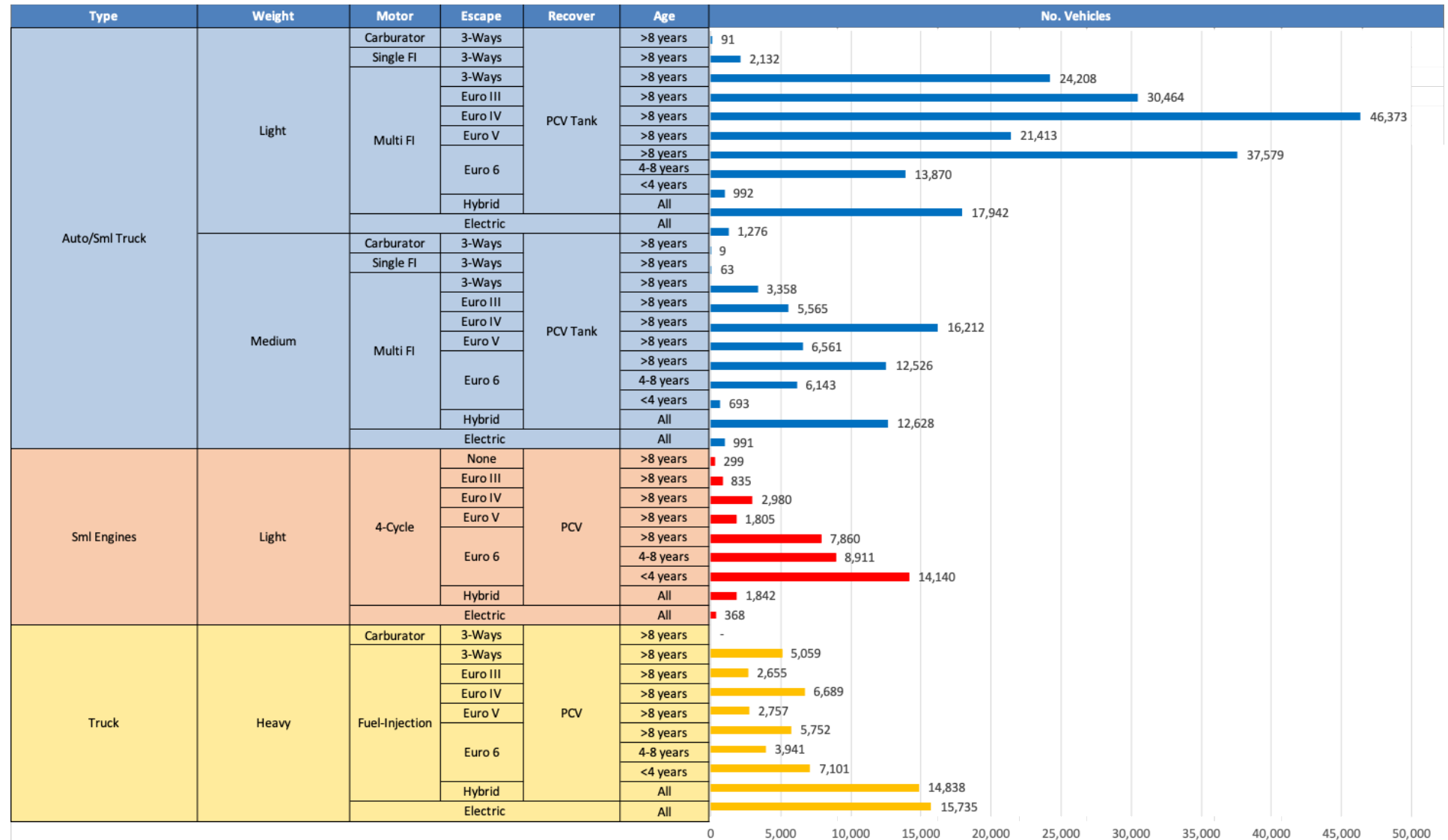


Average CIF 2024
Prices

Octane (RON)	98.0	98.8	100.1	101.3	102.6	104.1
Price (US\$/gal)	\$ 2.307	\$ 2.307	\$ 2.307	\$ 2.307	\$ 2.307	\$ 2.307

Source: Faro90

Gasoline Vehicle Fleet - Malta

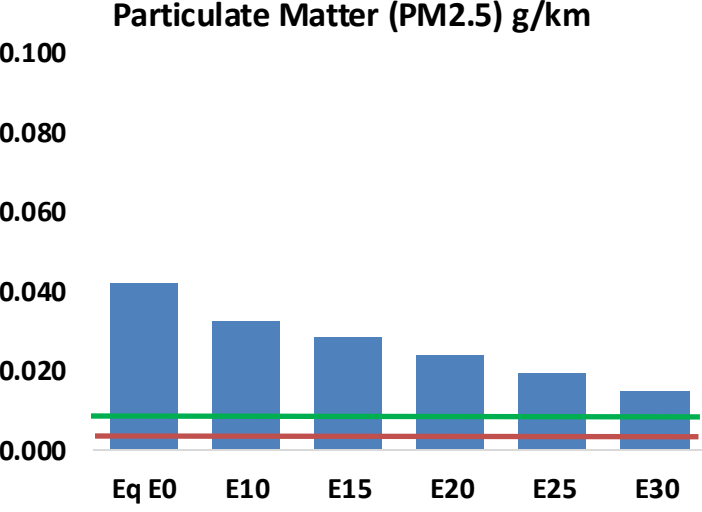
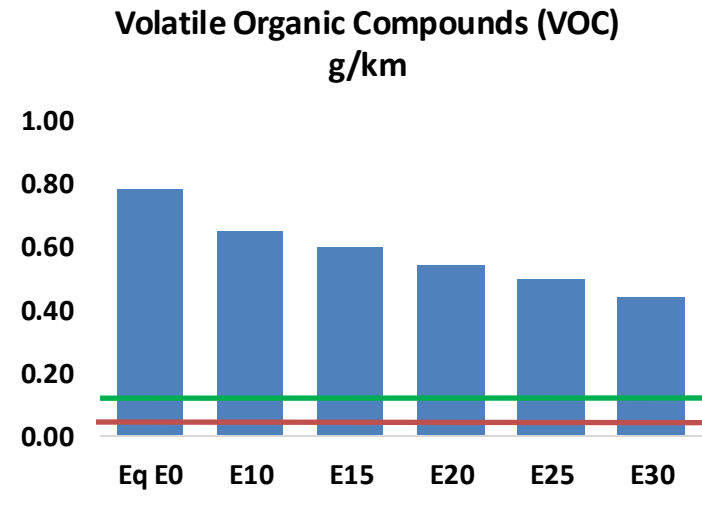
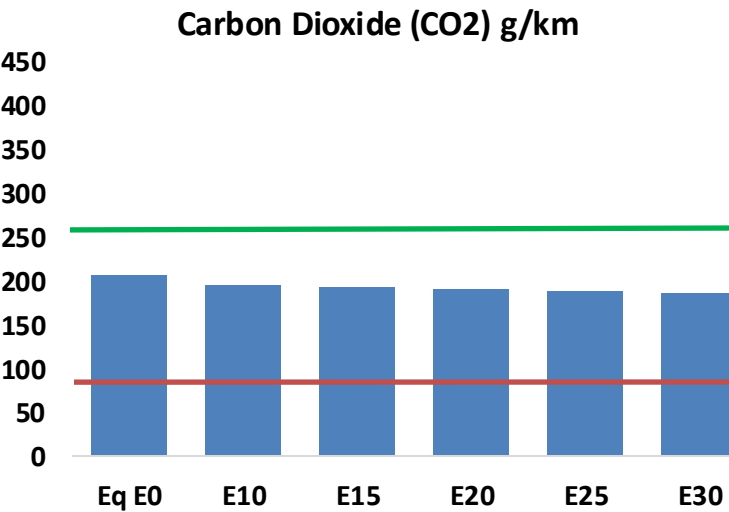
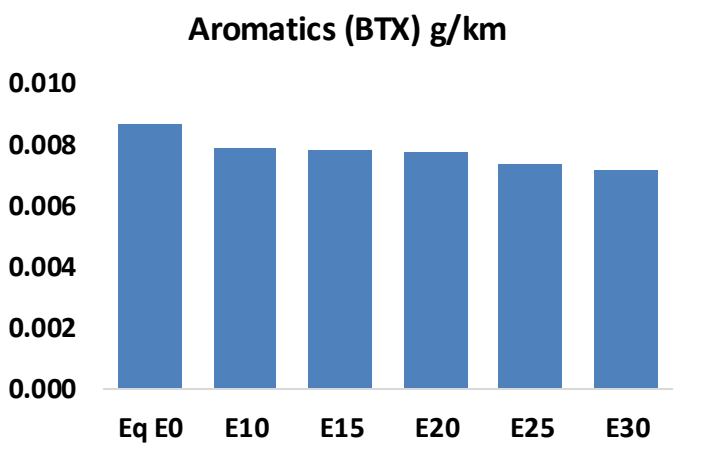
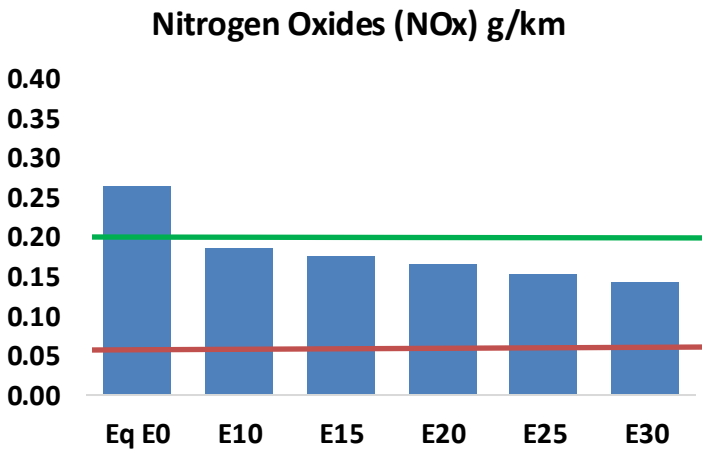
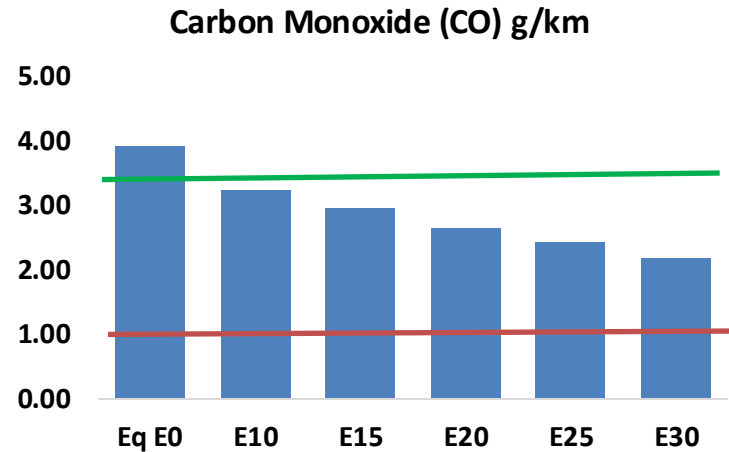
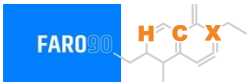


Vehicle Fleet: 364,656
Gasoline Vehicle Fleet 207,158

Source: Eurostat, ACEA

Malta – Vehicular Emissions

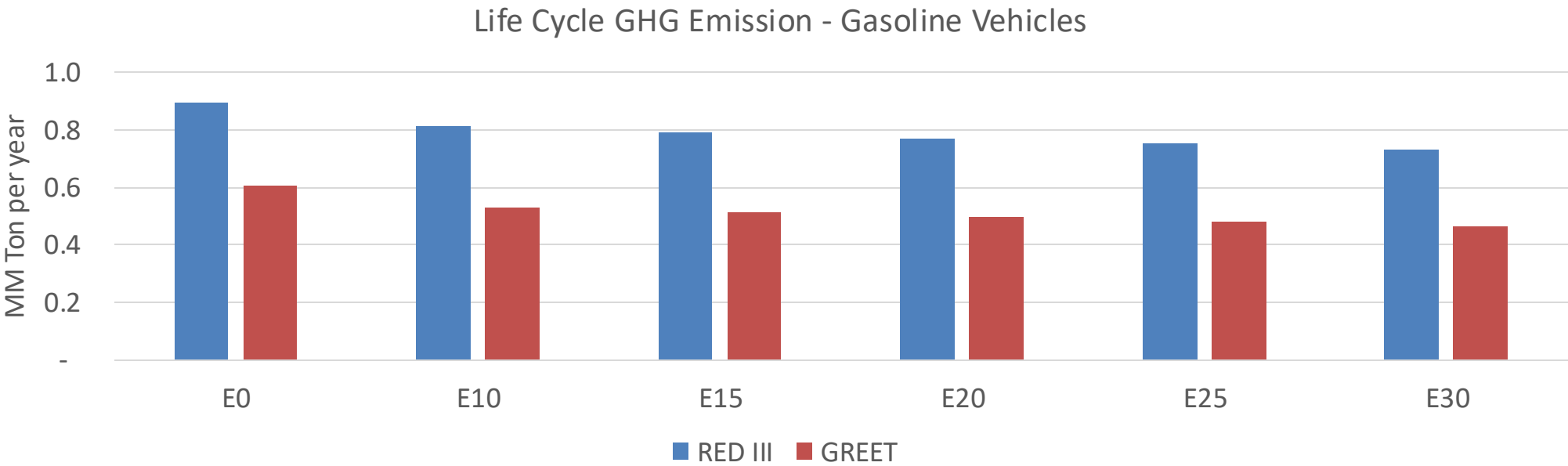
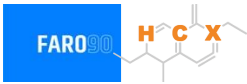
TIER III BIN 125 United States
Euro 6 Standard



Malta – Vehicular Emissions

Vehicular Fleet Emissions (kg/day)	Eq E0	E5	E10	E15	E20	E25	E30	Reduction (%)	
								E10	E20
CO	19,908	18,177	16,446	14,988	13,502	12,355	11,130	-17%	-32%
CO2	1,052,550	1,025,484	999,922	979,818	969,870	964,236	946,463	-5%	-8%
VOC	3,980	3,647	3,315	3,038	2,756	2,539	2,252	-17%	-31%
NOx	1,355	1,016	948	894	843	787	728	-30%	-38%
SOx	13	12	11	10	9	8	7	-15%	-28%
NH3	404	400	396	398	402	405	408	-2%	0%
N2O	15	15	15	15	16	16	16	-1%	2%
CH4	807	807	807	819	837	852	867	0%	4%
Lead	-	-	-	-	-	-	-	0%	0%
Butadiene	20	19	18	18	17	17	17	-8%	-13%
Acetaldehyde	66	76	111	169	230	263	310	68%	249%
Formaldehyde	245	238	277	326	341	378	411	13%	39%
Benzene	44	43	40	40	39	38	36	-9%	-11%
PM2.5	66	59	51	45	38	31	23	-22%	-43%
PM10	148	131	115	99	84	69	52	-22%	-43%

Malta – Gasoline Vehicle Fleet Life Cycle GHG Emissions



Country Emissions 2024 (MMT/año)	2.2
Target @ 2035 (MMT/year)	1.2
Delta	1.0

%Reduction vs E0	E0	E10	E15	E20	E25	E30
GREET 2024	0.0%	12.4%	15.6%	18.1%	20.5%	23.7%
RED II/III	0.0%	9.2%	11.7%	13.7%	15.6%	18.1%

% Target@2035	E0	E10	E15	E20	E25	E30
GREET 2024	0.0%	7.8%	9.8%	11.4%	12.9%	15.0%
RED II/III	0.0%	8.5%	10.9%	12.8%	14.6%	16.9%

Source: Greet, RED II/III, IPCC, climateactiontracker.org, F90

Ethanol Gasoline Blending in Netherlands



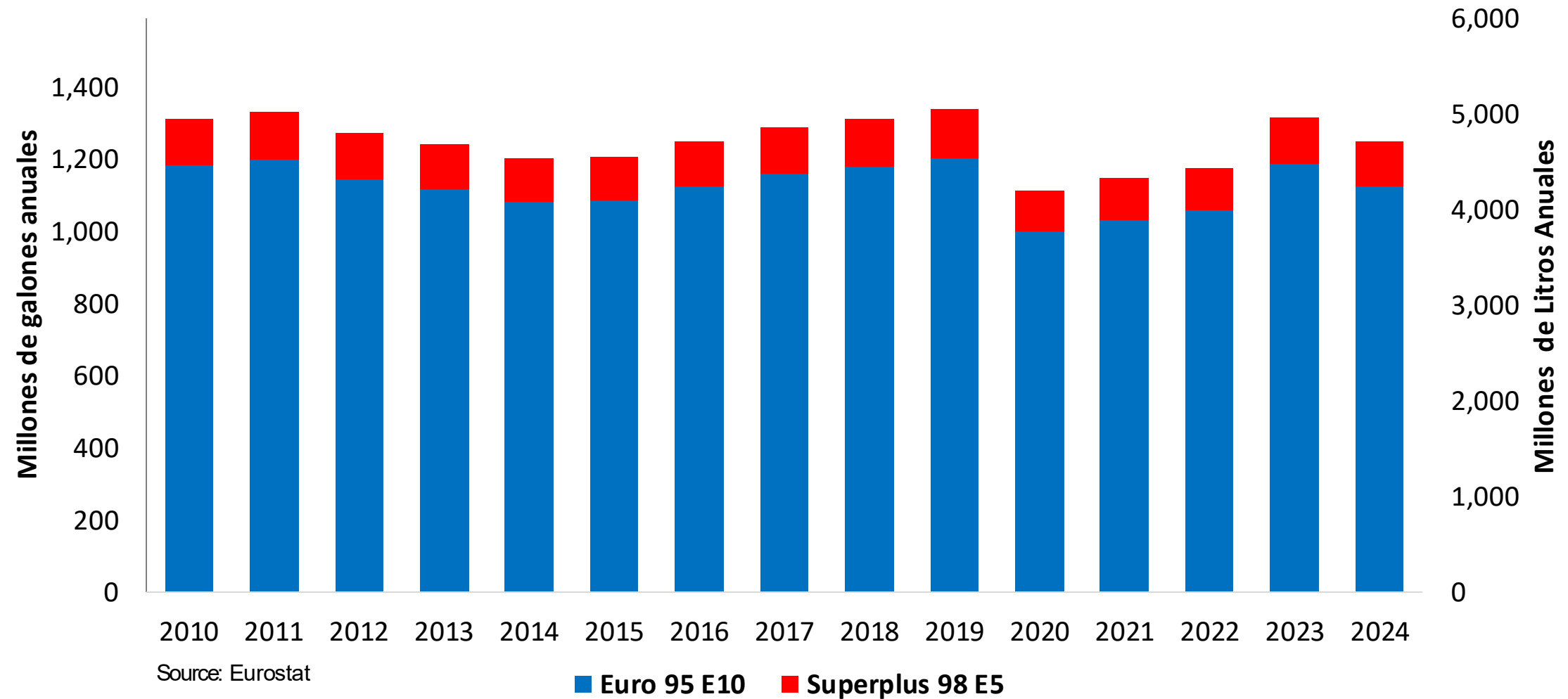
In 2024, gasoline consumption in Netherlands was 4,723 million liters (1,248 million gallons) and growth rate was -1.3%. Gasoline production and net imports were 6,295 and -14,373 million liters (1,663 and -3,797 million gallons) correspondingly. Netherlands complies with the European gasoline standards and offers two main grades: RON 95 and RON 98) with an average octane of 95.3 RON.

In 2024, ethanol consumption in Netherlands was 507 million liters (134 million gallons) representing 10.7% of gasoline demand. Ethanol production and Net Imports were 1,080 and -574 million liters (134 and -152 million gallons) correspondingly. Ethanol sources are Sugar Cane 5% and Cereals 95%.

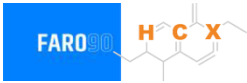
Source: Eurostat, European Commission

Gasoline Demand in Netherlands

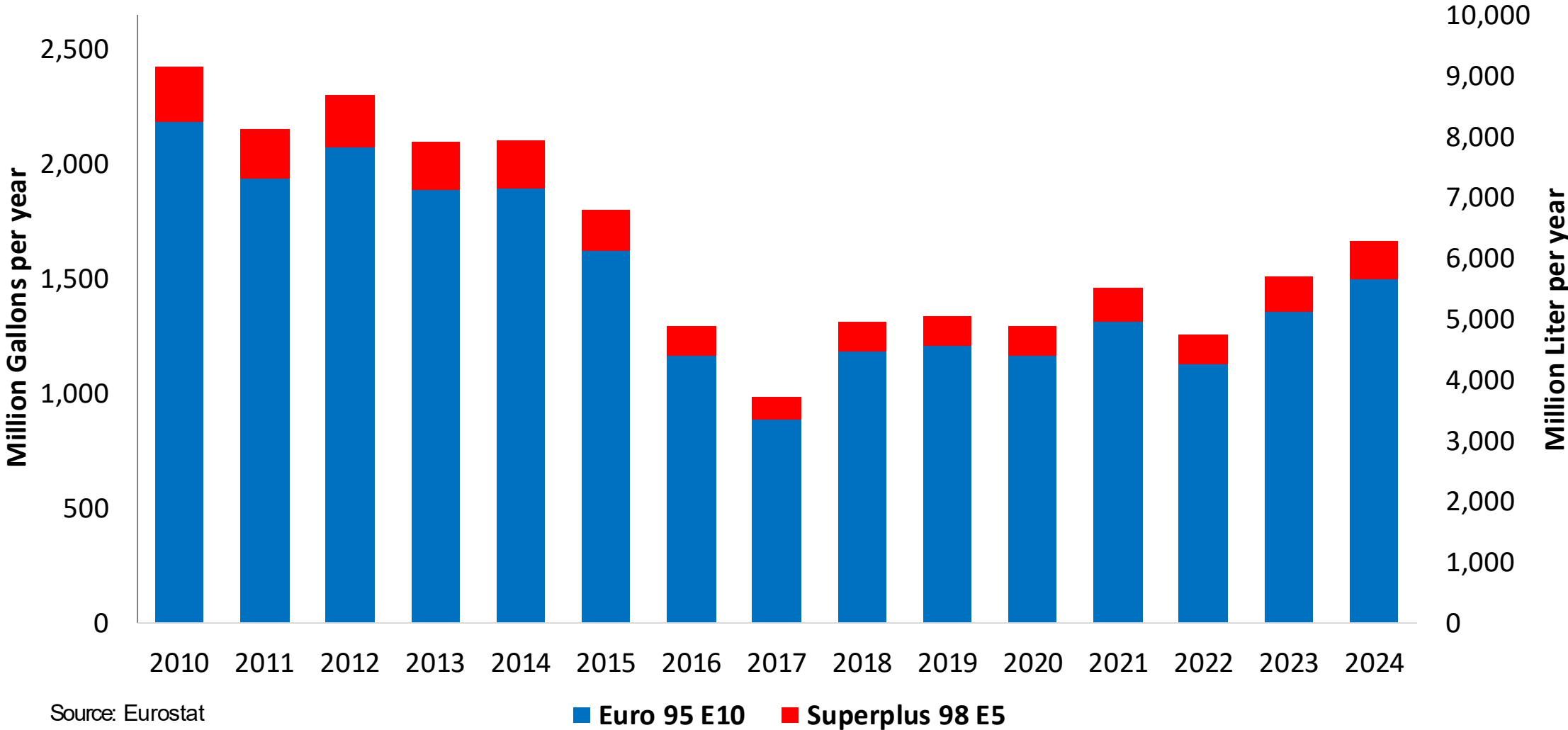
Gasoline Demand by Grade in Netherlands



Gasoline Production in Netherlands

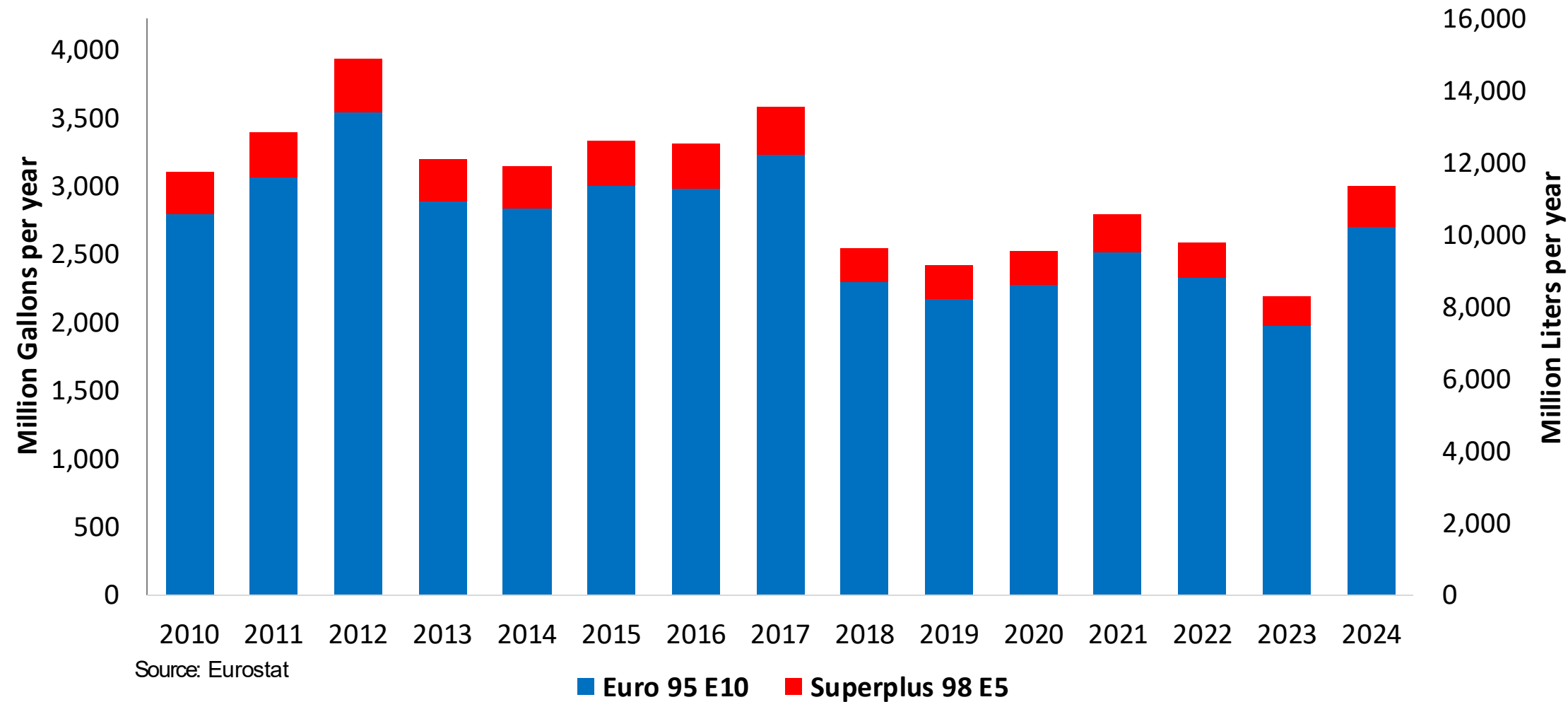


Gasoline Production by Grade in Netherlands



Gasoline Imports in Netherlands

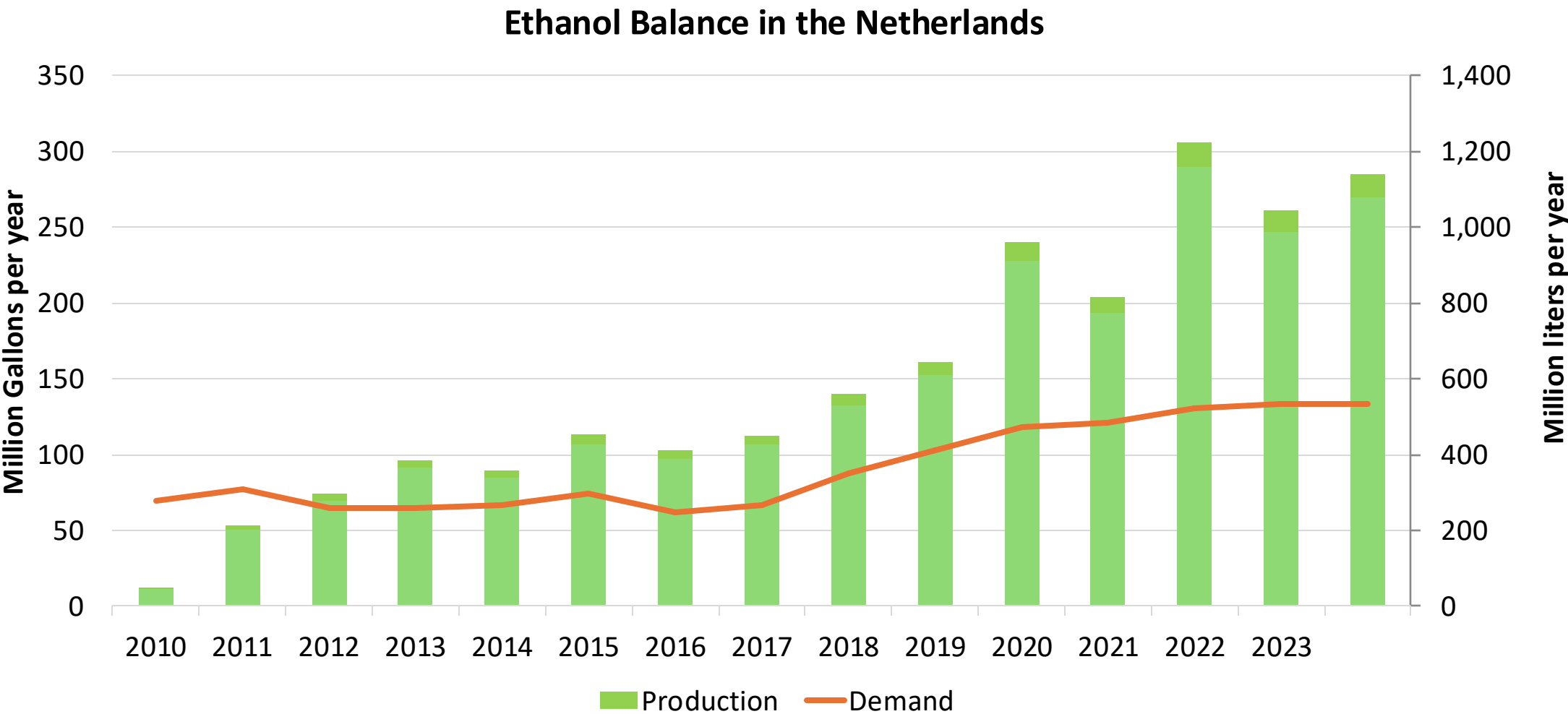
Gasoline Imports by Grade in Netherlands



The FARO90 logo is shown in a blue box, followed by a chemical structure of a substituted cyclohexadiene. The structure features a six-membered ring with two double bonds. Substituents include a methyl group, a propyl group, and a group labeled 'X'. The letters 'H', 'C', and 'X' are highlighted in orange in the original image.



Ethanol Balance in Netherlands



Source: Eurostat, Statistics Netherlands

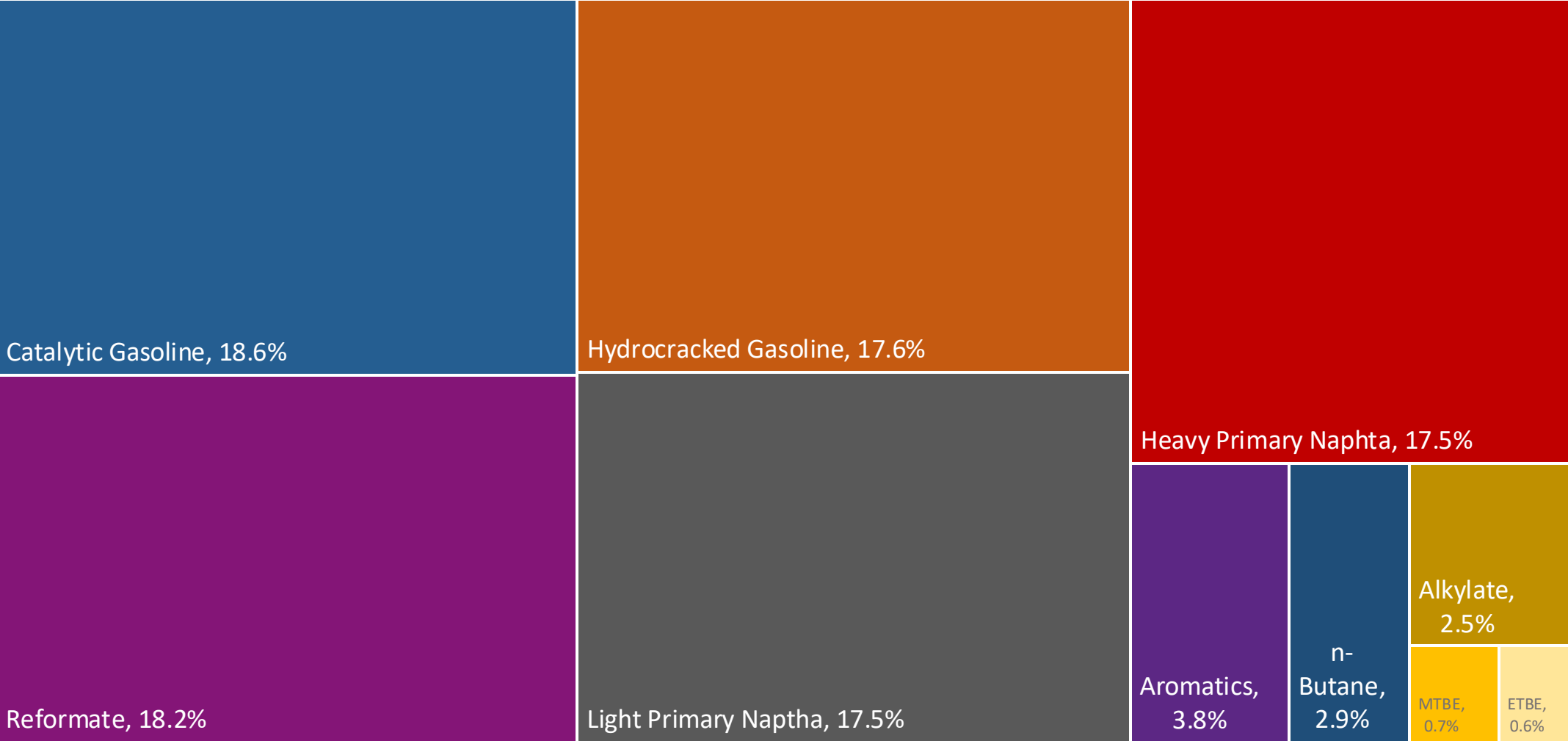
Gasoline Quality in Netherlands

Name	EN 228:2025 (Euro 6); Country Level Fuel Ordinances							
Implementation Date	2025							
Applicability	All EU Contries + UK							
Grade	RON 95 E5		RON 95 E10		RON 98 E5		RON 98 E10	
	Min	Max	Min	Max	Min	Max	Min	Max
RON	95.0		95.0		98.0		98.0	
MON	> 85	> 88	> 85	> 88	> 85	> 88	> 85	> 88
AKI								
Benzene (%vol)	< 1							
Aromatics (% vol)	< 35							
Olefins (%vol)	< 18							
Lead (g/L)	< 0.005							
Manganese (mg/L)	< 2,0							
Sulfur (ppm)	10.0							
Oxygen (%w)*	< 2,7	< 3,7	< 2,7	< 3,7	< 2,7	< 3,7	< 2,7	< 3,7
Ethanol (%vol)	< 5	< 10	< 5	< 10	< 5	< 10	< 5	< 10
PVR 37.8°C (psi)	<> 45 - 100 kPa in 6 Volatility Classes							
Overall Biofuels (%cal)	28.4							
Advanced Biofuels (%cal)	3.6							
Bioethanol in Gasoline Sales (%cal)	-							
GHG Emission Reduction (%)	-							

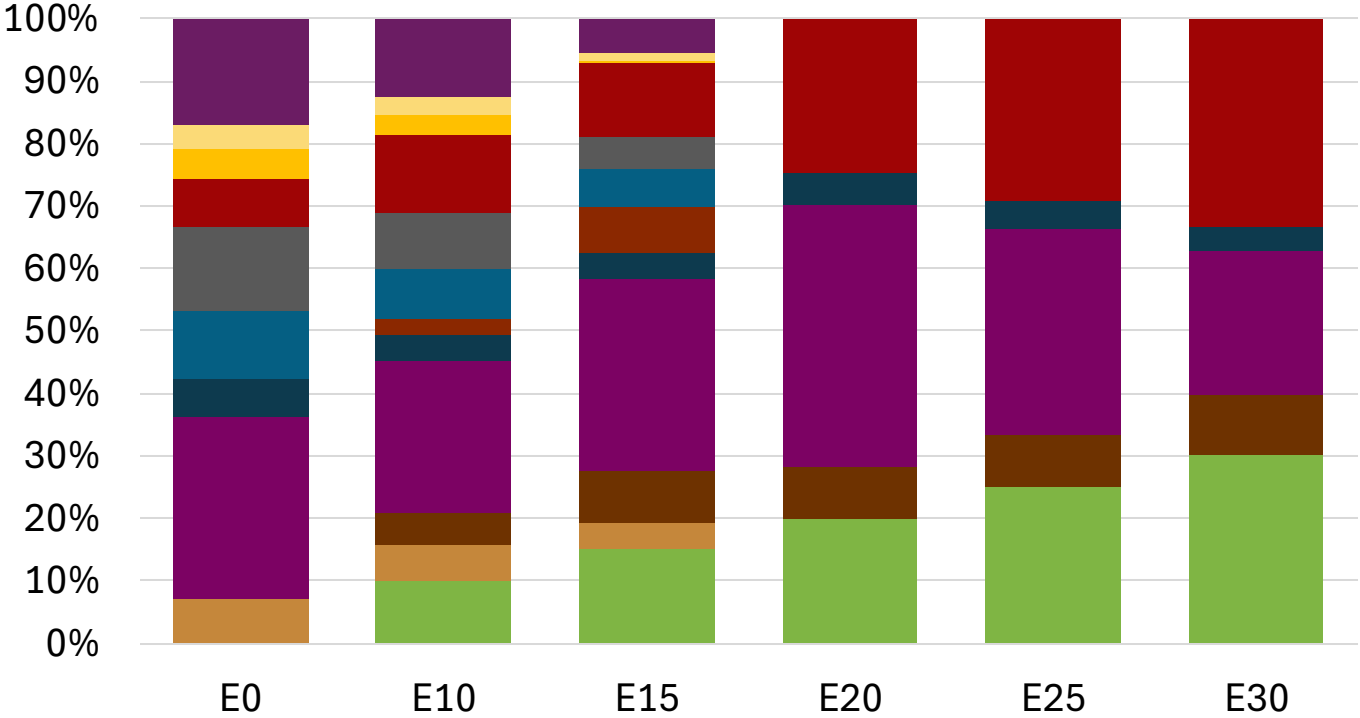
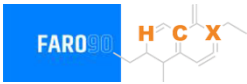
* Methanol 3%, Iso-propyl alcohol 12%, Ter-butyl alcohol 15%, Iso-butyl alcohol 15%, Ethers +5 Carbons 22%, Other Oxygenates 15%

Source: European Commision

Netherlands Available Blending Components



Netherlands – Gasoline 95 – Constant Octane

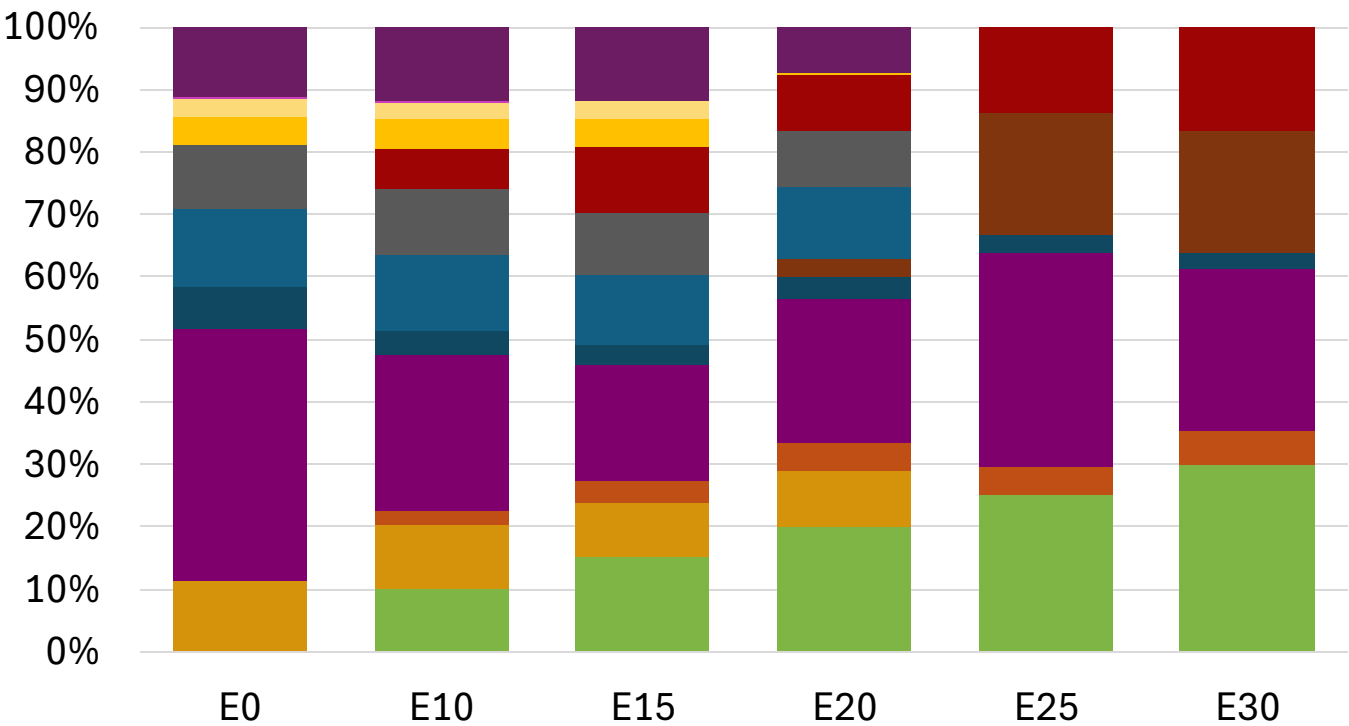
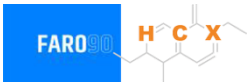


Average CIF 2024
Prices

Octane (RON)	95.0	95.0	95.0	95.1	95.3	95.3
Price (US\$/gal)	\$ 2.281	\$ 2.147	\$ 2.072	\$ 1.951	\$ 1.875	\$ 1.780

Source: Faro90

Netherlands - Gasoline 98 - Constant Octane

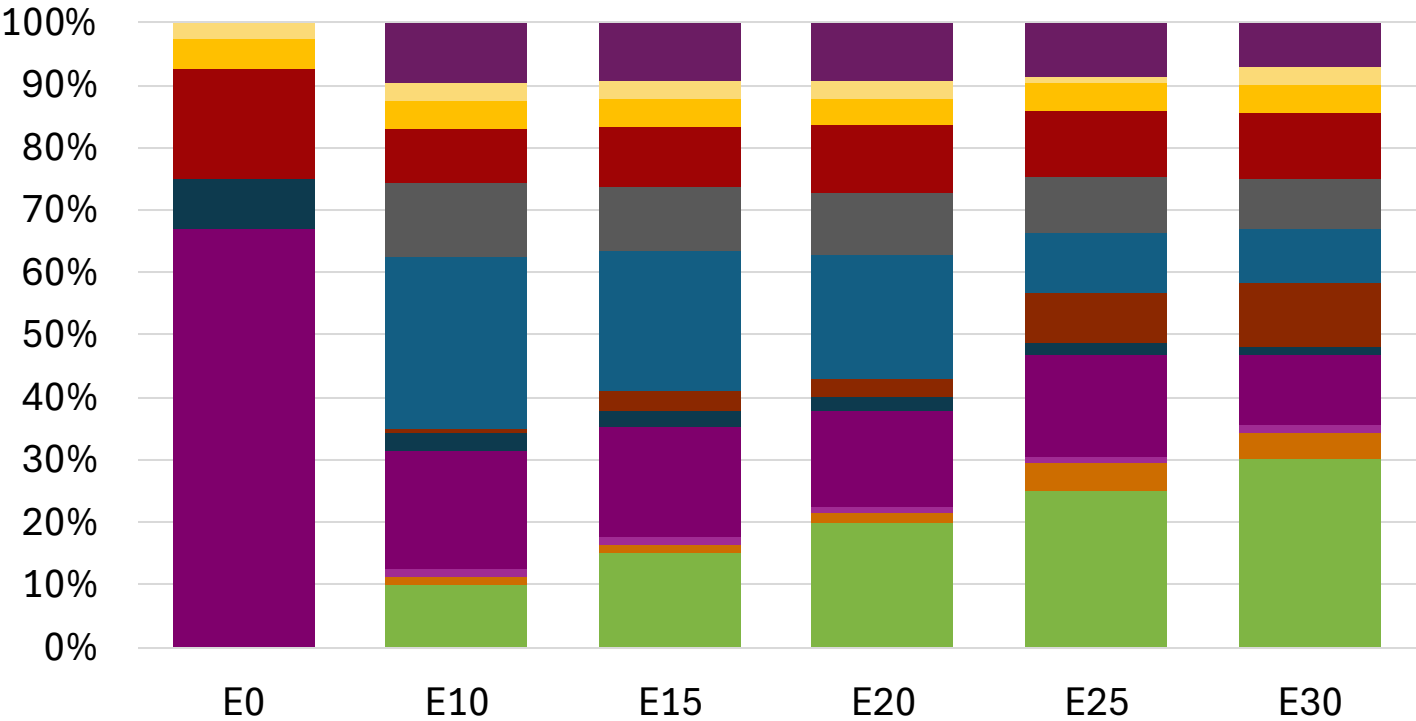
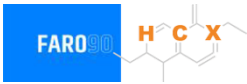


Average CIF 2024 Prices

Octane (RON)	98.0	98.0	98.0	98.0	98.1	98.5
Price (US\$/gal)	\$ 2.409	\$ 2.290	\$ 2.216	\$ 2.164	\$ 2.057	\$ 1.994

Source: Faro90

Netherlands – Gasoline 95 – Increased Octane

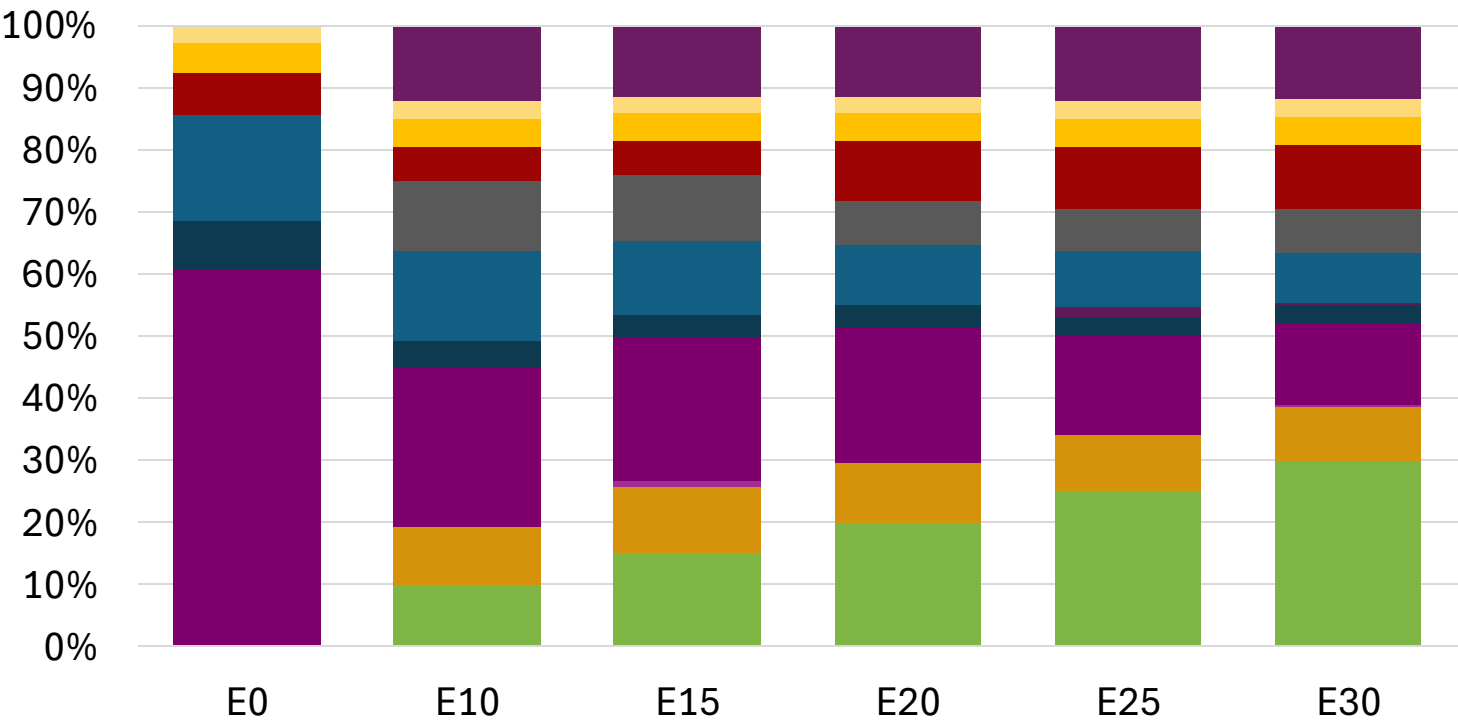


Average CIF 2024 Prices

Octane (RON)	95.0	96.5	97.9	99.4	100.5	102.1
Price (US\$/gal)	\$ 2.193	\$ 2.193	\$ 2.193	\$ 2.193	\$ 2.193	\$ 2.193

Source: Faro90

Netherlands – Gasoline 98 – Increased Octane

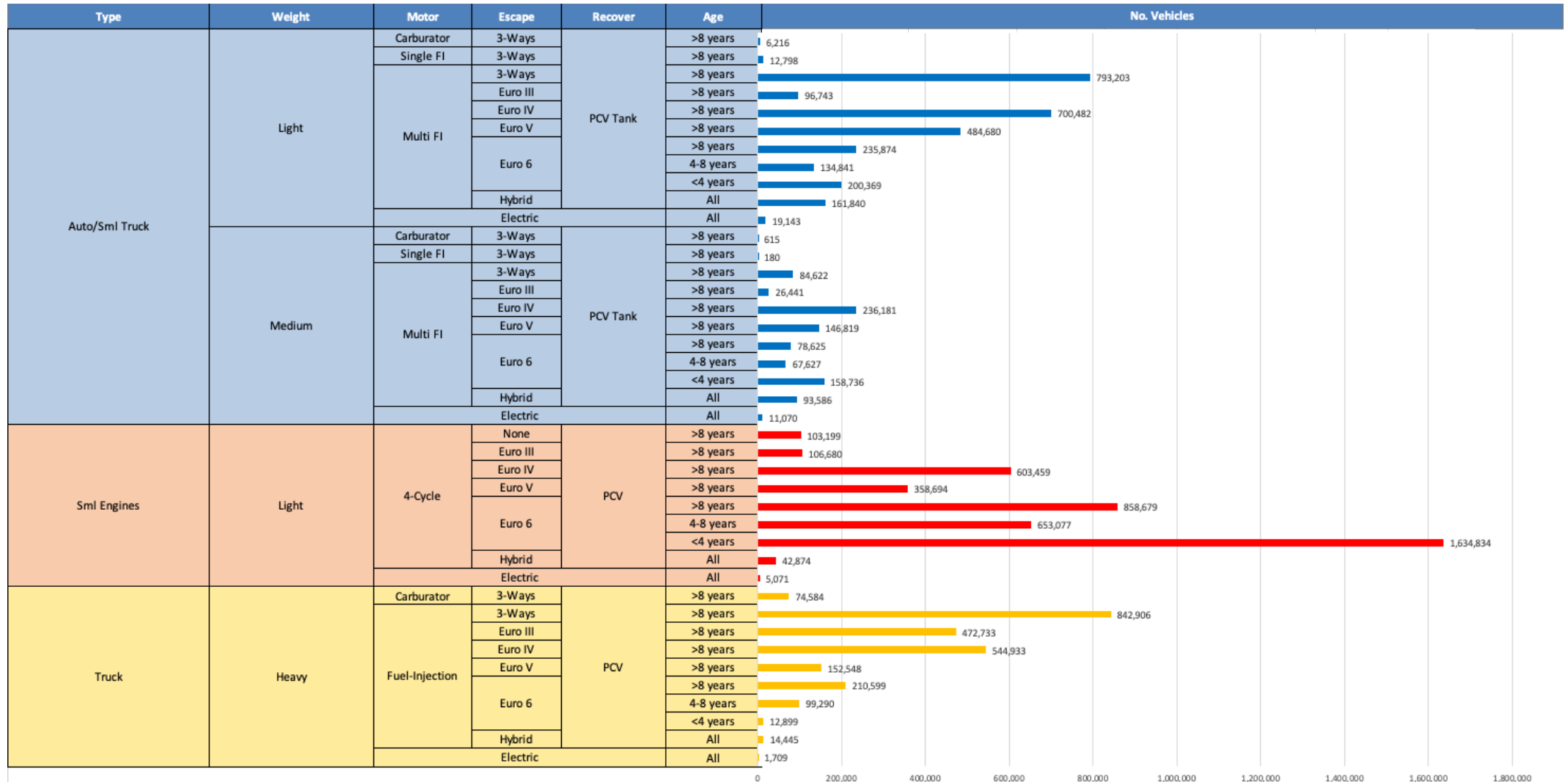


Average CIF 2024
Prices

Octane (RON)		98.0	98.8	100.1	101.6	102.9	104.2
Price (US\$/gal)	\$	2.329	\$ 2.329	\$ 2.329	\$ 2.329	\$ 2.329	\$ 2.329

Source: Faro90

Gasoline Vehicle Fleet - Netherlands

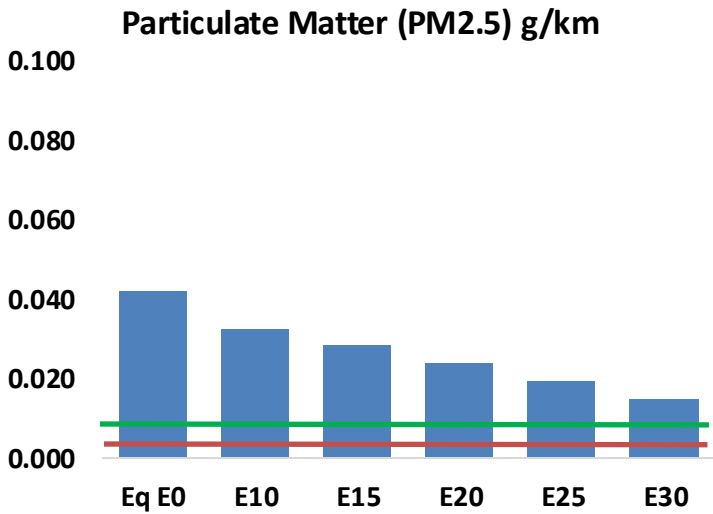
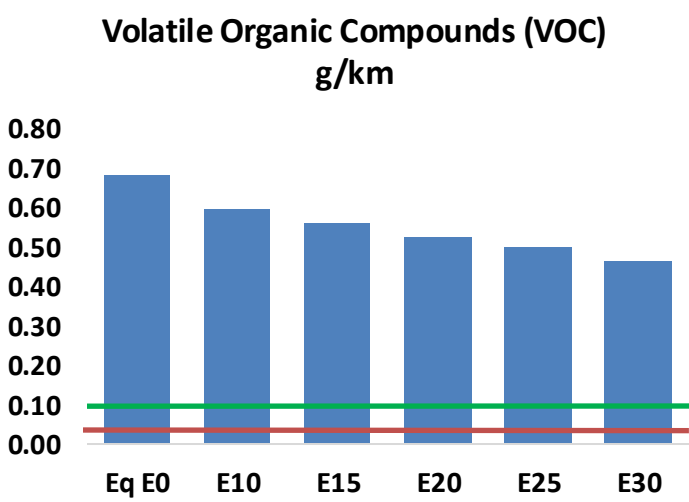
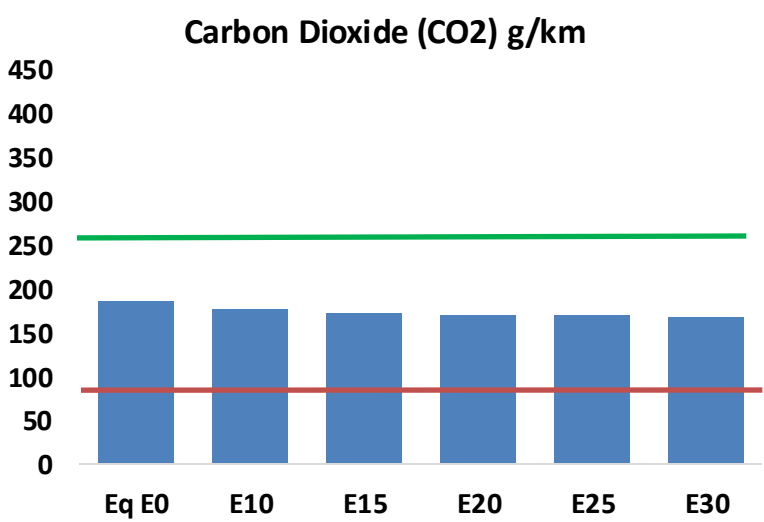
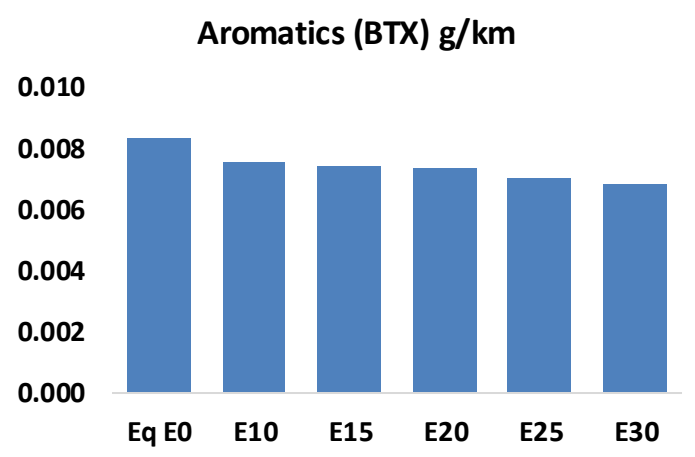
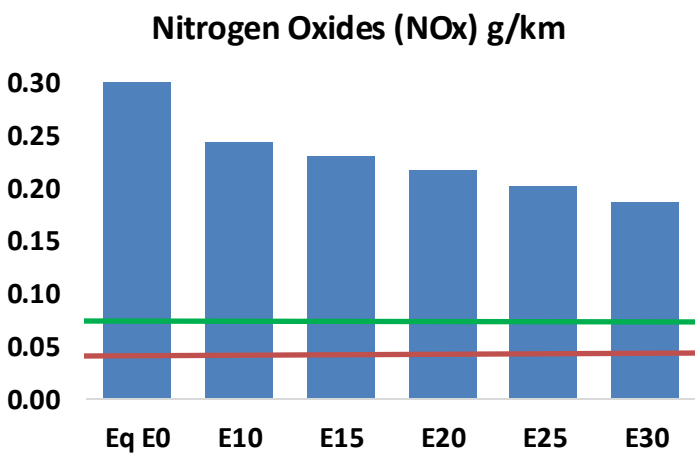
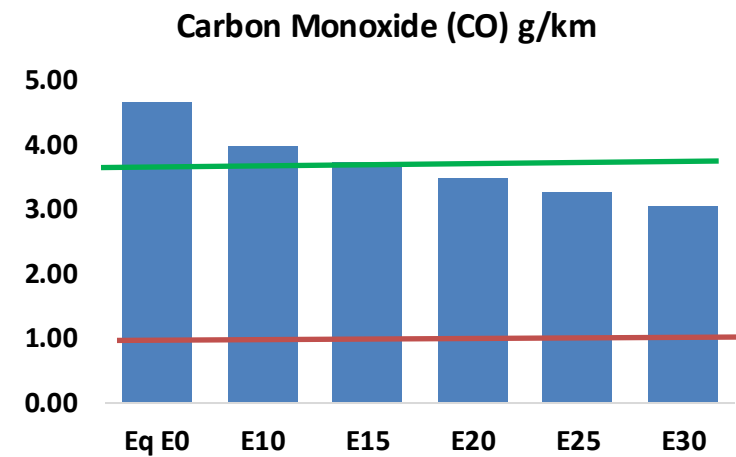
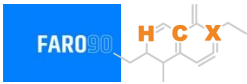


Vehicle Fleet: 10,593,903
Gasoline Vehicle Fleet: 7,721,374

Source: Eurostat, ACEA

Netherlands – Vehicular Emissions

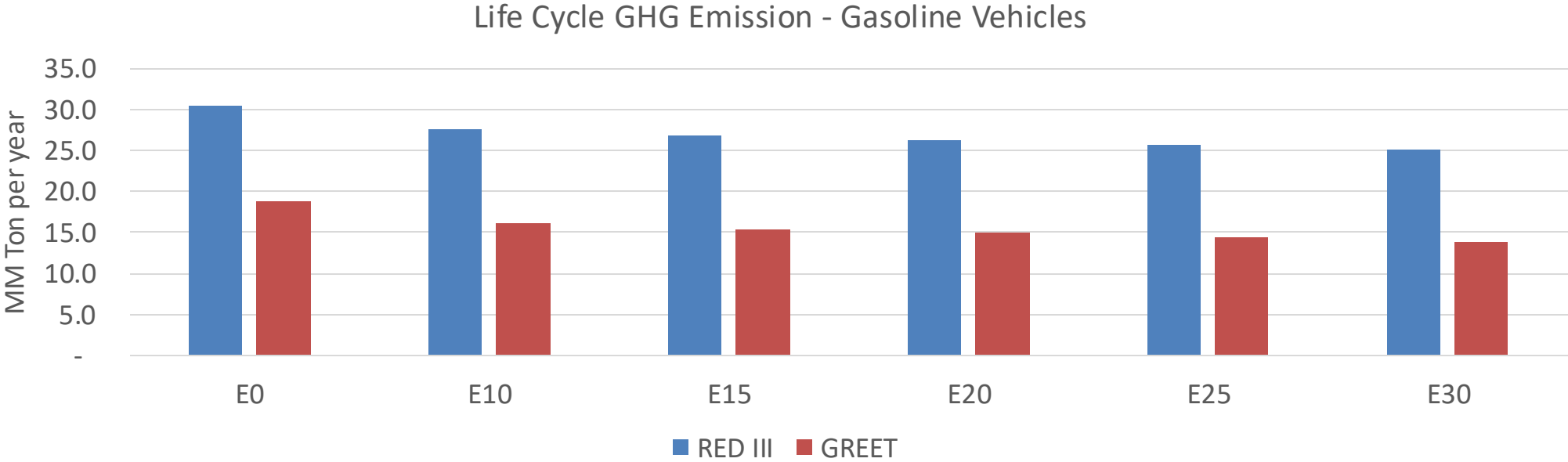
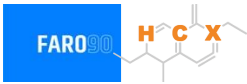
TIER III BIN 125 United States
Euro 6 Standard



Netherlands – Vehicular Emissions

Vehicular Fleet Emissions (kg/day)	Eq E0	E5	E10	E15	E20	E25	E30	Reduction (%)	
								E10	E20
CO	704,445	654,707	604,969	565,072	525,320	494,872	460,719	-14%	-25%
CO2	28,033,974	27,313,101	26,632,276	26,096,827	25,831,865	25,714,823	25,240,825	-5%	-8%
VOC	103,360	96,697	90,034	84,824	79,696	75,788	70,450	-13%	-23%
NOx	52,857	39,643	37,000	34,873	32,888	30,700	28,384	-30%	-38%
SOx	333	308	283	261	241	219	195	-15%	-28%
NH3	10,778	10,671	10,564	10,615	10,734	10,818	10,888	-2%	0%
N2O	473	470	468	470	480	485	491	-1%	2%
CH4	23,057	23,057	23,057	23,403	23,910	24,348	24,763	0%	4%
Lead	-	-	-	-	-	-	-	0%	0%
Butadiene	737	710	682	666	652	642	626	-7%	-12%
Acetaldehyde	2,047	2,349	3,447	5,250	7,152	8,173	9,657	68%	249%
Formaldehyde	7,461	7,253	8,455	9,928	10,401	11,507	12,526	13%	39%
Benzene	1,258	1,209	1,146	1,128	1,118	1,070	1,035	-9%	-11%
PM2.5	1,819	1,615	1,411	1,225	1,041	844	640	-22%	-43%
PM10	4,542	4,033	3,524	3,060	2,599	2,109	1,598	-22%	-43%

Netherlands – Gasoline Vehicle Fleet Life Cycle GHG Emissions



Country Emissions 2024 (MMT/año)	171.1
Target @ 2035 (MMT/year)	96.3
Delta	74.8

%Reduction vs E0	E0	E10	E15	E20	E25	E30
GREET 2024	0.0%	14.0%	18.6%	20.2%	22.9%	26.4%
RED II/III	0.0%	9.2%	11.5%	13.4%	15.3%	17.6%

% Target@2035	E0	E10	E15	E20	E25	E30
GREET 2024	0.0%	3.5%	4.7%	5.1%	5.8%	6.6%
RED II/III	0.0%	3.7%	4.7%	5.5%	6.2%	7.1%

Source: Greet, RED II/III, IPCC, climateactiontracker.org, F90

Ethanol Gasoline Blending in Poland

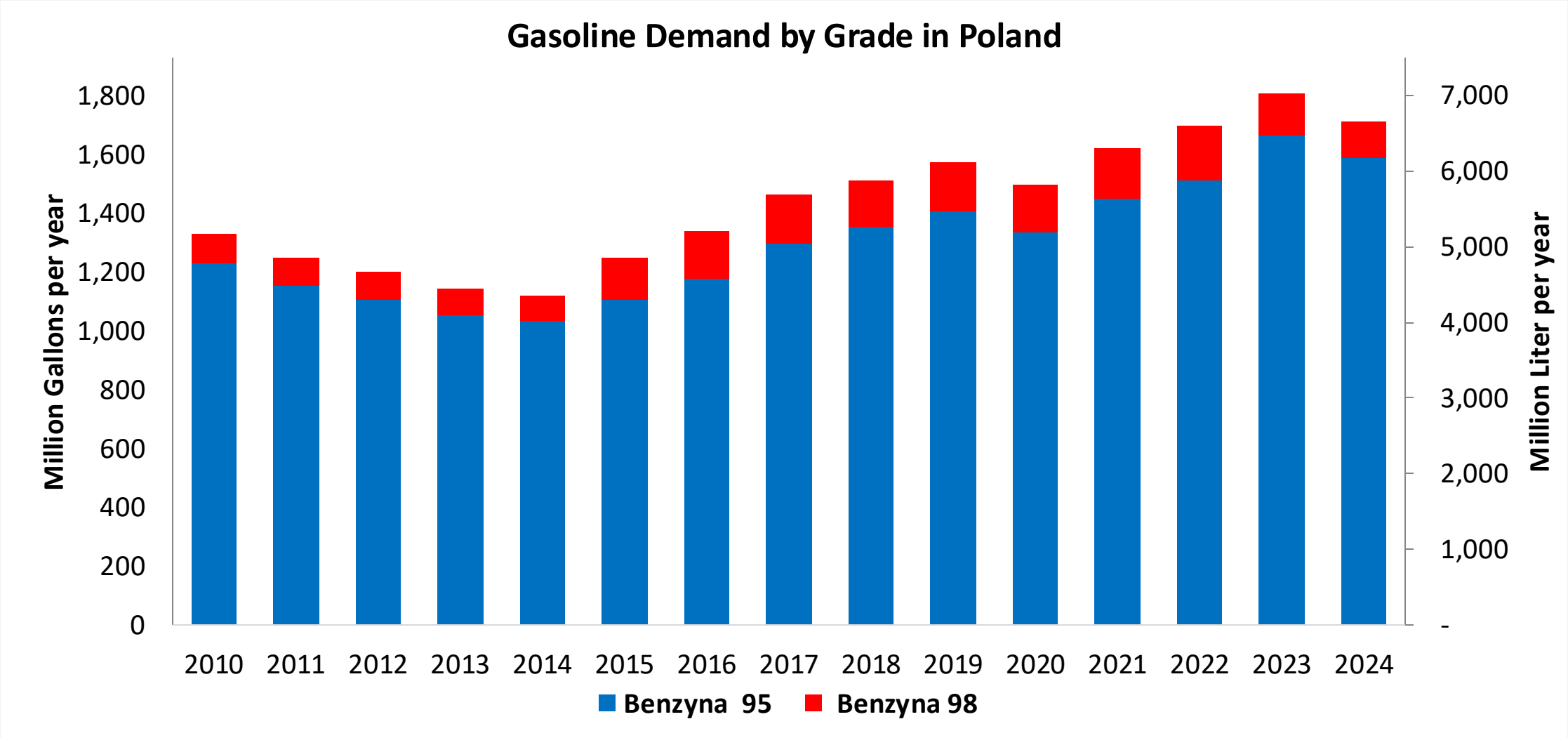


In 2024, gasoline consumption in Poland was 6,493 million liters (1,715 million gallons) and growth rate was 1.7%. Gasoline production and net imports were 5,606 and 913 million liters (1,481 and 241 million gallons) correspondingly. Poland complies with the European gasoline standards and offers two main grades: RON 95 and RON 98) with an average octane of 95.2 RON.

In 2024, ethanol consumption in Poland was 420 million liters (111 million gallons) representing 6.5% of gasoline demand. Ethanol production and Net Imports were 452 and -32 million liters (111 and -9 million gallons) correspondingly. Ethanol sources are Corn 57% and Cereals 43%.

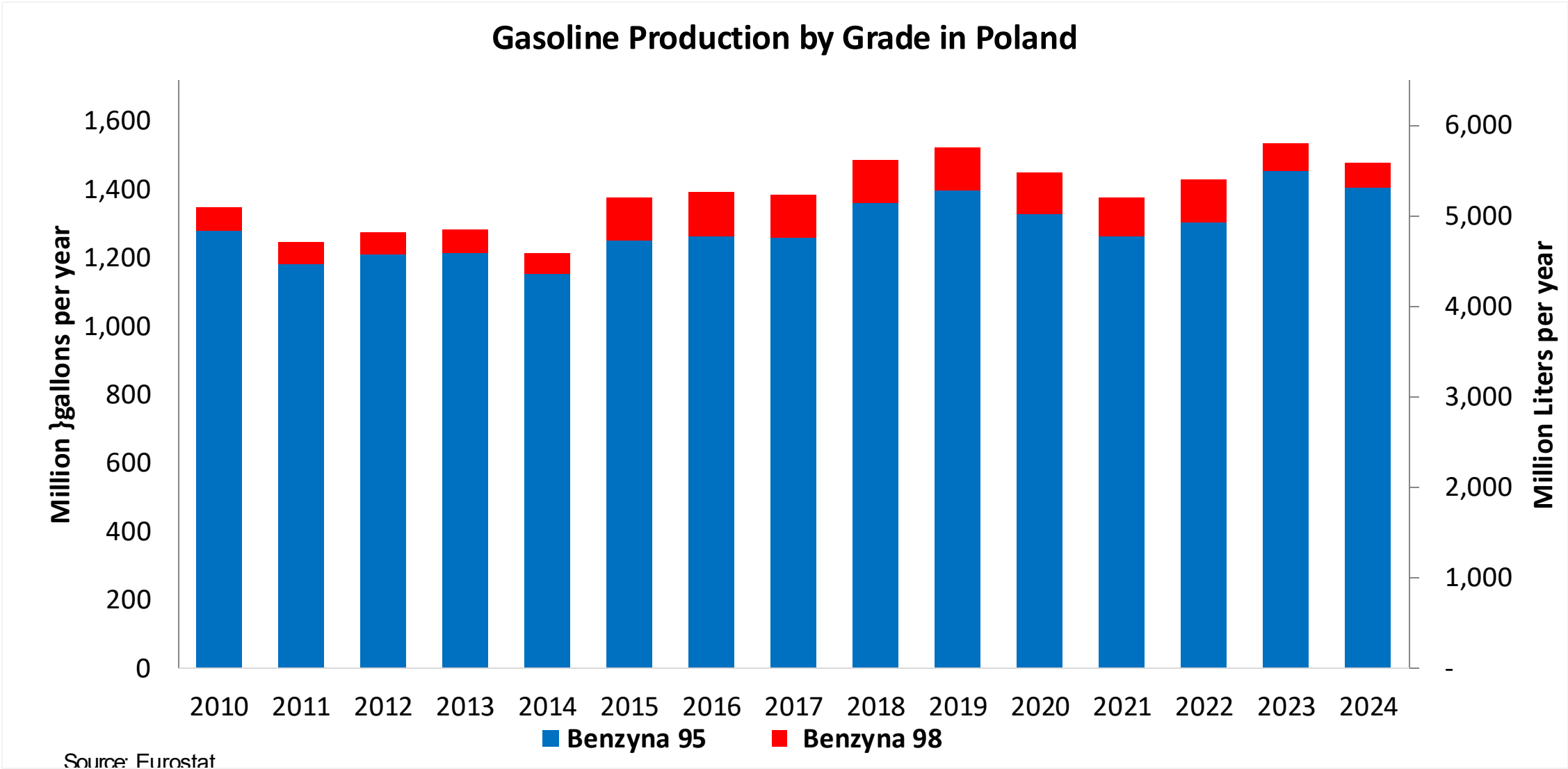
Source: Eurostat, European Commission

Gasoline Demand in Poland

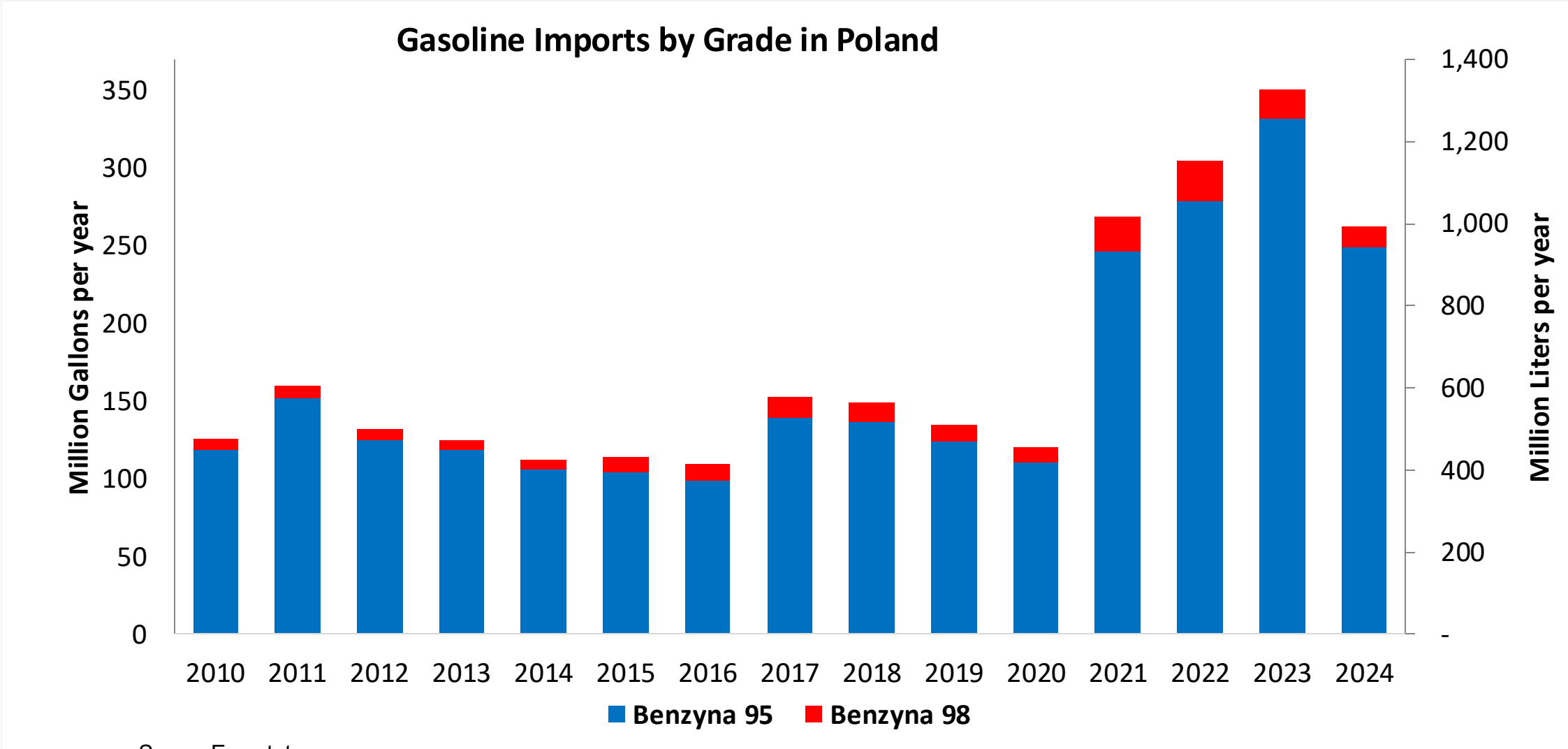


Source: Eurostat

Gasoline Production in Poland

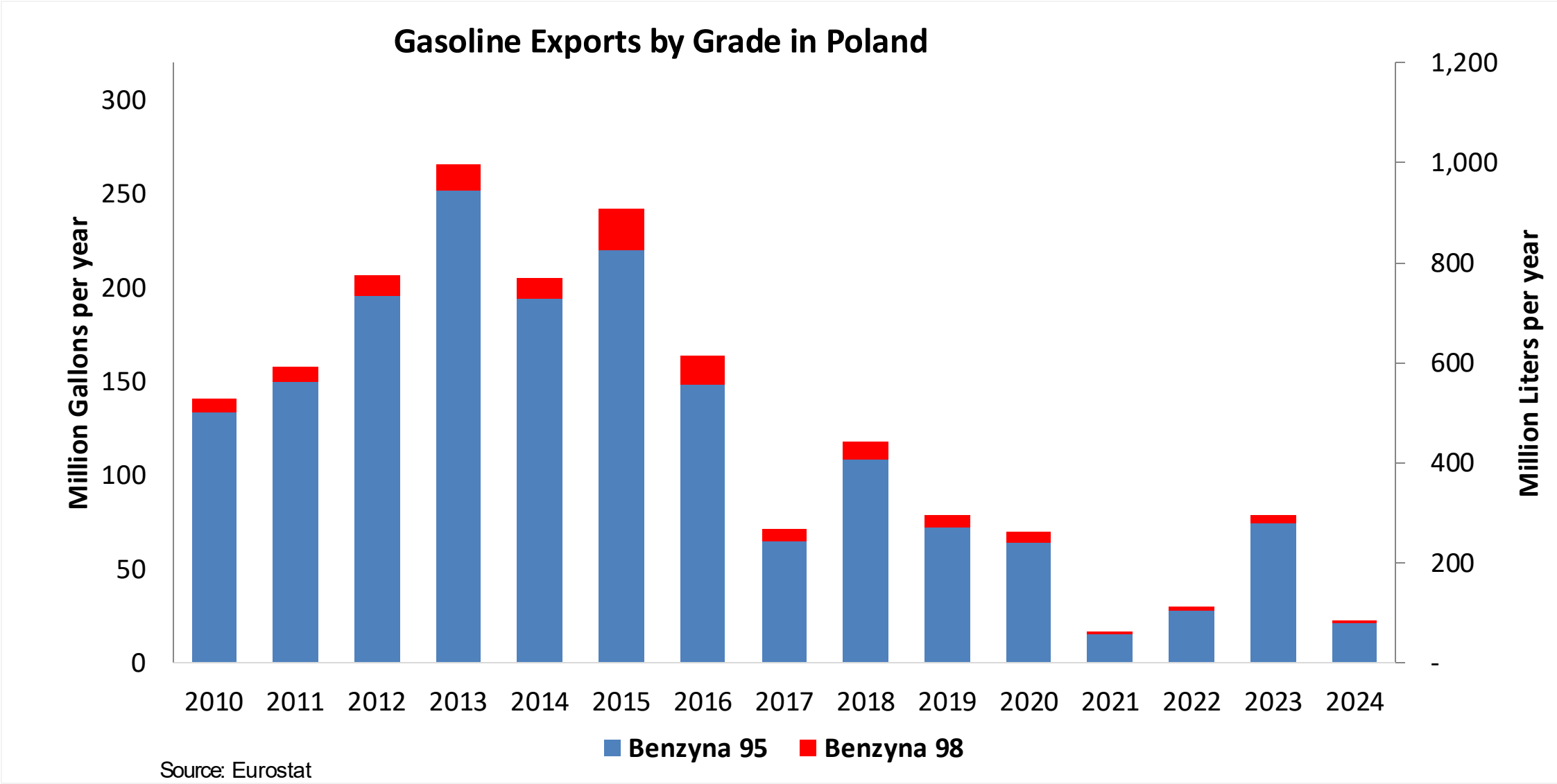


Gasoline Imports in Poland

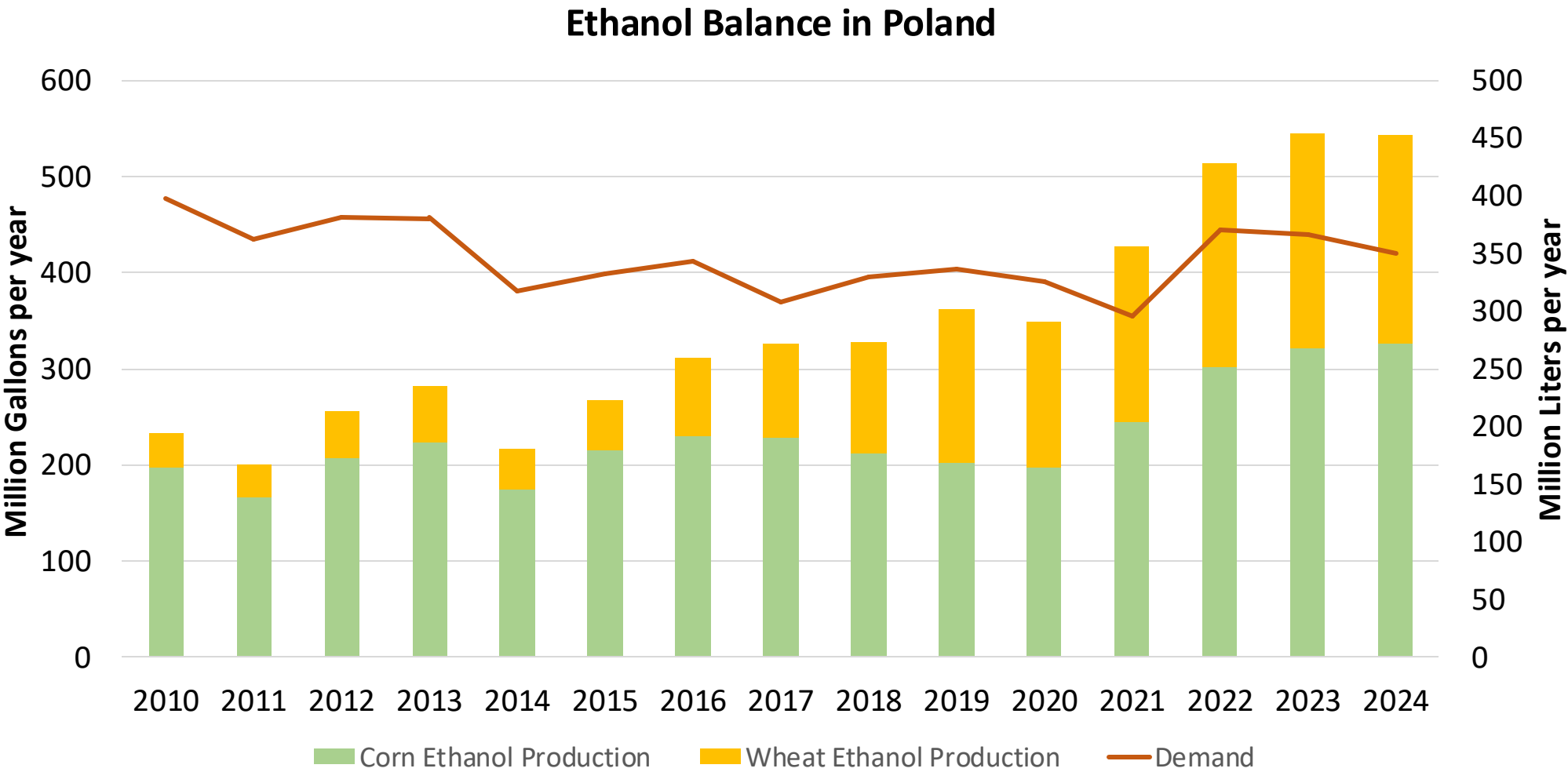


Source: Eurostat

Gasoline Exports in Poland

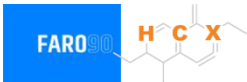


Ethanol Balance in Poland



Source: Eurostat

Gasoline Quality in Poland

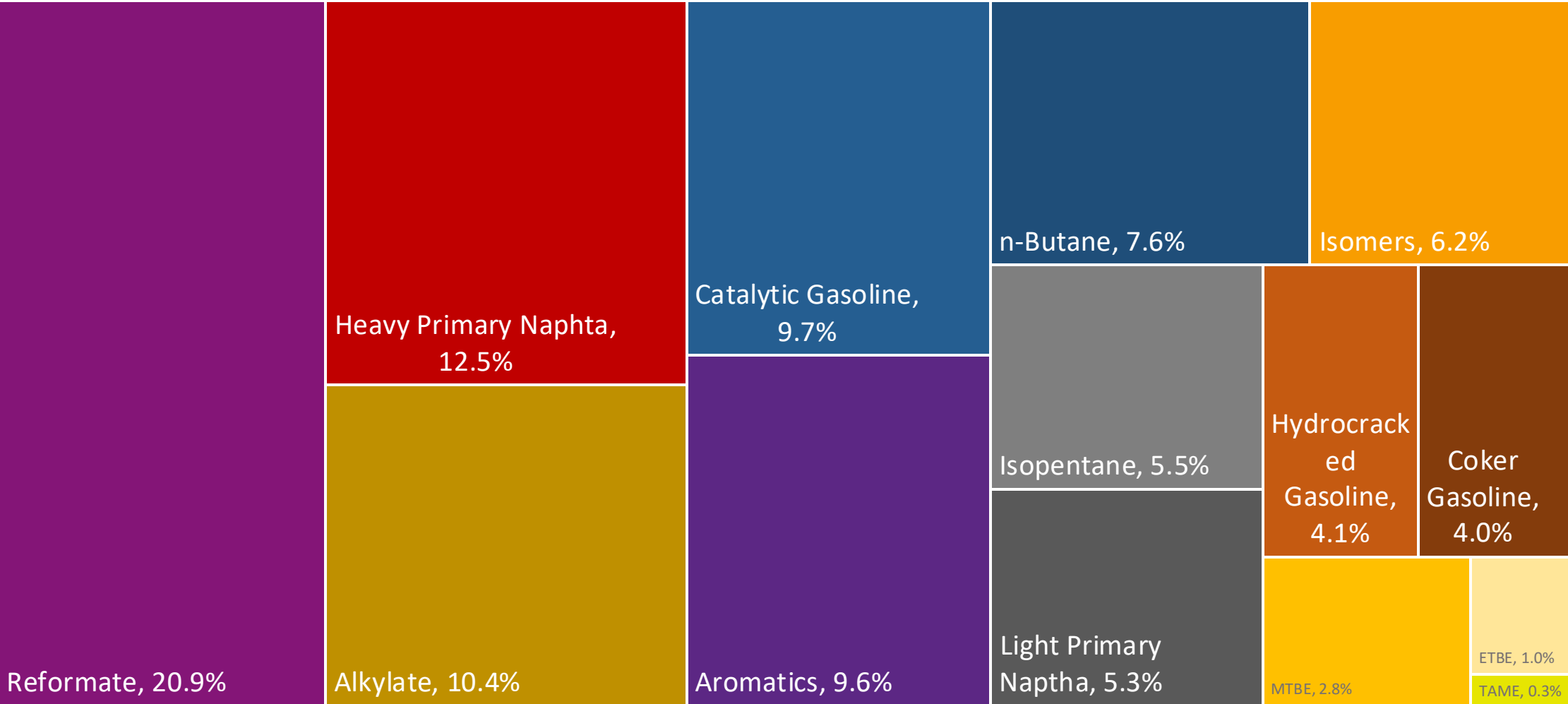
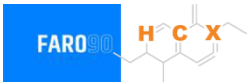


Name	EN 228:2025 (Euro 6); Country Level Fuel Ordinances							
Implementation Date	2025							
Applicability	All EU Contries + UK							
Grade	RON 95 E5		RON 95 E10		RON 98 E5		RON 98 E10	
	Min	Max	Min	Max	Min	Max	Min	Max
RON	95.0		95.0		98.0		98.0	
MON	> 85	> 88	> 85	> 88	> 85	> 88	> 85	> 88
AKI								
Benzene (%vol)	< 1							
Aromatics (% vol)	< 35							
Olefins (%vol)	< 18							
Lead (g/L)	< 0.005							
Manganese (mg/L)	< 2,0							
Sulfur (ppm)	10.0							
Oxygen (%w)*	< 2,7	< 3,7	< 2,7	< 3,7	< 2,7	< 3,7	< 2,7	< 3,7
Ethanol (%vol)	< 5	< 10	< 5	< 10	< 5	< 10	< 5	< 10
PVR 37.8°C (psi)	<> 45 - 100 kPa in 6 Volatility Classes							
Overall Biofuels (%cal)	9.2							
Advanced Biofuels (%cal)	-							
Bioethanol in Gasoline Sales (%cal)	4.6							
GHG Emission Reduction (%)	-							

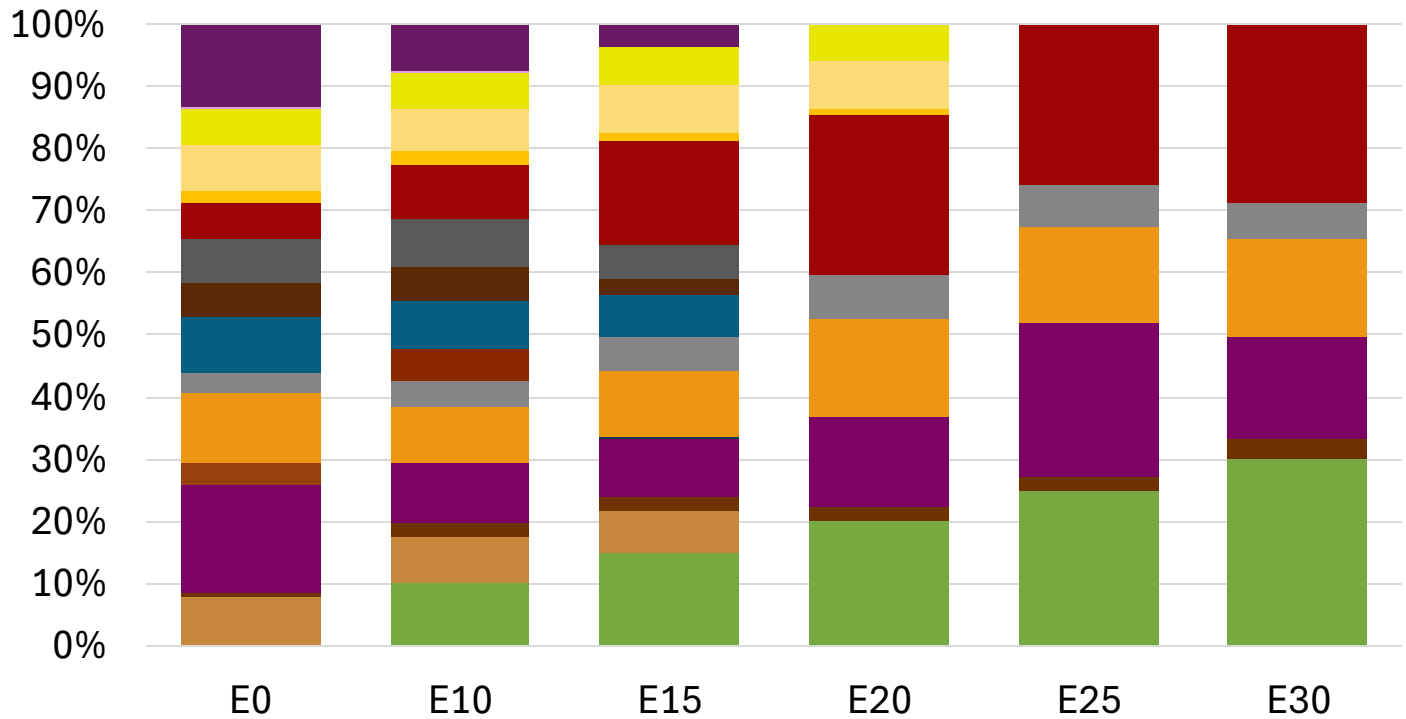
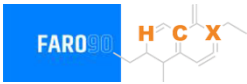
* Methanol 3%, Iso-propyl alcohol 12%, Ter-butyl alcohol 15%, Iso-butyl alcohol 15%, Ethers +5 Carbons 22%, Other Oxygenates 15%

Source: European Commision

Poland Available Blending Components



Poland – Gasoline 95 – Constant Octane

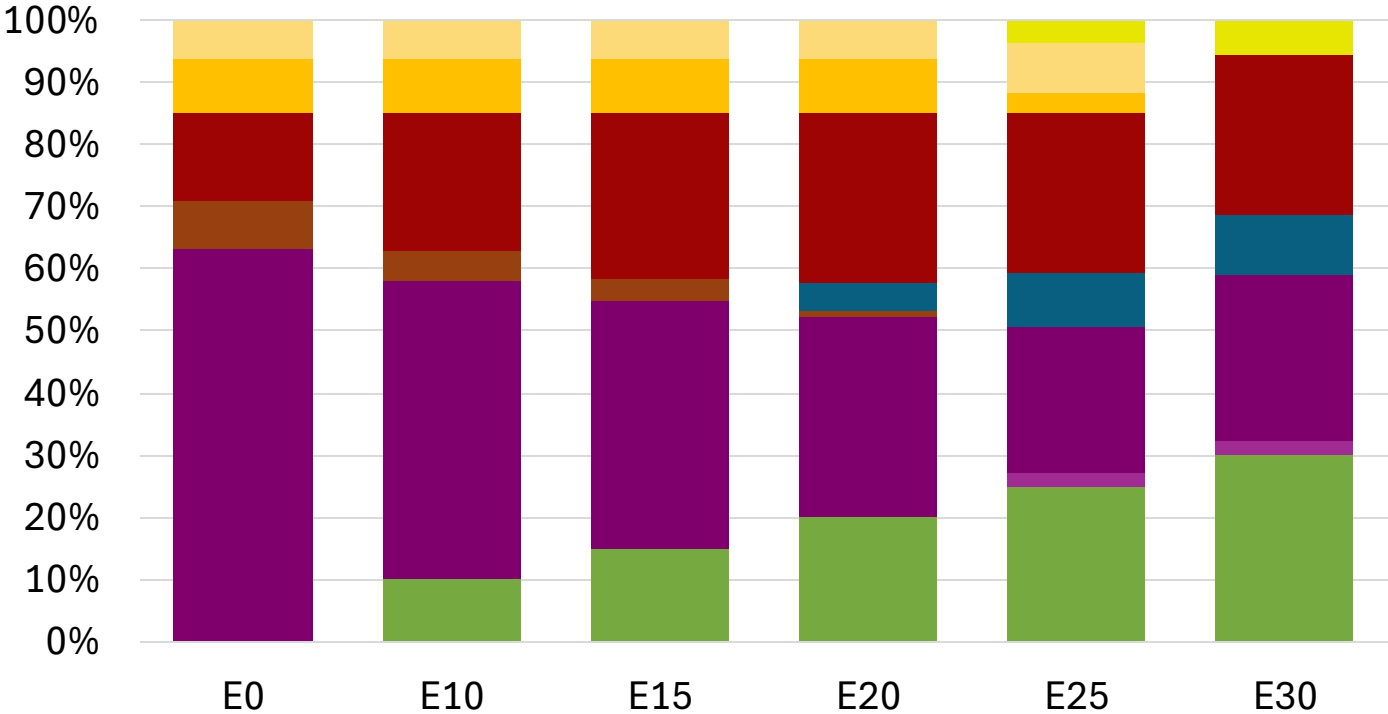
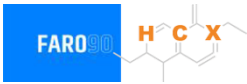


Average CIF 2024
Prices

Octane (RON)	95.0	95.0	95.0	95.0	95.0	95.8
Price (US\$/gal)	\$ 2.284	\$ 2.146	\$ 2.046	\$ 1.931	\$ 1.885	\$ 1.832

Source: Faro90

Poland - Gasoline 98 - Constant Octane

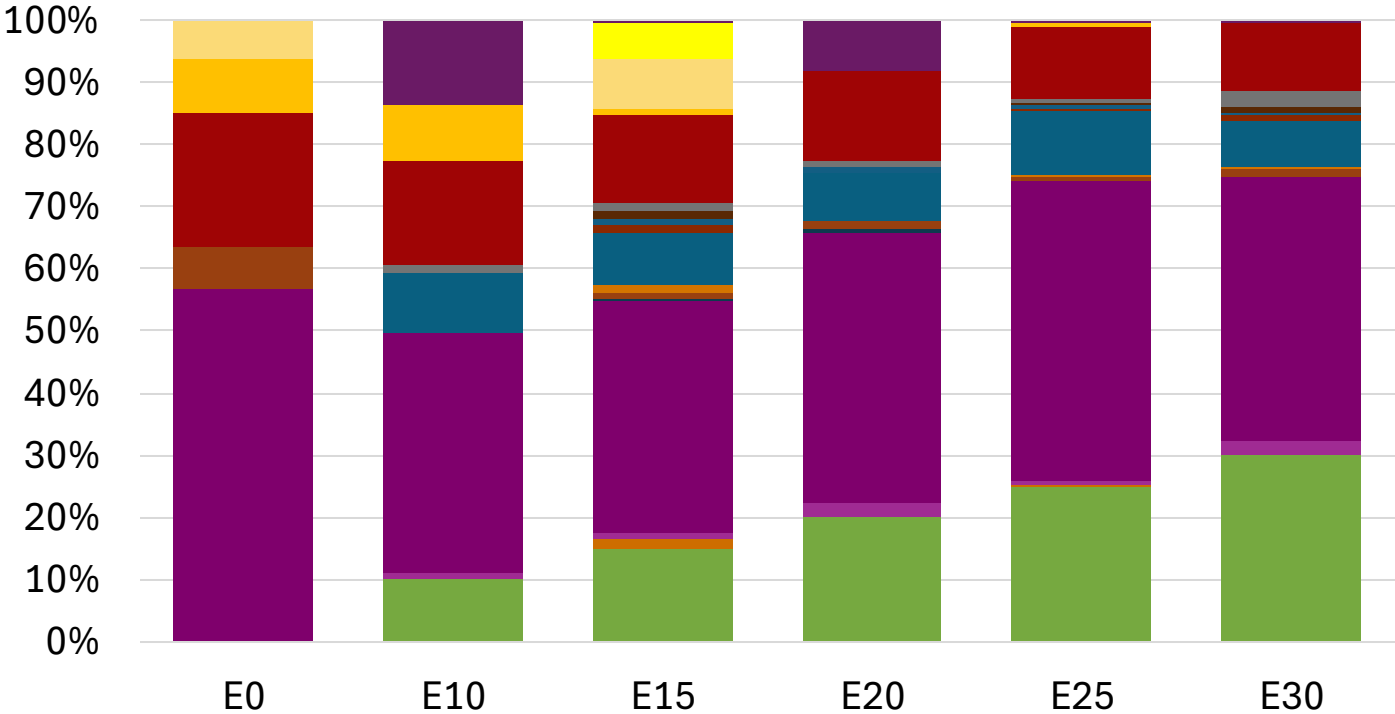
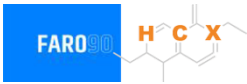


Average CIF 2024 Prices

Octane (RON)	98.0	98.0	98.0	98.0	98.0	98.0
Price (US\$/gal)	\$ 2.304	\$ 2.187	\$ 2.118	\$ 2.046	\$ 1.981	\$ 1.931

Source: Faro90

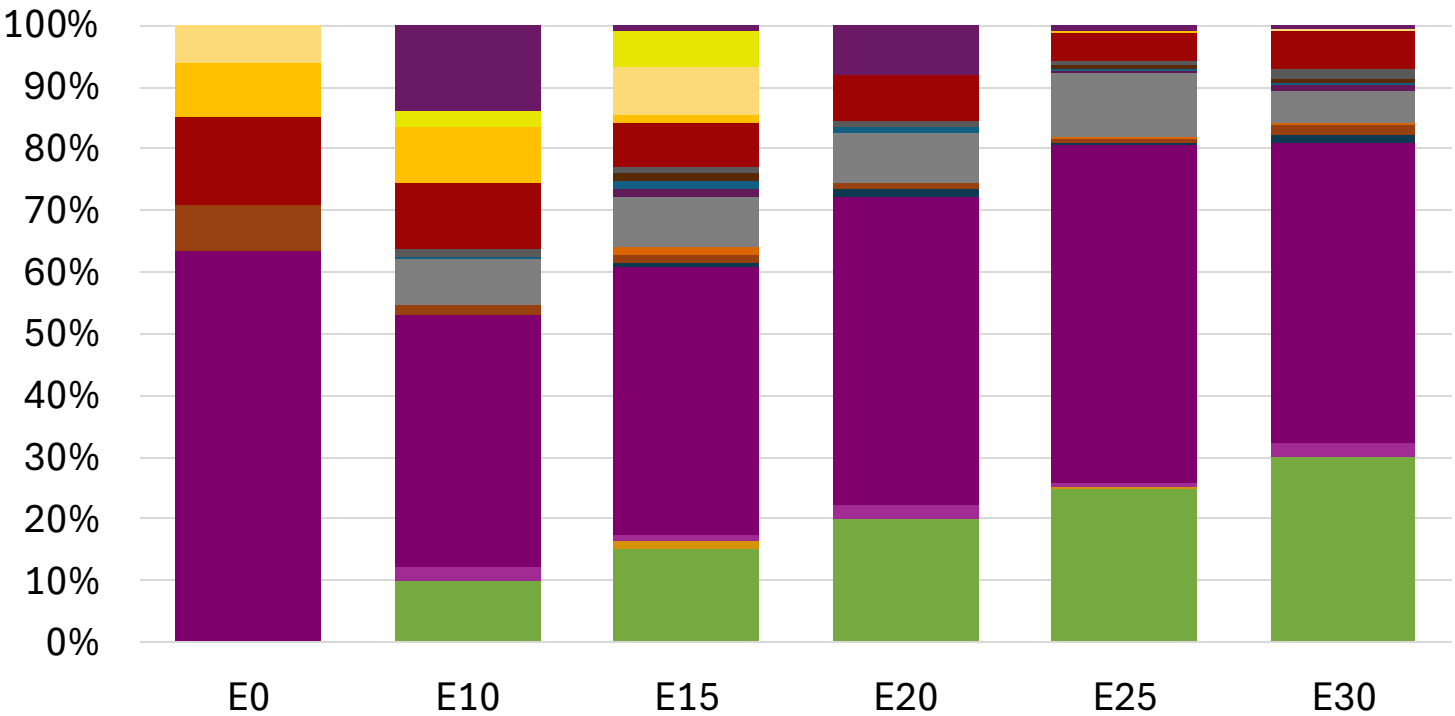
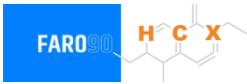
Poland – Gasoline 95 – Increased Octane



Average CIF 2024 Prices

Source: Faro90

Poland – Gasoline 98 – Increased Octane

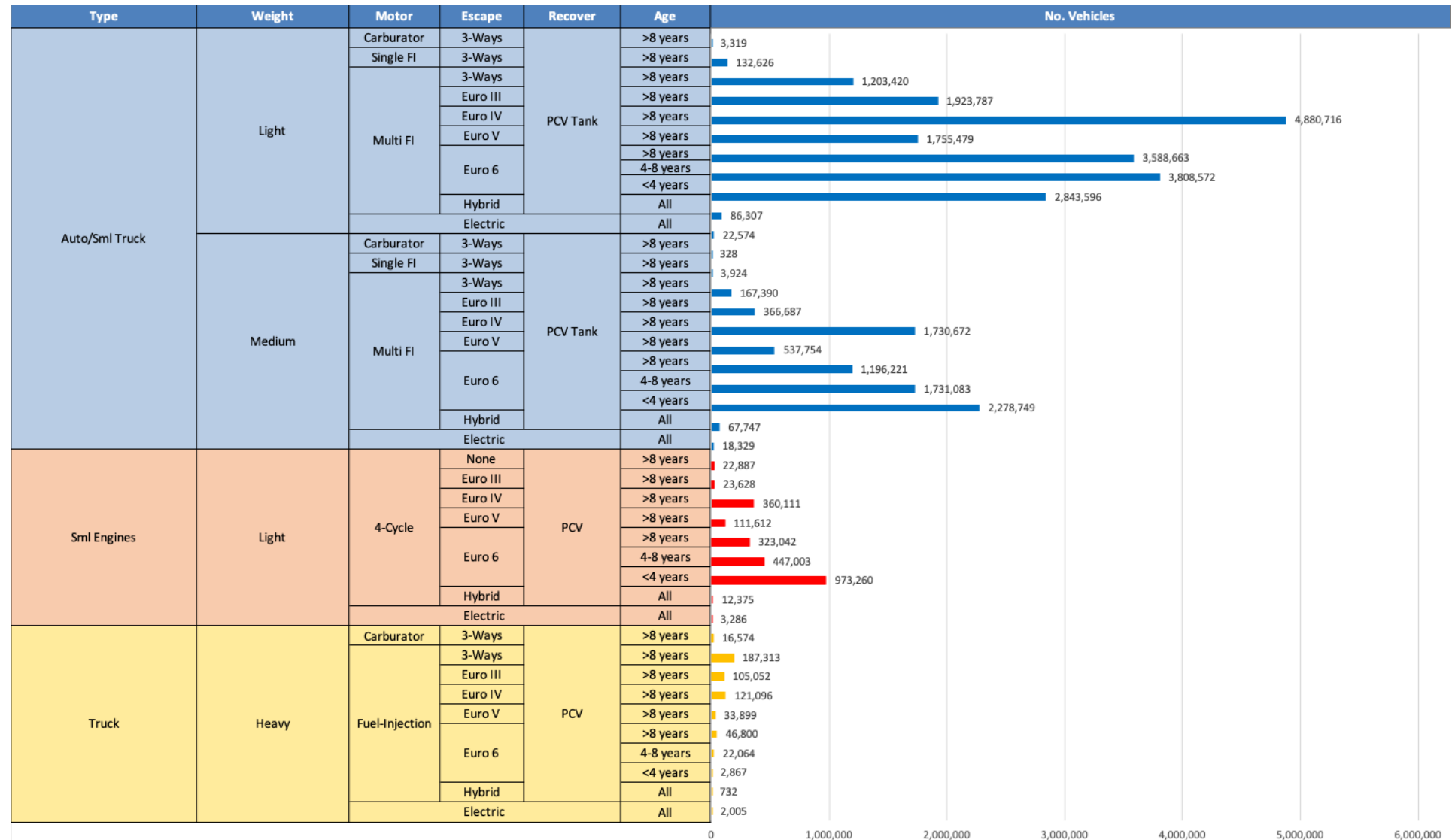


Average CIF 2024
Prices

Octane (RON)	98.0	99.0	100.9	101.2	102.8	104.1
Price (US\$/gal)	\$ 2.304	\$ 2.304	\$ 2.304	\$ 2.304	\$ 2.304	\$ 2.304

Source: Faro90

Gasoline Vehicle Fleet - Poland

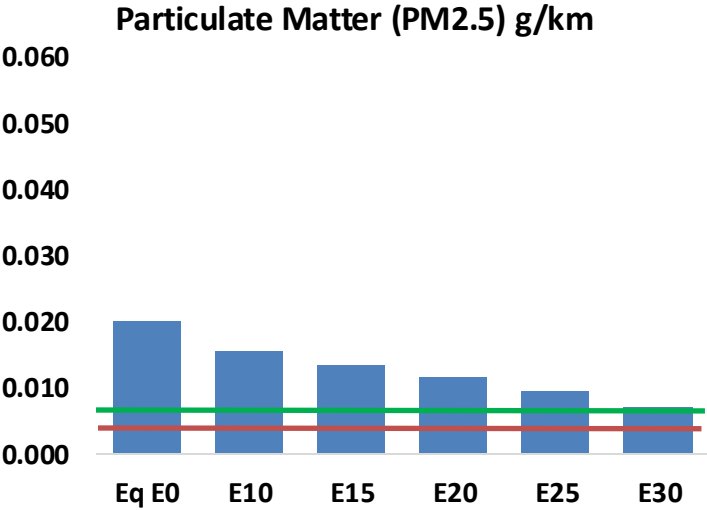
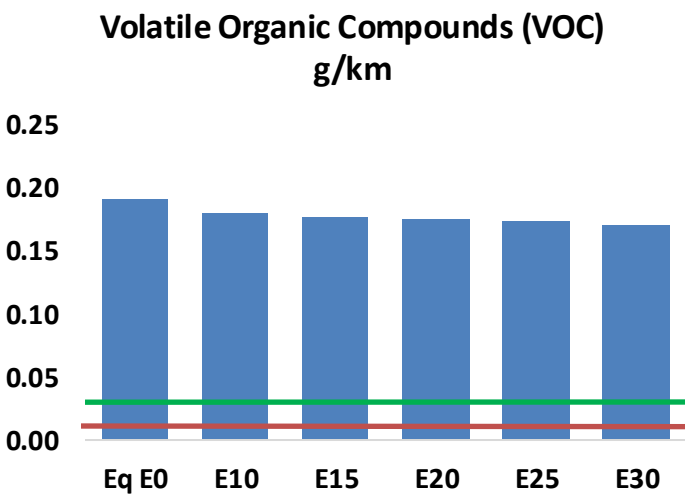
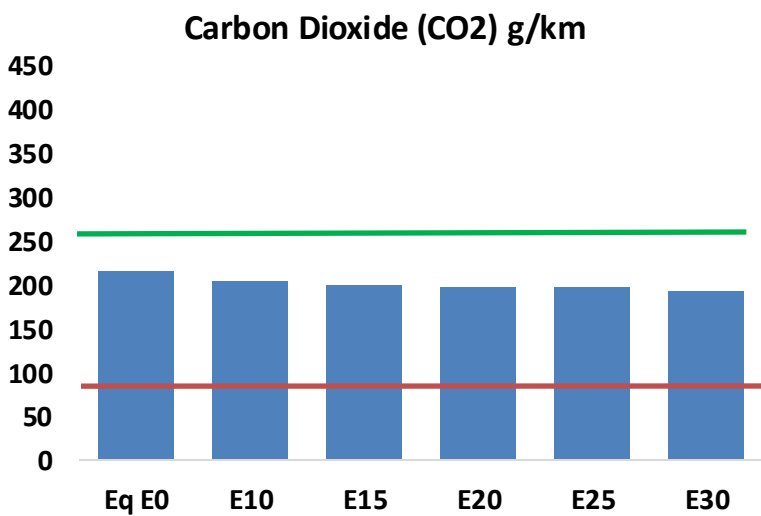
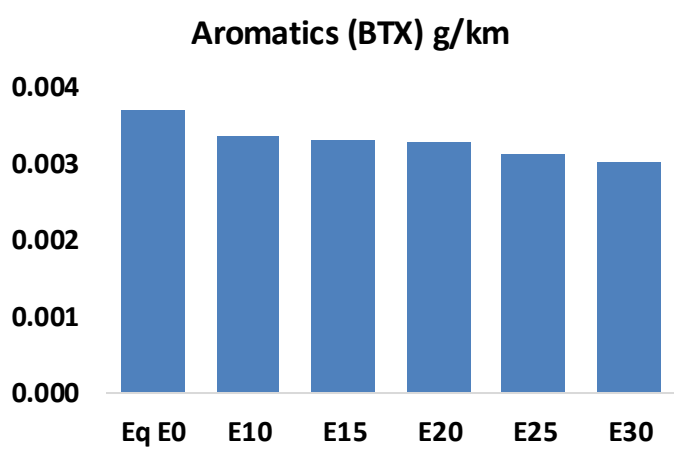
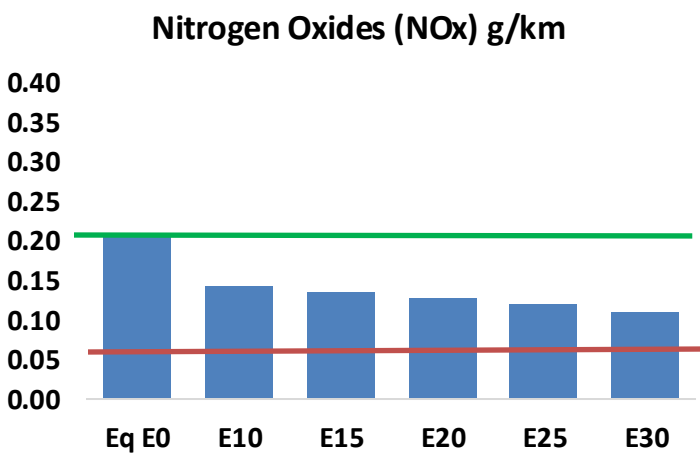
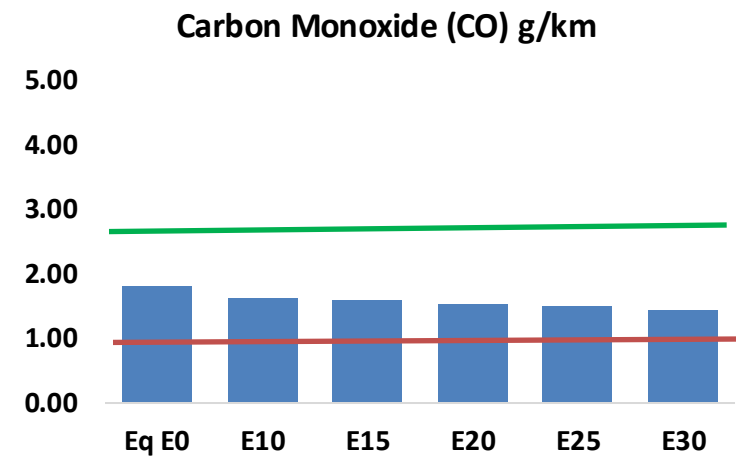
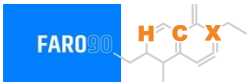


Vehicle Fleet: 31,163,548
Gasoline Vehicle Fleet: 13,217,757

Source: Eurostat, ACEA

Poland – Vehicular Emissions

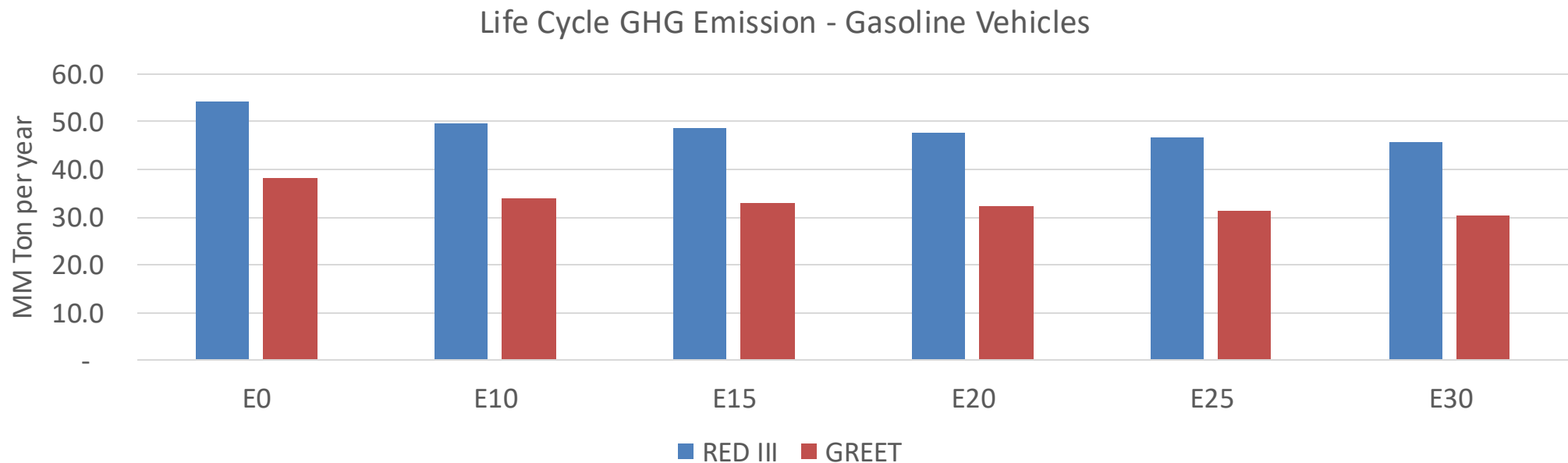
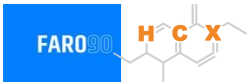
TIER III BIN 125 United States
Euro 6 Standard



Poland – Vehicular Emissions

Vehicular Fleet Emissions (kg/day)	Eq E0	E5	E10	E15	E20	E25	E30	Reduction (%)	
								E10	E20
CO	615,660	588,117	560,575	541,866	524,816	512,241	495,401	-9%	-15%
CO2	73,575,358	71,683,421	69,896,590	68,491,301	67,795,909	67,393,631	66,151,372	-5%	-8%
VOC	65,622	63,617	61,613	60,564	59,781	59,261	58,277	-6%	-9%
NOx	70,624	52,968	49,437	46,594	43,942	41,019	37,925	-30%	-38%
SOx	880	813	747	691	637	578	517	-15%	-28%
NH3	24,677	24,432	24,188	24,303	24,576	24,769	24,928	-2%	0%
N2O	1,147	1,140	1,136	1,141	1,165	1,178	1,191	-1%	2%
CH4	15,774	15,774	15,774	16,011	16,358	16,658	16,942	0%	4%
Lead	-	-	-	-	-	-	-	0%	0%
Butadiene	674	662	650	648	650	652	651	-4%	-4%
Acetaldehyde	1,351	1,550	2,275	3,464	4,720	5,393	6,373	68%	249%
Formaldehyde	4,074	3,961	4,617	5,421	5,680	6,283	6,840	13%	39%
Benzene	1,265	1,216	1,152	1,134	1,124	1,075	1,040	-9%	-11%
PM2.5	4,537	4,028	3,520	3,057	2,596	2,106	1,596	-22%	-43%
PM10	2,330	2,069	1,808	1,570	1,333	1,082	820	-22%	-43%

Poland – Gasoline Vehicle Fleet Life Cycle GHG Emissions



Country Emissions 2024 (MMT/año)	337.8
Target @ 2035 (MMT/year)	190.2
Delta	147.7

%Reduction vs E0	E0	E10	E15	E20	E25	E30
GREET 2024	0.0%	10.9%	13.8%	16.0%	18.0%	20.9%
RED II/III	0.0%	8.1%	10.3%	12.0%	13.6%	15.7%

% Target@2035	E0	E10	E15	E20	E25	E30
GREET 2024	0.0%	2.8%	3.6%	4.1%	4.7%	5.4%
RED II/III	0.0%	3.0%	3.8%	4.4%	5.0%	5.8%

Source: Greet, RED II/III, IPCC, climateactiontracker.org, F90

Ethanol Gasoline Blending in Portugal

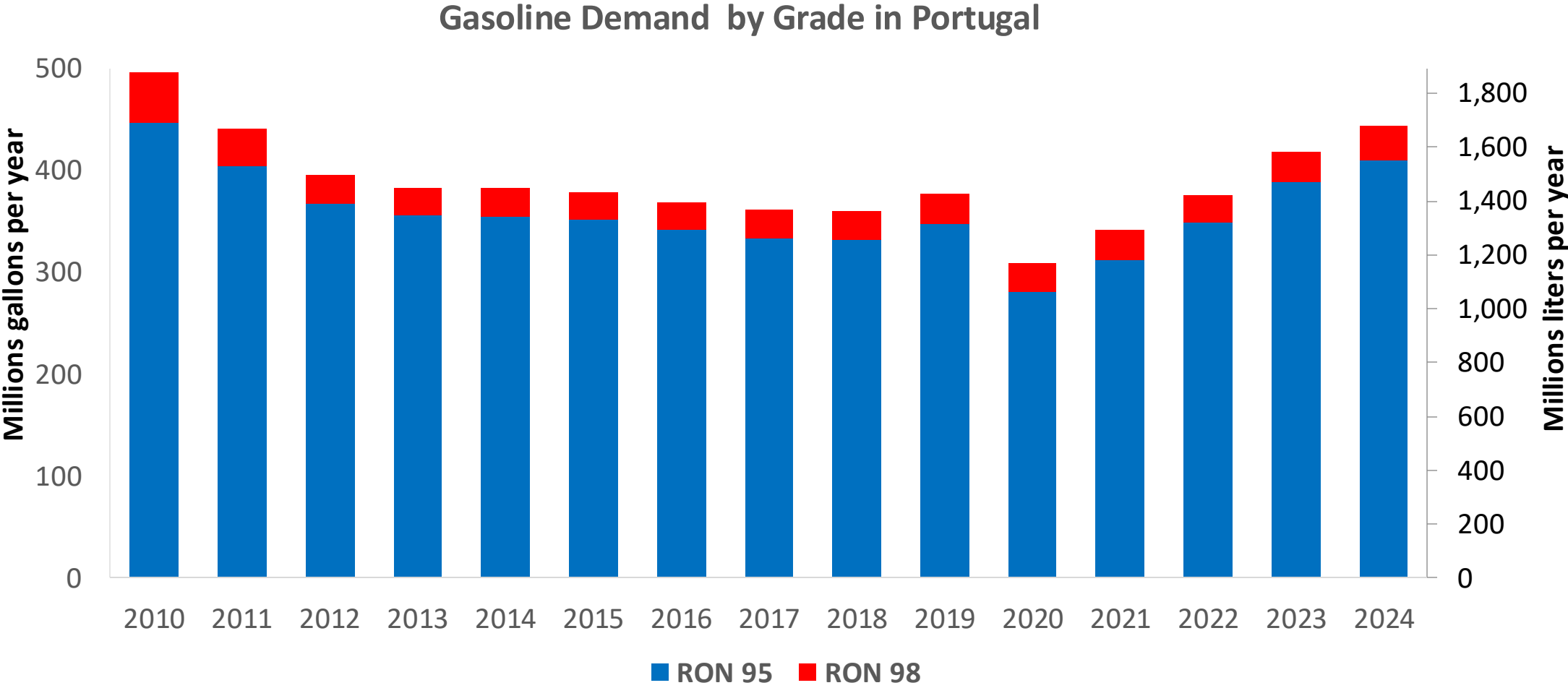


In 2024, gasoline consumption in Portugal was 1,688 million liters (446 million gallons) and growth rate was 0.2%. Gasoline production and net imports were 2,990 and -1,310 million liters (790 and -346 million gallons) correspondingly. Portugal complies with the European gasoline standards and offers two main grades: RON 95 and RON 98) with an average octane of 95.2 RON.

In 2024, ethanol consumption in Portugal was 64 million liters (17 million gallons) representing 3.8% of gasoline demand. Portugal does not produce ethanol, supply is done through imports. Local production plants are being built.

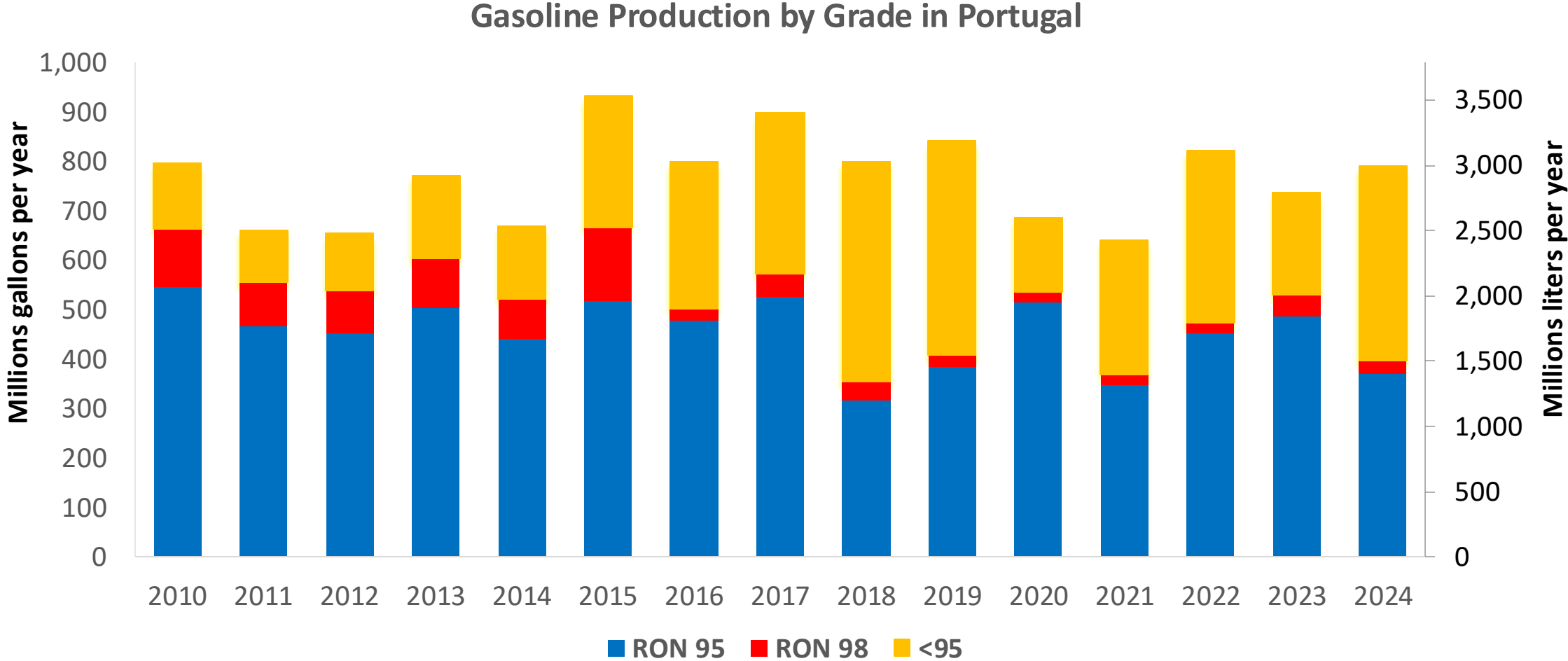
Source: Eurostat, European Commission

Gasoline Demand in Portugal



Source: Eurostat

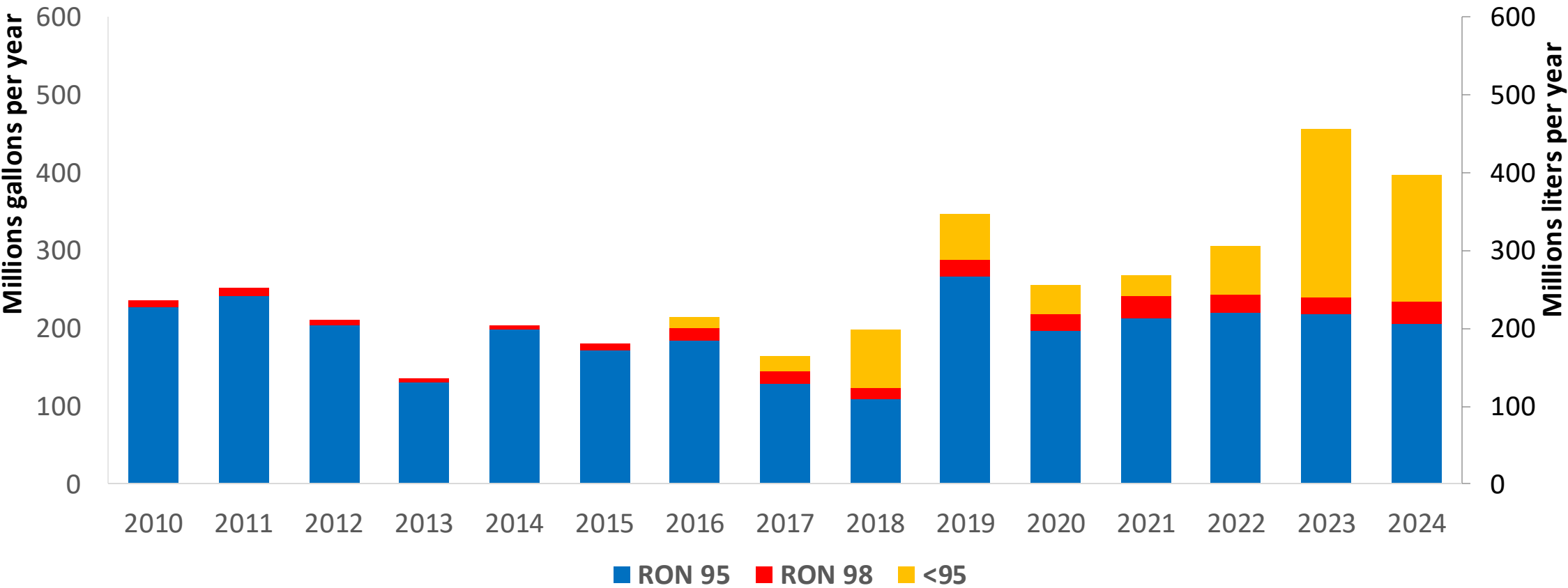
Gasoline Production in Portugal



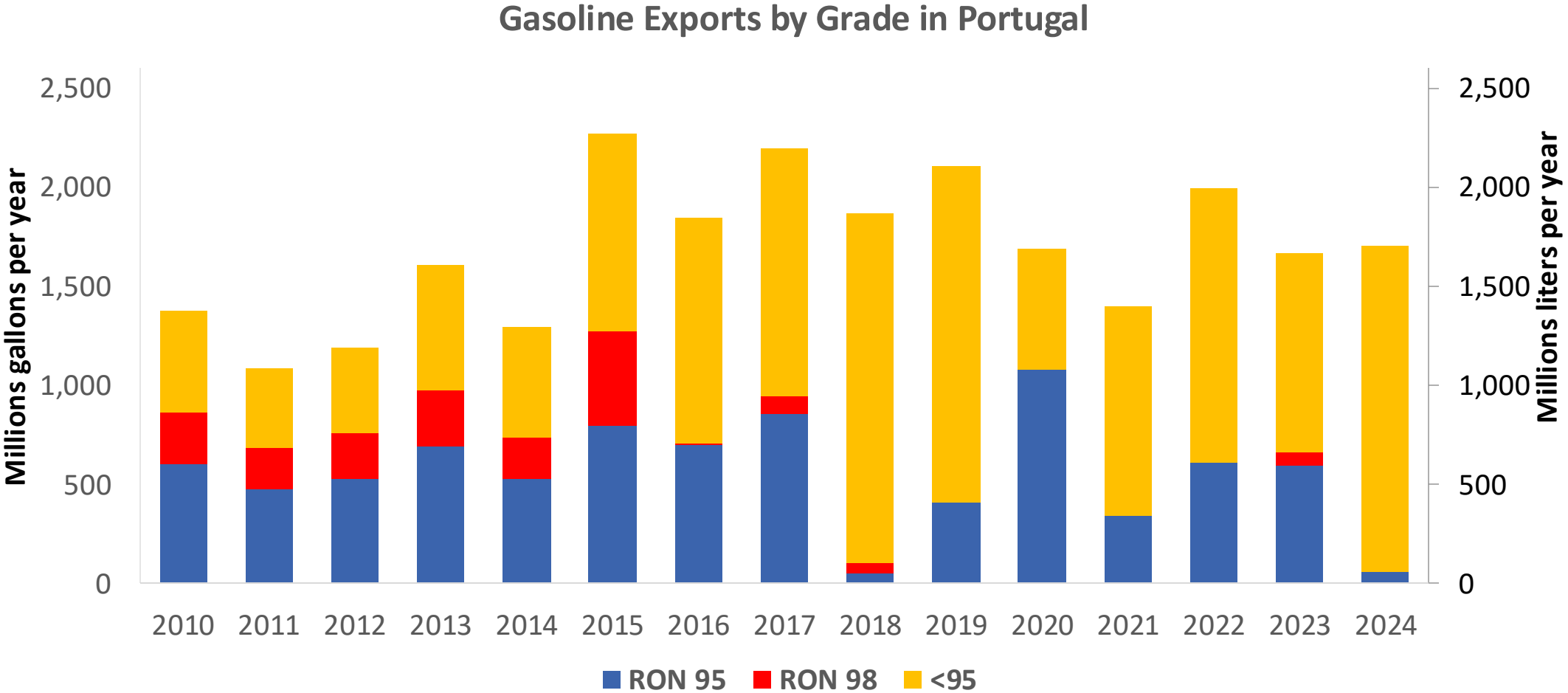
Source: Eurostat

Gasoline Imports in Portugal

Gasoline Imports by Grade in Portugal

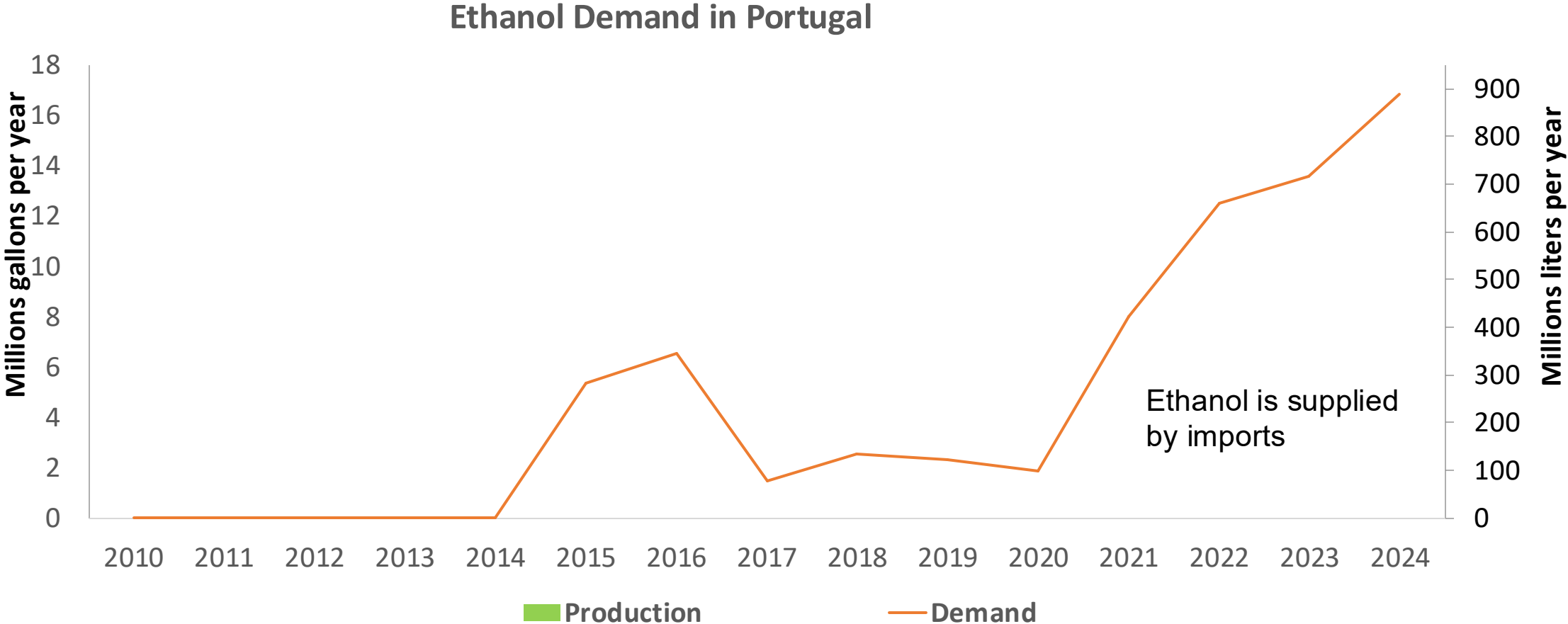


Source: Eurostat



Source: Eurostat

Ethanol Balance in Portugal



Source: Eurostat

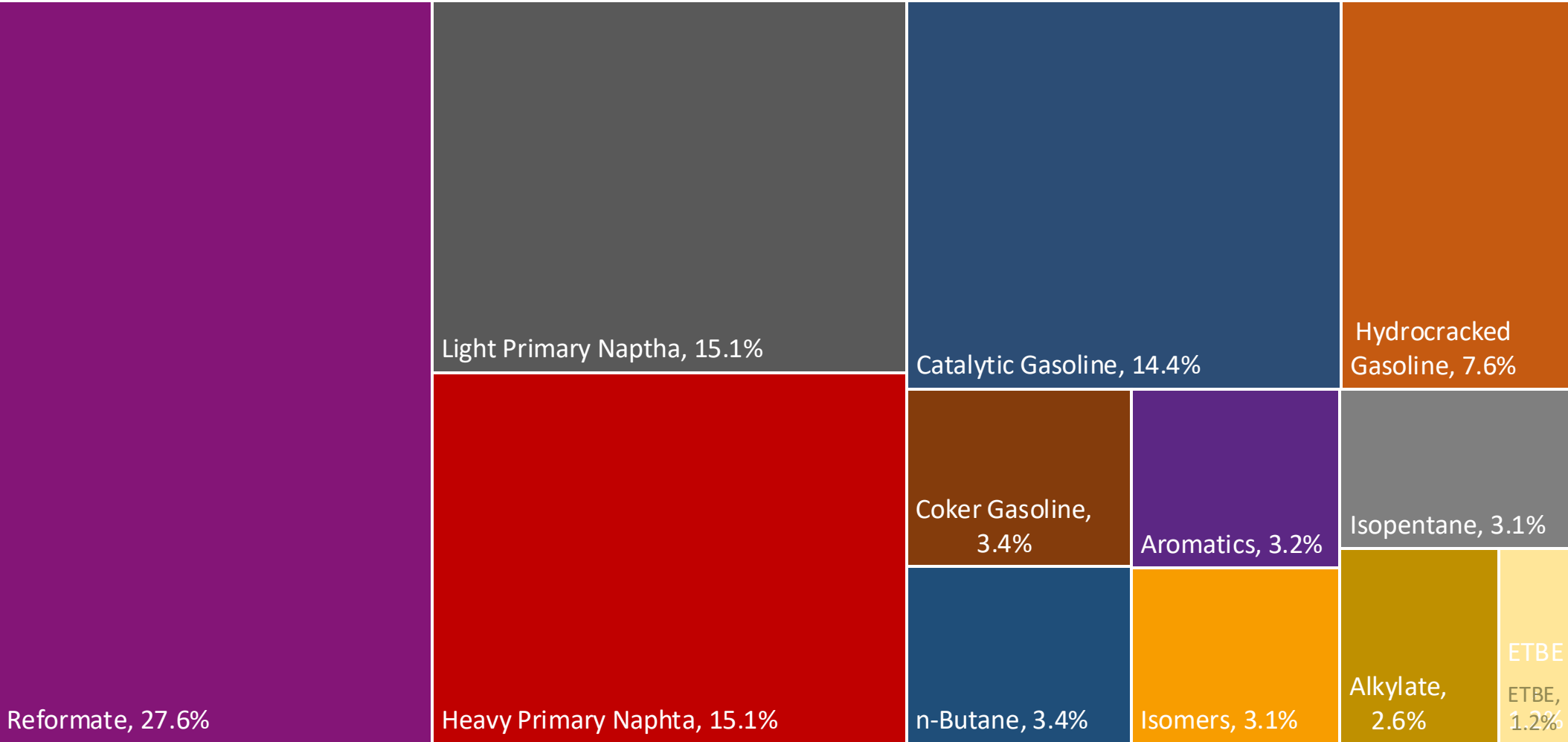
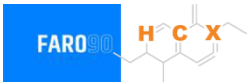
Gasoline Quality in Portugal

Name	EN 228:2025 (Euro 6); Country Level Fuel Ordinances							
Implementation Date	2025							
Applicability	All EU Contries + UK							
Grade	RON 95 E5		RON 95 E10		RON 98 E5		RON 98 E10	
	Min	Max	Min	Max	Min	Max	Min	Max
RON	95.0		95.0		98.0		98.0	
MON	> 85	> 88	> 85	> 88	> 85	> 88	> 85	> 88
AKI								
Benzene (%vol)	< 1							
Aromatics (% vol)	< 35							
Olefins (%vol)	< 18							
Lead (g/L)	< 0.005							
Manganese (mg/L)	< 2,0							
Sulfur (ppm)	10.0							
Oxygen (%w)*	< 2,7	< 3,7	< 2,7	< 3,7	< 2,7	< 3,7	< 2,7	< 3,7
Ethanol (%vol)	< 5	< 10	< 5	< 10	< 5	< 10	< 5	< 10
PVR 37.8°C (psi)	<> 45 - 100 kPa in 6 Volatility Classes							
Overall Biofuels (%cal)	13.0							
Advanced Biofuels (%cal)	2.0							
Bioethanol in Gasoline Sales (%cal)	-							
GHG Emission Reduction (%)	-							

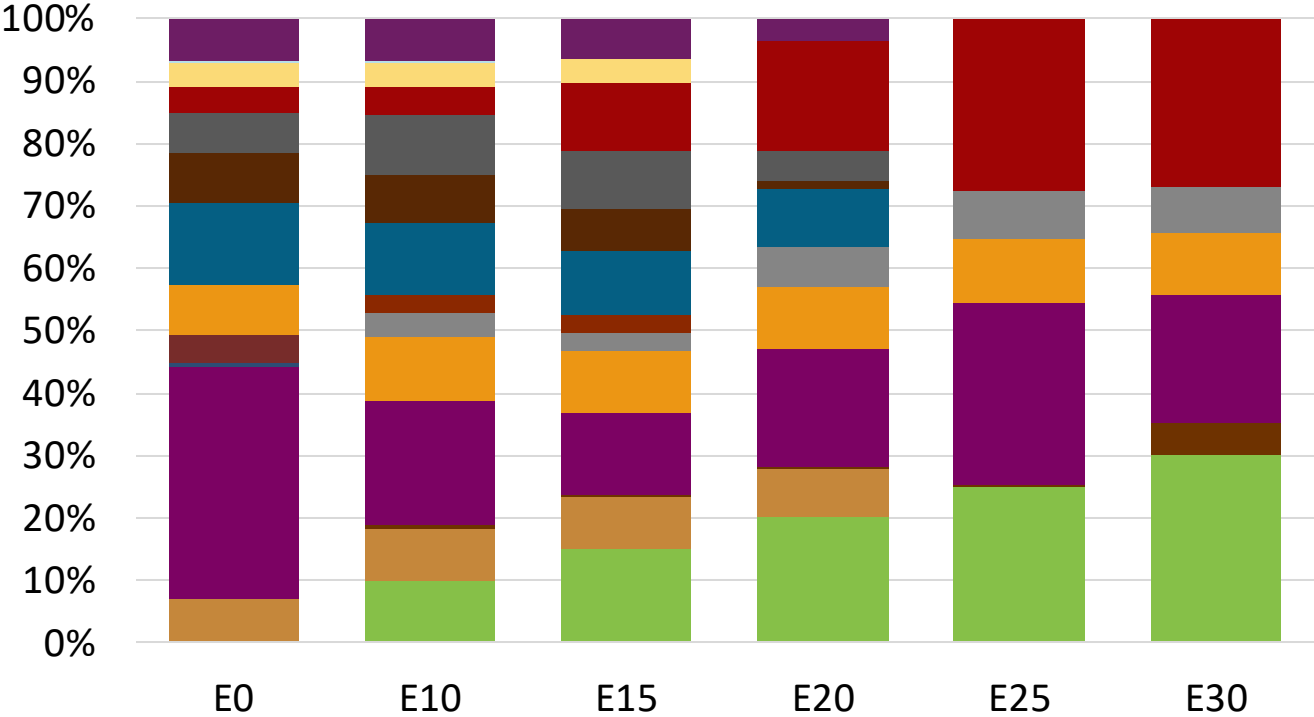
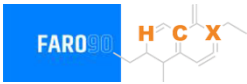
* Methanol 3%, Iso-propyl alcohol 12%, Ter-butyl alcohol 15%, Iso-butyl alcohol 15%, Ethers +5 Carbons 22%, Other Oxygenates 15%

Source: European Commision

Portugal Available Blending Components



Portugal – Gasoline 95 – Constant Octane

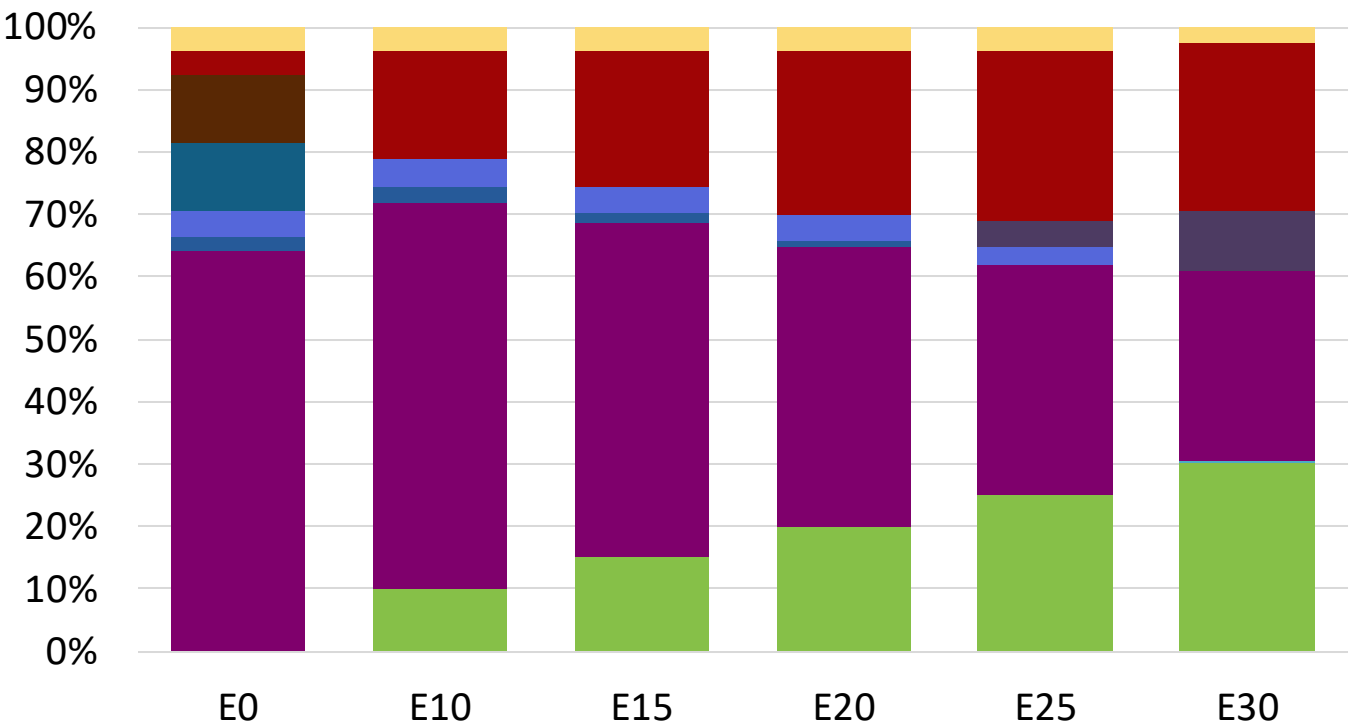
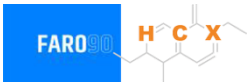


Average CIF 2024
Prices

Octane (RON)	95.0	95.0	95.0	95.0	95.3	95.8
Price (US\$/gal)	\$ 2.290	\$ 2.178	\$ 2.098	\$ 2.008	\$ 1.893	\$ 1.830

Source: Faro90

Portugal - Gasoline 98 - Constant Octane

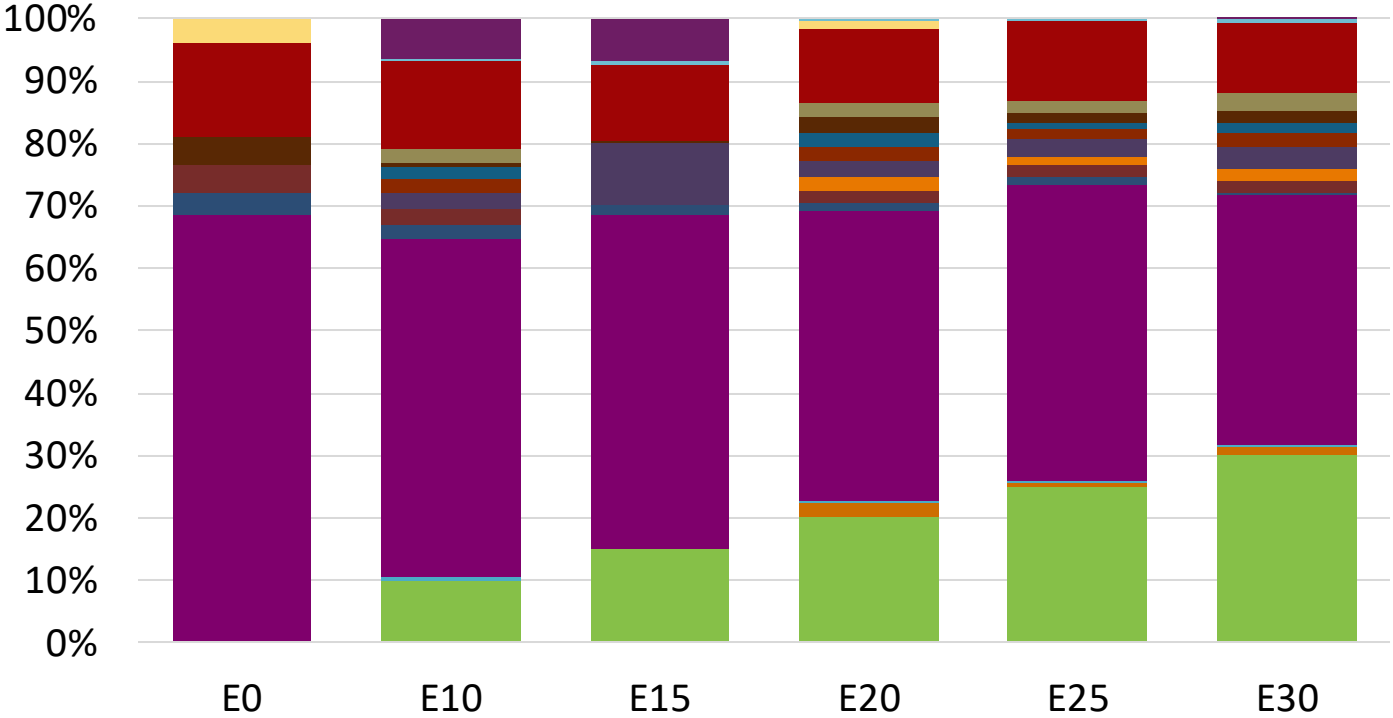
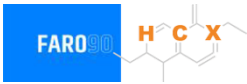


Average CIF 2024 Prices

Octane (RON)	98.0	98.0	98.0	98.0	98.0	98.0
Price (US\$/gal)	\$ 2.359	\$ 2.235	\$ 2.166	\$ 2.091	\$ 2.013	\$ 1.934

Source: Faro90

Portugal – Gasoline 95 – Increased Octane



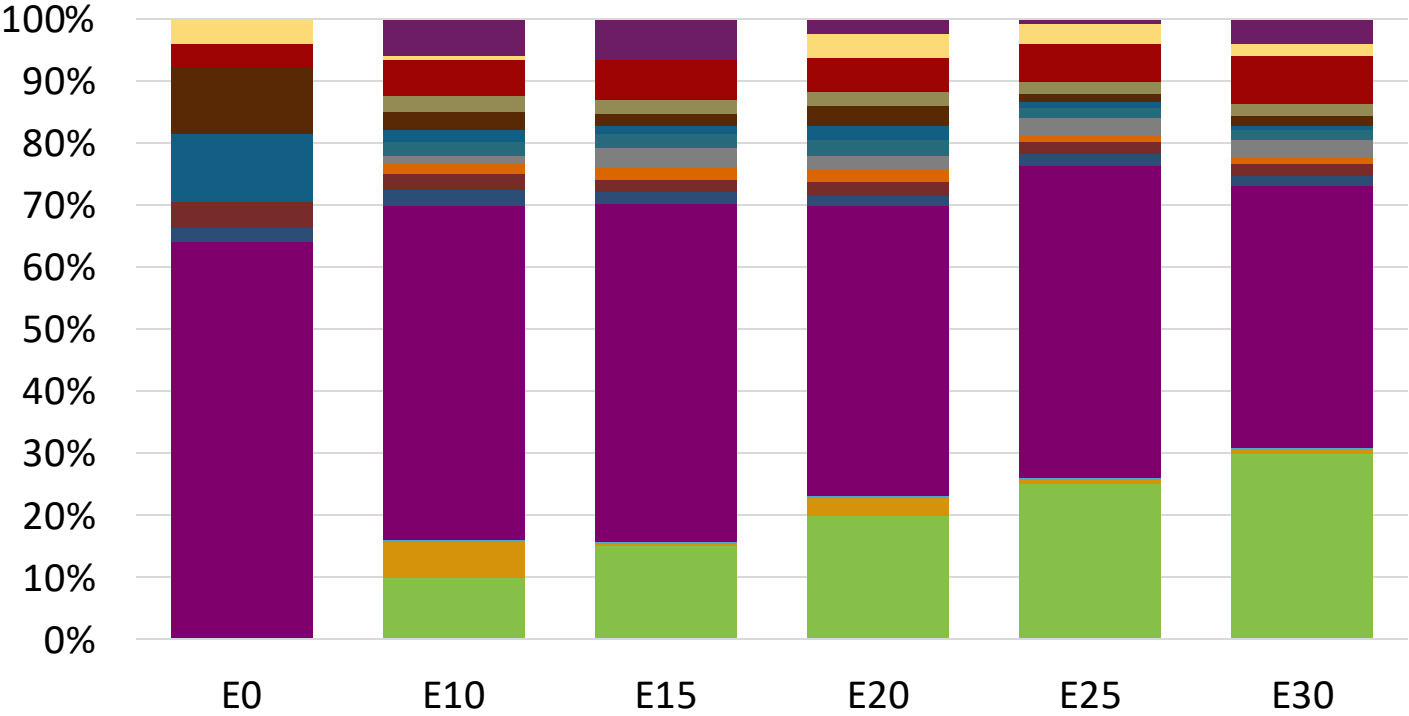
Ethanol
Alkylate
Raffinate
Reformate
n-Butane
Isomerate
Isopentane
Hydrocracked Gasoline
Catalytic Gasoline
Coker Naphtha
Light Primary Naphtha
Heavy Primary Naphtha
MTBE
ETBE
TAME
Aromatics

Average CIF 2024
Prices

Octane (RON)	95.0	96.8	98.2	99.9	101.4	102.6
Price (US\$/gal)	\$ 2.222	\$ 2.222	\$ 2.222	\$ 2.222	\$ 2.222	\$ 2.222

Source: Faro90

Portugal – Gasoline 98 – Increased Octane

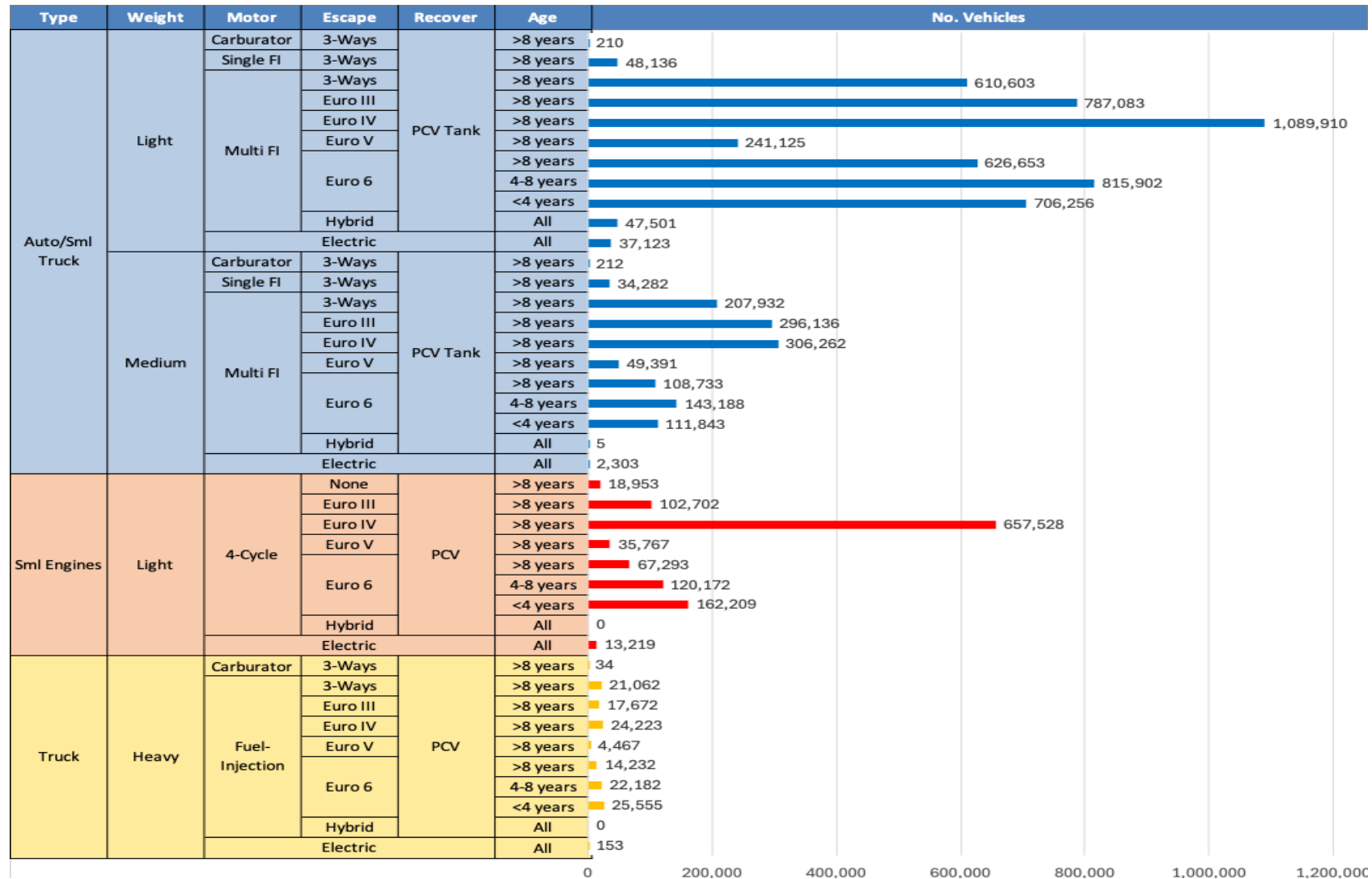


Average CIF 2024
Prices

Octane (RON)	98.0	99.3	100.9	102.5	104.0	105.0
Price (US\$/gal)	\$ 2.359	\$ 2.359	\$ 2.359	\$ 2.359	\$ 2.359	\$ 2.359

Source: Faro90

Gasoline Vehicle Fleet - Portugal

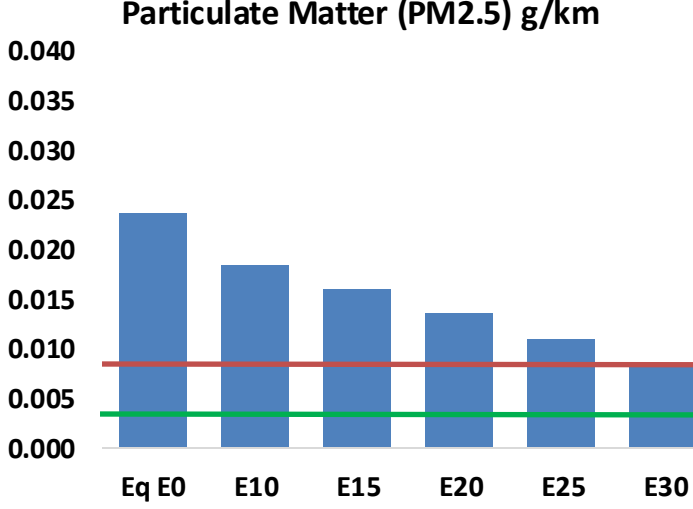
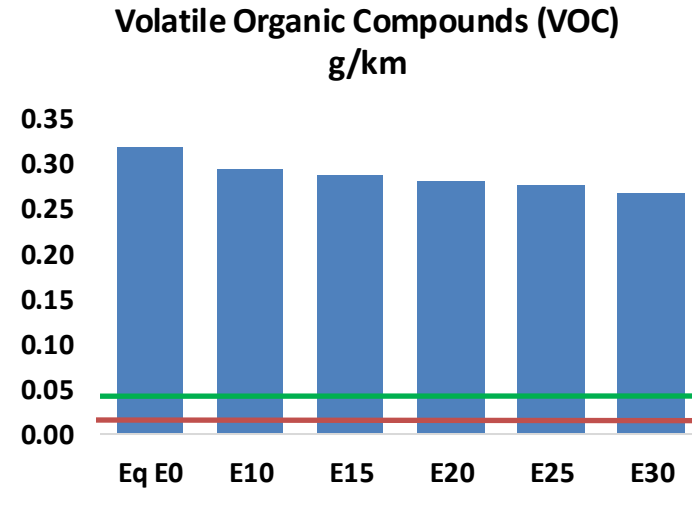
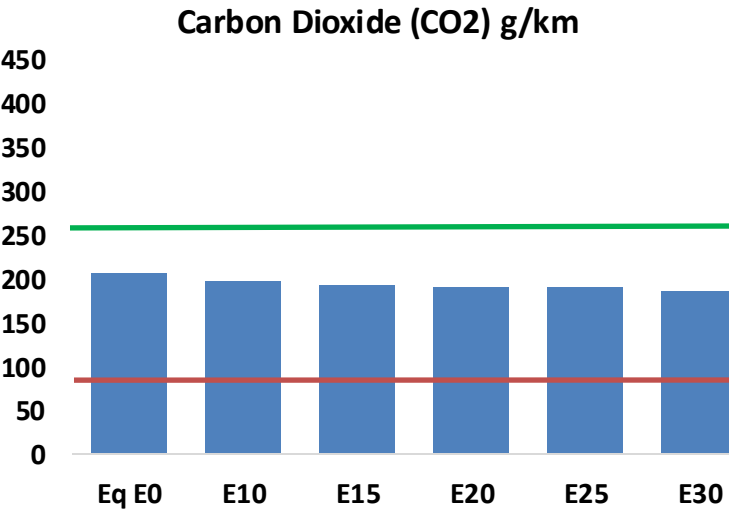
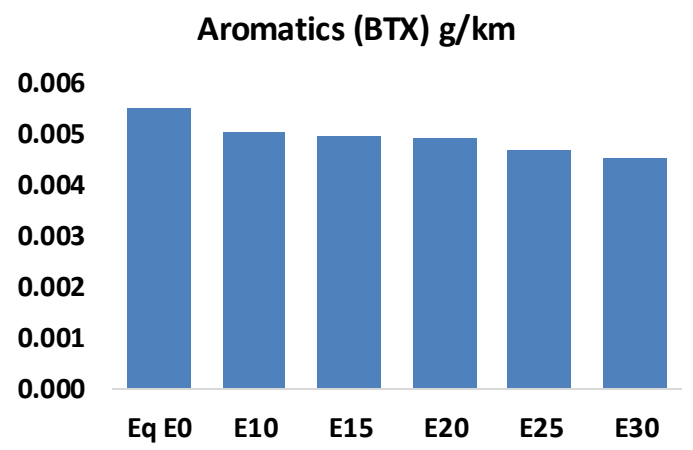
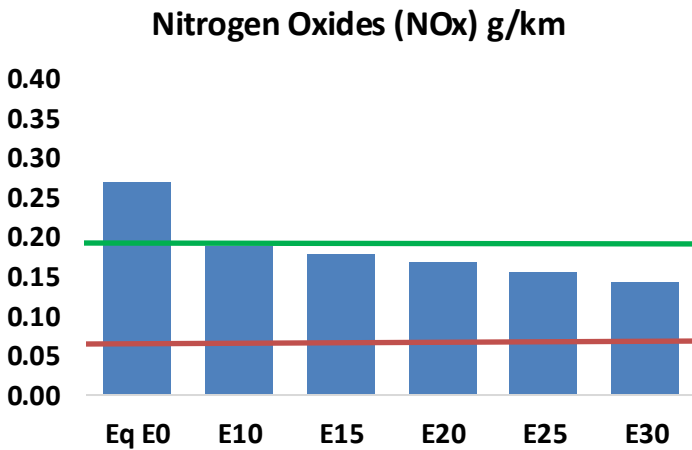
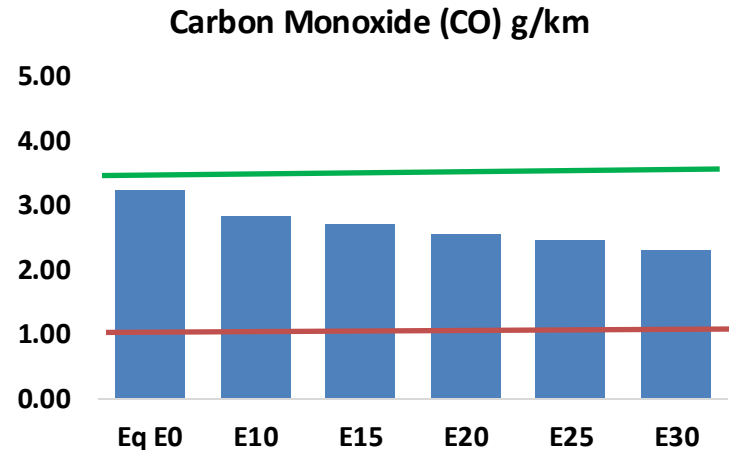
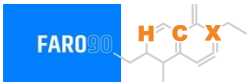


Vehicle Fleet: 7,578,214
Gasoline Vehicle Fleet: 2,654,720

Source: Eurostat, ACEA

Portugal – Vehicular Emissions

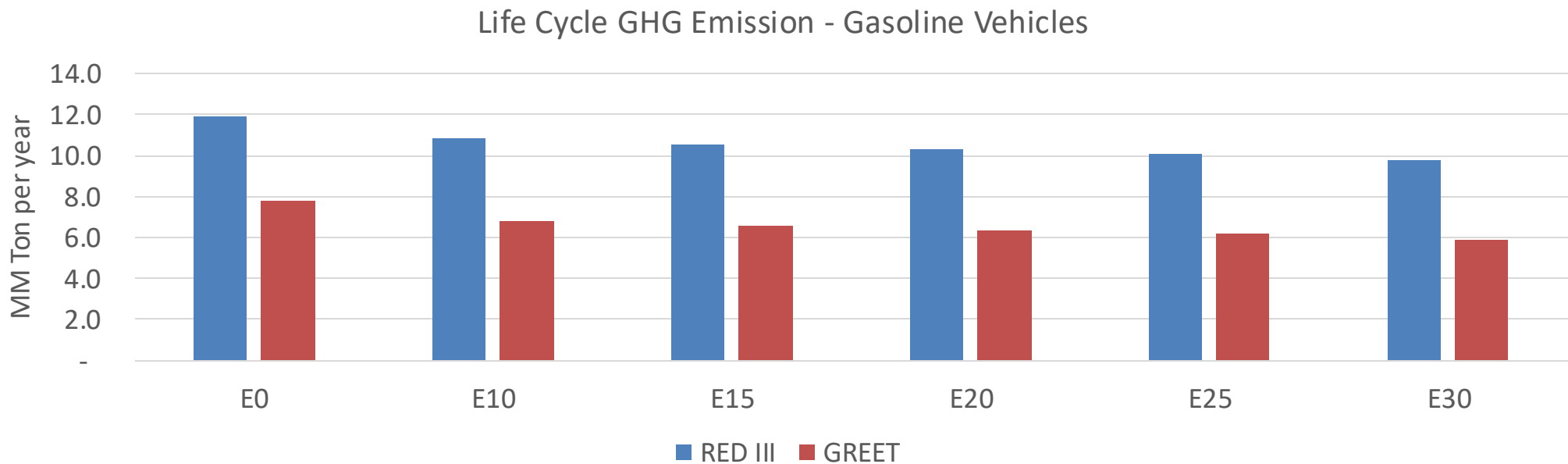
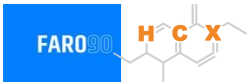
TIER III BIN 125 United States
Euro 6 Standard



Portugal – Vehicular Emissions

Vehicular Fleet Emissions (kg/day)	Eq E0	E5	E10	E15	E20	E25	E30	Reduction (%)	
								E10	E20
CO	209,074	196,659	184,244	174,785	165,594	158,626	150,367	-12%	-21%
CO2	13,460,947	13,114,808	12,787,900	12,530,796	12,403,570	12,331,260	12,103,959	-5%	-8%
VOC	20,658	19,868	19,077	18,586	18,162	17,858	17,377	-8%	-12%
NOx	17,465	13,098	12,225	11,522	10,866	10,143	9,379	-30%	-38%
SOx	160	148	136	125	116	105	94	-15%	-28%
NH3	4,599	4,553	4,508	4,529	4,580	4,616	4,646	-2%	0%
N2O	175	174	173	174	178	180	182	-1%	2%
CH4	4,900	4,900	4,900	4,974	5,082	5,175	5,263	0%	4%
Lead	-	-	-	-	-	-	-	0%	0%
Butadiene	198	193	187	184	182	181	178	-6%	-8%
Acetaldehyde	421	483	708	1,079	1,470	1,679	1,984	68%	249%
Formaldehyde	1,337	1,300	1,516	1,780	1,864	2,062	2,245	13%	39%
Benzene	358	344	326	321	319	305	295	-9%	-11%
PM2.5	839	745	651	566	480	390	295	-22%	-43%
PM10	690	613	536	465	395	320	243	-22%	-43%

Portugal – Gasoline Vehicle Fleet Life Cycle GHG Emissions



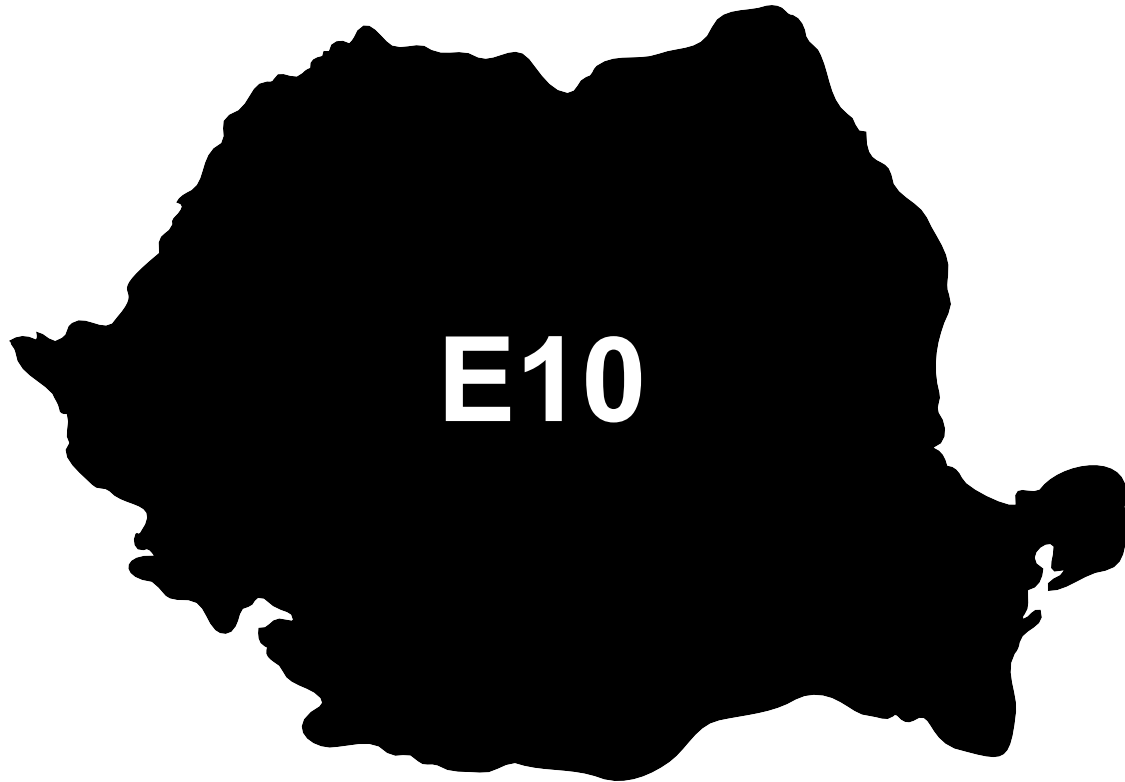
Country Emissions 2024 (MMT/año)	59.0
Target @ 2035 (MMT/year)	33.2
Delta	25.8

%Reduction vs E0	E0	E10	E15	E20	E25	E30
GREET 2024	0.0%	12.6%	15.8%	18.5%	21.0%	24.4%
RED II/III	0.0%	9.1%	11.6%	13.7%	15.7%	18.2%

% Target@2035	E0	E10	E15	E20	E25	E30
GREET 2024	0.0%	3.8%	4.8%	5.6%	6.4%	7.4%
RED II/III	0.0%	4.2%	5.4%	6.3%	7.3%	8.4%

Source: Greet, RED II/III, IPCC, climateactiontracker.org, F90

Ethanol Gasoline Blending in Romania

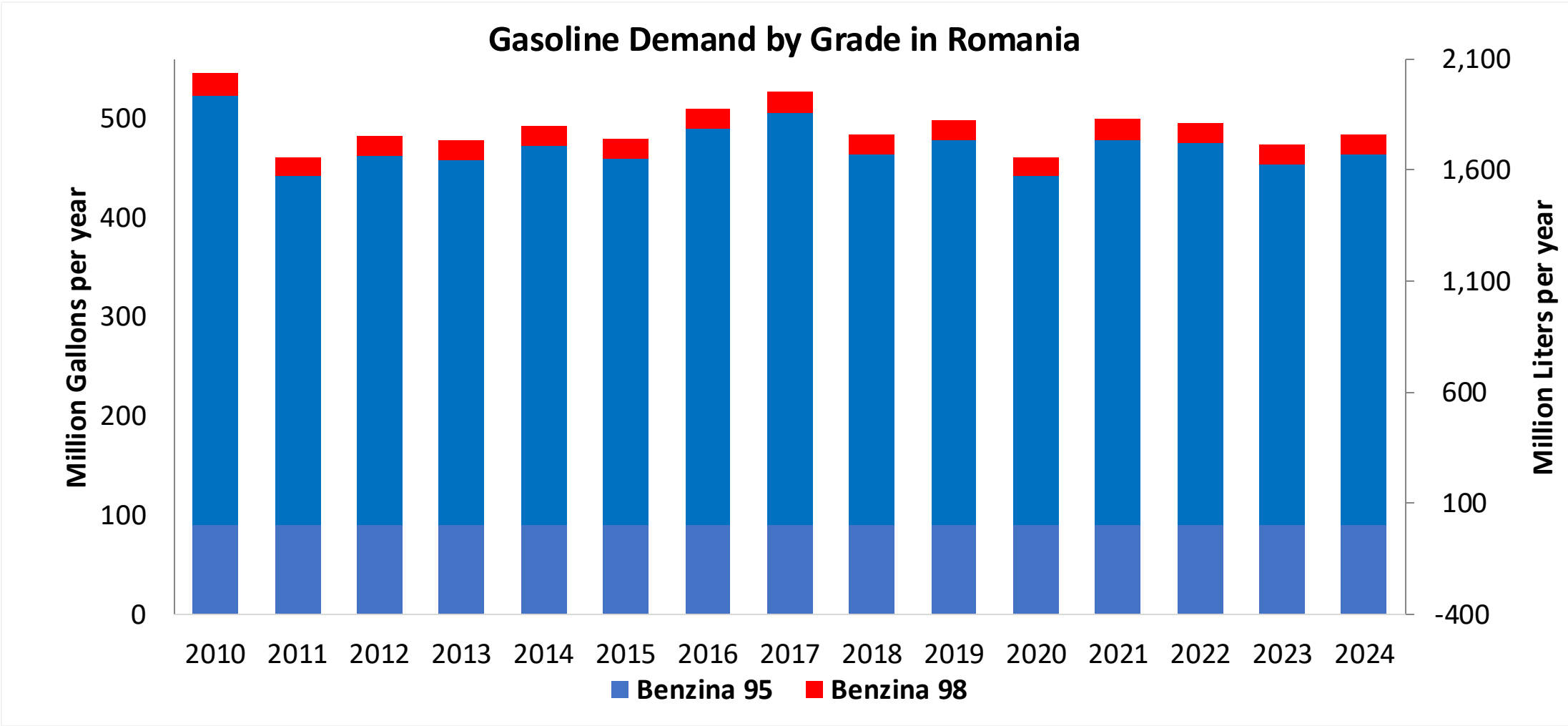


In 2024, gasoline consumption in Romania was 1,762 million liters (465 million gallons) and growth rate was -.7%. Gasoline production and net imports were 4,358 and -2,596 million liters (1,151 and -686 million gallons) correspondingly. Romania complies with the European gasoline standards and offers two main grades: RON 95 and RON 98) with an average octane of 95.2 RON.

In 2024, ethanol consumption in Romania was 144 million liters (38 million gallons) representing 8.2% of gasoline demand. Ethanol production and Net Imports were 107 and 37 million liters (38 and 10 million gallons) correspondingly. Ethanol sources are Corn 28% and Cereals 72%.

Source: Eurostat, European Commision

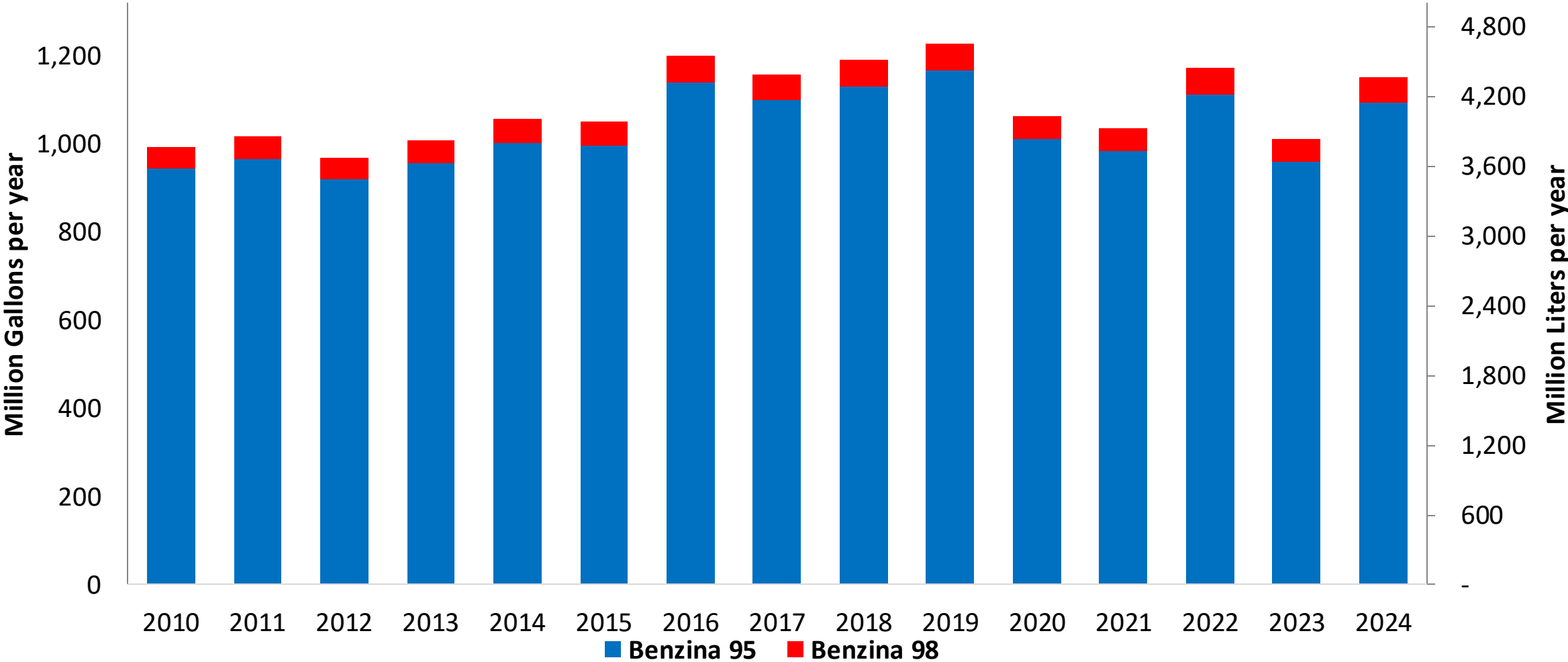
Gasoline Demand in Romania



Source: Eurostat

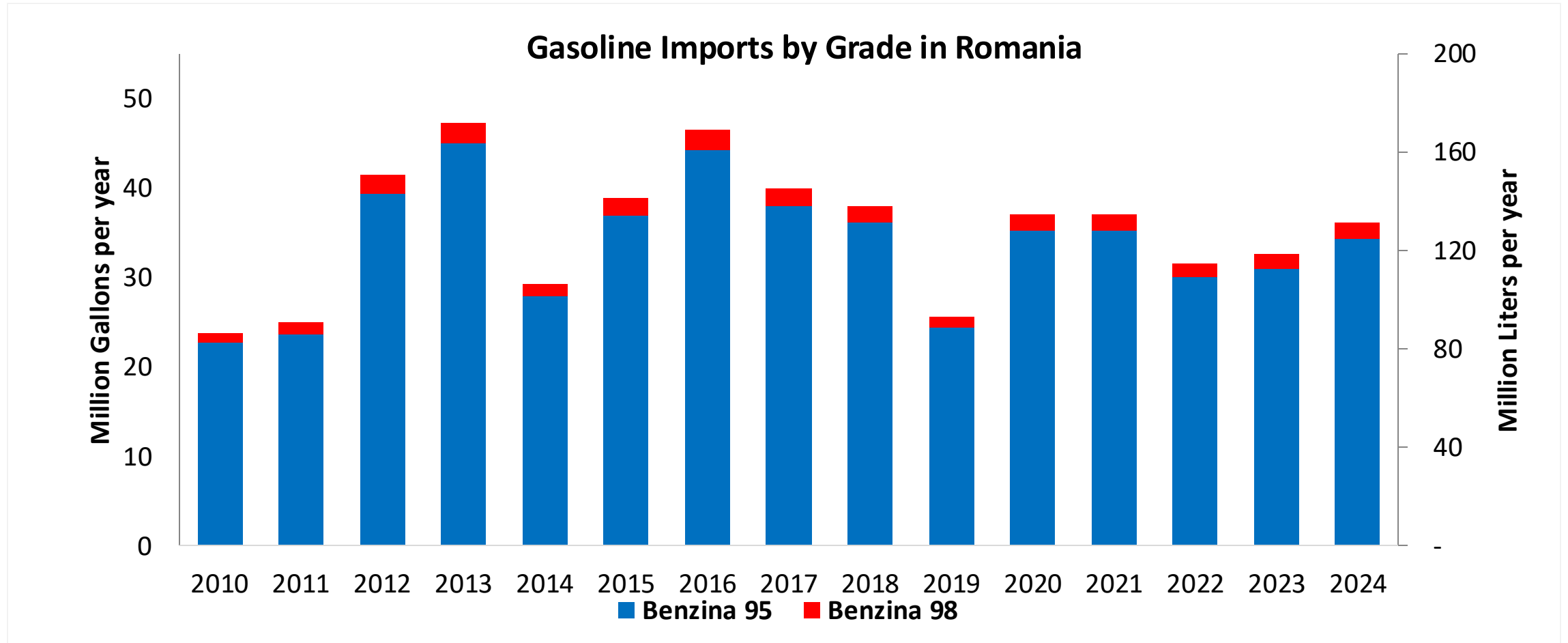
Gasoline Production in Romania

Gasoline Production by Grade in Romania



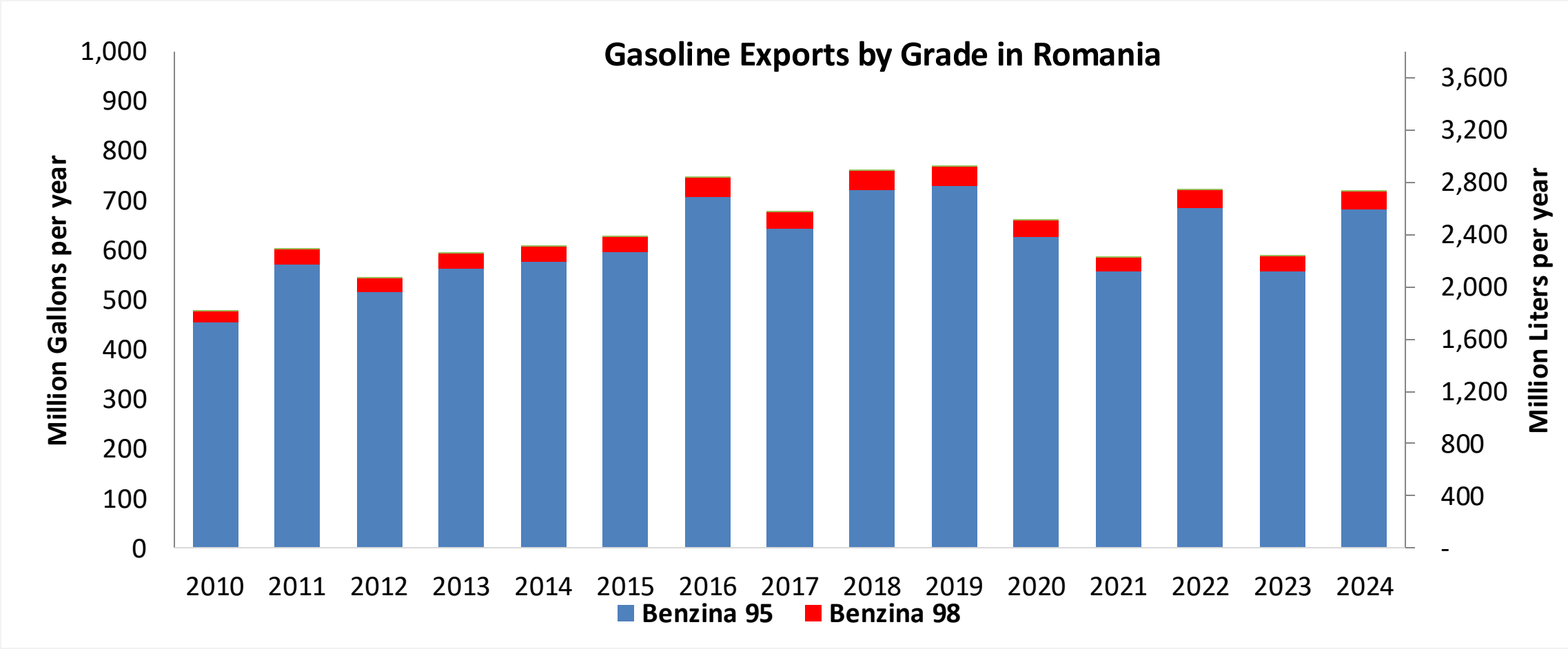
Source: Eurostat

The FARO90 logo is shown in a blue box, and a chemical structure is displayed next to it.



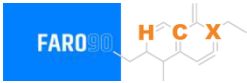
Source: Eurostat

Gasoline Exports in Romania

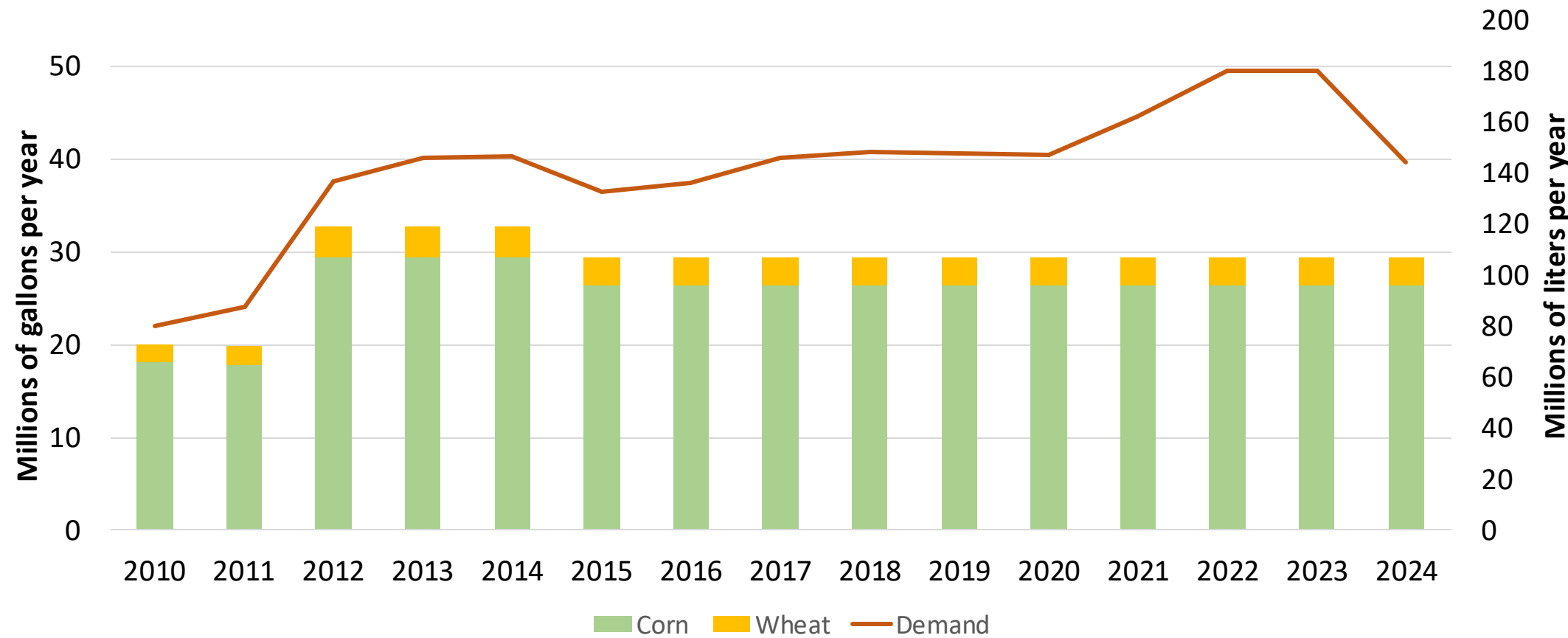


Source: Eurostat

Ethanol Balance in Romania



Ethanol Demand in Romania



Source: Eurostat

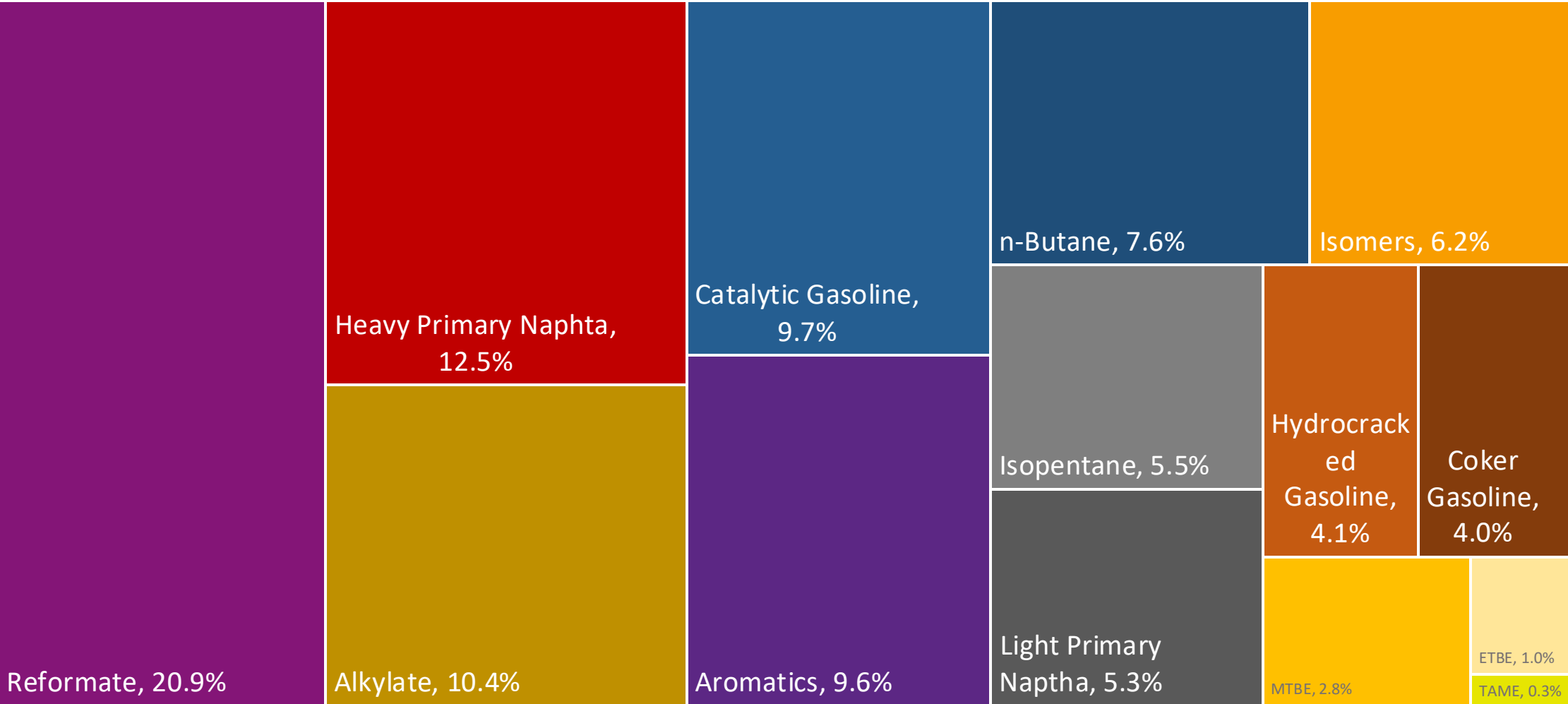
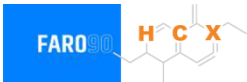
Gasoline Quality in Romania

Name	EN 228:2025 (Euro 6); Country Level Fuel Ordinances							
Implementation Date	2025							
Applicability	All EU Contries + UK							
Grade	RON 95 E5		RON 95 E10		RON 98 E5		RON 98 E10	
	Min	Max	Min	Max	Min	Max	Min	Max
RON	95.0		95.0		98.0		98.0	
MON	> 85	> 88	> 85	> 88	> 85	> 88	> 85	> 88
AKI								
Benzene (%vol)	< 1							
Aromatics (% vol)	< 35							
Olefins (%vol)	< 18							
Lead (g/L)	< 0.005							
Manganese (mg/L)	< 2,0							
Sulfur (ppm)	10.0							
Oxygen (%w)*	< 2,7	< 3,7	< 2,7	< 3,7	< 2,7	< 3,7	< 2,7	< 3,7
Ethanol (%vol)	< 5	< 10	< 5	< 10	< 5	< 10	< 5	< 10
PVR 37.8°C (psi)	<> 45 - 100 kPa in 6 Volatility Classes							
Overall Biofuels (%cal)	10.0							
Advanced Biofuels (%cal)	-							
Bioethanol in Gasoline Sales (%cal)	-							
GHG Emission Reduction (%)	-							

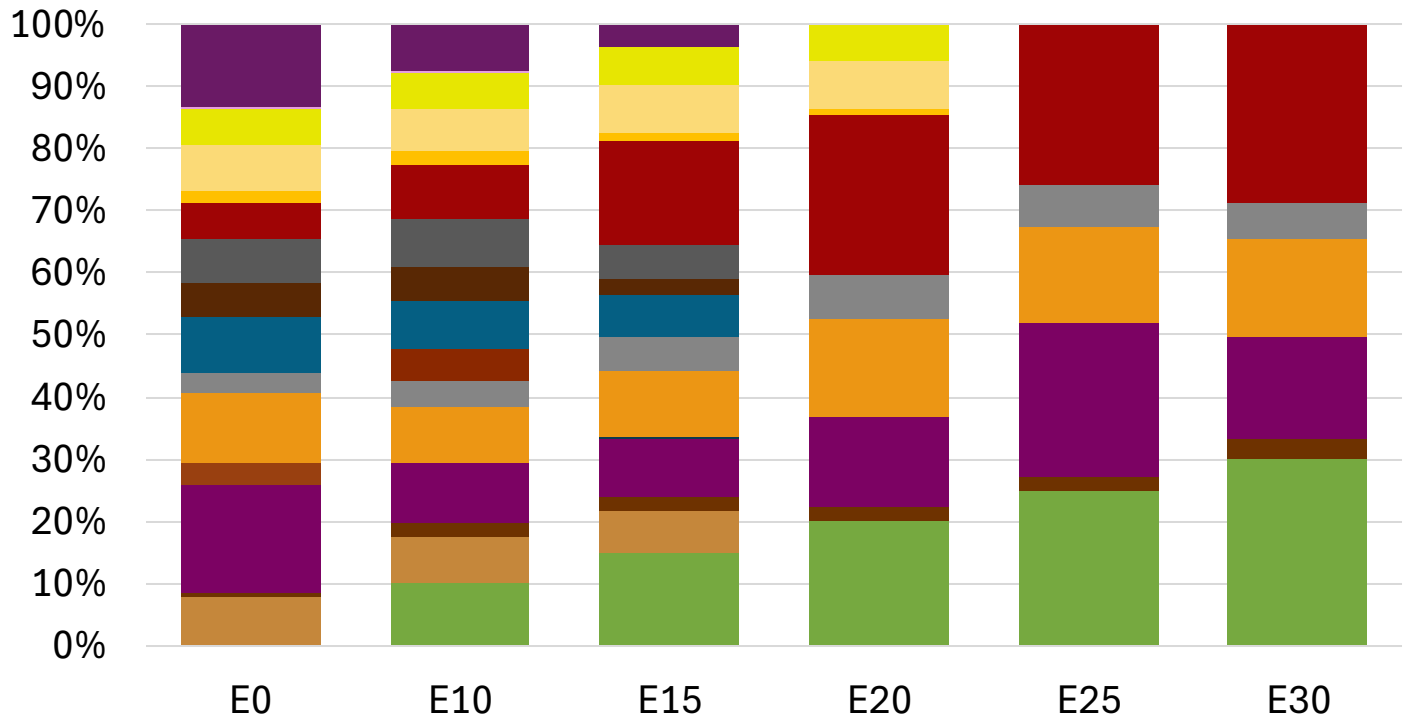
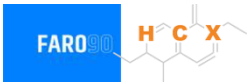
* Methanol 3%, Iso-propyl alcohol 12%, Ter-butyl alcohol 15%, Iso-butyl alcohol 15%, Ethers +5 Carbons 22%, Other Oxygenates 15%

Source: European Commision

Romania Available Blending Components



Romania – Gasoline 95 – Constant Octane

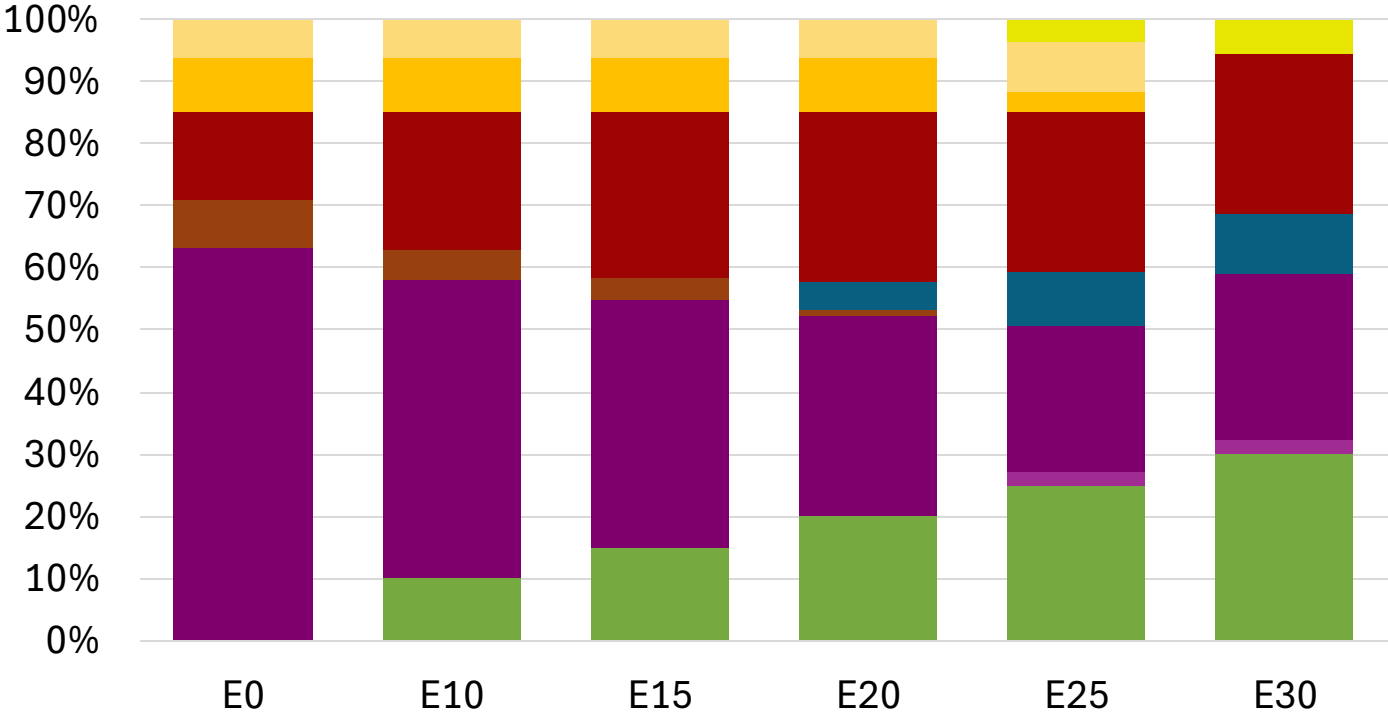
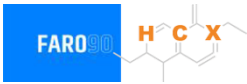


Average CIF 2024
Prices

Octane (RON)	95.0	95.0	95.0	95.0	95.0	95.8
Price (US\$/gal)	\$ 2.284	\$ 2.146	\$ 2.046	\$ 1.931	\$ 1.885	\$ 1.832

Source: Faro90

Romania - Gasoline 98 - Constant Octane



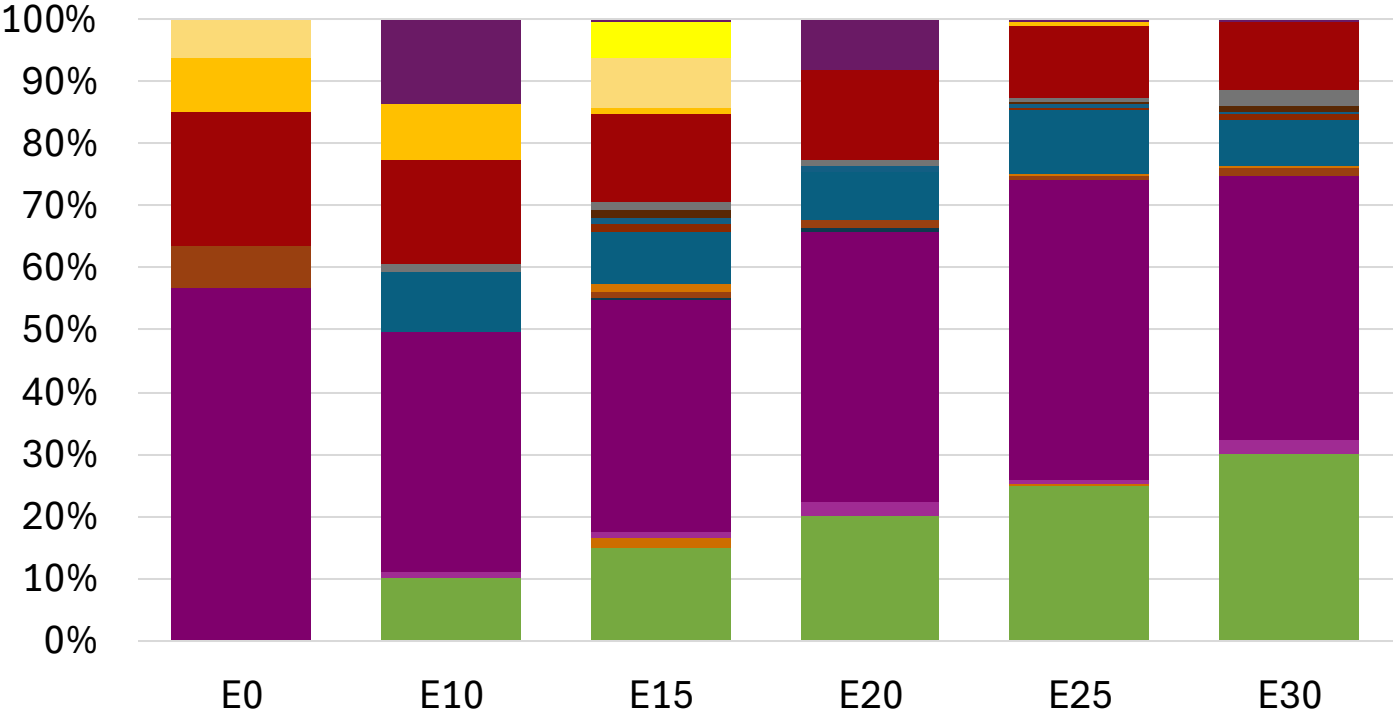
Ethanol
Alkylate
Raffinate
Reformate
n-Butane
Isomerate
Isopentane
Hydrocracked Gasoline
Catalytic Gasoline
Coker Naphtha
Light Primary Naphtha
Heavy Primary Naphtha
MTBE
ETBE
TAME
Aromatics

Average CIF 2024 Prices

Octane (RON)	98.0	98.0	98.0	98.0	98.0	98.0
Price (US\$/gal)	\$ 2.304	\$ 2.187	\$ 2.118	\$ 2.046	\$ 1.981	\$ 1.931

Source: Faro90

Romania – Gasoline 95 – Increased Octane

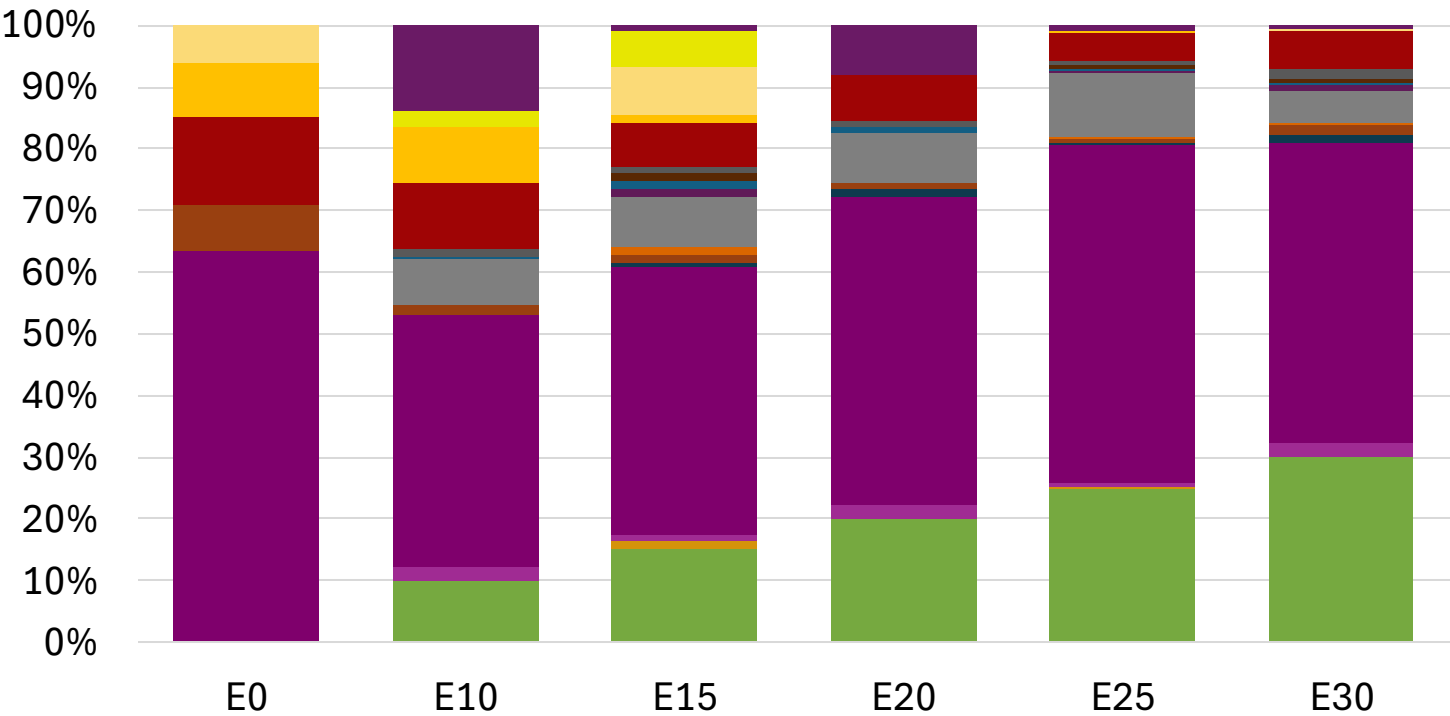


Average CIF 2024
Prices

Octane (RON)	95.0	96.2	98.3	98.7	100.5	101.8
Price (US\$/gal)	\$ 2.172	\$ 2.172	\$ 2.172	\$ 2.172	\$ 2.172	\$ 2.172

Source: Faro90

Romania – Gasoline 98 – Increased Octane

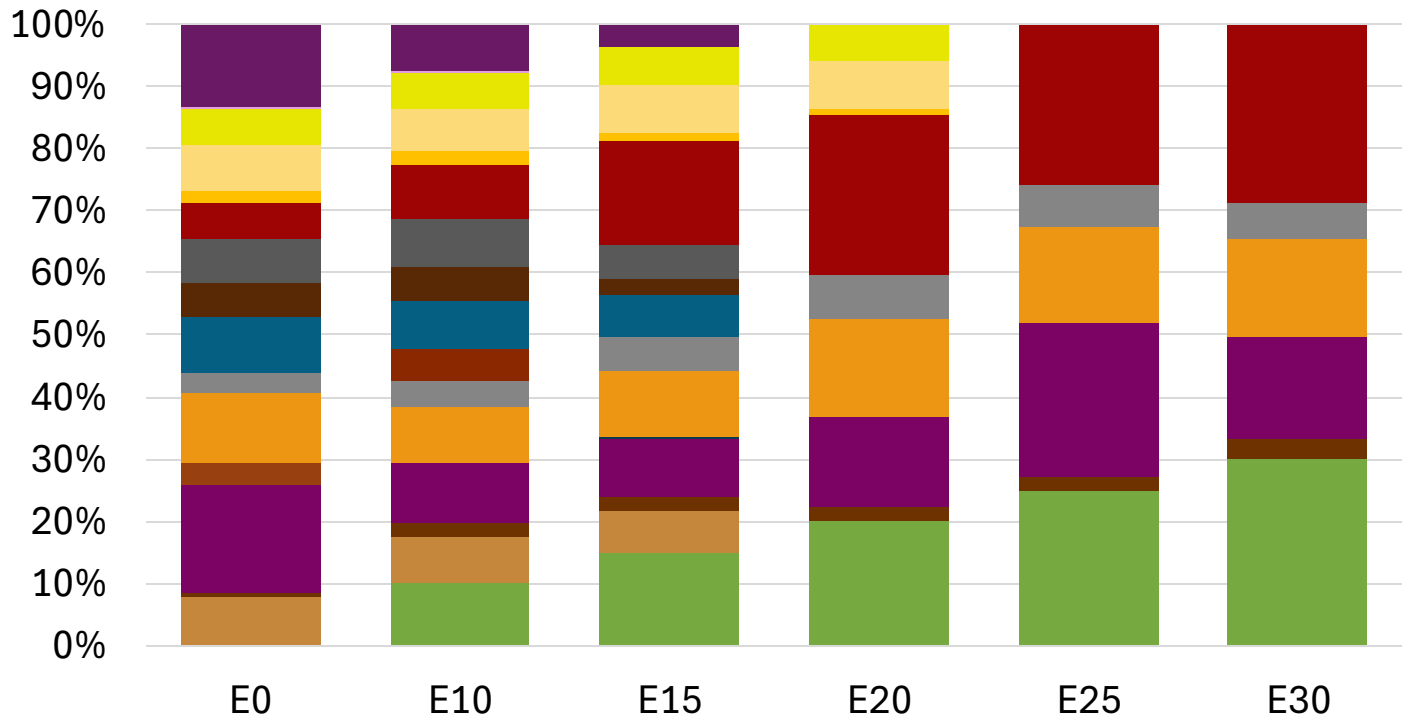
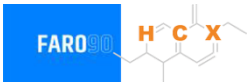


Average CIF 2024
Prices

Octane (RON)	98.0	99.0	100.9	101.2	102.8	104.1
Price (US\$/gal)	\$ 2.304	\$ 2.304	\$ 2.304	\$ 2.304	\$ 2.304	\$ 2.304

Source: Faro90

Romania – Gasoline 95 – Constant Octane

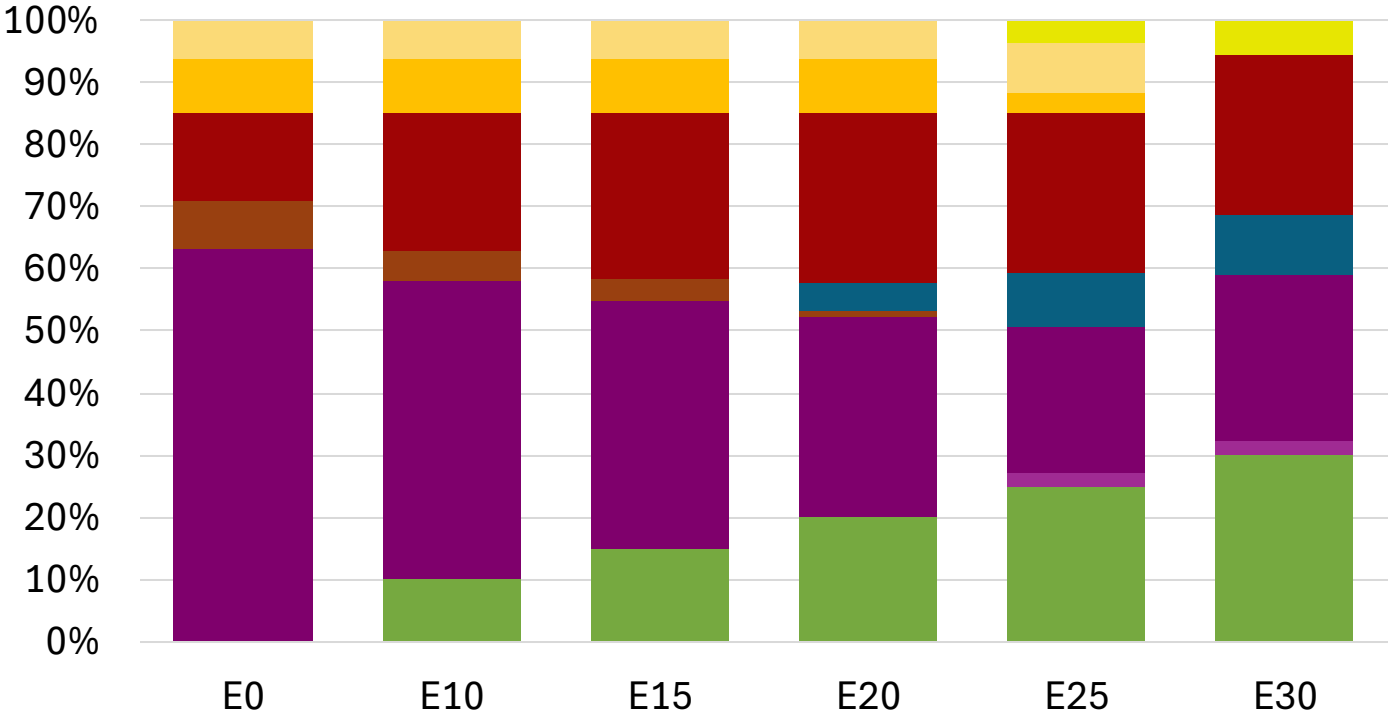
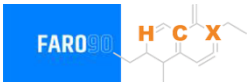


Average CIF 2024
Prices

Octane (RON)	95.0	95.0	95.0	95.0	95.0	95.8
Price (US\$/gal)	\$ 2.274	\$ 2.136	\$ 2.036	\$ 1.921	\$ 1.875	\$ 1.822

Source: Faro90

Romania - Gasoline 98 - Constant Octane

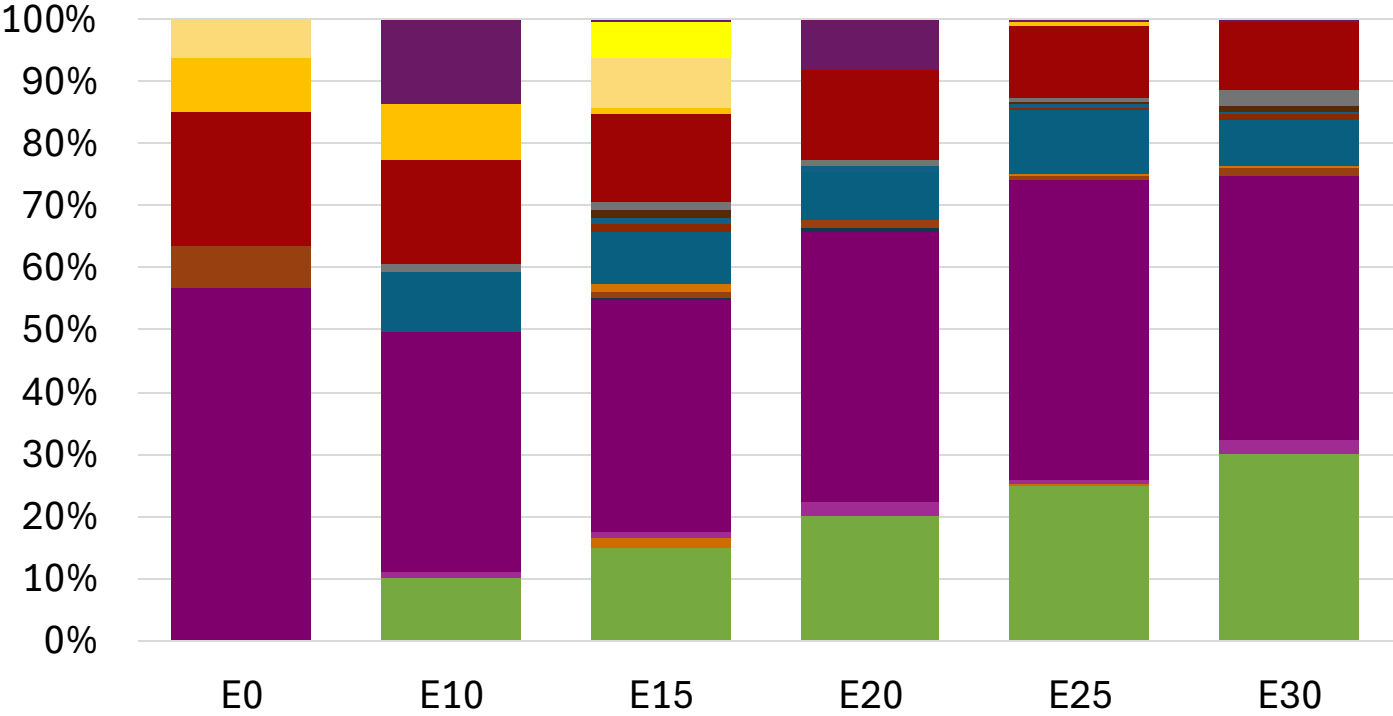


Average CIF 2024 Prices

Octane (RON)	98.0	98.0	98.0	98.0	98.0	98.0
Price (US\$/gal)	\$ 2.294	\$ 2.177	\$ 2.108	\$ 2.036	\$ 1.971	\$ 1.921

Source: Faro90

Romania – Gasoline 95 – Increased Octane

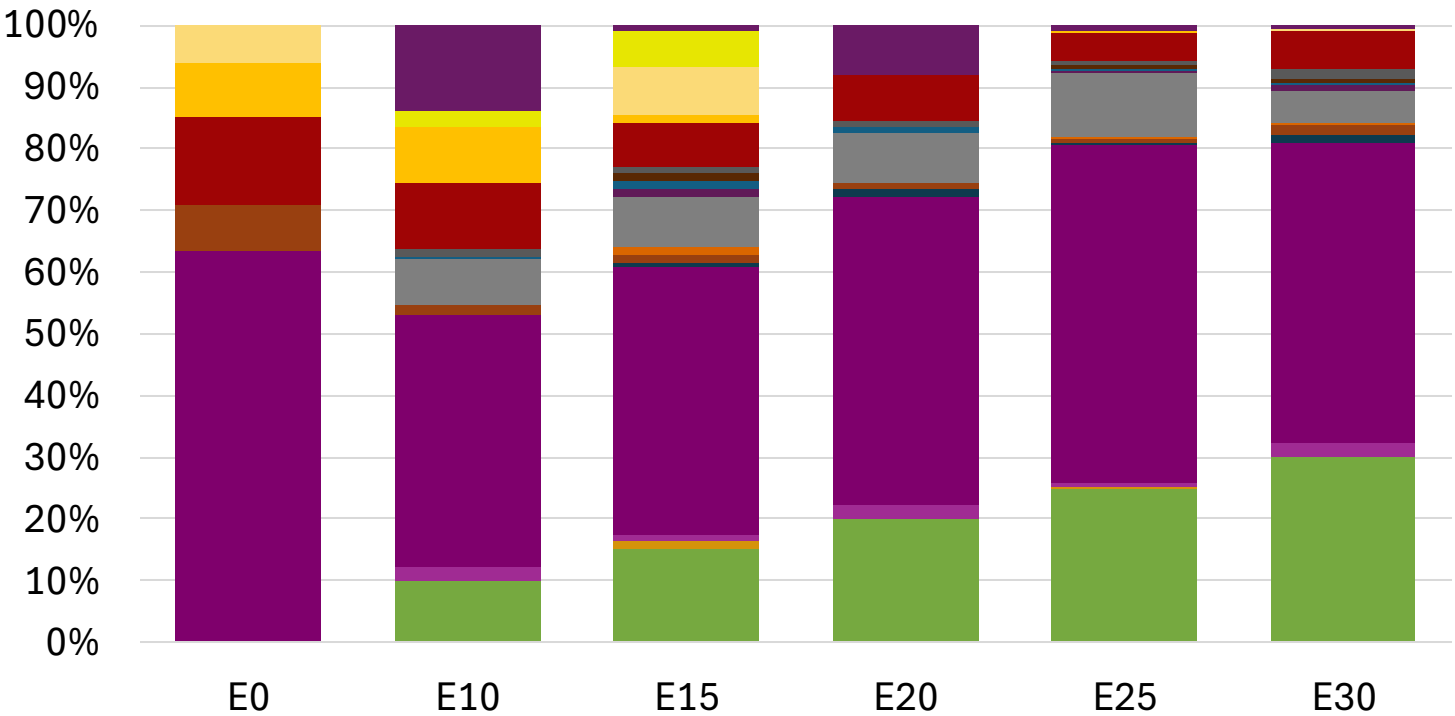
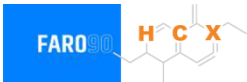


Average CIF 2024
Prices

Octane (RON)	95.0	96.2	98.3	98.7	100.5	101.8
Price (US\$/gal)	\$ 2.162	\$ 2.162	\$ 2.162	\$ 2.162	\$ 2.162	\$ 2.162

Source: Faro90

Romania – Gasoline 98 – Increased Octane

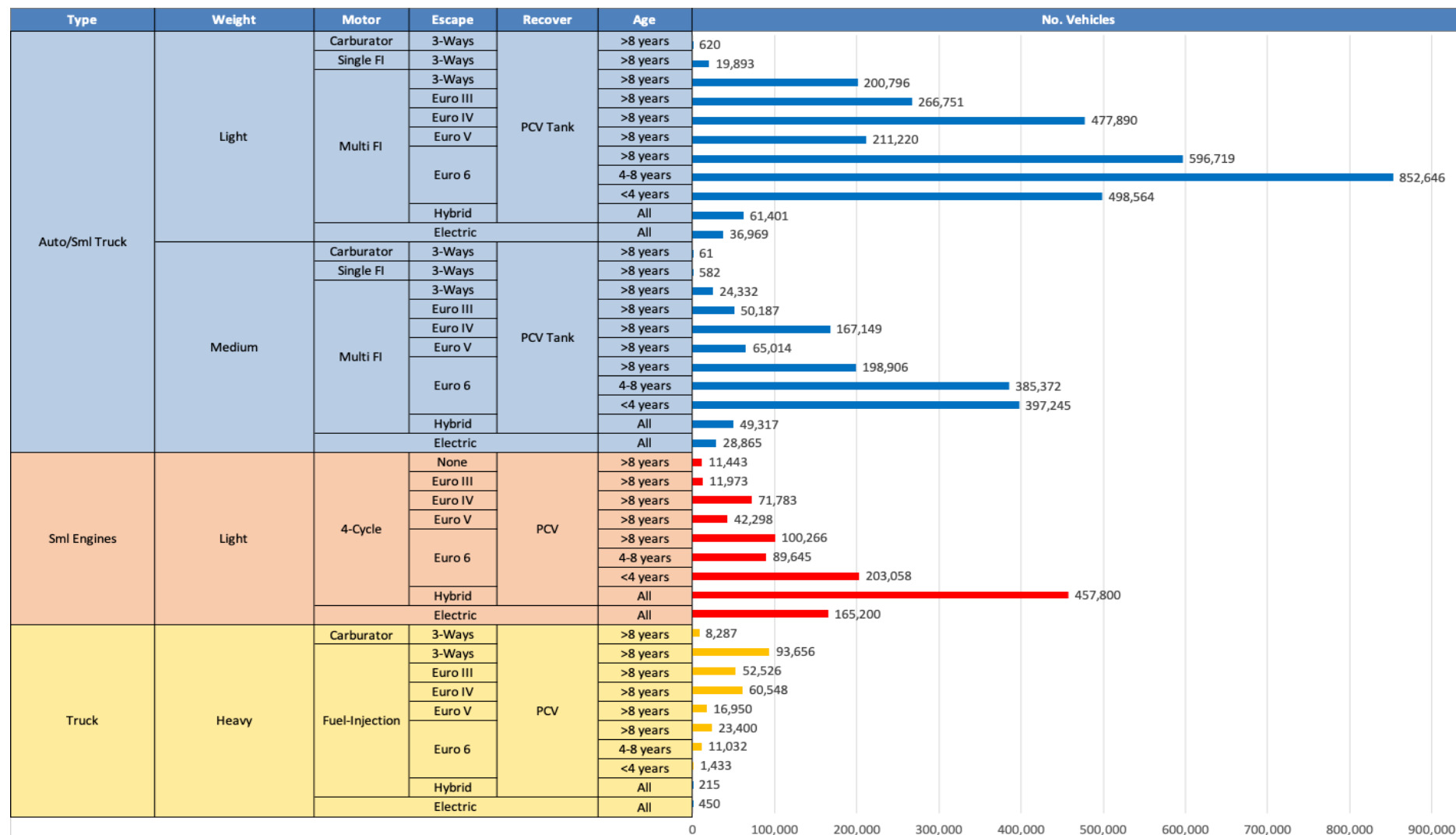


Average CIF 2024
Prices

Octane (RON)	98.0	99.0	100.9	101.2	102.8	104.1
Price (US\$/gal)	\$ 2.294	\$ 2.294	\$ 2.294	\$ 2.294	\$ 2.294	\$ 2.294

Source: Faro90

Gasoline Vehicle Fleet - Romania

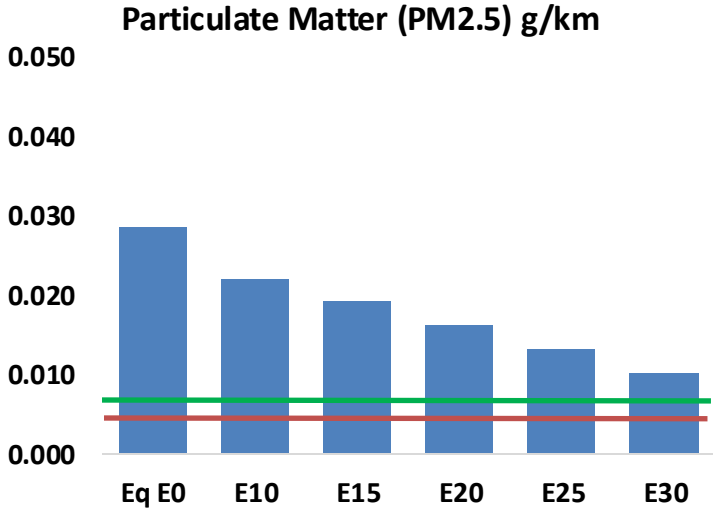
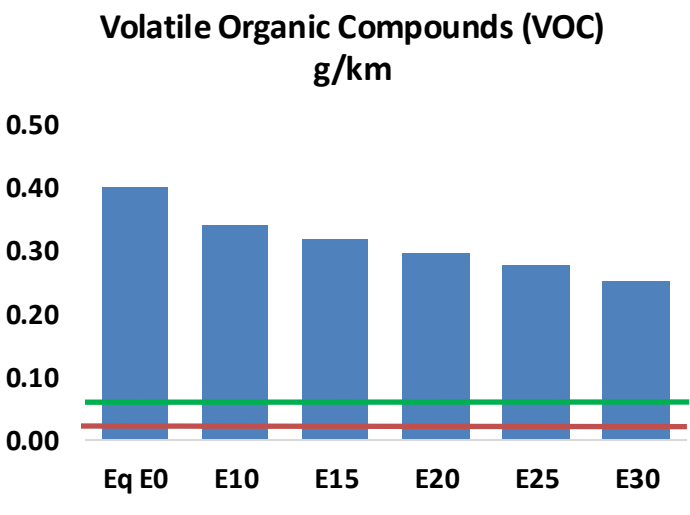
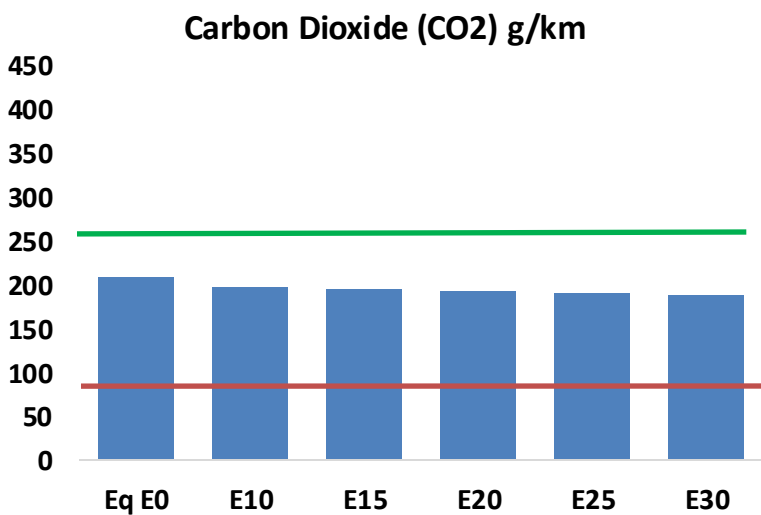
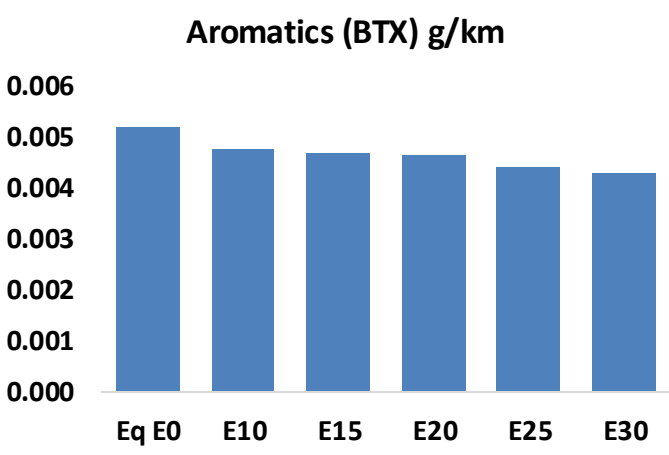
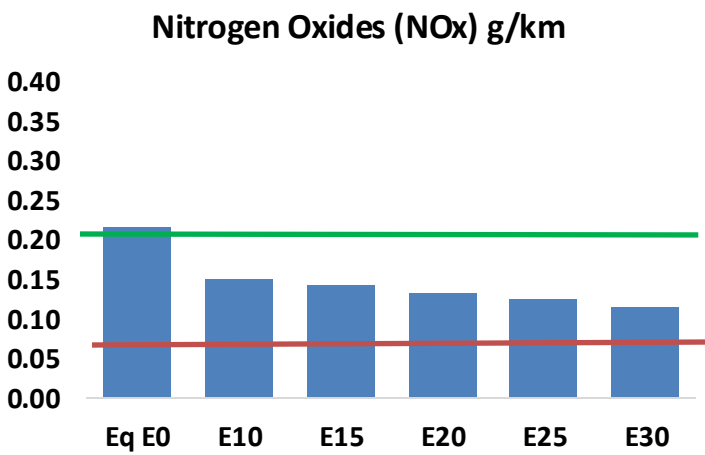
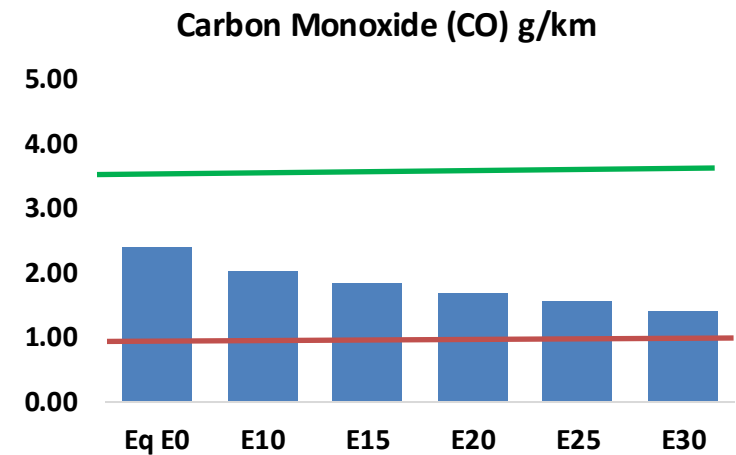
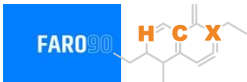


Vehicle Fleet: 6,012,464
Gasoline Vehicle Fleet: 2,699,932

Source: Eurostat, ACEA

Romania – Vehicular Emissions

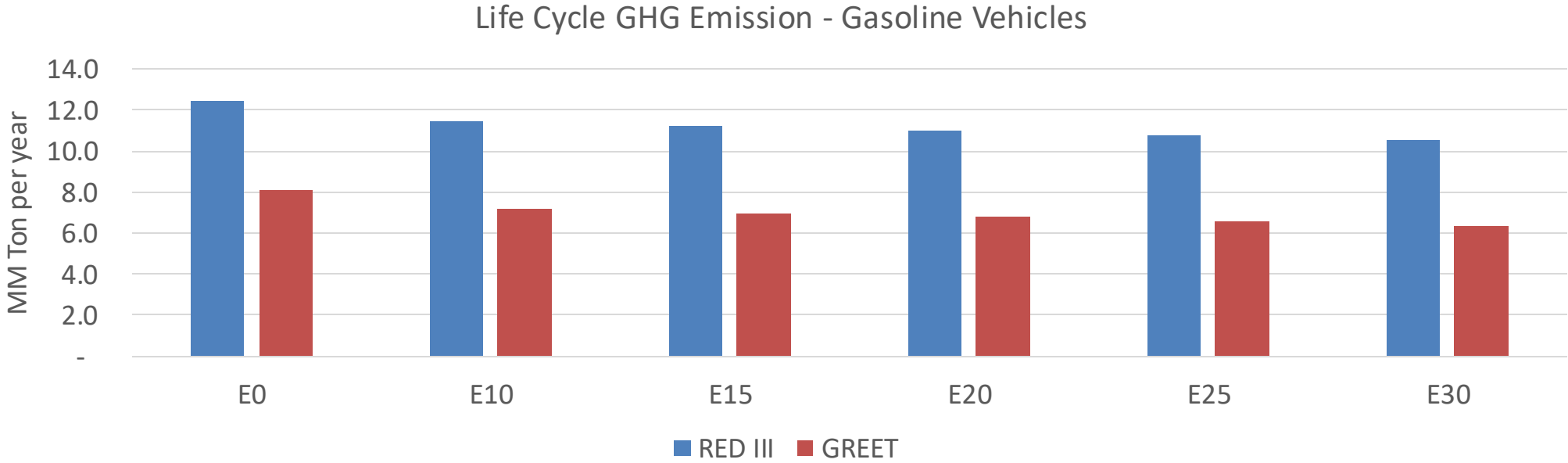
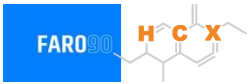
TIER III BIN 125 United States
Euro 6 Standard



Romania – Vehicular Emissions

Vehicular Fleet Emissions (kg/day)	Eq E0	E5	E10	E15	E20	E25	E30	Reduction (%)	
								E10	E20
CO	171,877	157,764	143,651	131,911	120,016	110,845	101,081	-16%	-30%
CO2	14,864,151	14,481,930	14,120,943	13,837,038	13,696,551	13,615,510	13,364,537	-5%	-8%
VOC	28,470	26,412	24,354	22,693	21,033	19,760	18,049	-14%	-26%
NOx	15,404	11,553	10,783	10,163	9,584	8,947	8,272	-30%	-38%
SOx	169	157	144	133	123	111	100	-15%	-28%
NH3	5,167	5,116	5,065	5,089	5,146	5,186	5,220	-2%	0%
N2O	247	246	245	246	251	254	257	-1%	2%
CH4	6,222	6,222	6,222	6,315	6,452	6,570	6,682	0%	4%
Lead	-	-	-	-	-	-	-	0%	0%
Butadiene	175	168	161	156	152	149	144	-8%	-14%
Acetaldehyde	489	561	823	1,253	1,707	1,950	2,305	68%	249%
Formaldehyde	1,766	1,717	2,002	2,350	2,462	2,724	2,965	13%	39%
Benzene	372	358	339	333	331	316	306	-9%	-11%
PM2.5	921	818	715	621	527	428	324	-22%	-43%
PM10	1,108	984	860	747	634	515	390	-22%	-43%

Romania – Gasoline Vehicle Fleet Life Cycle GHG Emissions



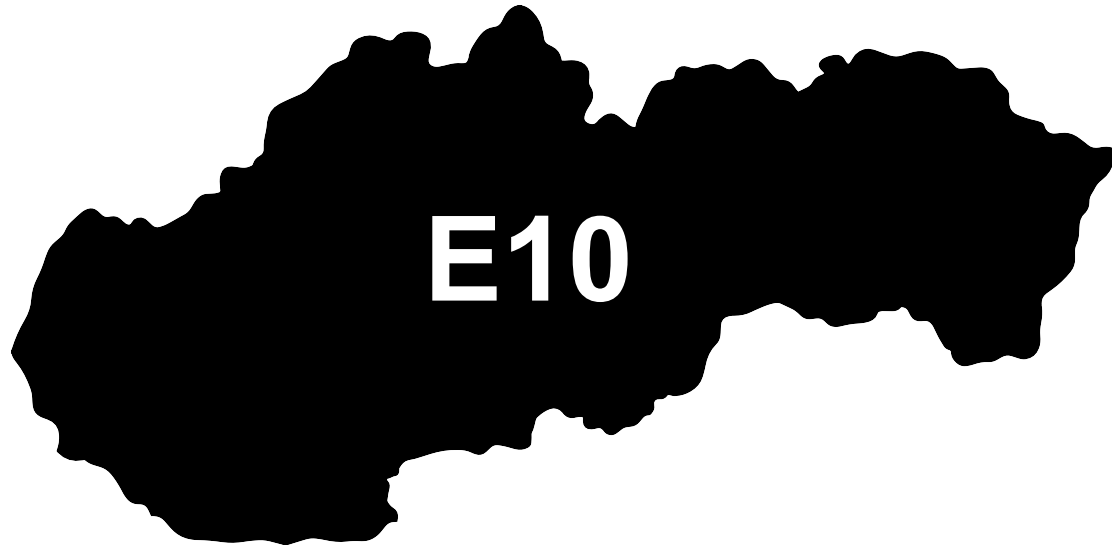
Country Emissions 2024 (MMT/año)	76.7
Target @ 2035 (MMT/year)	43.2
Delta	33.5

%Reduction vs E0	E0	E10	E15	E20	E25	E30
GREET 2024	0.0%	11.3%	14.4%	16.7%	18.9%	22.0%
RED II/III	0.0%	7.8%	10.0%	11.7%	13.3%	15.5%

% Target@2035	E0	E10	E15	E20	E25	E30
GREET 2024	0.0%	2.7%	3.5%	4.1%	4.6%	5.3%
RED II/III	0.0%	2.9%	3.7%	4.4%	5.0%	5.8%

Source: Greet, RED II/III, IPCC, climateactiontracker.org, F90

Ethanol Gasoline Blending in Slovakia



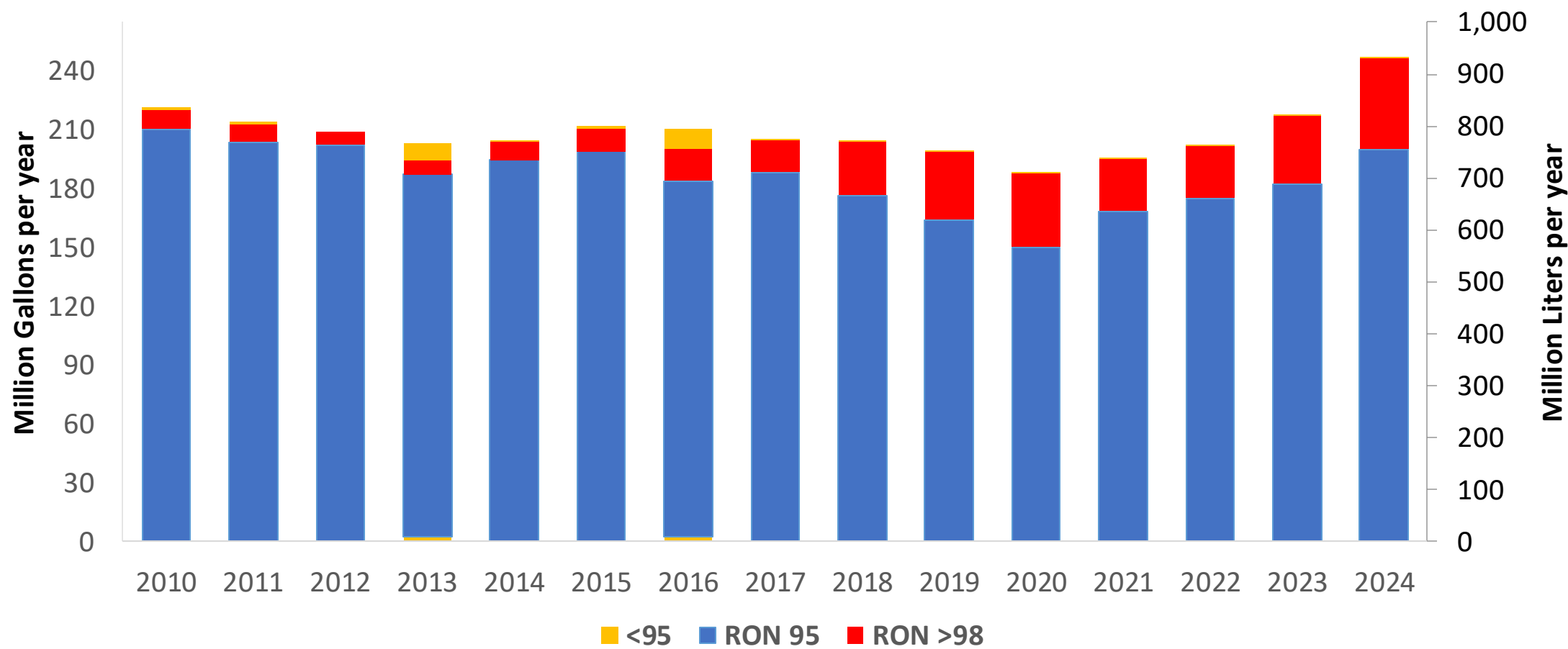
In 2024, gasoline consumption in Slovakia was 931 million liters (246 million gallons) and growth rate was 4.4%. Gasoline production and net imports were and 931 million liters (and 246 million gallons) correspondingly. Slovakia complies with the European gasoline standards and offers two main grades: RON 95 and RON 98) with an average octane of 95.6 RON.

In 2024, ethanol consumption in Slovakia was 80 million liters (21 million gallons) representing 8.6% of gasoline demand. Ethanol production and Net Imports were 80 and million liters (21 and million gallons) correspondingly. Ethanol source is Cereals 100%.

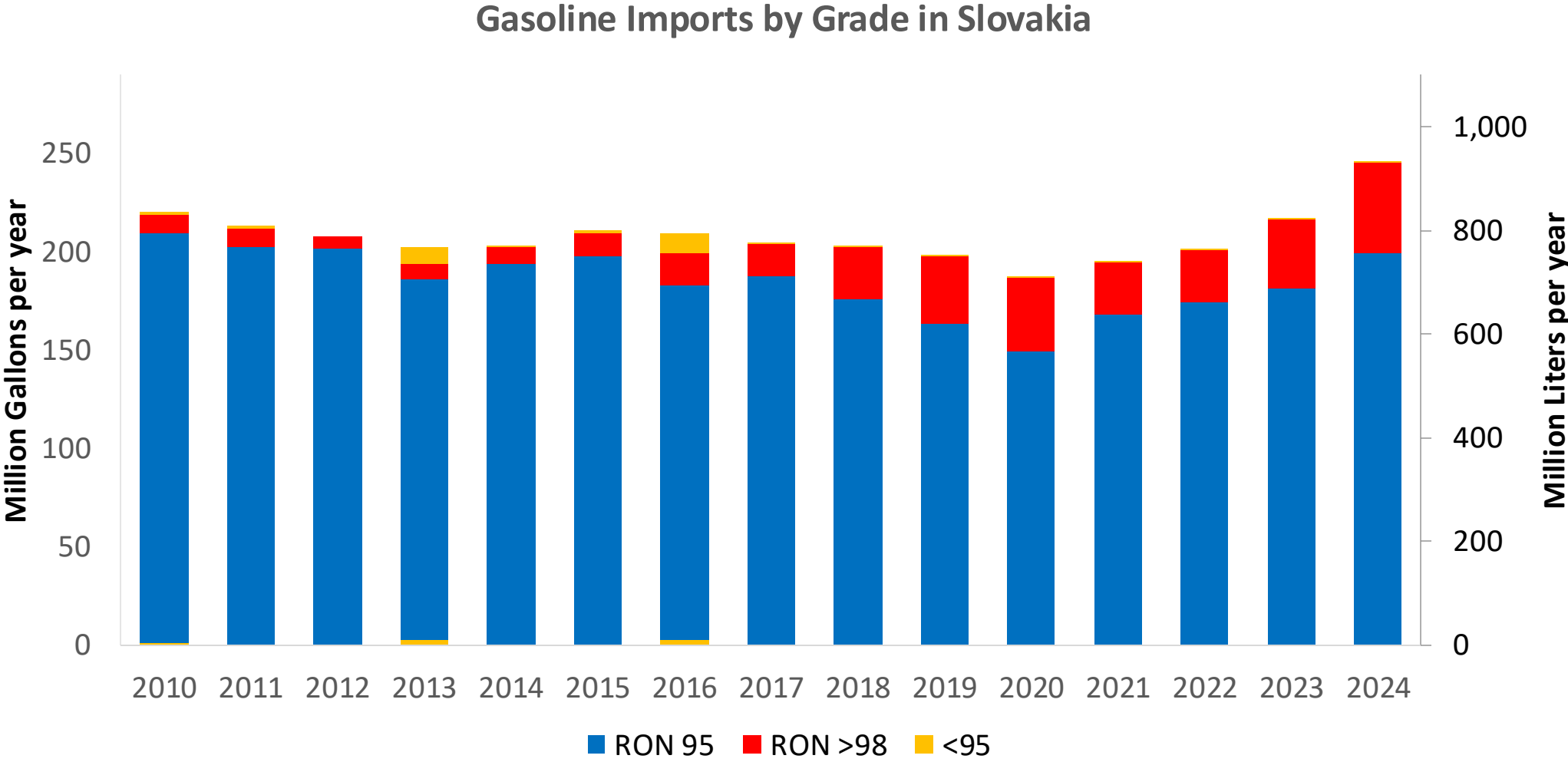
Source: Eurostat, European Commission

Gasoline Demand in Slovakia

Gasoline Demand by Grade in Slovakia

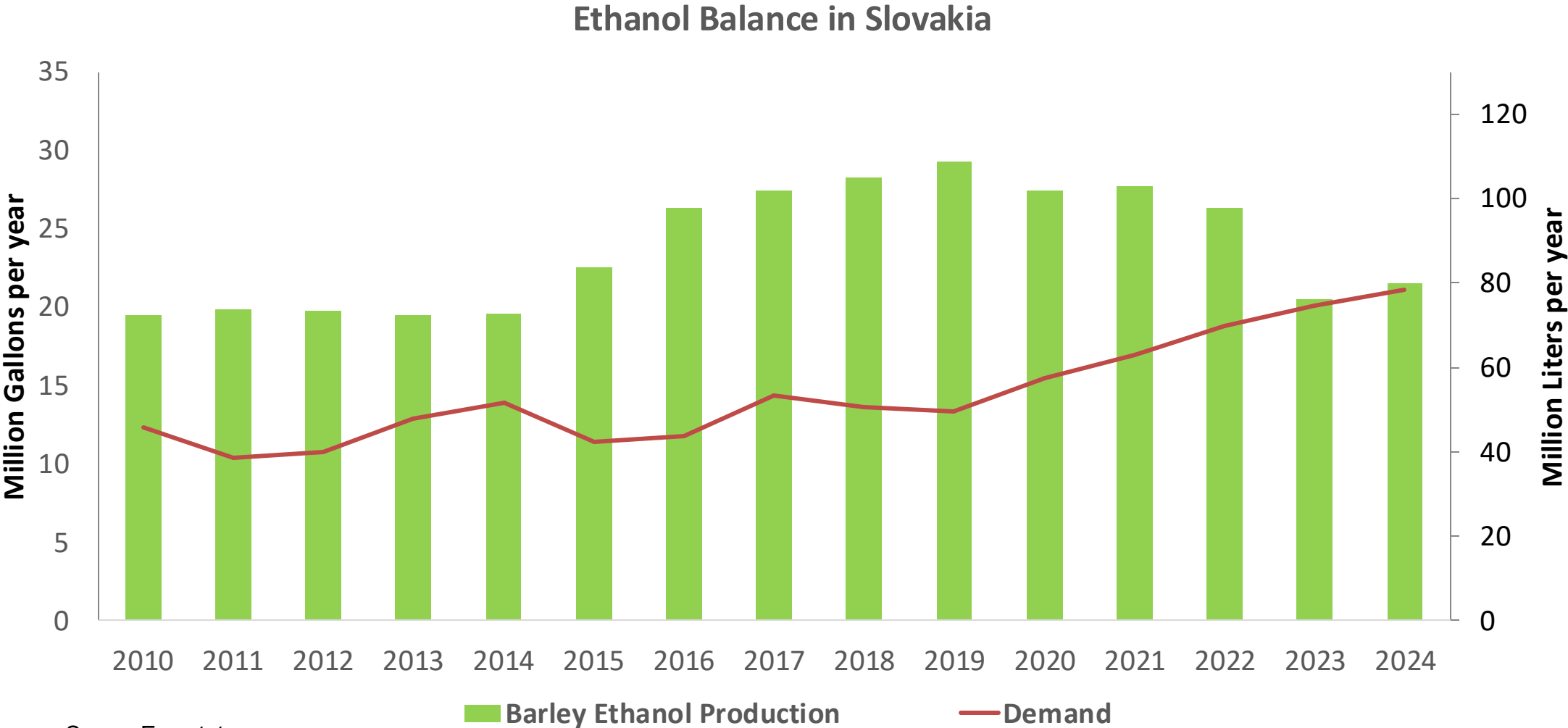


Source: Eurostat



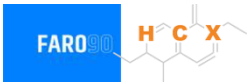
Source: Eurostat

Ethanol Balance in Slovakia



Source: Eurostat

Gasoline Quality in Slovakia

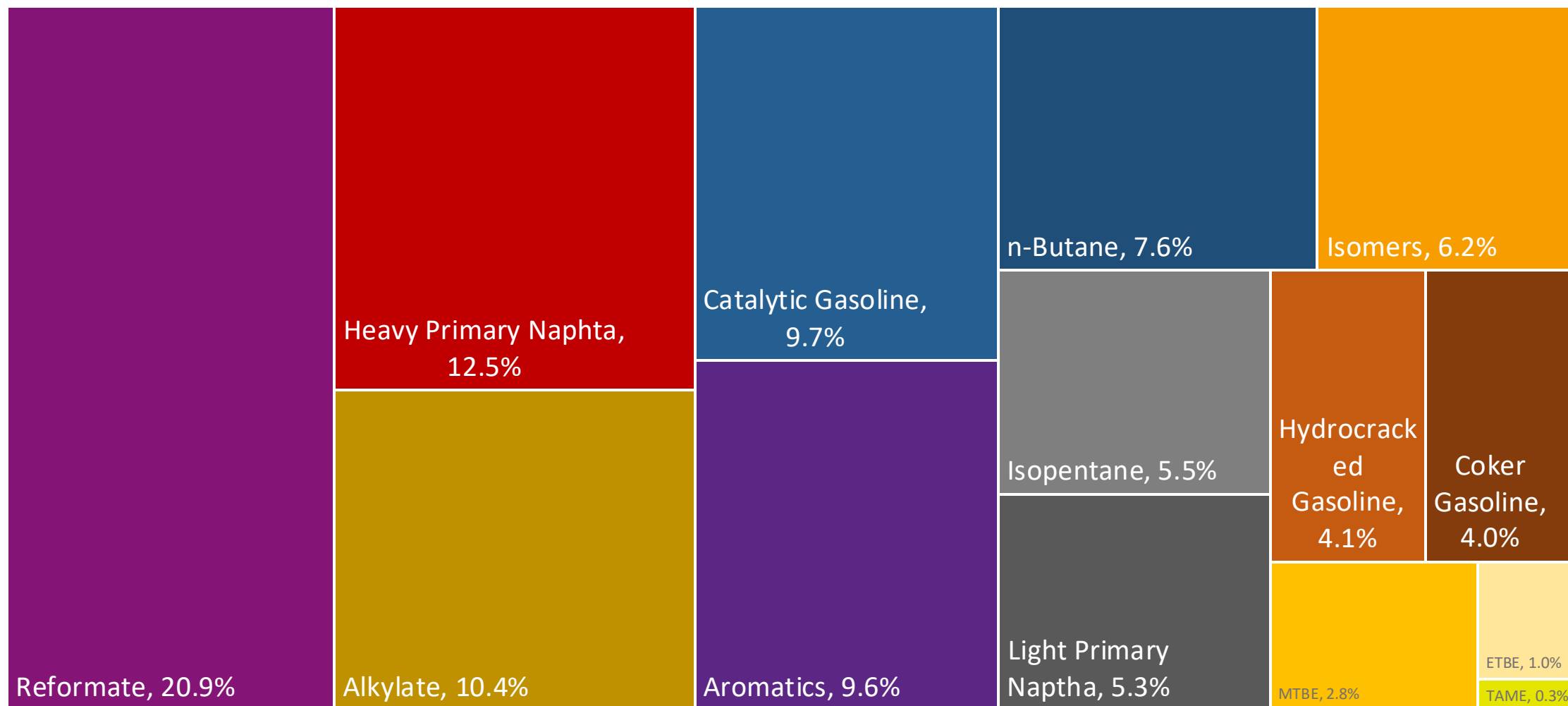


Name	EN 228:2025 (Euro 6); Country Level Fuel Ordinances							
Implementation Date	2025							
Applicability	All EU Contries + UK							
Grade	RON 95 E5		RON 95 E10		RON 98 E5		RON 98 E10	
	Min	Max	Min	Max	Min	Max	Min	Max
RON	95.0		95.0		98.0		98.0	
MON	> 85	> 88	> 85	> 88	> 85	> 88	> 85	> 88
AKI								
Benzene (%vol)	< 1							
Aromatics (% vol)	< 35							
Olefins (%vol)	< 18							
Lead (g/L)	< 0.005							
Manganese (mg/L)	< 2,0							
Sulfur (ppm)	10.0							
Oxygen (%w)*	< 2,7	< 3,7	< 2,7	< 3,7	< 2,7	< 3,7	< 2,7	< 3,7
Ethanol (%vol)	< 5	< 10	< 5	< 10	< 5	< 10	< 5	< 10
PVR 37.8°C (psi)	<> 45 - 100 kPa in 6 Volatility Classes							
Overall Biofuels (%cal)	9.2							
Advanced Biofuels (%cal)	1.1							
Bioethanol in Gasoline Sales (%cal)	-							
GHG Emission Reduction (%)	-							

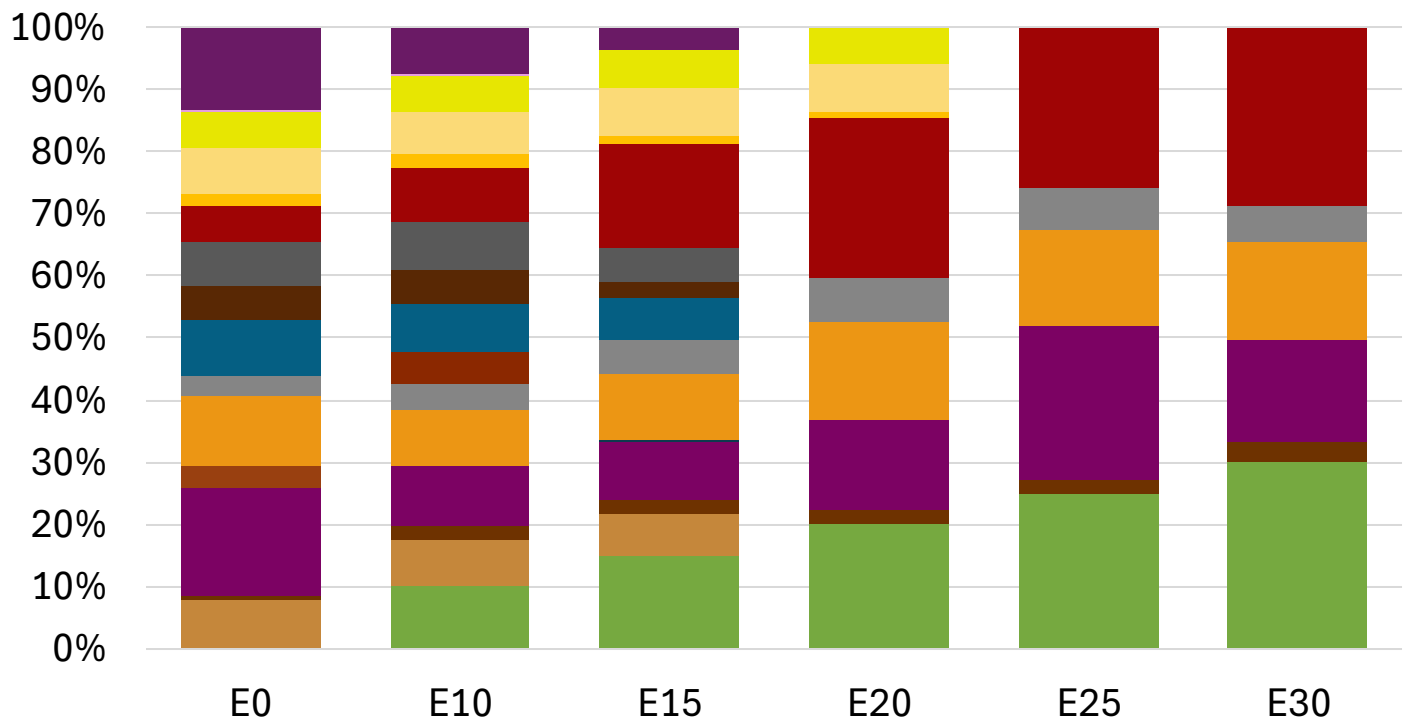
* Methanol 3%, Iso-propyl alcohol 12%, Ter-butyl alcohol 15%, Iso-butyl alcohol 15%, Ethers +5 Carbons 22%, Other Oxygenates 15%

Source: European Commision

Slovakia Available Blending Components



Slovakia – Gasoline 95 – Constant Octane

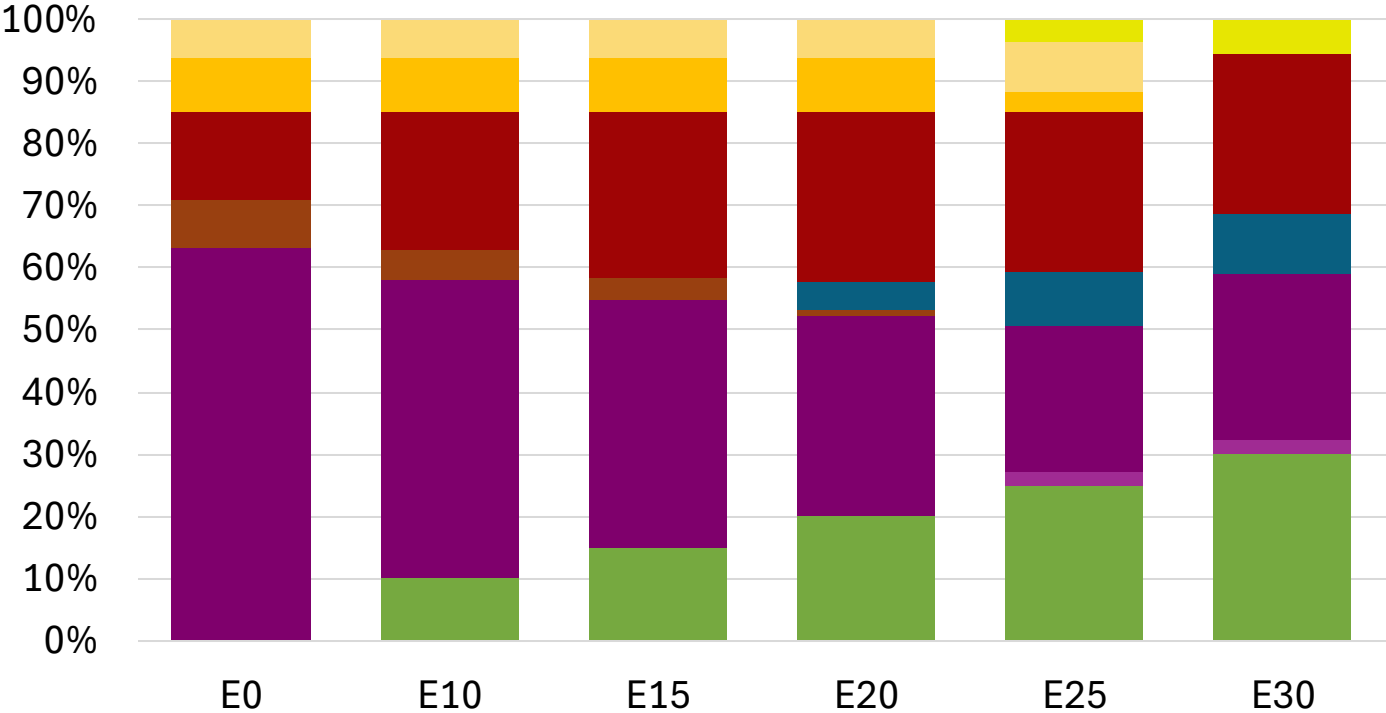
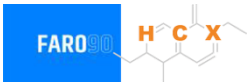


Average CIF 2024
Prices

Octane (RON)	95.0	95.0	95.0	95.0	95.0	95.8
Price (US\$/gal)	\$ 2.304	\$ 2.166	\$ 2.066	\$ 1.951	\$ 1.905	\$ 1.852

Source: Faro90

Slovakia - Gasoline 98 - Constant Octane

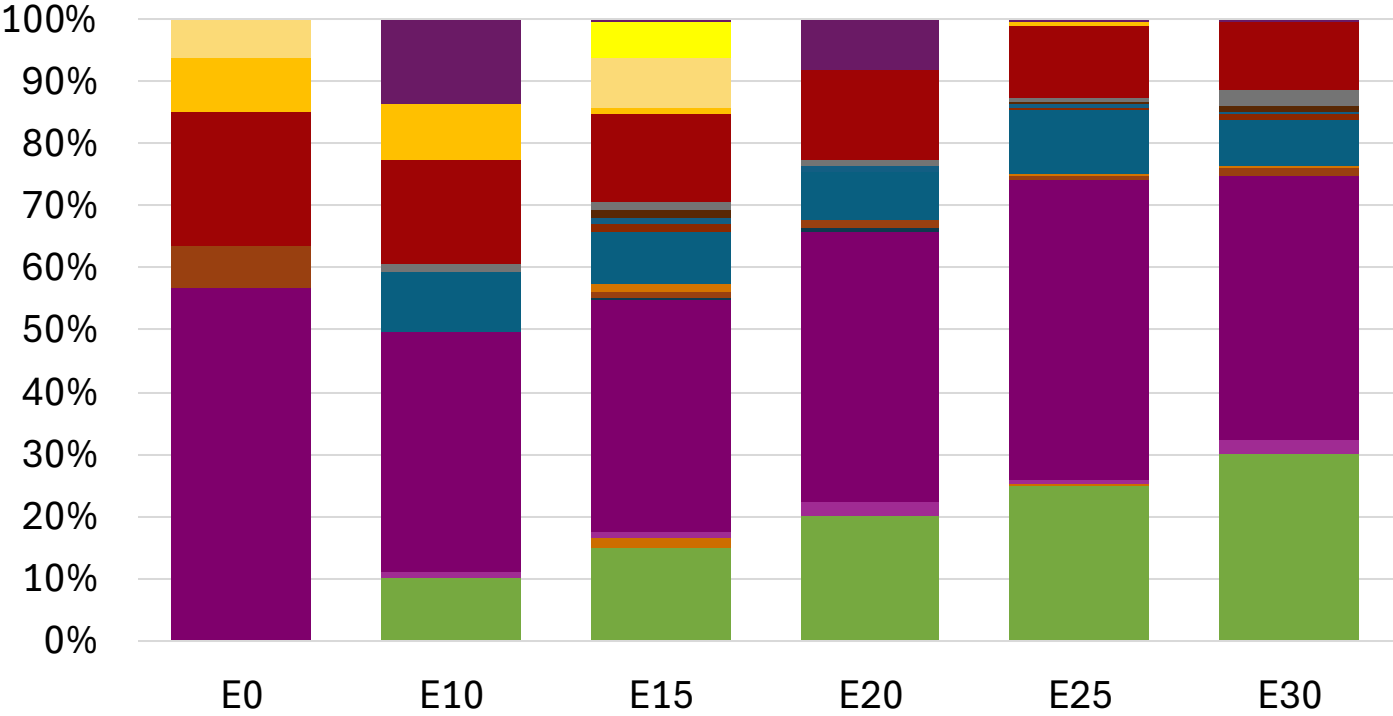


Average CIF 2024 Prices

Octane (RON)	98.0	98.0	98.0	98.0	98.0	98.0
Price (US\$/gal)	\$ 2.324	\$ 2.207	\$ 2.138	\$ 2.066	\$ 2.001	\$ 1.951

Source: Faro90

Slovakia – Gasoline 95 – Increased Octane

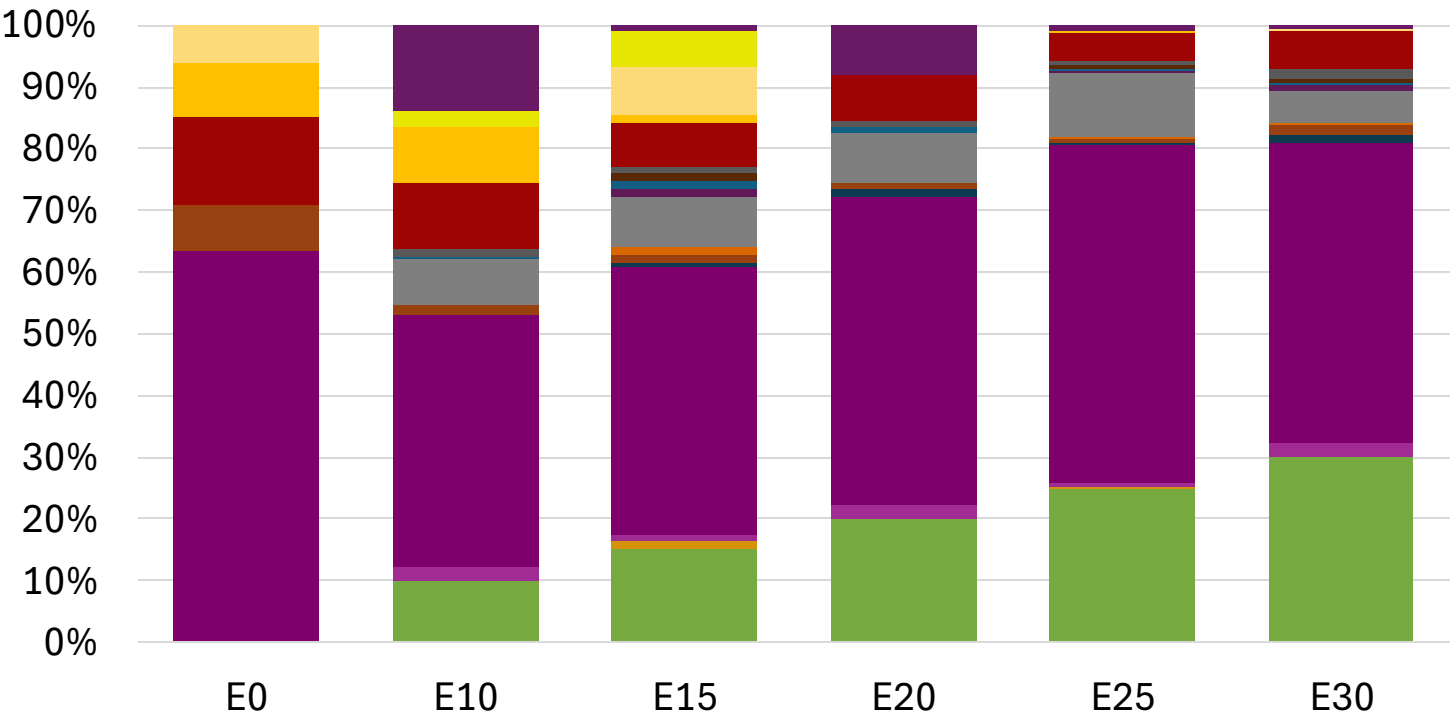


Average CIF 2024
Prices

Octane (RON)	95.0	96.2	98.3	98.7	100.5	101.8
Price (US\$/gal)	\$ 2.192	\$ 2.192	\$ 2.192	\$ 2.192	\$ 2.192	\$ 2.192

Source: Faro90

Slovakia – Gasoline 98 – Increased Octane

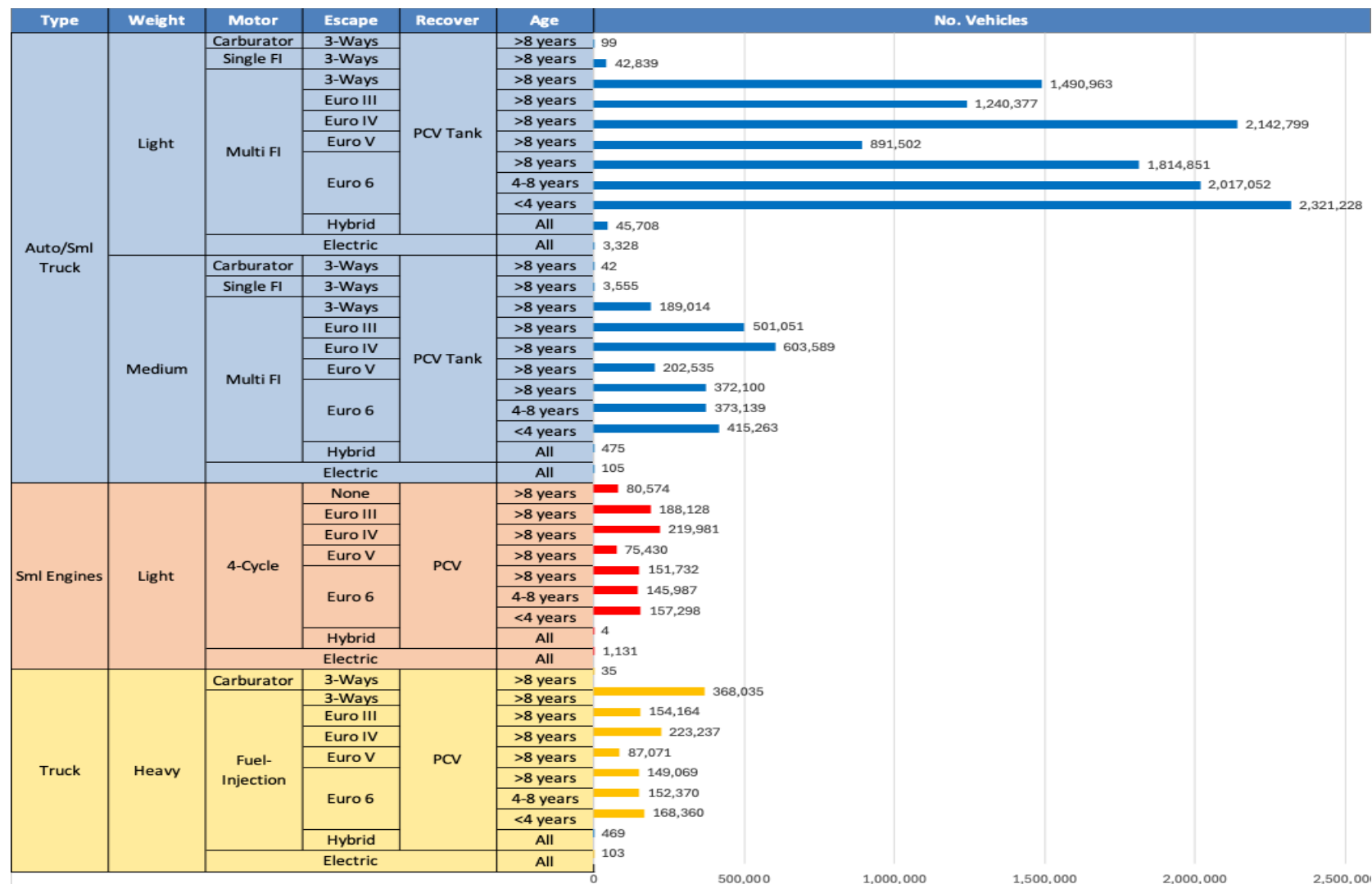


Average CIF 2024
Prices

Octane (RON)	98.0	99.0	100.9	101.2	102.8	104.1
Price (US\$/gal)	\$ 2.324	\$ 2.324	\$ 2.324	\$ 2.324	\$ 2.324	\$ 2.324

Source: Faro90

Gasoline Vehicle Fleet - Slovakia

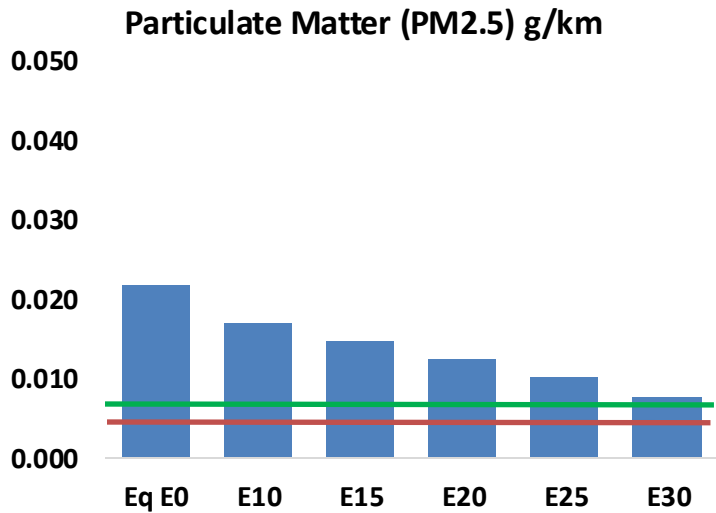
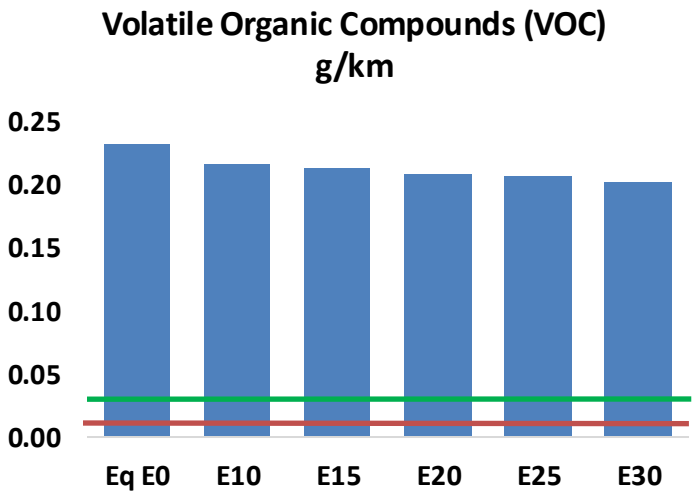
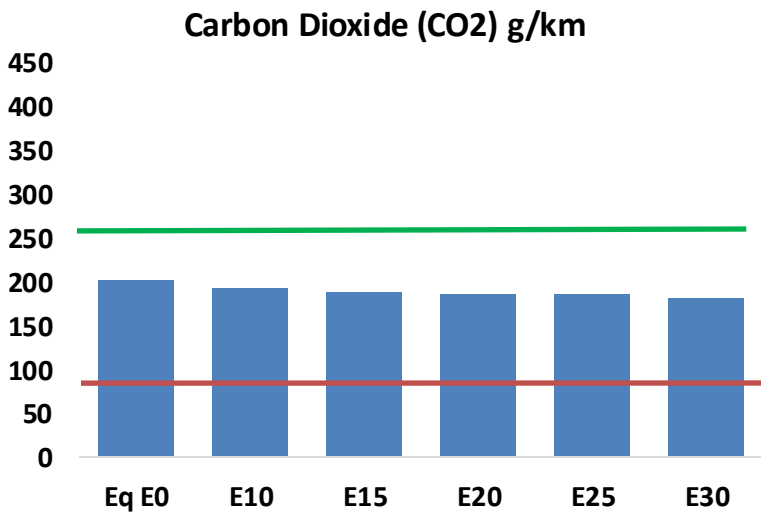
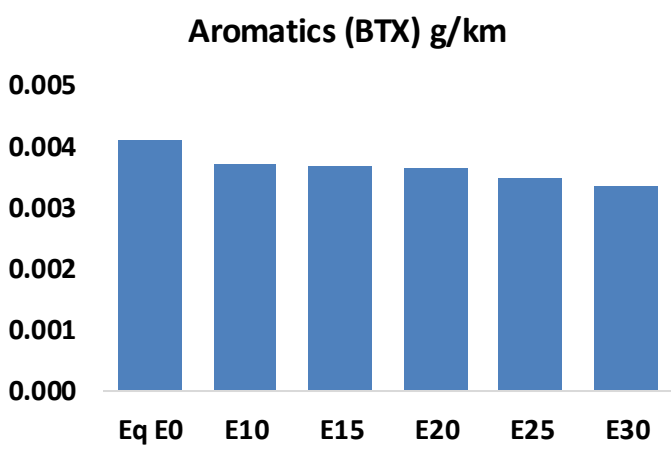
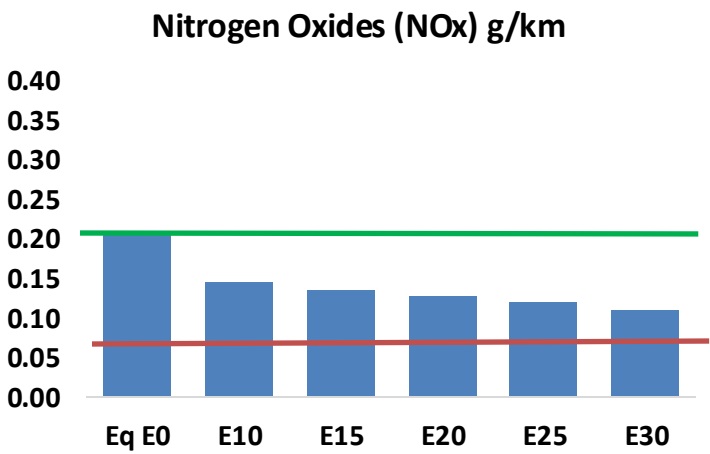
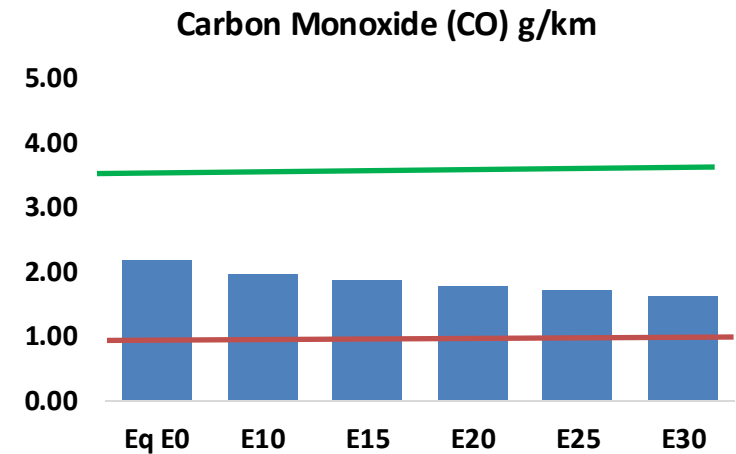
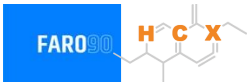


Vehicle Fleet: 5,792,410
Gasoline Vehicle Fleet: 1,843,374

Source: Eurostat, ACEA

Slovakia – Vehicular Emissions

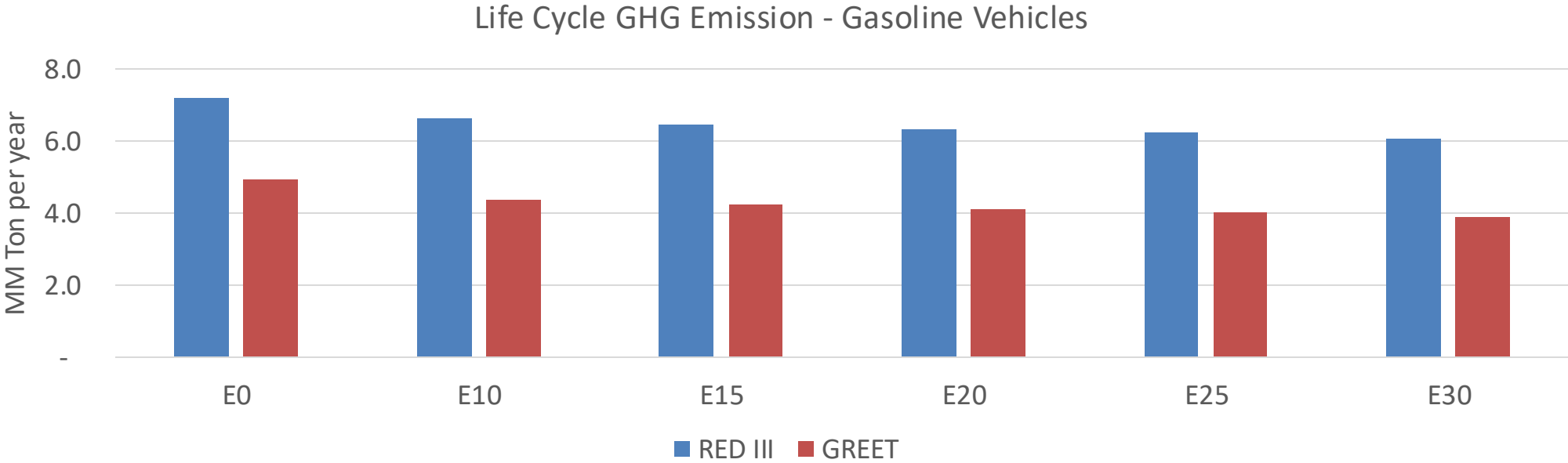
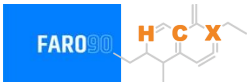
TIER III BIN 125 United States
Euro 6 Standard



Slovakia – Vehicular Emissions

Vehicular Fleet Emissions (kg/day)	Eq E0	E5	E10	E15	E20	E25	E30	Reduction (%)	
								E10	E20
CO	99,865	94,329	88,793	84,675	80,722	77,741	74,129	-11%	-19%
CO2	9,214,229	8,977,292	8,753,517	8,577,526	8,490,438	8,440,370	8,284,790	-5%	-8%
VOC	10,576	10,223	9,870	9,671	9,511	9,401	9,210	-7%	-10%
NOx	9,439	7,079	6,608	6,228	5,873	5,482	5,069	-30%	-38%
SOx	104	96	88	81	75	68	61	-15%	-28%
NH3	3,205	3,174	3,142	3,157	3,192	3,217	3,238	-2%	0%
N2O	103	102	102	103	105	106	107	-1%	2%
CH4	2,586	2,586	2,586	2,625	2,681	2,731	2,777	0%	4%
Lead	-	-	-	-	-	-	-	0%	0%
Butadiene	109	106	103	102	102	101	100	-5%	-7%
Acetaldehyde	233	267	392	597	813	929	1,098	68%	249%
Formaldehyde	744	724	844	991	1,038	1,148	1,250	13%	39%
Benzene	187	179	170	167	166	159	153	-9%	-11%
PM2.5	590	524	458	398	338	274	208	-22%	-43%
PM10	403	358	313	272	231	187	142	-22%	-43%

Slovakia – Gasoline Vehicle Fleet Life Cycle GHG Emissions



Country Emissions 2024 (MMT/año)	36.8
Target @ 2035 (MMT/year)	20.7
Delta	16.1

%Reduction vs E0	E0	E10	E15	E20	E25	E30
GREET 2024	0.0%	11.2%	14.2%	16.5%	18.6%	21.6%
RED II/III	0.0%	8.1%	10.3%	12.0%	13.5%	15.7%

% Target@2035	E0	E10	E15	E20	E25	E30
GREET 2024	0.0%	3.4%	4.4%	5.0%	5.7%	6.6%
RED II/III	0.0%	3.6%	4.6%	5.4%	6.1%	7.0%

Source: Greet, RED II/III, IPCC, climateactiontracker.org, F90

Ethanol Gasoline Blending in Slovenia

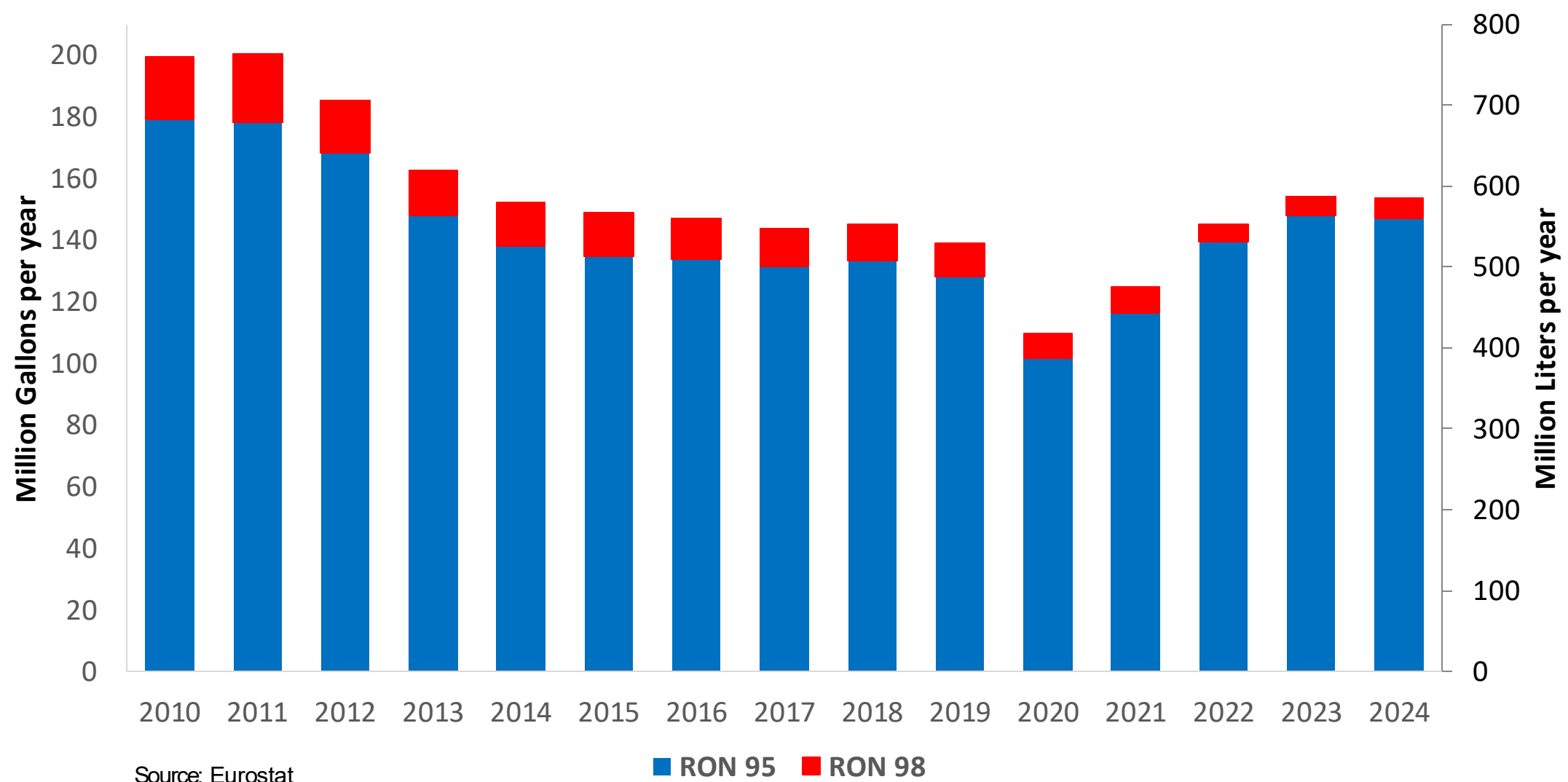


In 2024, gasoline consumption in Slovenia was 585 million liters (155 million gallons) and growth rate was 2.0%. Gasoline production and net imports were and 574 million liters (and 152 million gallons) correspondingly. Slovenia complies with the European gasoline standards and offers two main grades: RON 95 and RON 98) with an average octane of 95.1 RON.

In 2024, ethanol consumption in Slovenia was 3 million liters (1 million gallons) representing .5% of gasoline demand. Ethanol production and Net Imports were and 3 million liters (1 and 1 million gallons) correspondingly. Ethanol source is Cereals 100%.

Source: Eurostat, European Commission

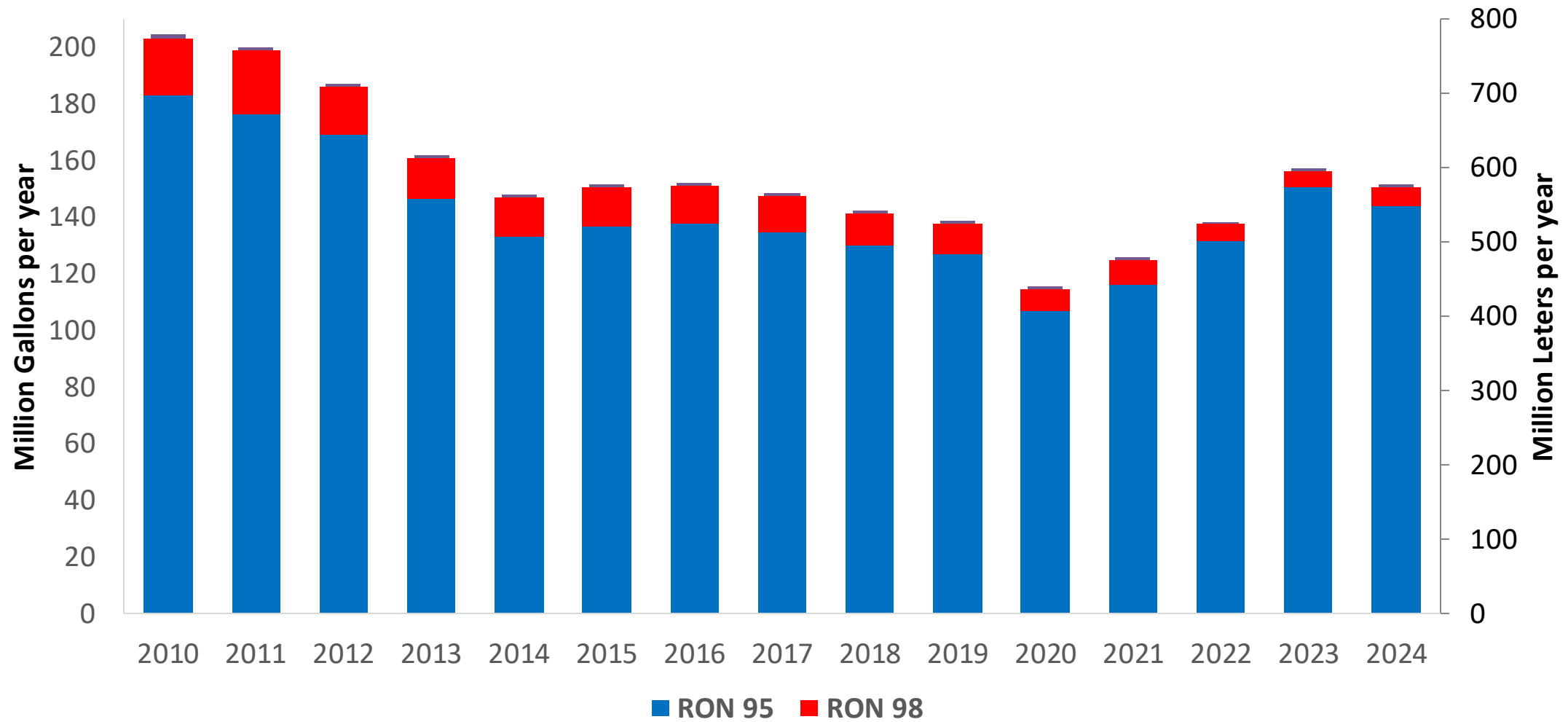
Gasoline Demand by Grade in Slovenia



Source: Eurostat

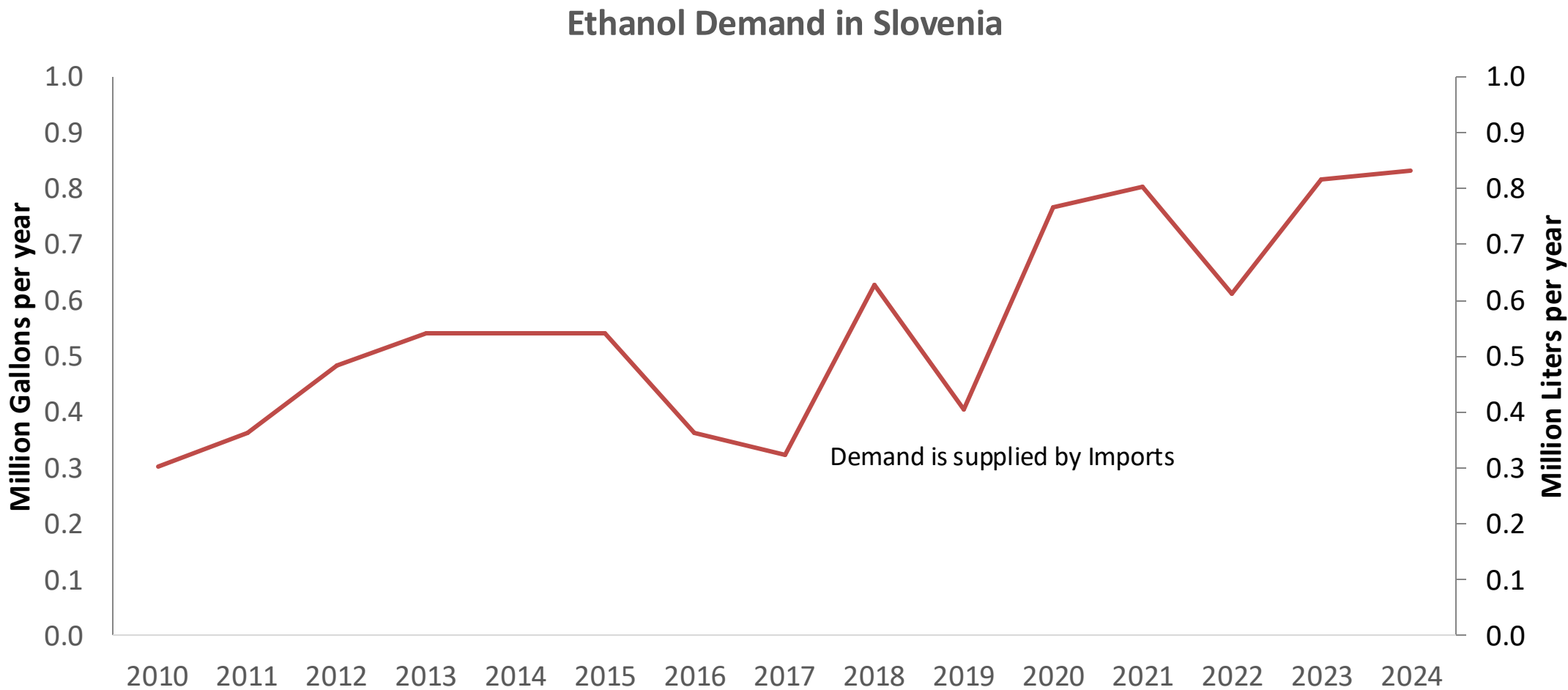
Gasoline Imports in Slovenia

Gasoline Imports by Grade in Slovenia



Source: Eurostat

Ethanol Balance in Slovenia



Source: Eurostat

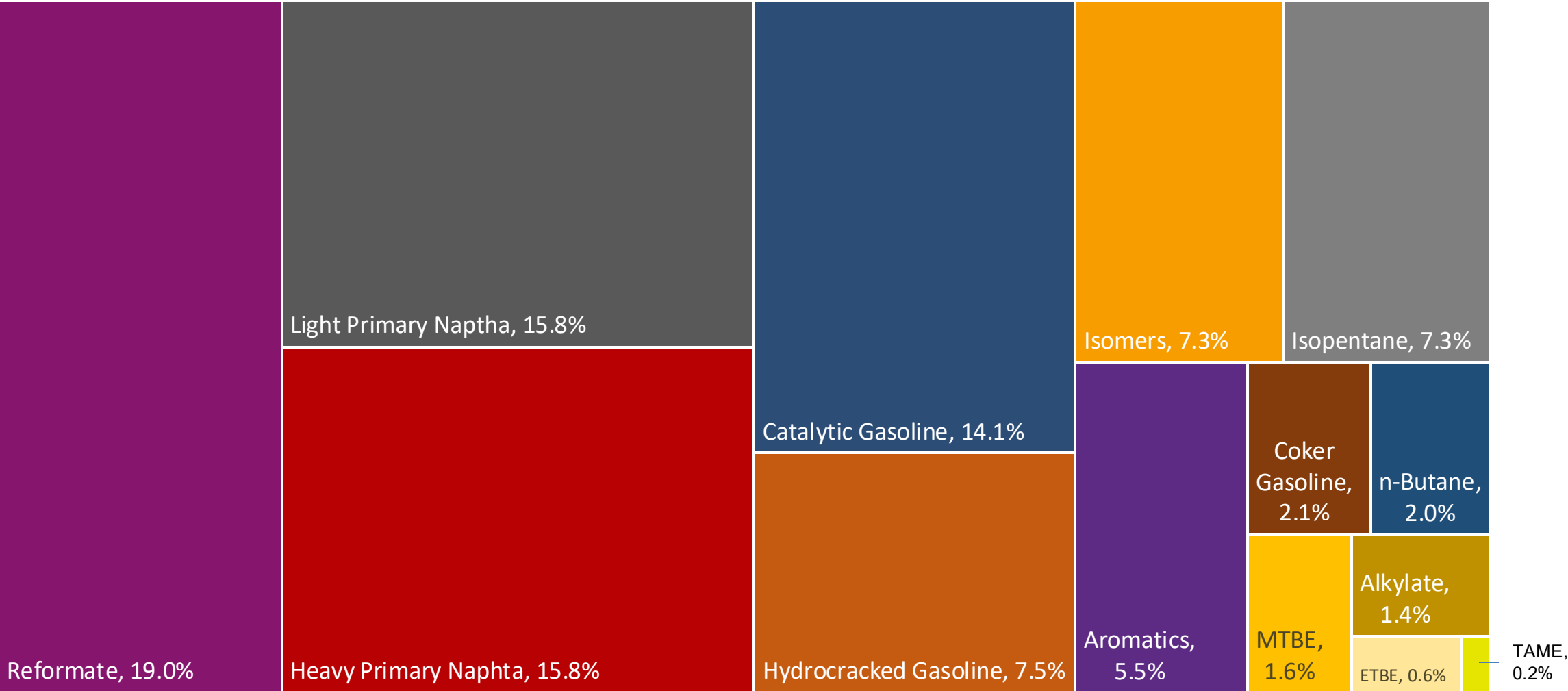
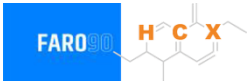
Gasoline Quality in Slovenia

Name	EN 228:2025 (Euro 6); Country Level Fuel Ordinances							
Implementation Date	2025							
Applicability	All EU Contries + UK							
Grade	RON 95 E5		RON 95 E10		RON 98 E5		RON 98 E10	
	Min	Max	Min	Max	Min	Max	Min	Max
RON	95.0		95.0		98.0		98.0	
MON	> 85	> 88	> 85	> 88	> 85	> 88	> 85	> 88
AKI								
Benzene (%vol)	< 1							
Aromatics (% vol)	< 35							
Olefins (%vol)	< 18							
Lead (g/L)	< 0.005							
Manganese (mg/L)	< 2,0							
Sulfur (ppm)	10.0							
Oxygen (%w)*	< 2,7	< 3,7	< 2,7	< 3,7	< 2,7	< 3,7	< 2,7	< 3,7
Ethanol (%vol)	< 5	< 10	< 5	< 10	< 5	< 10	< 5	< 10
PVR 37.8°C (psi)	<> 45 - 100 kPa in 6 Volatility Classes							
Overall Biofuels (%cal)	11.2							
Advanced Biofuels (%cal)	0.2							
Bioethanol in Gasoline Sales (%cal)	-							
GHG Emission Reduction (%)	-							

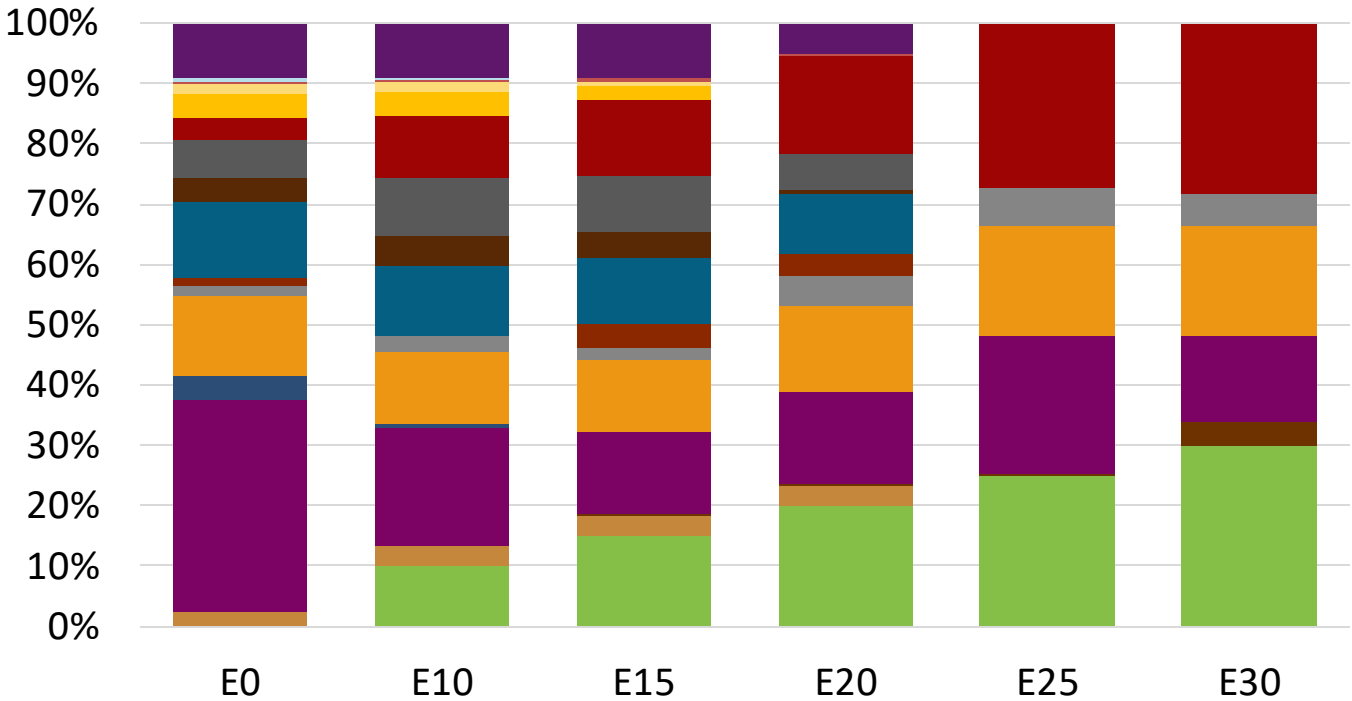
* Methanol 3%, Iso-propyl alcohol 12%, Ter-butyl alcohol 15%, Iso-butyl alcohol 15%, Ethers +5 Carbons 22%, Other Oxygenates 15%

Source: European Commision

Slovenia Available Blending Components



Slovenia – Gasoline 95 – Constant Octane

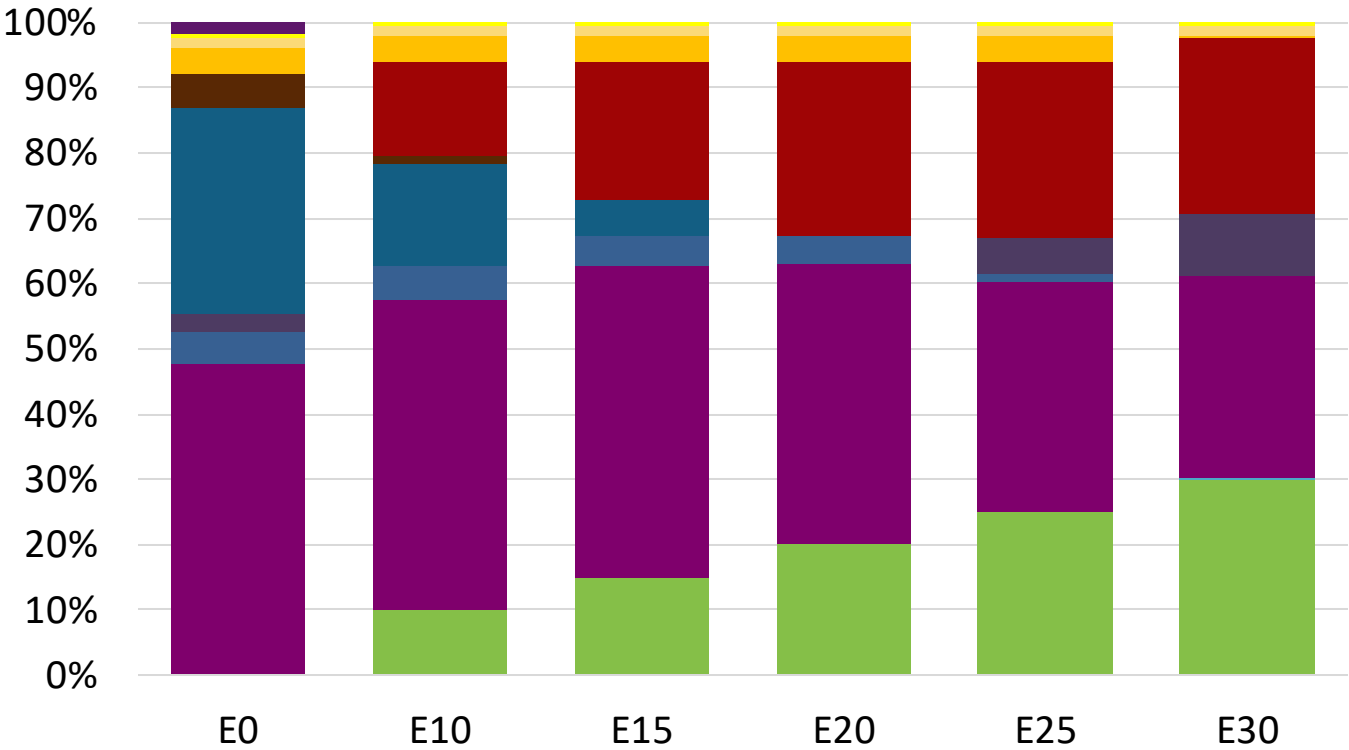


Average CIF 2024
Prices

Octane (RON)	95.0	95.0	95.0	95.0	95.1	95.6
Price (US\$/gal)	\$ 2.292	\$ 2.164	\$ 2.103	\$ 2.022	\$ 1.900	\$ 1.834

Source: Faro90

Slovenia - Gasoline 98 - Constant Octane

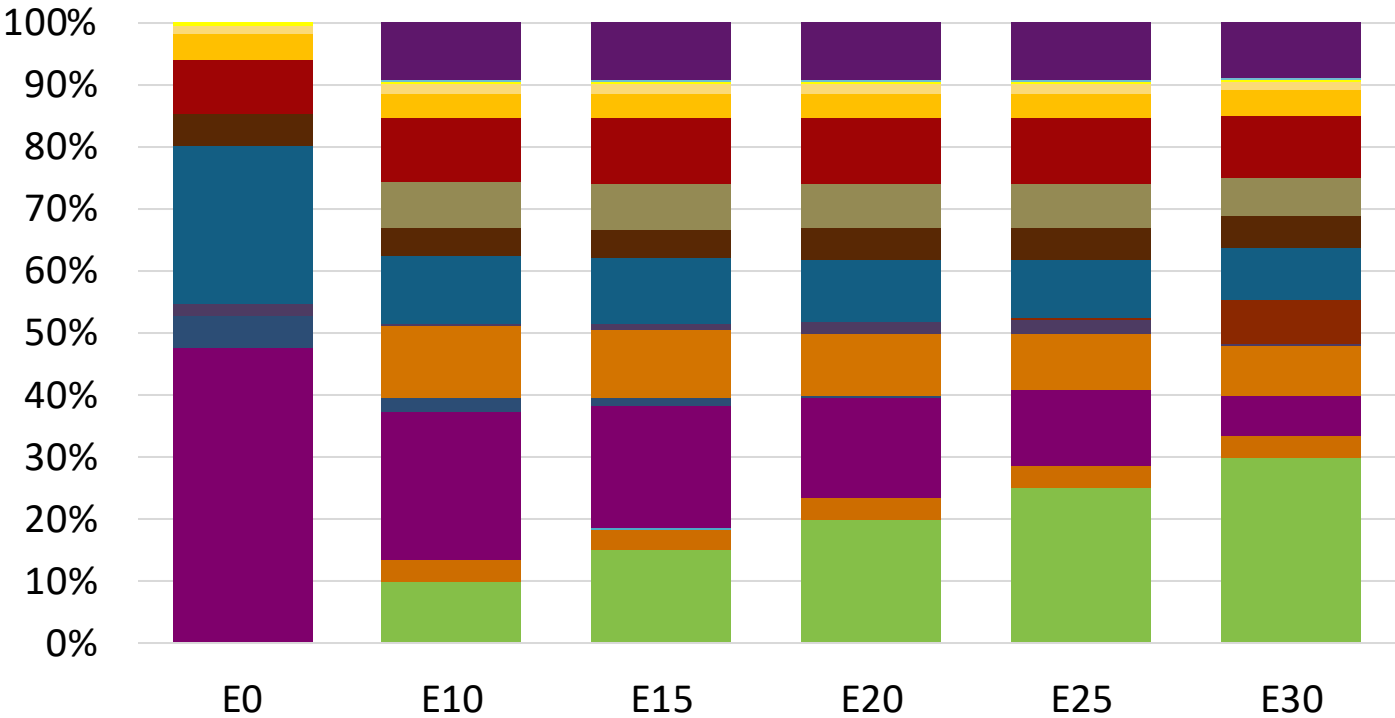
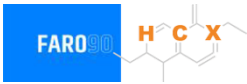


Average CIF 2024
Prices

Octane (RON)	98.0	98.0	98.0	98.0	98.0	98.0
Price (US\$/gal)	\$ 2.367	\$ 2.238	\$ 2.166	\$ 2.089	\$ 2.012	\$ 1.945

Source: Faro90

Slovenia – Gasoline 95 – Increased Octane

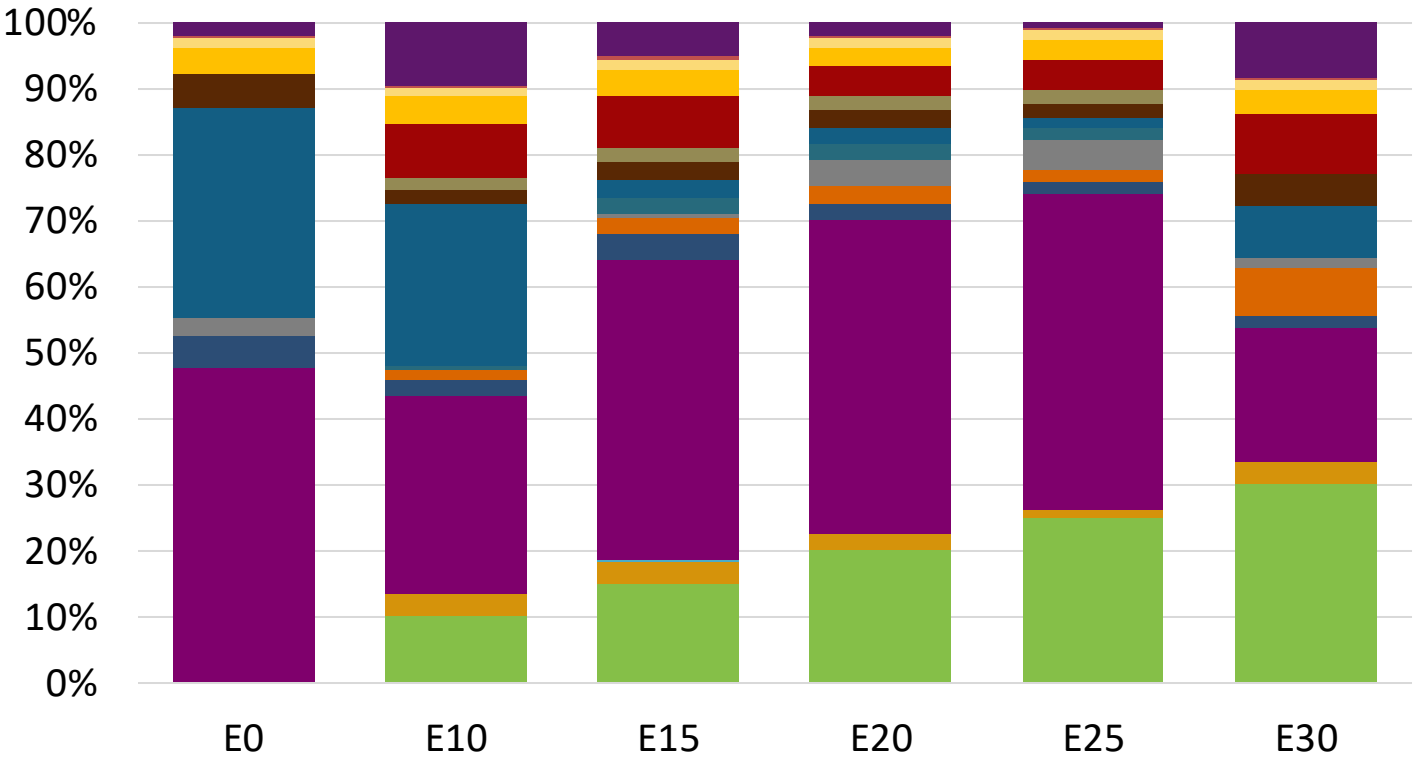
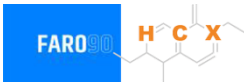


Average CIF 2024
Prices

Octane (RON)	95.0	96.5	97.9	99.4	100.8	102.0
Price (US\$/gal)	\$ 2.229	\$ 2.229	\$ 2.229	\$ 2.229	\$ 2.229	\$ 2.229

Source: Faro90

Slovenia – Gasoline 98 – Increased Octane

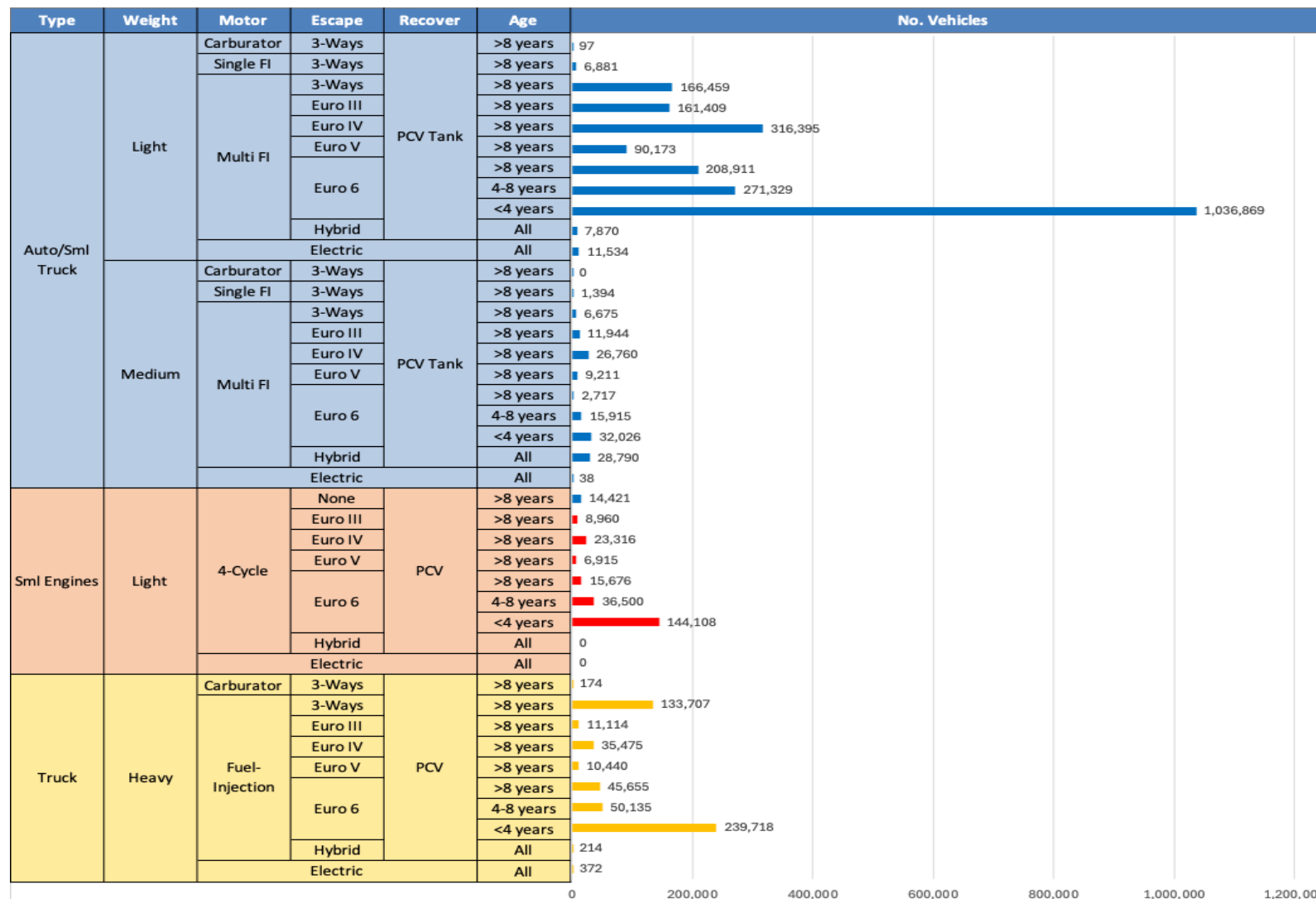


Average CIF 2024
Prices

Octane (RON)	98.0	99.2	101.3	102.6	104.0	104.7
Price (US\$/gal)	\$ 2.367	\$ 2.367	\$ 2.367	\$ 2.367	\$ 2.367	\$ 2.367

Source: Faro90

Gasoline Vehicle Fleet - Slovenia

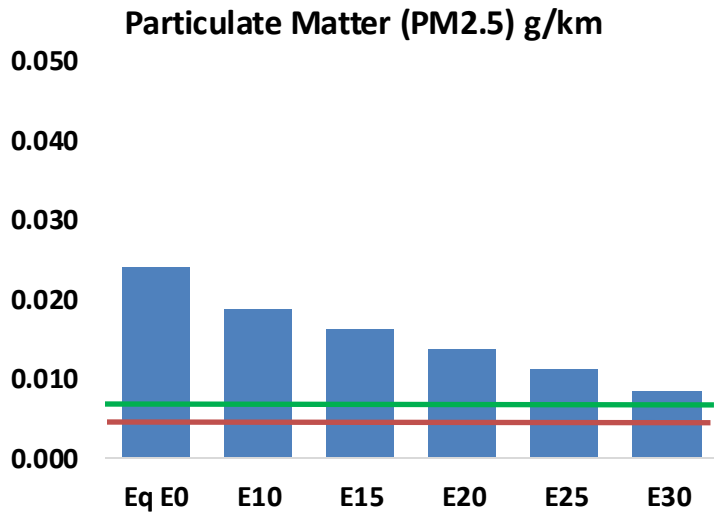
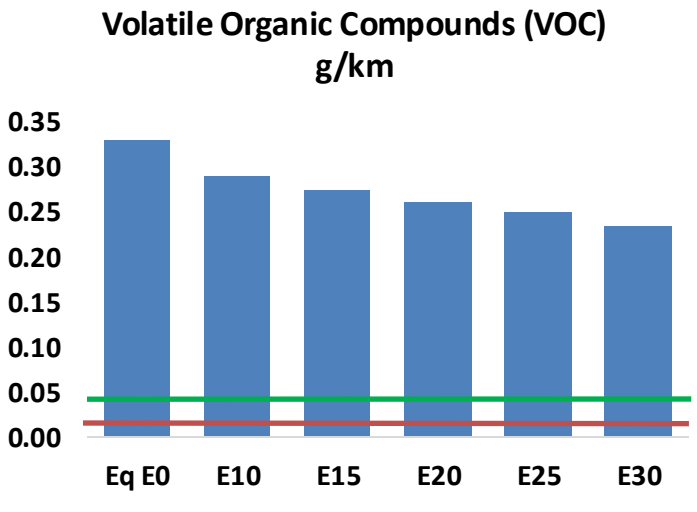
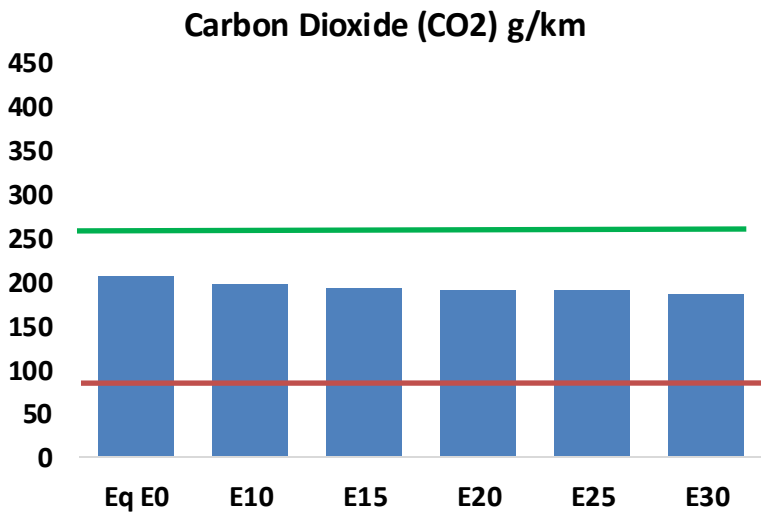
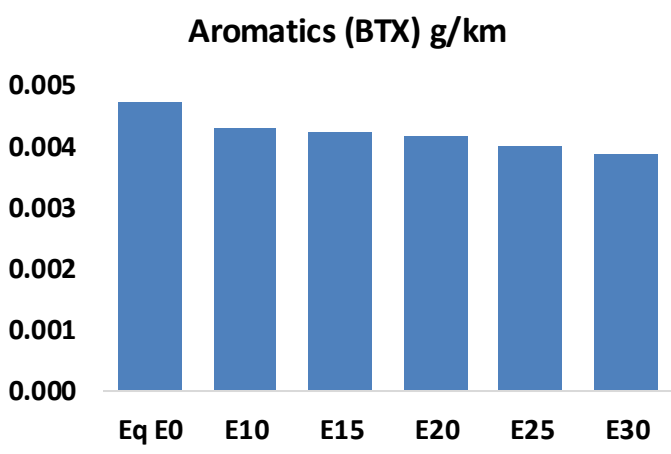
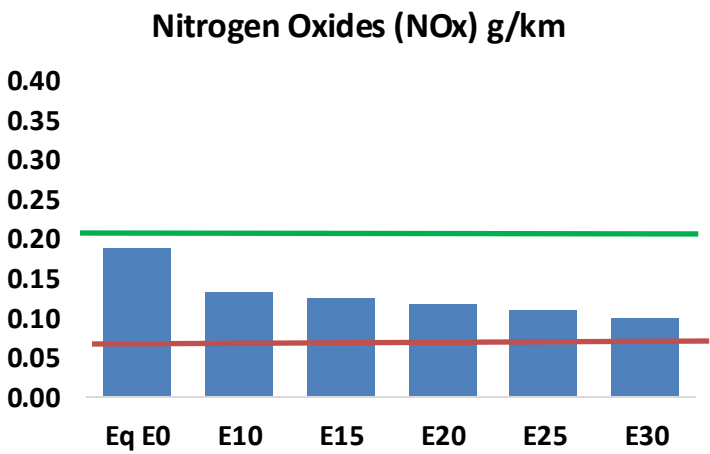
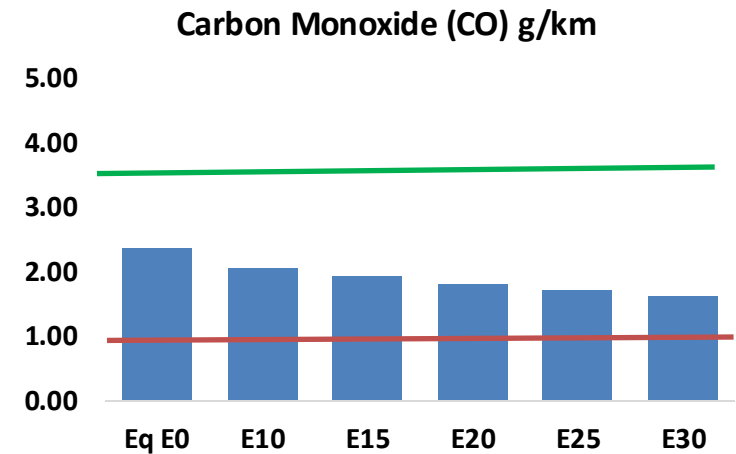
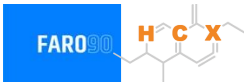


Vehicle Fleet: 3,190,297
Gasoline Vehicle Fleet: 1,163,863

Source: Eurostat, ACEA

Slovenia – Vehicular Emissions

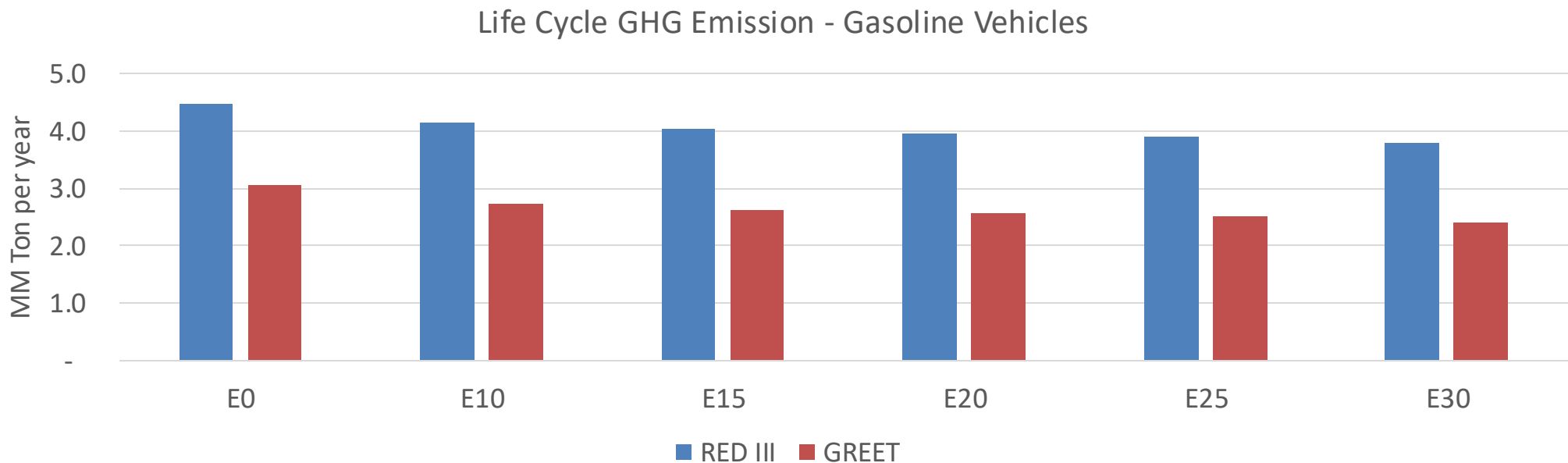
TIER III BIN 125 United States
Euro 6 Standard



Slovenia – Vehicular Emissions

Vehicular Fleet Emissions (kg/day)	Eq E0	E5	E10	E15	E20	E25	E30	Reduction (%)	
								E10	E20
CO	66,460	62,046	57,631	54,148	50,707	48,079	45,138	-13%	-24%
CO2	5,814,866	5,665,340	5,524,122	5,413,058	5,358,100	5,329,401	5,231,165	-5%	-8%
VOC	9,254	8,696	8,137	7,709	7,293	6,976	6,539	-12%	-21%
NOx	5,327	3,996	3,729	3,515	3,315	3,094	2,861	-30%	-38%
SOx	65	60	55	51	47	43	38	-15%	-28%
NH3	2,043	2,023	2,003	2,012	2,035	2,051	2,064	-2%	0%
N2O	61	61	60	61	62	63	63	-1%	2%
CH4	2,030	2,030	2,030	2,061	2,105	2,144	2,180	0%	4%
Lead	-	-	-	-	-	-	-	0%	0%
Butadiene	63	61	59	57	57	56	55	-6%	-9%
Acetaldehyde	160	183	269	410	558	638	754	68%	249%
Formaldehyde	545	529	617	725	759	840	914	13%	39%
Benzene	132	127	120	119	118	112	109	-9%	-11%
PM2.5	371	329	288	250	212	172	130	-22%	-43%
PM10	303	269	235	204	173	141	107	-22%	-43%

Slovenia – Gasoline Vehicle Fleet Life Cycle GHG Emissions



Country Emissions 2024 (MMT/año)	16.5
Target @ 2035 (MMT/year)	9.3
Delta	7.2

%Reduction vs E0	E0	E10	E15	E20	E25	E30
GREET 2024	0.0%	10.8%	13.7%	16.0%	18.0%	20.9%
RED II/III	0.0%	7.7%	9.9%	11.6%	13.1%	15.2%

% Target@2035	E0	E10	E15	E20	E25	E30
GREET 2024	0.0%	4.6%	5.8%	6.7%	7.6%	8.9%
RED II/III	0.0%	4.8%	6.2%	7.2%	8.1%	9.4%

Source: Greet, RED II/III, IPCC, climateactiontracker.org, F90

Ethanol Gasoline Blending in Spain

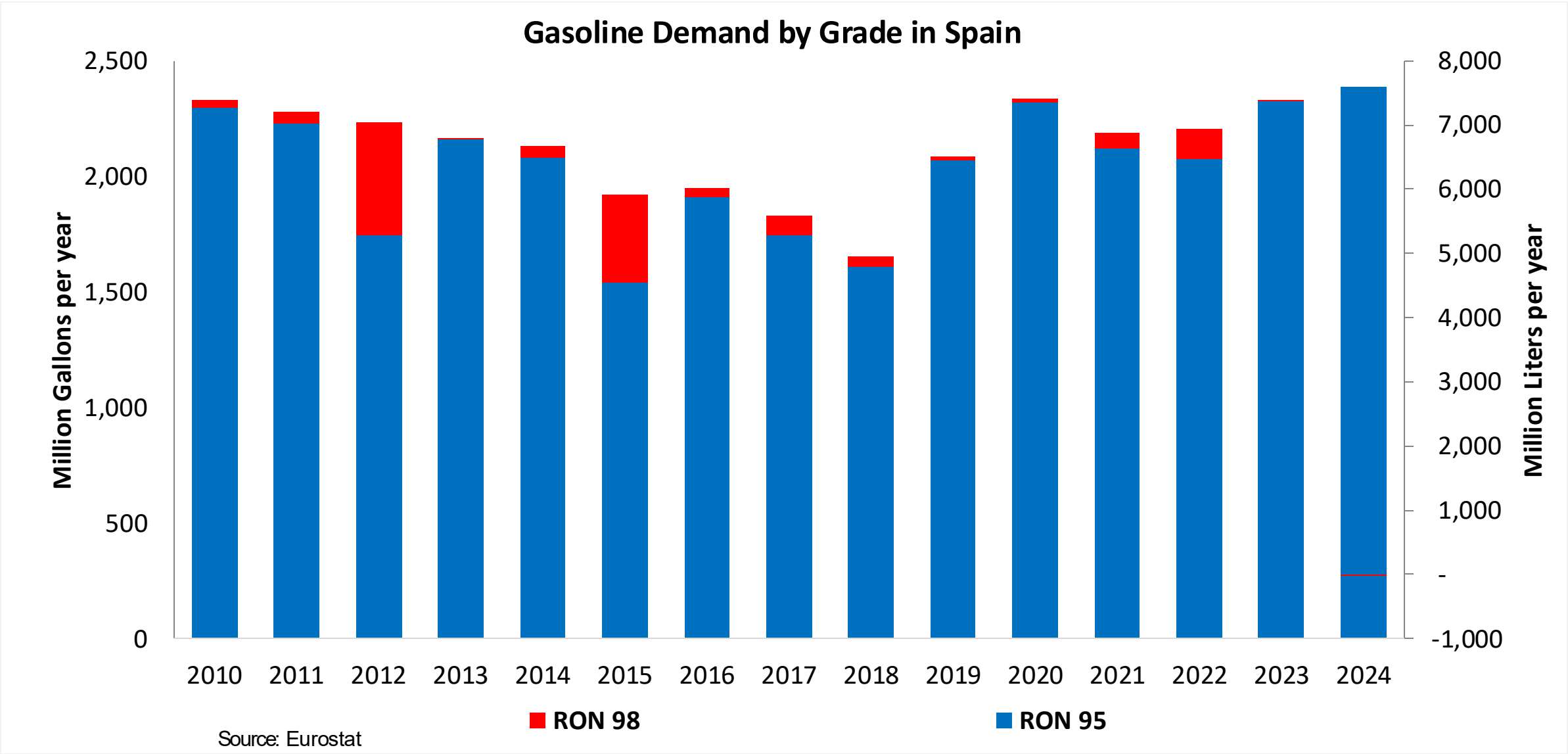


In 2024, gasoline consumption in Spain was 7,577 million liters (2,002 million gallons) and growth rate was 3.1%. Gasoline production and net imports were 11,890 and -3,071 million liters (3,141 and -811 million gallons) correspondingly. Spain complies with the European gasoline standards and offers two main grades: RON 95 and RON 98) with an average octane of 95.0 RON.

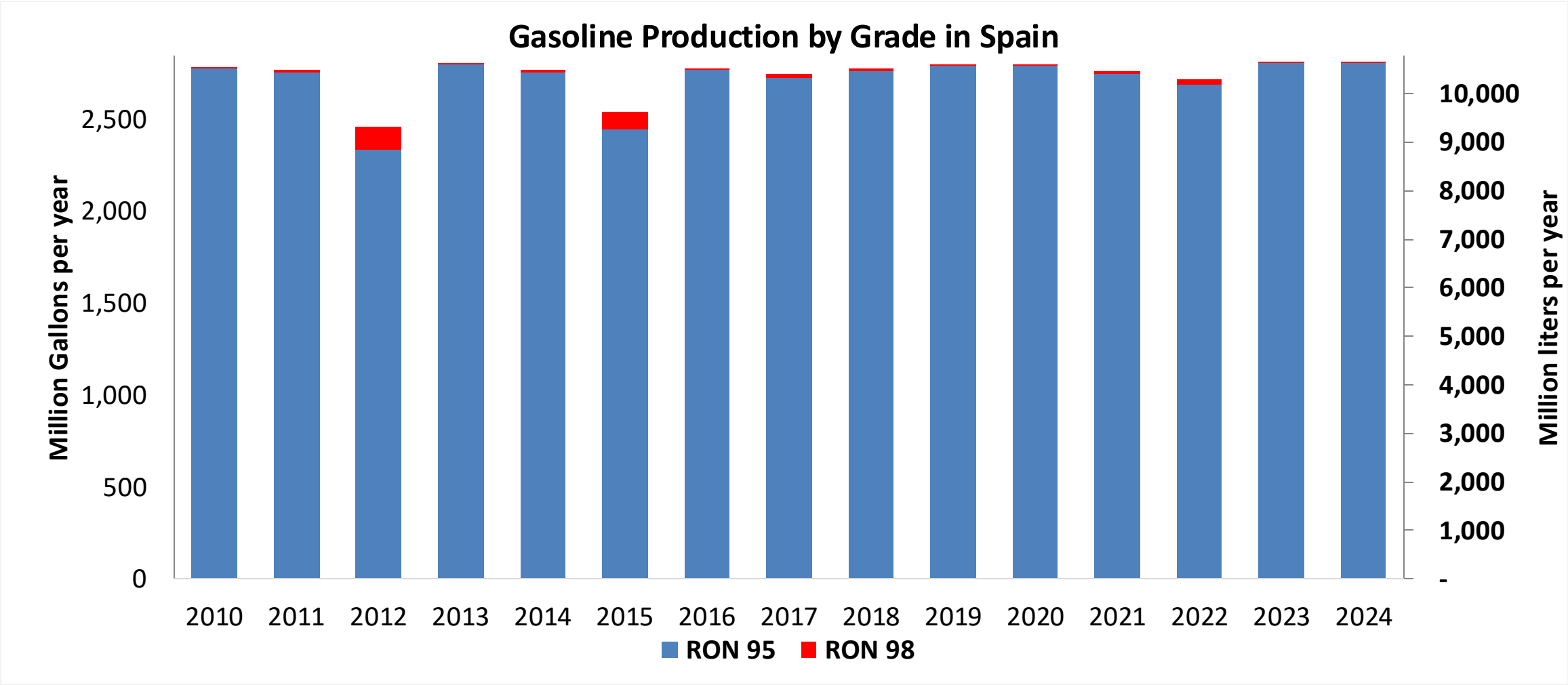
In 2024, ethanol consumption in Spain was 253 million liters (67 million gallons) representing 3.3% of gasoline demand. Ethanol production and Net Imports were 545 and -292 million liters (67 and -77 million gallons) correspondingly. Ethanol sources are Corn 60%, Cereals 25% and Sugar Beet/Wine 40%.

Source: Eurostat, European Commission

Gasoline Demand in Spain



Gasoline Production in Spain

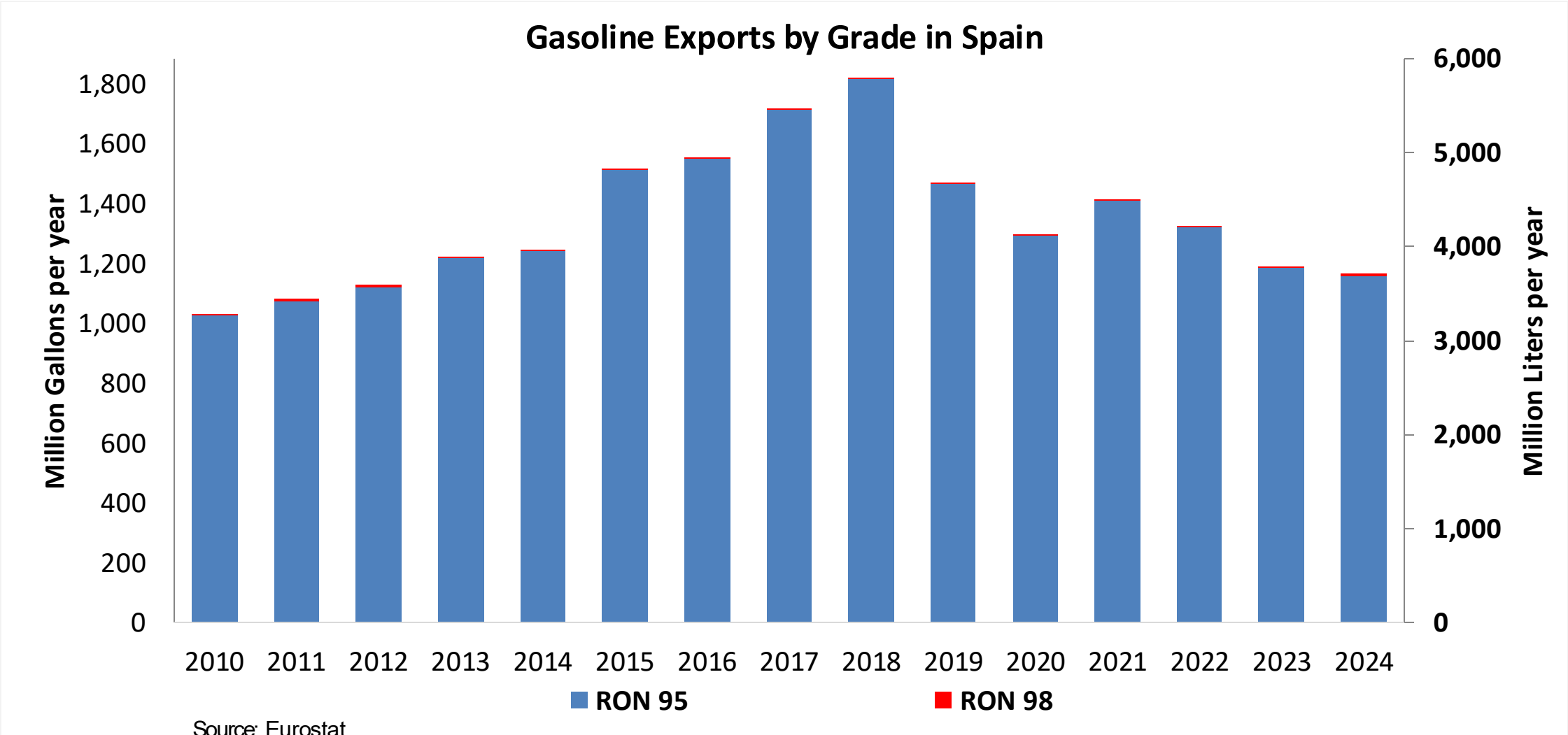


Source: Eurostat

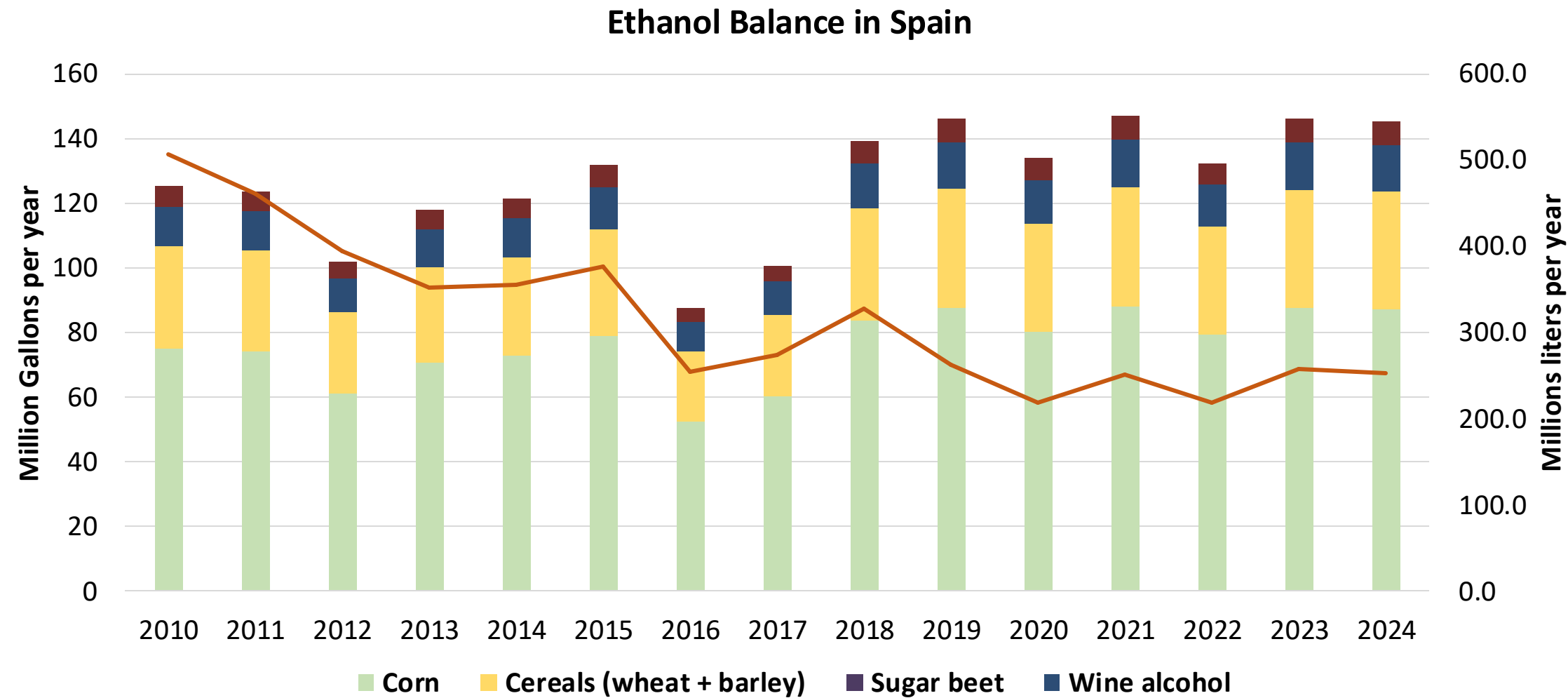
The FARO90 logo is shown in a blue box, followed by a chemical structure diagram of a substituted cyclohexadiene derivative. The structure features a six-membered ring with two double bonds. Substituents include a methyl group, a propyl group, and a group labeled 'X'. The letters 'H', 'C', and 'X' are highlighted in orange in the original image.



Gasoline Exports in Spain

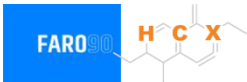


Ethanol Balance in Spain



Source: Eurostat

Gasoline Quality in Spain

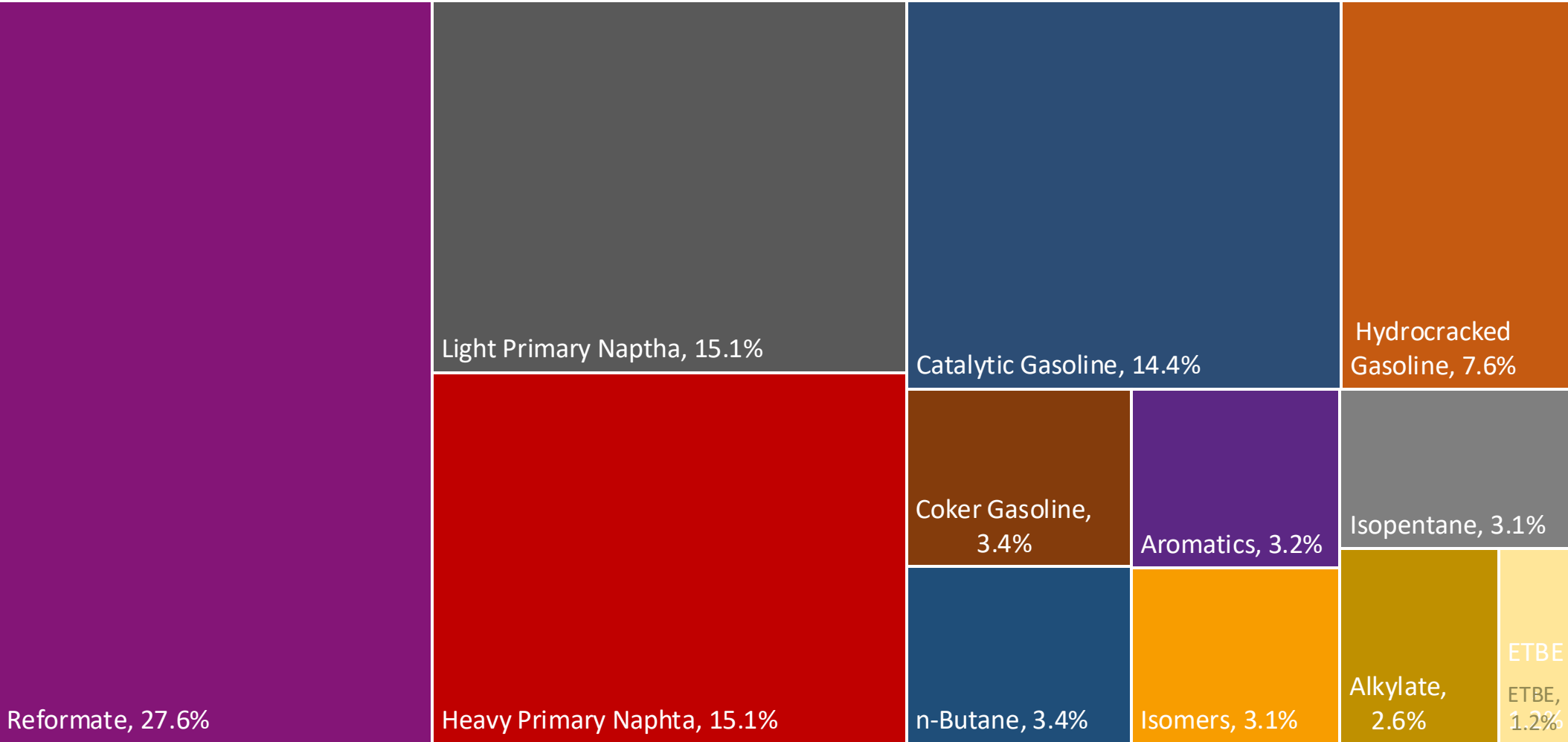


Name	EN 228:2025 (Euro 6); Country Level Fuel Ordinances							
Implementation Date	2025							
Applicability	All EU Contries + UK							
Grade	RON 95 E5		RON 95 E10		RON 98 E5		RON 98 E10	
	Min	Max	Min	Max	Min	Max	Min	Max
RON	95.0		95.0		98.0		98.0	
MON	> 85	> 88	> 85	> 88	> 85	> 88	> 85	> 88
AKI								
Benzene (%vol)	< 1							
Aromatics (% vol)	< 35							
Olefins (%vol)	< 18							
Lead (g/L)	< 0.005							
Manganese (mg/L)	< 2,0							
Sulfur (ppm)	10.0							
Oxygen (%w)*	< 2,7	< 3,7	< 2,7	< 3,7	< 2,7	< 3,7	< 2,7	< 3,7
Ethanol (%vol)	< 5	< 10	< 5	< 10	< 5	< 10	< 5	< 10
PVR 37.8°C (psi)	<> 45 - 100 kPa in 6 Volatility Classes							
Overall Biofuels (%cal)	11.5							
Advanced Biofuels (%cal)	1.0							
Bioethanol in Gasoline Sales (%cal)	-							
GHG Emission Reduction (%)	-							

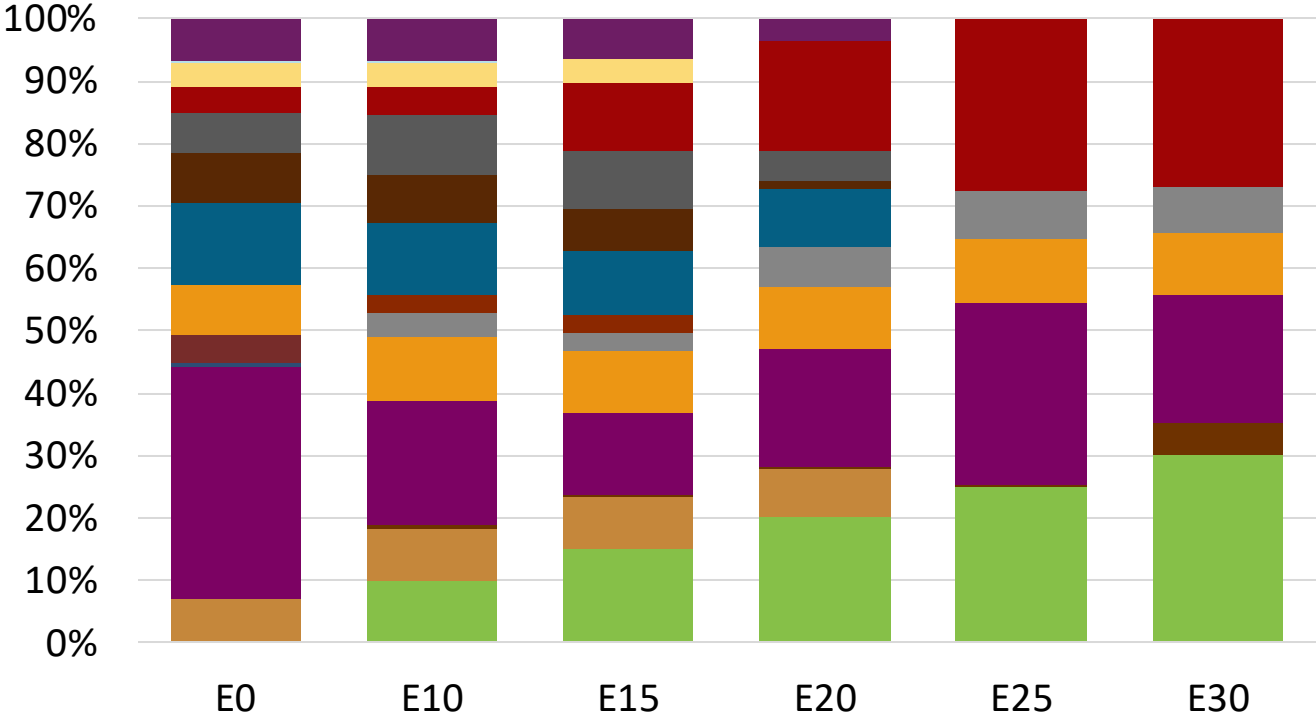
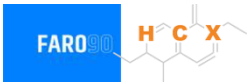
* Methanol 3%, Iso-propyl alcohol 12%, Ter-butyl alcohol 15%, Iso-butyl alcohol 15%, Ethers +5 Carbons 22%, Other Oxygenates 15%

Source: European Commision

Spain Available Blending Components



Spain – Gasoline 95 – Constant Octane

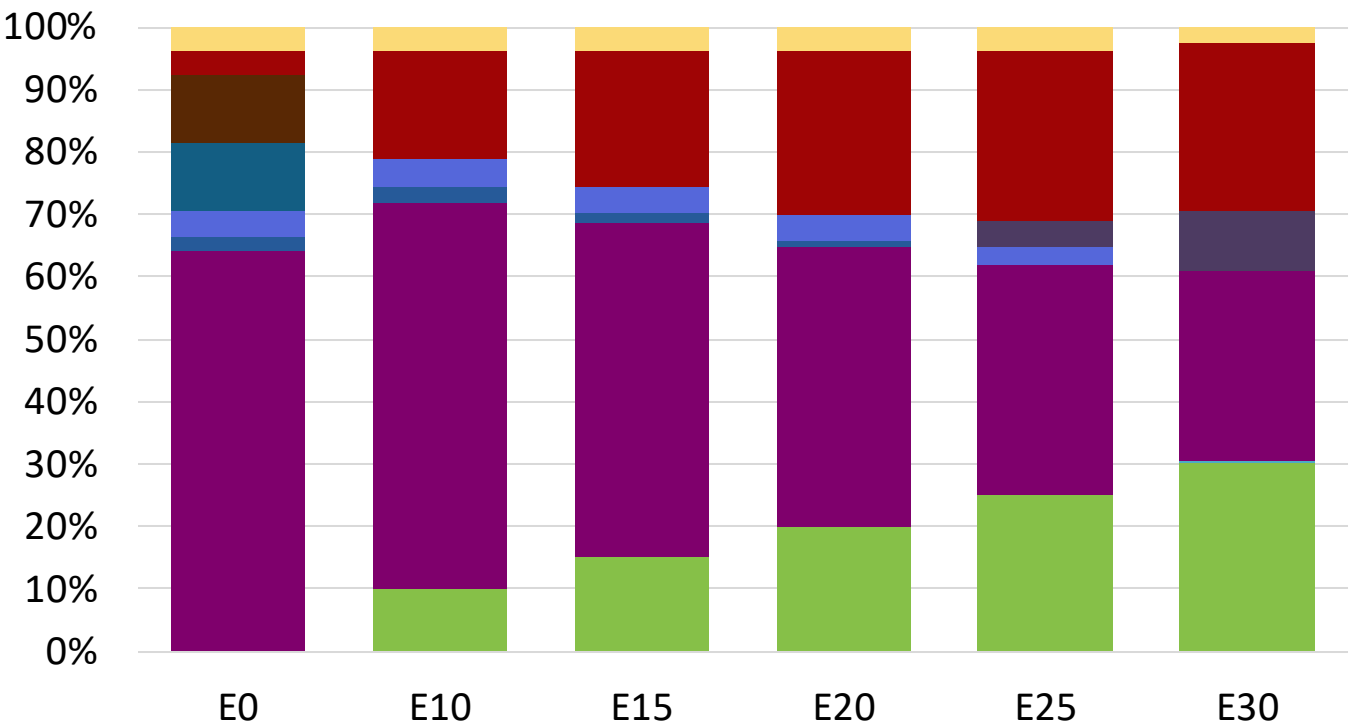
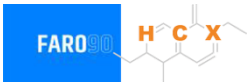


Average CIF 2024 Prices

Octane (RON)	95.0	95.0	95.0	95.0	95.3	95.8
Price (US\$/gal)	\$ 2.280	\$ 2.168	\$ 2.088	\$ 1.998	\$ 1.883	\$ 1.820

Source: Faro90

Spain - Gasoline 98 - Constant Octane

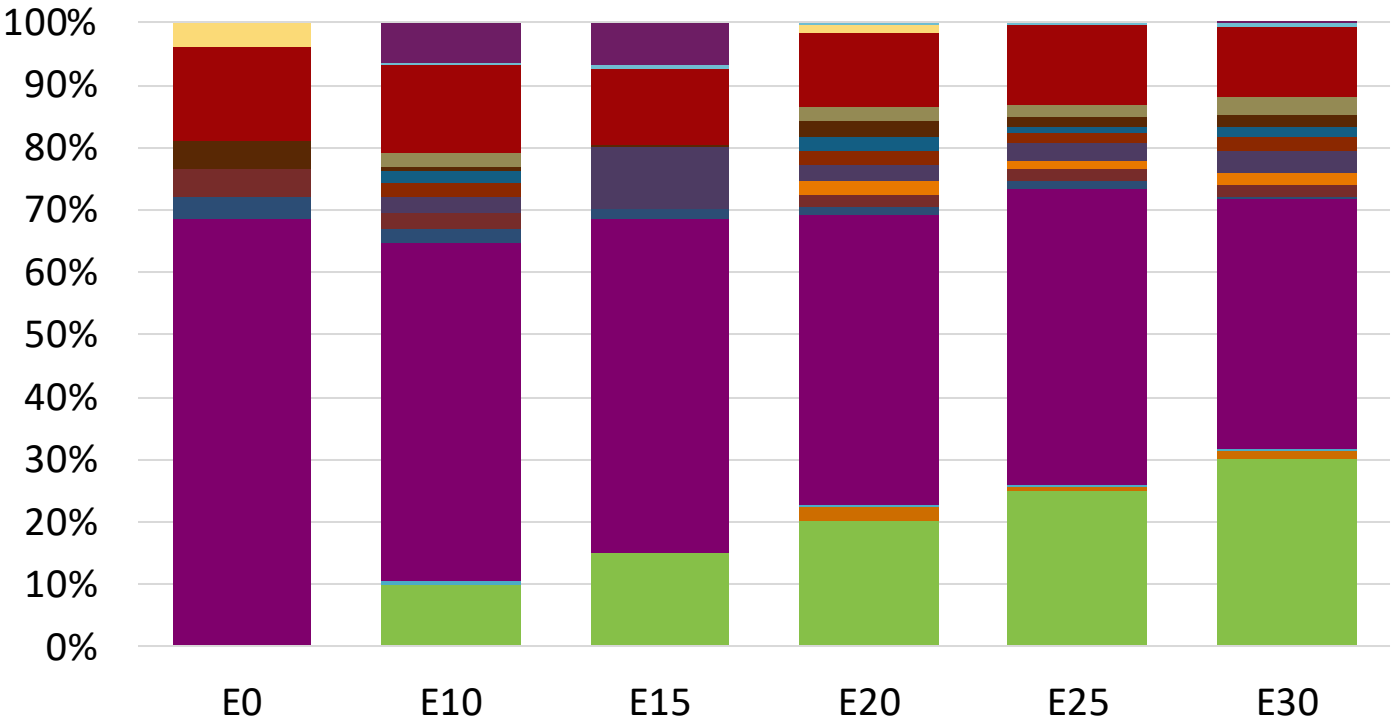


Average CIF 2024 Prices

Octane (RON)	98.0	98.0	98.0	98.0	98.0	98.0
Price (US\$/gal)	\$ 2.349	\$ 2.225	\$ 2.156	\$ 2.081	\$ 2.003	\$ 1.924

Source: Faro90

Spain – Gasoline 95 – Increased Octane

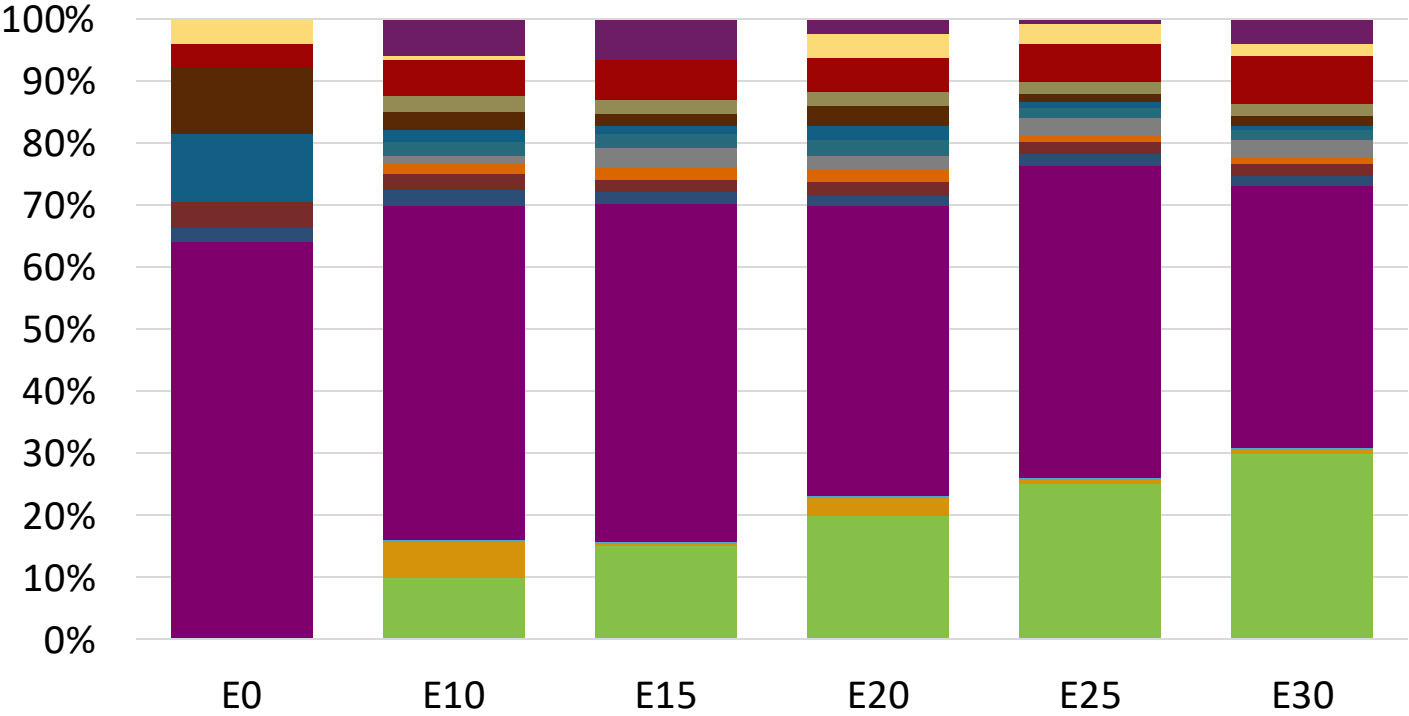


- Ethanol
- Alkylate
- Raffinate
- Reformate
- n-Butane
- Isomerate
- Isopentane
- Hydrocracked Gasoline
- Catalytic Gasoline
- Coker Naphtha
- Light Primary Naphtha
- Heavy Primary Naphtha
- MTBE
- ETBE
- TAME
- Aromatics

Average CIF 2024
Prices

Source: Faro90

Spain – Gasoline 98 – Increased Octane

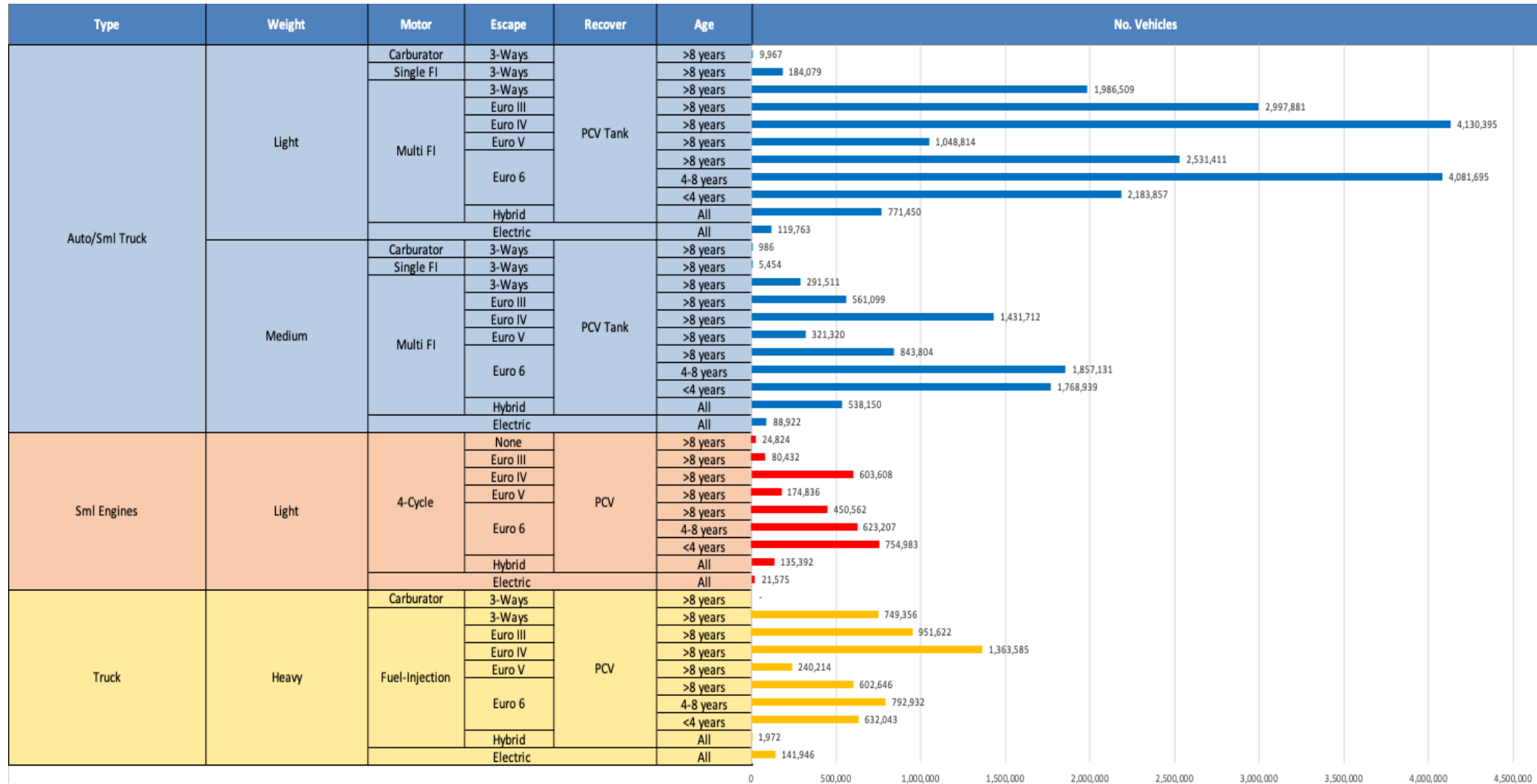


Average CIF 2024
Prices

Octane (RON)		98.0	99.3	100.9	102.5	104.0	105.0
Price (US\$/gal)	\$	2.349	\$ 2.349	\$ 2.349	\$ 2.349	\$ 2.349	\$ 2.349

Source: Faro90

Gasoline Vehicle Fleet - Spain

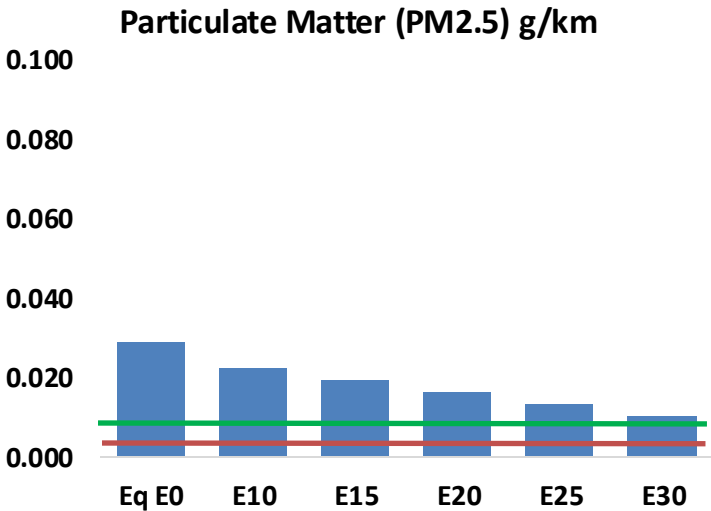
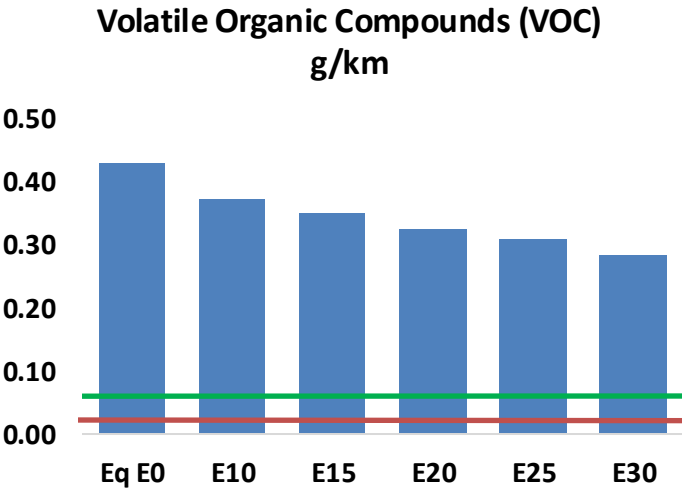
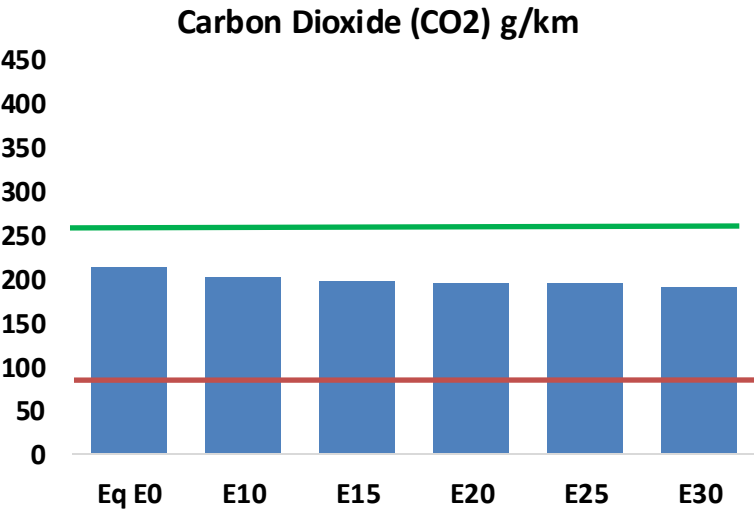
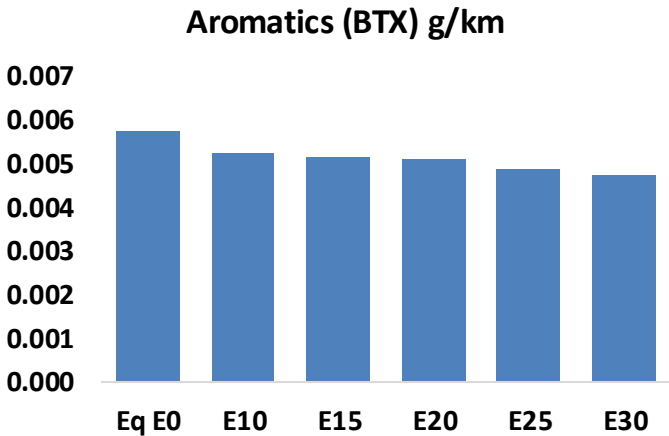
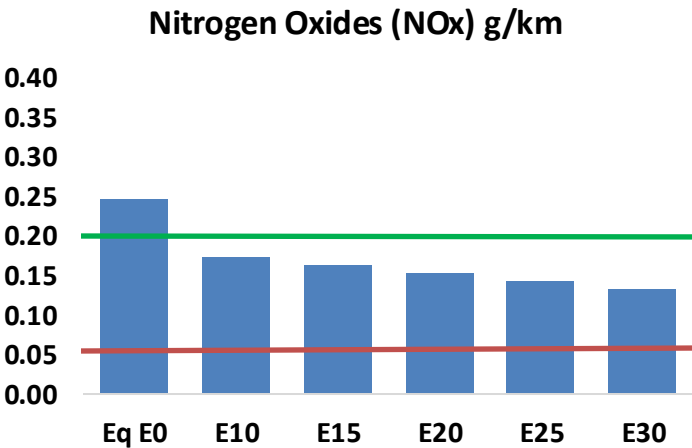
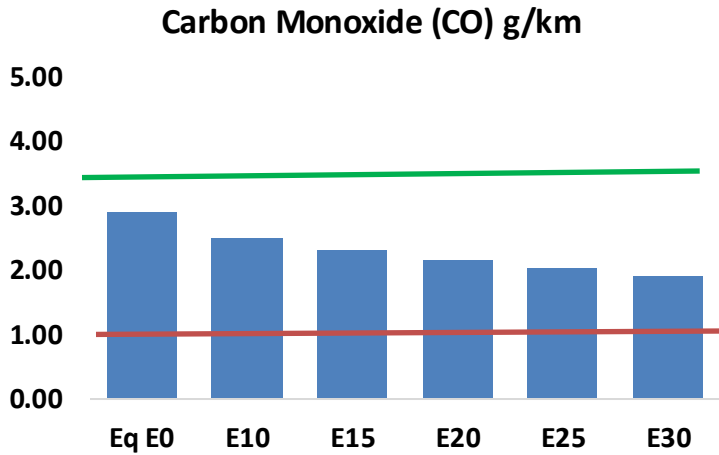
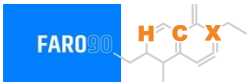


Vehicle Fleet: 36,100,582
Gasoline Vehicle Fleet: 11,736,365

Source: Eurostat, ACEA

Spain – Vehicular Emissions

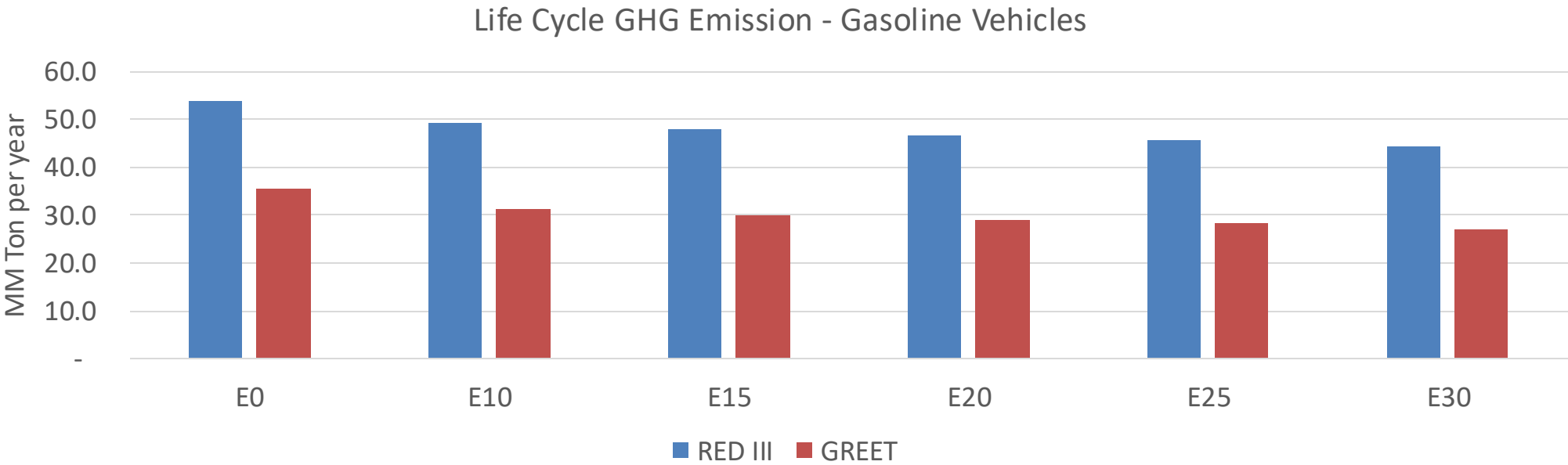
TIER III BIN 125 United States
Euro 6 Standard



Spain – Vehicular Emissions

Vehicular Fleet Emissions (kg/day)	Eq E0	E5	E10	E15	E20	E25	E30	Reduction (%)	
								E10	E20
CO	861,045	800,129	739,214	690,327	641,604	604,282	562,670	-14%	-25%
CO2	62,968,337	61,349,152	59,819,921	58,617,225	58,022,085	57,682,674	56,619,416	-5%	-8%
VOC	127,636	118,955	110,275	103,383	96,548	91,324	84,243	-14%	-24%
NOx	73,264	54,948	51,285	48,336	45,585	42,552	39,343	-30%	-38%
SOx	751	694	638	590	543	494	441	-15%	-28%
NH3	22,063	21,845	21,626	21,729	21,974	22,145	22,288	-2%	0%
N2O	1,015	1,010	1,006	1,011	1,031	1,043	1,055	-1%	2%
CH4	27,368	27,368	27,368	27,778	28,381	28,901	29,393	0%	4%
Lead	-	-	-	-	-	-	-	0%	0%
Butadiene	838	810	782	767	755	747	732	-7%	-10%
Acetaldehyde	2,231	2,561	3,757	5,722	7,796	8,908	10,526	68%	249%
Formaldehyde	7,772	7,556	8,808	10,342	10,835	11,987	13,049	13%	39%
Benzene	1,710	1,643	1,557	1,532	1,519	1,453	1,406	-9%	-11%
PM2.5	3,893	3,457	3,020	2,623	2,227	1,807	1,370	-22%	-43%
PM10	4,624	4,106	3,588	3,116	2,646	2,147	1,627	-22%	-43%

Spain – Gasoline Vehicle Fleet Life Cycle GHG Emissions



Country Emissions 2024 (MMT/año)	269.9
Target @ 2035 (MMT/year)	151.9
Delta	118.0

%Reduction vs E0	E0	E10	E15	E20	E25	E30
GREET 2024	0.0%	12.1%	15.3%	17.9%	20.3%	23.6%
RED II/III	0.0%	8.8%	11.3%	13.3%	15.3%	17.7%

% Target@2035	E0	E10	E15	E20	E25	E30
GREET 2024	0.0%	3.6%	4.6%	5.4%	6.1%	7.1%
RED II/III	0.0%	4.0%	5.2%	6.1%	7.0%	8.1%

Source: Greet, RED II/III, IPCC, climateactiontracker.org, F90

Ethanol Gasoline Blending in Sweden



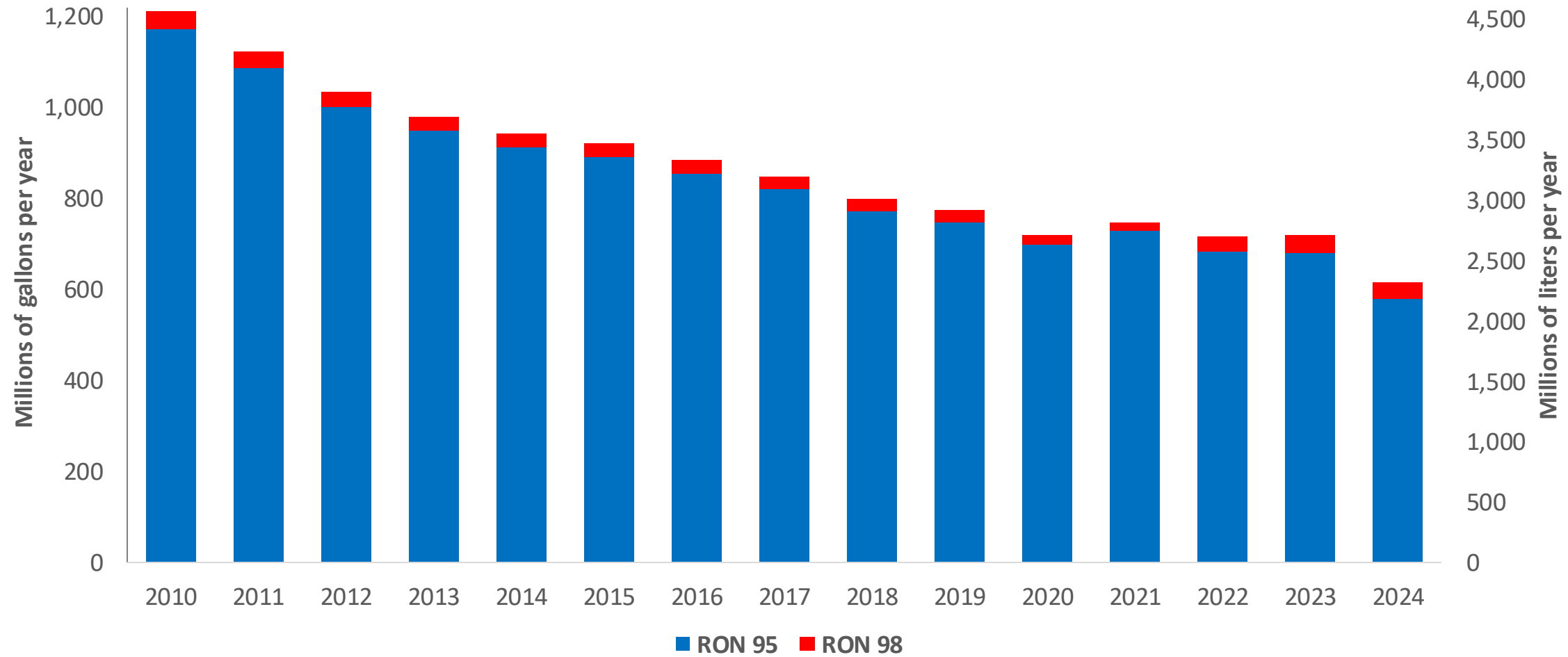
In 2024, gasoline consumption in Sweden was 2,316 million liters (612 million gallons) and growth rate was -4.5%. Gasoline production and net imports were 6,510 and -4,194 million liters (1,720 and -1,108 million gallons) correspondingly. Sweden complies with the European gasoline standards and offers two main grades: RON 95 and RON 98) with an average octane of 95.2 RON.

In 2024, ethanol consumption in Sweden was 178 million liters (47 million gallons) representing 7.7% of gasoline demand. Ethanol production and Net Imports were 133 and 45 million liters (47 and 12 million gallons) correspondingly. Ethanol sources are Wood 25% and Cereals 75%.

Source: Eurostat, European Commission

Gasoline Demand in Sweden

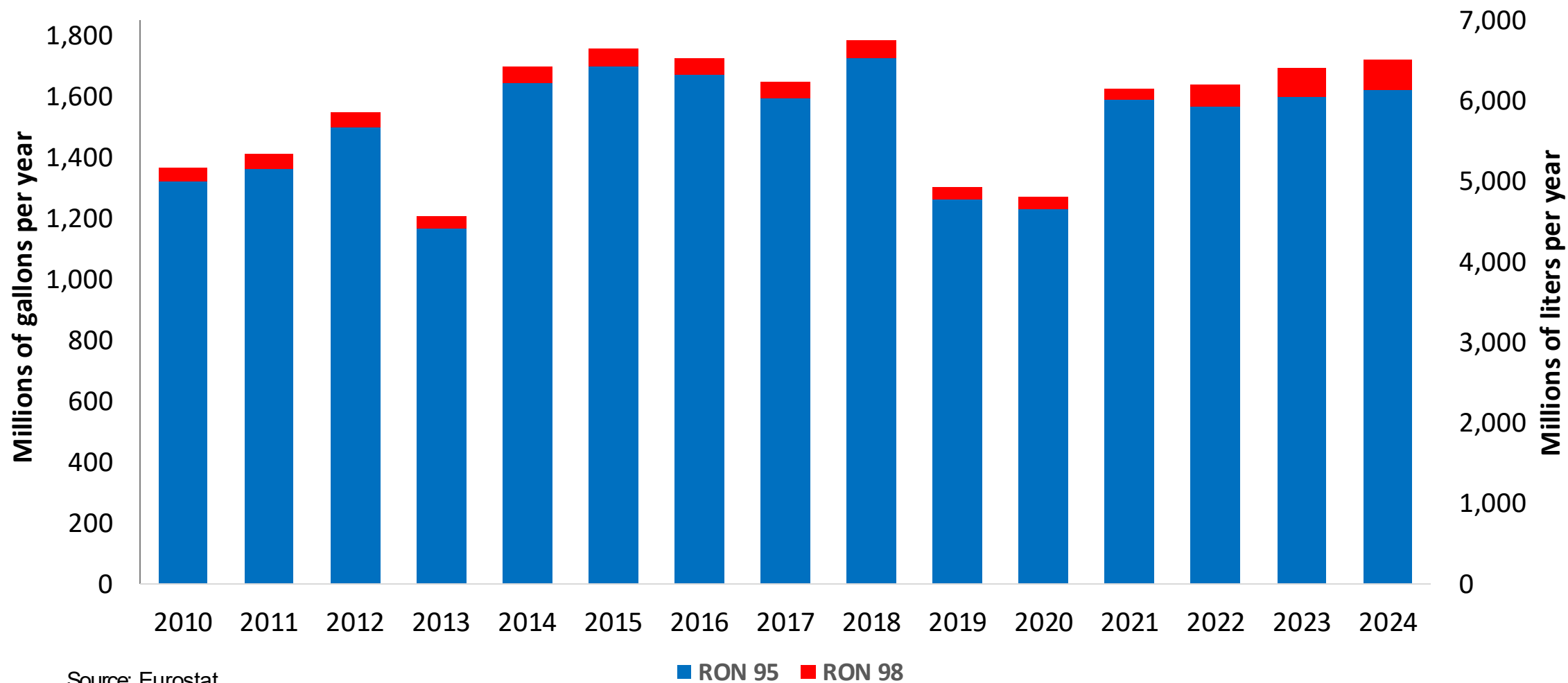
Gasoline Demand by Grade in Sweden



Source: Eurostat

Gasoline Production in Sweden

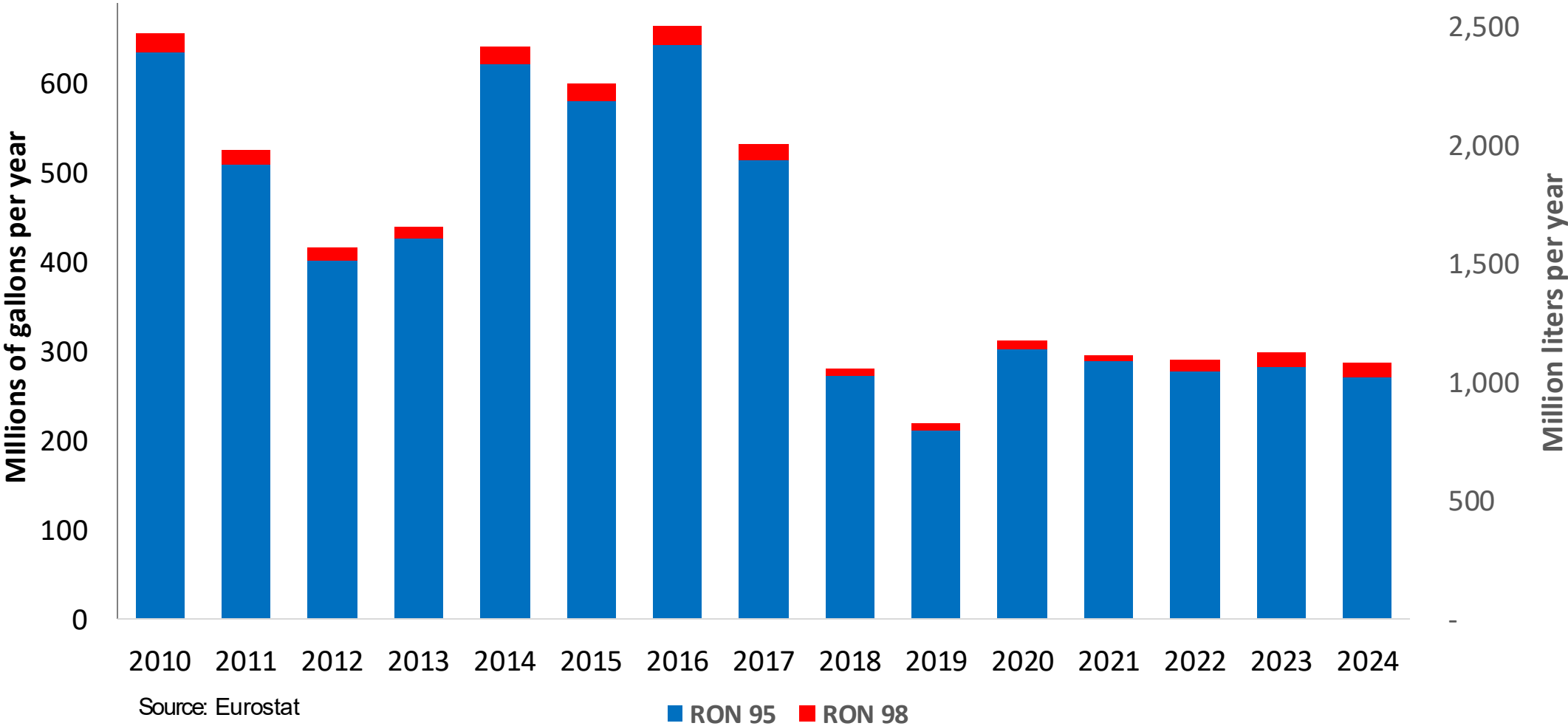
Gasoline Production by Grade in Sweden



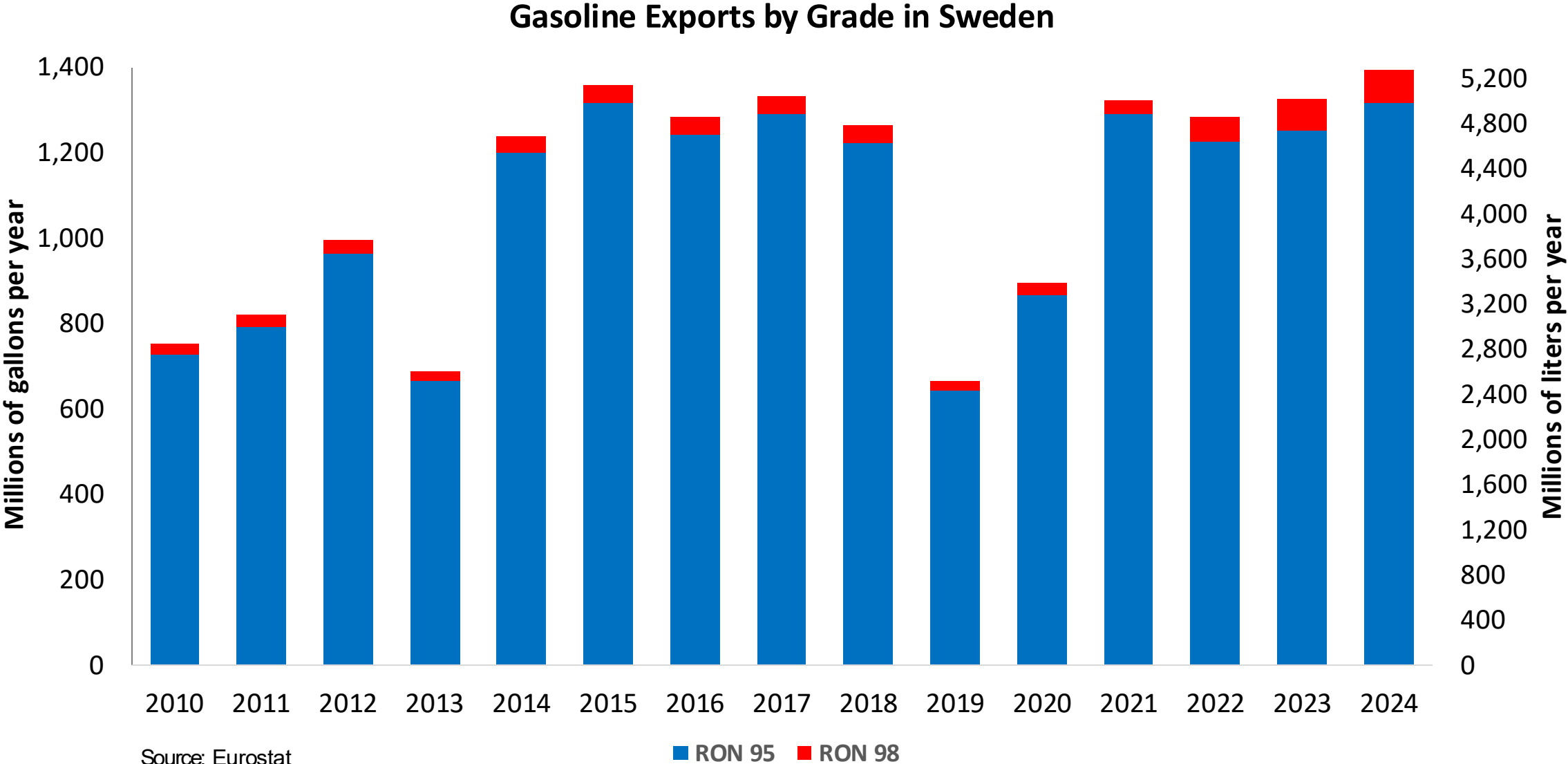
Source: Eurostat

Gasoline Imports in Sweden

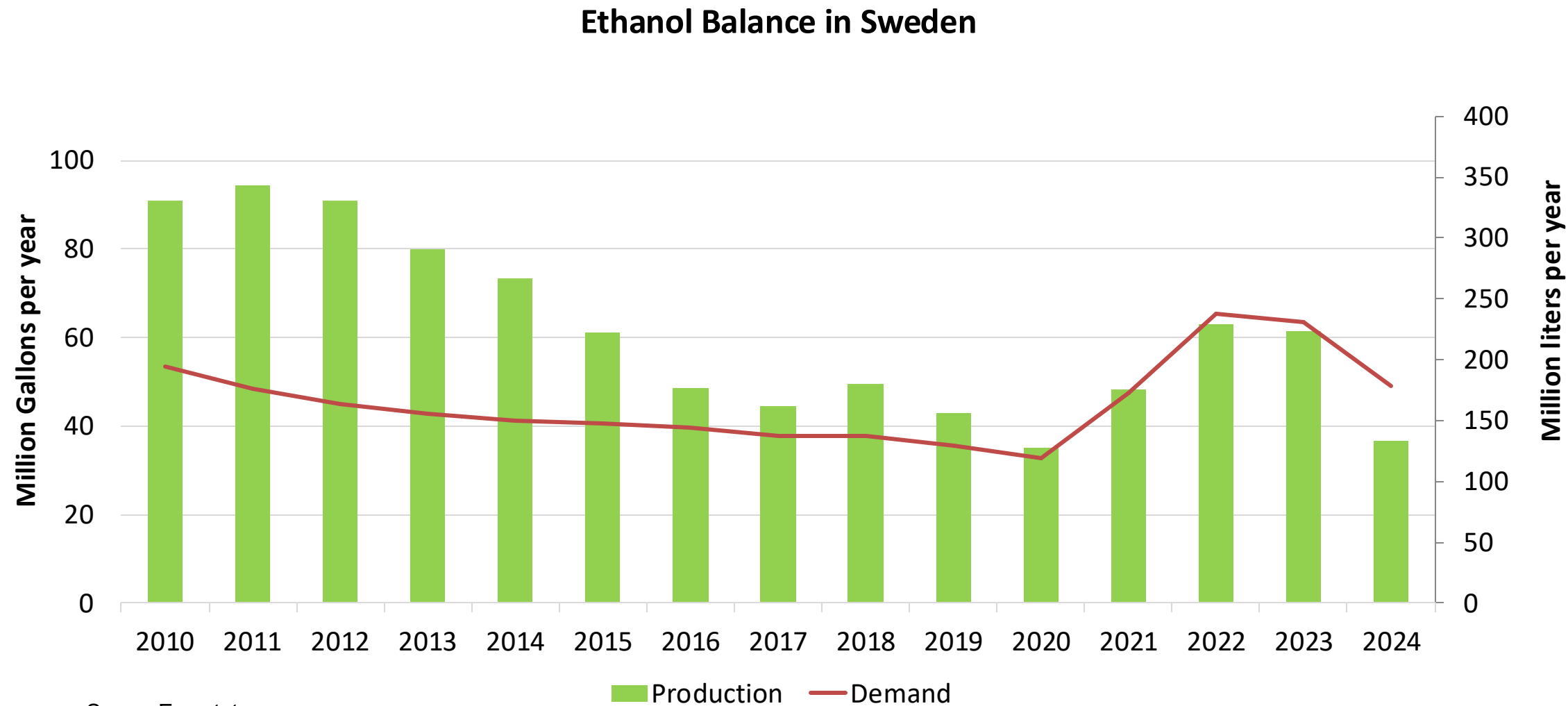
Gasoline Imports by Grade in Sweden



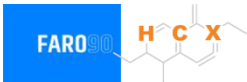
Gasoline Exports in Sweden



Ethanol Balance in Sweden



Gasoline Quality in Sweden

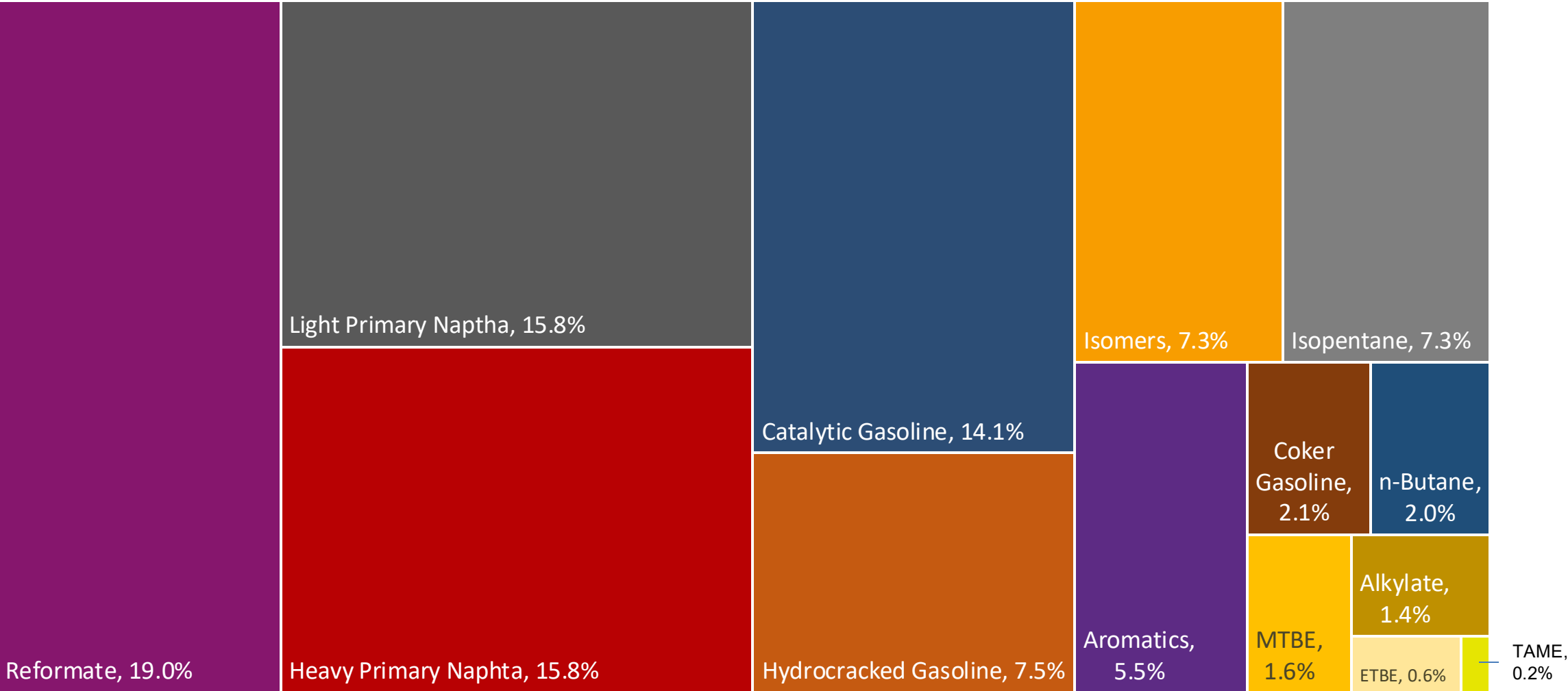
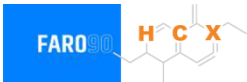


Name	EN 228:2025 (Euro 6); Country Level Fuel Ordinances							
Implementation Date	2025							
Applicability	All EU Contries + UK							
Grade	RON 95 E5		RON 95 E10		RON 98 E5		RON 98 E10	
	Min	Max	Min	Max	Min	Max	Min	Max
RON	95.0		95.0		98.0		98.0	
MON	> 85	> 88	> 85	> 88	> 85	> 88	> 85	> 88
AKI								
Benzene (%vol)	< 1							
Aromatics (% vol)	< 35							
Olefins (%vol)	< 18							
Lead (g/L)	< 0.005							
Manganese (mg/L)	< 2,0							
Sulfur (ppm)	10.0							
Oxygen (%w)*	< 2,7	< 3,7	< 2,7	< 3,7	< 2,7	< 3,7	< 2,7	< 3,7
Ethanol (%vol)	< 5	< 10	< 5	< 10	< 5	< 10	< 5	< 10
PVR 37.8°C (psi)	<> 45 - 100 kPa in 6 Volatility Classes							
Overall Biofuels (%cal)	-							
Advanced Biofuels (%cal)	-							
Bioethanol in Gasoline Sales (%cal)	-							
GHG Emission Reduction (%)	6.0							

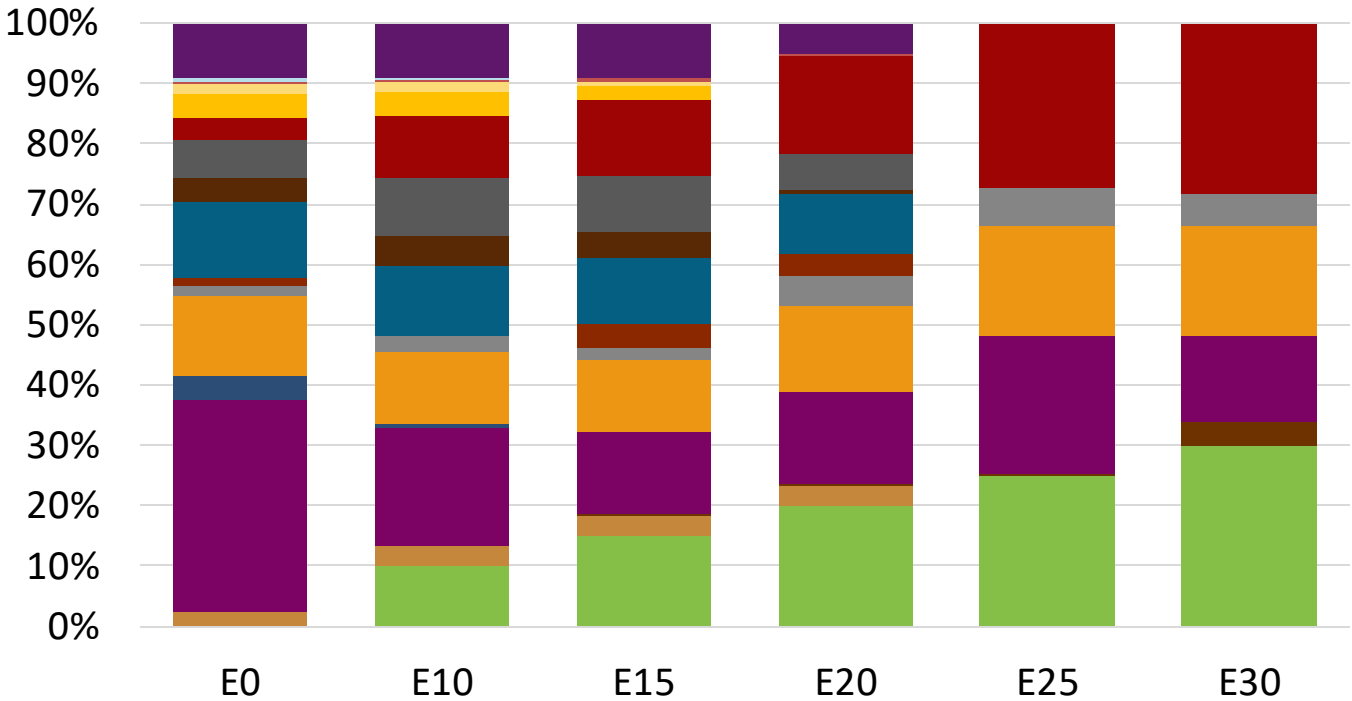
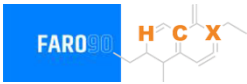
* Methanol 3%, Iso-propyl alcohol 12%, Ter-butyl alcohol 15%, Iso-butyl alcohol 15%, Ethers +5 Carbons 22%, Other Oxygenates 15%

Source: European Commision

Sweden Available Blending Components



Sweden – Gasoline 95 – Constant Octane

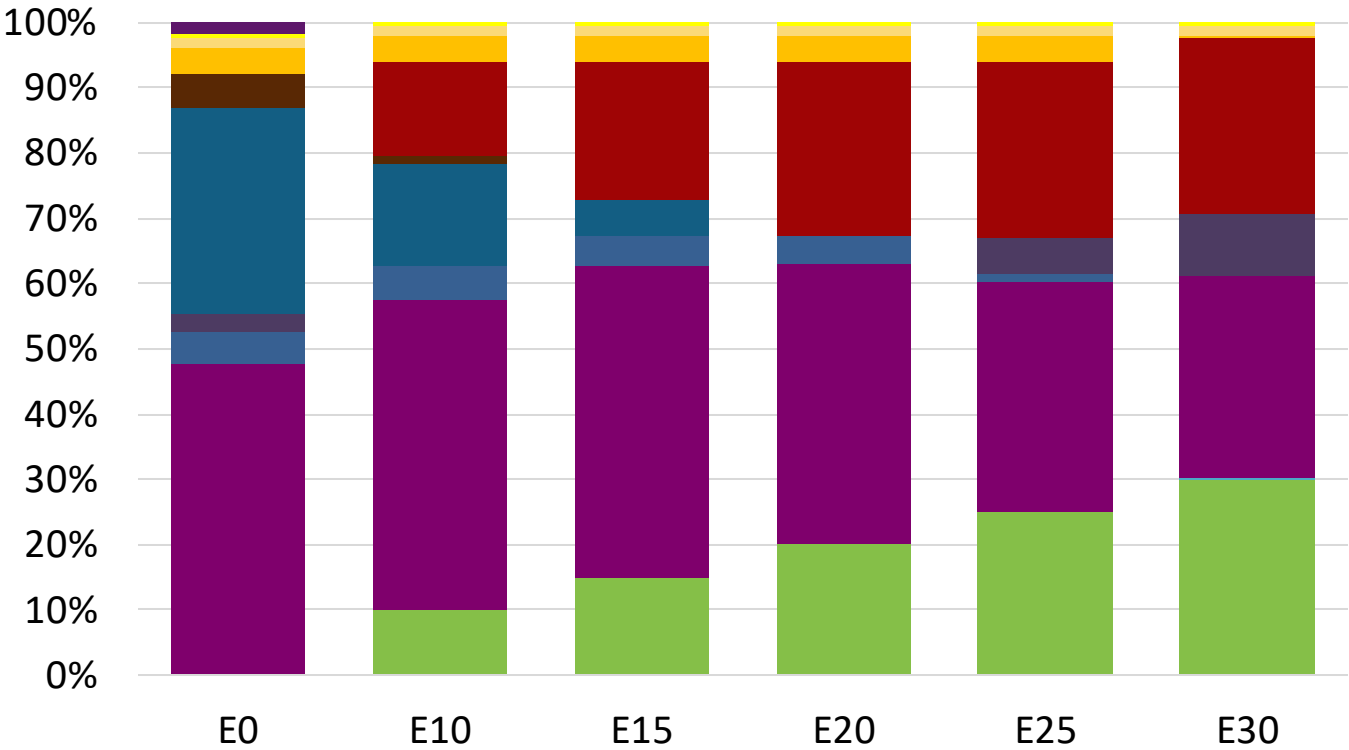
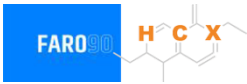


Average CIF 2024
Prices

Octane (RON)	95.0	95.0	95.0	95.0	95.1	95.6
Price (US\$/gal)	\$ 2.312	\$ 2.184	\$ 2.123	\$ 2.042	\$ 1.920	\$ 1.854

Source: Faro90

Sweden - Gasoline 98 - Constant Octane

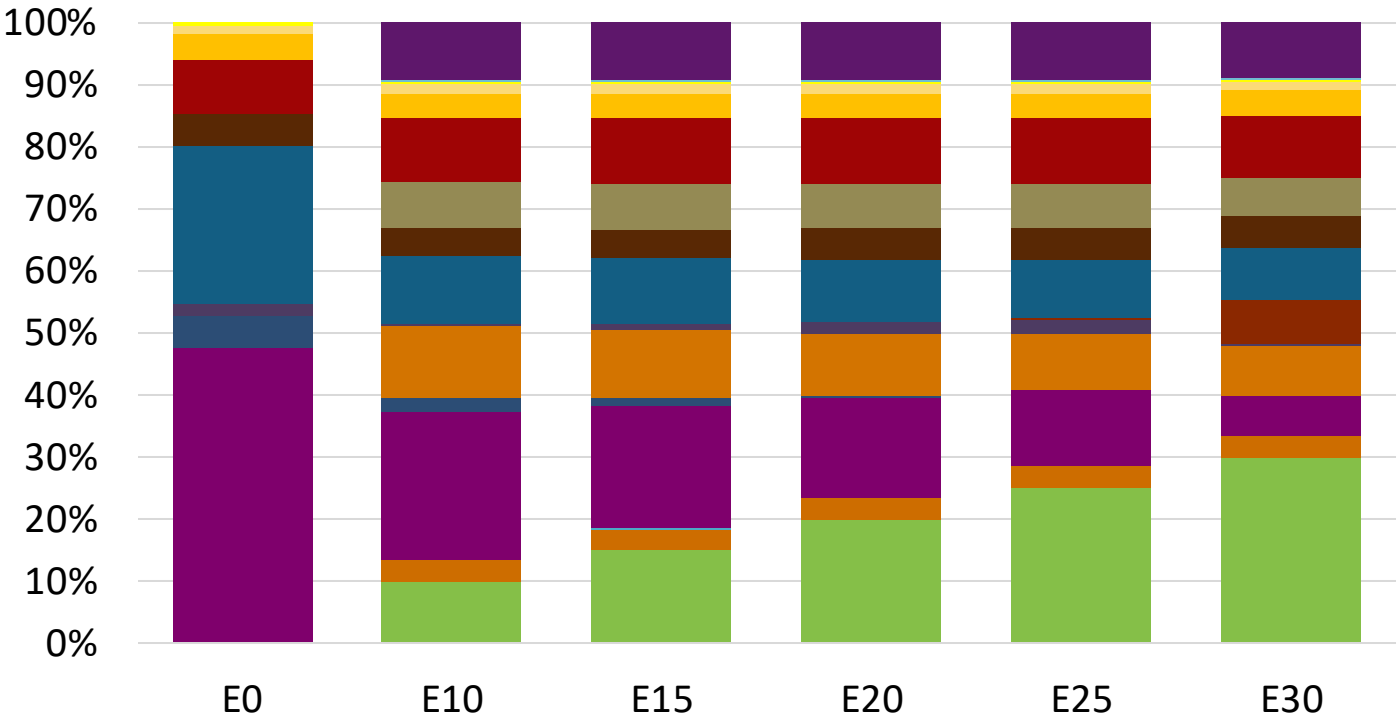


Average CIF 2024 Prices

Octane (RON)	98.0	98.0	98.0	98.0	98.0	98.0
Price (US\$/gal)	\$ 2.387	\$ 2.258	\$ 2.186	\$ 2.109	\$ 2.032	\$ 1.965

Source: Faro90

Sweden – Gasoline 95 – Increased Octane

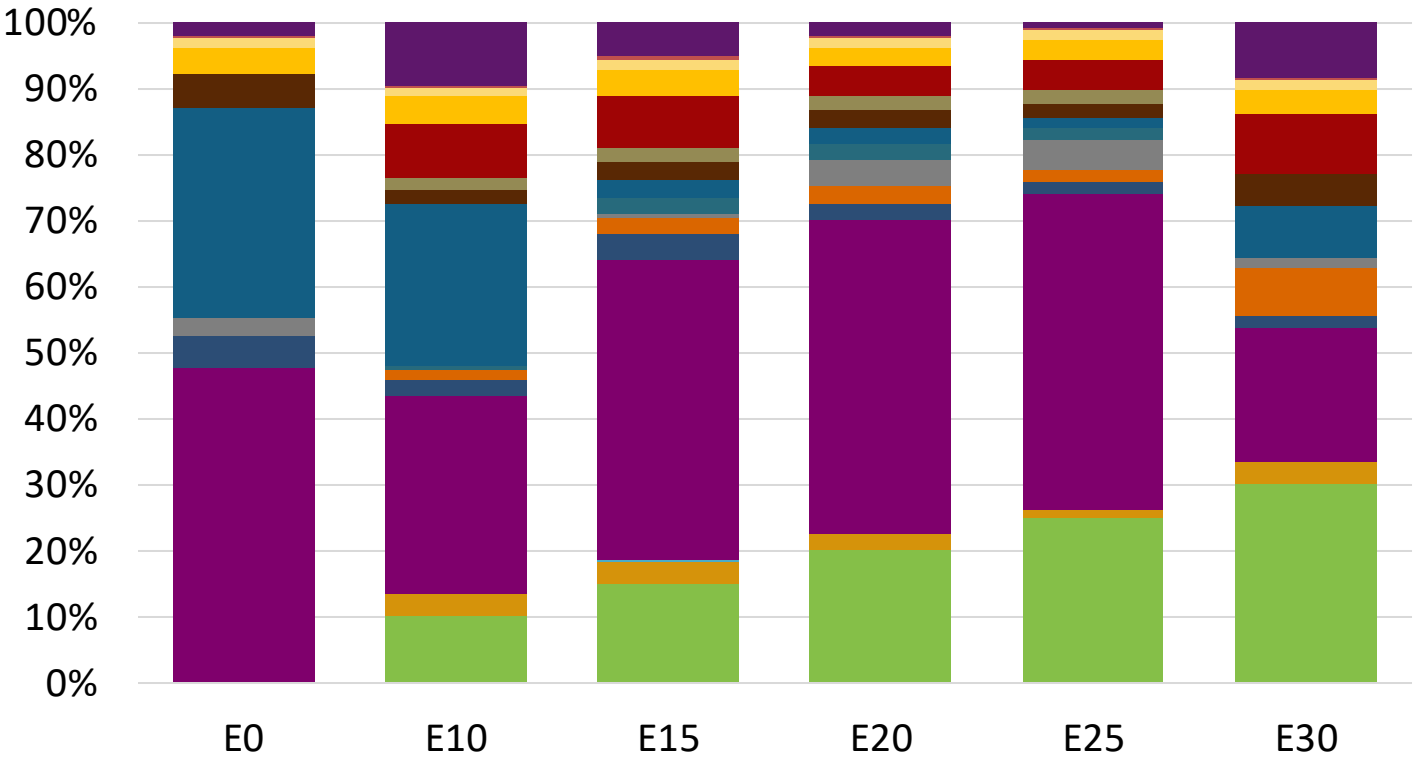


Average CIF 2024
Prices

Octane (RON)	95.0	96.5	97.9	99.4	100.8	102.0
Price (US\$/gal)	\$ 2.249	\$ 2.249	\$ 2.249	\$ 2.249	\$ 2.249	\$ 2.249

Source: Faro90

Sweden – Gasoline 98 – Increased Octane

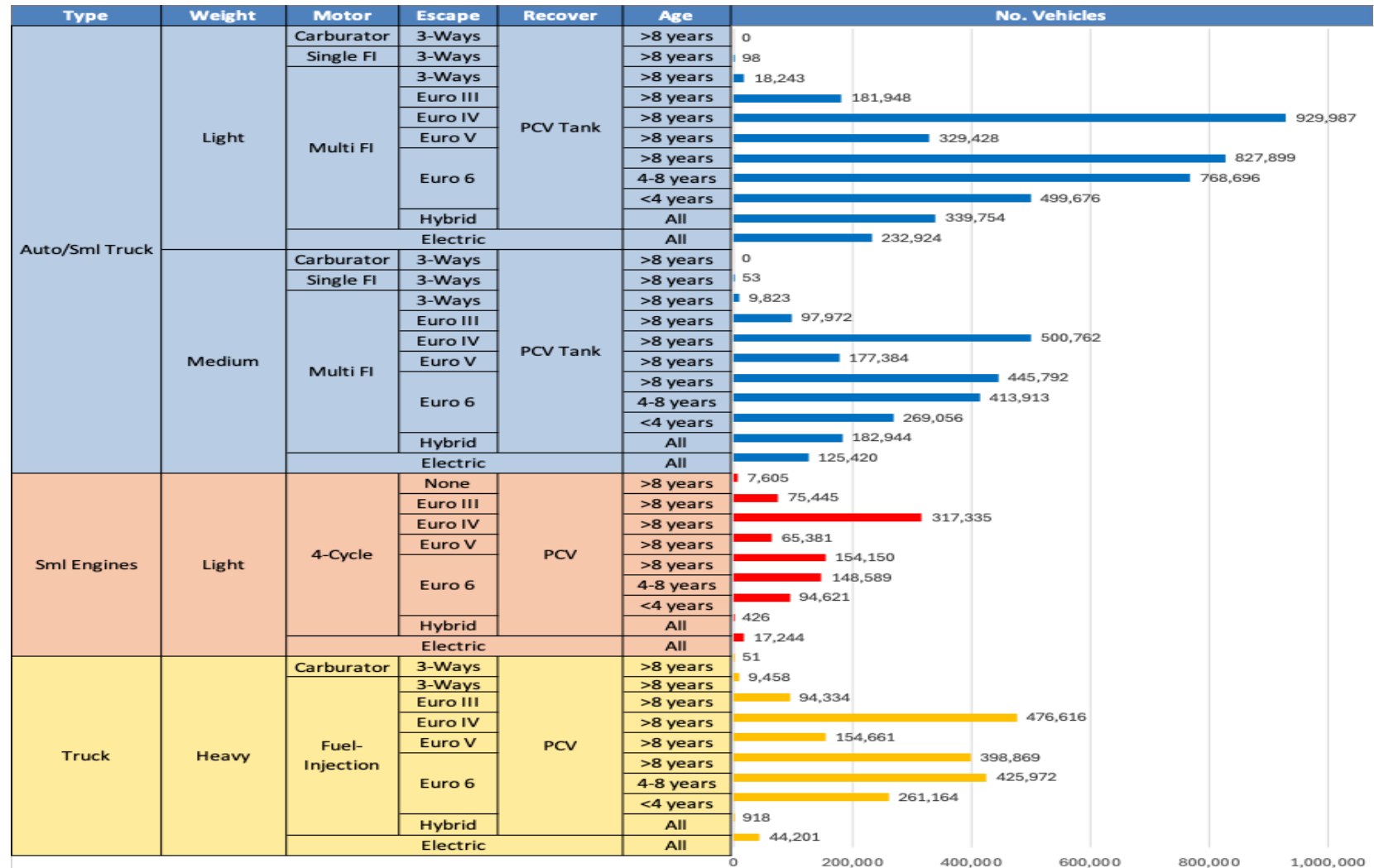


Average CIF 2024
Prices

Octane (RON)		98.0	99.2	101.3	102.6	104.0	104.7
Price (US\$/gal)	\$	2.387	\$ 2.387	\$ 2.387	\$ 2.387	\$ 2.387	\$ 2.387

Source: Faro90

Gasoline Vehicle Fleet - Sweden

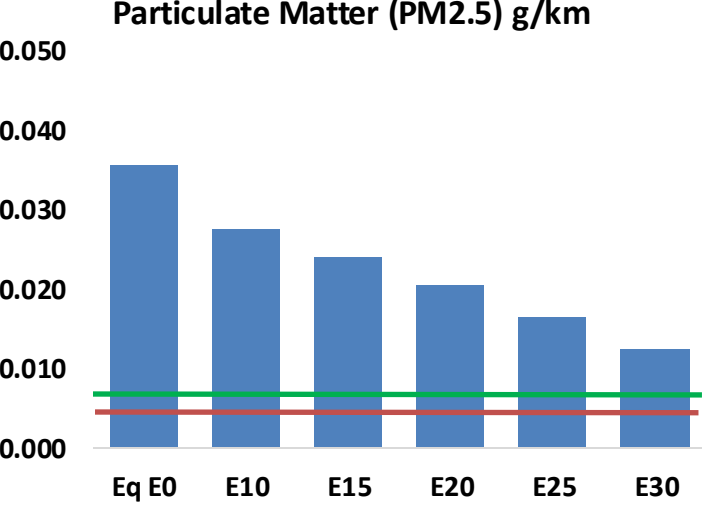
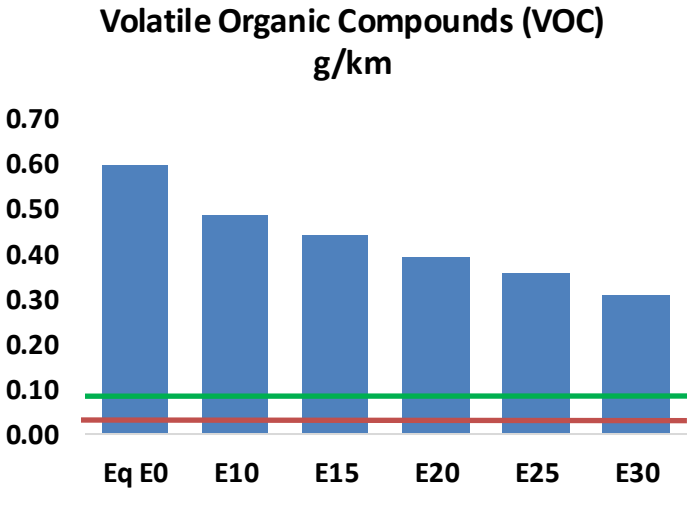
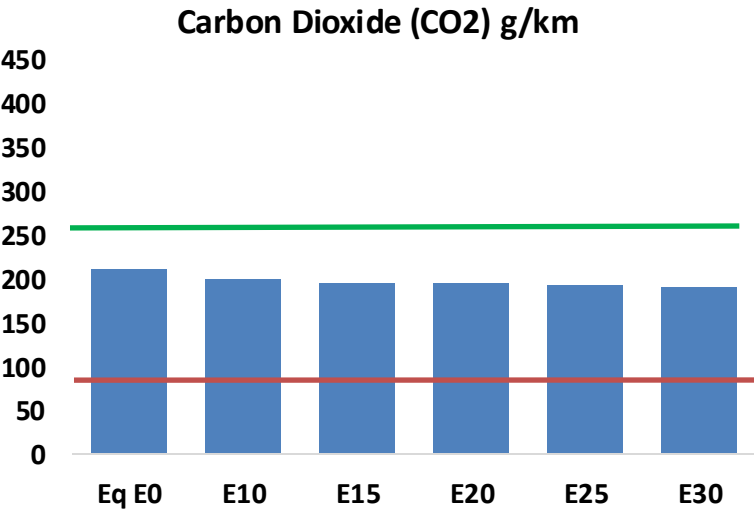
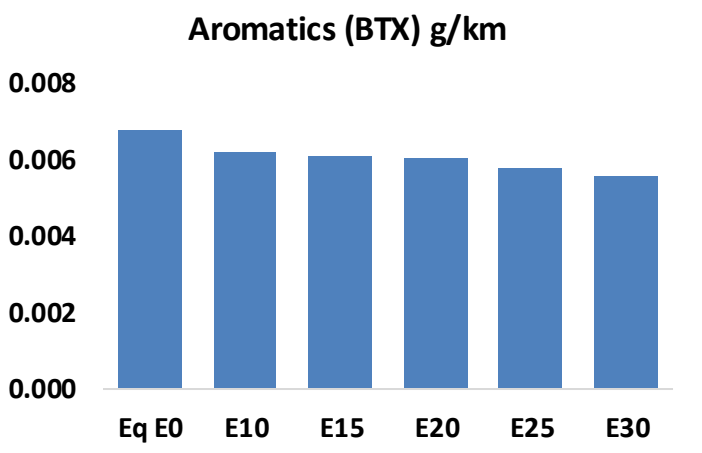
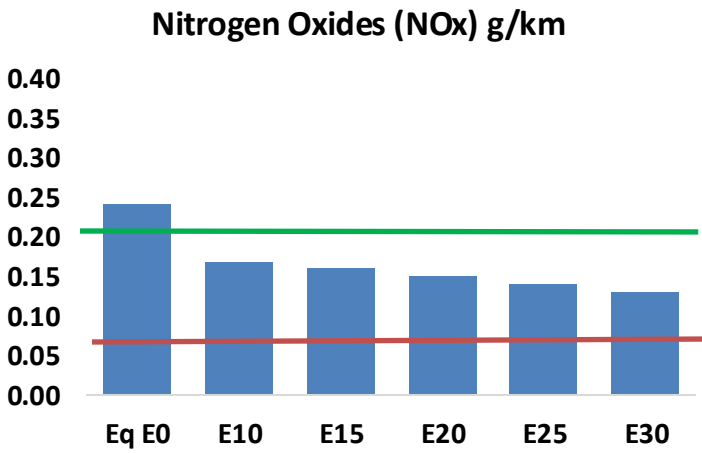
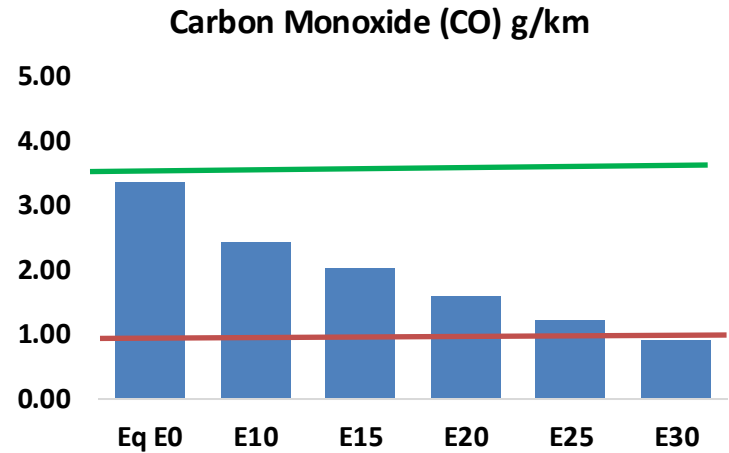
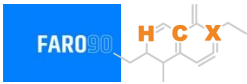


Vehicle Fleet: 9,098,811
Gasoline Vehicle Fleet: 3,553,018

Source: Eurostat, ACEA

Sweden – Vehicular Emissions

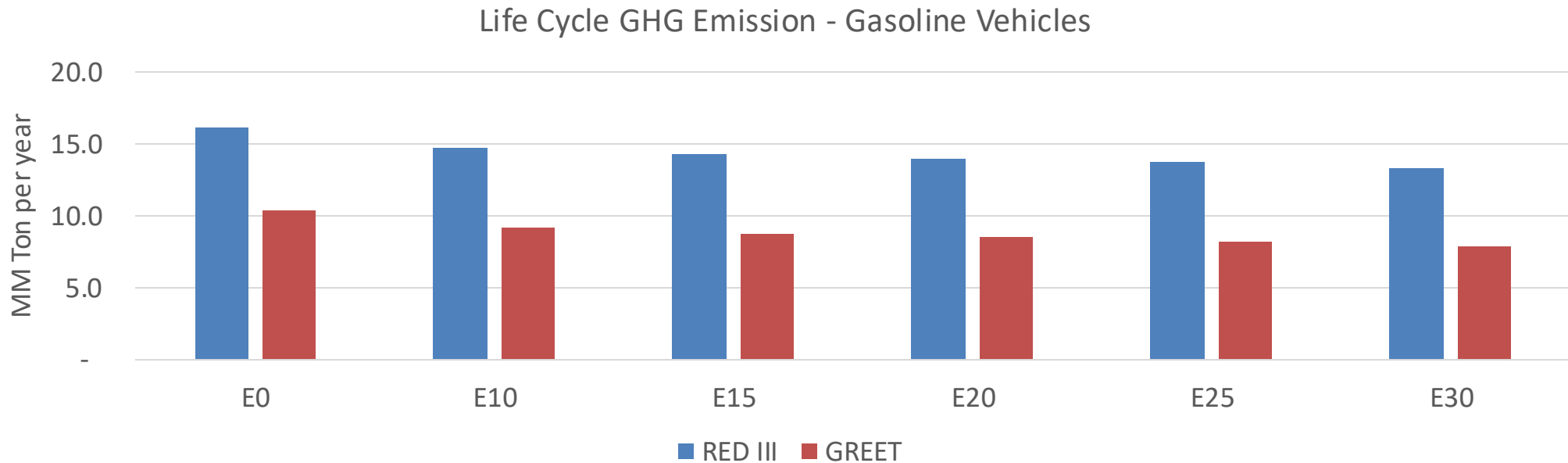
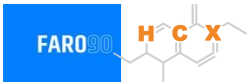
TIER III BIN 125 United States
Euro 6 Standard



Sweden – Vehicular Emissions

Vehicular Fleet Emissions (kg/day)	Eq E0	E5	E10	E15	E20	E25	E30	Reduction (%)	
								E10	E20
CO	294,225	253,630	213,035	176,261	137,640	107,455	77,868	-28%	-53%
CO2	18,387,085	17,914,274	17,467,730	17,116,537	16,942,753	16,846,077	16,535,555	-5%	-8%
VOC	51,886	47,146	42,405	38,374	34,251	31,059	26,872	-18%	-34%
NOx	21,156	15,867	14,809	13,958	13,163	12,288	11,361	-30%	-38%
SOx	212	196	180	166	153	139	124	-15%	-28%
NH3	6,619	6,553	6,488	6,519	6,592	6,644	6,686	-2%	0%
N2O	309	308	306	308	314	318	321	-1%	2%
CH4	10,736	10,736	10,736	10,897	11,133	11,337	11,530	0%	4%
Lead	-	-	-	-	-	-	-	0%	0%
Butadiene	261	242	224	210	195	184	169	-14%	-25%
Acetaldehyde	795	913	1,339	2,039	2,778	3,175	3,751	68%	249%
Formaldehyde	3,016	2,932	3,418	4,013	4,204	4,651	5,063	13%	39%
Benzene	594	570	541	532	528	505	488	-9%	-11%
PM2.5	1,127	1,001	874	759	645	523	397	-22%	-43%
PM10	1,979	1,757	1,535	1,333	1,132	919	696	-22%	-43%

Sweden – Gasoline Vehicle Fleet Life Cycle GHG Emissions



Country Emissions 2024 (MMT/año)	35.9
Target @ 2035 (MMT/year)	20.2
Delta	15.7

%Reduction vs E0	E0	E10	E15	E20	E25	E30
GREET 2024	0.0%	12.2%	15.5%	18.2%	20.8%	24.2%
RED II/III	0.0%	8.5%	10.9%	12.9%	14.8%	17.1%

% Target@2035	E0	E10	E15	E20	E25	E30
GREET 2024	0.0%	8.1%	10.3%	12.1%	13.7%	16.0%
RED II/III	0.0%	8.7%	11.2%	13.2%	15.1%	17.6%

Source: Greet, RED II/III, IPCC, climateactiontracker.org, F90

Ethanol Gasoline Blending in United Kingdom

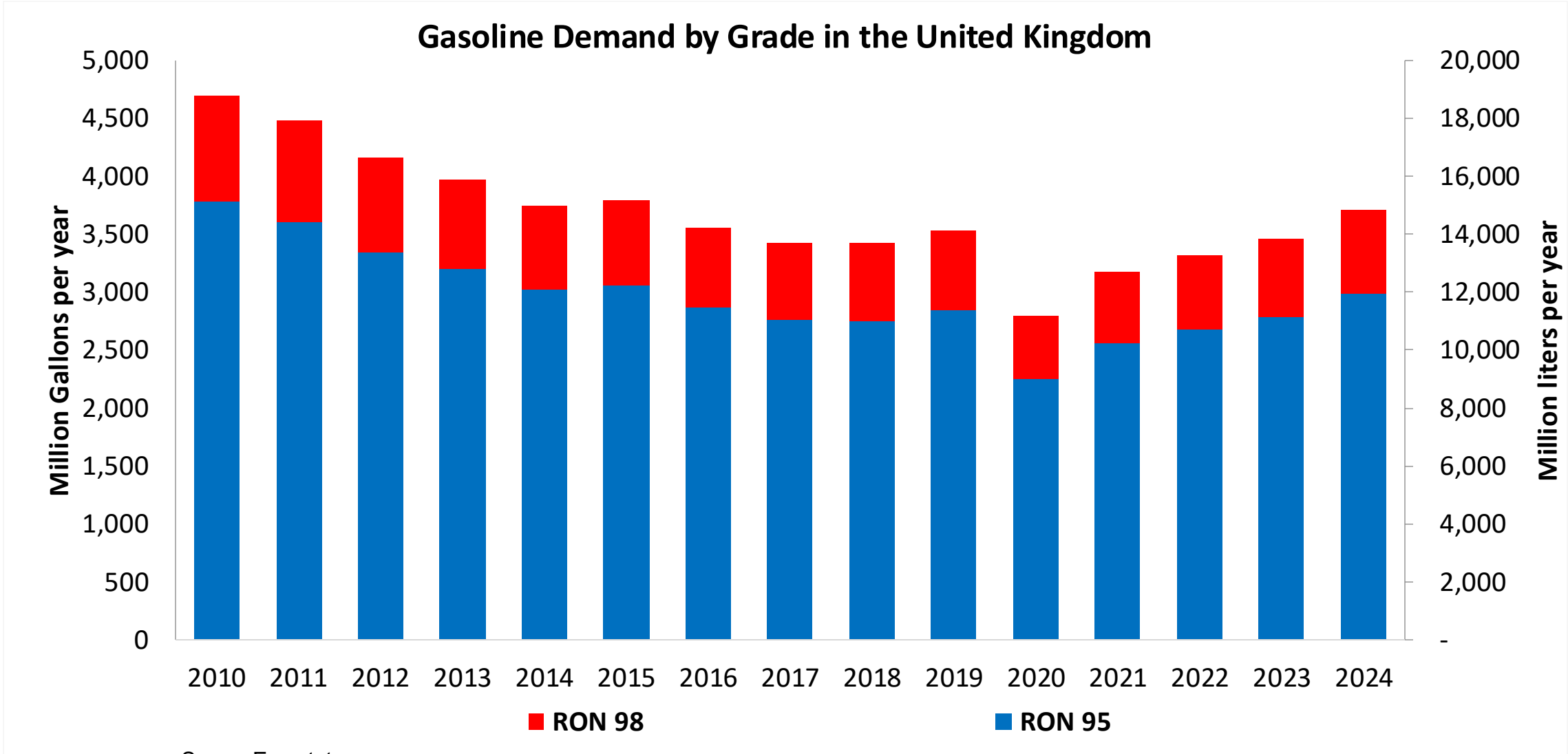


In 2024, gasoline consumption in United Kingdom was 14,052 million liters (3,712 million gallons) and growth rate was 1.0%. Gasoline production and net imports were 19,997 and -5,945 million liters (5,283 and -1,571 million gallons) correspondingly. United Kingdom complies with the European gasoline standards and offers two main grades: RON 95 and RON 98) with an average octane of 95.5 RON.

In 2024, ethanol consumption in United Kingdom was 1,105 million liters (292 million gallons) representing 7.9% of gasoline demand. Ethanol production and Net Imports were 177 and 929 million liters (292 and 245 million gallons) correspondingly. Ethanol sources are Corn 84%, Cereals 13% and Sugar Beet 3%.

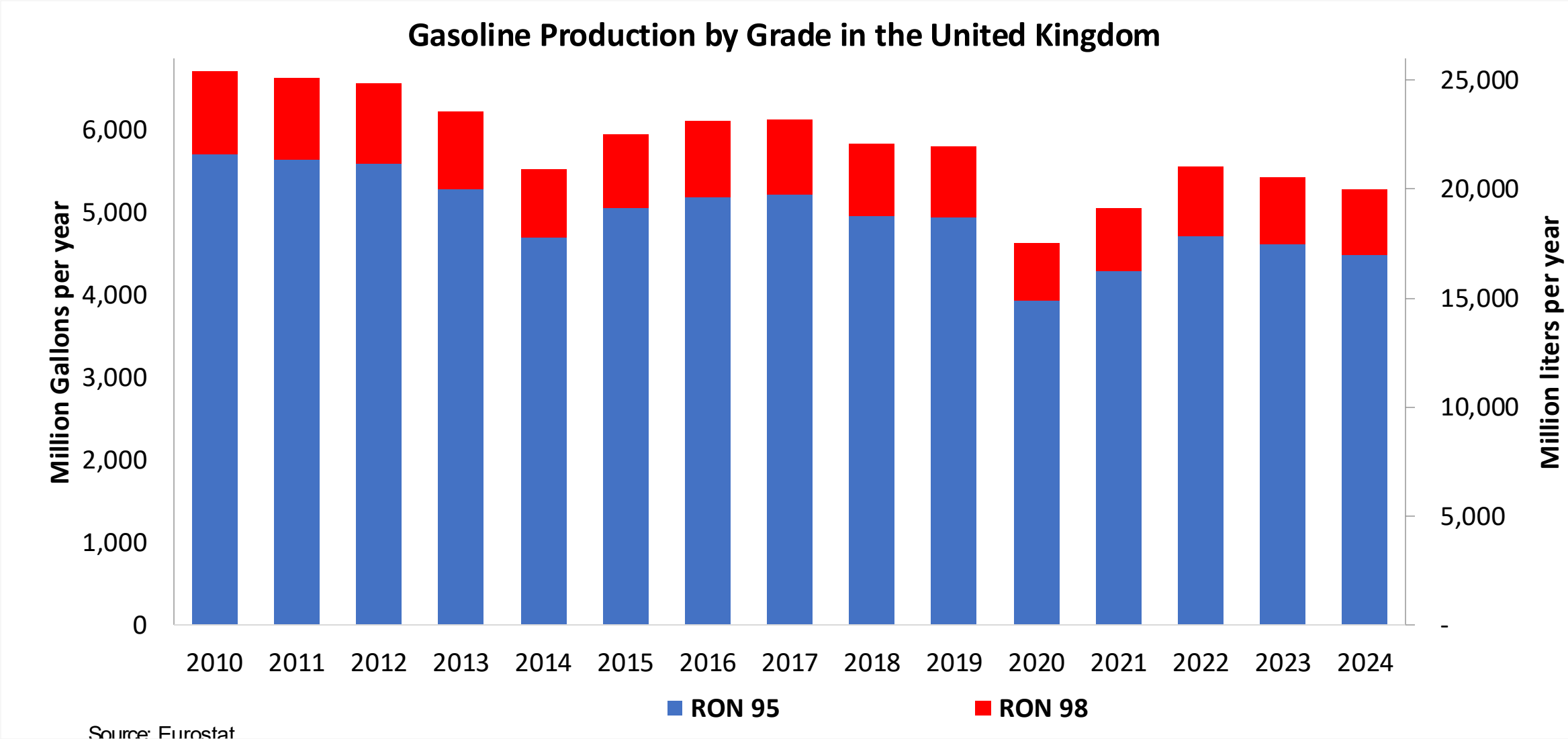
Source: Eurostat, European Commission

Gasoline Demand in United Kingdom

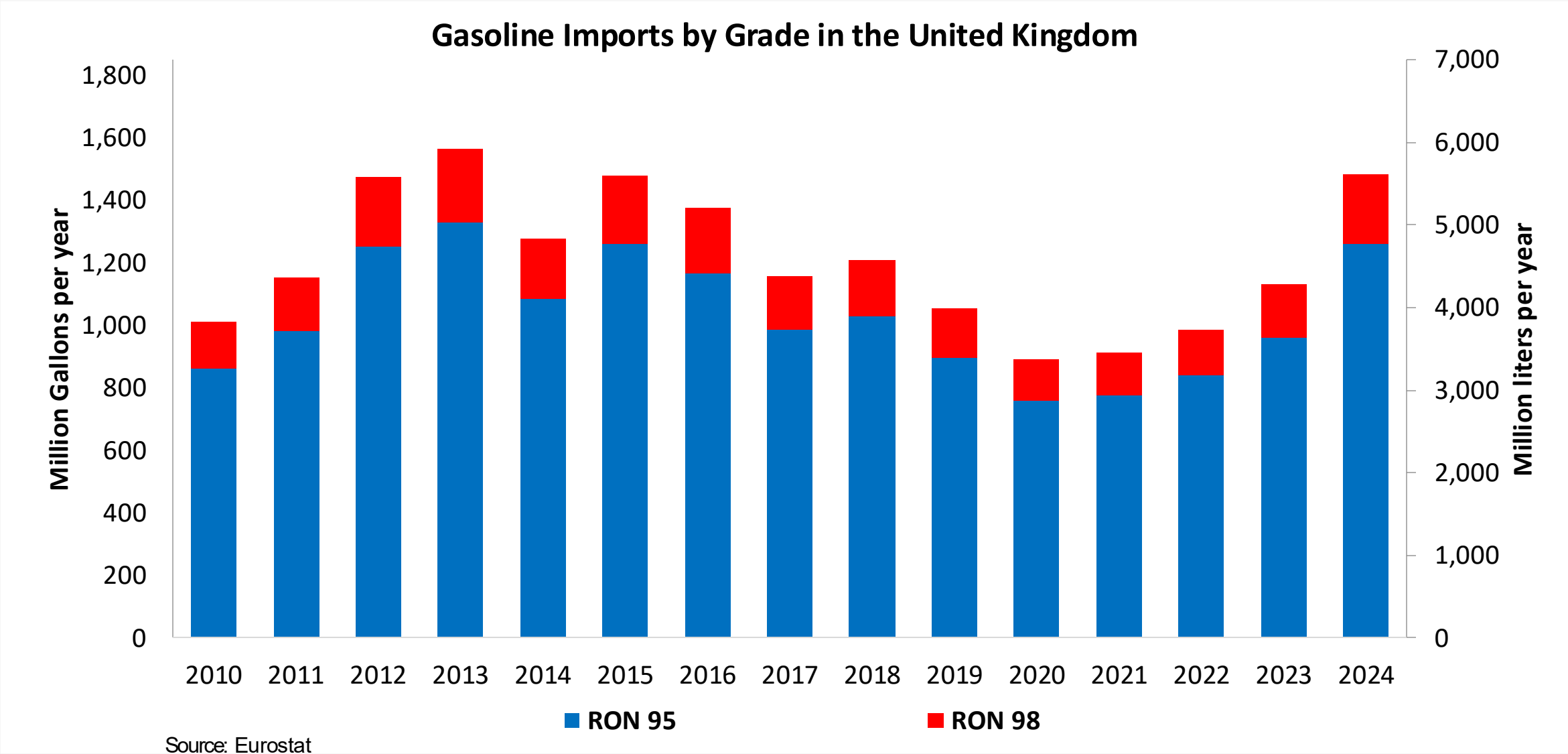


Source: Eurostat

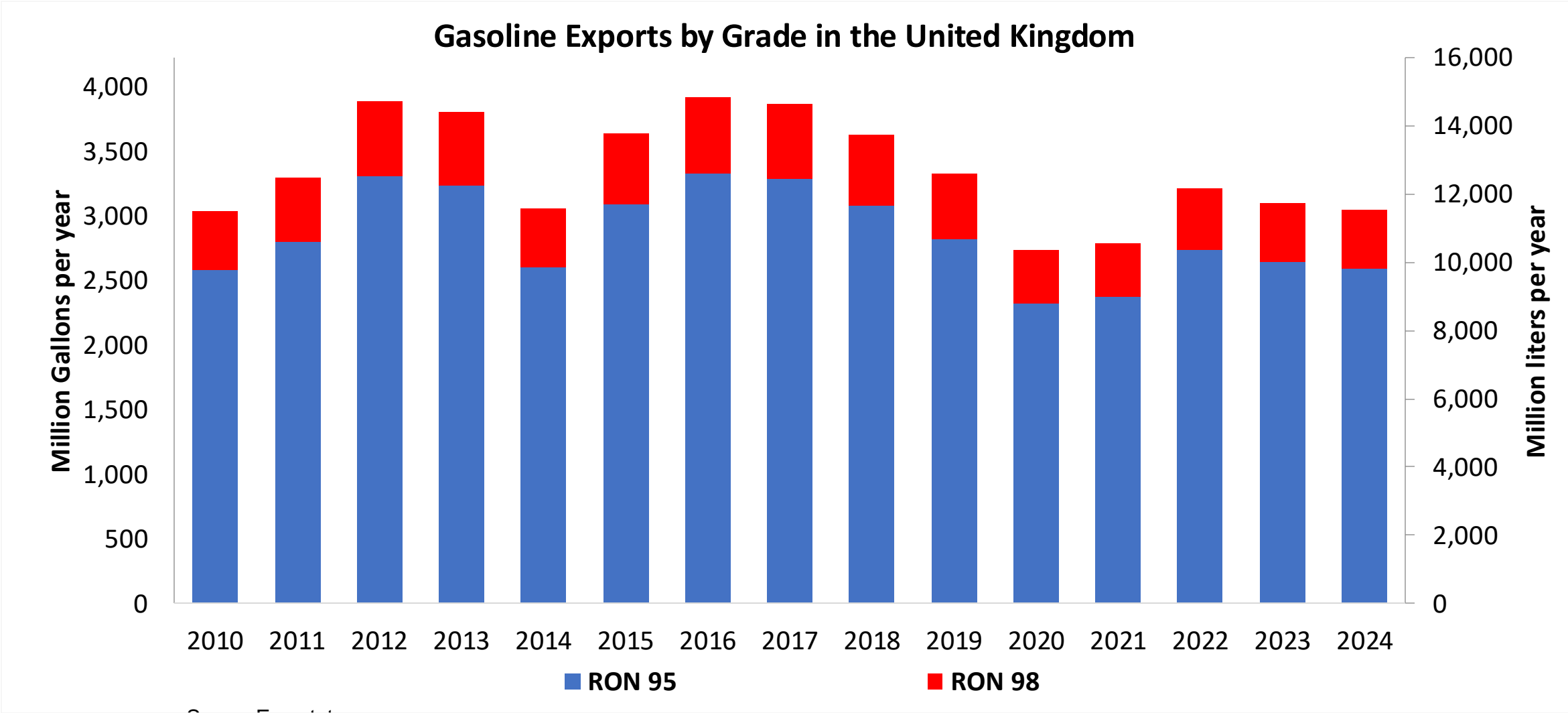
Gasoline Production in United Kingdom



Gasoline Imports in United Kingdom

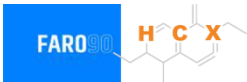


Gasoline Exports in United Kingdom

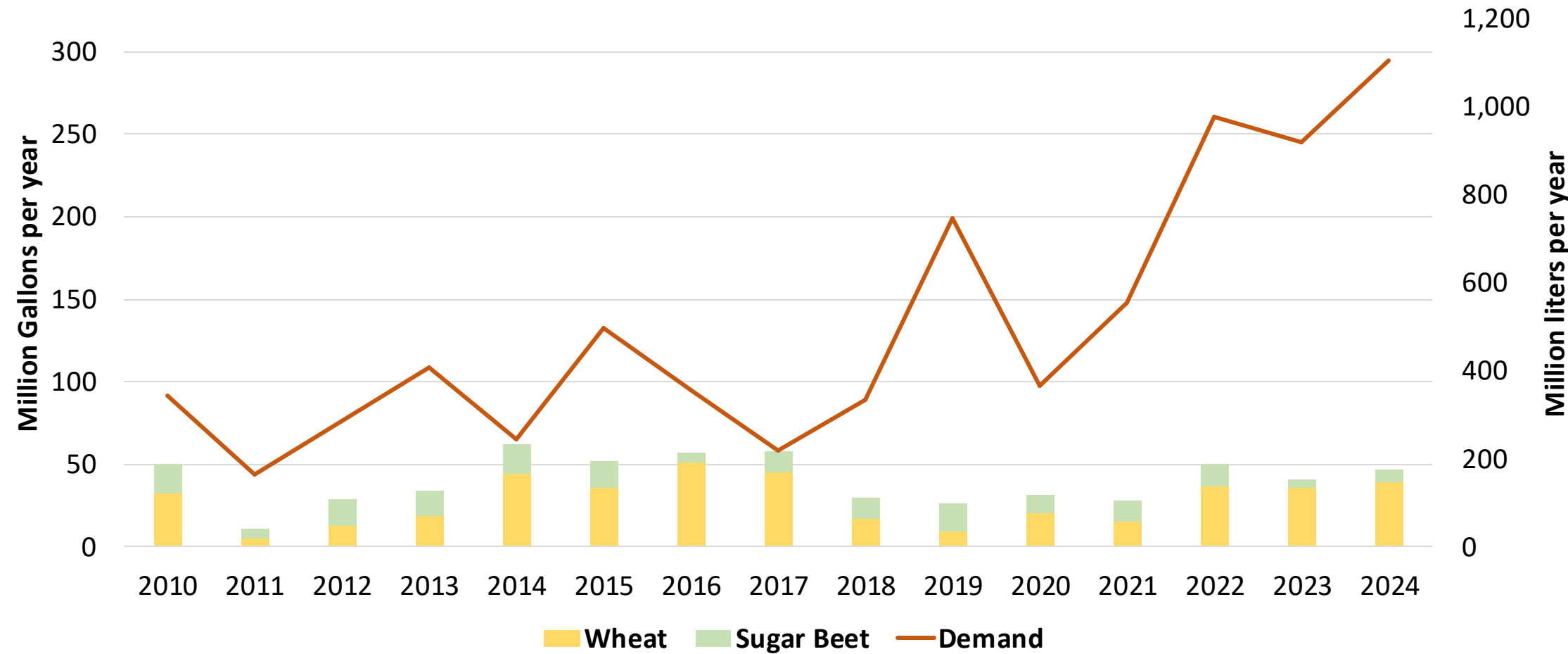


Source: Eurostat

Ethanol Balance in United Kingdom



Ethanol Demand in the United Kingdom



Source: Eurostat

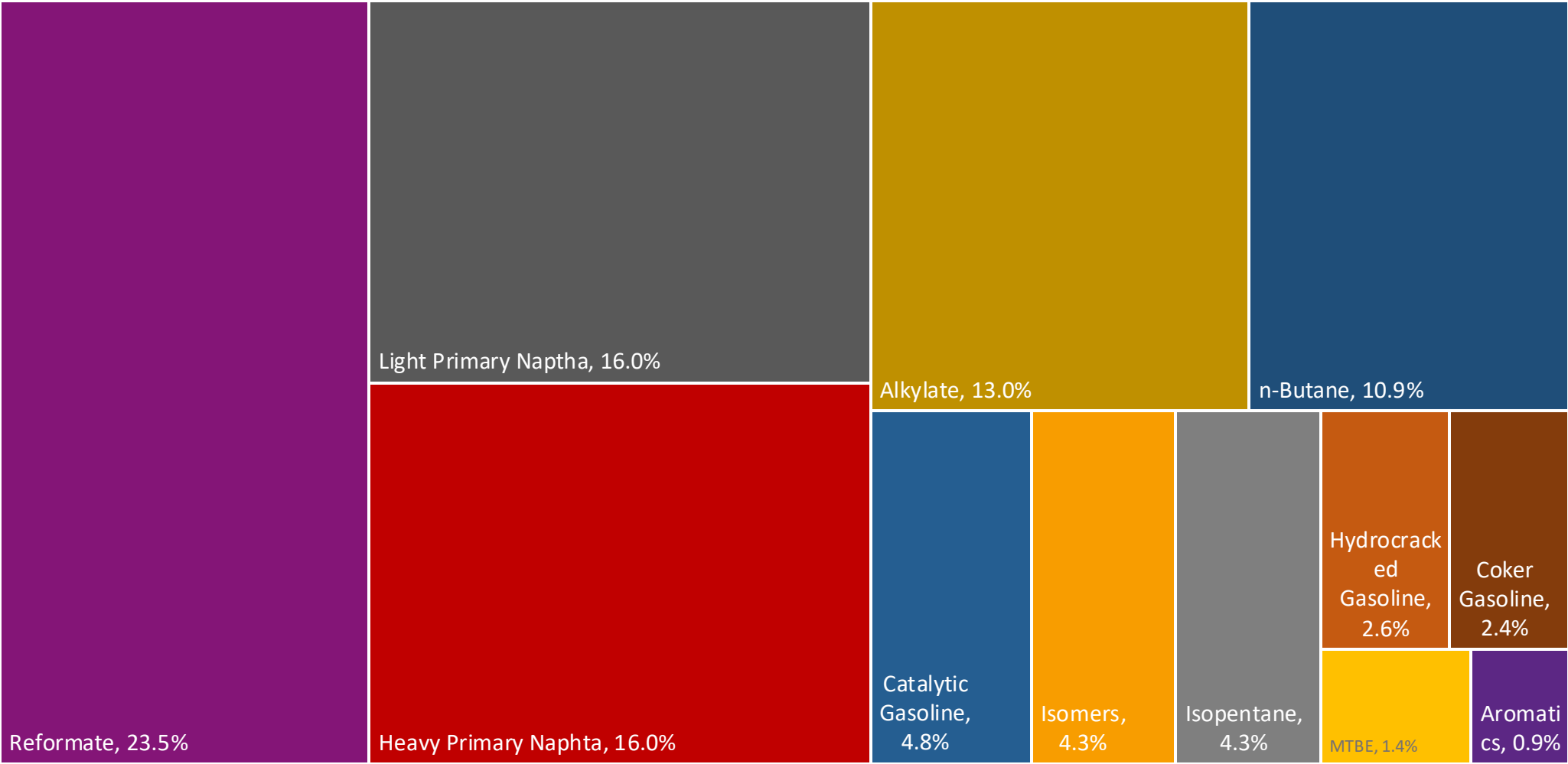
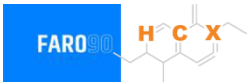
Gasoline Quality in United Kingdom

Name	EN 228:2025 (Euro 6); Country Level Fuel Ordinances							
Implementation Date	2025							
Applicability	All EU Contries + UK							
Grade	RON 95 E5		RON 95 E10		RON 98 E5		RON 98 E10	
	Min	Max	Min	Max	Min	Max	Min	Max
RON	95.0		95.0		98.0		98.0	
MON	> 85	> 88	> 85	> 88	> 85	> 88	> 85	> 88
AKI								
Benzene (%vol)	< 1							
Aromatics (% vol)	< 35							
Olefins (%vol)	< 18							
Lead (g/L)	< 0.005							
Manganese (mg/L)	< 2,0							
Sulfur (ppm)	10.0							
Oxygen (%w)*	< 2,7	< 3,7	< 2,7	< 3,7	< 2,7	< 3,7	< 2,7	< 3,7
Ethanol (%vol)	< 5	< 10	< 5	< 10	< 5	< 10	< 5	< 10
PVR 37.8°C (psi)	<> 45 - 100 kPa in 6 Volatility Classes							
Overall Biofuels (%cal)	-							
Advanced Biofuels (%cal)	-							
Bioethanol in Gasoline Sales (%cal)	-							
GHG Emission Reduction (%)	-							

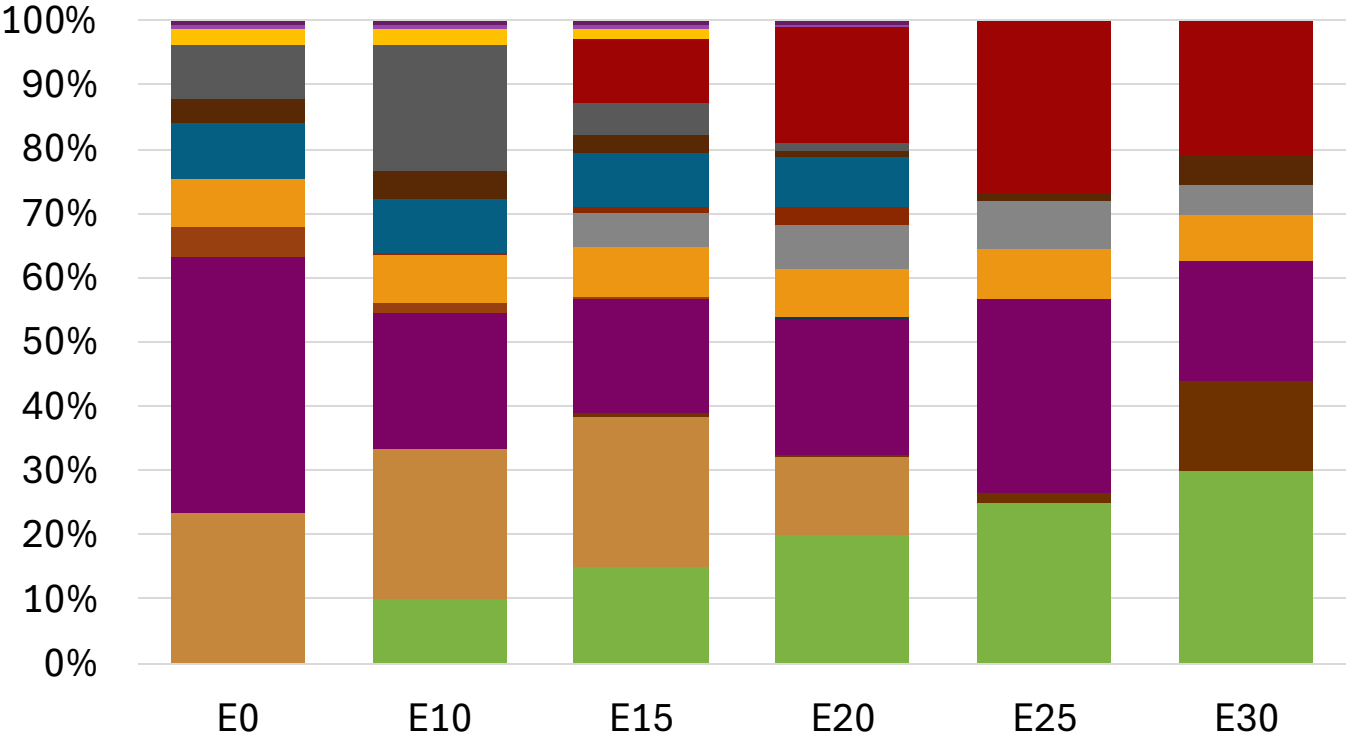
* Methanol 3%, Iso-propyl alcohol 12%, Ter-butyl alcohol 15%, Iso-butyl alcohol 15%, Ethers +5 Carbons 22%, Other Oxygenates 15%

Source: European Commision

United Kingdom Available Blending Components



United Kingdom – Gasoline 95 – Constant Octane

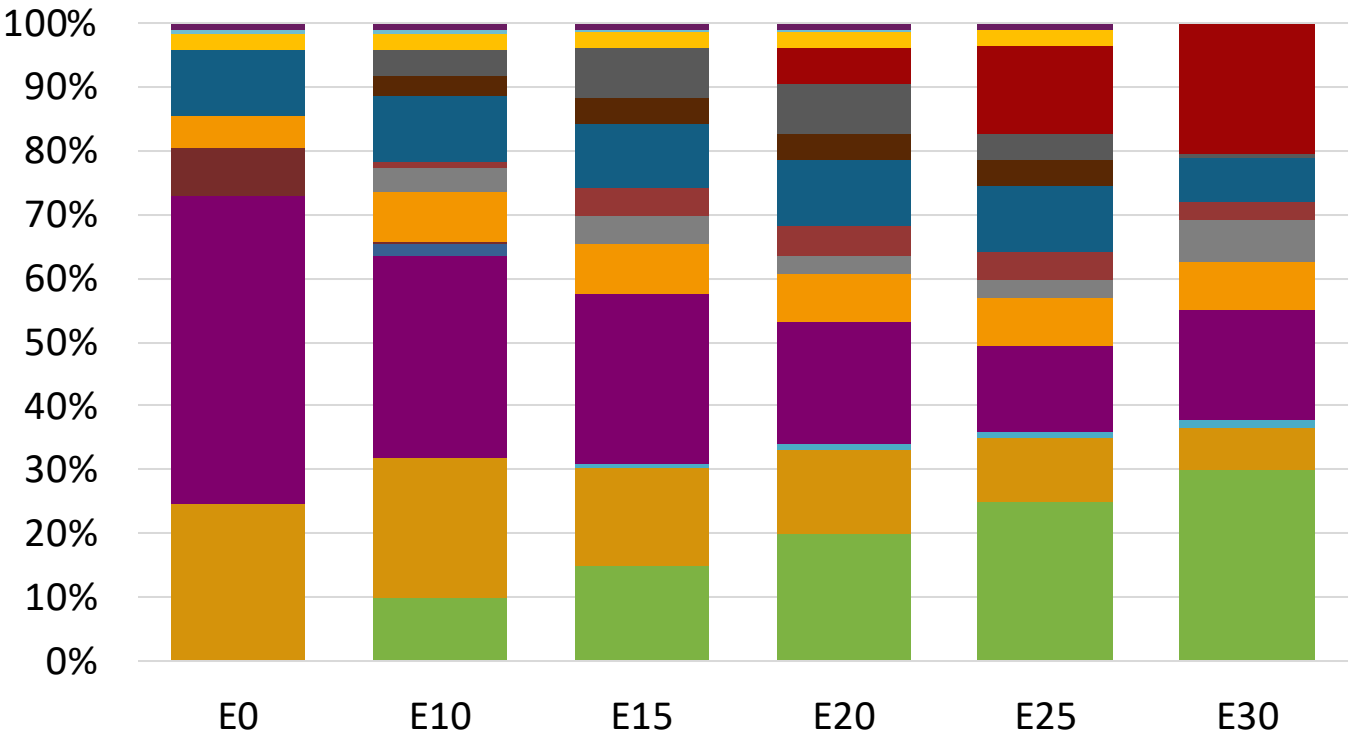
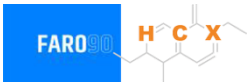


Average CIF 2024
Prices

Octane (RON)	95.0	95.0	95.0	95.0	95.4	95.8
Price (US\$/gal)	\$ 2.317	\$ 2.187	\$ 2.105	\$ 1.996	\$ 1.886	\$ 1.821

Source: Faro90

United Kingdom - Gasoline 98 - Constant Octane

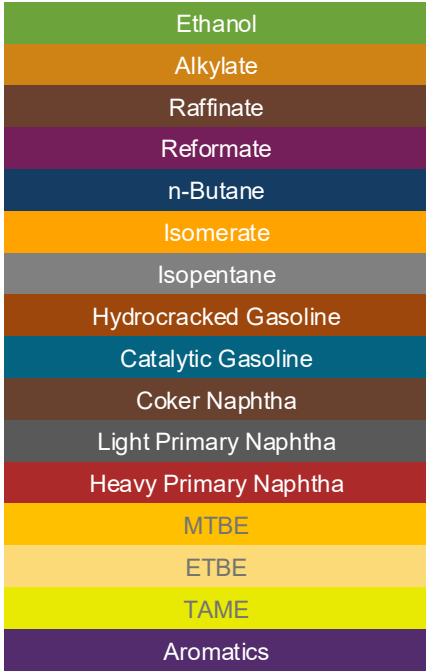
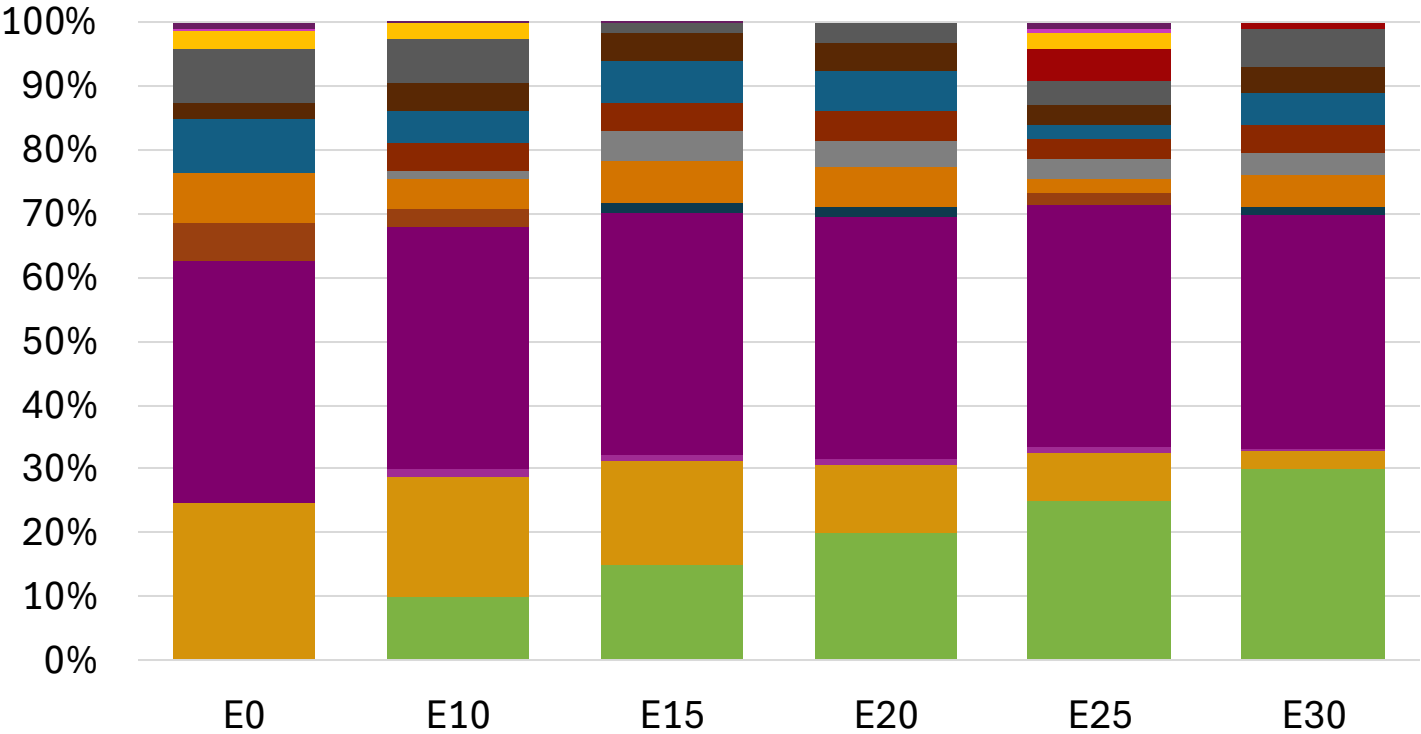
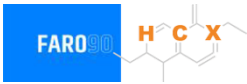


Average CIF 2024 Prices

Octane (RON)	98.0	98.0	98.0	98.0	98.0	98.0
Price (US\$/gal)	\$ 2.440	\$ 2.317	\$ 2.240	\$ 2.157	\$ 2.062	\$ 1.967

Source: Faro90

United Kingdom – Gasoline 95 – Increased Octane

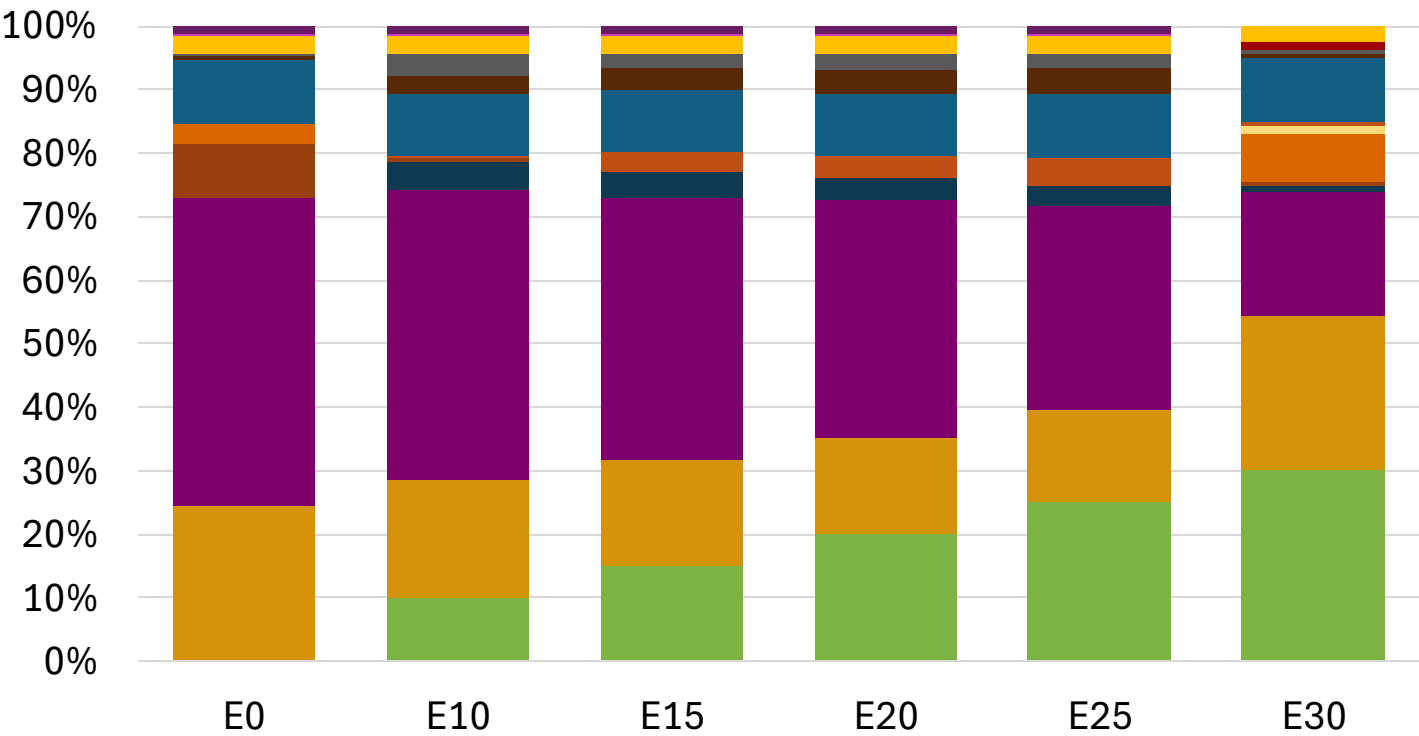
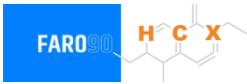


Average CIF 2024
Prices

Octane (RON)		95.0	97.9	99.3	100.9	102.6	103.9
Price (US\$/gal)	\$	2.301	\$ 2.301	\$ 2.301	\$ 2.301	\$ 2.301	\$ 2.301

Source: Faro90

United Kingdom – Gasoline 98 – Increased Octane

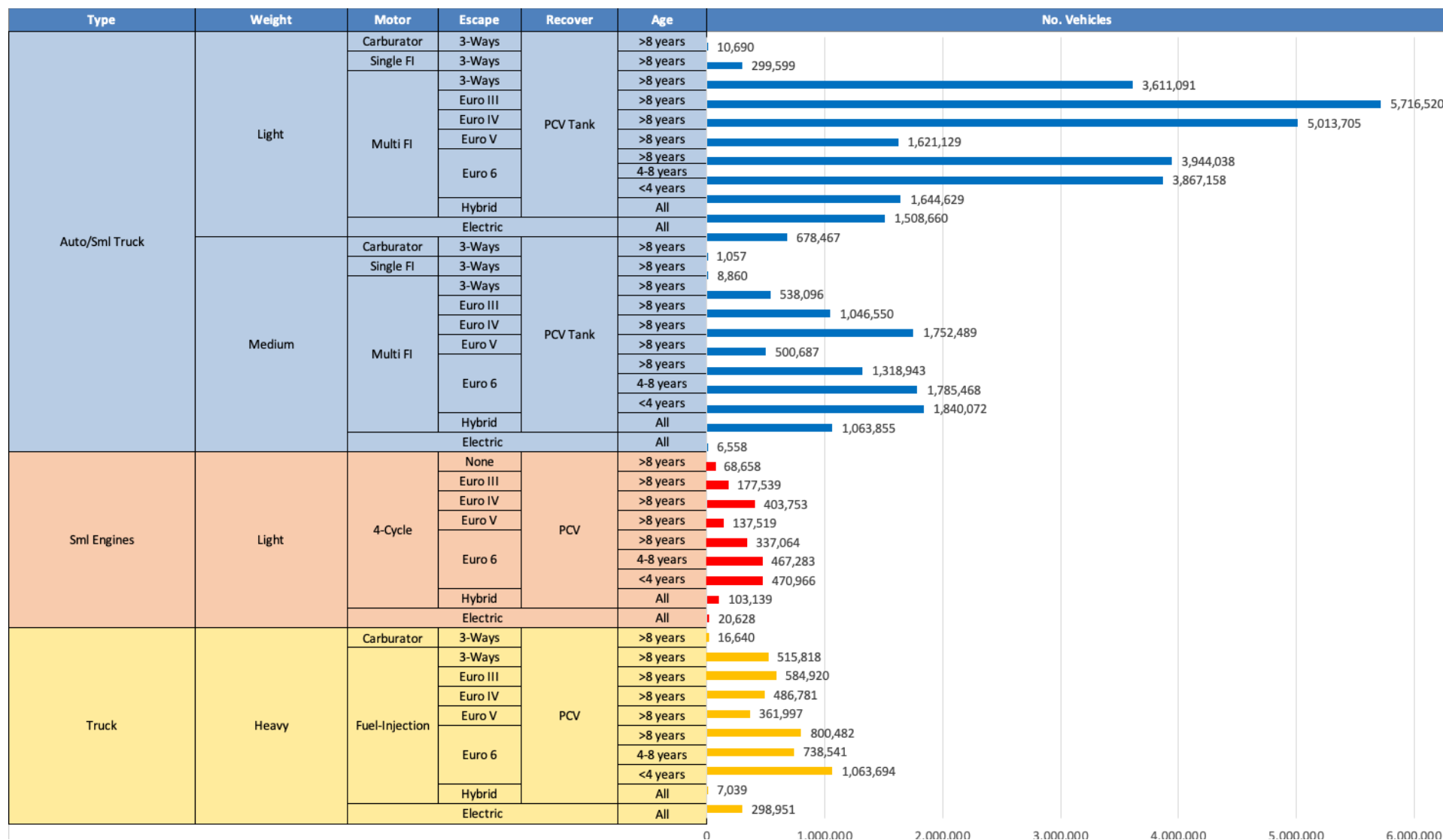


Average CIF 2024 Prices

Octane (RON)		98.0	100.7	102.0	103.3	104.5	105.0
Price (US\$/gal)	\$	2.425	\$ 2.425	\$ 2.425	\$ 2.425	\$ 2.425	\$ 2.425

Source: Faro90

Gasoline Vehicle Fleet - United Kingdom

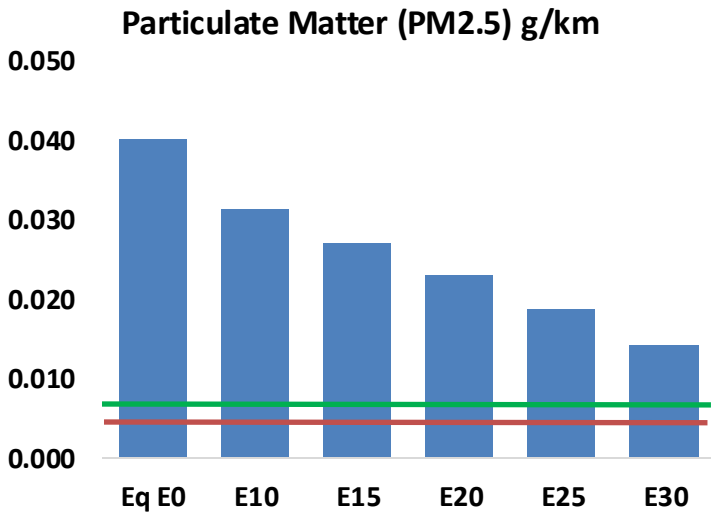
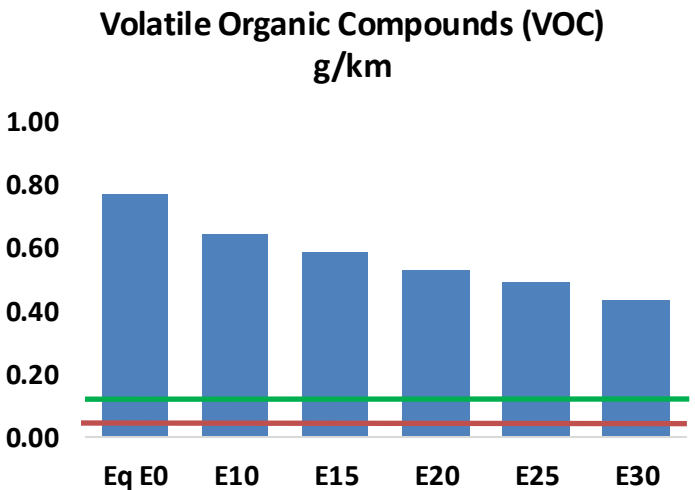
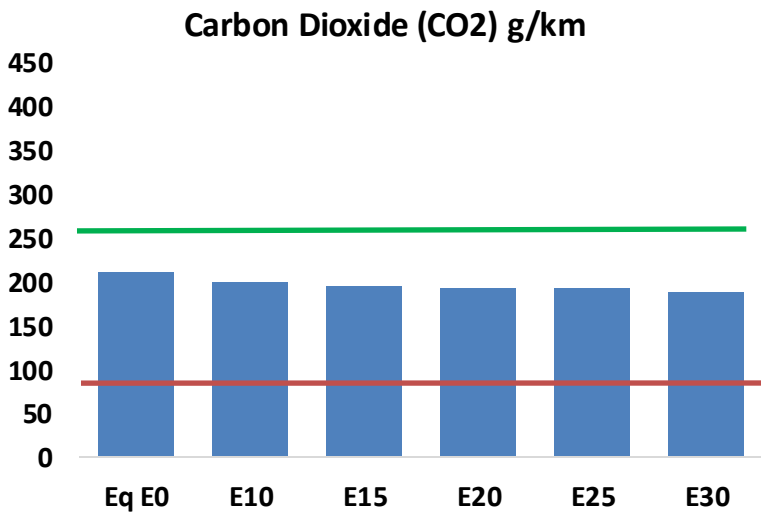
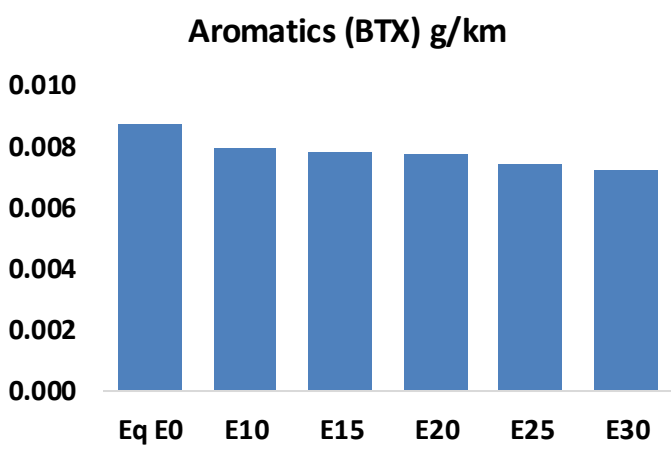
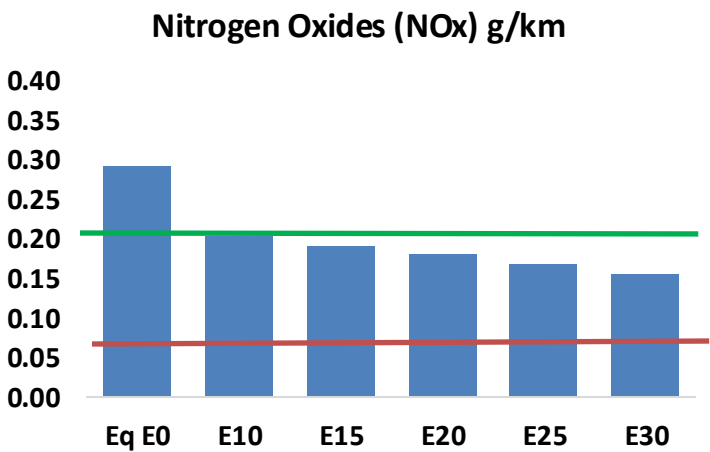
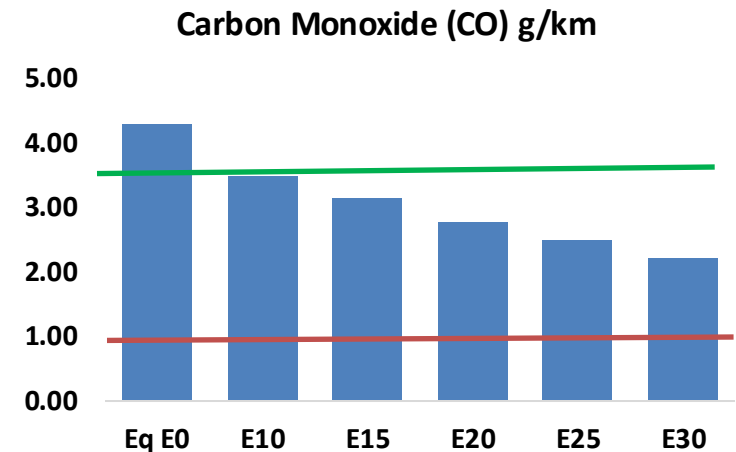
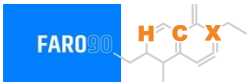


Vehicle Fleet: 44,839,734
Gasoline Vehicle Fleet: 22,834,680

Source: Eurostat, ACEA

United Kingdom – Vehicular Emissions

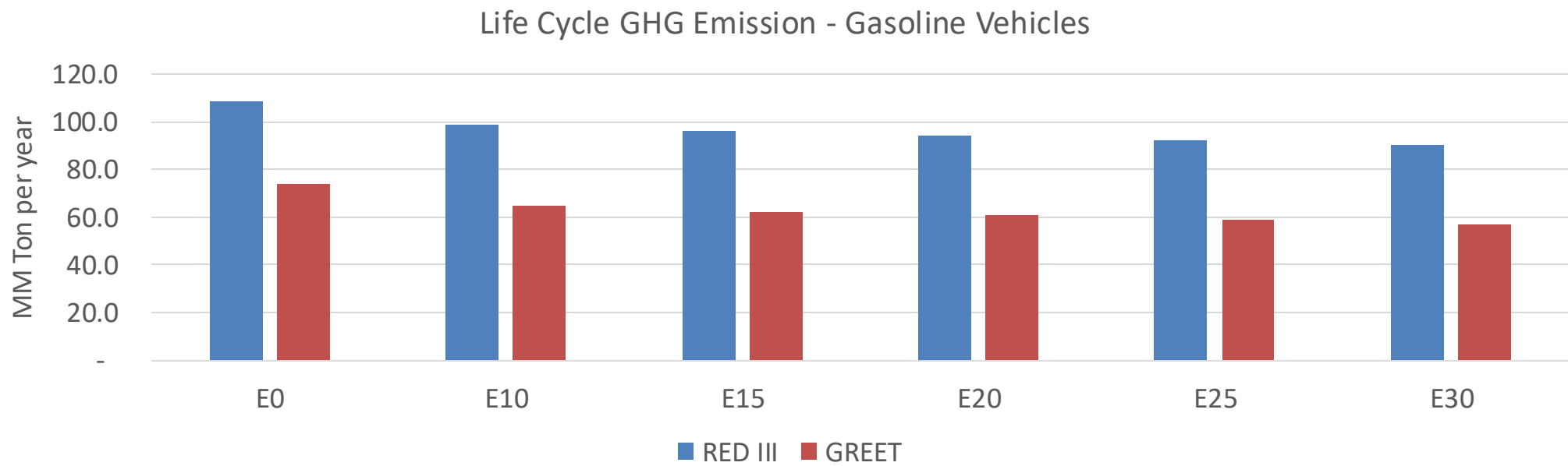
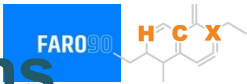
TIER III BIN 125 United States
Euro 6 Standard



United Kingdom – Vehicular Emissions

Vehicular Fleet Emissions (kg/day)	Eq E0	E5	E10	E15	E20	E25	E30	Reduction (%)	
								E10	E20
CO	2,592,464	2,347,600	2,102,737	1,893,191	1,678,226	1,511,697	1,337,359	-19%	-35%
CO2	126,780,565	123,520,493	120,441,537	118,020,028	116,821,771	116,113,706	113,973,396	-5%	-8%
VOC	464,487	425,388	386,288	353,587	320,373	294,742	260,863	-17%	-31%
NOx	176,401	132,301	123,481	116,381	109,757	102,454	94,728	-30%	-38%
SOx	1,449	1,339	1,230	1,137	1,048	952	850	-15%	-28%
NH3	46,556	46,095	45,634	45,851	46,366	46,729	47,031	-2%	0%
N2O	2,090	2,079	2,070	2,080	2,123	2,147	2,171	-1%	2%
CH4	93,659	93,659	93,659	95,064	97,125	98,904	100,590	0%	4%
Lead	-	-	-	-	-	-	-	0%	0%
Butadiene	2,440	2,325	2,211	2,131	2,058	2,003	1,924	-9%	-16%
Acetaldehyde	7,685	8,820	12,942	19,708	26,851	30,682	36,254	68%	249%
Formaldehyde	28,374	27,586	32,157	37,760	39,559	43,763	47,641	13%	39%
Benzene	5,291	5,084	4,818	4,741	4,702	4,497	4,352	-9%	-11%
PM2.5	7,836	6,958	6,080	5,280	4,483	3,638	2,757	-22%	-43%
PM10	16,450	14,606	12,763	11,083	9,411	7,637	5,787	-22%	-43%

United Kingdom – Gasoline Vehicle Fleet Life Cycle GHG Emissions



Country Emissions 2024 (MMT/año)	402.0
Target @ 2035 (MMT/year)	264.3
Delta	137.7

%Reduction vs E0	E0	E10	E15	E20	E25	E30
GREET 2024	0.0%	12.5%	16.5%	18.0%	20.3%	23.4%
RED II/III	0.0%	9.0%	11.3%	13.1%	14.9%	17.1%

% Target@2035	E0	E10	E15	E20	E25	E30
GREET 2024	0.0%	6.7%	8.9%	9.7%	10.9%	12.6%
RED II/III	0.0%	7.1%	8.9%	10.4%	11.7%	13.5%

Source: Greet, RED II/III, IPCC, climateactiontracker.org, F90